Ответы и официальные разборы

Том 9. Официальные оригиналы

Оригинальная вёрстка сохранена вместе с формулами, рисунками и авторскими пояснениями. В сборниках могут встречаться и условия, и ответы, и критерии оценивания.

Соответствие отдельной задаче проверяйте по олимпиаде, году, классу, номеру и варианту. Если этих данных недостаточно, документ не объявляется точным решением конкретной задачи.

Оглавление PDF содержит начала документов. Ссылки на источники и номера страниц находятся в каталоге библиотеки.

Shortlisted Problems with Solutions

56th International Mathematical Olympiad



Shortlisted Problems with Solutions

56th International Mathematical Olympiad

Chiang Mai, Thailand, 4–16



Note of Confidentiality

The shortlisted problems should be kept strictly confidential until IMO 2016.

Contributing Countries

The Organizing Committee and the Problem Selection Committee of IMO 2015 thank the following 53 countries for contributing 155 problem proposals:

Albania, Algeria, Armenia, Australia, Austria, Brazil, Bulgaria, Canada, Costa Rica, Croatia, Cyprus, Denmark, El Salvador, Estonia, Finland, France, Georgia, Germany, Greece, Hong Kong, Hungary, India, Iran, Ireland, Israel, Italy, Japan, Kazakhstan, Lithuania, Luxembourg, Montenegro, Morocco, Netherlands, Pakistan, Poland, Romania, Russia, Saudi Arabia, Serbia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Sweden, Turkey, Turkmenistan, Taiwan, Tanzania, Ukraine, United Kingdom, U.S.A., Uzbekistan

Problem Selection Committee



Dungjade Shiowattana, Ilya I. Bogdanov, Tirasan Khandhawit, Wittawat Kositwattanarerk, Géza Kós, Weerachai Neeranartvong, Nipun Pitimanaaree, Christian Reiher, Nat Sothanaphan, Warut Suksompong, Wuttisak Trongsiriwat, Wijit Yangjit

Assistants: Jirawat Anunrojwong, Pakawut Jiradilok

Problems

Algebra

A1. Suppose that a sequence $a_1, a_2, \ldots$ of positive real numbers satisfies

$$a_{k+1} \geqslant \frac{ka_k}{a_k^2 + (k-1)}$$

for every positive integer k. Prove that $a_1 + a_2 + \cdots + a_n \geqslant n$ for every $n \geqslant 2$.

(Serbia)

A2. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ with the property that

$$f\big(x - f(y)\big) = f\big(f(x)\big) - f(y) - 1$$

holds for all $x, y \in \mathbb{Z}$.

(Croatia)

A3. Let n be a fixed positive integer. Find the maximum possible value of

$$\sum_{1 \leqslant r < s \leqslant 2n} (s - r - n) x_r x_s \,,$$

where $-1 \leqslant x_i \leqslant 1$ for all $i = 1, 2, \ldots, 2n$.

(Austria)

A4. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ satisfying the equation

$$f\big(x + f(x+y)\big) + f(xy) = x + f(x+y) + yf(x)$$

for all real numbers x and y.

(Albania)

A5. Let $2\mathbb{Z} + 1$ denote the set of odd integers. Find all functions $f\colon \mathbb{Z} \to 2\mathbb{Z} + 1$ satisfying

$$f\big(x + f(x) + y\big) + f\big(x - f(x) - y\big) = f(x+y) + f(x-y)$$

for every $x, y \in \mathbb{Z}$.

(U.S.A.)

A6. Let n be a fixed integer with $n \geqslant 2$. We say that two polynomials P and Q with real coefficients are *block-similar* if for each $i \in \{1, 2, \ldots, n\}$ the sequences

$$P(2015i), P(2015i-1), \ldots, P(2015i-2014) \quad \text{and}$$
$$Q(2015i), Q(2015i-1), \ldots, Q(2015i-2014)$$

are permutations of each other.

(*a*) Prove that there exist distinct block-similar polynomials of degree $n+1$.

(*b*) Prove that there do not exist distinct block-similar polynomials of degree n.

(Canada)

Combinatorics

C1. In Lineland there are $n \geqslant 1$ towns, arranged along a road running from left to right. Each town has a *left bulldozer* (put to the left of the town and facing left) and a *right bulldozer* (put to the right of the town and facing right). The sizes of the $2n$ bulldozers are distinct. Every time when a right and a left bulldozer confront each other, the larger bulldozer pushes the smaller one off the road. On the other hand, the bulldozers are quite unprotected at their rears; so, if a bulldozer reaches the rear-end of another one, the first one pushes the second one off the road, regardless of their sizes.

Let A and B be two towns, with B being to the right of A. We say that town A can *sweep* town B *away* if the right bulldozer of A can move over to B pushing off all bulldozers it meets. Similarly, B can sweep A away if the left bulldozer of B can move to A pushing off all bulldozers of all towns on its way.

Prove that there is exactly one town which cannot be swept away by any other one.

(Estonia)

C2. Let $\mathcal{V}$ be a finite set of points in the plane. We say that $\mathcal{V}$ is *balanced* if for any two distinct points $A, B \in \mathcal{V}$, there exists a point $C \in \mathcal{V}$ such that $AC = BC$. We say that $\mathcal{V}$ is *center-free* if for any distinct points $A, B, C \in \mathcal{V}$, there does not exist a point $P \in \mathcal{V}$ such that $PA = PB = PC$.

(a) Show that for all $n \geqslant 3$, there exists a balanced set consisting of n points.

(b) For which $n \geqslant 3$ does there exist a balanced, center-free set consisting of n points?

(Netherlands)

C3. For a finite set A of positive integers, we call a partition of A into two disjoint nonempty subsets A_1 and A_2 *good* if the least common multiple of the elements in A_1 is equal to the greatest common divisor of the elements in A_2. Determine the minimum value of n such that there exists a set of n positive integers with exactly 2015 good partitions.

(Ukraine)

C4. Let n be a positive integer. Two players A and B play a game in which they take turns choosing positive integers $k \leqslant n$. The rules of the game are:

(i) A player cannot choose a number that has been chosen by either player on any previous turn.

(ii) A player cannot choose a number consecutive to any of those the player has already chosen on any previous turn.

(iii) The game is a draw if all numbers have been chosen; otherwise the player who cannot choose a number anymore loses the game.

The player A takes the first turn. Determine the outcome of the game, assuming that both players use optimal strategies.

(Finland)

C5. Consider an infinite sequence $a_1, a_2, \ldots$ of positive integers with $a_i \leqslant 2015$ for all $i \geqslant 1$. Suppose that for any two distinct indices i and j we have $i + a_i \neq j + a_j$.

Prove that there exist two positive integers b and N such that

$$\left| \sum_{i=m+1}^{n} (a_i - b) \right| \leqslant 1007^2$$

whenever $n > m \geqslant N$.

(Australia)

C6. Let S be a nonempty set of positive integers. We say that a positive integer n is *clean* if it has a unique representation as a sum of an odd number of distinct elements from S. Prove that there exist infinitely many positive integers that are not clean.

(U.S.A.)

C7. In a company of people some pairs are enemies. A group of people is called *unsociable* if the number of members in the group is odd and at least 3, and it is possible to arrange all its members around a round table so that every two neighbors are enemies. Given that there are at most 2015 unsociable groups, prove that it is possible to partition the company into 11 parts so that no two enemies are in the same part.

(Russia)

Geometry

G1. Let ABC be an acute triangle with orthocenter H. Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI. Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J. Prove that $IJ = AH$.

(Australia)

G2. Let ABC be a triangle inscribed into a circle Ω with center O. A circle Γ with center A meets the side BC at points D and E such that D lies between B and E. Moreover, let F and G be the common points of Γ and Ω. We assume that F lies on the arc AB of Ω not containing C, and G lies on the arc AC of Ω not containing B. The circumcircles of the triangles BDF and CEG meet the sides AB and AC again at K and L, respectively. Suppose that the lines FK and GL are distinct and intersect at X. Prove that the points A, X, and O are collinear.

(Greece)

G3. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C. A point D is chosen inside the triangle CBH so that CH bisects AD. Let P be the intersection point of the lines BD and CH. Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q. Prove that the lines CQ and AD meet on ω.

(Georgia)

G4. Let ABC be an acute triangle, and let M be the midpoint of AC. A circle ω passing through B and M meets the sides AB and BC again at P and Q, respectively. Let T be the point such that the quadrilateral $BPTQ$ is a parallelogram. Suppose that T lies on the circumcircle of the triangle ABC. Determine all possible values of BT/BM.

(Russia)

G5. Let ABC be a triangle with $CA \neq CB$. Let D, F, and G be the midpoints of the sides AB, AC, and BC, respectively. A circle Γ passing through C and tangent to AB at D meets the segments AF and BG at H and I, respectively. The points H' and I' are symmetric to H and I about F and G, respectively. The line $H'I'$ meets CD and FG at Q and M, respectively. The line CM meets Γ again at P. Prove that $CQ = QP$.

(El Salvador)

G6. Let ABC be an acute triangle with $AB > AC$, and let Γ be its circumcircle. Let H, M, and F be the orthocenter of the triangle, the midpoint of BC, and the foot of the altitude from A, respectively. Let Q and K be the two points on Γ that satisfy $\angle AQH = 90^\circ$ and $\angle QKH = 90^\circ$. Prove that the circumcircles of the triangles KQH and KFM are tangent to each other.

(Ukraine)

G7. Let $ABCD$ be a convex quadrilateral, and let P, Q, R, and S be points on the sides AB, BC, CD, and DA, respectively. Let the line segments PR and QS meet at O. Suppose that each of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$ has an incircle. Prove that the lines AC, PQ, and RS are either concurrent or parallel to each other.

(Bulgaria)

G8. A *triangulation* of a convex polygon Π is a partitioning of Π into triangles by diagonals having no common points other than the vertices of the polygon. We say that a triangulation is a *Thaiangulation* if all triangles in it have the same area.

Prove that any two different Thaiangulations of a convex polygon Π differ by exactly two triangles. (In other words, prove that it is possible to replace one pair of triangles in the first Thaiangulation with a different pair of triangles so as to obtain the second Thaiangulation.)

(Bulgaria)

Number Theory

N1. Determine all positive integers M for which the sequence $a_0, a_1, a_2, \ldots$, defined by $a_0 = \frac{2M+1}{2}$ and $a_{k+1} = a_k \lfloor a_k \rfloor$ for $k = 0, 1, 2, \ldots$, contains at least one integer term.

(Luxembourg)

N2. Let a and b be positive integers such that $a!b!$ is a multiple of $a! + b!$. Prove that $3a \geqslant 2b + 2$.

(United Kingdom)

N3. Let m and n be positive integers such that $m > n$. Define $x_k = (m+k)/(n+k)$ for $k = 1, 2, \ldots, n+1$. Prove that if all the numbers $x_1, x_2, \ldots, x_{n+1}$ are integers, then $x_1 x_2 \cdots x_{n+1} - 1$ is divisible by an odd prime.

(Austria)

N4. Suppose that $a_0, a_1, \ldots$ and $b_0, b_1, \ldots$ are two sequences of positive integers satisfying $a_0, b_0 \geqslant 2$ and

$$a_{n+1} = \gcd(a_n, b_n) + 1, \qquad b_{n+1} = \operatorname{lcm}(a_n, b_n) - 1$$

for all $n \geqslant 0$. Prove that the sequence (a_n) is eventually periodic; in other words, there exist integers $N \geqslant 0$ and $t > 0$ such that $a_{n+t} = a_n$ for all $n \geqslant N$.

(France)

N5. Determine all triples (a, b, c) of positive integers for which $ab - c$, $bc - a$, and $ca - b$ are powers of 2.

Explanation: A power of 2 is an integer of the form 2^n, where n denotes some nonnegative integer.

(Serbia)

N6. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Consider a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$. For any $m, n \in \mathbb{Z}_{>0}$ we write $f^n(m) = \underbrace{f(f(\ldots f}_{n}(m)\ldots))$. Suppose that f has the following two properties:

(*i*) If $m, n \in \mathbb{Z}_{>0}$, then $\dfrac{f^n(m) - m}{n} \in \mathbb{Z}_{>0}$;

(*ii*) The set $\mathbb{Z}_{>0} \setminus \{f(n) \mid n \in \mathbb{Z}_{>0}\}$ is finite.

Prove that the sequence $f(1) - 1, f(2) - 2, f(3) - 3, \ldots$ is periodic.

(Singapore)

N7. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. For any positive integer k, a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ is called *k-good* if $\gcd\big(f(m) + n, f(n) + m\big) \leqslant k$ for all $m \neq n$. Find all k such that there exists a k-good function.

(Canada)

N8. For every positive integer n with prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$, define

$$\mho(n) = \sum_{i\colon p_i > 10^{100}} \alpha_i.$$

That is, $\mho(n)$ is the number of prime factors of n greater than 10^{100}, counted with multiplicity.

Find all strictly increasing functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$\mho\big(f(a) - f(b)\big) \leqslant \mho(a - b) \qquad \text{for all integers } a \text{ and } b \text{ with } a > b.$$

(Brazil)

Solutions

Algebra

A1. Suppose that a sequence $a_1, a_2, \ldots$ of positive real numbers satisfies

$$a_{k+1} \geqslant \frac{ka_k}{a_k^2 + (k-1)} \tag{1}$$

for every positive integer k. Prove that $a_1 + a_2 + \cdots + a_n \geqslant n$ for every $n \geqslant 2$.

(Serbia)

Solution. From the constraint (1), it can be seen that

$$\frac{k}{a_{k+1}} \leqslant \frac{a_k^2 + (k-1)}{a_k} = a_k + \frac{k-1}{a_k},$$

and so

$$a_k \geqslant \frac{k}{a_{k+1}} - \frac{k-1}{a_k}.$$

Summing up the above inequality for $k = 1, \ldots, m$, we obtain

$$a_1 + a_2 + \cdots + a_m \geqslant \left(\frac{1}{a_2} - \frac{0}{a_1}\right) + \left(\frac{2}{a_3} - \frac{1}{a_2}\right) + \cdots + \left(\frac{m}{a_{m+1}} - \frac{m-1}{a_m}\right) = \frac{m}{a_{m+1}}. \tag{2}$$

Now we prove the problem statement by induction on n. The case $n = 2$ can be done by applying (1) to $k = 1$:

$$a_1 + a_2 \geqslant a_1 + \frac{1}{a_1} \geqslant 2.$$

For the induction step, assume that the statement is true for some $n \geqslant 2$. If $a_{n+1} \geqslant 1$, then the induction hypothesis yields

$$\big(a_1 + \cdots + a_n\big) + a_{n+1} \geqslant n + 1. \tag{3}$$

Otherwise, if $a_{n+1} < 1$ then apply (2) as

$$\big(a_1 + \cdots + a_n\big) + a_{n+1} \geqslant \frac{n}{a_{n+1}} + a_{n+1} = \frac{n-1}{a_{n+1}} + \left(\frac{1}{a_{n+1}} + a_{n+1}\right) > (n-1) + 2.$$

That completes the solution.

Comment 1. It can be seen easily that having equality in the statement requires $a_1 = a_2 = 1$ in the base case $n = 2$, and $a_{n+1} = 1$ in (3). So the equality $a_1 + \cdots + a_n = n$ is possible only in the trivial case $a_1 = \cdots = a_n = 1$.

Comment 2. After obtaining (2), there are many ways to complete the solution. We outline three such possibilities.

- With defining $s_n = a_1 + \cdots + a_n$, the induction step can be replaced by

$$s_{n+1} = s_n + a_{n+1} \geqslant s_n + \frac{n}{s_n} \geqslant n + 1,$$

because the function $x \mapsto x + \dfrac{n}{x}$ increases on $[n, \infty)$.

- By applying the AM–GM inequality to the numbers $a_1 + \cdots + a_k$ and ka_{k+1}, we can conclude

$$a_1 + \cdots + a_k + ka_{k+1} \geqslant 2k$$

  and sum it up for $k = 1, \ldots, n-1$.

- We can derive the symmetric estimate

$$\sum_{1 \leqslant i < j \leqslant n} a_i a_j = \sum_{j=2}^{n} (a_1 + \cdots + a_{j-1}) a_j \geqslant \sum_{j=2}^{n} (j-1) = \frac{n(n-1)}{2}$$

  and combine it with the AM–QM inequality.

A2. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ with the property that

$$f\big(x - f(y)\big) = f\big(f(x)\big) - f(y) - 1 \tag{1}$$

holds for all $x, y \in \mathbb{Z}$.

(Croatia)

Answer. There are two such functions, namely the constant function $x \mapsto -1$ and the successor function $x \mapsto x + 1$.

Solution 1. It is immediately checked that both functions mentioned in the answer are as desired.

Now let f denote any function satisfying (1) for all $x, y \in \mathbb{Z}$. Substituting $x = 0$ and $y = f(0)$ into (1) we learn that the number $z = -f\big(f(0)\big)$ satisfies $f(z) = -1$. So by plugging $y = z$ into (1) we deduce that

$$f(x + 1) = f\big(f(x)\big) \tag{2}$$

holds for all $x \in \mathbb{Z}$. Thereby (1) simplifies to

$$f\big(x - f(y)\big) = f(x + 1) - f(y) - 1\,. \tag{3}$$

We now work towards showing that f is linear by contemplating the difference $f(x+1)-f(x)$ for any $x \in \mathbb{Z}$. By applying (3) with $y = x$ and (2) in this order, we obtain

$$f(x + 1) - f(x) = f\big(x - f(x)\big) + 1 = f\big(f(x - 1 - f(x))\big) + 1\,.$$

Since (3) shows $f\big(x - 1 - f(x)\big) = f(x) - f(x) - 1 = -1$, this simplifies to

$$f(x + 1) = f(x) + A\,,$$

where $A = f(-1) + 1$ is some absolute constant.

Now a standard induction in both directions reveals that f is indeed linear and that in fact we have $f(x) = Ax + B$ for all $x \in \mathbb{Z}$, where $B = f(0)$. Substituting this into (2) we obtain that

$$Ax + (A + B) = A^2x + (AB + B)$$

holds for all $x \in \mathbb{Z}$; applying this to $x = 0$ and $x = 1$ we infer $A + B = AB + B$ and $A^2 = A$. The second equation leads to $A = 0$ or $A = 1$. In case $A = 1$, the first equation gives $B = 1$, meaning that f has to be the successor function. If $A = 0$, then f is constant and (1) shows that its constant value has to be -1. Thereby the solution is complete.

Comment. After (2) and (3) have been obtained, there are several other ways to combine them so as to obtain linearity properties of f. For instance, using (2) thrice in a row and then (3) with $x = f(y)$ one may deduce that

$$f(y + 2) = f\big(f(y + 1)\big) = f\big(f\big(f(y)\big)\big) = f\big(f(y) + 1\big) = f(y) + f(0) + 1$$

holds for all $y \in \mathbb{Z}$. It follows that f behaves linearly on the even numbers and on the odd numbers separately, and moreover that the slopes of these two linear functions coincide. From this point, one may complete the solution with some straightforward case analysis.

A different approach using the equations (2) and (3) will be presented in Solution 2. To show that it is also possible to start in a completely different way, we will also present a third solution that avoids these equations entirely.

Solution 2. We commence by deriving (2) and (3) as in the first solution. Now provided that f is injective, (2) tells us that f is the successor function. Thus we may assume from now on that f is not injective, i.e., that there are two integers $a > b$ with $f(a) = f(b)$. A straightforward induction using (2) in the induction step reveals that we have $f(a+n) = f(b+n)$ for all nonnegative integers n. Consequently, the sequence $\gamma_n = f(b+n)$ is periodic and thus in particular bounded, which means that the numbers

$$\varphi = \min_{n \geqslant 0} \gamma_n \quad \text{and} \quad \psi = \max_{n \geqslant 0} \gamma_n$$

exist.

Let us pick any integer y with $f(y) = \varphi$ and then an integer $x \geqslant a$ with $f\big(x - f(y)\big) = \varphi$. Due to the definition of φ and (3) we have

$$\varphi \leqslant f(x+1) = f\big(x - f(y)\big) + f(y) + 1 = 2\varphi + 1\,,$$

whence $\varphi \geqslant -1$. The same reasoning applied to ψ yields $\psi \leqslant -1$. Since $\varphi \leqslant \psi$ holds trivially, it follows that $\varphi = \psi = -1$, or in other words that we have $f(t) = -1$ for all integers $t \geqslant a$.

Finally, if any integer y is given, we may find an integer x which is so large that $x + 1 \geqslant a$ and $x - f(y) \geqslant a$ hold. Due to (3) and the result from the previous paragraph we get

$$f(y) = f(x+1) - f\big(x - f(y)\big) - 1 = (-1) - (-1) - 1 = -1\,.$$

Thereby the problem is solved.

Solution 3. Set $d = f(0)$. By plugging $x = f(y)$ into (1) we obtain

$$f^3(y) = f(y) + d + 1 \tag{4}$$

for all $y \in \mathbb{Z}$, where the left-hand side abbreviates $f\big(f(f(y))\big)$. When we replace x in (1) by $f(x)$ we obtain $f\big(f(x) - f(y)\big) = f^3(x) - f(y) - 1$ and as a consequence of (4) this simplifies to

$$f\big(f(x) - f(y)\big) = f(x) - f(y) + d\,. \tag{5}$$

Now we consider the set

$$E = \{f(x) - d \,|\, x \in \mathbb{Z}\}\,.$$

Given two integers a and b from E, we may pick some integers x and y with $f(x) = a + d$ and $f(y) = b + d$; now (5) tells us that $f(a-b) = (a-b) + d$, which means that $a - b$ itself exemplifies $a - b \in E$. Thus,

$$E \text{ is closed under taking differences.} \tag{6}$$

Also, the definitions of d and E yield $0 \in E$. If $E = \{0\}$, then f is a constant function and (1) implies that the only value attained by f is indeed -1.

So let us henceforth suppose that E contains some number besides zero. It is known that in this case (6) entails E to be the set of all integer multiples of some positive integer k. Indeed, this holds for

$$k = \min\big\{|x| \,\big|\, x \in E \text{ and } x \neq 0\big\}\,,$$

as one may verify by an argument based on division with remainder.

Thus we have

$$\{f(x) \,|\, x \in \mathbb{Z}\} = \{k \cdot t + d \,|\, t \in \mathbb{Z}\}\,. \tag{7}$$

Due to (5) and (7) we get

$$f(k \cdot t) = k \cdot t + d$$

for all $t \in \mathbb{Z}$, whence in particular $f(k) = k + d$. So by comparing the results of substituting $y = 0$ and $y = k$ into (1) we learn that

$$f(z + k) = f(z) + k \tag{8}$$

holds for all integers z. In plain English, this means that on any residue class modulo k the function f is linear with slope 1.

Now by (7) the set of all values attained by f is such a residue class. Hence, there exists an absolute constant c such that $f\big(f(x)\big) = f(x) + c$ holds for all $x \in \mathbb{Z}$. Thereby (1) simplifies to

$$f\big(x - f(y)\big) = f(x) - f(y) + c - 1\,. \tag{9}$$

On the other hand, considering (1) modulo k we obtain $d \equiv -1 \pmod{k}$ because of (7). So by (7) again, f attains the value -1.

Thus we may apply (9) to some integer y with $f(y) = -1$, which gives $f(x + 1) = f(x) + c$. So f is a linear function with slope c. Hence, (8) leads to $c = 1$, wherefore there is an absolute constant d' with $f(x) = x + d'$ for all $x \in \mathbb{Z}$. Using this for $x = 0$ we obtain $d' = d$ and finally (4) discloses $d = 1$, meaning that f is indeed the successor function.

A3. Let n be a fixed positive integer. Find the maximum possible value of

$$\sum_{1\leqslant r<s\leqslant 2n}(s-r-n)x_rx_s\,,$$

where $-1\leqslant x_i\leqslant 1$ for all $i=1,2,\ldots,2n$.

(Austria)

Answer. $n(n-1)$.

Solution 1. Let Z be the expression to be maximized. Since this expression is linear in every variable x_i and $-1\leqslant x_i\leqslant 1$, the maximum of Z will be achieved when $x_i=-1$ or 1. Therefore, it suffices to consider only the case when $x_i\in\{-1,1\}$ for all $i=1,2,\ldots,2n$.

For $i=1,2,\ldots,2n$, we introduce auxiliary variables

$$y_i=\sum_{r=1}^{i}x_r-\sum_{r=i+1}^{2n}x_r\,.$$

Taking squares of both sides, we have

$$\begin{aligned}y_i^2&=\sum_{r=1}^{2n}x_r^2+\sum_{r<s\leqslant i}2x_rx_s+\sum_{i<r<s}2x_rx_s-\sum_{r\leqslant i<s}2x_rx_s\\&=2n+\sum_{r<s\leqslant i}2x_rx_s+\sum_{i<r<s}2x_rx_s-\sum_{r\leqslant i<s}2x_rx_s\,,\end{aligned}\tag{1}$$

where the last equality follows from the fact that $x_r\in\{-1,1\}$. Notice that for every $r<s$, the coefficient of x_rx_s in (1) is 2 for each $i=1,\ldots,r-1,s,\ldots,2n$, and this coefficient is -2 for each $i=r,\ldots,s-1$. This implies that the coefficient of x_rx_s in $\sum_{i=1}^{2n}y_i^2$ is $2(2n-s+r)-2(s-r)=4(n-s+r)$. Therefore, summing (1) for $i=1,2,\ldots,2n$ yields

$$\sum_{i=1}^{2n}y_i^2=4n^2+\sum_{1\leqslant r<s\leqslant 2n}4(n-s+r)x_rx_s=4n^2-4Z.\tag{2}$$

Hence, it suffices to find the minimum of the left-hand side.

Since $x_r\in\{-1,1\}$, we see that y_i is an even integer. In addition, $y_i-y_{i-1}=2x_i=\pm 2$, and so y_{i-1} and y_i are consecutive even integers for every $i=2,3,\ldots,2n$. It follows that $y_{i-1}^2+y_i^2\geqslant 4$, which implies

$$\sum_{i=1}^{2n}y_i^2=\sum_{j=1}^{n}\left(y_{2j-1}^2+y_{2j}^2\right)\geqslant 4n.\tag{3}$$

Combining (2) and (3), we get

$$4n\leqslant\sum_{i=1}^{2n}y_i^2=4n^2-4Z.\tag{4}$$

Hence, $Z\leqslant n(n-1)$.

If we set $x_i=1$ for odd indices i and $x_i=-1$ for even indices i, then we obtain equality in (3) (and thus in (4)). Therefore, the maximum possible value of Z is $n(n-1)$, as desired.

Comment 1. $Z=n(n-1)$ can be achieved by several other examples. In particular, x_i needs not be ± 1. For instance, setting $x_i=(-1)^i$ for all $2\leqslant i\leqslant 2n$, we find that the coefficient of x_1 in Z is 0. Therefore, x_1 can be chosen arbitrarily in the interval $[-1,1]$.

Nevertheless, if $x_i\in\{-1,1\}$ for all $i=1,2,\ldots,2n$, then the equality $Z=n(n-1)$ holds only when $(y_1,y_2,\ldots,y_{2n})=(0,\pm 2,0,\pm 2,\ldots,0,\pm 2)$ or $(\pm 2,0,\pm 2,0,\ldots,\pm 2,0)$. In each case, we can reconstruct x_i accordingly. The sum $\sum_{i=1}^{2n}x_i$ in the optimal cases needs not be 0, but it must equal 0 or ± 2.

Comment 2. Several variations in setting up the auxiliary variables are possible. For instance, one may let $x_{2n+i} = -x_i$ and $y'_i = x_i + x_{i+1} + \cdots + x_{i+n-1}$ for any $1 \leqslant i \leqslant 2n$. Similarly to Solution 1, we obtain $Y := y'^2_1 + y'^2_2 + \cdots + y'^2_{2n} = 2n^2 - 2Z$. Then, it suffices to show that $Y \geqslant 2n$. If n is odd, then each y'_i is odd, and so $y'^2_i \geqslant 1$. If n is even, then each y'_i is even. We can check that at least one of y'_i, y'_{i+1}, y'_{n+i}, and y'_{n+i+1} is nonzero, so that $y'^2_i + y'^2_{i+1} + y'^2_{n+i} + y'^2_{n+i+1} \geqslant 4$; summing these up for $i = 1, 3, \ldots, n-1$ yields $Y \geqslant 2n$.

Solution 2. We present a different method of obtaining the bound $Z \leqslant n(n-1)$. As in the previous solution, we reduce the problem to the case $x_i \in \{-1, 1\}$. For brevity, we use the notation $[2n] = \{1, 2, \ldots, 2n\}$.

Consider any $x_1, x_2, \ldots, x_{2n} \in \{-1, 1\}$. Let

$$A = \{i \in [2n] : \ x_i = 1\} \quad \text{and} \quad B = \{i \in [2n] : \ x_i = -1\}\,.$$

For any subsets X and Y of $[2n]$ we define

$$e(X, Y) = \sum_{r<s,\ r\in X,\ s\in Y} (s - r - n)\,.$$

One may observe that

$$e(A,A)+e(A,B)+e(B,A)+e(B,B) = e([2n],[2n]) = \sum_{1\leqslant r<s\leqslant 2n} (s-r-n) = -\frac{(n-1)n(2n-1)}{3}\,.$$

Therefore, we have

$$Z = e(A,A) - e(A,B) - e(B,A) + e(B,B) = 2\big(e(A,A) + e(B,B)\big) + \frac{(n-1)n(2n-1)}{3}\,. \quad (5)$$

Thus, we need to maximize $e(A, A) + e(B, B)$, where A and B form a partition of $[2n]$.

Due to the symmetry, we may assume that $|A| = n - p$ and $|B| = n + p$, where $0 \leqslant p \leqslant n$. From now on, we fix the value of p and find an upper bound for Z in terms of n and p.

Let $a_1 < a_2 < \cdots < a_{n-p}$ and $b_1 < b_2 < \cdots < b_{n+p}$ list all elements of A and B, respectively. Then

$$e(A,A) = \sum_{1\leqslant i<j\leqslant n-p} (a_j - a_i - n) = \sum_{i=1}^{n-p} (2i - 1 - n + p)a_i - \binom{n-p}{2} \cdot n \quad (6)$$

and similarly

$$e(B,B) = \sum_{i=1}^{n+p} (2i - 1 - n - p)b_i - \binom{n+p}{2} \cdot n\,. \quad (7)$$

Thus, now it suffices to maximize the value of

$$M = \sum_{i=1}^{n-p} (2i - 1 - n + p)a_i + \sum_{i=1}^{n+p} (2i - 1 - n - p)b_i\,. \quad (8)$$

In order to get an upper bound, we will apply the rearrangement inequality to the sequence $a_1, a_2, \ldots, a_{n-p}, b_1, b_2, \ldots, b_{n+p}$ (which is a permutation of $1, 2, \ldots, 2n$), together with the sequence of coefficients of these numbers in (8). The coefficients of a_i form the sequence

$$n - p - 1, n - p - 3, \ldots, 1 - n + p\,,$$

and those of b_i form the sequence

$$n + p - 1, n + p - 3, \ldots, 1 - n - p\,.$$

Altogether, these coefficients are, in descending order:

- $n+p+1-2i$, for $i=1,2,\ldots,p$;
- $n-p+1-2i$, counted twice, for $i=1,2,\ldots,n-p$; and
- $-(n+p+1-2i)$, for $i=p,p-1,\ldots,1$.

Thus, the rearrangement inequality yields

$$\begin{aligned} M \leqslant \sum_{i=1}^{p}(n+p+1-2i)(2n+1-i) \\ + \sum_{i=1}^{n-p}(n-p+1-2i)\big((2n+2-p-2i)+(2n+1-p-2i)\big) \\ - \sum_{i=1}^{p}(n+p+1-2i)i\,. \end{aligned} \tag{9}$$

Finally, combining the information from (5), (6), (7), and (9), we obtain

$$\begin{aligned} Z \leqslant \frac{(n-1)n(2n-1)}{3} - 2n\left(\binom{n-p}{2}+\binom{n+p}{2}\right) \\ + 2\sum_{i=1}^{p}(n+p+1-2i)(2n+1-2i) + 2\sum_{i=1}^{n-p}(n-p+1-2i)(4n-2p+3-4i)\,, \end{aligned}$$

which can be simplified to

$$Z \leqslant n(n-1) - \frac{2}{3}p(p-1)(p+1)\,.$$

Since p is a nonnegative integer, this yields $Z \leqslant n(n-1)$.

A4. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ satisfying the equation

$$f\big(x + f(x+y)\big) + f(xy) = x + f(x+y) + yf(x) \tag{1}$$

for all real numbers x and y.

(Albania)

Answer. There are two such functions, namely the identity function and $x \mapsto 2 - x$.

Solution. Clearly, each of the functions $x \mapsto x$ and $x \mapsto 2 - x$ satisfies (1). It suffices now to show that they are the only solutions to the problem.

Suppose that f is any function satisfying (1). Then setting $y = 1$ in (1), we obtain

$$f\big(x + f(x+1)\big) = x + f(x+1); \tag{2}$$

in other words, $x + f(x+1)$ is a fixed point of f for every $x \in \mathbb{R}$.

We distinguish two cases regarding the value of $f(0)$.

Case 1. $f(0) \neq 0$.

By letting $x = 0$ in (1), we have

$$f\big(f(y)\big) + f(0) = f(y) + yf(0).$$

So, if y_0 is a fixed point of f, then substituting $y = y_0$ in the above equation we get $y_0 = 1$. Thus, it follows from (2) that $x + f(x+1) = 1$ for all $x \in \mathbb{R}$. That is, $f(x) = 2 - x$ for all $x \in \mathbb{R}$.

Case 2. $f(0) = 0$.

By letting $y = 0$ and replacing x by $x + 1$ in (1), we obtain

$$f\big(x + f(x+1) + 1\big) = x + f(x+1) + 1. \tag{3}$$

From (1), the substitution $x = 1$ yields

$$f\big(1 + f(y+1)\big) + f(y) = 1 + f(y+1) + yf(1). \tag{4}$$

By plugging $x = -1$ into (2), we see that $f(-1) = -1$. We then plug $y = -1$ into (4) and deduce that $f(1) = 1$. Hence, (4) reduces to

$$f\big(1 + f(y+1)\big) + f(y) = 1 + f(y+1) + y. \tag{5}$$

Accordingly, if both y_0 and $y_0 + 1$ are fixed points of f, then so is $y_0 + 2$. Thus, it follows from (2) and (3) that $x + f(x+1) + 2$ is a fixed point of f for every $x \in \mathbb{R}$; i.e.,

$$f\big(x + f(x+1) + 2\big) = x + f(x+1) + 2.$$

Replacing x by $x - 2$ simplifies the above equation to

$$f\big(x + f(x-1)\big) = x + f(x-1).$$

On the other hand, we set $y = -1$ in (1) and get

$$f\big(x + f(x-1)\big) = x + f(x-1) - f(x) - f(-x).$$

Therefore, $f(-x) = -f(x)$ for all $x \in \mathbb{R}$.

Finally, we substitute (x, y) by $(-1, -y)$ in (1) and use the fact that $f(-1) = -1$ to get

$$f\big(-1 + f(-y-1)\big) + f(y) = -1 + f(-y-1) + y.$$

Since f is an odd function, the above equation becomes

$$-f\big(1 + f(y+1)\big) + f(y) = -1 - f(y+1) + y.$$

By adding this equation to (5), we conclude that $f(y) = y$ for all $y \in \mathbb{R}$.

A5. Let $2\mathbb{Z}+1$ denote the set of odd integers. Find all functions $f\colon \mathbb{Z}\to 2\mathbb{Z}+1$ satisfying

$$f\big(x+f(x)+y\big)+f\big(x-f(x)-y\big)=f(x+y)+f(x-y) \tag{1}$$

for every $x,y\in\mathbb{Z}$.

(U.S.A.)

Answer. Fix an odd positive integer d, an integer k, and odd integers $\ell_0,\ell_1,\ldots,\ell_{d-1}$. Then the function defined as

$$f(md+i)=2kmd+\ell_i d \qquad (m\in\mathbb{Z},\quad i=0,1,\ldots,d-1)$$

satisfies the problem requirements, and these are all such functions.

Solution. Throughout the solution, all functions are assumed to map integers to integers.

For any function g and any nonzero integer t, define

$$\Delta_t g(x)=g(x+t)-g(x).$$

For any nonzero integers a and b, notice that $\Delta_a\Delta_b g=\Delta_b\Delta_a g$. Moreover, if $\Delta_a g=0$ and $\Delta_b g=0$, then $\Delta_{a+b}g=0$ and $\Delta_{at}g=0$ for all nonzero integers t. We say that g is *t-quasi-periodic* if $\Delta_t g$ is a constant function (in other words, if $\Delta_1\Delta_t g=0$, or $\Delta_1 g$ is t-periodic). In this case, we call t a *quasi-period* of g. We say that g is *quasi-periodic* if it is t-quasi-periodic for some nonzero integer t.

Notice that a quasi-period of g is a period of $\Delta_1 g$. So if g is quasi-periodic, then its minimal positive quasi-period t divides all its quasi-periods.

We now assume that f satisfies (1). First, by setting $a=x+y$, the problem condition can be rewritten as

$$\Delta_{f(x)}f(a)=\Delta_{f(x)}f\big(2x-a-f(x)\big) \qquad \text{for all } x,a\in\mathbb{Z}. \tag{2}$$

Let b be an arbitrary integer and let k be an arbitrary positive integer. Applying (2) when a is substituted by $b, b+f(x),\ldots,b+(k-1)f(x)$ and summing up all these equations, we get

$$\Delta_{kf(x)}f(b)=\Delta_{kf(x)}f\big(2x-b-kf(x)\big).$$

Notice that a similar argument works when k is negative, so that

$$\Delta_M f(b)=\Delta_M f(2x-b-M) \qquad \text{for any nonzero integer } M \text{ such that } f(x)\mid M. \tag{3}$$

We now prove two lemmas.

Lemma 1. For any distinct integers x and y, the function $\Delta_{\operatorname{lcm}(f(x),f(y))}f$ is $2(y-x)$-periodic.

Proof. Denote $L=\operatorname{lcm}\big(f(x),f(y)\big)$. Applying (3) twice, we obtain

$$\Delta_L f(b)=\Delta_L f(2x-b-L)=\Delta_L f\big(2y-(b+2(y-x))-L\big)=\Delta_L f\big(b+2(y-x)\big).$$

Thus, the function $\Delta_L f$ is $2(y-x)$-periodic, as required. □

Lemma 2. Let g be a function. If t and s are nonzero integers such that $\Delta_{ts}g=0$ and $\Delta_t\Delta_t g=0$, then $\Delta_t g=0$.

Proof. Assume, without loss of generality, that s is positive. Let a be an arbitrary integer. Since $\Delta_t\Delta_t g=0$, we have

$$\Delta_t g(a)=\Delta_t g(a+t)=\cdots=\Delta_t g\big(a+(s-1)t\big).$$

The sum of these s equal numbers is $\Delta_{ts}g(a)=0$, so each of them is zero, as required. □

We now return to the solution.

Step 1. We prove that f is quasi-periodic.

Let $Q = \operatorname{lcm}\big(f(0), f(1)\big)$. Applying Lemma 1, we get that the function $g = \Delta_Q f$ is 2-periodic. In other words, the values of g are constant on even numbers and on odd numbers separately. Moreover, setting $M = Q$ and $x = b = 0$ in (3), we get $g(0) = g(-Q)$. Since 0 and $-Q$ have different parities, the value of g at even numbers is the same as that at odd numbers. Thus, g is constant, which means that Q is a quasi-period of f.

Step 2. Denote the minimal positive quasi-period of f by T. We prove that $T \mid f(x)$ for all integers x.

Since an odd number Q is a quasi-period of f, the number T is also odd. Now suppose, to the contrary, that there exist an odd prime p, a positive integer α, and an integer u such that $p^\alpha \mid T$ but $p^\alpha \nmid f(u)$. Setting $x = u$ and $y = 0$ in (1), we have $2f(u) = f\big(u+f(u)\big)+f\big(u-f(u)\big)$, so p^α does not divide the value of f at one of the points $u+f(u)$ or $u-f(u)$. Denote this point by v.

Let $L = \operatorname{lcm}\big(f(u), f(v)\big)$. Since $|u - v| = f(u)$, from Lemma 1 we get $\Delta_{2f(u)}\Delta_L f = 0$. Hence the function $\Delta_L f$ is $2f(u)$-periodic as well as T-periodic, so it is $\gcd\big(T, 2f(u)\big)$-periodic, or $\Delta_{\gcd(T,2f(u))}\Delta_L f = 0$. Similarly, observe that the function $\Delta_{\gcd(T,2f(u))}f$ is L-periodic as well as T-periodic, so we may conclude that $\Delta_{\gcd(T,L)}\Delta_{\gcd(T,2f(u))}f = 0$. Since $p^\alpha \nmid L$, both $\gcd\big(T, 2f(u)\big)$ and $\gcd(T, L)$ divide T/p. We thus obtain $\Delta_{T/p}\Delta_{T/p}f = 0$, which yields

$$\Delta_{T/p}\Delta_{T/p}\Delta_1 f = 0.$$

Since $\Delta_T\Delta_1 f = 0$, we can apply Lemma 2 to the function $\Delta_1 f$, obtaining $\Delta_{T/p}\Delta_1 f = 0$. However, this means that f is (T/p)-quasi-periodic, contradicting the minimality of T. Our claim is proved.

Step 3. We describe all functions f.

Let d be the greatest common divisor of all values of f. Then d is odd. By Step 2, d is a quasi-period of f, so that $\Delta_d f$ is constant. Since the value of $\Delta_d f$ is even and divisible by d, we may denote this constant by $2dk$, where k is an integer. Next, for all $i = 0, 1, \ldots, d-1$, define $\ell_i = f(i)/d$; notice that ℓ_i is odd. Then

$$f(md + i) = \Delta_{md} f(i) + f(i) = 2kmd + \ell_i d \qquad \text{for all } m \in \mathbb{Z} \quad \text{and } i = 0, 1, \ldots, d-1.$$

This shows that all functions satisfying (1) are listed in the answer.

It remains to check that all such functions indeed satisfy (1). This is equivalent to checking (2), which is true because for every integer x, the value of $f(x)$ is divisible by d, so that $\Delta_{f(x)} f$ is constant.

Comment. After obtaining Lemmas 1 and 2, it is possible to complete the steps in a different order. Here we sketch an alternative approach.

For any function g and any nonzero integer t, we say that g is *t-pseudo-periodic* if $\Delta_t\Delta_t g = 0$. In this case, we call t a *pseudo-period* of g, and we say that g is *pseudo-periodic*.

Let us first prove a basic property: if a function g is pseudo-periodic, then its minimal positive pseudo-period divides all its pseudo-periods. To establish this, it suffices to show that if t and s are pseudo-periods of g with $t \neq s$, then so is $t - s$. Indeed, suppose that $\Delta_t\Delta_t g = \Delta_s\Delta_s g = 0$. Then $\Delta_t\Delta_t\Delta_s g = \Delta_{ts}\Delta_s g = 0$, so that $\Delta_t\Delta_s g = 0$ by Lemma 2. Taking differences, we obtain $\Delta_t\Delta_{t-s} g = \Delta_s\Delta_{t-s} g = 0$, and thus $\Delta_{t-s}\Delta_{t-s} g = 0$.

Now let f satisfy the problem condition. We will show that f is pseudo-periodic. When this is done, we will let T' be the minimal pseudo-period of f, and show that T' divides $2f(x)$ for every integer x, using arguments similar to Step 2 of the solution. Then we will come back to Step 1 by showing that T' is also a quasi-period of f.

First, Lemma 1 yields that $\Delta_{2(y-x)}\Delta_{\mathrm{lcm}(f(x),f(y))}f = 0$ for every distinct integers x and y. Hence f is pseudo-periodic with pseudo-period $L_{x,y} = \mathrm{lcm}\big(2(y-x), f(x), f(y)\big)$.

We now show that $T' \mid 2f(x)$ for every integer x. Suppose, to the contrary, that there exists an integer u, a prime p, and a positive integer α such that $p^\alpha \mid T'$ and $p^\alpha \nmid 2f(u)$. Choose v as in Step 2 and employ Lemma 1 to obtain $\Delta_{2f(u)}\Delta_{\mathrm{lcm}(f(u),f(v))}f = 0$. However, this implies that $\Delta_{T'/p}\Delta_{T'/p}f = 0$, a contradiction with the minimality of T'.

We now claim that $\Delta_{T'}\Delta_2 f = 0$. Indeed, Lemma 1 implies that there exists an integer s such that $\Delta_s\Delta_2 f = 0$. Hence $\Delta_{T's}\Delta_2 f = \Delta_{T'}\Delta_{T'}\Delta_2 f = 0$, which allows us to conclude that $\Delta_{T'}\Delta_2 f = 0$ by Lemma 2. (The last two paragraphs are similar to Step 2 of the solution.)

Now, it is not difficult to finish the solution, though more work is needed to eliminate the factors of 2 from the subscripts of $\Delta_{T'}\Delta_2 f = 0$. Once this is done, we will obtain an odd quasi-period of f that divides $f(x)$ for all integers x. Then we can complete the solution as in Step 3.

A6. Let n be a fixed integer with $n \geqslant 2$. We say that two polynomials P and Q with real coefficients are *block-similar* if for each $i \in \{1, 2, \ldots, n\}$ the sequences

$$P(2015i), P(2015i-1), \ldots, P(2015i-2014) \quad \text{and}$$
$$Q(2015i), Q(2015i-1), \ldots, Q(2015i-2014)$$

are permutations of each other.

(*a*) Prove that there exist distinct block-similar polynomials of degree $n+1$.

(*b*) Prove that there do not exist distinct block-similar polynomials of degree n.

(Canada)

Solution 1. For convenience, we set $k = 2015 = 2\ell + 1$.

Part (*a*). Consider the following polynomials of degree $n+1$:

$$P(x) = \prod_{i=0}^{n}(x - ik) \qquad \text{and} \qquad Q(x) = \prod_{i=0}^{n}(x - ik - 1).$$

Since $Q(x) = P(x-1)$ and $P(0) = P(k) = P(2k) = \cdots = P(nk)$, these polynomials are block-similar (and distinct).

Part (*b*). For every polynomial $F(x)$ and every nonnegative integer m, define $\Sigma_F(m) = \sum_{i=1}^{m} F(i)$; in particular, $\Sigma_F(0) = 0$. It is well-known that for every nonnegative integer d the sum $\sum_{i=1}^{m} i^d$ is a polynomial in m of degree $d+1$. Thus Σ_F may also be regarded as a real polynomial of degree $\deg F + 1$ (with the exception that if $F = 0$, then $\Sigma_F = 0$ as well). This allows us to consider the values of Σ_F at all real points (where the initial definition does not apply).

Assume for the sake of contradiction that there exist two distinct block-similar polynomials $P(x)$ and $Q(x)$ of degree n. Then both polynomials $\Sigma_{P-Q}(x)$ and $\Sigma_{P^2-Q^2}(x)$ have roots at the points $0, k, 2k, \ldots, nk$. This motivates the following lemma, where we use the special polynomial

$$T(x) = \prod_{i=0}^{n}(x - ik).$$

Lemma. Assume that $F(x)$ is a nonzero polynomial such that $0, k, 2k, \ldots, nk$ are among the roots of the polynomial $\Sigma_F(x)$. Then $\deg F \geqslant n$, and there exists a polynomial $G(x)$ such that $\deg G = \deg F - n$ and $F(x) = T(x)G(x) - T(x-1)G(x-1)$.

Proof. If $\deg F < n$, then $\Sigma_F(x)$ has at least $n+1$ roots, while its degree is less than $n+1$. Therefore, $\Sigma_F(x) = 0$ and hence $F(x) = 0$, which is impossible. Thus $\deg F \geqslant n$.

The lemma condition yields that $\Sigma_F(x) = T(x)G(x)$ for some polynomial $G(x)$ such that $\deg G = \deg \Sigma_F - (n+1) = \deg F - n$.

Now, let us define $F_1(x) = T(x)G(x) - T(x-1)G(x-1)$. Then for every positive integer n we have

$$\Sigma_{F_1}(n) = \sum_{i=1}^{n}\big(T(x)G(x) - T(x-1)G(x-1)\big) = T(n)G(n) - T(0)G(0) = T(n)G(n) = \Sigma_F(n),$$

so the polynomial $\Sigma_{F-F_1}(x) = \Sigma_F(x) - \Sigma_{F_1}(x)$ has infinitely many roots. This means that this polynomial is zero, which in turn yields $F(x) = F_1(x)$, as required. □

First, we apply the lemma to the nonzero polynomial $R_1(x) = P(x) - Q(x)$. Since the degree of $R_1(x)$ is at most n, we conclude that it is exactly n. Moreover, $R_1(x) = \alpha \cdot \big(T(x) - T(x-1)\big)$ for some nonzero constant α.

Our next aim is to prove that the polynomial $S(x) = P(x) + Q(x)$ is constant. Assume the contrary. Then, notice that the polynomial $R_2(x) = P(x)^2 - Q(x)^2 = R_1(x)S(x)$ is also nonzero and satisfies the lemma condition. Since $n < \deg R_1 + \deg S = \deg R_2 \leqslant 2n$, the lemma yields

$$R_2(x) = T(x)G(x) - T(x-1)G(x-1)$$

with some polynomial $G(x)$ with $0 < \deg G \leqslant n$.

Since the polynomial $R_1(x) = \alpha\big(T(x) - T(x-1)\big)$ divides the polynomial

$$R_2(x) = T(x)\big(G(x) - G(x-1)\big) + G(x-1)\big(T(x) - T(x-1)\big),$$

we get $R_1(x) \mid T(x)\big(G(x) - G(x-1)\big)$. On the other hand,

$$\gcd\big(T(x), R_1(x)\big) = \gcd\big(T(x), T(x) - T(x-1)\big) = \gcd\big(T(x), T(x-1)\big) = 1,$$

since both $T(x)$ and $T(x-1)$ are the products of linear polynomials, and their roots are distinct. Thus $R_1(x) \mid G(x) - G(x-1)$. However, this is impossible since $G(x) - G(x-1)$ is a nonzero polynomial of degree less than $n = \deg R_1$.

Thus, our assumption is wrong, and $S(x)$ is a constant polynomial, say $S(x) = \beta$. Notice that the polynomials $\big(2P(x) - \beta\big)/\alpha$ and $\big(2Q(x) - \beta\big)/\alpha$ are also block-similar and distinct. So we may replace the initial polynomials by these ones, thus obtaining two block-similar polynomials $P(x)$ and $Q(x)$ with $P(x) = -Q(x) = T(x) - T(x-1)$. It remains to show that this is impossible.

For every $i = 1, 2 \ldots, n$, the values $T(ik - k + 1)$ and $T(ik - 1)$ have the same sign. This means that the values $P(ik - k + 1) = T(ik - k + 1)$ and $P(ik) = -T(ik - 1)$ have opposite signs, so $P(x)$ has a root in each of the n segments $[ik - k + 1, ik]$. Since $\deg P = n$, it must have exactly one root in each of them.

Thus, the sequence $P(1), P(2), \ldots, P(k)$ should change sign exactly once. On the other hand, since $P(x)$ and $-P(x)$ are block-similar, this sequence must have as many positive terms as negative ones. Since $k = 2\ell + 1$ is odd, this shows that the middle term of the sequence above must be zero, so $P(\ell + 1) = 0$, or $T(\ell + 1) = T(\ell)$. However, this is not true since

$$|T(\ell+1)| = |\ell+1| \cdot |\ell| \cdot \prod_{i=2}^{n} |\ell + 1 - ik| < |\ell| \cdot |\ell+1| \cdot \prod_{i=2}^{n} |\ell - ik| = |T(\ell)|\,,$$

where the strict inequality holds because $n \geqslant 2$. We come to the final contradiction.

Comment 1. In the solution above, we used the fact that $k > 1$ is odd. One can modify the arguments of the last part in order to work for every (not necessarily odd) sufficiently large value of k; namely, when k is even, one may show that the sequence $P(1), P(2), \ldots, P(k)$ has different numbers of positive and negative terms.

On the other hand, the problem statement with k replaced by 2 is false, since the polynomials $P(x) = T(x) - T(x-1)$ and $Q(x) = T(x-1) - T(x)$ *are* block-similar in this case, due to the fact that $P(2i-1) = -P(2i) = Q(2i) = -Q(2i-1) = T(2i-1)$ for all $i = 1, 2, \ldots, n$. Thus, every complete solution should use the relation $k > 2$.

One may easily see that the condition $n \geqslant 2$ is also substantial, since the polynomials x and $k + 1 - x$ become block-similar if we set $n = 1$.

It is easily seen from the solution that the result still holds if we assume that the polynomials have degree *at most* n.

Solution 2. We provide an alternative argument for part (b).

Assume again that there exist two distinct block-similar polynomials $P(x)$ and $Q(x)$ of degree n. Let $R(x) = P(x) - Q(x)$ and $S(x) = P(x) + Q(x)$. For brevity, we also denote the segment $\big[(i-1)k+1, ik\big]$ by I_i, and the set $\big\{(i-1)k+1, (i-1)k+2, \ldots, ik\big\}$ of all integer points in I_i by Z_i.

Step 1. We prove that $R(x)$ has exactly one root in each segment I_i, $i = 1, 2, \ldots, n$, and all these roots are simple.

Indeed, take any $i \in \{1, 2, \ldots, n\}$ and choose some points $p^-, p^+ \in Z_i$ so that

$$P(p^-) = \min_{x \in Z_i} P(x) \quad \text{and} \qquad P(p^+) = \max_{x \in Z_i} P(x).$$

Since the sequences of values of P and Q in Z_i are permutations of each other, we have $R(p^-) = P(p^-) - Q(p^-) \leqslant 0$ and $R(p^+) = P(p^+) - Q(p^+) \geqslant 0$. Since $R(x)$ is continuous, there exists at least one root of $R(x)$ between p^- and p^+ — thus in I_i.

So, $R(x)$ has at least one root in each of the n disjoint segments I_i with $i = 1, 2, \ldots, n$. Since $R(x)$ is nonzero and its degree does not exceed n, it should have exactly one root in each of these segments, and all these roots are simple, as required.

Step 2. We prove that $S(x)$ is constant.

We start with the following claim.

Claim. For every $i = 1, 2, \ldots, n$, the sequence of values $S\big((i-1)k+1\big)$, $S\big((i-1)k+2\big)$, $\ldots$, $S(ik)$ cannot be strictly increasing.

Proof. Fix any $i \in \{1, 2, \ldots, n\}$. Due to the symmetry, we may assume that $P\big(ik\big) \leqslant Q(ik)$. Choose now p^- and p^+ as in Step 1. If we had $P(p^+) = P(p^-)$, then P would be constant on Z_i, so all the elements of Z_i would be the roots of $R(x)$, which is not the case. In particular, we have $p^+ \neq p^-$. If $p^- > p^+$, then $S(p^-) = P(p^-) + Q(p^-) \leqslant Q(p^+) + P(p^+) = S(p^+)$, so our claim holds.

We now show that the remaining case $p^- < p^+$ is impossible. Assume first that $P(p^+) > Q(p^+)$. Then, like in Step 1, we have $R(p^-) \leqslant 0$, $R(p^+) > 0$, and $R(ik) \leqslant 0$, so $R(x)$ has a root in each of the intervals $[p^-, p^+)$ and $(p^+, ik]$. This contradicts the result of Step 1.

We are left only with the case $p^- < p^+$ and $P(p^+) = Q(p^+)$ (thus p^+ is the unique root of $R(x)$ in I_i). If $p^+ = ik$, then the values of $R(x)$ on $Z_i \setminus \{ik\}$ are all of the same sign, which is absurd since their sum is zero. Finally, if $p^- < p^+ < ik$, then $R(p^-)$ and $R(ik)$ are both negative. This means that $R(x)$ should have an even number of roots in $[p^-, ik]$, counted with multiplicity. This also contradicts the result of Step 1. □

In a similar way, one may prove that for every $i = 1, 2, \ldots, n$, the sequence $S\big((i-1)k+1\big)$, $S\big((i-1)k+2\big)$, $\ldots$, $S(ik)$ cannot be strictly decreasing. This means that the polynomial $\Delta S(x) = S(x) - S(x-1)$ attains at least one nonnegative value, as well as at least one nonpositive value, on the set Z_i (and even on $Z_i \setminus \big\{(i-1)k+1\big\}$); so ΔS has a root in I_i.

Thus ΔS has at least n roots; however, its degree is less than n, so ΔS should be identically zero. This shows that $S(x)$ is a constant, say $S(x) \equiv \beta$.

Step 3. Notice that the polynomials $P(x) - \beta/2$ and $Q(x) - \beta/2$ are also block-similar and distinct. So we may replace the initial polynomials by these ones, thus reaching $P(x) = -Q(x)$.

Then $R(x) = 2P(x)$, so $P(x)$ has exactly one root in each of the segments I_i, $i = 1, 2, \ldots, n$. On the other hand, $P(x)$ and $-P(x)$ should attain the same number of positive values on Z_i. Since k is odd, this means that Z_i contains exactly one root of $P(x)$; moreover, this root should be at the center of Z_i, because $P(x)$ has the same number of positive and negative values on Z_i.

Thus we have found all n roots of $P(x)$, so

$$P(x) = c \prod_{i=1}^{n} (x - ik + \ell) \qquad \text{for some } c \in \mathbb{R} \setminus \{0\},$$

where $\ell = (k-1)/2$. It remains to notice that for every $t \in Z_1 \setminus \{1\}$ we have

$$|P(t)| = |c| \cdot |t - \ell - 1| \cdot \prod_{i=2}^{n} |t - ik + \ell| < |c| \cdot \ell \cdot \prod_{i=2}^{n} |1 - ik + \ell| = |P(1)|,$$

so $P(1) \neq -P(t)$ for all $t \in Z_1$. This shows that $P(x)$ is not block-similar to $-P(x)$. The final contradiction.

Comment 2. One may merge Steps 1 and 2 in the following manner. As above, we set $R(x) = P(x) - Q(x)$ and $S(x) = P(x) + Q(x)$.

We aim to prove that the polynomial $S(x) = 2P(x) - R(x) = 2Q(x) + R(x)$ is constant. Since the degrees of $R(x)$ and $S(x)$ do not exceed n, it suffices to show that the total number of roots of $R(x)$ and $\Delta S(x) = S(x) - S(x-1)$ is at least $2n$. For this purpose, we prove the following claim.

Claim. For every $i = 1, 2, \ldots, n$, either each of R and ΔS has a root in I_i, or R has at least two roots in I_i.

Proof. Fix any $i \in \{1, 2, \ldots, n\}$. Let $r \in Z_i$ be a point such that $|R(r)| = \max_{x \in Z_i} |R(x)|$; we may assume that $R(r) > 0$. Next, let $p^-, q^+ \in I_i$ be some points such that $P(p^-) = \min_{x \in Z_i} P(x)$ and $Q(q^+) = \max_{x \in Z_i} Q(x)$. Notice that $P(p^-) \leqslant Q(r) < P(r)$ and $Q(q^+) \geqslant P(r) > Q(r)$, so r is different from p^- and q^+.

Without loss of generality, we may assume that $p^- < r$. Then we have $R(p^-) = P(p^-) - Q(p^-) \leqslant 0 < R(r)$, so $R(x)$ has a root in $[p^-, r)$. If $q^+ > r$, then, similarly, $R(q^+) \leqslant 0 < R(r)$, and $R(x)$ also has a root in $(r, q^+]$; so $R(x)$ has two roots in I_i, as required.

In the remaining case we have $q^+ < r$; it suffices now to show that in this case ΔS has a root in I_i. Since $P(p^-) \leqslant Q(r)$ and $|R(p^-)| \leqslant R(r)$, we have $S(p^-) = 2P(p^-) - R(p^-) \leqslant 2Q(r) + R(r) = S(r)$. Similarly, we get $S(q^+) = 2Q(q^+) + R(q^+) \geqslant 2P(r) - R(r) = S(r)$. Therefore, the sequence of values of S on Z_i is neither strictly increasing nor strictly decreasing, which shows that ΔS has a root in I_i. □

Comment 3. After finding the relation $P(x) - Q(x) = \alpha\big(T(x) - T(x-1)\big)$ from Solution 1, one may also follow the approach presented in Solution 2. Knowledge of the difference of polynomials may simplify some steps; e.g., it is clear now that $P(x) - Q(x)$ has exactly one root in each of the segments I_i.

Combinatorics

C1. In Lineland there are $n \geqslant 1$ towns, arranged along a road running from left to right. Each town has a *left bulldozer* (put to the left of the town and facing left) and a *right bulldozer* (put to the right of the town and facing right). The sizes of the $2n$ bulldozers are distinct. Every time when a right and a left bulldozer confront each other, the larger bulldozer pushes the smaller one off the road. On the other hand, the bulldozers are quite unprotected at their rears; so, if a bulldozer reaches the rear-end of another one, the first one pushes the second one off the road, regardless of their sizes.

Let A and B be two towns, with B being to the right of A. We say that town A can *sweep* town B *away* if the right bulldozer of A can move over to B pushing off all bulldozers it meets. Similarly, B can sweep A away if the left bulldozer of B can move to A pushing off all bulldozers of all towns on its way.

Prove that there is exactly one town which cannot be swept away by any other one.

(Estonia)

Solution 1. Let $T_1, T_2, \ldots, T_n$ be the towns enumerated from left to right. Observe first that, if town T_i can sweep away town T_j, then T_i also can sweep away every town located between T_i and T_j.

We prove the problem statement by strong induction on n. The base case $n = 1$ is trivial.

For the induction step, we first observe that the left bulldozer in T_1 and the right bulldozer in T_n are completely useless, so we may forget them forever. Among the other $2n-2$ bulldozers, we choose the largest one. Without loss of generality, it is the right bulldozer of some town T_k with $k < n$.

Surely, with this large bulldozer T_k can sweep away all the towns to the right of it. Moreover, none of these towns can sweep T_k away; so they also cannot sweep away any town to the left of T_k. Thus, if we remove the towns $T_{k+1}, T_{k+2}, \ldots, T_n$, none of the remaining towns would change its status of being (un)sweepable away by the others.

Applying the induction hypothesis to the remaining towns, we find a unique town among $T_1, T_2, \ldots, T_k$ which cannot be swept away. By the above reasons, it is also the unique such town in the initial situation. Thus the induction step is established.

Solution 2. We start with the same enumeration and the same observation as in Solution 1. We also denote by ℓ_i and r_i the sizes of the left and the right bulldozers belonging to T_i, respectively. One may easily see that no two towns T_i and T_j with $i < j$ can sweep each other away, for this would yield $r_i > \ell_j > r_i$.

Clearly, there is no town which can sweep T_n away from the right. Then we may choose the leftmost town T_k which cannot be swept away from the right. One can observe now that no town T_i with $i > k$ may sweep away some town T_j with $j < k$, for otherwise T_i would be able to sweep T_k away as well.

Now we prove two claims, showing together that T_k is the unique town which cannot be swept away, and thus establishing the problem statement.

Claim 1. T_k also cannot be swept away from the left.

Proof. Let T_m be some town to the left of T_k. By the choice of T_k, town T_m can be swept away from the right by some town T_p with $p > m$. As we have already observed, p cannot be greater than k. On the other hand, T_m cannot sweep T_p away, so *a fortiori* it cannot sweep T_k away. □

Claim 2. Any town T_m with $m \neq k$ can be swept away by some other town.

Proof. If $m < k$, then T_m can be swept away from the right due to the choice of T_k. In the remaining case we have $m > k$.

Let T_p be a town among $T_k, T_{k+1}, \ldots, T_{m-1}$ having the largest right bulldozer. We claim that T_p can sweep T_m away. If this is not the case, then $r_p < \ell_q$ for some q with $p < q \leqslant m$. But this means that ℓ_q is greater than all the numbers r_i with $k \leqslant i \leqslant m-1$, so T_q can sweep T_k away. This contradicts the choice of T_k. □

Comment 1. One may employ the same ideas within the inductive approach. Here we sketch such a solution.

Assume that the problem statement holds for the collection of towns $T_1, T_2, \ldots, T_{n-1}$, so that there is a unique town T_i among them which cannot be swept away by any other of them. Thus we need to prove that in the full collection $T_1, T_2, \ldots, T_n$, exactly one of the towns T_i and T_n cannot be swept away.

If T_n cannot sweep T_i away, then it remains to prove that T_n can be swept away by some other town. This can be established as in the second paragraph of the proof of Claim 2.

If T_n can sweep T_i away, then it remains to show that T_n cannot be swept away by any other town. Since T_n can sweep T_i away, it also can sweep all the towns $T_i, T_{i+1}, \ldots, T_{n-1}$ away, so T_n cannot be swept away by any of those. On the other hand, none of the remaining towns $T_1, T_2, \ldots, T_{i-1}$ can sweep T_i away, so that they cannot sweep T_n away as well.

Comment 2. Here we sketch yet another inductive approach. Assume that $n > 1$. Firstly, we find a town which can be swept away by each of its neighbors (each town has two neighbors, except for the bordering ones each of which has one); we call such town a *loser*. Such a town exists, because there are $n-1$ pairs of neighboring towns, and in each of them there is only one which can sweep the other away; so there exists a town which is a winner in none of these pairs.

Notice that a loser can be swept away, but it cannot sweep any other town away (due to its neighbors' protection). Now we remove a loser, and *suggest* its left bulldozer to its *right* neighbor (if it exists), and its right bulldozer to a *left* one (if it exists). Surely, a town accepts a suggestion if a suggested bulldozer is larger than the town's one of the same orientation.

Notice that suggested bulldozers are useless in attack (by the definition of a loser), but may serve for defensive purposes. Moreover, each suggested bulldozer's protection works for the same pairs of remaining towns as before the removal.

By the induction hypothesis, the new configuration contains exactly one town which cannot be swept away. The arguments above show that the initial one also satisfies this property.

Solution 3. We separately prove that (i) there exists a town which cannot be swept away, and that (ii) there is at most one such town. We also make use of the two observations from the previous solutions.

To prove (i), assume contrariwise that every town can be swept away. Let t_1 be the leftmost town; next, for every $k = 1, 2, \ldots$ we inductively choose t_{k+1} to be some town which can sweep t_k away. Now we claim that for every $k = 1, 2, \ldots$, the town t_{k+1} is to the right of t_k; this leads to the contradiction, since the number of towns is finite.

Induction on k. The base case $k = 1$ is clear due to the choice of t_1. Assume now that for all j with $1 \leqslant j < k$, the town t_{j+1} is to the right of t_j. Suppose that t_{k+1} is situated to the left of t_k; then it lies between t_j and t_{j+1} (possibly coinciding with t_j) for some $j < k$. Therefore, t_{k+1} can be swept away by t_{j+1}, which shows that it cannot sweep t_{j+1} away — so t_{k+1} also cannot sweep t_k away. This contradiction proves the induction step.

To prove (ii), we also argue indirectly and choose two towns A and B neither of which can be swept away, with A being to the left of B. Consider the largest bulldozer b between them (taking into consideration the right bulldozer of A and the left bulldozer of B). Without loss of generality, b is a left bulldozer; then it is situated in some town to the right of A, and this town may sweep A away since nothing prevents it from doing that. A contradiction.

Comment 3. The Problem Selection Committee decided to reformulate this problem. The original formulation was as follows.

Let n be a positive integer. There are n cards in a deck, enumerated from bottom to top with numbers $1, 2, \ldots, n$. For each $i = 1, 2, \ldots, n$, an even number a_i is printed on the lower side and an odd number b_i is printed on the upper side of the ith card. We say that the ith card opens *the jth card, if $i < j$ and $b_i < a_k$ for every $k = i + 1, i + 2, \ldots, j$. Similarly, we say that the ith card* closes *the jth card, if $i > j$ and $a_i < b_k$ for every $k = i - 1, i - 2, \ldots, j$. Prove that the deck contains exactly one card which is neither opened nor closed by any other card.*

C2. Let $\mathcal{V}$ be a finite set of points in the plane. We say that $\mathcal{V}$ is *balanced* if for any two distinct points $A, B \in \mathcal{V}$, there exists a point $C \in \mathcal{V}$ such that $AC = BC$. We say that $\mathcal{V}$ is *center-free* if for any distinct points $A, B, C \in \mathcal{V}$, there does not exist a point $P \in \mathcal{V}$ such that $PA = PB = PC$.

(a) Show that for all $n \geqslant 3$, there exists a balanced set consisting of n points.

(b) For which $n \geqslant 3$ does there exist a balanced, center-free set consisting of n points?

(Netherlands)

Answer for part (b). All odd integers $n \geqslant 3$.

Solution.

Part (a). Assume that n is odd. Consider a regular n-gon. Label the vertices of the n-gon as $A_1, A_2, \ldots, A_n$ in counter-clockwise order, and set $\mathcal{V} = \{A_1, \ldots, A_n\}$. We check that $\mathcal{V}$ is balanced. For any two distinct vertices A_i and A_j, let $k \in \{1, 2, \ldots, n\}$ be the solution of $2k \equiv i + j \pmod{n}$. Then, since $k - i \equiv j - k \pmod{n}$, we have $A_iA_k = A_jA_k$, as required.

Now assume that n is even. Consider a regular $(3n - 6)$-gon, and let O be its circumcenter. Again, label its vertices as $A_1, \ldots, A_{3n-6}$ in counter-clockwise order, and choose $\mathcal{V} = \{O, A_1, A_2, \ldots, A_{n-1}\}$. We check that $\mathcal{V}$ is balanced. For any two distinct vertices A_i and A_j, we always have $OA_i = OA_j$. We now consider the vertices O and A_i. First note that the triangle $OA_iA_{n/2-1+i}$ is equilateral for all $i \leqslant \frac{n}{2}$. Hence, if $i \leqslant \frac{n}{2}$, then we have $OA_{n/2-1+i} = A_iA_{n/2-1+i}$; otherwise, if $i > \frac{n}{2}$, then we have $OA_{i-n/2+1} = A_iA_{i-n/2+1}$. This completes the proof.

An example of such a construction when $n = 10$ is shown in Figure 1.

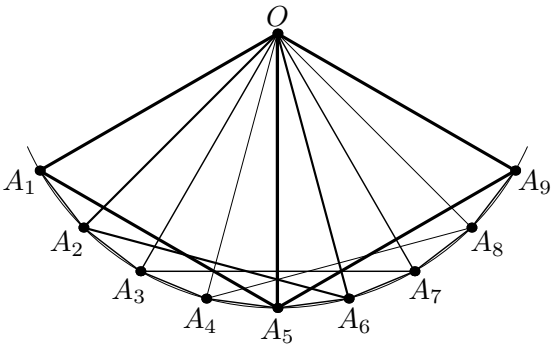


Figure 1

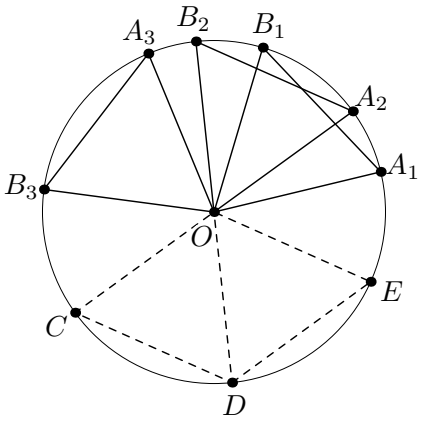


Figure 2

Comment (a). There are many ways to construct an example by placing equilateral triangles in a circle. Here we present one general method.

Let O be the center of a circle and let $A_1, B_1, \ldots, A_k, B_k$ be distinct points on the circle such that the triangle OA_iB_i is equilateral for each i. Then $\mathcal{V} = \{O, A_1, B_1, \ldots, A_k, B_k\}$ is balanced. To construct a set of even cardinality, put extra points C, D, E on the circle such that triangles OCD and ODE are equilateral (see Figure 2). Then $\mathcal{V} = \{O, A_1, B_1, \ldots, A_k, B_k, C, D, E\}$ is balanced.

Part (b). We now show that there exists a balanced, center-free set containing n points for all odd $n \geqslant 3$, and that one does not exist for any even $n \geqslant 3$.

If n is odd, then let $\mathcal{V}$ be the set of vertices of a regular n-gon. We have shown in part (a) that $\mathcal{V}$ is balanced. We claim that $\mathcal{V}$ is also center-free. Indeed, if P is a point such that

$PA = PB = PC$ for some three distinct vertices A, B and C, then P is the circumcenter of the n-gon, which is not contained in $\mathcal{V}$.

Now suppose that $\mathcal{V}$ is a balanced, center-free set of even cardinality n. We will derive a contradiction. For a pair of distinct points $A, B \in \mathcal{V}$, we say that a point $C \in \mathcal{V}$ is *associated* with the pair $\{A, B\}$ if $AC = BC$. Since there are $\frac{n(n-1)}{2}$ pairs of points, there exists a point $P \in \mathcal{V}$ which is associated with at least $\left\lceil \frac{n(n-1)}{2} / n \right\rceil = \frac{n}{2}$ pairs. Note that none of these $\frac{n}{2}$ pairs can contain P, so that the union of these $\frac{n}{2}$ pairs consists of at most $n-1$ points. Hence there exist two such pairs that share a point. Let these two pairs be $\{A, B\}$ and $\{A, C\}$. Then $PA = PB = PC$, which is a contradiction.

Comment (*b*). We can rephrase the argument in graph theoretic terms as follows. Let $\mathcal{V}$ be a balanced, center-free set consisting of n points. For any pair of distinct vertices $A, B \in \mathcal{V}$ and for any $C \in \mathcal{V}$ such that $AC = BC$, draw directed edges $A \to C$ and $B \to C$. Then all pairs of vertices generate altogether at least $n(n-1)$ directed edges; since the set is center-free, these edges are distinct. So we must obtain a graph in which any two vertices are connected in both directions. Now, each vertex has exactly $n-1$ incoming edges, which means that $n-1$ is even. Hence n is odd.

C3. For a finite set A of positive integers, we call a partition of A into two disjoint nonempty subsets A_1 and A_2 *good* if the least common multiple of the elements in A_1 is equal to the greatest common divisor of the elements in A_2. Determine the minimum value of n such that there exists a set of n positive integers with exactly 2015 good partitions.

(Ukraine)

Answer. 3024.

Solution. Let $A = \{a_1, a_2, \ldots, a_n\}$, where $a_1 < a_2 < \cdots < a_n$. For a finite nonempty set B of positive integers, denote by $\operatorname{lcm} B$ and $\gcd B$ the least common multiple and the greatest common divisor of the elements in B, respectively.

Consider any good partition (A_1, A_2) of A. By definition, $\operatorname{lcm} A_1 = d = \gcd A_2$ for some positive integer d. For any $a_i \in A_1$ and $a_j \in A_2$, we have $a_i \leqslant d \leqslant a_j$. Therefore, we have $A_1 = \{a_1, a_2, \ldots, a_k\}$ and $A_2 = \{a_{k+1}, a_{k+2}, \ldots, a_n\}$ for some k with $1 \leqslant k < n$. Hence, each good partition is determined by an element a_k, where $1 \leqslant k < n$. We call such a_k *partitioning*.

It is convenient now to define $\ell_k = \operatorname{lcm}(a_1, a_2, \ldots, a_k)$ and $g_k = \gcd(a_{k+1}, a_{k+2}, \ldots, a_n)$ for $1 \leqslant k \leqslant n-1$. So a_k is partitioning exactly when $\ell_k = g_k$.

We proceed by proving some properties of partitioning elements, using the following claim.

Claim. If a_{k-1} and a_k are partitioning where $2 \leqslant k \leqslant n-1$, then $g_{k-1} = g_k = a_k$.

Proof. Assume that a_{k-1} and a_k are partitioning. Since $\ell_{k-1} = g_{k-1}$, we have $\ell_{k-1} \mid a_k$. Therefore, $g_k = \ell_k = \operatorname{lcm}(\ell_{k-1}, a_k) = a_k$, and $g_{k-1} = \gcd(a_k, g_k) = a_k$, as desired. □

Property 1. For every $k = 2, 3, \ldots, n-2$, at least one of a_{k-1}, a_k, and a_{k+1} is not partitioning.

Proof. Suppose, to the contrary, that all three numbers a_{k-1}, a_k, and a_{k+1} are partitioning. The claim yields that $a_{k+1} = g_k = a_k$, a contradiction. □

Property 2. The elements a_1 and a_2 cannot be simultaneously partitioning. Also, a_{n-2} and a_{n-1} cannot be simultaneously partitioning

Proof. Assume that a_1 and a_2 are partitioning. By the claim, it follows that $a_2 = g_1 = \ell_1 = \operatorname{lcm}(a_1) = a_1$, a contradiction.

Similarly, assume that a_{n-2} and a_{n-1} are partitioning. The claim yields that $a_{n-1} = g_{n-1} = \gcd(a_n) = a_n$, a contradiction. □

Now let A be an n-element set with exactly 2015 good partitions. Clearly, we have $n \geqslant 5$. Using Property 2, we find that there is at most one partitioning element in each of $\{a_1, a_2\}$ and $\{a_{n-2}, a_{n-1}\}$. By Property 1, there are at least $\left\lfloor \frac{n-5}{3} \right\rfloor$ non-partitioning elements in $\{a_3, a_4, \ldots, a_{n-3}\}$. Therefore, there are at most $(n-1) - 2 - \left\lfloor \frac{n-5}{3} \right\rfloor = \left\lceil \frac{2(n-2)}{3} \right\rceil$ partitioning elements in A. Thus, $\left\lceil \frac{2(n-2)}{3} \right\rceil \geqslant 2015$, which implies that $n \geqslant 3024$.

Finally, we show that there exists a set of 3024 positive integers with exactly 2015 partitioning elements. Indeed, in the set $A = \{2 \cdot 6^i, 3 \cdot 6^i, 6^{i+1} \mid 0 \leqslant i \leqslant 1007\}$, each element of the form $3 \cdot 6^i$ or 6^i, except 6^{1008}, is partitioning.

Therefore, the minimum possible value of n is 3024.

Comment. Here we will work out the general case when 2015 is replaced by an arbitrary positive integer m. Note that the bound $\left\lceil \frac{2(n-2)}{3} \right\rceil \geqslant m$ obtained in the solution is, in fact, true for any positive integers m and n. Using this bound, one can find that $n \geqslant \left\lceil \frac{3m}{2} \right\rceil + 1$.

To show that the bound is sharp, one constructs a set of $\left\lceil \frac{3m}{2} \right\rceil + 1$ elements with exactly m good partitions. Indeed, the minimum is attained on the set $\{6^i, 2 \cdot 6^i, 3 \cdot 6^i \mid 0 \leqslant i \leqslant t-1\} \cup \{6^t\}$ for every even $m = 2t$, and $\{2 \cdot 6^i, 3 \cdot 6^i, 6^{i+1} \mid 0 \leqslant i \leqslant t-1\}$ for every odd $m = 2t-1$.

C4. Let n be a positive integer. Two players A and B play a game in which they take turns choosing positive integers $k \leqslant n$. The rules of the game are:

(*i*) A player cannot choose a number that has been chosen by either player on any previous turn.

(*ii*) A player cannot choose a number consecutive to any of those the player has already chosen on any previous turn.

(*iii*) The game is a draw if all numbers have been chosen; otherwise the player who cannot choose a number anymore loses the game.

The player A takes the first turn. Determine the outcome of the game, assuming that both players use optimal strategies.

(Finland)

Answer. The game ends in a draw when $n = 1, 2, 4, 6$; otherwise B wins.

Solution. For brevity, we denote by $[n]$ the set $\{1, 2, \ldots, n\}$.

Firstly, we show that B wins whenever $n \neq 1, 2, 4, 6$. For this purpose, we provide a strategy which guarantees that B can always make a move after A's move, and also guarantees that the game does not end in a draw.

We begin with an important observation.

Lemma. Suppose that B's first pick is n and that A has made the kth move where $k \geqslant 2$. Then B can also make the kth move.

Proof. Let $\mathcal{S}$ be the set of the first k numbers chosen by A. Since $\mathcal{S}$ does not contain consecutive integers, we see that the set $[n] \setminus \mathcal{S}$ consists of k "contiguous components" if $1 \in \mathcal{S}$, and $k+1$ components otherwise. Since B has chosen only $k-1$ numbers, there is at least one component of $[n] \setminus \mathcal{S}$ consisting of numbers not yet picked by B. Hence, B can choose a number from this component. □

We will now describe a winning strategy for B, when $n \neq 1, 2, 4, 6$. By symmetry, we may assume that A's first choice is a number not exceeding $\frac{n+1}{2}$. So B can pick the number n in B's first turn. We now consider two cases.

Case 1. n is odd and $n \geqslant 3$. The only way the game ends in a draw is that A eventually picks all the odd numbers from the set $[n]$. However, B has already chosen n, so this cannot happen. Thus B can continue to apply the lemma until A cannot make a move.

Case 2. n is even and $n \geqslant 8$. Since B has picked n, the game is a draw only if A can eventually choose all the odd numbers from the set $[n-1]$. So B picks a number from the set $\{1, 3, 5, \ldots, n-3\}$ not already chosen by A, on B's second move. This is possible since the set consists of $\frac{n-2}{2} \geqslant 3$ numbers and A has chosen only 2 numbers. Hereafter B can apply the lemma until A cannot make a move.

Hence, in both cases A loses.

We are left with the cases $n = 1, 2, 4, 6$. The game is trivially a draw when $n = 1, 2$. When $n = 4$, A has to first pick 1 to avoid losing. Similarly, B has to choose 4 as well. It then follows that the game ends in a draw.

When $n = 6$, B gets at least a draw by the lemma or by using a mirror strategy. On the other hand, A may also get at least a draw in the following way. In the first turn, A chooses 1. After B's response by a number b, A finds a neighbor c of b which differs from 1 and 2, and reserves c for A's third move. Now, clearly A can make the second move by choosing a number different from $1, 2, c-1, c, c+1$. Therefore A will not lose.

Comment 1. We present some explicit winning strategies for B.

We start with the case n is odd and $n \geqslant 3$. B starts by picking n in the first turn. On the kth move for $k \geqslant 2$, B chooses the number exactly 1 less than A's k^{th} pick. The only special case is when A's k^{th} choice is 1. In this situation, A's first pick was a number $a > 1$ and B can respond by choosing $a - 1$ on the kth move instead.

We now give an alternative winning strategy in the case n is even and $n \geqslant 8$. We first present a winning strategy for the case when A's first pick is 1. We consider two cases depending on A's second move.

Case 1. A's second pick is 3. Then B chooses $n-3$ on the second move. On the kth move, B chooses the number exactly 1 less than A's k^{th} pick except that B chooses 2 if A's k^{th} pick is $n-2$ or $n-1$.

Case 2. A's second pick is $a > 3$. Then B chooses $a - 2$ on the second move. Afterwards on the kth move, B picks the number exactly 1 less than A's k^{th} pick.

One may easily see that this strategy guarantees B's victory, when A's first pick is 1.

The following claim shows how to extend the strategy to the general case.

Claim. Assume that B has an explicit strategy leading to a victory after A picks 1 on the first move. Then B also has an explicit strategy leading to a victory after any first moves of A.

Proof. Let S be an optimal strategy of B after A picks 1 on the first move. Assume that A picks some number $a > 1$ on this move; we show how B can make use of S in order to win in this case.

In parallel to the real play, B starts an *imaginary* play. The positions in these plays differ by flipping the segment $[1, a]$; so, if a player chooses some number x in the real play, then the same player chooses a number x or $a + 1 - x$ in the imaginary play, depending on whether $x > a$ or $x \leqslant a$. Thus A's first pick in the imaginary play is 1.

Clearly, a number is chosen in the real play exactly if the corresponding number is chosen in the imaginary one. Next, if an unchosen number is neighboring to one chosen by A in the imaginary play, then the corresponding number also has this property in the real play, so A also cannot choose it. One can easily see that a similar statement with real and imaginary plays interchanged holds for B instead of A.

Thus, when A makes some move in the real play, B may imagine the corresponding legal move in the imaginary one. Then B chooses the response according to S in the imaginary game and makes the corresponding legal move in the real one. Acting so, B wins the imaginary game, thus B will also win the real one. □

Hence, B has a winning strategy for all even n greater or equal to 8.

Notice that the claim can also be used to simplify the argument when n is odd.

Comment 2. One may also employ symmetry when n is odd. In particular, B could use a mirror strategy. However, additional ideas are required to modify the strategy after A picks $\frac{n+1}{2}$.

C5. Consider an infinite sequence $a_1, a_2, \ldots$ of positive integers with $a_i \leqslant 2015$ for all $i \geqslant 1$. Suppose that for any two distinct indices i and j we have $i + a_i \neq j + a_j$.

Prove that there exist two positive integers b and N such that

$$\left| \sum_{i=m+1}^{n} (a_i - b) \right| \leqslant 1007^2$$

whenever $n > m \geqslant N$.

(Australia)

Solution 1. We visualize the set of positive integers as a sequence of points. For each n we draw an arrow emerging from n that points to $n + a_n$; so the *length* of this arrow is a_n. Due to the condition that $m + a_m \neq n + a_n$ for $m \neq n$, each positive integer receives at most one arrow. There are some positive integers, such as 1, that receive no arrows; these will be referred to as *starting points* in the sequel. When one starts at any of the starting points and keeps following the arrows, one is led to an infinite path, called its *ray*, that visits a strictly increasing sequence of positive integers. Since the length of any arrow is at most 2015, such a ray, say with starting point s, meets every interval of the form $[n, n + 2014]$ with $n \geqslant s$ at least once.

Suppose for the sake of contradiction that there would be at least 2016 starting points. Then we could take an integer n that is larger than the first 2016 starting points. But now the interval $[n, n + 2014]$ must be met by at least 2016 rays in distinct points, which is absurd. We have thereby shown that the number b of starting points satisfies $1 \leqslant b \leqslant 2015$. Let N denote any integer that is larger than all starting points. We contend that b and N are as required.

To see this, let any two integers m and n with $n > m \geqslant N$ be given. The sum $\sum_{i=m+1}^{n} a_i$ gives the total length of the arrows emerging from $m + 1, \ldots, n$. Taken together, these arrows form b subpaths of our rays, some of which may be empty. Now on each ray we look at the first number that is larger than m; let $x_1, \ldots, x_b$ denote these numbers, and let $y_1, \ldots, y_b$ enumerate in corresponding order the numbers defined similarly with respect to n. Then the list of differences $y_1 - x_1, \ldots, y_b - x_b$ consists of the lengths of these paths and possibly some zeros corresponding to empty paths. Consequently, we obtain

$$\sum_{i=m+1}^{n} a_i = \sum_{j=1}^{b} (y_j - x_j)\,,$$

whence

$$\sum_{i=m+1}^{n} (a_i - b) = \sum_{j=1}^{b} (y_j - n) - \sum_{j=1}^{b} (x_j - m)\,.$$

Now each of the b rays meets the interval $[m + 1, m + 2015]$ at some point and thus $x_1 - m, \ldots, x_b - m$ are b distinct members of the set $\{1, 2, \ldots, 2015\}$. Moreover, since $m + 1$ is not a starting point, it must belong to some ray; so 1 has to appear among these numbers, wherefore

$$1 + \sum_{j=1}^{b-1} (j + 1) \leqslant \sum_{j=1}^{b} (x_j - m) \leqslant 1 + \sum_{j=1}^{b-1} (2016 - b + j)\,.$$

The same argument applied to n and $y_1, \ldots, y_b$ yields

$$1 + \sum_{j=1}^{b-1} (j + 1) \leqslant \sum_{j=1}^{b} (y_j - n) \leqslant 1 + \sum_{j=1}^{b-1} (2016 - b + j)\,.$$

So altogether we get

$$\left|\sum_{i=m+1}^{n}(a_i-b)\right| \leqslant \sum_{j=1}^{b-1}\big((2016-b+j)-(j+1)\big) = (b-1)(2015-b)$$
$$\leqslant \left(\frac{(b-1)+(2015-b)}{2}\right)^2 = 1007^2\,,$$

as desired.

Solution 2. Set $s_n = n + a_n$ for all positive integers n. By our assumptions, we have

$$n+1 \leqslant s_n \leqslant n+2015$$

for all $n \in \mathbb{Z}_{>0}$. The members of the sequence $s_1, s_2, \ldots$ are distinct. We shall investigate the set

$$M = \mathbb{Z}_{>0} \setminus \{s_1, s_2, \ldots\}\,.$$

Claim. At most 2015 numbers belong to M.

Proof. Otherwise let $m_1 < m_2 < \cdots < m_{2016}$ be any 2016 distinct elements from M. For $n = m_{2016}$ we have

$$\{s_1, \ldots, s_n\} \cup \{m_1, \ldots, m_{2016}\} \subseteq \{1, 2, \ldots, n+2015\}\,,$$

where on the left-hand side we have a disjoint union containing altogether $n+2016$ elements. But the set on the right-hand side has only $n+2015$ elements. This contradiction proves our claim. □

Now we work towards proving that the positive integers $b = |M|$ and $N = \max(M)$ are as required. Recall that we have just shown $b \leqslant 2015$.

Let us consider any integer $r \geqslant N$. As in the proof of the above claim, we see that

$$B_r = M \cup \{s_1, \ldots, s_r\} \tag{1}$$

is a subset of $[1, r+2015] \cap \mathbb{Z}$ with precisely $b+r$ elements. Due to the definitions of M and N, we also know $[1, r+1] \cap \mathbb{Z} \subseteq B_r$. It follows that there is a set $C_r \subseteq \{1, 2, \ldots, 2014\}$ with $|C_r| = b-1$ and

$$B_r = \big([1, r+1] \cap \mathbb{Z}\big) \cup \big\{r+1+x \,\big|\, x \in C_r\big\}\,. \tag{2}$$

For any finite set of integers J we denote the sum of its elements by $\sum J$. Now the equations (1) and (2) give rise to two ways of computing $\sum B_r$ and the comparison of both methods leads to

$$\sum M + \sum_{i=1}^{r} s_i = \sum_{i=1}^{r} i + b(r+1) + \sum C_r\,,$$

or in other words to

$$\sum M + \sum_{i=1}^{r}(a_i - b) = b + \sum C_r\,. \tag{3}$$

After this preparation, we consider any two integers m and n with $n > m \geqslant N$. Plugging $r = n$ and $r = m$ into (3) and subtracting the estimates that result, we deduce

$$\sum_{i=m+1}^{n}(a_i - b) = \sum C_n - \sum C_m\,.$$

Since C_n and C_m are subsets of $\{1, 2, \ldots, 2014\}$ with $|C_n| = |C_m| = b - 1$, it is clear that the absolute value of the right-hand side of the above inequality attains its largest possible value if either $C_m = \{1, 2, \ldots, b-1\}$ and $C_n = \{2016 - b, \ldots, 2014\}$, or the other way around. In these two cases we have

$$\left|\sum C_n - \sum C_m\right| = (b-1)(2015-b)\,,$$

so in the general case we find

$$\left|\sum_{i=m+1}^{n} (a_i - b)\right| \leqslant (b-1)(2015-b) \leqslant \left(\frac{(b-1)+(2015-b)}{2}\right)^2 = 1007^2\,,$$

as desired.

Comment. The sets C_n may be visualized by means of the following process: Start with an empty blackboard. For $n \geqslant 1$, the following happens during the nth step. The number a_n gets written on the blackboard, then all numbers currently on the blackboard are decreased by 1, and finally all zeros that have arisen get swept away.

It is not hard to see that the numbers present on the blackboard after n steps are distinct and form the set C_n. Moreover, it is possible to complete a solution based on this idea.

C6. Let S be a nonempty set of positive integers. We say that a positive integer n is *clean* if it has a unique representation as a sum of an odd number of distinct elements from S. Prove that there exist infinitely many positive integers that are not clean.

(U.S.A.)

Solution 1. Define an *odd* (respectively, *even*) *representation* of n to be a representation of n as a sum of an odd (respectively, even) number of distinct elements of S. Let $\mathbb{Z}_{>0}$ denote the set of all positive integers.

Suppose, to the contrary, that there exist only finitely many positive integers that are not clean. Therefore, there exists a positive integer N such that every integer $n > N$ has exactly one odd representation.

Clearly, S is infinite. We now claim the following properties of odd and even representations.

Property 1. Any positive integer n has at most one odd and at most one even representation.

Proof. We first show that every integer n has at most one even representation. Since S is infinite, there exists $x \in S$ such that $x > \max\{n, N\}$. Then, the number $n+x$ must be clean, and x does not appear in any even representation of n. If n has more than one even representation, then we obtain two distinct odd representations of $n + x$ by adding x to the even representations of n, which is impossible. Therefore, n can have at most one even representation.

Similarly, there exist two distinct elements $y, z \in S$ such that $y, z > \max\{n, N\}$. If n has more than one odd representation, then we obtain two distinct odd representations of $n+y+z$ by adding y and z to the odd representations of n. This is again a contradiction. □

Property 2. Fix $s \in S$. Suppose that a number $n > N$ has no even representation. Then $n + 2as$ has an even representation containing s for all integers $a \geqslant 1$.

Proof. It is sufficient to prove the following statement: If n has no even representation without s, then $n+2s$ has an even representation containing s (and hence no even representation without s by Property 1).

Notice that the odd representation of $n + s$ does not contain s; otherwise, we have an even representation of n without s. Then, adding s to this odd representation of $n + s$, we get that $n + 2s$ has an even representation containing s, as desired. □

Property 3. Every sufficiently large integer has an even representation.

Proof. Fix any $s \in S$, and let r be an arbitrary element in $\{1, 2, \ldots, 2s\}$. Then, Property 2 implies that the set $Z_r = \{r + 2as \colon a \geqslant 0\}$ contains at most one number exceeding N with no even representation. Therefore, Z_r contains finitely many positive integers with no even representation, and so does $\mathbb{Z}_{>0} = \bigcup_{r=1}^{2s} Z_r$. □

In view of Properties 1 and 3, we may assume that N is chosen such that every $n > N$ has exactly one odd and exactly one even representation. In particular, each element $s > N$ of S has an even representation.

Property 4. For any $s, t \in S$ with $N < s < t$, the even representation of t contains s.

Proof. Suppose the contrary. Then, $s+t$ has at least two odd representations: one obtained by adding s to the even representation of t and one obtained by adding t to the even representation of s. Since the latter does not contain s, these two odd representations of $s + t$ are distinct, a contradiction. □

Let $s_1 < s_2 < \cdots$ be all the elements of S, and set $\sigma_n = \sum_{i=1}^{n} s_i$ for each nonnegative integer n. Fix an integer k such that $s_k > N$. Then, Property 4 implies that for every $i > k$ the even representation of s_i contains all the numbers $s_k, s_{k+1}, \ldots, s_{i-1}$. Therefore,

$$s_i = s_k + s_{k+1} + \cdots + s_{i-1} + R_i = \sigma_{i-1} - \sigma_{k-1} + R_i, \tag{1}$$

where R_i is a sum of some of $s_1, \ldots, s_{k-1}$. In particular, $0 \leqslant R_i \leqslant s_1 + \cdots + s_{k-1} = \sigma_{k-1}$.

Let j_0 be an integer satisfying $j_0 > k$ and $\sigma_{j_0} > 2\sigma_{k-1}$. Then (1) shows that, for every $j > j_0$,

$$s_{j+1} \geqslant \sigma_j - \sigma_{k-1} > \sigma_j/2. \tag{2}$$

Next, let $p > j_0$ be an index such that $R_p = \min_{i>j_0} R_i$. Then,

$$s_{p+1} = s_k + s_{k+1} + \cdots + s_p + R_{p+1} = (s_p - R_p) + s_p + R_{p+1} \geqslant 2s_p.$$

Therefore, there is no element of S larger than s_p but smaller than $2s_p$. It follows that the even representation τ of $2s_p$ does not contain any element larger than s_p. On the other hand, inequality (2) yields $2s_p > s_1 + \cdots + s_{p-1}$, so τ must contain a term larger than s_{p-1}. Thus, it must contain s_p. After removing s_p from τ, we have that s_p has an odd representation not containing s_p, which contradicts Property 1 since s_p itself also forms an odd representation of s_p.

Solution 2. We will also use Property 1 from Solution 1.

We first define some terminology and notations used in this solution. Let $\mathbb{Z}_{\geqslant 0}$ denote the set of all nonnegative integers. All sums mentioned are regarded as sums of distinct elements of S. Moreover, a sum is called *even* or *odd* depending on the parity of the number of terms in it. All closed or open intervals refer to sets of all integers inside them, e.g., $[a, b] = \{x \in \mathbb{Z} \colon a \leqslant x \leqslant b\}$.

Again, let $s_1 < s_2 < \cdots$ be all elements of S, and denote $\sigma_n = \sum_{i=1}^{n} s_i$ for each positive integer n. Let O_n (respectively, E_n) be the set of numbers representable as an odd (respectively, even) sum of elements of $\{s_1, \ldots, s_n\}$. Set $E = \bigcup_{n=1}^{\infty} E_n$ and $O = \bigcup_{n=1}^{\infty} O_n$. We assume that $0 \in E_n$ since 0 is representable as a sum of 0 terms.

We now proceed to our proof. Assume, to the contrary, that there exist only finitely many positive integers that are not clean and denote the number of non-clean positive integers by $m - 1$. Clearly, S is infinite. By Property 1 from Solution 1, every positive integer n has at most one odd and at most one even representation.

Step 1. We estimate s_{n+1} and σ_{n+1}.

Upper bounds: Property 1 yields $|O_n| = |E_n| = 2^{n-1}$, so $\left|[1, 2^{n-1} + m] \setminus O_n\right| \geqslant m$. Hence, there exists a clean integer $x_n \in [1, 2^{n-1} + m] \setminus O_n$. The definition of O_n then yields that the odd representation of x_n contains a term larger than s_n. Therefore, $s_{n+1} \leqslant x_n \leqslant 2^{n-1} + m$ for every positive integer n. Moreover, since s_1 is the smallest clean number, we get $\sigma_1 = s_1 \leqslant m$. Then,

$$\sigma_{n+1} = \sum_{i=2}^{n+1} s_i + s_1 \leqslant \sum_{i=2}^{n+1} (2^{i-2} + m) + m = 2^n - 1 + (n+1)m$$

for every positive integer n. Notice that this estimate also holds for $n = 0$.

Lower bounds: Since $O_{n+1} \subseteq [1, \sigma_{n+1}]$, we have $\sigma_{n+1} \geqslant |O_{n+1}| = 2^n$ for all positive integers n. Then,

$$s_{n+1} = \sigma_{n+1} - \sigma_n \geqslant 2^n - (2^{n-1} - 1 + nm) = 2^{n-1} + 1 - nm$$

for every positive integer n.

Combining the above inequalities, we have

$$2^{n-1} + 1 - nm \leqslant s_{n+1} \leqslant 2^{n-1} + m \qquad \text{and} \qquad 2^n \leqslant \sigma_{n+1} \leqslant 2^n - 1 + (n+1)m, \tag{3}$$

for every positive integer n.

Step 2. We prove Property 3 from Solution 1.

For every integer x and set of integers Y, define $x \pm Y = \{x \pm y \colon y \in Y\}$.

In view of Property 1, we get

$$E_{n+1} = E_n \sqcup (s_{n+1} + O_n) \qquad \text{and} \qquad O_{n+1} = O_n \sqcup (s_{n+1} + E_n),$$

where $\sqcup$ denotes the disjoint union operator. Notice also that $s_{n+2} \geqslant 2^n + 1 - (n+1)m > 2^{n-1} - 1 + nm \geqslant \sigma_n$ for every sufficiently large n. We now claim the following.

Claim 1. $(\sigma_n - s_{n+1}, s_{n+2} - s_{n+1}) \subseteq E_n$ for every sufficiently large n.

Proof. For sufficiently large n, all elements of (σ_n, s_{n+2}) are clean. Clearly, the elements of (σ_n, s_{n+2}) can be in neither O_n nor $O \setminus O_{n+1}$. So, $(\sigma_n, s_{n+2}) \subseteq O_{n+1} \setminus O_n = s_{n+1} + E_n$, which yields the claim. □

Now, Claim 1 together with inequalities (3) implies that, for all sufficiently large n,

$$E \supseteq E_n \supseteq (\sigma_n - s_{n+1}, s_{n+2} - s_{n+1}) \supseteq \big(2nm, 2^{n-1} - (n+2)m\big).$$

This easily yields that $\mathbb{Z}_{\geqslant 0} \setminus E$ is also finite. Since $\mathbb{Z}_{\geqslant 0} \setminus O$ is also finite, by Property 1, there exists a positive integer N such that every integer $n > N$ has exactly one even and one odd representation.

Step 3. We investigate the structures of E_n and O_n.

Suppose that $z \in E_{2n}$. Since z can be represented as an even sum using $\{s_1, s_2, \ldots, s_{2n}\}$, so can its complement $\sigma_{2n} - z$. Thus, we get $E_{2n} = \sigma_{2n} - E_{2n}$. Similarly, we have

$$E_{2n} = \sigma_{2n} - E_{2n}, \quad O_{2n} = \sigma_{2n} - O_{2n}, \qquad E_{2n+1} = \sigma_{2n+1} - O_{2n+1}, \quad O_{2n+1} = \sigma_{2n+1} - E_{2n+1}. \tag{4}$$

Claim 2. For every sufficiently large n, we have

$$[0, \sigma_n] \supseteq O_n \supseteq (N, \sigma_n - N) \qquad \text{and} \qquad [0, \sigma_n] \supseteq E_n \supseteq (N, \sigma_n - N).$$

Proof. Clearly $O_n, E_n \subseteq [0, \sigma_n]$ for every positive integer n. We now prove $O_n, E_n \supseteq (N, \sigma_n - N)$. Taking n sufficiently large, we may assume that $s_{n+1} \geqslant 2^{n-1} + 1 - nm > \frac{1}{2}(2^{n-1} - 1 + nm) \geqslant \sigma_n/2$. Therefore, the odd representation of every element of $(N, \sigma_n/2]$ cannot contain a term larger than s_n. Thus, $(N, \sigma_n/2] \subseteq O_n$. Similarly, since $s_{n+1} + s_1 > \sigma_n/2$, we also have $(N, \sigma_n/2] \subseteq E_n$. Equations (4) then yield that, for sufficiently large n, the interval $(N, \sigma_n - N)$ is a subset of both O_n and E_n, as desired. □

Step 4. We obtain a final contradiction.

Notice that $0 \in \mathbb{Z}_{\geqslant 0} \setminus O$ and $1 \in \mathbb{Z}_{\geqslant 0} \setminus E$. Therefore, the sets $\mathbb{Z}_{\geqslant 0} \setminus O$ and $\mathbb{Z}_{\geqslant 0} \setminus E$ are nonempty. Denote $o = \max(\mathbb{Z}_{\geqslant 0} \setminus O)$ and $e = \max(\mathbb{Z}_{\geqslant 0} \setminus E)$. Observe also that $e, o \leqslant N$.

Taking k sufficiently large, we may assume that $\sigma_{2k} > 2N$ and that Claim 2 holds for all $n \geqslant 2k$. Due to (4) and Claim 2, we have that $\sigma_{2k} - e$ is the minimal number greater than N which is not in E_{2k}, i.e., $\sigma_{2k} - e = s_{2k+1} + s_1$. Similarly,

$$\sigma_{2k} - o = s_{2k+1}, \qquad \sigma_{2k+1} - e = s_{2k+2}, \qquad \text{and} \qquad \sigma_{2k+1} - o = s_{2k+2} + s_1.$$

Therefore, we have

$$\begin{aligned} s_1 &= (s_{2k+1} + s_1) - s_{2k+1} = (\sigma_{2k} - e) - (\sigma_{2k} - o) = o - e \\ &= (\sigma_{2k+1} - e) - (\sigma_{2k+1} - o) = s_{2k+2} - (s_{2k+2} + s_1) = -s_1, \end{aligned}$$

which is impossible since $s_1 > 0$.

C7. In a company of people some pairs are enemies. A group of people is called *unsociable* if the number of members in the group is odd and at least 3, and it is possible to arrange all its members around a round table so that every two neighbors are enemies. Given that there are at most 2015 unsociable groups, prove that it is possible to partition the company into 11 parts so that no two enemies are in the same part.

(Russia)

Solution 1. Let $G = (V, E)$ be a graph where V is the set of people in the company and E is the set of the enemy pairs — the edges of the graph. In this language, partitioning into 11 disjoint enemy-free subsets means properly coloring the vertices of this graph with 11 colors.

We will prove the following more general statement.

Claim. Let G be a graph with chromatic number $k \geqslant 3$. Then G contains at least $2^{k-1} - k$ unsociable groups.

Recall that the chromatic number of G is the least k such that a proper coloring

$$V = V_1 \sqcup \cdots \sqcup V_k \tag{1}$$

exists. In view of $2^{11} - 12 > 2015$, the claim implies the problem statement.

Let G be a graph with chromatic number k. We say that a proper coloring (1) of G is *leximinimal*, if the k-tuple $(|V_1|, |V_2|, \ldots, |V_k|)$ is lexicographically minimal; in other words, the following conditions are satisfied: the number $n_1 = |V_1|$ is minimal; the number $n_2 = |V_2|$ is minimal, subject to the previously chosen value of n_1; …; the number $n_{k-1} = |V_{k-1}|$ is minimal, subject to the previously chosen values of $n_1, \ldots, n_{k-2}$.

The following lemma is the core of the proof.

Lemma 1. Suppose that $G = (V, E)$ is a graph with odd chromatic number $k \geqslant 3$, and let (1) be one of its leximinimal colorings. Then G contains an odd cycle which visits all color classes $V_1, V_2, \ldots, V_k$.

Proof of Lemma 1. Let us call a cycle *colorful* if it visits all color classes.

Due to the definition of the chromatic number, V_1 is nonempty. Choose an arbitrary vertex $v \in V_1$. We construct a colorful odd cycle that has only one vertex in V_1, and this vertex is v.

We draw a subgraph of G as follows. Place v in the center, and arrange the sets $V_2, V_3, \ldots, V_k$ in counterclockwise circular order around it. For convenience, let $V_{k+1} = V_2$. We will draw arrows to add direction to some edges of G, and mark the vertices these arrows point to. First we draw arrows from v to all its neighbors in V_2, and mark all those neighbors. If some vertex $u \in V_i$ with $i \in \{2, 3, \ldots, k\}$ is already marked, we draw arrows from u to all its neighbors in V_{i+1} which are not marked yet, and we mark all of them. We proceed doing this as long as it is possible. The process of marking is exemplified in Figure 1.

Notice that by the rules of our process, in the final state, marked vertices in V_i cannot have unmarked neighbors in V_{i+1}. Moreover, v is connected to all marked vertices by directed paths.

Now move each marked vertex to the next color class in circular order (see an example in Figure 3). In view of the arguments above, the obtained coloring $V_1 \sqcup W_2 \sqcup \cdots \sqcup W_k$ is proper. Notice that v has a neighbor $w \in W_2$, because otherwise

$$\big(V_1 \setminus \{v\}\big) \sqcup \big(W_2 \cup \{v\}\big) \sqcup W_3 \sqcup \cdots \sqcup W_k$$

would be a proper coloring lexicographically smaller than (1). If w was unmarked, i.e., w was an element of V_2, then it would be marked at the beginning of the process and thus moved to V_3, which did not happen. Therefore, w is marked and $w \in V_k$.

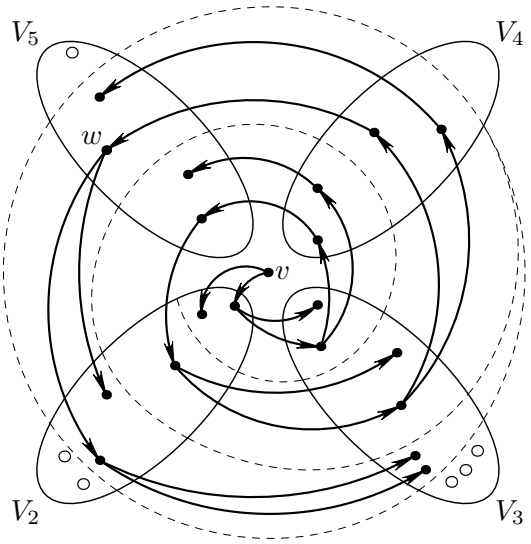


Figure 1

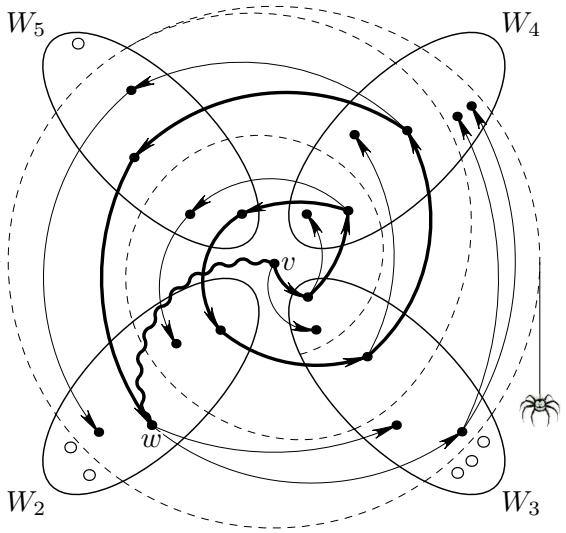


Figure 2

Since w is marked, there exists a directed path from v to w. This path moves through the sets $V_2, \ldots, V_k$ in circular order, so the number of edges in it is divisible by $k-1$ and thus even. Closing this path by the edge $w \to v$, we get a colorful odd cycle, as required. □

Proof of the claim. Let us choose a leximinimal coloring (1) of G. For every set $C \subseteq \{1, 2, \ldots, k\}$ such that $|C|$ is odd and greater than 1, we will provide an odd cycle visiting exactly those color classes whose indices are listed in the set C. This property ensures that we have different cycles for different choices of C, and it proves the claim because there are $2^{k-1} - k$ choices for the set C.

Let $V_C = \bigcup_{c \in C} V_c$, and let G_C be the induced subgraph of G on the vertex set V_C. We also have the induced coloring of V_C with $|C|$ colors; this coloring is of course proper. Notice further that the induced coloring is leximinimal: if we had a lexicographically smaller coloring $(W_c)_{c \in C}$ of G_C, then these classes, together the original color classes V_i for $i \notin C$, would provide a proper coloring which is lexicographically smaller than (1). Hence Lemma 1, applied to the subgraph G_C and its leximinimal coloring $(V_c)_{c \in C}$, provides an odd cycle that visits exactly those color classes that are listed in the set C. □

Solution 2. We provide a different proof of the claim from the previous solution.

We say that a graph is *critical* if deleting any vertex from the graph decreases the graph's chromatic number. Obviously every graph contains a critical induced subgraph with the same chromatic number.

Lemma 2. Suppose that $G = (V, E)$ is a critical graph with chromatic number $k \geqslant 3$. Then every vertex v of G is contained in at least $2^{k-2} - 1$ unsociable groups.

Proof. For every set $X \subseteq V$, denote by $n(X)$ the number of neighbors of v in the set X.

Since G is critical, there exists a proper coloring of $G \setminus \{v\}$ with $k-1$ colors, so there exists a proper coloring $V = V_1 \sqcup V_2 \sqcup \cdots \sqcup V_k$ of G such that $V_1 = \{v\}$. Among such colorings, take one for which the sequence $\big(n(V_2), n(V_3), \ldots, n(V_k)\big)$ is lexicographically minimal. Clearly, $n(V_i) > 0$ for every $i = 2, 3, \ldots, k$; otherwise $V_2 \sqcup \ldots \sqcup V_{i-1} \sqcup (V_i \cup V_1) \sqcup V_{i+1} \sqcup \ldots V_k$ would be a proper coloring of G with $k-1$ colors.

We claim that for every $C \subseteq \{2, 3, \ldots, k\}$ with $|C| \geqslant 2$ being even, G contains an unsociable group so that the set of its members' colors is precisely $C \cup \{1\}$. Since the number of such sets C is $2^{k-2} - 1$, this proves the lemma. Denote the elements of C by $c_1, \ldots, c_{2\ell}$ in increasing order. For brevity, let $U_i = V_{c_i}$. Denote by N_i the set of neighbors of v in U_i.

We show that for every $i = 1, \ldots, 2\ell - 1$ and $x \in N_i$, the subgraph induced by $U_i \cup U_{i+1}$ contains a path that connects x with another point in N_{i+1}. For the sake of contradiction, suppose that no such path exists. Let S be the set of vertices that lie in the connected component of x in the subgraph induced by $U_i \cup U_{i+1}$, and let $P = U_i \cap S$, and $Q = U_{i+1} \cap S$ (see Figure 3). Since x is separated from N_{i+1}, the sets Q and N_{i+1} are disjoint. So, if we re-color G by replacing U_i and U_{i+1} by $(U_i \cup Q) \setminus P$ and $(U_{i+1} \cup P) \setminus Q$, respectively, we obtain a proper coloring such that $n(U_i) = n(V_{c_i})$ is decreased and only $n(U_{i+1}) = n(V_{c_{i+1}})$ is increased. That contradicts the lexicographical minimality of $\big(n(V_2), n(V_3), \ldots, n(V_k)\big)$.

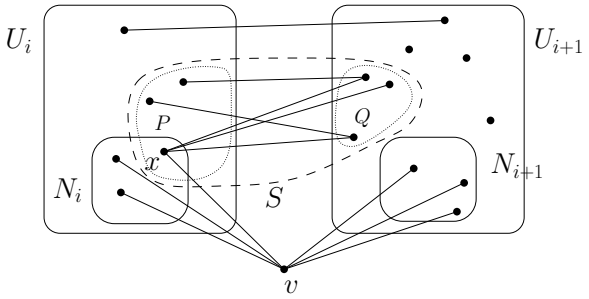


Figure 3

Next, we build a path through $U_1, U_2, \ldots, U_{2\ell}$ as follows. Let the starting point of the path be an arbitrary vertex v_1 in the set N_1. For $i \leqslant 2\ell - 1$, if the vertex $v_i \in N_i$ is already defined, connect v_i to some vertex in N_{i+1} in the subgraph induced by $U_i \cup U_{i+1}$, and add these edges to the path. Denote the new endpoint of the path by v_{i+1}; by the construction we have $v_{i+1} \in N_{i+1}$ again, so the process can be continued. At the end we have a path that starts at $v_1 \in N_1$ and ends at some $v_{2\ell} \in N_{2\ell}$. Moreover, all edges in this path connect vertices in neighboring classes: if a vertex of the path lies in U_i, then the next vertex lies in U_{i+1} or U_{i-1}. Notice that the path is not necessary simple, so take a minimal subpath of it. The minimal subpath is simple and connects the same endpoints v_1 and $v_{2\ell}$. The property that every edge steps to a neighboring color class (i.e., from U_i to U_{i+1} or U_{i-1}) is preserved. So the resulting path also visits all of $U_1, \ldots, U_{2\ell}$, and its length must be odd. Closing the path with the edges vv_1 and $v_{2\ell}v$ we obtain the desired odd cycle (see Figure 4). □

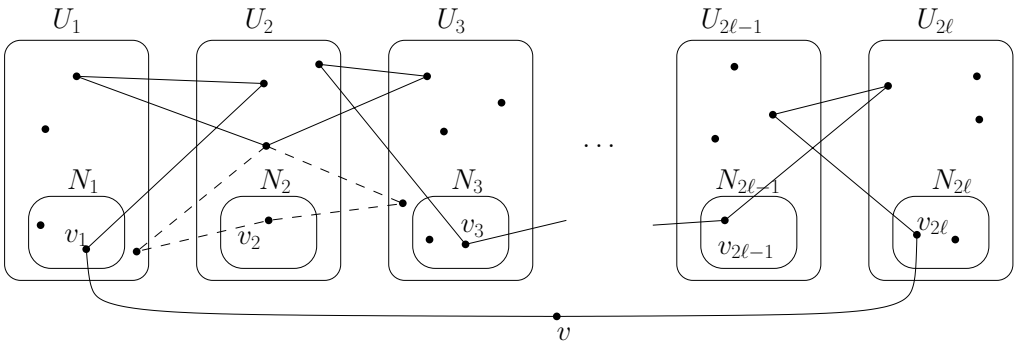


Figure 4

Now we prove the claim by induction on $k \geqslant 3$. The base case $k = 3$ holds by applying Lemma 2 to a critical subgraph. For the induction step, let G_0 be a critical k-chromatic subgraph of G, and let v be an arbitrary vertex of G_0. By Lemma 2, G_0 has at least $2^{k-2} - 1$ unsociable groups containing v. On the other hand, the graph $G_0 \setminus \{v\}$ has chromatic number $k - 1$, so it contains at least $2^{k-2} - (k - 1)$ unsociable groups by the induction hypothesis. Altogether, this gives $2^{k-2} - 1 + 2^{k-2} - (k - 1) = 2^{k-1} - k$ distinct unsociable groups in G_0 (and thus in G).

Comment 1. The claim we proved is sharp. The complete graph with k vertices has chromatic number k and contains exactly $2^{k-1} - k$ unsociable groups.

Comment 2. The proof of Lemma 2 works for odd values of $|C| \geqslant 3$ as well. Hence, the second solution shows the analogous statement that the number of even sized unsociable groups is at least $2^k - 1 - \binom{k}{2}$.

This page is intentionally left blank

Geometry

G1. Let ABC be an acute triangle with orthocenter H. Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI. Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J. Prove that $IJ = AH$.

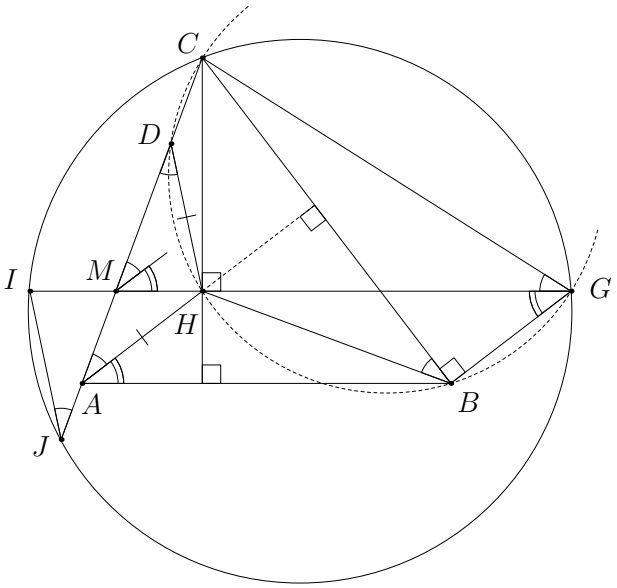
(Australia)

Solution 1. Since $HG \parallel AB$ and $BG \parallel AH$, we have $BG \perp BC$ and $CH \perp GH$. Therefore, the quadrilateral $BGCH$ is cyclic. Since H is the orthocenter of the triangle ABC, we have $\angle HAC = 90^\circ - \angle ACB = \angle CBH$. Using that $BGCH$ and $CGJI$ are cyclic quadrilaterals, we get

$$\angle CJI = \angle CGH = \angle CBH = \angle HAC.$$

Let M be the intersection of AC and GH, and let $D \neq A$ be the point on the line AC such that $AH = HD$. Then $\angle MJI = \angle HAC = \angle MDH$.

Since $\angle MJI = \angle MDH$, $\angle IMJ = \angle HMD$, and $IM = MH$, the triangles IMJ and HMD are congruent, and thus $IJ = HD = AH$.



Comment. Instead of introducing the point D, one can complete the solution by using the law of sines in the triangles IJM and AMH, yielding

$$\frac{IJ}{IM} = \frac{\sin\angle IMJ}{\sin\angle MJI} = \frac{\sin\angle AMH}{\sin\angle HAM} = \frac{AH}{MH} = \frac{AH}{IM}.$$

Solution 2. Obtain $\angle CGH = \angle HAC$ as in the previous solution. In the parallelogram $ABGH$ we have $\angle BAH = \angle HGB$. It follows that

$$\angle HMC = \angle BAC = \angle BAH + \angle HAC = \angle HGB + \angle CGH = \angle CGB.$$

So the right triangles CMH and CGB are similar. Also, in the circumcircle of triangle GCI we have similar triangles MIJ and MCG. Therefore,

$$\frac{IJ}{CG} = \frac{MI}{MC} = \frac{MH}{MC} = \frac{GB}{GC} = \frac{AH}{CG}.$$

Hence $IJ = AH$.

G2. Let ABC be a triangle inscribed into a circle Ω with center O. A circle Γ with center A meets the side BC at points D and E such that D lies between B and E. Moreover, let F and G be the common points of Γ and Ω. We assume that F lies on the arc AB of Ω not containing C, and G lies on the arc AC of Ω not containing B. The circumcircles of the triangles BDF and CEG meet the sides AB and AC again at K and L, respectively. Suppose that the lines FK and GL are distinct and intersect at X. Prove that the points A, X, and O are collinear.
(Greece)

Solution 1. It suffices to prove that the lines FK and GL are symmetric about AO. Now the segments AF and AG, being chords of Ω with the same length, are clearly symmetric with respect to AO. Hence it is enough to show

$$\angle KFA = \angle AGL\,. \tag{1}$$

Let us denote the circumcircles of BDF and CEG by ω_B and ω_C, respectively. To prove (1), we start from

$$\angle KFA = \angle DFG + \angle GFA - \angle DFK\,.$$

In view of the circles ω_B, Γ, and Ω, this may be rewritten as

$$\angle KFA = \angle CEG + \angle GBA - \angle DBK = \angle CEG - \angle CBG\,.$$

Due to the circles ω_C and Ω, we obtain $\angle KFA = \angle CLG - \angle CAG = \angle AGL$. Thereby the problem is solved.

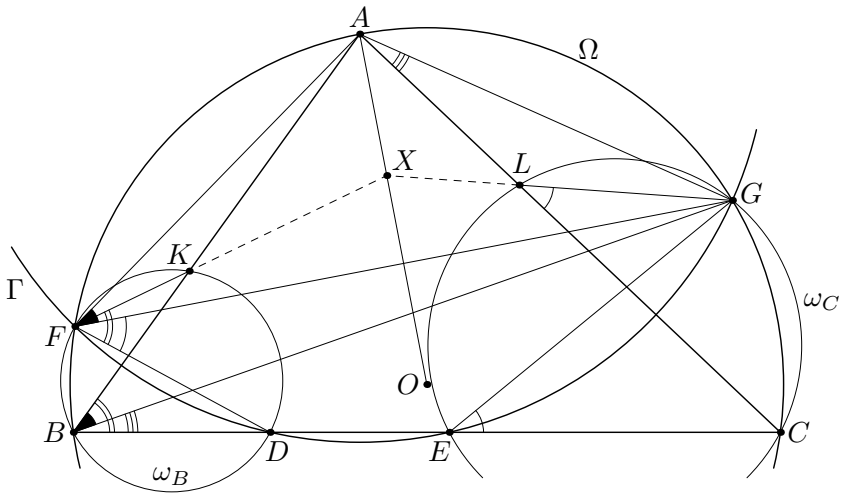


Figure 1

Solution 2. Again, we denote the circumcircle of $BDKF$ by ω_B. In addition, we set $\alpha = \angle BAC$, $\varphi = \angle ABF$, and $\psi = \angle EDA = \angle AED$ (see Figure 2). Notice that $AF = AG$ entails $\varphi = \angle GCA$, so all three of α, φ, and ψ respect the "symmetry" between B and C of our configuration. Again, we reduce our task to proving (1).

This time, we start from

$$2\angle KFA = 2(\angle DFA - \angle DFK)\,.$$

Since the triangle AFD is isosceles, we have

$$\angle DFA = \angle ADF = \angle EDF - \psi = \angle BFD + \angle EBF - \psi\,.$$

Moreover, because of the circle ω_B we have $\angle DFK = \angle CBA$. Altogether, this yields

$$2\angle KFA = \angle DFA + \big(\angle BFD + \angle EBF - \psi\big) - 2\angle CBA\,,$$

which simplifies to

$$2\angle KFA = \angle BFA + \varphi - \psi - \angle CBA\,.$$

Now the quadrilateral $AFBC$ is cyclic, so this entails $2\angle KFA = \alpha + \varphi - \psi$.

Due to the "symmetry" between B and C alluded to above, this argument also shows that $2\angle AGL = \alpha + \varphi - \psi$. This concludes the proof of (1).

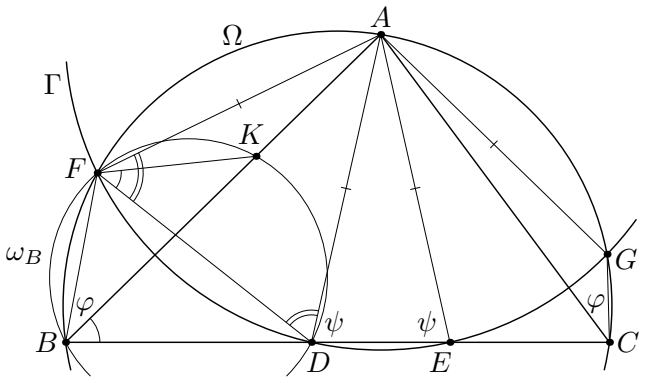


Figure 2

Comment 1. As the first solution shows, the assumption that A be the center of Γ may be weakened to the following one: The center of Γ lies on the line OA. The second solution may be modified to yield the same result.

Comment 2. It might be interesting to remark that $\angle GDK = 90^\circ$. To prove this, let G' denote the point on Γ diametrically opposite to G. Because of $\angle KDF = \angle KBF = \angle AGF = \angle G'DF$, the points D, K, and G' are collinear, which leads to the desired result. Notice that due to symmetry we also have $\angle LEF = 90^\circ$.

Moreover, a standard argument shows that the triangles AGL and BGE are similar. By symmetry again, also the triangles AFK and CDF are similar.

There are several ways to derive a solution from these facts. For instance, one may argue that

$$\begin{aligned}\angle KFA &= \angle BFA - \angle BFK = \angle BFA - \angle EDG' = (180^\circ - \angle AGB) - (180^\circ - \angle G'GE)\\ &= \angle AGE - \angle AGB = \angle BGE = \angle AGL\,.\end{aligned}$$

Comment 3. The original proposal did not contain the point X in the assumption and asked instead to prove that the lines FK, GL, and AO are concurrent. This differs from the version given above only insofar as it also requires to show that these lines cannot be parallel. The Problem Selection Committee removed this part from the problem intending to make it thus more suitable for the Olympiad.

For the sake of completeness, we would still like to sketch one possibility for proving $FK \nparallel AO$ here. As the points K and O lie in the angular region $\angle FAG$, it suffices to check $\angle KFA + \angle FAO < 180^\circ$. Multiplying by 2 and making use of the formulae from the second solution, we see that this is equivalent to $(\alpha + \varphi - \psi) + (180^\circ - 2\varphi) < 360^\circ$, which in turn is an easy consequence of $\alpha < 180^\circ$.

G3. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C. A point D is chosen inside the triangle CBH so that CH bisects AD. Let P be the intersection point of the lines BD and CH. Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q. Prove that the lines CQ and AD meet on ω.

(Georgia)

Solution 1. Let K be the projection of D onto AB; then $AH = HK$ (see Figure 1). Since $PH \parallel DK$, we have

$$\frac{PD}{PB} = \frac{HK}{HB} = \frac{AH}{HB}. \tag{1}$$

Let L be the projection of Q onto DB. Since PQ is tangent to ω and $\angle DQB = \angle BLQ = 90^\circ$, we have $\angle PQD = \angle QBP = \angle DQL$. Therefore, QD and QB are respectively the internal and the external bisectors of $\angle PQL$. By the angle bisector theorem, we obtain

$$\frac{PD}{DL} = \frac{PQ}{QL} = \frac{PB}{BL}. \tag{2}$$

The relations (1) and (2) yield $\dfrac{AH}{HB} = \dfrac{PD}{PB} = \dfrac{DL}{LB}$. So, the spiral similarity τ centered at B and sending A to D maps H to L. Moreover, τ sends the semicircle with diameter AB passing through C to ω. Due to $CH \perp AB$ and $QL \perp DB$, it follows that $\tau(C) = Q$.

Hence, the triangles ABD and CBQ are similar, so $\angle ADB = \angle CQB$. This means that the lines AD and CQ meet at some point T, and this point satisfies $\angle BDT = \angle BQT$. Therefore, T lies on ω, as needed.

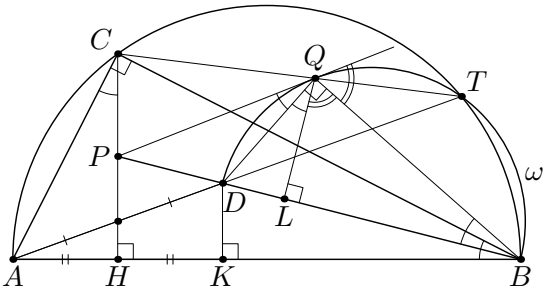


Figure 1

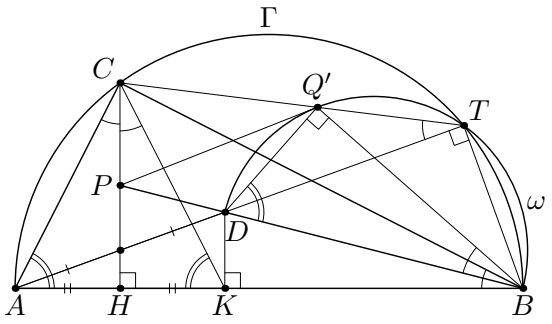


Figure 2

Comment 1. Since $\angle BAD = \angle BCQ$, the point T lies also on the circumcircle of the triangle ABC.

Solution 2. Let Γ be the circumcircle of ABC, and let AD meet ω at T. Then $\angle ATB = \angle ACB = 90^\circ$, so T lies on Γ as well. As in the previous solution, let K be the projection of D onto AB; then $AH = HK$ (see Figure 2).

Our goal now is to prove that the points C, Q, and T are collinear. Let CT meet ω again at Q'. Then, it suffices to show that PQ' is tangent to ω, or that $\angle PQ'D = \angle Q'BD$.

Since the quadrilateral $BDQ'T$ is cyclic and the triangles AHC and KHC are congruent, we have $\angle Q'BD = \angle Q'TD = \angle CTA = \angle CBA = \angle ACH = \angle HCK$. Hence, the right triangles CHK and $BQ'D$ are similar. This implies that $\dfrac{HK}{CK} = \dfrac{Q'D}{BD}$, and thus $HK \cdot BD = CK \cdot Q'D$.

Notice that $PH \parallel DK$; therefore, we have $\dfrac{PD}{BD} = \dfrac{HK}{BK}$, and so $PD \cdot BK = HK \cdot BD$. Consequently, $PD \cdot BK = HK \cdot BD = CK \cdot Q'D$, which yields $\dfrac{PD}{Q'D} = \dfrac{CK}{BK}$.

Since $\angle CKA = \angle KAC = \angle BDQ'$, the triangles CKB and PDQ' are similar, so $\angle PQ'D = \angle CBA = \angle Q'BD$, as required.

Comment 2. There exist several other ways to prove that PQ' is tangent to ω. For instance, one may compute $\dfrac{PD}{PB}$ and $\dfrac{PQ'}{PB}$ in terms of AH and HB to verify that $PQ'^2 = PD \cdot PB$, concluding that PQ' is tangent to ω.

Another possible approach is the following. As in Solution 2, we introduce the points T and Q' and mention that the triangles ABC and DBQ' are similar (see Figure 3).

Let M be the midpoint of AD, and let L be the projection of Q' onto AB. Construct E on the line AB so that EP is parallel to AD. Projecting from P, we get $(A, B; H, E) = (A, D; M, \infty) = -1$.

Since $\dfrac{EA}{AB} = \dfrac{PD}{DB}$, the point P is the image of E under the similarity transform mapping ABC to DBQ'. Therefore, we have $(D, B; L, P) = (A, B; H, E) = -1$, which means that $Q'D$ and $Q'B$ are respectively the internal and the external bisectors of $\angle PQ'L$. This implies that PQ' is tangent to ω, as required.

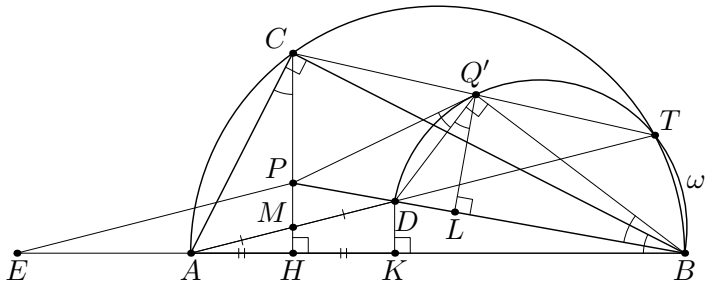


Figure 3

Solution 3. Introduce the points T and Q' as in the previous solution. Note that T lies on the circumcircle of ABC. Here we present yet another proof that PQ' is tangent to ω.

Let Ω be the circle completing the semicircle ω. Construct a point F symmetric to C with respect to AB. Let $S \neq T$ be the second intersection point of FT and Ω (see Figure 4).

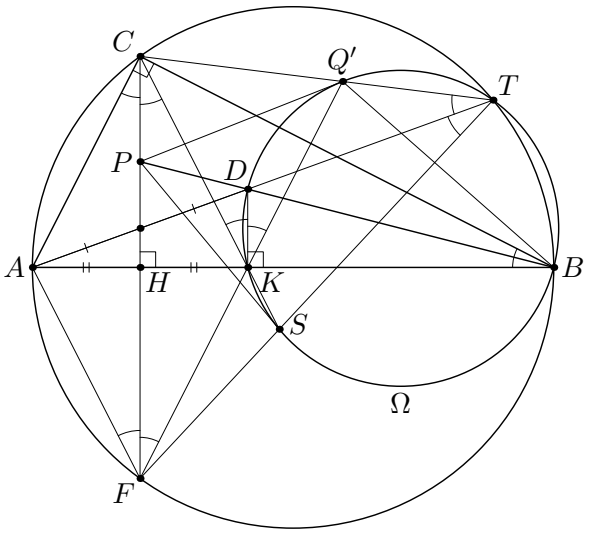


Figure 4

Since $AC = AF$, we have $\angle DKC = \angle HCK = \angle CBA = \angle CTA = \angle DTS = 180^\circ - \angle SKD$. Thus, the points C, K, and S are collinear. Notice also that $\angle Q'KD = \angle Q'TD = \angle HCK = \angle KFH = 180^\circ - \angle DKF$. This implies that the points F, K, and Q' are collinear.

Applying Pascal's theorem to the degenerate hexagon $KQ'Q'TSS$, we get that the tangents to Ω passing through Q' and S intersect on CF. The relation $\angle Q'TD = \angle DTS$ yields that Q' and S are symmetric with respect to BD. Therefore, the two tangents also intersect on BD. Thus, the two tangents pass through P. Hence, PQ' is tangent to ω, as needed.

G4. Let ABC be an acute triangle, and let M be the midpoint of AC. A circle ω passing through B and M meets the sides AB and BC again at P and Q, respectively. Let T be the point such that the quadrilateral $BPTQ$ is a parallelogram. Suppose that T lies on the circumcircle of the triangle ABC. Determine all possible values of BT/BM.

(Russia)

Answer. $\sqrt{2}$.

Solution 1. Let S be the center of the parallelogram $BPTQ$, and let $B' \neq B$ be the point on the ray BM such that $BM = MB'$ (see Figure 1). It follows that $ABCB'$ is a parallelogram. Then, $\angle ABB' = \angle PQM$ and $\angle BB'A = \angle B'BC = \angle MPQ$, and so the triangles ABB' and MQP are similar. It follows that AM and MS are corresponding medians in these triangles. Hence,

$$\angle SMP = \angle B'AM = \angle BCA = \angle BTA. \tag{1}$$

Since $\angle ACT = \angle PBT$ and $\angle TAC = \angle TBC = \angle BTP$, the triangles TCA and PBT are similar. Again, as TM and PS are corresponding medians in these triangles, we have

$$\angle MTA = \angle TPS = \angle BQP = \angle BMP. \tag{2}$$

Now we deal separately with two cases.

Case 1. S does not lie on BM. Since the configuration is symmetric between A and C, we may assume that S and A lie on the same side with respect to the line BM.

Applying (1) and (2), we get

$$\angle BMS = \angle BMP - \angle SMP = \angle MTA - \angle BTA = \angle MTB,$$

and so the triangles BSM and BMT are similar. We now have $BM^2 = BS \cdot BT = BT^2/2$, so $BT = \sqrt{2}BM$.

Case 2. S lies on BM. It follows from (2) that $\angle BCA = \angle MTA = \angle BQP = \angle BMP$ (see Figure 2). Thus, $PQ \parallel AC$ and $PM \parallel AT$. Hence, $BS/BM = BP/BA = BM/BT$, so $BT^2 = 2BM^2$ and $BT = \sqrt{2}BM$.

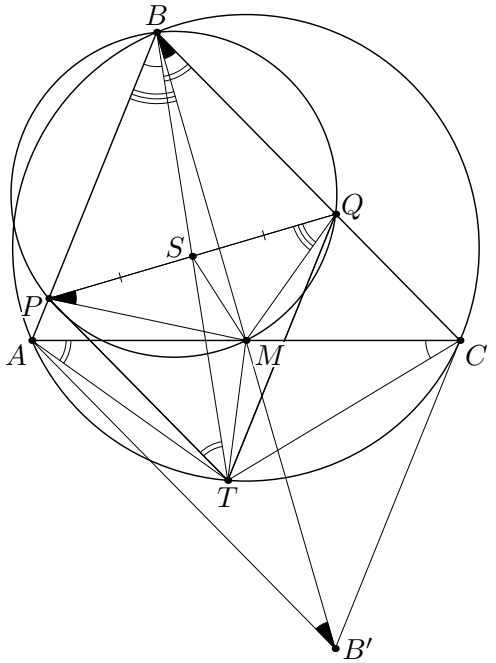


Figure 1

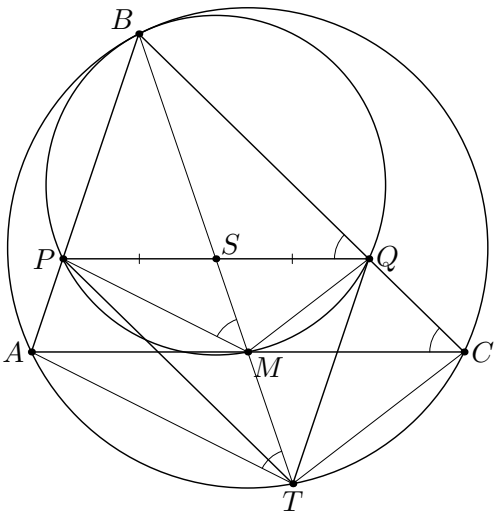


Figure 2

Comment 1. Here is another way to show that the triangles BSM and BMT are similar. Denote by Ω the circumcircle of the triangle ABC. Let R be the second point of intersection of ω and Ω, and let τ be the spiral similarity centered at R mapping ω to Ω. Then, one may show that τ maps each point X on ω to a point Y on Ω such that B, X, and Y are collinear (see Figure 3). If we let K and L be the second points of intersection of BM with Ω and of BT with ω, respectively, then it follows that the triangle MKT is the image of SML under τ. We now obtain $\angle BSM = \angle TMB$, which implies the desired result.

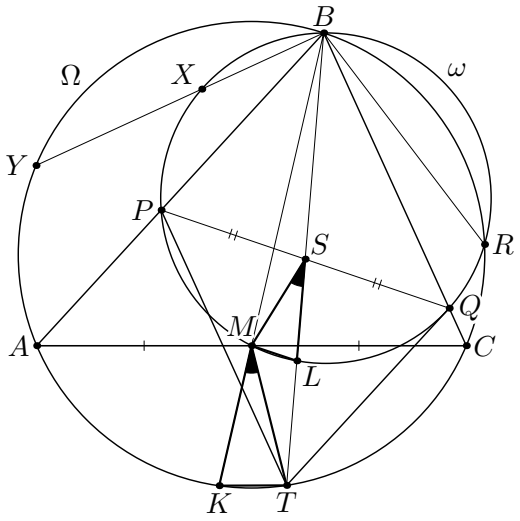


Figure 3

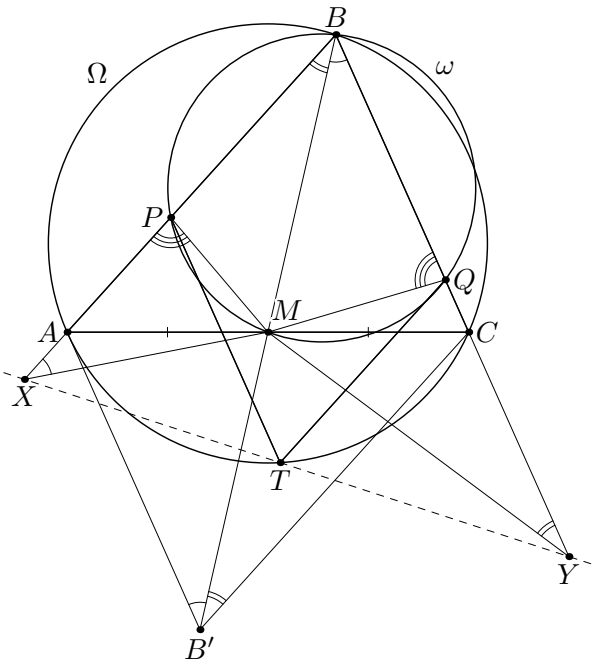


Figure 4

Solution 2. Again, we denote by Ω the circumcircle of the triangle ABC.

Choose the points X and Y on the rays BA and BC respectively, so that $\angle MXB = \angle MBC$ and $\angle BYM = \angle ABM$ (see Figure 4). Then the triangles BMX and YMB are similar. Since $\angle XPM = \angle BQM$, the points P and Q correspond to each other in these triangles. So, if $\overrightarrow{BP} = \mu \cdot \overrightarrow{BX}$, then $\overrightarrow{BQ} = (1-\mu) \cdot \overrightarrow{BY}$. Thus

$$\overrightarrow{BT} = \overrightarrow{BP} + \overrightarrow{BQ} = \overrightarrow{BY} + \mu \cdot (\overrightarrow{BX} - \overrightarrow{BY}) = \overrightarrow{BY} + \mu \cdot \overrightarrow{YX},$$

which means that T lies on the line XY.

Let $B' \neq B$ be the point on the ray BM such that $BM = MB'$. Then $\angle MB'A = \angle MBC = \angle MXB$ and $\angle CB'M = \angle ABM = \angle BYM$. This means that the triangles BMX, BAB', YMB, and $B'CB$ are all similar; hence $BA \cdot BX = BM \cdot BB' = BC \cdot BY$. Thus there exists an inversion centered at B which swaps A with X, M with B', and C with Y. This inversion then swaps Ω with the line XY, and hence it preserves T. Therefore, we have $BT^2 = BM \cdot BB' = 2BM^2$, and $BT = \sqrt{2}BM$.

Solution 3. We begin with the following lemma.

Lemma. Let $ABCT$ be a cyclic quadrilateral. Let P and Q be points on the sides BA and BC respectively, such that $BPTQ$ is a parallelogram. Then $BP \cdot BA + BQ \cdot BC = BT^2$.

Proof. Let the circumcircle of the triangle QTC meet the line BT again at J (see Figure 5). The power of B with respect to this circle yields

$$BQ \cdot BC = BJ \cdot BT. \tag{3}$$

We also have $\angle TJQ = 180^\circ - \angle QCT = \angle TAB$ and $\angle QTJ = \angle ABT$, and so the triangles TJQ and BAT are similar. We now have $TJ/TQ = BA/BT$. Therefore,

$$TJ \cdot BT = TQ \cdot BA = BP \cdot BA. \tag{4}$$

Combining (3) and (4) now yields the desired result. □

Let X and Y be the midpoints of BA and BC respectively (see Figure 6). Applying the lemma to the cyclic quadrilaterals $PBQM$ and $ABCT$, we obtain

$$BX \cdot BP + BY \cdot BQ = BM^2$$

and

$$BP \cdot BA + BQ \cdot BC = BT^2.$$

Since $BA = 2BX$ and $BC = 2BY$, we have $BT^2 = 2BM^2$, and so $BT = \sqrt{2}BM$.

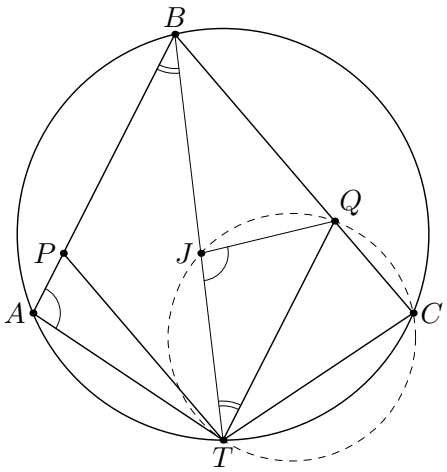


Figure 5

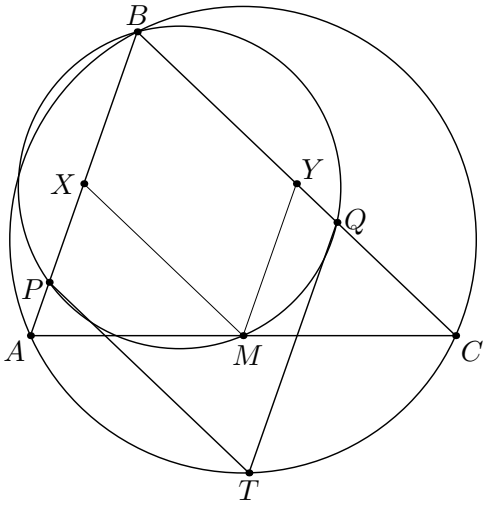


Figure 6

Comment 2. Here we give another proof of the lemma using Ptolemy's theorem. We readily have

$$TC \cdot BA + TA \cdot BC = AC \cdot BT.$$

The lemma now follows from

$$\frac{BP}{TC} = \frac{BQ}{TA} = \frac{BT}{AC} = \frac{\sin \angle BCT}{\sin \angle ABC}.$$

G5. Let ABC be a triangle with $CA \neq CB$. Let D, F, and G be the midpoints of the sides AB, AC, and BC, respectively. A circle Γ passing through C and tangent to AB at D meets the segments AF and BG at H and I, respectively. The points H' and I' are symmetric to H and I about F and G, respectively. The line $H'I'$ meets CD and FG at Q and M, respectively. The line CM meets Γ again at P. Prove that $CQ = QP$.

(El Salvador)

Solution 1. We may assume that $CA > CB$. Observe that H' and I' lie inside the segments CF and CG, respectively. Therefore, M lies outside $\triangle ABC$ (see Figure 1).

Due to the powers of points A and B with respect to the circle Γ, we have

$$CH' \cdot CA = AH \cdot AC = AD^2 = BD^2 = BI \cdot BC = CI' \cdot CB.$$

Therefore, $CH' \cdot CF = CI' \cdot CG$. Hence, the quadrilateral $H'I'GF$ is cyclic, and so $\angle I'H'C = \angle CGF$.

Let DF and DG meet Γ again at R and S, respectively. We claim that the points R and S lie on the line $H'I'$.

Observe that $FH' \cdot FA = FH \cdot FC = FR \cdot FD$. Thus, the quadrilateral $ADH'R$ is cyclic, and hence $\angle RH'F = \angle FDA = \angle CGF = \angle I'H'C$. Therefore, the points R, H', and I' are collinear. Similarly, the points S, H', and I' are also collinear, and so all the points R, H', Q, I', S, and M are all collinear.

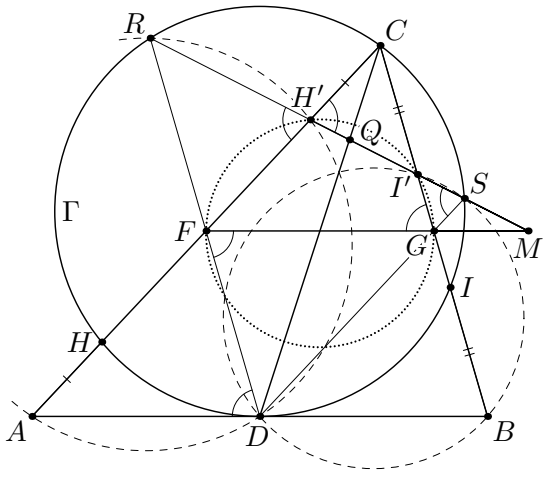


Figure 1

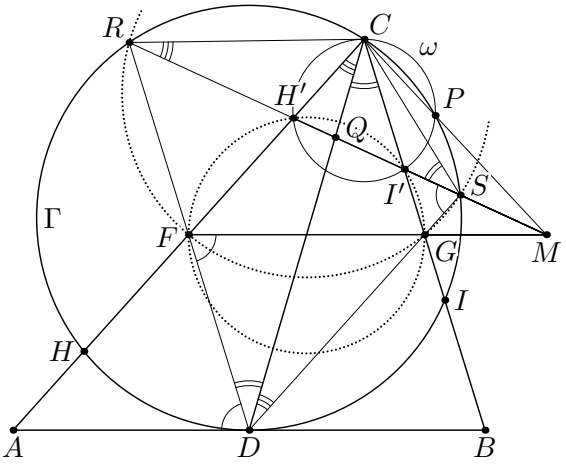


Figure 2

Then, $\angle RSD = \angle RDA = \angle DFG$. Hence, the quadrilateral $RSGF$ is cyclic (see Figure 2). Therefore, $MH' \cdot MI' = MF \cdot MG = MR \cdot MS = MP \cdot MC$. Thus, the quadrilateral $CPI'H'$ is also cyclic. Let ω be its circumcircle.

Notice that $\angle H'CQ = \angle SDC = \angle SRC$ and $\angle QCI' = \angle CDR = \angle CSR$. Hence, $\triangle CH'Q \sim \triangle RCQ$ and $\triangle CI'Q \sim \triangle SCQ$, and therefore $QH' \cdot QR = QC^2 = QI' \cdot QS$.

We apply the inversion with center Q and radius QC. Observe that the points R, C, and S are mapped to H', C, and I', respectively. Therefore, the circumcircle Γ of $\triangle RCS$ is mapped to the circumcircle ω of $\triangle H'CI'$. Since P and C belong to both circles and the point C is preserved by the inversion, we have that P is also mapped to itself. We then get $QP^2 = QC^2$. Hence, $QP = QC$.

Comment 1. The problem statement still holds when Γ intersects the sides CA and CB outside segments AF and BG, respectively.

Solution 2. Let $X = HI \cap AB$, and let the tangent to Γ at C meet AB at Y. Let XC meet Γ again at X' (see Figure 3). Projecting from C, X, and C again, we have $(X, A; D, B) = (X', H; D, I) = (C, I; D, H) = (Y, B; D, A)$. Since A and B are symmetric about D, it follows that X and Y are also symmetric about D.

Now, Menelaus' theorem applied to $\triangle ABC$ with the line HIX yields

$$1 = \frac{CH}{HA} \cdot \frac{BI}{IC} \cdot \frac{AX}{XB} = \frac{AH'}{H'C} \cdot \frac{CI'}{I'B} \cdot \frac{BY}{YA}.$$

By the converse of Menelaus' theorem applied to $\triangle ABC$ with points H', I', Y, we get that the points H', I', Y are collinear.

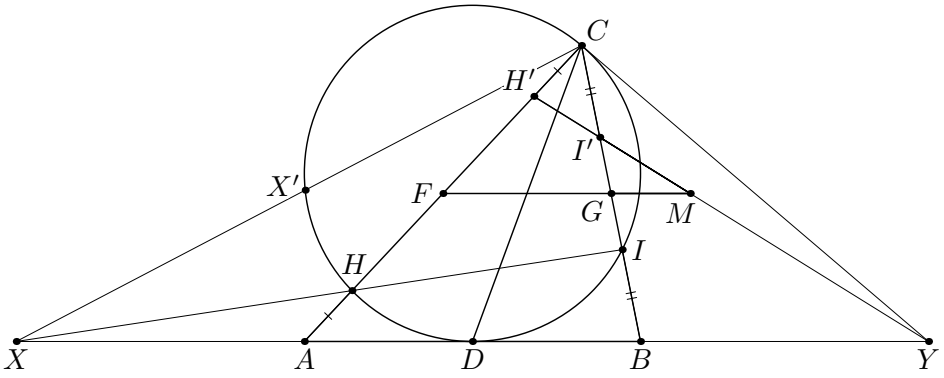


Figure 3

Let T be the midpoint of CD, and let O be the center of Γ. Let CM meet TY at N. To avoid confusion, we clean some superfluous details out of the picture (see Figure 4).

Let $V = MT \cap CY$. Since $MT \parallel YD$ and $DT = TC$, we get $CV = VY$. Then Ceva's theorem applied to $\triangle CTY$ with the point M yields

$$1 = \frac{TQ}{QC} \cdot \frac{CV}{VY} \cdot \frac{YN}{NT} = \frac{TQ}{QC} \cdot \frac{YN}{NT}.$$

Therefore, $\frac{TQ}{QC} = \frac{TN}{NY}$. So, $NQ \parallel CY$, and thus $NQ \perp OC$.

Note that the points O, N, T, and Y are collinear. Therefore, $CQ \perp ON$. So, Q is the orthocenter of $\triangle OCN$, and hence $OQ \perp CP$. Thus, Q lies on the perpendicular bisector of CP, and therefore $CQ = QP$, as required.

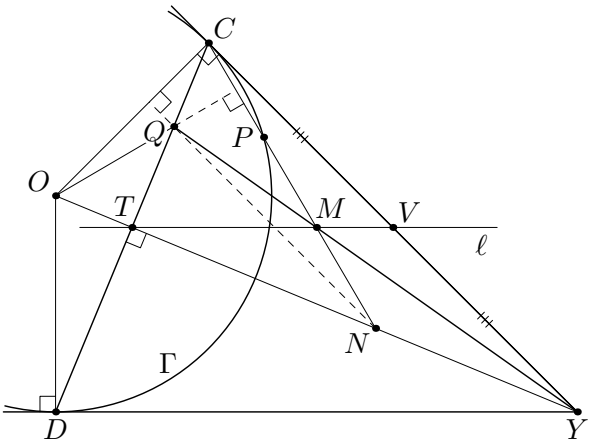


Figure 4

Comment 2. The second part of Solution 2 provides a proof of the following more general statement, which does not involve a specific choice of Q on CD.

Let YC and YD be two tangents to a circle Γ with center O (see Figure 4). Let ℓ be the midline of $\triangle YCD$ parallel to YD. Let Q and M be two points on CD and ℓ, respectively, such that the line QM passes through Y. Then $OQ \perp CM$.

G6. Let ABC be an acute triangle with $AB > AC$, and let Γ be its circumcircle. Let H, M, and F be the orthocenter of the triangle, the midpoint of BC, and the foot of the altitude from A, respectively. Let Q and K be the two points on Γ that satisfy $\angle AQH = 90^\circ$ and $\angle QKH = 90^\circ$. Prove that the circumcircles of the triangles KQH and KFM are tangent to each other.

(Ukraine)

Solution 1. Let A' be the point diametrically opposite to A on Γ. Since $\angle AQA' = 90^\circ$ and $\angle AQH = 90^\circ$, the points Q, H, and A' are collinear. Similarly, if Q' denotes the point on Γ diametrically opposite to Q, then K, H, and Q' are collinear. Let the line AHF intersect Γ again at E; it is known that M is the midpoint of the segment HA' and that F is the midpoint of HE. Let J be the midpoint of HQ'.

Consider any point T such that TK is tangent to the circle KQH at K with Q and T lying on different sides of KH (see Figure 1). Then $\angle HKT = \angle HQK$ and we are to prove that $\angle MKT = \angle CFK$. Thus it remains to show that $\angle HQK = \angle CFK + \angle HKM$. Due to $\angle HQK = 90^\circ - \angle Q'HA'$ and $\angle CFK = 90^\circ - \angle KFA$, this means the same as $\angle Q'HA' = \angle KFA - \angle HKM$. Now, since the triangles KHE and AHQ' are similar with F and J being the midpoints of corresponding sides, we have $\angle KFA = \angle HJA$, and analogously one may obtain $\angle HKM = \angle JQH$. Thereby our task is reduced to verifying

$$\angle Q'HA' = \angle HJA - \angle JQH\,.$$

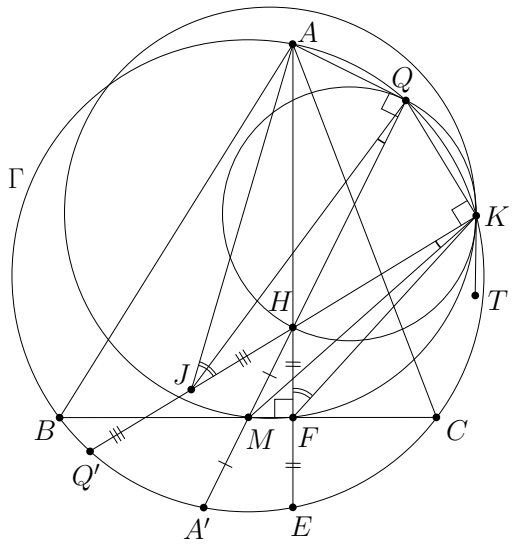


Figure 1

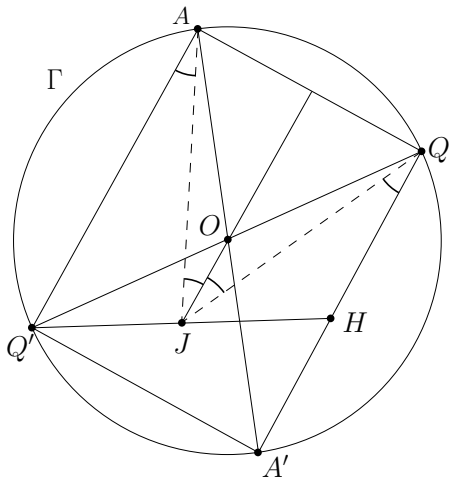


Figure 2

To avoid confusion, let us draw a new picture at this moment (see Figure 2). Owing to $\angle Q'HA' = \angle JQH + \angle HJQ$ and $\angle HJA = \angle QJA + \angle HJQ$, we just have to show that $2\angle JQH = \angle QJA$. To this end, it suffices to remark that $AQA'Q'$ is a rectangle and that J, being defined to be the midpoint of HQ', has to lie on the mid parallel of QA' and $Q'A$.

Solution 2. We define the points A' and E and prove that the ray MH passes through Q in the same way as in the first solution. Notice that the points A' and E can play analogous roles to the points Q and K, respectively: point A' is the second intersection of the line MH with Γ, and E is the point on Γ with the property $\angle HEA' = 90^\circ$ (see Figure 3).

In the circles KQH and $EA'H$, the line segments HQ and HA' are diameters, respectively; so, these circles have a common tangent t at H, perpendicular to MH. Let R be the radical center of the circles ABC, KQH and $EA'H$. Their pairwise radical axes are the lines QK, $A'E$ and the line t; they all pass through R. Let S be the midpoint of HR; by $\angle QKH =$

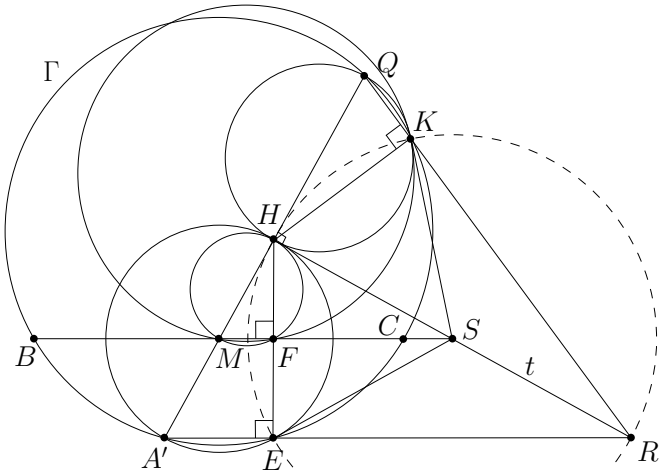


Figure 3

$\angle HEA' = 90^\circ$, the quadrilateral $HERK$ is cyclic and its circumcenter is S; hence we have $SK = SE = SH$. The line BC, being the perpendicular bisector of HE, passes through S.

The circle HMF also is tangent to t at H; from the power of S with respect to the circle HMF we have

$$SM \cdot SF = SH^2 = SK^2.$$

So, the power of S with respect to the circles KQH and KFM is SK^2. Therefore, the line segment SK is tangent to both circles at K.

G7. Let $ABCD$ be a convex quadrilateral, and let P, Q, R, and S be points on the sides AB, BC, CD, and DA, respectively. Let the line segments PR and QS meet at O. Suppose that each of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$ has an incircle. Prove that the lines AC, PQ, and RS are either concurrent or parallel to each other.

(Bulgaria)

Solution 1. Denote by γ_A, γ_B, γ_C, and γ_D the incircles of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$, respectively.

We start with proving that the quadrilateral $ABCD$ also has an incircle which will be referred to as Ω. Denote the points of tangency as in Figure 1. It is well-known that $QQ_1 = OO_1$ (if $BC \parallel PR$, this is obvious; otherwise, one may regard the two circles involved as the incircle and an excircle of the triangle formed by the lines OQ, PR, and BC). Similarly, $OO_1 = PP_1$. Hence we have $QQ_1 = PP_1$. The other equalities of segment lengths marked in Figure 1 can be proved analogously. These equalities, together with $AP_1 = AS_1$ and similar ones, yield $AB + CD = AD + BC$, as required.

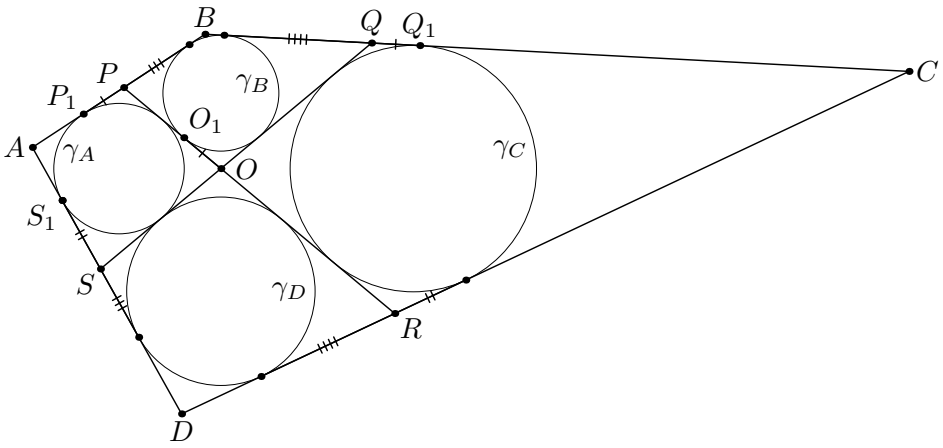


Figure 1

Next, let us draw the lines parallel to QS through P and R, and also draw the lines parallel to PR through Q and S. These lines form a parallelogram; denote its vertices by A', B', C', and D' as shown in Figure 2.

Since the quadrilateral $APOS$ has an incircle, we have $AP - AS = OP - OS = A'S - A'P$. It is well-known that in this case there also exists a circle ω_A tangent to the four rays AP, AS, $A'P$, and $A'S$. It is worth mentioning here that in case when, say, the lines AB and $A'B'$ coincide, the circle ω_A is just tangent to AB at P. We introduce the circles ω_B, ω_C, and ω_D in a similar manner.

Assume that the radii of the circles ω_A and ω_C are different. Let X be the center of the homothety having a positive scale factor and mapping ω_A to ω_C.

Now, Monge's theorem applied to the circles ω_A, Ω, and ω_C shows that the points A, C, and X are collinear. Applying the same theorem to the circles ω_A, ω_B, and ω_C, we see that the points P, Q, and X are also collinear. Similarly, the points R, S, and X are collinear, as required.

If the radii of ω_A and ω_C are equal but these circles do not coincide, then the degenerate version of the same theorem yields that the three lines AC, PQ, and RS are parallel to the line of centers of ω_A and ω_C.

Finally, we need to say a few words about the case when ω_A and ω_C coincide (and thus they also coincide with Ω, ω_B, and ω_D). It may be regarded as the limit case in the following manner.

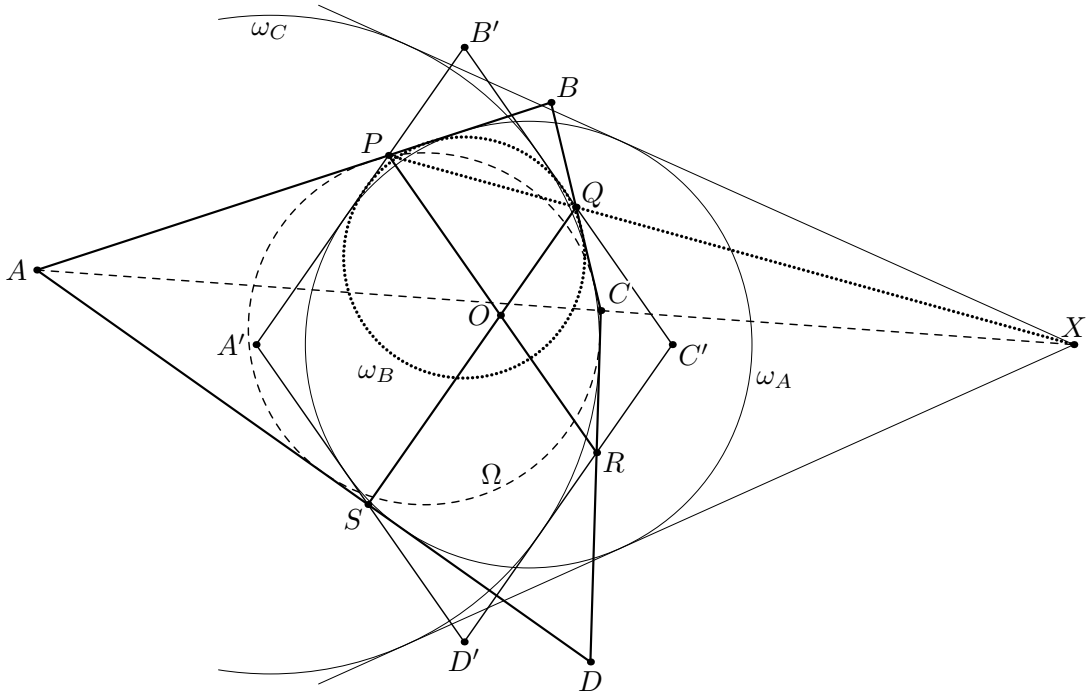


Figure 2

Let us fix the positions of A, P, O, and S (thus we also fix the circles ω_A, γ_A, γ_B, and γ_D). Now we vary the circle γ_C inscribed into $\angle QOR$; for each of its positions, one may reconstruct the lines BC and CD as the external common tangents to γ_B, γ_C and γ_C, γ_D different from PR and QS, respectively. After such variation, the circle Ω changes, so the result obtained above may be applied.

Solution 2. Applying Menelaus' theorem to $\triangle ABC$ with the line PQ and to $\triangle ACD$ with the line RS, we see that the line AC meets PQ and RS at the same point (possibly at infinity) if and only if

$$\frac{AP}{PB}\cdot\frac{BQ}{QC}\cdot\frac{CR}{RD}\cdot\frac{DS}{SA}=1. \tag{1}$$

So, it suffices to prove (1).

We start with the following result.

Lemma 1. Let $EFGH$ be a circumscribed quadrilateral, and let M be its incenter. Then

$$\frac{EF\cdot FG}{GH\cdot HE}=\frac{FM^2}{HM^2}.$$

Proof. Notice that $\angle EMH+\angle GMF=\angle FME+\angle HMG=180^\circ$, $\angle FGM=\angle MGH$, and $\angle HEM=\angle MEF$ (see Figure 3). By the law of sines, we get

$$\frac{EF}{FM}\cdot\frac{FG}{FM}=\frac{\sin\angle FME\cdot\sin\angle GMF}{\sin\angle MEF\cdot\sin\angle FGM}=\frac{\sin\angle HMG\cdot\sin\angle EMH}{\sin\angle MGH\cdot\sin\angle HEM}=\frac{GH}{HM}\cdot\frac{HE}{HM}. \qquad \square$$

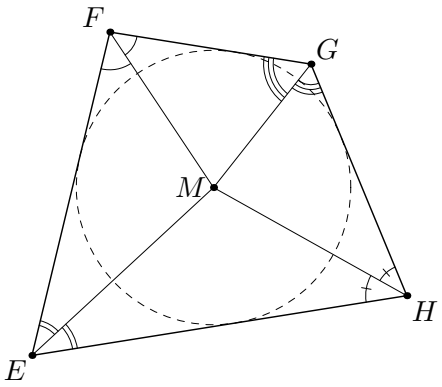


Figure 3

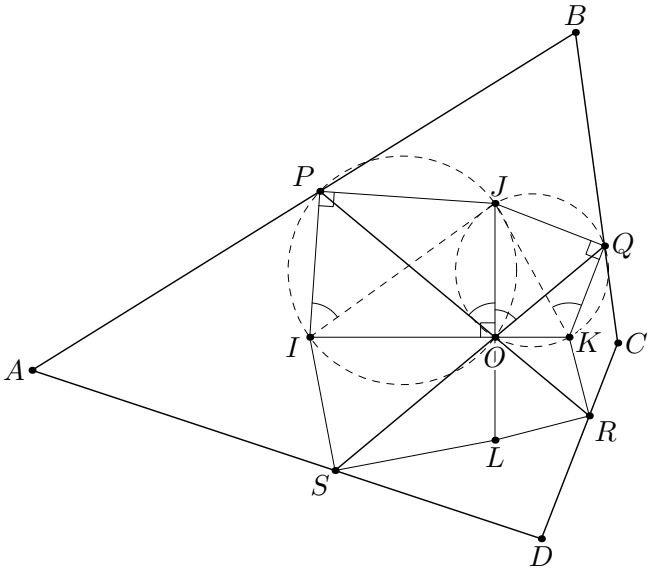


Figure 4

We denote by I, J, K, and L the incenters of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$, respectively. Applying Lemma 1 to these four quadrilaterals we get

$$\frac{AP \cdot PO}{OS \cdot SA} \cdot \frac{BQ \cdot QO}{OP \cdot PB} \cdot \frac{CR \cdot RO}{OQ \cdot QC} \cdot \frac{DS \cdot SO}{OR \cdot RD} = \frac{PI^2}{SI^2} \cdot \frac{QJ^2}{PJ^2} \cdot \frac{RK^2}{QK^2} \cdot \frac{SL^2}{RL^2},$$

which reduces to

$$\frac{AP}{PB} \cdot \frac{BQ}{QC} \cdot \frac{CR}{RD} \cdot \frac{DS}{SA} = \frac{PI^2}{PJ^2} \cdot \frac{QJ^2}{QK^2} \cdot \frac{RK^2}{RL^2} \cdot \frac{SL^2}{SI^2}. \tag{2}$$

Next, we have $\angle IPJ = \angle JOI = 90^\circ$, and the line OP separates I and J (see Figure 4). This means that the quadrilateral $IPJO$ is cyclic. Similarly, we get that the quadrilateral $JQKO$ is cyclic with $\angle JQK = 90^\circ$. Thus, $\angle QKJ = \angle QOJ = \angle JOP = \angle JIP$. Hence, the right triangles IPJ and KQJ are similar. Therefore, $\dfrac{PI}{PJ} = \dfrac{QK}{QJ}$. Likewise, we obtain $\dfrac{RK}{RL} = \dfrac{SI}{SL}$. These two equations together with (2) yield (1).

Comment. Instead of using the sine law, one may prove Lemma 1 by the following approach.

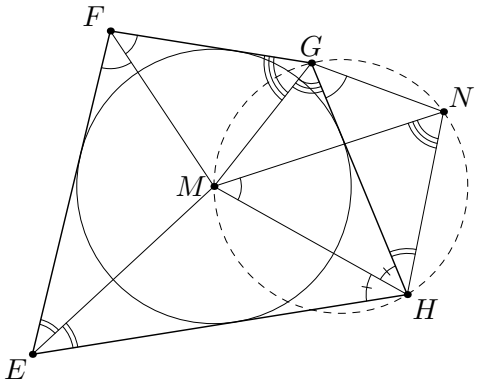


Figure 5

Let N be the point such that $\triangle NHG \sim \triangle MEF$ and such that N and M lie on different sides of the line GH, as shown in Figure 5. Then $\angle GNH + \angle HMG = \angle FME + \angle HMG = 180^\circ$. So,

the quadrilateral $GNHM$ is cyclic. Thus, $\angle MNH = \angle MGH = \angle FGM$ and $\angle HMN = \angle HGN = \angle EFM = \angle MFG$. Hence, $\triangle HMN \sim \triangle MFG$. Therefore, $\dfrac{HM}{HG} = \dfrac{HM}{HN} \cdot \dfrac{HN}{HG} = \dfrac{MF}{MG} \cdot \dfrac{EM}{EF}$. Similarly, we obtain $\dfrac{HM}{HE} = \dfrac{MF}{ME} \cdot \dfrac{GM}{GF}$. By multiplying these two equations, we complete the proof.

Solution 3. We present another approach for showing (1) from Solution 2.

Lemma 2. Let $EFGH$ and $E'F'G'H'$ be circumscribed quadrilaterals such that $\angle E + \angle E' = \angle F + \angle F' = \angle G + \angle G' = \angle H + \angle H' = 180^\circ$. Then

$$\frac{EF \cdot GH}{FG \cdot HE} = \frac{E'F' \cdot G'H'}{F'G' \cdot H'E'}.$$

Proof. Let M and M' be the incenters of $EFGH$ and $E'F'G'H'$, respectively. We use the notation $[XYZ]$ for the area of a triangle XYZ.

Taking into account the relation $\angle FME + \angle F'M'E' = 180^\circ$ together with the analogous ones, we get

$$\begin{aligned}\frac{EF \cdot GH}{FG \cdot HE} &= \frac{[MEF] \cdot [MGH]}{[MFG] \cdot [MHE]} = \frac{ME \cdot MF \cdot \sin\angle FME \cdot MG \cdot MH \cdot \sin\angle HMG}{MF \cdot MG \cdot \sin\angle GMF \cdot MH \cdot ME \cdot \sin\angle EMH} \\ &= \frac{M'E' \cdot M'F' \cdot \sin\angle F'M'E' \cdot M'G' \cdot M'H' \cdot \sin\angle H'M'G'}{M'F' \cdot M'G' \cdot \sin\angle G'M'F' \cdot M'H' \cdot M'E' \cdot \sin\angle E'M'H'} = \frac{E'F' \cdot G'H'}{F'G' \cdot H'E'}.\end{aligned}$$

□

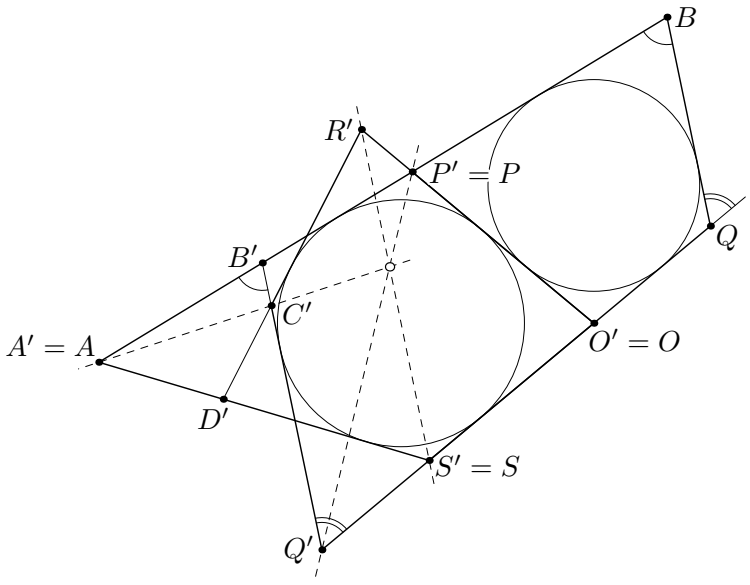


Figure 6

Denote by h the homothety centered at O that maps the incircle of $CROQ$ to the incircle of $APOS$. Let $Q' = h(Q)$, $C' = h(C)$, $R' = h(R)$, $O' = O$, $S' = S$, $A' = A$, and $P' = P$. Furthermore, define $B' = A'P' \cap C'Q'$ and $D' = A'S' \cap C'R'$ as shown in Figure 6. Then

$$\frac{AP \cdot OS}{PO \cdot SA} = \frac{A'P' \cdot O'S'}{P'O' \cdot S'A'}$$

holds trivially. We also have

$$\frac{CR \cdot OQ}{RO \cdot QC} = \frac{C'R' \cdot O'Q'}{R'O' \cdot Q'C'}$$

by the similarity of the quadrilaterals $CROQ$ and $C'R'O'Q'$.

Next, consider the circumscribed quadrilaterals $BQOP$ and $B'Q'O'P'$ whose incenters lie on different sides of the quadrilaterals' shared side line $OP = O'P'$. Observe that $BQ \parallel B'Q'$ and that B' and Q' lie on the lines BP and QO, respectively. It is now easy to see that the two quadrilaterals satisfy the hypotheses of Lemma 2. Thus, we deduce

$$\frac{BQ \cdot OP}{QO \cdot PB} = \frac{B'Q' \cdot O'P'}{Q'O' \cdot P'B'}.$$

Similarly, we get

$$\frac{DS \cdot OR}{SO \cdot RD} = \frac{D'S' \cdot O'R'}{S'O' \cdot R'D'}.$$

Multiplying these four equations, we obtain

$$\frac{AP}{PB} \cdot \frac{BQ}{QC} \cdot \frac{CR}{RD} \cdot \frac{DS}{SA} = \frac{A'P'}{P'B'} \cdot \frac{B'Q'}{Q'C'} \cdot \frac{C'R'}{R'D'} \cdot \frac{D'S'}{S'A'}. \tag{3}$$

Finally, we apply Brianchon's theorem to the circumscribed hexagon $A'P'R'C'Q'S'$ and deduce that the lines $A'C'$, $P'Q'$, and $R'S'$ are either concurrent or parallel to each other. So, by Menelaus' theorem, we obtain

$$\frac{A'P'}{P'B'} \cdot \frac{B'Q'}{Q'C'} \cdot \frac{C'R'}{R'D'} \cdot \frac{D'S'}{S'A'} = 1.$$

This equation together with (3) yield (1).

G8. A *triangulation* of a convex polygon Π is a partitioning of Π into triangles by diagonals having no common points other than the vertices of the polygon. We say that a triangulation is a *Thaiangulation* if all triangles in it have the same area.

Prove that any two different Thaiangulations of a convex polygon Π differ by exactly two triangles. (In other words, prove that it is possible to replace one pair of triangles in the first Thaiangulation with a different pair of triangles so as to obtain the second Thaiangulation.)

(Bulgaria)

Solution 1. We denote by $[S]$ the area of a polygon S.

Recall that each triangulation of a convex n-gon has exactly $n-2$ triangles. This means that all triangles in any two Thaiangulations of a convex polygon Π have the same area.

Let $\mathcal{T}$ be a triangulation of a convex polygon Π. If four vertices A, B, C, and D of Π form a parallelogram, and $\mathcal{T}$ contains two triangles whose union is this parallelogram, then we say that $\mathcal{T}$ *contains* parallelogram $ABCD$. Notice here that if two Thaiangulations $\mathcal{T}_1$ and $\mathcal{T}_2$ of Π differ by two triangles, then the union of these triangles is a quadrilateral each of whose diagonals bisects its area, i.e., a parallelogram.

We start with proving two properties of triangulations.

Lemma 1. A triangulation of a convex polygon Π cannot contain two parallelograms.

Proof. Arguing indirectly, assume that P_1 and P_2 are two parallelograms contained in some triangulation $\mathcal{T}$. If they have a common triangle in $\mathcal{T}$, then we may assume that P_1 consists of triangles ABC and ADC of $\mathcal{T}$, while P_2 consists of triangles ADC and CDE (see Figure 1). But then $BC \parallel AD \parallel CE$, so the three vertices B, C, and E of Π are collinear, which is absurd.

Assume now that P_1 and P_2 contain no common triangle. Let $P_1 = ABCD$. The sides AB, BC, CD, and DA partition Π into several parts, and P_2 is contained in one of them; we may assume that this part is cut off from P_1 by AD. Then one may label the vertices of P_2 by X, Y, Z, and T so that the polygon $ABCDXYZT$ is convex (see Figure 2; it may happen that $D = X$ and/or $T = A$, but still this polygon has at least six vertices). But the sum of the external angles of this polygon at B, C, Y, and Z is already 360°, which is impossible. A final contradiction. □

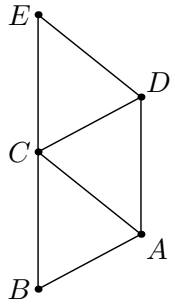


Figure 1

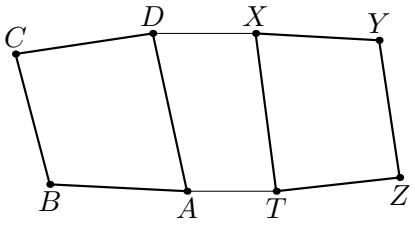


Figure 2

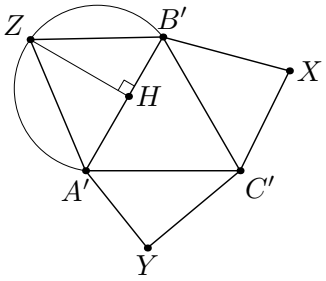


Figure 3

Lemma 2. Every triangle in a Thaiangulation $\mathcal{T}$ of Π contains a side of Π.

Proof. Let ABC be a triangle in $\mathcal{T}$. Apply an affine transform such that ABC maps to an equilateral triangle; let $A'B'C'$ be the image of this triangle, and Π' be the image of Π. Clearly, $\mathcal{T}$ maps into a Thaiangulation $\mathcal{T}'$ of Π'.

Assume that none of the sides of $\triangle A'B'C'$ is a side of Π'. Then $\mathcal{T}'$ contains some other triangles with these sides, say, $A'B'Z$, $C'A'Y$, and $B'C'X$; notice that $A'ZB'XC'Y$ is a convex hexagon (see Figure 3). The sum of its external angles at X, Y, and Z is less than 360°. So one of these angles (say, at Z) is less than 120°, hence $\angle A'ZB' > 60^\circ$. Then Z lies on a circular arc subtended by $A'B'$ and having angular measure less than 240°; consequently, the altitude ZH of $\triangle A'B'Z$ is less than $\sqrt{3}\,A'B'/2$. Thus $[A'B'Z] < [A'B'C']$, and $\mathcal{T}'$ is not a Thaiangulation. A contradiction. □

Now we pass to the solution. We say that a triangle in a triangulation of Π is an *ear* if it contains two sides of Π. Note that each triangulation of a polygon contains some ear.

Arguing indirectly, we choose a convex polygon Π with the least possible number of sides such that some two Thaiangulations $\mathcal{T}_1$ and $\mathcal{T}_2$ of Π violate the problem statement (thus Π has at least five sides). Consider now any ear ABC in $\mathcal{T}_1$, with AC being a diagonal of Π. If $\mathcal{T}_2$ also contains $\triangle ABC$, then one may cut $\triangle ABC$ off from Π, getting a polygon with a smaller number of sides which also violates the problem statement. This is impossible; thus $\mathcal{T}_2$ does not contain $\triangle ABC$.

Next, $\mathcal{T}_1$ contains also another triangle with side AC, say $\triangle ACD$. By Lemma 2, this triangle contains a side of Π, so D is adjacent to either A or C on the boundary of Π. We may assume that D is adjacent to C.

Assume that $\mathcal{T}_2$ does not contain the triangle BCD. Then it contains two different triangles BCX and CDY (possibly, with $X = Y$); since these triangles have no common interior points, the polygon $ABCDYX$ is convex (see Figure 4). But, since $[ABC] = [BCX] = [ACD] = [CDY]$, we get $AX \parallel BC$ and $AY \parallel CD$ which is impossible. Thus $\mathcal{T}_2$ contains $\triangle BCD$.

Therefore, $[ABD] = [ABC] + [ACD] - [BCD] = [ABC]$, and $ABCD$ is a parallelogram contained in $\mathcal{T}_1$. Let $\mathcal{T}'$ be the Thaiangulation of Π obtained from $\mathcal{T}_1$ by replacing the diagonal AC with BD; then $\mathcal{T}'$ is distinct from $\mathcal{T}_2$ (otherwise $\mathcal{T}_1$ and $\mathcal{T}_2$ would differ by two triangles). Moreover, $\mathcal{T}'$ shares a common ear BCD with $\mathcal{T}_2$. As above, cutting this ear away we obtain that $\mathcal{T}_2$ and $\mathcal{T}'$ differ by two triangles forming a parallelogram different from $ABCD$. Thus $\mathcal{T}'$ contains two parallelograms, which contradicts Lemma 1.

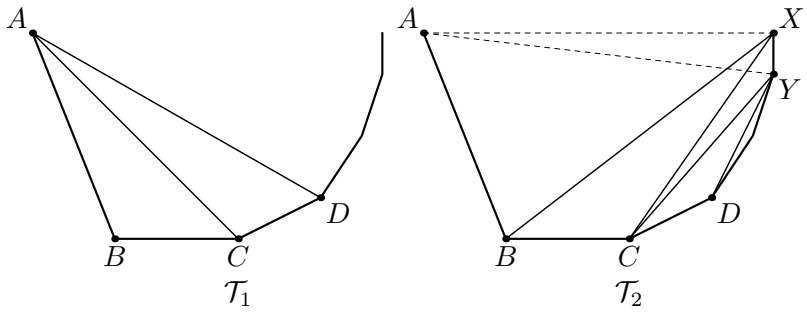


Figure 4

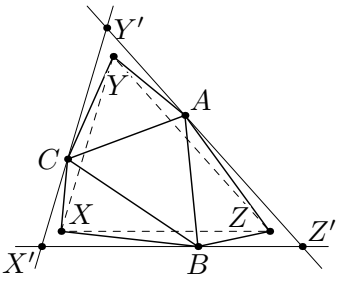


Figure 5

Comment 1. Lemma 2 is equivalent to the well-known Erdős–Debrunner inequality stating that for any triangle PQR and any points A, B, C lying on the sides QR, RP, and PQ, respectively, we have

$$[ABC] \geqslant \min\{[ABR], [BCP], [CAQ]\}. \tag{1}$$

To derive this inequality from Lemma 2, one may assume that (1) does not hold, and choose some points X, Y, and Z inside the triangles BCP, CAQ, and ABR, respectively, so that $[ABC] = [ABZ] = [BCX] = [CAY]$. Then a convex hexagon $AZBXCY$ has a Thaiangulation containing $\triangle ABC$, which contradicts Lemma 2.

Conversely, assume that a Thaiangulation $\mathcal{T}$ of Π contains a triangle ABC none of whose sides is a side of Π, and let ABZ, AYC, and XBC be other triangles in $\mathcal{T}$ containing the corresponding sides. Then $AZBXCY$ is a convex hexagon.

Consider the lines through A, B, and C parallel to YZ, ZX, and XY, respectively. They form a triangle $X'Y'Z'$ similar to $\triangle XYZ$ (see Figure 5). By (1) we have

$$[ABC] \geqslant \min\{[ABZ'], [BCX'], [CAY']\} > \min\{[ABZ], [BCX], [CAY]\},$$

so $\mathcal{T}$ is not a Thaiangulation.

Solution 2. We will make use of the preliminary observations from Solution 1, together with Lemma 1.

Arguing indirectly, we choose a convex polygon Π with the least possible number of sides such that some two Thaiangulations $\mathcal{T}_1$ and $\mathcal{T}_2$ of Π violate the statement (thus Π has at least five sides). Assume that $\mathcal{T}_1$ and $\mathcal{T}_2$ share a diagonal d splitting Π into two smaller polygons Π_1 and Π_2. Since the problem statement holds for any of them, the induced Thaiangulations of each of Π_i differ by two triangles forming a parallelogram (the Thaiangulations induced on Π_i by $\mathcal{T}_1$ and T_2 may not coincide, otherwise $\mathcal{T}_1$ and $\mathcal{T}_2$ would differ by at most two triangles). But both these parallelograms are contained in $\mathcal{T}_1$; this contradicts Lemma 1. Therefore, $\mathcal{T}_1$ and $\mathcal{T}_2$ share no diagonal. Hence they also share no triangle.

We consider two cases.

Case 1. Assume that some vertex B of Π is an endpoint of some diagonal in $\mathcal{T}_1$, as well as an endpoint of some diagonal in $\mathcal{T}_2$.

Let A and C be the vertices of Π adjacent to B. Then $\mathcal{T}_1$ contains some triangles ABX and BCY, while $\mathcal{T}_2$ contains some triangles ABX' and BCY'. Here, some of the points X, X', Y, and Y' may coincide; however, in view of our assumption together with the fact that $\mathcal{T}_1$ and $\mathcal{T}_2$ share no triangle, all four triangles ABX, BCY, ABX', and BCY' are distinct.

Since $[ABX] = [BCY] = [ABX'] = [BCY']$, we have $XX' \parallel AB$ and $YY' \parallel BC$. Now, if $X = Y$, then X' and Y' lie on different lines passing through X and are distinct from that point, so that $X' \neq Y'$. In this case, we may switch the two Thaiangulations. So, hereafter we assume that $X \neq Y$.

In the convex pentagon $ABCYX$ we have either $\angle BAX + \angle AXY > 180^\circ$ or $\angle XYC + \angle YCB > 180^\circ$ (or both); due to the symmetry, we may assume that the first inequality holds. Let r be the ray emerging from X and co-directed with $\overrightarrow{AB}$; our inequality shows that r points to the interior of the pentagon (and thus to the interior of Π). Therefore, the ray opposite to r points outside Π, so X' lies on r; moreover, X' lies on the "arc" CY of Π not containing X. So the segments XX' and YB intersect (see Figure 6).

Let O be the intersection point of the rays r and BC. Since the triangles ABX' and BCY' have no common interior points, Y' must lie on the "arc" CX' which is situated inside the triangle XBO. Therefore, the line YY' meets two sides of $\triangle XBO$, none of which may be XB (otherwise the diagonals XB and YY' would share a common point). Thus YY' intersects BO, which contradicts $YY' \parallel BC$.

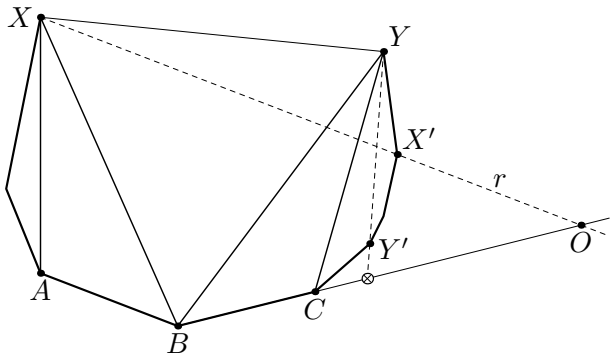


Figure 6

Case 2. In the remaining case, each vertex of Π is an endpoint of a diagonal in at most one of $\mathcal{T}_1$ and $\mathcal{T}_2$. On the other hand, a triangulation cannot contain two consecutive vertices with no diagonals from each. Therefore, the vertices of Π alternatingly emerge diagonals in $\mathcal{T}_1$ and in $\mathcal{T}_2$. In particular, Π has an even number of sides.

Next, we may choose five consecutive vertices A, B, C, D, and E of Π in such a way that

$$\angle ABC + \angle BCD > 180^\circ \quad \text{and} \quad \angle BCD + \angle CDE > 180^\circ. \tag{2}$$

In order to do this, it suffices to choose three consecutive vertices B, C, and D of Π such that the sum of their external angles is at most 180°. This is possible, since Π has at least six sides.

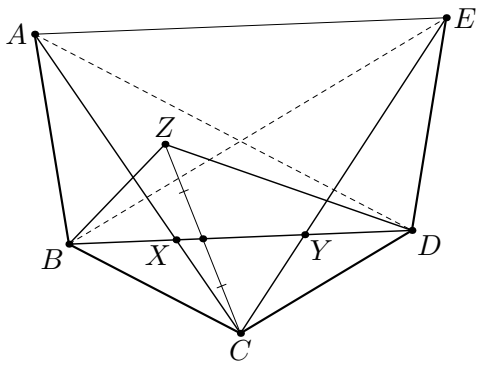


Figure 7

We may assume that $\mathcal{T}_1$ has no diagonals from B and D (and thus contains the triangles ABC and CDE), while $\mathcal{T}_2$ has no diagonals from A, C, and E (and thus contains the triangle BCD). Now, since $[ABC] = [BCD] = [CDE]$, we have $AD \parallel BC$ and $BE \parallel CD$ (see Figure 7). By (2) this yields that $AD > BC$ and $BE > CD$. Let $X = AC \cap BD$ and $Y = CE \cap BD$; then the inequalities above imply that $AX > CX$ and $EY > CY$.

Finally, $\mathcal{T}_2$ must also contain some triangle BDZ with $Z \neq C$; then the ray CZ lies in the angle ACE. Since $[BCD] = [BDZ]$, the diagonal BD bisects CZ. Together with the inequalities above, this yields that Z lies inside the triangle ACE (but Z is distinct from A and E), which is impossible. The final contradiction.

Comment 2. Case 2 may also be accomplished with the use of Lemma 2. Indeed, since each triangulation of an n-gon contains $n - 2$ triangles neither of which may contain three sides of Π, Lemma 2 yields that each Thaiangulation contains exactly two ears. But each vertex of Π is a vertex of an ear either in $\mathcal{T}_1$ or in $\mathcal{T}_2$, so Π cannot have more than four vertices.

Number Theory

N1. Determine all positive integers M for which the sequence $a_0, a_1, a_2, \ldots$, defined by $a_0 = \frac{2M+1}{2}$ and $a_{k+1} = a_k\lfloor a_k \rfloor$ for $k = 0, 1, 2, \ldots$, contains at least one integer term.

(Luxembourg)

Answer. All integers $M \geqslant 2$.

Solution 1. Define $b_k = 2a_k$ for all $k \geqslant 0$. Then

$$b_{k+1} = 2a_{k+1} = 2a_k\lfloor a_k \rfloor = b_k \left\lfloor \frac{b_k}{2} \right\rfloor.$$

Since b_0 is an integer, it follows that b_k is an integer for all $k \geqslant 0$.

Suppose that the sequence $a_0, a_1, a_2, \ldots$ does not contain any integer term. Then b_k must be an odd integer for all $k \geqslant 0$, so that

$$b_{k+1} = b_k \left\lfloor \frac{b_k}{2} \right\rfloor = \frac{b_k(b_k - 1)}{2}. \tag{1}$$

Hence

$$b_{k+1} - 3 = \frac{b_k(b_k - 1)}{2} - 3 = \frac{(b_k - 3)(b_k + 2)}{2} \tag{2}$$

for all $k \geqslant 0$.

Suppose that $b_0 - 3 > 0$. Then equation (2) yields $b_k - 3 > 0$ for all $k \geqslant 0$. For each $k \geqslant 0$, define c_k to be the highest power of 2 that divides $b_k - 3$. Since $b_k - 3$ is even for all $k \geqslant 0$, the number c_k is positive for every $k \geqslant 0$.

Note that $b_k + 2$ is an odd integer. Therefore, from equation (2), we have that $c_{k+1} = c_k - 1$. Thus, the sequence $c_0, c_1, c_2, \ldots$ of positive integers is strictly decreasing, a contradiction. So, $b_0 - 3 \leqslant 0$, which implies $M = 1$.

For $M = 1$, we can check that the sequence is constant with $a_k = \frac{3}{2}$ for all $k \geqslant 0$. Therefore, the answer is $M \geqslant 2$.

Solution 2. We provide an alternative way to show $M = 1$ once equation (1) has been reached. We claim that $b_k \equiv 3 \pmod{2^m}$ for all $k \geqslant 0$ and $m \geqslant 1$. If this is true, then we would have $b_k = 3$ for all $k \geqslant 0$ and hence $M = 1$.

To establish our claim, we proceed by induction on m. The base case $b_k \equiv 3 \pmod 2$ is true for all $k \geqslant 0$ since b_k is odd. Now suppose that $b_k \equiv 3 \pmod{2^m}$ for all $k \geqslant 0$. Hence $b_k = 2^m d_k + 3$ for some integer d_k. We have

$$3 \equiv b_{k+1} \equiv (2^m d_k + 3)(2^{m-1} d_k + 1) \equiv 3 \cdot 2^{m-1} d_k + 3 \pmod{2^m},$$

so that d_k must be even. This implies that $b_k \equiv 3 \pmod{2^{m+1}}$, as required.

Comment. The reason the number 3 which appears in both solutions is important, is that it is a nontrivial fixed point of the recurrence relation for b_k.

N2. Let a and b be positive integers such that $a!b!$ is a multiple of $a! + b!$. Prove that $3a \geqslant 2b + 2$.

(United Kingdom)

Solution 1. If $a > b$, we immediately get $3a \geqslant 2b + 2$. In the case $a = b$, the required inequality is equivalent to $a \geqslant 2$, which can be checked easily since $(a, b) = (1, 1)$ does not satisfy $a! + b! \mid a!b!$. We now assume $a < b$ and denote $c = b - a$. The required inequality becomes $a \geqslant 2c + 2$.

Suppose, to the contrary, that $a \leqslant 2c + 1$. Define $M = \frac{b!}{a!} = (a + 1)(a + 2) \cdots (a + c)$. Since $a! + b! \mid a!b!$ implies $1 + M \mid a!M$, we obtain $1 + M \mid a!$. Note that we must have $c < a$; otherwise $1 + M > a!$, which is impossible. We observe that $c! \mid M$ since M is a product of c consecutive integers. Thus $\gcd(1 + M, c!) = 1$, which implies

$$1 + M \,\Big|\, \frac{a!}{c!} = (c + 1)(c + 2) \cdots a. \tag{1}$$

If $a \leqslant 2c$, then $\frac{a!}{c!}$ is a product of $a - c \leqslant c$ integers not exceeding a whereas M is a product of c integers exceeding a. Therefore, $1 + M > \frac{a!}{c!}$, which is a contradiction.

It remains to exclude the case $a = 2c + 1$. Since $a + 1 = 2(c + 1)$, we have $c + 1 \mid M$. Hence, we can deduce from (1) that $1 + M \mid (c + 2)(c + 3) \cdots a$. Now $(c + 2)(c + 3) \cdots a$ is a product of $a - c - 1 = c$ integers not exceeding a; thus it is smaller than $1 + M$. Again, we arrive at a contradiction.

Comment 1. One may derive a weaker version of (1) and finish the problem as follows. After assuming $a \leqslant 2c + 1$, we have $\left\lfloor \frac{a}{2} \right\rfloor \leqslant c$, so $\left\lfloor \frac{a}{2} \right\rfloor! \mid M$. Therefore,

$$1 + M \,\Big|\, \left(\left\lfloor \frac{a}{2} \right\rfloor + 1\right)\left(\left\lfloor \frac{a}{2} \right\rfloor + 2\right) \cdots a.$$

Observe that $\left(\left\lfloor \frac{a}{2} \right\rfloor + 1\right)\left(\left\lfloor \frac{a}{2} \right\rfloor + 2\right) \cdots a$ is a product of $\left\lceil \frac{a}{2} \right\rceil$ integers not exceeding a. This leads to a contradiction when a is even since $\left\lceil \frac{a}{2} \right\rceil = \frac{a}{2} \leqslant c$ and M is a product of c integers exceeding a.

When a is odd, we can further deduce that $1 + M \mid \left(\frac{a+3}{2}\right)\left(\frac{a+5}{2}\right) \cdots a$ since $\left\lfloor \frac{a}{2} \right\rfloor + 1 = \frac{a+1}{2} \,\Big|\, a + 1$. Now $\left(\frac{a+3}{2}\right)\left(\frac{a+5}{2}\right) \cdots a$ is a product of $\frac{a-1}{2} \leqslant c$ numbers not exceeding a, and we get a contradiction.

Solution 2. As in Solution 1, we may assume that $a < b$ and let $c = b - a$. Suppose, to the contrary, that $a \leqslant 2c + 1$. From $a! + b! \mid a!b!$, we have

$$N = 1 + (a + 1)(a + 2) \cdots (a + c) \;\Big|\; (a + c)!,$$

which implies that all prime factors of N are at most $a + c$.

Let p be a prime factor of N. If $p \leqslant c$ or $p \geqslant a + 1$, then p divides one of $a + 1, \ldots, a + c$ which is impossible. Hence $a \geqslant p \geqslant c + 1$. Furthermore, we must have $2p > a + c$; otherwise, $a + 1 \leqslant 2c + 2 \leqslant 2p \leqslant a + c$ so $p \mid N - 1$, again impossible. Thus, we have $p \in \left(\frac{a+c}{2}, a\right]$, and $p^2 \nmid (a + c)!$ since $2p > a + c$. Therefore, $p^2 \nmid N$ as well.

If $a \leqslant c + 2$, then the interval $\left(\frac{a+c}{2}, a\right]$ contains at most one integer and hence at most one prime number, which has to be a. Since $p^2 \nmid N$, we must have $N = p = a$ or $N = 1$, which is absurd since $N > a \geqslant 1$. Thus, we have $a \geqslant c + 3$, and so $\frac{a+c+1}{2} \geqslant c + 2$. It follows that p lies in the interval $[c + 2, a]$.

Thus, every prime appearing in the prime factorization of N lies in the interval $[c+2, a]$, and its exponent is exactly 1. So we must have $N \mid (c+2)(c+3) \cdots a$. However, $(c+2)(c+3) \cdots a$ is a product of $a-c-1 \leqslant c$ numbers not exceeding a, so it is less than N. This is a contradiction.

Comment 2. The original problem statement also asks to determine when the equality $3a = 2b + 2$ holds. It can be checked that the answer is $(a, b) = (2, 2), (4, 5)$.

N3. Let m and n be positive integers such that $m > n$. Define $x_k = (m+k)/(n+k)$ for $k = 1, 2, \ldots, n+1$. Prove that if all the numbers $x_1, x_2, \ldots, x_{n+1}$ are integers, then $x_1 x_2 \cdots x_{n+1} - 1$ is divisible by an odd prime.

(Austria)

Solution. Assume that $x_1, x_2, \ldots, x_{n+1}$ are integers. Define the integers

$$a_k = x_k - 1 = \frac{m+k}{n+k} - 1 = \frac{m-n}{n+k} > 0$$

for $k = 1, 2, \ldots, n+1$.

Let $P = x_1 x_2 \cdots x_{n+1} - 1$. We need to prove that P is divisible by an odd prime, or in other words, that P is not a power of 2. To this end, we investigate the powers of 2 dividing the numbers a_k.

Let 2^d be the largest power of 2 dividing $m-n$, and let 2^c be the largest power of 2 not exceeding $2n+1$. Then $2n+1 \leqslant 2^{c+1}-1$, and so $n+1 \leqslant 2^c$. We conclude that 2^c is one of the numbers $n+1, n+2, \ldots, 2n+1$, and that it is the only multiple of 2^c appearing among these numbers. Let ℓ be such that $n+\ell = 2^c$. Since $\frac{m-n}{n+\ell}$ is an integer, we have $d \geqslant c$. Therefore, $2^{d-c+1} \nmid a_\ell = \frac{m-n}{n+\ell}$, while $2^{d-c+1} \mid a_k$ for all $k \in \{1, \ldots, n+1\} \setminus \{\ell\}$.

Computing modulo 2^{d-c+1}, we get

$$P = (a_1+1)(a_2+1)\cdots(a_{n+1}+1) - 1 \equiv (a_\ell + 1)\cdot 1^n - 1 \equiv a_\ell \not\equiv 0 \pmod{2^{d-c+1}}.$$

Therefore, $2^{d-c+1} \nmid P$.

On the other hand, for any $k \in \{1, \ldots, n+1\} \setminus \{\ell\}$, we have $2^{d-c+1} \mid a_k$. So $P \geqslant a_k \geqslant 2^{d-c+1}$, and it follows that P is not a power of 2.

Comment. Instead of attempting to show that P is not a power of 2, one may try to find an odd factor of P (greater than 1) as follows:

From $a_k = \frac{m-n}{n+k} \in \mathbb{Z}_{>0}$, we get that $m-n$ is divisible by $n+1, n+2, \ldots, 2n+1$, and thus it is also divisible by their least common multiple L. So $m-n = qL$ for some positive integer q; hence $x_k = q \cdot \frac{L}{n+k} + 1$.

Then, since $n+1 \leqslant 2^c = n+\ell \leqslant 2n+1 \leqslant 2^{c+1}-1$, we have $2^c \mid L$, but $2^{c+1} \nmid L$. So $\frac{L}{n+\ell}$ is odd, while $\frac{L}{n+k}$ is even for $k \neq \ell$. Computing modulo $2q$ yields

$$x_1 x_2 \cdots x_{n+1} - 1 \equiv (q+1)\cdot 1^n - 1 \equiv q \pmod{2q}.$$

Thus, $x_1 x_2 \cdots x_{n+1} - 1 = 2qr + q = q(2r+1)$ for some integer r.

Since $x_1 x_2 \cdots x_{n+1} - 1 \geqslant x_1 x_2 - 1 \geqslant (q+1)^2 - 1 > q$, we have $r \geqslant 1$. This implies that $x_1 x_2 \cdots x_{n+1} - 1$ is divisible by an odd prime.

N4. Suppose that $a_0, a_1, \ldots$ and $b_0, b_1, \ldots$ are two sequences of positive integers satisfying $a_0, b_0 \geqslant 2$ and

$$a_{n+1} = \gcd(a_n, b_n) + 1, \qquad b_{n+1} = \operatorname{lcm}(a_n, b_n) - 1$$

for all $n \geqslant 0$. Prove that the sequence (a_n) is eventually periodic; in other words, there exist integers $N \geqslant 0$ and $t > 0$ such that $a_{n+t} = a_n$ for all $n \geqslant N$.

(France)

Solution 1. Let $s_n = a_n + b_n$. Notice that if $a_n \mid b_n$, then $a_{n+1} = a_n + 1$, $b_{n+1} = b_n - 1$ and $s_{n+1} = s_n$. So, a_n increases by 1 and s_n does not change until the first index is reached with $a_n \nmid s_n$. Define

$$W_n = \{m \in \mathbb{Z}_{>0} \colon m \geqslant a_n \text{ and } m \nmid s_n\} \quad \text{and} \quad w_n = \min W_n.$$

Claim 1. The sequence (w_n) is non-increasing.

Proof. If $a_n \mid b_n$ then $a_{n+1} = a_n + 1$. Due to $a_n \mid s_n$, we have $a_n \notin W_n$. Moreover $s_{n+1} = s_n$; therefore, $W_{n+1} = W_n$ and $w_{n+1} = w_n$.

Otherwise, if $a_n \nmid b_n$, then $a_n \nmid s_n$, so $a_n \in W_n$ and thus $w_n = a_n$. We show that $a_n \in W_{n+1}$; this implies $w_{n+1} \leqslant a_n = w_n$. By the definition of W_{n+1}, we need that $a_n \geqslant a_{n+1}$ and $a_n \nmid s_{n+1}$. The first relation holds because of $\gcd(a_n, b_n) < a_n$. For the second relation, observe that in $s_{n+1} = \gcd(a_n, b_n) + \operatorname{lcm}(a_n, b_n)$, the second term is divisible by a_n, but the first term is not. So $a_n \nmid s_{n+1}$; that completes the proof of the claim. □

Let $w = \min\limits_n w_n$ and let N be an index with $w = w_N$. Due to Claim 1, we have $w_n = w$ for all $n \geqslant N$.

Let $g_n = \gcd(w, s_n)$. As we have seen, starting from an arbitrary index $n \geqslant N$, the sequence $a_n, a_{n+1}, \ldots$ increases by 1 until it reaches w, which is the first value not dividing s_n; then it drops to $\gcd(w, s_n) + 1 = g_n + 1$.

Claim 2. The sequence (g_n) is constant for $n \geqslant N$.

Proof. If $a_n \mid b_n$, then $s_{n+1} = s_n$ and hence $g_{n+1} = g_n$. Otherwise we have $a_n = w$,

$$\gcd(a_n, b_n) = \gcd(a_n, s_n) = \gcd(w, s_n) = g_n,$$

$$s_{n+1} = \gcd(a_n, b_n) + \operatorname{lcm}(a_n, b_n) = g_n + \frac{a_n b_n}{g_n} = g_n + \frac{w(s_n - w)}{g_n}, \tag{1}$$

$$\text{and} \quad g_{n+1} = \gcd(w, s_{n+1}) = \gcd\left(w,\ g_n + \frac{s_n - w}{g_n} w\right) = \gcd(w, g_n) = g_n. \qquad \square$$

Let $g = g_N$. We have proved that the sequence (a_n) eventually repeats the following cycle:

$$g + 1 \mapsto g + 2 \mapsto \ldots \mapsto w \mapsto g + 1.$$

Solution 2. By Claim 1 in the first solution, we have $a_n \leqslant w_n \leqslant w_0$, so the sequence (a_n) is bounded, and hence it has only finitely many values.

Let $M = \operatorname{lcm}(a_1, a_2, \ldots)$, and consider the sequence b_n modulo M. Let r_n be the remainder of b_n, divided by M. For every index n, since $a_n \mid M \mid b_n - r_n$, we have $\gcd(a_n, b_n) = \gcd(a_n, r_n)$, and therefore

$$a_{n+1} = \gcd(a_n, r_n) + 1.$$

Moreover,

$$r_{n+1} \equiv b_{n+1} = \operatorname{lcm}(a_n, b_n) - 1 = \frac{a_n}{\gcd(a_n, b_n)} b_n - 1$$
$$= \frac{a_n}{\gcd(a_n, r_n)} b_n - 1 \equiv \frac{a_n}{\gcd(a_n, r_n)} r_n - 1 \pmod{M}.$$

Hence, the pair (a_n, r_n) uniquely determines the pair (a_{n+1}, r_{n+1}). Since there are finitely many possible pairs, the sequence of pairs (a_n, r_n) is eventually periodic; in particular, the sequence (a_n) is eventually periodic.

Comment. We show that there are only four possibilities for g and w (as defined in Solution 1), namely

$$(w, g) \in \big\{(2,1), (3,1), (4,2), (5,1)\big\}. \tag{2}$$

This means that the sequence (a_n) eventually repeats one of the following cycles:

$$(2), \quad (2,3), \quad (3,4), \quad \text{or} \quad (2,3,4,5). \tag{3}$$

Using the notation of Solution 1, for $n \geqslant N$ the sequence (a_n) has a cycle $(g+1, g+2, \ldots, w)$ such that $g = \gcd(w, s_n)$. By the observations in the proof of Claim 2, the numbers $g+1, \ldots, w-1$ all divide s_n; so the number $L = \operatorname{lcm}(g+1, g+2, \ldots, w-1)$ also divides s_n. Moreover, g also divides w.

Now choose any $n \geqslant N$ such that $a_n = w$. By (1), we have

$$s_{n+1} = g + \frac{w(s_n - w)}{g} = s_n \cdot \frac{w}{g} - \frac{w^2 - g^2}{g}.$$

Since L divides both s_n and s_{n+1}, it also divides the number $T = \frac{w^2-g^2}{g}$.

Suppose first that $w \geqslant 6$, which yields $g + 1 \leqslant \frac{w}{2} + 1 \leqslant w - 2$. Then $(w-2)(w-1) \mid L \mid T$, so we have either $w^2 - g^2 \geqslant 2(w-1)(w-2)$, or $g = 1$ and $w^2 - g^2 = (w-1)(w-2)$. In the former case we get $(w-1)(w-5) + (g^2 - 1) \leqslant 0$ which is false by our assumption. The latter equation rewrites as $3w = 3$, so $w = 1$, which is also impossible.

Now we are left with the cases when $w \leqslant 5$ and $g \mid w$. The case $(w, g) = (4, 1)$ violates the condition $L \mid \frac{w^2-g^2}{g}$; all other such pairs are listed in (2).

In the table below, for each pair (w, g), we provide possible sequences (a_n) and (b_n). That shows that the cycles shown in (3) are indeed possible.

$w = 2$	$g = 1$	$a_n = 2$	$b_n = 2 \cdot 2^n + 1$
$w = 3$	$g = 1$	$(a_{2k}, a_{2k+1}) = (2, 3)$	$(b_{2k}, b_{2k+1}) = (6 \cdot 3^k + 2, 6 \cdot 3^k + 1)$
$w = 4$	$g = 2$	$(a_{2k}, a_{2k+1}) = (3, 4)$	$(b_{2k}, b_{2k+1}) = (12 \cdot 2^k + 3, 12 \cdot 2^k + 2)$
$w = 5$	$g = 1$	$(a_{4k}, \ldots, a_{4k+3}) = (2, 3, 4, 5)$	$(b_{4k}, \ldots, b_{4k+3}) = (6 \cdot 5^k + 4, \ldots, 6 \cdot 5^k + 1)$

N5. Determine all triples (a, b, c) of positive integers for which $ab - c$, $bc - a$, and $ca - b$ are powers of 2.

Explanation: A power of 2 *is an integer of the form* 2^n*, where* n *denotes some nonnegative integer.*

(Serbia)

Answer. There are sixteen such triples, namely $(2, 2, 2)$, the three permutations of $(2, 2, 3)$, and the six permutations of each of $(2, 6, 11)$ and $(3, 5, 7)$.

Solution 1. It can easily be verified that these sixteen triples are as required. Now let (a, b, c) be any triple with the desired property. If we would have $a = 1$, then both $b - c$ and $c - b$ were powers of 2, which is impossible since their sum is zero; because of symmetry, this argument shows $a, b, c \geqslant 2$.

Case 1. Among a*,* b*, and* c *there are at least two equal numbers.*

Without loss of generality we may suppose that $a = b$. Then $a^2 - c$ and $a(c - 1)$ are powers of 2. The latter tells us that actually a and $c - 1$ are powers of 2. So there are nonnegative integers α and γ with $a = 2^\alpha$ and $c = 2^\gamma + 1$. Since $a^2 - c = 2^{2\alpha} - 2^\gamma - 1$ is a power of 2 and thus incongruent to -1 modulo 4, we must have $\gamma \leqslant 1$. Moreover, each of the terms $2^{2\alpha} - 2$ and $2^{2\alpha} - 3$ can only be a power of 2 if $\alpha = 1$. It follows that the triple (a, b, c) is either $(2, 2, 2)$ or $(2, 2, 3)$.

Case 2. The numbers a*,* b*, and* c *are distinct.*

Due to symmetry we may suppose that

$$2 \leqslant a < b < c\,. \tag{1}$$

We are to prove that the triple (a, b, c) is either $(2, 6, 11)$ or $(3, 5, 7)$. By our hypothesis, there exist three nonnegative integers α, β, and γ such that

$$bc - a = 2^\alpha\,, \tag{2}$$

$$ac - b = 2^\beta\,, \tag{3}$$

$$\text{and} \quad ab - c = 2^\gamma\,. \tag{4}$$

Evidently we have

$$\alpha > \beta > \gamma\,. \tag{5}$$

Depending on how large a is, we divide the argument into two further cases.

Case 2.1. $a = 2$.

We first prove that $\gamma = 0$. Assume for the sake of contradiction that $\gamma > 0$. Then c is even by (4) and, similarly, b is even by (5) and (3). So the left-hand side of (2) is congruent to 2 modulo 4, which is only possible if $bc = 4$. As this contradicts (1), we have thereby shown that $\gamma = 0$, i.e., that $c = 2b - 1$.

Now (3) yields $3b - 2 = 2^\beta$. Due to $b > 2$ this is only possible if $\beta \geqslant 4$. If $\beta = 4$, then we get $b = 6$ and $c = 2 \cdot 6 - 1 = 11$, which is a solution. It remains to deal with the case $\beta \geqslant 5$. Now (2) implies

$$9 \cdot 2^\alpha = 9b(2b - 1) - 18 = (3b - 2)(6b + 1) - 16 = 2^\beta(2^{\beta+1} + 5) - 16\,,$$

and by $\beta \geqslant 5$ the right-hand side is not divisible by 32. Thus $\alpha \leqslant 4$ and we get a contradiction to (5).

Case 2.2. $a \geqslant 3$.

Pick an integer $\vartheta \in \{-1, +1\}$ such that $c - \vartheta$ is not divisible by 4. Now

$$2^\alpha + \vartheta \cdot 2^\beta = (bc - a\vartheta^2) + \vartheta(ca - b) = (b + a\vartheta)(c - \vartheta)$$

is divisible by 2^β and, consequently, $b+a\vartheta$ is divisible by $2^{\beta-1}$. On the other hand, $2^\beta = ac-b > (a-1)c \geqslant 2c$ implies in view of (1) that a and b are smaller than $2^{\beta-1}$. All this is only possible if $\vartheta = 1$ and $a + b = 2^{\beta-1}$. Now (3) yields

$$ac - b = 2(a + b)\,, \tag{6}$$

whence $4b > a + 3b = a(c - 1) \geqslant ab$, which in turn yields $a = 3$.

So (6) simplifies to $c = b + 2$ and (2) tells us that $b(b + 2) - 3 = (b - 1)(b + 3)$ is a power of 2. Consequently, the factors $b-1$ and $b+3$ are powers of 2 themselves. Since their difference is 4, this is only possible if $b = 5$ and thus $c = 7$. Thereby the solution is complete.

Solution 2. As in the beginning of the first solution, we observe that $a, b, c \geqslant 2$. Depending on the parities of a, b, and c we distinguish three cases.

Case 1. The numbers a, b, and c are even.

Let 2^A, 2^B, and 2^C be the largest powers of 2 dividing a, b, and c respectively. We may assume without loss of generality that $1 \leqslant A \leqslant B \leqslant C$. Now 2^B is the highest power of 2 dividing $ac - b$, whence $ac - b = 2^B \leqslant b$. Similarly, we deduce $bc - a = 2^A \leqslant a$. Adding both estimates we get $(a + b)c \leqslant 2(a + b)$, whence $c \leqslant 2$. So $c = 2$ and thus $A = B = C = 1$; moreover, we must have had equality throughout, i.e., $a = 2^A = 2$ and $b = 2^B = 2$. We have thereby found the solution $(a, b, c) = (2, 2, 2)$.

Case 2. The numbers a, b, and c are odd.

If any two of these numbers are equal, say $a = b$, then $ac - b = a(c - 1)$ has a nontrivial odd divisor and cannot be a power of 2. Hence a, b, and c are distinct. So we may assume without loss of generality that $a < b < c$.

Let α and β denote the nonnegative integers for which $bc - a = 2^\alpha$ and $ac - b = 2^\beta$ hold. Clearly, we have $\alpha > \beta$, and thus 2^β divides

$$a \cdot 2^\alpha - b \cdot 2^\beta = a(bc - a) - b(ac - b) = b^2 - a^2 = (b + a)(b - a)\,.$$

Since a is odd, it is not possible that both factors $b+a$ and $b-a$ are divisible by 4. Consequently, one of them has to be a multiple of $2^{\beta-1}$. Hence one of the numbers $2(b + a)$ and $2(b - a)$ is divisible by 2^β and in either case we have

$$ac - b = 2^\beta \leqslant 2(a + b)\,. \tag{7}$$

This in turn yields $(a - 1)b < ac - b < 4b$ and thus $a = 3$ (recall that a is odd and larger than 1). Substituting this back into (7) we learn $c \leqslant b + 2$. But due to the parity $b < c$ entails that $b + 2 \leqslant c$ holds as well. So we get $c = b + 2$ and from $bc - a = (b - 1)(b + 3)$ being a power of 2 it follows that $b = 5$ and $c = 7$.

Case 3. Among a, b, and c both parities occur.

Without loss of generality, we suppose that c is odd and that $a \leqslant b$. We are to show that (a, b, c) is either $(2, 2, 3)$ or $(2, 6, 11)$. As at least one of a and b is even, the expression $ab - c$ is odd; since it is also a power of 2, we obtain

$$ab - c = 1\,. \tag{8}$$

If $a = b$, then $c = a^2 - 1$, and from $ac - b = a(a^2 - 2)$ being a power of 2 it follows that both a and $a^2 - 2$ are powers of 2, whence $a = 2$. This gives rise to the solution $(2, 2, 3)$.

We may suppose $a < b$ from now on. As usual, we let $\alpha > \beta$ denote the integers satisfying

$$2^\alpha = bc - a \quad \text{and} \quad 2^\beta = ac - b\,. \tag{9}$$

If $\beta = 0$ it would follow that $ac - b = ab - c = 1$ and hence that $b = c = 1$, which is absurd. So β and α are positive and consequently a and b are even. Substituting $c = ab - 1$ into (9) we obtain

$$2^\alpha = ab^2 - (a + b)\,, \tag{10}$$

$$\text{and} \quad 2^\beta = a^2b - (a + b)\,. \tag{11}$$

The addition of both equation yields $2^\alpha + 2^\beta = (ab - 2)(a + b)$. Now $ab - 2$ is even but not divisible by 4, so the highest power of 2 dividing $a + b$ is $2^{\beta-1}$. For this reason, the equations (10) and (11) show that the highest powers of 2 dividing either of the numbers ab^2 and a^2b is likewise $2^{\beta-1}$. Thus there is an integer $\tau \geqslant 1$ together with odd integers A, B, and C such that $a = 2^\tau A$, $b = 2^\tau B$, $a + b = 2^{3\tau}C$, and $\beta = 1 + 3\tau$.

Notice that $A + B = 2^{2\tau}C \geqslant 4C$. Moreover, (11) entails $A^2B - C = 2$. Thus $8 = 4A^2B - 4C \geqslant 4A^2B - A - B \geqslant A^2(3B - 1)$. Since A and B are odd with $A < B$, this is only possible if $A = 1$ and $B = 3$. Finally, one may conclude $C = 1$, $\tau = 1$, $a = 2$, $b = 6$, and $c = 11$. We have thereby found the triple $(2, 6, 11)$. This completes the discussion of the third case, and hence the solution.

Comment. In both solutions, there are many alternative ways to proceed in each of its cases. Here we present a different treatment of the part "$a < b$" of Case 3 in Solution 2, assuming that (8) and (9) have already been written down:

Put $d = \gcd(a, b)$ and define the integers p and q by $a = dp$ and $b = dq$; notice that $p < q$ and $\gcd(p, q) = 1$. Now (8) implies $c = d^2pq - 1$ and thus we have

$$2^\alpha = d(d^2pq^2 - p - q)$$

$$\text{and} \quad 2^\beta = d(d^2p^2q - p - q)\,. \tag{12}$$

Now 2^β divides $2^\alpha - 2^\beta = d^3pq(q - p)$ and, as p and q are easily seen to be coprime to $d^2p^2q - p - q$, it follows that

$$(d^2p^2q - p - q) \mid d^2(q - p)\,. \tag{13}$$

In particular, we have $d^2p^2q - p - q \leqslant d^2(q - p)$, i.e., $d^2(p^2q + p - q) \leqslant p + q$. As $p^2q + p - q > 0$, this may be weakened to $p^2q + p - q \leqslant p + q$. Hence $p^2q \leqslant 2q$, which is only possible if $p = 1$.

Going back to (13), we get

$$(d^2q - q - 1) \mid d^2(q - 1)\,. \tag{14}$$

Now $2(d^2q - q - 1) \leqslant d^2(q - 1)$ would entail $d^2(q + 1) \leqslant 2(q + 1)$ and thus $d = 1$. But this would tell us that $a = dp = 1$, which is absurd. This argument proves $2(d^2q - q - 1) > d^2(q - 1)$ and in the light of (14) it follows that $d^2q - q - 1 = d^2(q - 1)$, i.e., $q = d^2 - 1$. Plugging this together with $p = 1$ into (12) we infer $2^\beta = d^3(d^2 - 2)$. Hence d and $d^2 - 2$ are powers of 2. Consequently, $d = 2$, $q = 3$, $a = 2$, $b = 6$, and $c = 11$, as desired.

N6. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Consider a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$. For any $m, n \in \mathbb{Z}_{>0}$ we write $f^n(m) = \underbrace{f(f(\ldots f}_{n}(m)\ldots))$. Suppose that f has the following two properties:

(*i*) If $m, n \in \mathbb{Z}_{>0}$, then $\dfrac{f^n(m) - m}{n} \in \mathbb{Z}_{>0}$;

(*ii*) The set $\mathbb{Z}_{>0} \setminus \{f(n) \mid n \in \mathbb{Z}_{>0}\}$ is finite.

Prove that the sequence $f(1) - 1, f(2) - 2, f(3) - 3, \ldots$ is periodic.

(Singapore)

Solution. We split the solution into three steps. In the first of them, we show that the function f is injective and explain how this leads to a useful visualization of f. Then comes the second step, in which most of the work happens: its goal is to show that for any $n \in \mathbb{Z}_{>0}$ the sequence $n, f(n), f^2(n), \ldots$ is an arithmetic progression. Finally, in the third step we put everything together, thus solving the problem.

Step 1. We commence by checking that f is injective. For this purpose, we consider any $m, k \in \mathbb{Z}_{>0}$ with $f(m) = f(k)$. By (*i*), every positive integer n has the property that

$$\frac{k - m}{n} = \frac{f^n(m) - m}{n} - \frac{f^n(k) - k}{n}$$

is a difference of two integers and thus integral as well. But for $n = |k - m| + 1$ this is only possible if $k = m$. Thereby, the injectivity of f is established.

Now recall that due to condition (*ii*) there are finitely many positive integers $a_1, \ldots, a_k$ such that $\mathbb{Z}_{>0}$ is the disjoint union of $\{a_1, \ldots, a_k\}$ and $\{f(n) \mid n \in \mathbb{Z}_{>0}\}$. Notice that by plugging $n = 1$ into condition (*i*) we get $f(m) > m$ for all $m \in \mathbb{Z}_{>0}$.

We contend that every positive integer n may be expressed uniquely in the form $n = f^j(a_i)$ for some $j \geqslant 0$ and $i \in \{1, \ldots, k\}$. The uniqueness follows from the injectivity of f. The existence can be proved by induction on n in the following way. If $n \in \{a_1, \ldots, a_k\}$, then we may take $j = 0$; otherwise there is some $n' < n$ with $f(n') = n$ to which the induction hypothesis may be applied.

The result of the previous paragraph means that every positive integer appears exactly once in the following infinite picture, henceforth referred to as "the Table":

a_1	$f(a_1)$	$f^2(a_1)$	$f^3(a_1)$	$\ldots$
a_2	$f(a_2)$	$f^2(a_2)$	$f^3(a_2)$	$\ldots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
a_k	$f(a_k)$	$f^2(a_k)$	$f^3(a_k)$	$\ldots$

The Table

Step 2. Our next goal is to prove that each row of the Table is an arithmetic progression. Assume contrariwise that the number t of rows which are arithmetic progressions would satisfy $0 \leqslant t < k$. By permuting the rows if necessary we may suppose that precisely the first t rows are arithmetic progressions, say with steps $T_1, \ldots, T_t$. Our plan is to find a further row that is "not too sparse" in an asymptotic sense, and then to prove that such a row has to be an arithmetic progression as well.

Let us write $T = \operatorname{lcm}(T_1, T_2, \ldots, T_t)$ and $A = \max\{a_1, a_2, \ldots, a_t\}$ if $t > 0$; and $T = 1$ and $A = 0$ if $t = 0$. For every integer $n \geqslant A$, the interval $\Delta_n = [n + 1, n + T]$ contains exactly T/T_i

elements of the ith row ($1 \leqslant i \leqslant t$). Therefore, the number of elements from the last $(k-t)$ rows of the Table contained in Δ_n does not depend on $n \geqslant A$. It is not possible that none of these intervals Δ_n contains an element from the $k-t$ last rows, because infinitely many numbers appear in these rows. It follows that for each $n \geqslant A$ the interval Δ_n contains at least one member from these rows.

This yields that for every positive integer d, the interval $\big[A+1, A+(d+1)(k-t)T\big]$ contains at least $(d+1)(k-t)$ elements from the last $k-t$ rows; therefore, there exists an index x with $t+1 \leqslant x \leqslant k$, possibly depending on d, such that our interval contains at least $d+1$ elements from the xth row. In this situation we have

$$f^d(a_x) \leqslant A + (d+1)(k-t)T\,.$$

Finally, since there are finitely many possibilities for x, there exists an index $x \geqslant t+1$ such that the set

$$X = \big\{d \in \mathbb{Z}_{>0} \,\big|\, f^d(a_x) \leqslant A + (d+1)(k-t)T\big\}$$

is infinite. Thereby we have found the "dense row" promised above.

By assumption (i), for every $d \in X$ the number

$$\beta_d = \frac{f^d(a_x) - a_x}{d}$$

is a positive integer not exceeding

$$\frac{A + (d+1)(k-t)T}{d} \leqslant \frac{Ad + 2d(k-t)T}{d} = A + 2(k-t)T\,.$$

This leaves us with finitely many choices for β_d, which means that there exists a number T_x such that the set

$$Y = \big\{d \in X \,\big|\, \beta_d = T_x\big\}$$

is infinite. Notice that we have $f^d(a_x) = a_x + d \cdot T_x$ for all $d \in Y$.

Now we are prepared to prove that the numbers in the xth row form an arithmetic progression, thus coming to a contradiction with our assumption. Let us fix any positive integer j. Since the set Y is infinite, we can choose a number $y \in Y$ such that $y - j > \big|f^j(a_x) - (a_x + jT_x)\big|$. Notice that both numbers

$$f^y(a_x) - f^j(a_x) = f^{y-j}\big(f^j(a_x)\big) - f^j(a_x) \quad \text{and} \quad f^y(a_x) - (a_x + jT_x) = (y-j)T_x$$

are divisible by $y-j$. Thus, the difference between these numbers is also divisible by $y-j$. Since the absolute value of this difference is less than $y-j$, it has to vanish, so we get $f^j(a_x) = a_x + j \cdot T_x$.

Hence, it is indeed true that all rows of the Table are arithmetic progressions.

Step 3. Keeping the above notation in force, we denote the step of the ith row of the table by T_i. Now we claim that we have $f(n) - n = f(n+T) - (n+T)$ for all $n \in \mathbb{Z}_{>0}$, where

$$T = \operatorname{lcm}(T_1, \ldots, T_k)\,.$$

To see this, let any $n \in \mathbb{Z}_{>0}$ be given and denote the index of the row in which it appears in the Table by i. Then we have $f^j(n) = n + j \cdot T_i$ for all $j \in \mathbb{Z}_{>0}$, and thus indeed

$$f(n+T) - f(n) = f^{1+T/T_i}(n) - f(n) = (n + T + T_i) - (n + T_i) = T\,.$$

This concludes the solution.

Comment 1. There are some alternative ways to complete the second part once the index x corresponding to a "dense row" is found. For instance, one may show that for some integer T_x^* the set

$$Y^* = \left\{ j \in \mathbb{Z}_{>0} \,\middle|\, f^{j+1}(a_x) - f^j(a_x) = T_x^* \right\}$$

is infinite, and then one may conclude with a similar divisibility argument.

Comment 2. It may be checked that, conversely, any way to fill out the Table with finitely many arithmetic progressions so that each positive integer appears exactly once, gives rise to a function f satisfying the two conditions mentioned in the problem. For example, we may arrange the positive integers as follows:

2	4	6	8	10	...
1	5	9	13	17	...
3	7	11	15	19	...

This corresponds to the function

$$f(n) = \begin{cases} n+2 & \text{if } n \text{ is even;} \\ n+4 & \text{if } n \text{ is odd.} \end{cases}$$

As this example shows, it is not true that the function $n \mapsto f(n) - n$ has to be constant.

N7. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. For any positive integer k, a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ is called *k-good* if $\gcd\big(f(m)+n, f(n)+m\big) \leqslant k$ for all $m \neq n$. Find all k such that there exists a k-good function.

(Canada)

Answer. $k \geqslant 2$.

Solution 1. For any function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$, let $G_f(m,n) = \gcd\big(f(m)+n, f(n)+m\big)$. Note that a k-good function is also $(k+1)$-good for any positive integer k. Hence, it suffices to show that there does not exist a 1-good function and that there exists a 2-good function.

We first show that there is no 1-good function. Suppose that there exists a function f such that $G_f(m,n) = 1$ for all $m \neq n$. Now, if there are two distinct even numbers m and n such that $f(m)$ and $f(n)$ are both even, then $2 \mid G_f(m,n)$, a contradiction. A similar argument holds if there are two distinct odd numbers m and n such that $f(m)$ and $f(n)$ are both odd. Hence we can choose an even m and an odd n such that $f(m)$ is odd and $f(n)$ is even. This also implies that $2 \mid G_f(m,n)$, a contradiction.

We now construct a 2-good function. Define $f(n) = 2^{g(n)+1} - n - 1$, where g is defined recursively by $g(1) = 1$ and $g(n+1) = (2^{g(n)+1})!$.

For any positive integers $m > n$, set

$$A = f(m) + n = 2^{g(m)+1} - m + n - 1, \qquad B = f(n) + m = 2^{g(n)+1} - n + m - 1.$$

We need to show that $\gcd(A,B) \leqslant 2$. First, note that $A + B = 2^{g(m)+1} + 2^{g(n)+1} - 2$ is not divisible by 4, so that $4 \nmid \gcd(A,B)$. Now we suppose that there is an odd prime p for which $p \mid \gcd(A,B)$ and derive a contradiction.

We first claim that $2^{g(m-1)+1} \geqslant B$. This is a rather weak bound; one way to prove it is as follows. Observe that $g(k+1) > g(k)$ and hence $2^{g(k+1)+1} \geqslant 2^{g(k)+1} + 1$ for every positive integer k. By repeatedly applying this inequality, we obtain $2^{g(m-1)+1} \geqslant 2^{g(n)+1} + (m-1) - n = B$.

Now, since $p \mid B$, we have $p - 1 < B \leqslant 2^{g(m-1)+1}$, so that $p - 1 \mid (2^{g(m-1)+1})! = g(m)$. Hence $2^{g(m)} \equiv 1 \pmod{p}$, which yields $A + B \equiv 2^{g(n)+1} \pmod{p}$. However, since $p \mid A + B$, this implies that $p = 2$, a contradiction.

Solution 2. We provide an alternative construction of a 2-good function f.

Let $\mathcal{P}$ be the set consisting of 4 and all odd primes. For every $p \in \mathcal{P}$, we say that a number $a \in \{0, 1, \ldots, p-1\}$ is *p-useful* if $a \not\equiv -a \pmod{p}$. Note that a residue modulo p which is neither 0 nor 2 is p-useful (the latter is needed only when $p = 4$).

We will construct f recursively; in some steps, we will also define a p-useful number a_p. After the mth step, the construction will satisfy the following conditions:

(i) The values of $f(n)$ have already been defined for all $n \leqslant m$, and p-useful numbers a_p have already been defined for all $p \leqslant m + 2$;

(ii) If $n \leqslant m$ and $p \leqslant m + 2$, then $f(n) + n \not\equiv a_p \pmod{p}$;

(iii) $\gcd\big(f(n_1) + n_2, f(n_2) + n_1\big) \leqslant 2$ for all $n_1 < n_2 \leqslant m$.

If these conditions are satisfied, then f will be a 2-good function.

Step 1. Set $f(1) = 1$ and $a_3 = 1$. Clearly, all the conditions are satisfied.

Step m, for $m \geqslant 2$. We need to determine $f(m)$ and, if $m + 2 \in \mathcal{P}$, the number a_{m+2}.

Defining $f(m)$. Let $X_m = \{p \in \mathcal{P}\colon p \mid f(n) + m \text{ for some } n < m\}$. We will determine $f(m) \mod p$ for all $p \in X_m$ and then choose $f(m)$ using the Chinese Remainder Theorem.

Take any $p \in X_m$. If $p \leqslant m+1$, then we define $f(m) \equiv -a_p - m \pmod p$. Otherwise, if $p \geqslant m+2$, then we define $f(m) \equiv 0 \pmod p$.

Defining a_{m+2}. Now let $p = m+2$ and suppose that $p \in \mathcal{P}$. We choose a_p to be a residue modulo p that is not congruent to 0, 2, or $f(n)+n$ for any $n \leqslant m$. Since $f(1)+1 = 2$, there are at most $m+1 < p$ residues to avoid, so we can always choose a remaining residue.

We first check that (*ii*) is satisfied. We only need to check it if $p = m+2$ or $n = m$. In the former case, we have $f(n)+n \not\equiv a_p \pmod p$ by construction. In the latter case, if $n = m$ and $p \leqslant m+1$, then we have $f(m)+m \equiv -a_p \not\equiv a_p \pmod p$, where we make use of the fact that a_p is p-useful.

Now we check that (*iii*) holds. Suppose, to the contrary, that $p \mid \gcd\big(f(n)+m, f(m)+n\big)$ for some $n < m$. Then $p \in X_m$ and $p \mid f(m)+n$. If $p \geqslant m+2$, then $0 \equiv f(m)+n \equiv n \pmod p$, which is impossible since $n < m < p$.

Otherwise, if $p \leqslant m+1$, then

$$0 \equiv \big(f(m)+n\big) + \big(f(n)+m\big) \equiv \big(f(n)+n\big) + \big(f(m)+m\big) \equiv \big(f(n)+n\big) - a_p \pmod p.$$

This implies that $f(n)+n \equiv a_p \pmod p$, a contradiction with (*ii*).

Comment 1. For any $p \in \mathcal{P}$, we may also define a_p at step m for an arbitrary $m \leqslant p-2$. The construction will work as long as we define a finite number of a_p at each step.

Comment 2. When attempting to construct a 2-good function f recursively, the following way seems natural. Start with setting $f(1) = 1$. Next, for each integer $m > 1$, introduce the set X_m like in Solution 2 and define $f(m)$ so as to satisfy

$$\begin{aligned} f(m) &\equiv f(m-p) \pmod p \qquad \text{for all } p \in X_m \text{ with } p < m, \quad \text{and} \\ f(m) &\equiv 0 \pmod p \qquad \text{for all } p \in X_m \text{ with } p \geqslant m. \end{aligned}$$

This construction might seem to work. Indeed, consider a fixed $p \in \mathcal{P}$, and suppose that p divides $\gcd\big(f(n)+m, f(m)+n\big)$ for some $n < m$. Choose such m and n so that $\max(m,n)$ is minimal. Then $p \in X_m$. We can check that $p < m$, so that the construction implies that p divides $\gcd\big(f(n)+(m-p), f(m-p)+n\big)$. Since $\max(n, m-p) < \max(m,n)$, this *almost* leads to a contradiction—the only trouble is the possibility that $n = m-p$. However, this flaw may happen to be not so easy to fix.

We will present one possible way to repair this argument in the next comment.

Comment 3. There are many recursive constructions for a 2-good function f. Here we sketch one general approach which may be specified in different ways. For convenience, we denote by $\mathbb{Z}_p$ the set of residues modulo p; all operations on elements of $\mathbb{Z}_p$ are also performed modulo p.

The general structure is the same as in Solution 2, i.e. using the Chinese Remainder Theorem to successively determine $f(m)$. But instead of designating a common "safe" residue a_p for future steps, we act as follows.

For every $p \in \mathcal{P}$, in some step of the process we define p subsets $B_p^{(1)}, B_p^{(2)}, \ldots, B_p^{(p)} \subset \mathbb{Z}_p$. The meaning of these sets is that

$$f(m)+m \text{ should be congruent to some element in } B_p^{(i)} \text{ whenever } m \equiv i \pmod p \text{ for } i \in \mathbb{Z}_p. \quad (1)$$

Moreover, in every such subset we specify a *safe* element $b_p^{(i)} \in B_p^{(i)}$. The meaning now is that in future steps, it is safe to set $f(m)+m \equiv b_p^{(i)} \pmod p$ whenever $m \equiv i \pmod p$. In view of (1), this safety will follow from the condition that $p \nmid \gcd\big(b_p^{(i)} + (j-i), c^{(j)} - (j-i)\big)$ for all $j \in \mathbb{Z}_p$ and all $c^{(j)} \in B_p^{(j)}$. In turn, this condition can be rewritten as

$$-b_p^{(i)} \notin B_p^{(j)}, \qquad \text{where} \quad j \equiv i - b_p^{(i)} \pmod p. \quad (2)$$

The construction in Solution 2 is equivalent to setting $b_p^{(i)} = -a_p$ and $B_p^{(i)} = \mathbb{Z}_p \setminus \{a_p\}$ for all i. However, there are different, more technical specifications of our approach.

One may view the (incomplete) construction in Comment 2 as defining $B_p^{(i)}$ and $b_p^{(i)}$ at step $p-1$ by setting $B_p^{(0)} = \{b_p^{(0)}\} = \{0\}$ and $B_p^{(i)} = \{b_p^{(i)}\} = \{f(i) + i \mod p\}$ for every $i = 1, 2, \ldots, p-1$. However, this construction violates (2) as soon as some number of the form $f(i) + i$ is divisible by some p with $i + 2 \leqslant p \in \mathcal{P}$, since then $-b_p^{(i)} = b_p^{(i)} \in B_p^{(i)}$.

Here is one possible way to repair this construction. For all $p \in \mathcal{P}$, we define the sets $B_p^{(i)}$ and the elements $b_p^{(i)}$ at step $(p-2)$ as follows. Set $B_p^{(1)} = \{b_p^{(1)}\} = \{2\}$ and $B_p^{(-1)} = B_p^{(0)} = \{b_p^{(-1)}\} = \{b_p^{(0)}\} = \{-1\}$. Next, for all $i = 2, \ldots, p-2$, define $B_p^{(i)} = \{i, f(i) + i \mod p\}$ and $b_p^{(i)} = i$. One may see that these definitions agree with both (1) and (2).

N8. For every positive integer n with prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$, define

$$\mho(n) = \sum_{i:\, p_i > 10^{100}} \alpha_i.$$

That is, $\mho(n)$ is the number of prime factors of n greater than 10^{100}, counted with multiplicity.

Find all strictly increasing functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$\mho\big(f(a) - f(b)\big) \leqslant \mho(a-b) \qquad \text{for all integers } a \text{ and } b \text{ with } a > b. \tag{1}$$

(Brazil)

Answer. $f(x) = ax + b$, where b is an arbitrary integer, and a is an arbitrary positive integer with $\mho(a) = 0$.

Solution. A straightforward check shows that all the functions listed in the answer satisfy the problem condition. It remains to show the converse.

Assume that f is a function satisfying the problem condition. Notice that the function $g(x) = f(x) - f(0)$ also satisfies this condition. Replacing f by g, we assume from now on that $f(0) = 0$; then $f(n) > 0$ for any positive integer n. Thus, we aim to prove that there exists a positive integer a with $\mho(a) = 0$ such that $f(n) = an$ for all $n \in \mathbb{Z}$.

We start by introducing some notation. Set $N = 10^{100}$. We say that a prime p is *large* if $p > N$, and p is *small* otherwise; let $\mathcal{S}$ be the set of all small primes. Next, we say that a positive integer is *large* or *small* if all its prime factors are such (thus, the number 1 is the unique number which is both large and small). For a positive integer k, we denote the greatest large divisor of k and the greatest small divisor of k by $L(k)$ and $S(k)$, respectively; thus, $k = L(k)S(k)$.

We split the proof into three steps.

Step 1. We prove that for every large k, we have $k \mid f(a) - f(b) \iff k \mid a - b$. In other words, $L\big(f(a) - f(b)\big) = L(a-b)$ for all integers a and b with $a > b$.

We use induction on k. The base case $k = 1$ is trivial. For the induction step, assume that k_0 is a large number, and that the statement holds for all large numbers k with $k < k_0$.

Claim 1. For any integers x and y with $0 < x - y < k_0$, the number k_0 does not divide $f(x) - f(y)$.

Proof. Assume, to the contrary, that $k_0 \mid f(x) - f(y)$. Let $\ell = L(x-y)$; then $\ell \leqslant x - y < k_0$. By the induction hypothesis, $\ell \mid f(x) - f(y)$, and thus $\operatorname{lcm}(k_0, \ell) \mid f(x) - f(y)$. Notice that $\operatorname{lcm}(k_0, \ell)$ is large, and $\operatorname{lcm}(k_0, \ell) \geqslant k_0 > \ell$. But then

$$\mho\big(f(x) - f(y)\big) \geqslant \mho\big(\operatorname{lcm}(k_0, \ell)\big) > \mho(\ell) = \mho(x-y),$$

which is impossible. □

Now we complete the induction step. By Claim 1, for every integer a each of the sequences

$$f(a), f(a+1), \ldots, f(a+k_0-1) \quad \text{and} \quad f(a+1), f(a+2), \ldots, f(a+k_0)$$

forms a complete residue system modulo k_0. This yields $f(a) \equiv f(a+k_0) \pmod{k_0}$. Thus, $f(a) \equiv f(b) \pmod{k_0}$ whenever $a \equiv b \pmod{k_0}$.

Finally, if $a \not\equiv b \pmod{k_0}$ then there exists an integer b' such that $b' \equiv b \pmod{k_0}$ and $|a - b'| < k_0$. Then $f(b) \equiv f(b') \not\equiv f(a) \pmod{k_0}$. The induction step is proved.

Step 2. We prove that for some small integer a there exist infinitely many integers n such that $f(n) = an$. In other words, f is linear on some infinite set.

We start with the following general statement.

Claim 2. There exists a constant c such that $f(t) < ct$ for every positive integer $t > N$.

Proof. Let d be the product of all small primes, and let α be a positive integer such that $2^\alpha > f(N)$. Then, for every $p \in \mathcal{S}$ the numbers $f(0), f(1), \ldots, f(N)$ are distinct modulo p^α. Set $P = d^\alpha$ and $c = P + f(N)$.

Choose any integer $t > N$. Due to the choice of α, for every $p \in \mathcal{S}$ there exists at most one nonnegative integer $i \leqslant N$ with $p^\alpha \mid f(t) - f(i)$. Since $|\mathcal{S}| < N$, we can choose a nonnegative integer $j \leqslant N$ such that $p^\alpha \nmid f(t) - f(j)$ for all $p \in \mathcal{S}$. Therefore, $S\big(f(t) - f(j)\big) < P$.

On the other hand, Step 1 shows that $L\big(f(t) - f(j)\big) = L(t - j) \leqslant t - j$. Since $0 \leqslant j \leqslant N$, this yields

$$f(t) = f(j) + L\big(f(t) - f(j)\big) \cdot S\big(f(t) - f(j)\big) < f(N) + (t - j)P \leqslant \big(P + f(N)\big)t = ct. \quad \square$$

Now let $\mathcal{T}$ be the set of large primes. For every $t \in \mathcal{T}$, Step 1 implies $L\big(f(t)\big) = t$, so the ratio $f(t)/t$ is an integer. Now Claim 2 leaves us with only finitely many choices for this ratio, which means that there exists an infinite subset $\mathcal{T}' \subseteq \mathcal{T}$ and a positive integer a such that $f(t) = at$ for all $t \in \mathcal{T}'$, as required.

Since $L(t) = L\big(f(t)\big) = L(a)L(t)$ for all $t \in \mathcal{T}'$, we get $L(a) = 1$, so the number a is small.

Step 3. We show that $f(x) = ax$ for all $x \in \mathbb{Z}$.

Let $R_i = \big\{x \in \mathbb{Z} \colon x \equiv i \pmod{N!}\big\}$ denote the residue class of i modulo $N!$.

Claim 3. Assume that for some r, there are infinitely many $n \in R_r$ such that $f(n) = an$. Then $f(x) = ax$ for all $x \in R_{r+1}$.

Proof. Choose any $x \in R_{r+1}$. By our assumption, we can select $n \in R_r$ such that $f(n) = an$ and $|n - x| > \big|f(x) - ax\big|$. Since $n - x \equiv r - (r + 1) = -1 \pmod{N!}$, the number $|n - x|$ is large. Therefore, by Step 1 we have $f(x) \equiv f(n) = an \equiv ax \pmod{n - x}$, so $n - x \mid f(x) - ax$. Due to the choice of n, this yields $f(x) = ax$. $\square$

To complete Step 3, notice that the set $\mathcal{T}'$ found in Step 2 contains infinitely many elements of some residue class R_i. Applying Claim 3, we successively obtain that $f(x) = ax$ for all $x \in R_{i+1}, R_{i+2}, \ldots, R_{i+N!} = R_i$. This finishes the solution.

Comment 1. As the proposer also mentions, one may also consider the version of the problem where the condition (1) is replaced by the condition that $L\big(f(a) - f(b)\big) = L(a - b)$ for all integers a and b with $a > b$. This allows to remove of Step 1 from the solution.

Comment 2. Step 2 is the main step of the solution. We sketch several different approaches allowing to perform this step using statements which are weaker than Claim 2.

Approach 1. Let us again denote the product of all small primes by d. We focus on the values $f(d^i)$, $i \geqslant 0$. In view of Step 1, we have $L\big(f(d^i) - f(d^k)\big) = L(d^i - d^k) = d^{i-k} - 1$ for all $i > k \geqslant 0$.

Acting similarly to the beginning of the proof of Claim 2, one may choose a number $\alpha \geqslant 0$ such that the residues of the numbers $f(d^i)$, $i = 0, 1, \ldots, N$, are distinct modulo p^α for each $p \in \mathcal{S}$. Then, for every $i > N$, there exists an exponent $k = k(i) \leqslant N$ such that $S\big(f(d^i) - f(d^k)\big) < P = d^\alpha$.

Since there are only finitely many options for $k(i)$, as well as for the corresponding numbers $S\big(f(d^i) - f(d^k)\big)$, there exists an infinite set I of exponents $i > N$ such that $k(i)$ attains the same value k_0 for all $i \in I$, and such that, moreover, $S\big(f(d^i) - f(d^{k_0})\big)$ attains the same value s_0 for all $i \in I$. Therefore, for all such i we have

$$f(d^i) = f(d^{k_0}) + L\big(f(d^i) - f(d^{k_0})\big) \cdot S\big(f(d^i) - f(d^{k_0})\big) = f(d^{k_0}) + \big(d^{i-k_0} - 1\big)s_0,$$

which means that f is linear on the infinite set $\{d^i \colon i \in I\}$ (although with rational coefficients).

Finally, one may implement the relation $f(d^i) \equiv f(1) \pmod{d^i - 1}$ in order to establish that in fact $f(d^i)/d^i$ is a (small and fixed) integer for all $i \in I$.

Approach 2. Alternatively, one may start with the following lemma.

Lemma. There exists a positive constant c such that

$$L\left(\prod_{i=1}^{3N}\big(f(k)-f(i)\big)\right)=\prod_{i=1}^{3N}L\big(f(k)-f(i)\big)\geqslant c\big(f(k)\big)^{2N}$$

for all $k>3N$.

Proof. Let k be an integer with $k>3N$. Set $\Pi=\prod_{i=1}^{3N}\big(f(k)-f(i)\big)$.

Notice that for every prime $p\in\mathcal{S}$, at most one of the numbers in the set

$$\mathcal{H}=\big\{f(k)-f(i)\colon 1\leqslant i\leqslant 3N\big\}$$

is divisible by a power of p which is greater than $f(3N)$; we say that such elements of $\mathcal{H}$ are *bad*. Now, for each element $h\in\mathcal{H}$ which is not bad we have $S(h)\leqslant f(3N)^N$, while the bad elements do not exceed $f(k)$. Moreover, there are less than N bad elements in $\mathcal{H}$. Therefore,

$$S(\Pi)=\prod_{h\in\mathcal{H}}S\left(h\right)\leqslant\big(f(3N)\big)^{3N^2}\cdot\big(f(k)\big)^N.$$

This easily yields the lemma statement in view of the fact that $L(\Pi)S(\Pi)=\Pi\geqslant\mu\big(f(k)\big)^{3N}$ for some absolute constant μ. □

As a corollary of the lemma, one may get a weaker version of Claim 2 stating that there exists a positive constant C such that $f(k)\leqslant Ck^{3/2}$ for all $k>3N$. Indeed, from Step 1 we have

$$k^{3N}\geqslant\prod_{i=1}^{3N}L(k-i)=\prod_{i=1}^{3N}L\big(f(k)-f(i)\big)\geqslant c\big(f(k)\big)^{2N},$$

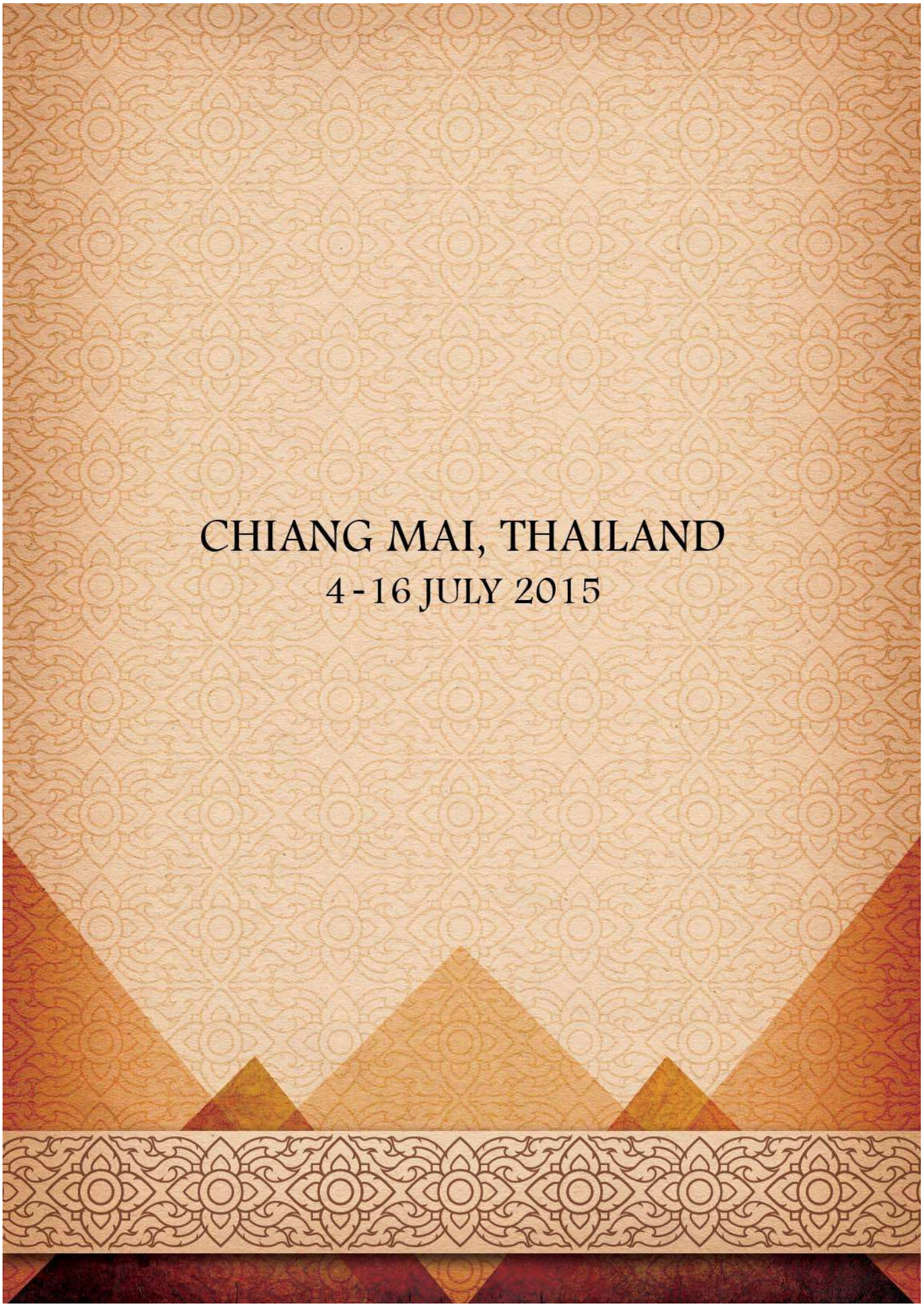
so $f(k)\leqslant c^{-1/(2N)}k^{3/2}$.

To complete Step 2 now, set $a=f(1)$. Due to the estimates above, we may choose a positive integer n_0 such that $\big|f(n)-an\big|<\frac{n(n-1)}{2}$ for all $n\geqslant n_0$.

Take any $n\geqslant n_0$ with $n\equiv 2\pmod{N!}$. Then $L\big(f(n)-f(0)\big)=L(n)=n/2$ and $L\big(f(n)-f(1)\big)=L(n-1)=n-1$; these relations yield $f(n)\equiv f(0)=0\equiv an\pmod{n/2}$ and $f(n)\equiv f(1)=a\equiv an\pmod{n-1}$, respectively. Thus, $\frac{n(n-1)}{2}\mid f(n)-an$, which shows that $f(n)=an$ in view of the estimate above.

Comment 3. In order to perform Step 3, it suffices to establish the equality $f(n)=an$ for any infinite set of values of n. However, if this set has some good structure, then one may find easier ways to complete this step.

For instance, after showing, as in Approach 2, that $f(n)=an$ for all $n\geqslant n_0$ with $n\equiv 2\pmod{N!}$, one may proceed as follows. Pick an arbitrary integer x and take any large prime p which is greater than $|f(x)-ax|$. By the Chinese Remainder Theorem, there exists a positive integer $n>\max(x,n_0)$ such that $n\equiv 2\pmod{N!}$ and $n\equiv x\pmod{p}$. By Step 1, we have $f(x)\equiv f(n)=an\equiv ax\pmod{p}$. Due to the choice of p, this is possible only if $f(x)=ax$.


CHIANG MAI, THAILAND
4-16 JULY 2015

55th
International
Mathematical
Olympiad

PROBLEMS SHORT LIST

WITH SOLUTIONS

IMO 2014
Cape Town - South Africa

Shortlisted Problems with Solutions

55th International Mathematical Olympiad

Cape Town, South Africa, 2014

Note of Confidentiality

The shortlisted problems should be kept strictly confidential until IMO 2015.

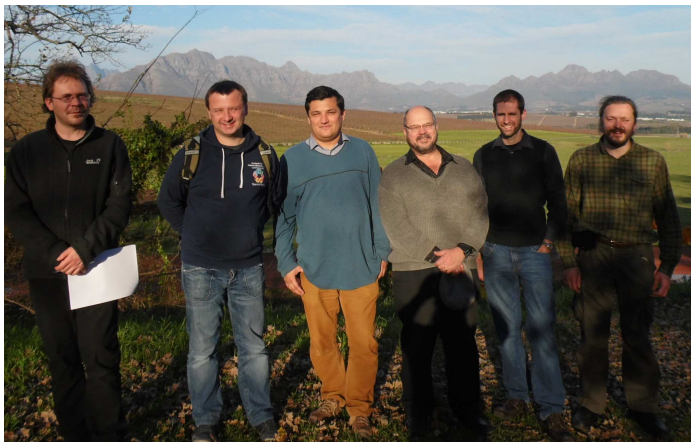
Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2014 thank the following 43 countries for contributing 141 problem proposals.

Australia, Austria, Belgium, Benin, Bulgaria, Colombia, Croatia, Cyprus, Czech Republic, Denmark, Ecuador, Estonia, Finland, France, Georgia, Germany, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Iran, Ireland, Japan, Lithuania, Luxembourg, Malaysia, Mongolia, Netherlands, Nigeria, Pakistan, Russia, Saudi Arabia, Serbia, Slovakia, Slovenia, South Korea, Thailand, Turkey, Ukraine, United Kingdom, U.S.A.

Problem Selection Committee

Johan Meyer
Ilya I. Bogdanov
Géza Kós
Waldemar Pompe
Christian Reiher
Stephan Wagner



Problems

Algebra

A1. Let $z_0 < z_1 < z_2 < \cdots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geqslant 1$ such that

$$z_n < \frac{z_0 + z_1 + \cdots + z_n}{n} \leqslant z_{n+1}.$$

(Austria)

A2. Define the function $f\colon (0,1) \to (0,1)$ by

$$f(x) = \begin{cases} x + \frac{1}{2} & \text{if } x < \frac{1}{2}, \\ x^2 & \text{if } x \geqslant \frac{1}{2}. \end{cases}$$

Let a and b be two real numbers such that $0 < a < b < 1$. We define the sequences a_n and b_n by $a_0 = a$, $b_0 = b$, and $a_n = f(a_{n-1})$, $b_n = f(b_{n-1})$ for $n > 0$. Show that there exists a positive integer n such that

$$(a_n - a_{n-1})(b_n - b_{n-1}) < 0.$$

(Denmark)

A3. For a sequence $x_1, x_2, \ldots, x_n$ of real numbers, we define its *price* as

$$\max_{1 \leqslant i \leqslant n} |x_1 + \cdots + x_i|.$$

Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D. Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the ith step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \cdots + x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G.

Find the least possible constant c such that for every positive integer n, for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leqslant cD$.

(Georgia)

A4. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f\big(f(m) + n\big) + f(m) = f(n) + f(3m) + 2014$$

for all integers m and n.

(Netherlands)

A5. Consider all polynomials $P(x)$ with real coefficients that have the following property: for any two real numbers x and y one has

$$|y^2 - P(x)| \leqslant 2\,|x| \quad \text{if and only if} \quad |x^2 - P(y)| \leqslant 2\,|y|\,.$$

Determine all possible values of $P(0)$.

(Belgium)

A6. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$n^2 + 4f(n) = f(f(n))^2$$

for all $n \in \mathbb{Z}$.

(United Kingdom)

Combinatorics

C1. Let n points be given inside a rectangle R such that no two of them lie on a line parallel to one of the sides of R. The rectangle R is to be dissected into smaller rectangles with sides parallel to the sides of R in such a way that none of these rectangles contains any of the given points in its interior. Prove that we have to dissect R into at least $n + 1$ smaller rectangles.

(Serbia)

C2. We have 2^m sheets of paper, with the number 1 written on each of them. We perform the following operation. In every step we choose two distinct sheets; if the numbers on the two sheets are a and b, then we erase these numbers and write the number $a + b$ on both sheets. Prove that after $m2^{m-1}$ steps, the sum of the numbers on all the sheets is at least 4^m.

(Iran)

C3. Let $n \geqslant 2$ be an integer. Consider an $n \times n$ chessboard divided into n^2 unit squares. We call a configuration of n rooks on this board *happy* if every row and every column contains exactly one rook. Find the greatest positive integer k such that for every happy configuration of rooks, we can find a $k \times k$ square without a rook on any of its k^2 unit squares.

(Croatia)

C4. Construct a tetromino by attaching two 2×1 dominoes along their longer sides such that the midpoint of the longer side of one domino is a corner of the other domino. This construction yields two kinds of tetrominoes with opposite orientations. Let us call them S- and Z-tetrominoes, respectively.



S-tetrominoes Z-tetrominoes

Assume that a lattice polygon P can be tiled with S-tetrominoes. Prove than no matter how we tile P using only S- and Z-tetrominoes, we always use an even number of Z-tetrominoes.

(Hungary)

C5. Consider $n \geqslant 3$ lines in the plane such that no two lines are parallel and no three have a common point. These lines divide the plane into polygonal regions; let $\mathcal{F}$ be the set of regions having finite area. Prove that it is possible to colour $\left\lceil \sqrt{n/2} \right\rceil$ of the lines blue in such a way that no region in $\mathcal{F}$ has a completely blue boundary. (For a real number x, $\lceil x \rceil$ denotes the least integer which is not smaller than x.)

(Austria)

C6. We are given an infinite deck of cards, each with a real number on it. For every real number x, there is exactly one card in the deck that has x written on it. Now two players draw disjoint sets A and B of 100 cards each from this deck. We would like to define a rule that declares one of them a winner. This rule should satisfy the following conditions:

1. The winner only depends on the relative order of the 200 cards: if the cards are laid down in increasing order face down and we are told which card belongs to which player, but not what numbers are written on them, we can still decide the winner.

2. If we write the elements of both sets in increasing order as $A = \{a_1, a_2, \ldots, a_{100}\}$ and $B = \{b_1, b_2, \ldots, b_{100}\}$, and $a_i > b_i$ for all i, then A beats B.

3. If three players draw three disjoint sets A, B, C from the deck, A beats B and B beats C, then A also beats C.

How many ways are there to define such a rule? Here, we consider two rules as different if there exist two sets A and B such that A beats B according to one rule, but B beats A according to the other.

(Russia)

C7. Let M be a set of $n \geqslant 4$ points in the plane, no three of which are collinear. Initially these points are connected with n segments so that each point in M is the endpoint of exactly two segments. Then, at each step, one may choose two segments AB and CD sharing a common interior point and replace them by the segments AC and BD if none of them is present at this moment. Prove that it is impossible to perform $n^3/4$ or more such moves.

(Russia)

C8. A card deck consists of 1024 cards. On each card, a set of distinct decimal digits is written in such a way that no two of these sets coincide (thus, one of the cards is empty). Two players alternately take cards from the deck, one card per turn. After the deck is empty, each player checks if he can throw out one of his cards so that each of the ten digits occurs on an even number of his remaining cards. If one player can do this but the other one cannot, the one who can is the winner; otherwise a draw is declared.

Determine all possible first moves of the first player after which he has a winning strategy.

(Russia)

C9. There are n circles drawn on a piece of paper in such a way that any two circles intersect in two points, and no three circles pass through the same point. Turbo the snail slides along the circles in the following fashion. Initially he moves on one of the circles in clockwise direction. Turbo always keeps sliding along the current circle until he reaches an intersection with another circle. Then he continues his journey on this new circle and also changes the direction of moving, i.e. from clockwise to anticlockwise or *vice versa*.

Suppose that Turbo's path entirely covers all circles. Prove that n must be odd.

(India)

Geometry

G1. The points P and Q are chosen on the side BC of an acute-angled triangle ABC so that $\angle PAB = \angle ACB$ and $\angle QAC = \angle CBA$. The points M and N are taken on the rays AP and AQ, respectively, so that $AP = PM$ and $AQ = QN$. Prove that the lines BM and CN intersect on the circumcircle of the triangle ABC.

(Georgia)

G2. Let ABC be a triangle. The points K, L, and M lie on the segments BC, CA, and AB, respectively, such that the lines AK, BL, and CM intersect in a common point. Prove that it is possible to choose two of the triangles ALM, BMK, and CKL whose inradii sum up to at least the inradius of the triangle ABC.

(Estonia)

G3. Let Ω and O be the circumcircle and the circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM. The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q, respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that $BR \parallel AC$. (Here we always assume that an angle bisector is a ray.)

(Russia)

G4. Consider a fixed circle Γ with three fixed points A, B, and C on it. Also, let us fix a real number $\lambda \in (0, 1)$. For a variable point $P \notin \{A, B, C\}$ on Γ, let M be the point on the segment CP such that $CM = \lambda \cdot CP$. Let Q be the second point of intersection of the circumcircles of the triangles AMP and BMC. Prove that as P varies, the point Q lies on a fixed circle.

(United Kingdom)

G5. Let $ABCD$ be a convex quadrilateral with $\angle B = \angle D = 90^\circ$. Point H is the foot of the perpendicular from A to BD. The points S and T are chosen on the sides AB and AD, respectively, in such a way that H lies inside triangle SCT and

$$\angle SHC - \angle BSC = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ .$$

Prove that the circumcircle of triangle SHT is tangent to the line BD.

(Iran)

G6. Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB, respectively, and let M be the midpoint of EF. Let the perpendicular bisector of EF intersect the line BC at K, and let the perpendicular bisector of MK intersect the lines AC and AB at S and T, respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic.

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that

$$\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC} .$$

(Iran)

G7. Let ABC be a triangle with circumcircle Ω and incentre I. Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at points U and V, respectively. Let the line passing through U and parallel to AI intersect AV at X, and let the line passing through V and parallel to AI intersect AB at Y. Let W and Z be the midpoints of AX and BC, respectively. Prove that if the points I, X, and Y are collinear, then the points I, W, and Z are also collinear.

(U.S.A.)

Number Theory

N1. Let $n \geqslant 2$ be an integer, and let A_n be the set

$$A_n = \{2^n - 2^k \mid k \in \mathbb{Z},\ 0 \leqslant k < n\}.$$

Determine the largest positive integer that cannot be written as the sum of one or more (not necessarily distinct) elements of A_n.

(Serbia)

N2. Determine all pairs (x, y) of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1\,.$$

(U.S.A.)

N3. A coin is called a *Cape Town coin* if its value is $1/n$ for some positive integer n. Given a collection of Cape Town coins of total value at most $99 + \frac{1}{2}$, prove that it is possible to split this collection into at most 100 groups each of total value at most 1.

(Luxembourg)

N4. Let $n > 1$ be a given integer. Prove that infinitely many terms of the sequence $(a_k)_{k \geqslant 1}$, defined by

$$a_k = \left\lfloor \frac{n^k}{k} \right\rfloor,$$

are odd. (For a real number x, $\lfloor x \rfloor$ denotes the largest integer not exceeding x.)

(Hong Kong)

N5. Find all triples (p, x, y) consisting of a prime number p and two positive integers x and y such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p.

(Belgium)

N6. Let $a_1 < a_2 < \cdots < a_n$ be pairwise coprime positive integers with a_1 being prime and $a_1 \geqslant n + 2$. On the segment $I = [0, a_1 a_2 \cdots a_n]$ of the real line, mark all integers that are divisible by at least one of the numbers $a_1, \ldots, a_n$. These points split I into a number of smaller segments. Prove that the sum of the squares of the lengths of these segments is divisible by a_1.

(Serbia)

N7. Let $c \geqslant 1$ be an integer. Define a sequence of positive integers by $a_1 = c$ and

$$a_{n+1} = a_n^3 - 4c \cdot a_n^2 + 5c^2 \cdot a_n + c$$

for all $n \geqslant 1$. Prove that for each integer $n \geqslant 2$ there exists a prime number p dividing a_n but none of the numbers $a_1, \ldots, a_{n-1}$.

(Austria)

N8. For every real number x, let $\|x\|$ denote the distance between x and the nearest integer. Prove that for every pair (a, b) of positive integers there exist an odd prime p and a positive integer k satisfying

$$\left\| \frac{a}{p^k} \right\| + \left\| \frac{b}{p^k} \right\| + \left\| \frac{a+b}{p^k} \right\| = 1.$$

(Hungary)

Solutions

Algebra

A1. Let $z_0 < z_1 < z_2 < \cdots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geqslant 1$ such that

$$z_n < \frac{z_0 + z_1 + \cdots + z_n}{n} \leqslant z_{n+1}. \tag{1}$$

(Austria)

Solution. For $n = 1, 2, \ldots$ define

$$d_n = (z_0 + z_1 + \cdots + z_n) - nz_n.$$

The sign of d_n indicates whether the first inequality in (1) holds; i.e., it is satisfied if and only if $d_n > 0$.

Notice that

$$nz_{n+1} - (z_0 + z_1 + \cdots + z_n) = (n+1)z_{n+1} - (z_0 + z_1 + \cdots + z_n + z_{n+1}) = -d_{n+1},$$

so the second inequality in (1) is equivalent to $d_{n+1} \leqslant 0$. Therefore, we have to prove that there is a unique index $n \geqslant 1$ that satisfies $d_n > 0 \geqslant d_{n+1}$.

By its definition the sequence $d_1, d_2, \ldots$ consists of integers and we have

$$d_1 = (z_0 + z_1) - 1 \cdot z_1 = z_0 > 0.$$

From

$$d_{n+1} - d_n = \big((z_0 + \cdots + z_n + z_{n+1}) - (n+1)z_{n+1}\big) - \big((z_0 + \cdots + z_n) - nz_n\big) = n(z_n - z_{n+1}) < 0$$

we can see that $d_{n+1} < d_n$ and thus the sequence strictly decreases.

Hence, we have a decreasing sequence $d_1 > d_2 > \ldots$ of integers such that its first element d_1 is positive. The sequence must drop below 0 at some point, and thus there is a unique index n, that is the index of the last positive term, satisfying $d_n > 0 \geqslant d_{n+1}$.

Comment. Omitting the assumption that $z_1, z_2, \ldots$ are integers allows the numbers d_n to be all positive. In such cases the desired n does not exist. This happens for example if $z_n = 2 - \frac{1}{2^n}$ for all integers $n \geqslant 0$.

A2. Define the function $f\colon (0,1) \to (0,1)$ by

$$f(x) = \begin{cases} x + \frac{1}{2} & \text{if } x < \frac{1}{2}, \\ x^2 & \text{if } x \geqslant \frac{1}{2}. \end{cases}$$

Let a and b be two real numbers such that $0 < a < b < 1$. We define the sequences a_n and b_n by $a_0 = a$, $b_0 = b$, and $a_n = f(a_{n-1})$, $b_n = f(b_{n-1})$ for $n > 0$. Show that there exists a positive integer n such that

$$(a_n - a_{n-1})(b_n - b_{n-1}) < 0.$$

(Denmark)

Solution. Note that

$$f(x) - x = \tfrac{1}{2} > 0$$

if $x < \frac{1}{2}$ and

$$f(x) - x = x^2 - x < 0$$

if $x \geqslant \frac{1}{2}$. So if we consider $(0,1)$ as being divided into the two subintervals $I_1 = (0, \frac{1}{2})$ and $I_2 = [\frac{1}{2}, 1)$, the inequality

$$(a_n - a_{n-1})(b_n - b_{n-1}) = \big(f(a_{n-1}) - a_{n-1}\big)\big(f(b_{n-1}) - b_{n-1}\big) < 0$$

holds if and only if a_{n-1} and b_{n-1} lie in distinct subintervals.

Let us now assume, to the contrary, that a_k and b_k always lie in the same subinterval. Consider the distance $d_k = |a_k - b_k|$. If both a_k and b_k lie in I_1, then

$$d_{k+1} = |a_{k+1} - b_{k+1}| = \big|a_k + \tfrac{1}{2} - b_k - \tfrac{1}{2}\big| = d_k.$$

If, on the other hand, a_k and b_k both lie in I_2, then $\min(a_k, b_k) \geqslant \frac{1}{2}$ and $\max(a_k, b_k) = \min(a_k, b_k) + d_k \geqslant \frac{1}{2} + d_k$, which implies

$$d_{k+1} = |a_{k+1} - b_{k+1}| = \big|a_k^2 - b_k^2\big| = \big|(a_k - b_k)(a_k + b_k)\big| \geqslant |a_k - b_k|\big(\tfrac{1}{2} + \tfrac{1}{2} + d_k\big) = d_k(1 + d_k) \geqslant d_k.$$

This means that the difference d_k is non-decreasing, and in particular $d_k \geqslant d_0 > 0$ for all k.

We can even say more. If a_k and b_k lie in I_2, then

$$d_{k+2} \geqslant d_{k+1} \geqslant d_k(1 + d_k) \geqslant d_k(1 + d_0).$$

If a_k and b_k both lie in I_1, then a_{k+1} and b_{k+1} both lie in I_2, and so we have

$$d_{k+2} \geqslant d_{k+1}(1 + d_{k+1}) \geqslant d_{k+1}(1 + d_0) = d_k(1 + d_0).$$

In either case, $d_{k+2} \geqslant d_k(1 + d_0)$, and inductively we get

$$d_{2m} \geqslant d_0(1 + d_0)^m.$$

For sufficiently large m, the right-hand side is greater than 1, but since a_{2m}, b_{2m} both lie in $(0,1)$, we must have $d_{2m} < 1$, a contradiction.

Thus there must be a positive integer n such that a_{n-1} and b_{n-1} do not lie in the same subinterval, which proves the desired statement.

A3. For a sequence $x_1, x_2, \ldots, x_n$ of real numbers, we define its *price* as

$$\max_{1 \leqslant i \leqslant n} |x_1 + \cdots + x_i|.$$

Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D. Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the ith step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \cdots + x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G.

Find the least possible constant c such that for every positive integer n, for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leqslant cD$.

(Georgia)

Answer. $c = 2$.

Solution. If the initial numbers are 1, -1, 2, and -2, then Dave may arrange them as $1, -2, 2, -1$, while George may get the sequence $1, -1, 2, -2$, resulting in $D = 1$ and $G = 2$. So we obtain $c \geqslant 2$.

Therefore, it remains to prove that $G \leqslant 2D$. Let $x_1, x_2, \ldots, x_n$ be the numbers Dave and George have at their disposal. Assume that Dave and George arrange them into sequences $d_1, d_2, \ldots, d_n$ and $g_1, g_2, \ldots, g_n$, respectively. Put

$$M = \max_{1 \leqslant i \leqslant n} |x_i|, \quad S = |x_1 + \cdots + x_n|, \quad \text{and} \quad N = \max\{M, S\}.$$

We claim that

$$D \geqslant S, \tag{1}$$

$$D \geqslant \frac{M}{2}, \qquad \text{and} \tag{2}$$

$$G \leqslant N = \max\{M, S\}. \tag{3}$$

These inequalities yield the desired estimate, as $G \leqslant \max\{M, S\} \leqslant \max\{M, 2S\} \leqslant 2D$.

The inequality (1) is a direct consequence of the definition of the price.

To prove (2), consider an index i with $|d_i| = M$. Then we have

$$M = |d_i| = \big|(d_1 + \cdots + d_i) - (d_1 + \cdots + d_{i-1})\big| \leqslant |d_1 + \cdots + d_i| + |d_1 + \cdots + d_{i-1}| \leqslant 2D,$$

as required.

It remains to establish (3). Put $h_i = g_1 + g_2 + \cdots + g_i$. We will prove by induction on i that $|h_i| \leqslant N$. The base case $i = 1$ holds, since $|h_1| = |g_1| \leqslant M \leqslant N$. Notice also that $|h_n| = S \leqslant N$.

For the induction step, assume that $|h_{i-1}| \leqslant N$. We distinguish two cases.

Case 1. Assume that no two of the numbers g_i, g_{i+1}, …, g_n have opposite signs.

Without loss of generality, we may assume that they are all nonnegative. Then one has $h_{i-1} \leqslant h_i \leqslant \cdots \leqslant h_n$, thus

$$|h_i| \leqslant \max\big\{|h_{i-1}|, |h_n|\big\} \leqslant N.$$

Case 2. Among the numbers $g_i, g_{i+1}, \ldots, g_n$ there are positive and negative ones.

Then there exists some index $j \geqslant i$ such that $h_{i-1}g_j \leqslant 0$. By the definition of George's sequence we have

$$|h_i| = |h_{i-1} + g_i| \leqslant |h_{i-1} + g_j| \leqslant \max\{|h_{i-1}|, |g_j|\} \leqslant N.$$

Thus, the induction step is established.

Comment 1. One can establish the weaker inequalities $D \geqslant \frac{M}{2}$ and $G \leqslant D + \frac{M}{2}$ from which the result also follows.

Comment 2. One may ask a more specific question to find the maximal suitable c if the number n is fixed. For $n = 1$ or 2, the answer is $c = 1$. For $n = 3$, the answer is $c = \frac{3}{2}$, and it is reached e.g., for the collection $1, 2, -4$. Finally, for $n \geqslant 4$ the answer is $c = 2$. In this case the arguments from the solution above apply, and the answer is reached e.g., for the same collection $1, -1, 2, -2$, augmented by several zeroes.

A4. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f\big(f(m)+n\big)+f(m)=f(n)+f(3m)+2014 \tag{1}$$

for all integers m and n.

(Netherlands)

Answer. There is only one such function, namely $n \longmapsto 2n+1007$.

Solution. Let f be a function satisfying (1). Set $C=1007$ and define the function $g\colon \mathbb{Z} \to \mathbb{Z}$ by $g(m)=f(3m)-f(m)+2C$ for all $m \in \mathbb{Z}$; in particular, $g(0)=2C$. Now (1) rewrites as

$$f\big(f(m)+n\big)=g(m)+f(n)$$

for all $m,n \in \mathbb{Z}$. By induction in both directions it follows that

$$f\big(tf(m)+n\big)=tg(m)+f(n) \tag{2}$$

holds for all $m,n,t \in \mathbb{Z}$. Applying this, for any $r \in \mathbb{Z}$, to the triples $\big(r,0,f(0)\big)$ and $\big(0,0,f(r)\big)$ in place of (m,n,t) we obtain

$$f(0)g(r)=f\big(f(r)f(0)\big)-f(0)=f(r)g(0)\,.$$

Now if $f(0)$ vanished, then $g(0)=2C>0$ would entail that f vanishes identically, contrary to (1). Thus $f(0)\neq 0$ and the previous equation yields $g(r)=\alpha\, f(r)$, where $\alpha=\frac{g(0)}{f(0)}$ is some nonzero constant.

So the definition of g reveals $f(3m)=(1+\alpha)f(m)-2C$, i.e.,

$$f(3m)-\beta=(1+\alpha)\big(f(m)-\beta\big) \tag{3}$$

for all $m \in \mathbb{Z}$, where $\beta=\frac{2C}{\alpha}$. By induction on k this implies

$$f(3^k m)-\beta=(1+\alpha)^k\big(f(m)-\beta\big) \tag{4}$$

for all integers $k \geqslant 0$ and m.

Since $3 \nmid 2014$, there exists by (1) some value $d=f(a)$ attained by f that is not divisible by 3. Now by (2) we have $f(n+td)=f(n)+tg(a)=f(n)+\alpha\cdot tf(a)$, i.e.,

$$f(n+td)=f(n)+\alpha\cdot td \tag{5}$$

for all $n,t \in \mathbb{Z}$.

Let us fix any positive integer k with $d \mid (3^k-1)$, which is possible, since $\gcd(3,d)=1$. E.g., by the Euler–Fermat theorem, we may take $k=\varphi(|d|)$. Now for each $m \in \mathbb{Z}$ we get

$$f(3^k m)=f(m)+\alpha(3^k-1)m$$

from (5), which in view of (4) yields $\big((1+\alpha)^k-1\big)\big(f(m)-\beta\big)=\alpha(3^k-1)m$. Since $\alpha\neq 0$, the right hand side does not vanish for $m\neq 0$, wherefore the first factor on the left hand side cannot vanish either. It follows that

$$f(m)=\frac{\alpha(3^k-1)}{(1+\alpha)^k-1}\cdot m+\beta\,.$$

So f is a linear function, say $f(m) = Am + \beta$ for all $m \in \mathbb{Z}$ with some constant $A \in \mathbb{Q}$. Plugging this into (1) one obtains $(A^2 - 2A)m + (A\beta - 2C) = 0$ for all m, which is equivalent to the conjunction of

$$A^2 = 2A \qquad \text{and} \qquad A\beta = 2C\,. \tag{6}$$

The first equation is equivalent to $A \in \{0, 2\}$, and as $C \neq 0$ the second one gives

$$A = 2 \qquad \text{and} \qquad \beta = C\,. \tag{7}$$

This shows that f is indeed the function mentioned in the answer and as the numbers found in (7) do indeed satisfy the equations (6) this function is indeed as desired.

Comment 1. One may see that $\alpha = 2$. A more pedestrian version of the above solution starts with a direct proof of this fact, that can be obtained by substituting some special values into (1), e.g., as follows.

Set $D = f(0)$. Plugging $m = 0$ into (1) and simplifying, we get

$$f(n + D) = f(n) + 2C \tag{8}$$

for all $n \in \mathbb{Z}$. In particular, for $n = 0, D, 2D$ we obtain $f(D) = 2C + D$, $f(2D) = f(D) + 2C = 4C + D$, and $f(3D) = f(2D) + 2C = 6C + D$. So substituting $m = D$ and $n = r - D$ into (1) and applying (8) with $n = r - D$ afterwards we learn

$$f(r + 2C) + 2C + D = \big(f(r) - 2C\big) + (6C + D) + 2C\,,$$

i.e., $f(r + 2C) = f(r) + 4C$. By induction in both directions it follows that

$$f(n + 2Ct) = f(n) + 4Ct \tag{9}$$

holds for all $n, t \in \mathbb{Z}$.

Claim. If a and b denote two integers with the property that $f(n + a) = f(n) + b$ holds for all $n \in \mathbb{Z}$, then $b = 2a$.

Proof. Applying induction in both directions to the assumption we get $f(n + ta) = f(n) + tb$ for all $n, t \in \mathbb{Z}$. Plugging $(n, t) = (0, 2C)$ into this equation and $(n, t) = (0, a)$ into (9) we get $f(2aC) - f(0) = 2bC = 4aC$, and, as $C \neq 0$, the claim follows. □

Now by (1), for any $m \in \mathbb{Z}$, the numbers $a = f(m)$ and $b = f(3m) - f(m) + 2C$ have the property mentioned in the claim, whence we have

$$f(3m) - C = 3\big(f(m) - C\big)\,.$$

In view of (3) this tells us indeed that $\alpha = 2$.

Now the solution may be completed as above, but due to our knowledge of $\alpha = 2$ we get the desired formula $f(m) = 2m + C$ directly without having the need to go through all linear functions. Now it just remains to check that this function does indeed satisfy (1).

Comment 2. It is natural to wonder what happens if one replaces the number 2014 appearing in the statement of the problem by some arbitrary integer B.

If B is odd, there is no such function, as can be seen by using the same ideas as in the above solution.

If $B \neq 0$ is even, however, then the only such function is given by $n \longmapsto 2n + B/2$. In case $3 \nmid B$ this was essentially proved above, but for the general case one more idea seems to be necessary. Writing $B = 3^\nu \cdot k$ with some integers ν and k such that $3 \nmid k$ one can obtain $f(n) = 2n + B/2$ for all n that are divisible by 3^ν in the same manner as usual; then one may use the formula $f(3n) = 3f(n) - B$ to establish the remaining cases.

Finally, in case $B = 0$ there are more solutions than just the function $n \longmapsto 2n$. It can be shown that all these other functions are periodic; to mention just one kind of example, for any even integers r and s the function

$$f(n) = \begin{cases} r & \text{if } n \text{ is even,} \\ s & \text{if } n \text{ is odd,} \end{cases}$$

also has the property under discussion.

A5. Consider all polynomials $P(x)$ with real coefficients that have the following property: for any two real numbers x and y one has

$$|y^2 - P(x)| \leqslant 2|x| \quad \text{if and only if} \quad |x^2 - P(y)| \leqslant 2|y|\,. \tag{1}$$

Determine all possible values of $P(0)$.

(Belgium)

Answer. The set of possible values of $P(0)$ is $(-\infty, 0) \cup \{1\}$.

Solution.

Part I. We begin by verifying that these numbers are indeed possible values of $P(0)$. To see that each negative real number $-C$ can be $P(0)$, it suffices to check that for every $C > 0$ the polynomial $P(x) = -\left(\frac{2x^2}{C} + C\right)$ has the property described in the statement of the problem. Due to symmetry it is enough for this purpose to prove $|y^2 - P(x)| > 2|x|$ for any two real numbers x and y. In fact we have

$$|y^2 - P(x)| = y^2 + \frac{x^2}{C} + \frac{(|x| - C)^2}{C} + 2|x| \geqslant \frac{x^2}{C} + 2|x| \geqslant 2|x|\,,$$

where in the first estimate equality can only hold if $|x| = C$, whilst in the second one it can only hold if $x = 0$. As these two conditions cannot be met at the same time, we have indeed $|y^2 - P(x)| > 2|x|$.

To show that $P(0) = 1$ is possible as well, we verify that the polynomial $P(x) = x^2 + 1$ satisfies (1). Notice that for all real numbers x and y we have

$$\begin{aligned} |y^2 - P(x)| \leqslant 2|x| &\iff (y^2 - x^2 - 1)^2 \leqslant 4x^2 \\ &\iff 0 \leqslant \left((y^2 - (x-1)^2\right)\left((x+1)^2 - y^2\right) \\ &\iff 0 \leqslant (y - x + 1)(y + x - 1)(x + 1 - y)(x + 1 + y) \\ &\iff 0 \leqslant \left((x+y)^2 - 1\right)\left(1 - (x-y)^2\right)\,. \end{aligned}$$

Since this inequality is symmetric in x and y, we are done.

Part II. Now we show that no values other than those mentioned in the answer are possible for $P(0)$. To reach this we let P denote any polynomial satisfying (1) and $P(0) \geqslant 0$; as we shall see, this implies $P(x) = x^2 + 1$ for all real x, which is actually more than what we want.

First step: We prove that P is even.

By (1) we have

$$|y^2 - P(x)| \leqslant 2|x| \iff |x^2 - P(y)| \leqslant 2|y| \iff |y^2 - P(-x)| \leqslant 2|x|$$

for all real numbers x and y. Considering just the equivalence of the first and third statement and taking into account that y^2 may vary through $\mathbb{R}_{\geqslant 0}$ we infer that

$$\left[P(x) - 2|x|, P(x) + 2|x|\right] \cap \mathbb{R}_{\geqslant 0} = \left[P(-x) - 2|x|, P(-x) + 2|x|\right] \cap \mathbb{R}_{\geqslant 0}$$

holds for all $x \in \mathbb{R}$. We claim that there are infinitely many real numbers x such that $P(x) + 2|x| \geqslant 0$. This holds in fact for any real polynomial with $P(0) \geqslant 0$; in order to see this, we may assume that the coefficient of P appearing in front of x is nonnegative. In this case the desired inequality holds for all sufficiently small positive real numbers.

For such numbers x satisfying $P(x) + 2|x| \geqslant 0$ we have $P(x) + 2|x| = P(-x) + 2|x|$ by the previous displayed formula, and hence also $P(x) = P(-x)$. Consequently the polynomial $P(x) - P(-x)$ has infinitely many zeros, wherefore it has to vanish identically. Thus P is indeed even.

Second step: We prove that $P(t) > 0$ for all $t \in \mathbb{R}$.

Let us assume for a moment that there exists a real number $t \neq 0$ with $P(t) = 0$. Then there is some open interval I around t such that $|P(y)| \leqslant 2\,|y|$ holds for all $y \in I$. Plugging $x = 0$ into (1) we learn that $y^2 = P(0)$ holds for all $y \in I$, which is clearly absurd. We have thus shown $P(t) \neq 0$ for all $t \neq 0$.

In combination with $P(0) \geqslant 0$ this informs us that our claim could only fail if $P(0) = 0$. In this case there is by our first step a polynomial $Q(x)$ such that $P(x) = x^2 Q(x)$. Applying (1) to $x = 0$ and an arbitrary $y \neq 0$ we get $|y\,Q(y)| > 2$, which is surely false when y is sufficiently small.

Third step: We prove that P is a quadratic polynomial.

Notice that P cannot be constant, for otherwise if $x = \sqrt{P(0)}$ and y is sufficiently large, the first part of (1) is false whilst the second part is true. So the degree n of P has to be at least 1. By our first step n has to be even as well, whence in particular $n \geqslant 2$.

Now assume that $n \geqslant 4$. Plugging $y = \sqrt{P(x)}$ into (1) we get $\bigl|x^2 - P\bigl(\sqrt{P(x)}\,\bigr)\bigr| \leqslant 2\sqrt{P(x)}$ and hence

$$P\bigl(\sqrt{P(x)}\,\bigr) \leqslant x^2 + 2\sqrt{P(x)}$$

for all real x. Choose positive real numbers x_0, a, and b such that if $x \in (x_0, \infty)$, then $ax^n < P(x) < bx^n$; this is indeed possible, for if $d > 0$ denotes the leading coefficient of P, then $\lim\limits_{x\to\infty} \frac{P(x)}{x^n} = d$, whence for instance the numbers $a = \frac{d}{2}$ and $b = 2d$ work provided that x_0 is chosen large enough.

Now for all sufficiently large real numbers x we have

$$a^{n/2+1} x^{n^2/2} < aP(x)^{n/2} < P\bigl(\sqrt{P(x)}\,\bigr) \leqslant x^2 + 2\sqrt{P(x)} < x^{n/2} + 2b^{1/2} x^{n/2}\,,$$

i.e.

$$x^{(n^2-n)/2} < \frac{1 + 2b^{1/2}}{a^{n/2+1}}\,,$$

which is surely absurd. Thus P is indeed a quadratic polynomial.

Fourth step: We prove that $P(x) = x^2 + 1$.

In the light of our first three steps there are two real numbers $a > 0$ and b such that $P(x) = ax^2 + b$. Now if x is large enough and $y = \sqrt{a}\,x$, the left part of (1) holds and the right part reads $|(1 - a^2)x^2 - b| \leqslant 2\sqrt{a}\,x$. In view of the fact that $a > 0$ this is only possible if $a = 1$. Finally, substituting $y = x + 1$ with $x > 0$ into (1) we get

$$|2x + 1 - b| \leqslant 2x \iff |2x + 1 + b| \leqslant 2x + 2\,,$$

i.e.,

$$b \in [1, 4x + 1] \iff b \in [-4x - 3, 1]$$

for all $x > 0$. Choosing x large enough, we can achieve that at least one of these two statements holds; then both hold, which is only possible if $b = 1$, as desired.

Comment 1. There are some issues with this problem in that its most natural solutions seem to use some basic facts from analysis, such as the continuity of polynomials or the intermediate value theorem. Yet these facts are intuitively obvious and implicitly clear to the students competing at this level of difficulty, so that the Problem Selection Committee still thinks that the problem is suitable for the IMO.

Comment 2. It seems that most solutions will in the main case, where $P(0)$ is nonnegative, contain an argument that is somewhat asymptotic in nature showing that P is quadratic, and some part narrowing that case down to $P(x) = x^2 + 1$.

Comment 3. It is also possible to skip the first step and start with the second step directly, but then one has to work a bit harder to rule out the case $P(0) = 0$. Let us sketch one possibility of doing this: Take the auxiliary polynomial $Q(x)$ such that $P(x) = xQ(x)$. Applying (1) to $x = 0$ and an arbitrary $y \neq 0$ we get $|Q(y)| > 2$. Hence we either have $Q(z) \geqslant 2$ for all real z or $Q(z) \leqslant -2$ for all real z. In particular there is some $\eta \in \{-1, +1\}$ such that $P(\eta) \geqslant 2$ and $P(-\eta) \leqslant -2$. Substituting $x = \pm\eta$ into (1) we learn

$$|y^2 - P(\eta)| \leqslant 2 \iff |1 - P(y)| \leqslant 2\,|y| \iff |y^2 - P(-\eta)| \leqslant 2\,.$$

But for $y = \sqrt{P(\eta)}$ the first statement is true, whilst the third one is false.

Also, if one has not obtained the evenness of P before embarking on the fourth step, one needs to work a bit harder there, but not in a way that is likely to cause major difficulties.

Comment 4. Truly curious people may wonder about the set of all polynomials having property (1). As explained in the solution above, $P(x) = x^2 + 1$ is the only one with $P(0) = 1$. On the other hand, it is not hard to notice that for negative $P(0)$ there are more possibilities than those mentioned above. E.g., as remarked by the proposer, if a and b denote two positive real numbers with $ab > 1$ and Q denotes a polynomial attaining nonnegative values only, then $P(x) = -\left(ax^2 + b + Q(x)\right)$ works.

More generally, it may be proved that if $P(x)$ satisfies (1) and $P(0) < 0$, then $-P(x) > 2\,|x|$ holds for all $x \in \mathbb{R}$ so that one just considers the equivalence of two false statements. One may generate all such polynomials P by going through all combinations of a solution of the polynomial equation

$$x = A(x)B(x) + C(x)D(x)$$

and a real $E > 0$, and setting

$$P(x) = -\left(A(x)^2 + B(x)^2 + C(x)^2 + D(x)^2 + E\right)$$

for each of them.

A6. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$n^2 + 4f(n) = f(f(n))^2 \tag{1}$$

for all $n \in \mathbb{Z}$.

(United Kingdom)

Answer. The possibilities are:

- $f(n) = n + 1$ for all n;
- or, for some $a \geqslant 1$, $f(n) = \begin{cases} n+1, & n > -a, \\ -n+1, & n \leqslant -a; \end{cases}$
- or $f(n) = \begin{cases} n+1, & n > 0, \\ 0, & n = 0, \\ -n+1, & n < 0. \end{cases}$

Solution 1.

Part I. Let us first check that each of the functions above really satisfies the given functional equation. If $f(n) = n + 1$ for all n, then we have

$$n^2 + 4f(n) = n^2 + 4n + 4 = (n + 2)^2 = f(n + 1)^2 = f(f(n))^2.$$

If $f(n) = n + 1$ for $n > -a$ and $f(n) = -n + 1$ otherwise, then we have the same identity for $n > -a$ and

$$n^2 + 4f(n) = n^2 - 4n + 4 = (2 - n)^2 = f(1 - n)^2 = f(f(n))^2$$

otherwise. The same applies to the third solution (with $a = 0$), where in addition one has

$$0^2 + 4f(0) = 0 = f(f(0))^2.$$

Part II. It remains to prove that these are really the only functions that satisfy our functional equation. We do so in three steps:

Step 1: We prove that $f(n) = n + 1$ for $n > 0$.

Consider the sequence (a_k) given by $a_k = f^k(1)$ for $k \geqslant 0$. Setting $n = a_k$ in (1), we get

$$a_k^2 + 4a_{k+1} = a_{k+2}^2.$$

Of course, $a_0 = 1$ by definition. Since $a_2^2 = 1 + 4a_1$ is odd, a_2 has to be odd as well, so we set $a_2 = 2r + 1$ for some $r \in \mathbb{Z}$. Then $a_1 = r^2 + r$ and consequently

$$a_3^2 = a_1^2 + 4a_2 = (r^2 + r)^2 + 8r + 4.$$

Since $8r + 4 \neq 0$, $a_3^2 \neq (r^2 + r)^2$, so the difference between a_3^2 and $(r^2 + r)^2$ is at least the distance from $(r^2 + r)^2$ to the nearest even square (since $8r + 4$ and $r^2 + r$ are both even). This implies that

$$|8r + 4| = \left|a_3^2 - (r^2 + r)^2\right| \geqslant (r^2 + r)^2 - (r^2 + r - 2)^2 = 4(r^2 + r - 1),$$

(for $r = 0$ and $r = -1$, the estimate is trivial, but this does not matter). Therefore, we ave

$$4r^2 \leqslant |8r + 4| - 4r + 4.$$

If $|r| \geqslant 4$, then

$$4r^2 \geqslant 16|r| \geqslant 12|r| + 16 > 8|r| + 4 + 4|r| + 4 \geqslant |8r + 4| - 4r + 4,$$

a contradiction. Thus $|r| < 4$. Checking all possible remaining values of r, we find that $(r^2 + r)^2 + 8r + 4$ is only a square in three cases: $r = -3$, $r = 0$ and $r = 1$. Let us now distinguish these three cases:

- $r = -3$, thus $a_1 = 6$ and $a_2 = -5$. For each $k \geqslant 1$, we have

  $$a_{k+2} = \pm\sqrt{a_k^2 + 4a_{k+1}},$$

  and the sign needs to be chosen in such a way that $a_{k+1}^2 + 4a_{k+2}$ is again a square. This yields $a_3 = -4$, $a_4 = -3$, $a_5 = -2$, $a_6 = -1$, $a_7 = 0$, $a_8 = 1$, $a_9 = 2$. At this point we have reached a contradiction, since $f(1) = f(a_0) = a_1 = 6$ and at the same time $f(1) = f(a_8) = a_9 = 2$.

- $r = 0$, thus $a_1 = 0$ and $a_2 = 1$. Then $a_3^2 = a_1^2 + 4a_2 = 4$, so $a_3 = \pm 2$. This, however, is a contradiction again, since it gives us $f(1) = f(a_0) = a_1 = 0$ and at the same time $f(1) = f(a_2) = a_3 = \pm 2$.

- $r = 1$, thus $a_1 = 2$ and $a_2 = 3$. We prove by induction that $a_k = k + 1$ for all $k \geqslant 0$ in this case, which we already know for $k \leqslant 2$ now. For the induction step, assume that $a_{k-1} = k$ and $a_k = k + 1$. Then

  $$a_{k+1}^2 = a_{k-1}^2 + 4a_k = k^2 + 4k + 4 = (k + 2)^2,$$

  so $a_{k+1} = \pm(k + 2)$. If $a_{k+1} = -(k + 2)$, then

  $$a_{k+2}^2 = a_k^2 + 4a_{k+1} = (k + 1)^2 - 4k - 8 = k^2 - 2k - 7 = (k - 1)^2 - 8.$$

  The latter can only be a square if $k = 4$ (since 1 and 9 are the only two squares whose difference is 8). Then, however, $a_4 = 5$, $a_5 = -6$ and $a_6 = \pm 1$, so

  $$a_7^2 = a_5^2 + 4a_6 = 36 \pm 4,$$

  but neither 32 nor 40 is a perfect square. Thus $a_{k+1} = k + 2$, which completes our induction. This also means that $f(n) = f(a_{n-1}) = a_n = n + 1$ for all $n \geqslant 1$.

Step 2: We prove that either $f(0) = 1$, or $f(0) = 0$ and $f(n) \neq 0$ for $n \neq 0$.

Set $n = 0$ in (1) to get

$$4f(0) = f(f(0))^2.$$

This means that $f(0) \geqslant 0$. If $f(0) = 0$, then $f(n) \neq 0$ for all $n \neq 0$, since we would otherwise have

$$n^2 = n^2 + 4f(n) = f(f(n))^2 = f(0)^2 = 0.$$

If $f(0) > 0$, then we know that $f(f(0)) = f(0) + 1$ from the first step, so

$$4f(0) = \big(f(0) + 1\big)^2,$$

which yields $f(0) = 1$.

Step 3: We discuss the values of $f(n)$ for $n < 0$.

Lemma. For every $n \geqslant 1$, we have $f(-n) = -n + 1$ or $f(-n) = n + 1$. Moreover, if $f(-n) = -n + 1$ for some $n \geqslant 1$, then also $f(-n + 1) = -n + 2$.

Proof. We prove this statement by strong induction on n. For $n = 1$, we get

$$1 + 4f(-1) = f(f(-1))^2.$$

Thus $f(-1)$ needs to be nonnegative. If $f(-1) = 0$, then $f(f(-1)) = f(0) = \pm 1$, so $f(0) = 1$ (by our second step). Otherwise, we know that $f(f(-1)) = f(-1) + 1$, so

$$1 + 4f(-1) = \big(f(-1) + 1\big)^2,$$

which yields $f(-1) = 2$ and thus establishes the base case. For the induction step, we consider two cases:

- If $f(-n) \leqslant -n$, then

  $$f(f(-n))^2 = (-n)^2 + 4f(-n) \leqslant n^2 - 4n < (n - 2)^2,$$

  so $|f(f(-n))| \leqslant n - 3$ (for $n = 2$, this case cannot even occur). If $f(f(-n)) \geqslant 0$, then we already know from the first two steps that $f(f(f(-n))) = f(f(-n)) + 1$, unless perhaps if $f(0) = 0$ and $f(f(-n)) = 0$. However, the latter would imply $f(-n) = 0$ (as shown in Step 2) and thus $n = 0$, which is impossible. If $f(f(-n)) < 0$, we can apply the induction hypothesis to $f(f(-n))$. In either case, $f(f(f(-n))) = \pm f(f(-n)) + 1$. Therefore,

  $$f(-n)^2 + 4f(f(-n)) = f(f(f(-n)))^2 = \big(\pm f(f(-n)) + 1\big)^2,$$

  which gives us

  $$\begin{aligned} n^2 \leqslant f(-n)^2 &= \big(\pm f(f(-n)) + 1\big)^2 - 4f(f(-n)) \leqslant f(f(-n))^2 + 6|f(f(-n))| + 1 \\ &\leqslant (n - 3)^2 + 6(n - 3) + 1 = n^2 - 8, \end{aligned}$$

  a contradiction.

- Thus, we are left with the case that $f(-n) > -n$. Now we argue as in the previous case: if $f(-n) \geqslant 0$, then $f(f(-n)) = f(-n) + 1$ by the first two steps, since $f(0) = 0$ and $f(-n) = 0$ would imply $n = 0$ (as seen in Step 2) and is thus impossible. If $f(-n) < 0$, we can apply the induction hypothesis, so in any case we can infer that $f(f(-n)) = \pm f(-n) + 1$. We obtain

  $$(-n)^2 + 4f(-n) = \big(\pm f(-n) + 1\big)^2,$$

  so either

  $$n^2 = f(-n)^2 - 2f(-n) + 1 = \big(f(-n) - 1\big)^2,$$

  which gives us $f(-n) = \pm n + 1$, or

  $$n^2 = f(-n)^2 - 6f(-n) + 1 = \big(f(-n) - 3\big)^2 - 8.$$

  Since 1 and 9 are the only perfect squares whose difference is 8, we must have $n = 1$, which we have already considered.

Finally, suppose that $f(-n) = -n + 1$ for some $n \geqslant 2$. Then

$$f(-n + 1)^2 = f(f(-n))^2 = (-n)^2 + 4f(-n) = (n - 2)^2,$$

so $f(-n+1) = \pm(n-2)$. However, we already know that $f(-n+1) = -n+2$ or $f(-n+1) = n$, so $f(-n + 1) = -n + 2$. □

Combining everything we know, we find the solutions as stated in the answer:

- One solution is given by $f(n) = n + 1$ for all n.
- If $f(n)$ is not always equal to $n + 1$, then there is a largest integer m (which cannot be positive) for which this is not the case. In view of the lemma that we proved, we must then have $f(n) = -n+1$ for any integer $n < m$. If $m = -a < 0$, we obtain $f(n) = -n+1$ for $n \leqslant -a$ (and $f(n) = n + 1$ otherwise). If $m = 0$, we have the additional possibility that $f(0) = 0$, $f(n) = -n + 1$ for negative n and $f(n) = n + 1$ for positive n.

Solution 2. Let us provide an alternative proof for Part II, which also proceeds in several steps.

Step 1. Let a be an arbitrary integer and $b = f(a)$. We first concentrate on the case where $|a|$ is sufficiently large.

1. If $b = 0$, then (1) applied to a yields $a^2 = f(f(a))^2$, thus
$$f(a) = 0 \quad \Rightarrow \quad a = \pm f(0). \tag{2}$$
From now on, we set $D = |f(0)|$. Throughout Step 1, we will assume that $a \notin \{-D, 0, D\}$, thus $b \neq 0$.

2. From (1), noticing that $f(f(a))$ and a have the same parity, we get
$$0 \neq 4|b| = \left|f(f(a))^2 - a^2\right| \geqslant a^2 - \big(|a| - 2\big)^2 = 4|a| - 4.$$
Hence we have
$$|b| = |f(a)| \geqslant |a| - 1 \qquad \text{for } a \notin \{-D, 0, D\}. \tag{3}$$
For the rest of Step 1, we also assume that $|a| \geqslant E = \max\{D + 2, 10\}$. Then by (3) we have $|b| \geqslant D + 1$ and thus $|f(b)| \geqslant D$.

3. Set $c = f(b)$; by (3), we have $|c| \geqslant |b| - 1$. Thus (1) yields
$$a^2 + 4b = c^2 \geqslant \big(|b| - 1\big)^2,$$
which implies
$$a^2 \geqslant \big(|b| - 1\big)^2 - 4|b| = \big(|b| - 3\big)^2 - 8 > \big(|b| - 4\big)^2$$
because $|b| \geqslant |a| - 1 \geqslant 9$. Thus (3) can be refined to
$$|a| + 3 \geqslant |f(a)| \geqslant |a| - 1 \qquad \text{for } |a| \geqslant E.$$
Now, from $c^2 = a^2 + 4b$ with $|b| \in [|a| - 1, |a| + 3]$ we get $c^2 = (a \pm 2)^2 + d$, where $d \in \{-16, -12, -8, -4, 0, 4, 8\}$. Since $|a \pm 2| \geqslant 8$, this can happen only if $c^2 = (a \pm 2)^2$, which in turn yields $b = \pm a + 1$. To summarise,
$$f(a) = 1 \pm a \qquad \text{for } |a| \geqslant E. \tag{4}$$

We have shown that, with at most finitely many exceptions, $f(a) = 1 \pm a$. Thus it will be convenient for our second step to introduce the sets

$$Z_+ = \big\{a \in \mathbb{Z}\colon f(a) = a + 1\big\}, \quad Z_- = \big\{a \in \mathbb{Z}\colon f(a) = 1 - a\big\}, \quad \text{and} \quad Z_0 = \mathbb{Z}\backslash\big(Z_+ \cup Z_-\big).$$

Step 2. Now we investigate the structure of the sets Z_+, Z_-, and Z_0.

4. Note that $f(E+1) = 1 \pm (E+1)$. If $f(E+1) = E+2$, then $E+1 \in Z_+$. Otherwise we have $f(1+E) = -E$; then the original equation (1) with $n = E+1$ gives us $(E-1)^2 = f(-E)^2$, so $f(-E) = \pm(E-1)$. By (4) this may happen only if $f(-E) = 1-E$, so in this case $-E \in Z_+$. In any case we find that $Z_+ \neq \varnothing$.

5. Now take any $a \in Z_+$. We claim that every integer $x \geqslant a$ also lies in Z_+. We proceed by induction on x, the base case $x = a$ being covered by our assumption. For the induction step, assume that $f(x-1) = x$ and plug $n = x-1$ into (1). We get $f(x)^2 = (x+1)^2$, so either $f(x) = x+1$ or $f(x) = -(x+1)$.

   Assume that $f(x) = -(x+1)$ and $x \neq -1$, since otherwise we already have $f(x) = x+1$. Plugging $n = x$ into (1), we obtain $f(-x-1)^2 = (x-2)^2 - 8$, which may happen only if $x-2 = \pm 3$ and $f(-x-1) = \pm 1$. Plugging $n = -x-1$ into (1), we get $f(\pm 1)^2 = (x+1)^2 \pm 4$, which in turn may happen only if $x+1 \in \{-2, 0, 2\}$.

   Thus $x \in \{-1, 5\}$ and at the same time $x \in \{-3, -1, 1\}$, which gives us $x = -1$. Since this has already been excluded, we must have $f(x) = x+1$, which completes our induction.

6. Now we know that either $Z_+ = \mathbb{Z}$ (if Z_+ is not bounded below), or $Z_+ = \{a \in \mathbb{Z} \colon a \geqslant a_0\}$, where a_0 is the smallest element of Z_+. In the former case, $f(n) = n+1$ for all $n \in \mathbb{Z}$, which is our first solution. So we assume in the following that Z_+ is bounded below and has a smallest element a_0.

   If $Z_0 = \varnothing$, then we have $f(x) = x+1$ for $x \geqslant a_0$ and $f(x) = 1-x$ for $x < a_0$. In particular, $f(0) = 1$ in any case, so $0 \in Z_+$ and thus $a_0 \leqslant 0$. Thus we end up with the second solution listed in the answer. It remains to consider the case where $Z_0 \neq \varnothing$.

7. Assume that there exists some $a \in Z_0$ with $b = f(a) \notin Z_0$, so that $f(b) = 1 \pm b$. Then we have $a^2 + 4b = (1 \pm b)^2$, so either $a^2 = (b-1)^2$ or $a^2 = (b-3)^2 - 8$. In the former case we have $b = 1 \pm a$, which is impossible by our choice of a. So we get $a^2 = (b-3)^2 - 8$, which implies $f(b) = 1-b$ and $|a| = 1$, $|b-3| = 3$.

   If $b = 0$, then we have $f(b) = 1$, so $b \in Z_+$ and therefore $a_0 \leqslant 0$; hence $a = -1$. But then $f(a) = 0 = a+1$, so $a \in Z_+$, which is impossible.

   If $b = 6$, then we have $f(6) = -5$, so $f(-5)^2 = 16$ and $f(-5) \in \{-4, 4\}$. Then $f(f(-5))^2 = 25 + 4f(-5) \in \{9, 41\}$, so $f(-5) = -4$ and $-5 \in Z_+$. This implies $a_0 \leqslant -5$, which contradicts our assumption that $\pm 1 = a \notin Z_+$.

8. Thus we have shown that $f(Z_0) \subseteq Z_0$, and Z_0 is finite. Take any element $c \in Z_0$, and consider the sequence defined by $c_i = f^i(c)$. All elements of the sequence (c_i) lie in Z_0, hence it is bounded. Choose an index k for which $|c_k|$ is maximal, so that in particular $|c_{k+1}| \leqslant |c_k|$ and $|c_{k+2}| \leqslant |c_k|$. Our functional equation (1) yields

$$(|c_k| - 2)^2 - 4 = |c_k|^2 - 4|c_k| \leqslant c_k^2 + 4c_{k+1} = c_{k+2}^2.$$

   Since c_k and c_{k+2} have the same parity and $|c_{k+2}| \leqslant |c_k|$, this leaves us with three possibilities: $|c_{k+2}| = |c_k|$, $|c_{k+2}| = |c_k| - 2$, and $|c_k| - 2 = \pm 2$, $c_{k+2} = 0$.

   If $|c_{k+2}| = |c_k| - 2$, then $f(c_k) = c_{k+1} = 1 - |c_k|$, which means that $c_k \in Z_-$ or $c_k \in Z_+$, and we reach a contradiction.

   If $|c_{k+2}| = |c_k|$, then $c_{k+1} = 0$, thus $c_{k+3}^2 = 4c_{k+2}$. So either $c_{k+3} \neq 0$ or (by maximality of $|c_{k+2}| = |c_k|$) $c_i = 0$ for all i. In the former case, we can repeat the entire argument

with c_{k+2} in the place of c_k. Now $|c_{k+4}| = |c_{k+2}|$ is not possible any more since $c_{k+3} \neq 0$, leaving us with the only possibility $|c_k| - 2 = |c_{k+2}| - 2 = \pm 2$.

Thus we know now that either all c_i are equal to 0, or $|c_k| = 4$. If $c_k = \pm 4$, then either $c_{k+1} = 0$ and $|c_{k+2}| = |c_k| = 4$, or $c_{k+2} = 0$ and $c_{k+1} = -4$. From this point onwards, all elements of the sequence are either 0 or ± 4.

Let c_r be the last element of the sequence that is not equal to 0 or ± 4 (if such an element exists). Then $c_{r+1}, c_{r+2} \in \{-4, 0, 4\}$, so

$$c_r^2 = c_{r+2}^2 - 4c_{r+1} \in \{-16, 0, 16, 32\},$$

which gives us a contradiction. Thus all elements of the sequence are equal to 0 or ± 4, and since the choice of $c_0 = c$ was arbitrary, $Z_0 \subseteq \{-4, 0, 4\}$.

9. Finally, we show that $4 \notin Z_0$ and $-4 \notin Z_0$. Suppose that $4 \in Z_0$. Then in particular a_0 (the smallest element of Z_+) cannot be less than 4, since this would imply $4 \in Z_+$. So $-3 \in Z_-$, which means that $f(-3) = 4$. Then $25 = (-3)^2 + 4f(-3) = f(f(-3))^2 = f(4)^2$, so $f(4) = \pm 5 \notin Z_0$, and we reach a contradiction.

   Suppose that $-4 \in Z_0$. The only possible values for $f(-4)$ that are left are 0 and -4. Note that $4f(0) = f(f(0))^2$, so $f(0) \geqslant 0$. If $f(-4) = 0$, then we get $16 = (-4)^2 + 0 = f(0)^2$, thus $f(0) = 4$. But then $f(f(-4)) \notin Z_0$, which is impossible. Thus $f(-4) = -4$, which gives us $0 = (-4)^2 + 4f(-4) = f(f(-4))^2 = 16$, and this is clearly absurd.

   Now we are left with $Z_0 = \{0\}$ and $f(0) = 0$ as the only possibility. If $1 \in Z_-$, then $f(1) = 0$, so $1 = 1^2 + 4f(1) = f(f(1))^2 = f(0)^2 = 0$, which is another contradiction. Thus $1 \in Z_+$, meaning that $a_0 \leqslant 1$. On the other hand, $a_0 \leqslant 0$ would imply $0 \in Z_+$, so we can only have $a_0 = 1$. Thus Z_+ comprises all positive integers, and Z_- comprises all negative integers. This gives us the third solution.

Comment. All solutions known to the Problem Selection Committee are quite lengthy and technical, as the two solutions presented here show. It is possible to make the problem easier by imposing additional assumptions, such as $f(0) \neq 0$ or $f(n) \geqslant 1$ for all $n \geqslant 0$, to remove some of the technicalities.

Combinatorics

C1. Let n points be given inside a rectangle R such that no two of them lie on a line parallel to one of the sides of R. The rectangle R is to be dissected into smaller rectangles with sides parallel to the sides of R in such a way that none of these rectangles contains any of the given points in its interior. Prove that we have to dissect R into at least $n + 1$ smaller rectangles.

(Serbia)

Solution 1. Let k be the number of rectangles in the dissection. The set of all points that are corners of one of the rectangles can be divided into three disjoint subsets:

- A, which consists of the four corners of the original rectangle R, each of which is the corner of exactly one of the smaller rectangles,
- B, which contains points where exactly two of the rectangles have a common corner (T-junctions, see the figure below),
- C, which contains points where four of the rectangles have a common corner (crossings, see the figure below).

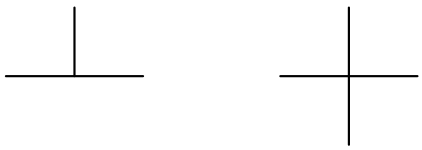


Figure 1: A T-junction and a crossing

We denote the number of points in B by b and the number of points in C by c. Since each of the k rectangles has exactly four corners, we get

$$4k = 4 + 2b + 4c.$$

It follows that $2b \leqslant 4k - 4$, so $b \leqslant 2k - 2$.

Each of the n given points has to lie on a side of one of the smaller rectangles (but not of the original rectangle R). If we extend this side as far as possible along borders between rectangles, we obtain a line segment whose ends are T-junctions. Note that every point in B can only be an endpoint of at most one such segment containing one of the given points, since it is stated that no two of them lie on a common line parallel to the sides of R. This means that

$$b \geqslant 2n.$$

Combining our two inequalities for b, we get

$$2k - 2 \geqslant b \geqslant 2n,$$

thus $k \geqslant n + 1$, which is what we wanted to prove.

Solution 2. Let k denote the number of rectangles. In the following, we refer to the directions of the sides of R as 'horizontal' and 'vertical' respectively. Our goal is to prove the inequality $k \geqslant n+1$ for fixed n. Equivalently, we can prove the inequality $n \leqslant k-1$ for each k, which will be done by induction on k. For $k = 1$, the statement is trivial.

Now assume that $k > 1$. If none of the line segments that form the borders between the rectangles is horizontal, then we have $k-1$ vertical segments dividing R into k rectangles. On each of them, there can only be one of the n points, so $n \leqslant k-1$, which is exactly what we want to prove.

Otherwise, consider the lowest horizontal line h that contains one or more of these line segments. Let R' be the rectangle that results when everything that lies below h is removed from R (see the example in the figure below).

The rectangles that lie entirely below h form blocks of rectangles separated by vertical line segments. Suppose there are r blocks and k_i rectangles in the ith block. The left and right border of each block has to extend further upwards beyond h. Thus we can move any points that lie on these borders upwards, so that they now lie inside R'. This can be done without violating the conditions, one only needs to make sure that they do not get to lie on a common horizontal line with one of the other given points.

All other borders between rectangles in the ith block have to lie entirely below h. There are $k_i - 1$ such line segments, each of which can contain at most one of the given points. Finally, there can be one point that lies on h. All other points have to lie in R' (after moving some of them as explained in the previous paragraph).

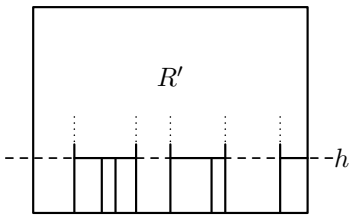


Figure 2: Illustration of the inductive argument

We see that R' is divided into $k - \sum_{i=1}^{r} k_i$ rectangles. Applying the induction hypothesis to R', we find that there are at most

$$\Big(k - \sum_{i=1}^{r} k_i\Big) - 1 + \sum_{i=1}^{r} (k_i - 1) + 1 = k - r$$

points. Since $r \geqslant 1$, this means that $n \leqslant k-1$, which completes our induction.

C2. We have 2^m sheets of paper, with the number 1 written on each of them. We perform the following operation. In every step we choose two distinct sheets; if the numbers on the two sheets are a and b, then we erase these numbers and write the number $a+b$ on both sheets. Prove that after $m2^{m-1}$ steps, the sum of the numbers on all the sheets is at least 4^m.

(Iran)

Solution. Let P_k be the product of the numbers on the sheets after k steps.

Suppose that in the $(k+1)^{\text{th}}$ step the numbers a and b are replaced by $a+b$. In the product, the number ab is replaced by $(a+b)^2$, and the other factors do not change. Since $(a+b)^2 \geqslant 4ab$, we see that $P_{k+1} \geqslant 4P_k$. Starting with $P_0 = 1$, a straightforward induction yields

$$P_k \geqslant 4^k$$

for all integers $k \geqslant 0$; in particular

$$P_{m\cdot 2^{m-1}} \geqslant 4^{m\cdot 2^{m-1}} = (2^m)^{2^m} ,$$

so by the AM–GM inequality, the sum of the numbers written on the sheets after $m2^{m-1}$ steps is at least

$$2^m \cdot \sqrt[2^m]{P_{m\cdot 2^{m-1}}} \geqslant 2^m \cdot 2^m = 4^m .$$

Comment 1. It is possible to achieve the sum 4^m in $m2^{m-1}$ steps. For example, starting from 2^m equal numbers on the sheets, in 2^{m-1} consecutive steps we can double all numbers. After m such doubling rounds we have the number 2^m on every sheet.

Comment 2. There are several versions of the solution above. E.g., one may try to assign to each positive integer n a *weight* w_n in such a way that the sum of the weights of the numbers written on the sheets increases, say, by at least 2 in each step. For this purpose, one needs the inequality

$$2w_{a+b} \geqslant w_a + w_b + 2 \tag{1}$$

to be satisfied for all positive integers a and b.

Starting from $w_1 = 1$ and trying to choose the weights as small as possible, one may find that these weights can be defined as follows: For every positive integer n, one chooses k to be the maximal integer such that $n \geqslant 2^k$, and puts

$$w_n = k + \frac{n}{2^k} = \min_{d\in\mathbb{Z}_{\geqslant 0}} \left(d + \frac{n}{2^d}\right). \tag{2}$$

Now, in order to prove that these weights satisfy (1), one may take arbitrary positive integers a and b, and choose an integer $d \geqslant 0$ such that $w_{a+b} = d + \frac{a+b}{2^d}$. Then one has

$$2w_{a+b} = 2d + 2\cdot\frac{a+b}{2^d} = \left((d-1) + \frac{a}{2^{d-1}}\right) + \left((d-1) + \frac{b}{2^{d-1}}\right) + 2 \geqslant w_a + w_b + 2.$$

Since the initial sum of the weights was 2^m, after $m2^{m-1}$ steps the sum is at least $(m+1)2^m$. To finish the solution, one may notice that by (2) for every positive integer a one has

$$w_a \leqslant m + \frac{a}{2^m}, \qquad \text{i.e.,} \quad a \geqslant 2^m(-m + w_a). \tag{3}$$

So the sum of the numbers $a_1, a_2, \ldots, a_{2^m}$ on the sheets can be estimated as

$$\sum_{i=1}^{2^m} a_i \geqslant \sum_{i=1}^{2^m} 2^m(-m + w_{a_i}) = -m2^m\cdot 2^m + 2^m \sum_{i=1}^{2^m} w_{a_i} \geqslant -m4^m + (m+1)4^m = 4^m,$$

as required.

For establishing the inequalities (1) and (3), one may also use the convexity argument, instead of the second definition of w_n in (2).

One may check that $\log_2 n \leqslant w_n \leqslant \log_2 n + 1$; thus, in some rough sense, this approach is obtained by "taking the logarithm" of the solution above.

Comment 3. An intuitive strategy to minimise the sum of numbers is that in every step we choose the two smallest numbers. We may call this the *greedy strategy*. In the following paragraphs we prove that the greedy strategy indeed provides the least possible sum of numbers.

Claim. Starting from any sequence $x_1, \ldots, x_N$ of positive real numbers on N sheets, for any number k of steps, the greedy strategy achieves the lowest possible sum of numbers.

Proof. We apply induction on k; for $k = 1$ the statement is obvious. Let $k \geqslant 2$, and assume that the claim is true for smaller values.

Every sequence of k steps can be encoded as $S = \big((i_1, j_1), \ldots, (i_k, j_k)\big)$, where, for $r = 1, 2, \ldots, k$, the numbers i_r and j_r are the indices of the two sheets that are chosen in the rth step. The resulting final sum will be some linear combination of $x_1, \ldots, x_N$, say, $c_1 x_1 + \cdots + c_N x_N$ with positive integers $c_1, \ldots, c_N$ that depend on S only. Call the numbers $(c_1, \ldots, c_N)$ the *characteristic vector* of S.

Choose a sequence $S_0 = \big((i_1, j_1), \ldots, (i_k, j_k)\big)$ of steps that produces the minimal sum, starting from $x_1, \ldots, x_N$, and let $(c_1, \ldots, c_N)$ be the characteristic vector of S. We may assume that the sheets are indexed in such an order that $c_1 \geqslant c_2 \geqslant \cdots \geqslant c_N$. If the sheets (and the numbers) are permuted by a permutation π of the indices $(1, 2, \ldots, N)$ and then the same steps are performed, we can obtain the sum $\sum_{t=1}^{N} c_t x_{\pi(t)}$. By the rearrangement inequality, the smallest possible sum can be achieved when the numbers $(x_1, \ldots, x_N)$ are in non-decreasing order. So we can assume that also $x_1 \leqslant x_2 \leqslant \cdots \leqslant x_N$.

Let ℓ be the largest index with $c_1 = \cdots = c_\ell$, and let the rth step be the first step for which $c_{i_r} = c_1$ or $c_{j_r} = c_1$. The role of i_r and j_r is symmetrical, so we can assume $c_{i_r} = c_1$ and thus $i_r \leqslant \ell$. We show that $c_{j_r} = c_1$ and $j_r \leqslant \ell$ hold, too.

Before the rth step, on the ${i_r}^{\text{th}}$ sheet we had the number x_{i_r}. On the ${j_r}^{\text{th}}$ sheet there was a linear combination that contains the number x_{j_r} with a positive integer coefficient, and possibly some other terms. In the rth step, the number x_{i_r} joins that linear combination. From this point, each sheet contains a linear combination of $x_1, \ldots, x_N$, with the coefficient of x_{j_r} being not smaller than the coefficient of x_{i_r}. This is preserved to the end of the procedure, so we have $c_{j_r} \geqslant c_{i_r}$. But $c_{i_r} = c_1$ is maximal among the coefficients, so we have $c_{j_r} = c_{i_r} = c_1$ and thus $j_r \leqslant \ell$.

Either from $c_{j_r} = c_{i_r} = c_1$ or from the arguments in the previous paragraph we can see that none of the ${i_r}^{\text{th}}$ and the ${j_r}^{\text{th}}$ sheets were used before step r. Therefore, the final linear combination of the numbers does not change if the step (i_r, j_r) is performed first: the sequence of steps

$$S_1 = \big((i_r, j_r), (i_1, j_1), \ldots, (i_{r-1}, j_{r-1}), (i_{r+1}, j_{r+1}), \ldots, (i_N, j_N)\big)$$

also produces the same minimal sum at the end. Therefore, we can replace S_0 by S_1 and we may assume that $r = 1$ and $c_{i_1} = c_{j_1} = c_1$.

As $i_1 \neq j_1$, we can see that $\ell \geqslant 2$ and $c_1 = c_2 = c_{i_1} = c_{j_1}$. Let π be such a permutation of the indices $(1, 2, \ldots, N)$ that exchanges $1, 2$ with i_r, j_r and does not change the remaining indices. Let

$$S_2 = \big((\pi(i_1), \pi(j_1)), \ldots, (\pi(i_N), \pi(j_N))\big).$$

Since $c_{\pi(i)} = c_i$ for all indices i, this sequence of steps produces the same, minimal sum. Moreover, in the first step we chose $x_{\pi(i_1)} = x_1$ and $x_{\pi(j_1)} = x_2$, the two smallest numbers.

Hence, it is possible to achieve the optimal sum if we follow the greedy strategy in the first step. By the induction hypothesis, following the greedy strategy in the remaining steps we achieve the optimal sum.

C3. Let $n \geqslant 2$ be an integer. Consider an $n \times n$ chessboard divided into n^2 unit squares. We call a configuration of n rooks on this board *happy* if every row and every column contains exactly one rook. Find the greatest positive integer k such that for every happy configuration of rooks, we can find a $k \times k$ square without a rook on any of its k^2 unit squares.

(Croatia)

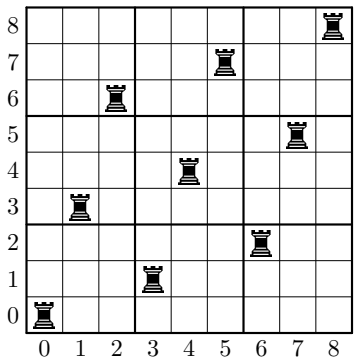
Answer. $\lfloor \sqrt{n-1} \rfloor$.

Solution. Let ℓ be a positive integer. We will show that **(i)** if $n > \ell^2$ then each happy configuration contains an empty $\ell \times \ell$ square, but **(ii)** if $n \leqslant \ell^2$ then there exists a happy configuration not containing such a square. These two statements together yield the answer.

(i). Assume that $n > \ell^2$. Consider any happy configuration. There exists a row R containing a rook in its leftmost square. Take ℓ consecutive rows with R being one of them. Their union U contains exactly ℓ rooks. Now remove the $n - \ell^2 \geqslant 1$ leftmost columns from U (thus at least one rook is also removed). The remaining part is an $\ell^2 \times \ell$ rectangle, so it can be split into ℓ squares of size $\ell \times \ell$, and this part contains at most $\ell - 1$ rooks. Thus one of these squares is empty.

(ii). Now we assume that $n \leqslant \ell^2$. Firstly, we will construct a happy configuration with no empty $\ell \times \ell$ square for the case $n = \ell^2$. After that we will modify it to work for smaller values of n.

Let us enumerate the rows from bottom to top as well as the columns from left to right by the numbers $0, 1, \ldots, \ell^2 - 1$. Every square will be denoted, as usual, by the pair (r, c) of its row and column numbers. Now we put the rooks on all squares of the form $(i\ell + j, j\ell + i)$ with $i, j = 0, 1, \ldots, \ell - 1$ (the picture below represents this arrangement for $\ell = 3$). Since each number from 0 to $\ell^2 - 1$ has a unique representation of the form $i\ell + j$ $(0 \leqslant i, j \leqslant \ell - 1)$, each row and each column contains exactly one rook.



Next, we show that each $\ell \times \ell$ square A on the board contains a rook. Consider such a square A, and consider ℓ consecutive rows the union of which contains A. Let the lowest of these rows have number $p\ell + q$ with $0 \leqslant p, q \leqslant \ell - 1$ (notice that $p\ell + q \leqslant \ell^2 - \ell$). Then the rooks in this union are placed in the columns with numbers $q\ell + p$, $(q+1)\ell + p$, $\ldots$, $(\ell-1)\ell + p$, $p + 1$, $\ell + (p + 1)$, $\ldots$, $(q - 1)\ell + p + 1$, or, putting these numbers in increasing order,

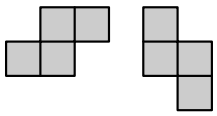
$$p + 1,\ \ell + (p + 1),\ \ldots,\ (q - 1)\ell + (p + 1),\ q\ell + p,\ (q + 1)\ell + p,\ \ldots,\ (\ell - 1)\ell + p.$$

One readily checks that the first number in this list is at most $\ell - 1$ (if $p = \ell - 1$, then $q = 0$, and the first listed number is $q\ell + p = \ell - 1$), the last one is at least $(\ell - 1)\ell$, and the difference between any two consecutive numbers is at most ℓ. Thus, one of the ℓ consecutive columns intersecting A contains a number listed above, and the rook in this column is inside A, as required. The construction for $n = \ell^2$ is established.

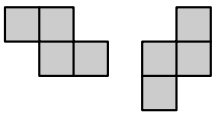
It remains to construct a happy configuration of rooks not containing an empty $\ell \times \ell$ square for $n < \ell^2$. In order to achieve this, take the construction for an $\ell^2 \times \ell^2$ square described above and remove the $\ell^2 - n$ bottom rows together with the $\ell^2 - n$ rightmost columns. We will have a rook arrangement with no empty $\ell \times \ell$ square, but several rows and columns may happen to be empty. Clearly, the number of empty rows is equal to the number of empty columns, so one can find a bijection between them, and put a rook on any crossing of an empty row and an empty column corresponding to each other.

Comment. Part **(i)** allows several different proofs. E.g., in the last paragraph of the solution, it suffices to deal only with the case $n = \ell^2 + 1$. Notice now that among the four corner squares, at least one is empty. So the rooks in its row and in its column are distinct. Now, deleting this row and column we obtain an $\ell^2 \times \ell^2$ square with $\ell^2 - 1$ rooks in it. This square can be partitioned into ℓ^2 squares of size $\ell \times \ell$, so one of them is empty.

C4. Construct a tetromino by attaching two 2×1 dominoes along their longer sides such that the midpoint of the longer side of one domino is a corner of the other domino. This construction yields two kinds of tetrominoes with opposite orientations. Let us call them S- and Z-tetrominoes, respectively.



S-tetrominoes

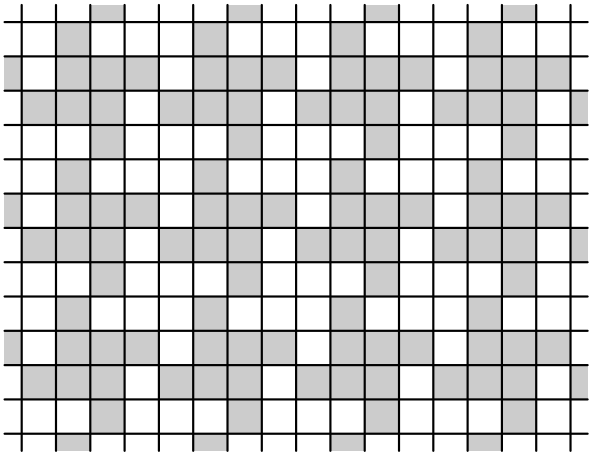


Z-tetrominoes

Assume that a lattice polygon P can be tiled with S-tetrominoes. Prove than no matter how we tile P using only S- and Z-tetrominoes, we always use an even number of Z-tetrominoes.

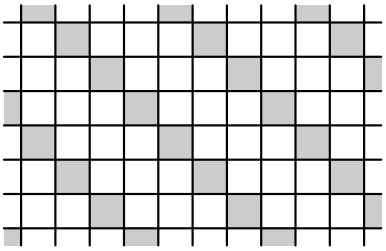
(Hungary)

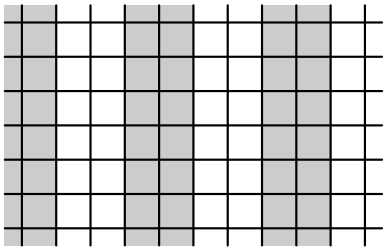
Solution 1. We may assume that polygon P is the union of some squares of an infinite chessboard. Colour the squares of the chessboard with two colours as the figure below illustrates.



Observe that no matter how we tile P, any S-tetromino covers an even number of black squares, whereas any Z-tetromino covers an odd number of them. As P can be tiled exclusively by S-tetrominoes, it contains an even number of black squares. But if some S-tetrominoes and some Z-tetrominoes cover an even number of black squares, then the number of Z-tetrominoes must be even.

Comment. An alternative approach makes use of the following two colourings, which are perhaps somewhat more natural:





Let s_1 and s_2 be the number of S-tetrominoes of the first and second type (as shown in the figure above) respectively that are used in a tiling of P. Likewise, let z_1 and z_2 be the number of Z-tetrominoes of the first and second type respectively. The first colouring shows that $s_1 + z_2$ is invariant modulo 2, the second colouring shows that $s_1 + z_1$ is invariant modulo 2. Adding these two conditions, we find that $z_1 + z_2$ is invariant modulo 2, which is what we have to prove. Indeed, the sum of the two colourings (regarding white as 0 and black as 1 and adding modulo 2) is the colouring shown in the solution.

Solution 2. Let us assign coordinates to the squares of the infinite chessboard in such a way that the squares of P have nonnegative coordinates only, and that the first coordinate increases as one moves to the right, while the second coordinate increases as one moves upwards. Write the integer $3^i \cdot (-3)^j$ into the square with coordinates (i, j), as in the following figure:

$\vdots$					
81	$\vdots$				
-27	-81	$\vdots$			
9	27	81	$\cdots$		
-3	-9	-27	-81	$\cdots$	
1	3	9	27	81	$\cdots$

The sum of the numbers written into four squares that can be covered by an S-tetromino is either of the form

$$3^i \cdot (-3)^j \cdot \big(1 + 3 + 3 \cdot (-3) + 3^2 \cdot (-3)\big) = -32 \cdot 3^i \cdot (-3)^j$$

(for the first type of S-tetrominoes), or of the form

$$3^i \cdot (-3)^j \cdot \big(3 + 3 \cdot (-3) + (-3) + (-3)^2\big) = 0,$$

and thus divisible by 32. For this reason, the sum of the numbers written into the squares of P, and thus also the sum of the numbers covered by Z-tetrominoes in the second covering, is likewise divisible by 32. Now the sum of the entries of a Z-tetromino is either of the form

$$3^i \cdot (-3)^j \cdot \big(3 + 3^2 + (-3) + 3 \cdot (-3)\big) = 0$$

(for the first type of Z-tetrominoes), or of the form

$$3^i \cdot (-3)^j \cdot \big(1 + (-3) + 3 \cdot (-3) + 3 \cdot (-3)^2\big) = 16 \cdot 3^i \cdot (-3)^j,$$

i.e., 16 times an odd number. Thus in order to obtain a total that is divisible by 32, an even number of the latter kind of Z-tetrominoes needs to be used. Rotating everything by 90°, we find that the number of Z-tetrominoes of the first kind is even as well. So we have even proven slightly more than necessary.

Comment 1. In the second solution, 3 and -3 can be replaced by other combinations as well. For example, for any positive integer $a \equiv 3 \pmod 4$, we can write $a^i \cdot (-a)^j$ into the square with coordinates (i, j) and apply the same argument.

Comment 2. As the second solution shows, we even have the stronger result that the parity of the number of each of the four types of tetrominoes in a tiling of P by S- and Z-tetrominoes is an invariant of P. This also remains true if there is no tiling of P that uses only S-tetrominoes.

C5. Consider $n \geqslant 3$ lines in the plane such that no two lines are parallel and no three have a common point. These lines divide the plane into polygonal regions; let $\mathcal{F}$ be the set of regions having finite area. Prove that it is possible to colour $\lceil \sqrt{n/2} \rceil$ of the lines blue in such a way that no region in $\mathcal{F}$ has a completely blue boundary. (For a real number x, $\lceil x \rceil$ denotes the least integer which is not smaller than x.)

(Austria)

Solution. Let L be the given set of lines. Choose a maximal (by inclusion) subset $B \subseteq L$ such that when we colour the lines of B blue, no region in $\mathcal{F}$ has a completely blue boundary. Let $|B| = k$. We claim that $k \geqslant \lceil \sqrt{n/2} \rceil$.

Let us colour all the lines of $L \backslash B$ red. Call a point *blue* if it is the intersection of two blue lines. Then there are $\binom{k}{2}$ blue points.

Now consider any red line ℓ. By the maximality of B, there exists at least one region $A \in \mathcal{F}$ whose only red side lies on ℓ. Since A has at least three sides, it must have at least one blue vertex. Let us take one such vertex and associate it to ℓ.

Since each blue point belongs to four regions (some of which may be unbounded), it is associated to at most four red lines. Thus the total number of red lines is at most $4\binom{k}{2}$. On the other hand, this number is $n - k$, so

$$n - k \leqslant 2k(k-1), \quad \text{thus} \quad n \leqslant 2k^2 - k \leqslant 2k^2,$$

and finally $k \geqslant \lceil \sqrt{n/2} \rceil$, which gives the desired result.

Comment 1. The constant factor in the estimate can be improved in different ways; we sketch two of them below. On the other hand, the Problem Selection Committee is not aware of any results showing that it is sometimes impossible to colour k lines satisfying the desired condition for $k \gg \sqrt{n}$. In this situation we find it more suitable to keep the original formulation of the problem.

1. Firstly, we show that in the proof above one has in fact $k = |B| \geqslant \lceil \sqrt{2n/3} \rceil$.

Let us make weighted associations as follows. Let a region A whose only red side lies on ℓ have k vertices, so that $k-2$ of them are blue. We associate each of these blue vertices to ℓ, and put the weight $\frac{1}{k-2}$ on each such association. So the sum of the weights of all the associations is exactly $n-k$.

Now, one may check that among the four regions adjacent to a blue vertex v, at most two are triangles. This means that the sum of the weights of all associations involving v is at most $1 + 1 + \frac{1}{2} + \frac{1}{2} = 3$. This leads to the estimate

$$n - k \leqslant 3\binom{k}{2},$$

or

$$2n \leqslant 3k^2 - k < 3k^2,$$

which yields $k \geqslant \lceil \sqrt{2n/3} \rceil$.

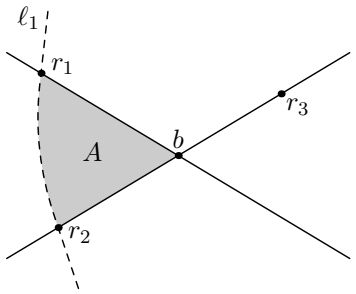
2. Next, we even show that $k = |B| \geqslant \lceil \sqrt{n} \rceil$. For this, we specify the process of associating points to red lines in one more different way.

Call a point *red* if it lies on a red line as well as on a blue line. Consider any red line ℓ, and take an arbitrary region $A \in \mathcal{F}$ whose only red side lies on ℓ. Let r', r, b_1, $\ldots$, b_k be its vertices in clockwise order with $r', r \in \ell$; then the points r', r are red, while all the points b_1, $\ldots$, b_k are blue. Let us associate to ℓ the red point r and the blue point b_1. One may notice that to each pair of a red point r and a blue point b, at most one red line can be associated, since there is at most one region A having r and b as two clockwise consecutive vertices.

We claim now that at most two red lines are associated to each blue point b; this leads to the desired bound

$$n - k \leqslant 2\binom{k}{2} \quad \Longleftrightarrow \quad n \leqslant k^2.$$

Assume, to the contrary, that three red lines ℓ_1, ℓ_2, and ℓ_3 are associated to the same blue point b. Let r_1, r_2, and r_3 respectively be the red points associated to these lines; all these points are distinct. The point b defines four blue rays, and each point r_i is the red point closest to b on one of these rays. So we may assume that the points r_2 and r_3 lie on one blue line passing through b, while r_1 lies on the other one.



Now consider the region A used to associate r_1 and b with ℓ_1. Three of its clockwise consecutive vertices are r_1, b, and either r_2 or r_3 (say, r_2). Since A has only one red side, it can only be the triangle r_1br_2; but then both ℓ_1 and ℓ_2 pass through r_2, as well as some blue line. This is impossible by the problem assumptions.

Comment 2. The condition that the lines be non-parallel is essentially not used in the solution, nor in the previous comment; thus it may be omitted.

C6. We are given an infinite deck of cards, each with a real number on it. For every real number x, there is exactly one card in the deck that has x written on it. Now two players draw disjoint sets A and B of 100 cards each from this deck. We would like to define a rule that declares one of them a winner. This rule should satisfy the following conditions:

1. The winner only depends on the relative order of the 200 cards: if the cards are laid down in increasing order face down and we are told which card belongs to which player, but not what numbers are written on them, we can still decide the winner.
2. If we write the elements of both sets in increasing order as $A = \{a_1, a_2, \ldots, a_{100}\}$ and $B = \{b_1, b_2, \ldots, b_{100}\}$, and $a_i > b_i$ for all i, then A beats B.
3. If three players draw three disjoint sets A, B, C from the deck, A beats B and B beats C, then A also beats C.

How many ways are there to define such a rule? Here, we consider two rules as different if there exist two sets A and B such that A beats B according to one rule, but B beats A according to the other.

(Russia)

Answer. 100.

Solution 1. We prove a more general statement for sets of cardinality n (the problem being the special case $n = 100$, then the answer is n). In the following, we write $A \succ B$ or $B \prec A$ for "A beats B".

Part I. Let us first define n different rules that satisfy the conditions. To this end, fix an index $k \in \{1, 2, \ldots, n\}$. We write both A and B in increasing order as $A = \{a_1, a_2, \ldots, a_n\}$ and $B = \{b_1, b_2, \ldots, b_n\}$ and say that A beats B if and only if $a_k > b_k$. This rule clearly satisfies all three conditions, and the rules corresponding to different k are all different. Thus there are at least n different rules.

Part II. Now we have to prove that there is no other way to define such a rule. Suppose that our rule satisfies the conditions, and let $k \in \{1, 2, \ldots, n\}$ be minimal with the property that

$$A_k = \{1, 2, \ldots, k, n+k+1, n+k+2, \ldots, 2n\} \prec B_k = \{k+1, k+2, \ldots, n+k\}.$$

Clearly, such a k exists, since this holds for $k = n$ by assumption. Now consider two disjoint sets $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$, both in increasing order (i.e., $x_1 < x_2 < \cdots < x_n$ and $y_1 < y_2 < \cdots < y_n$). We claim that $X \prec Y$ if (and only if – this follows automatically) $x_k < y_k$.

To prove this statement, pick arbitrary real numbers $u_i, v_i, w_i \notin X \cup Y$ such that

$$u_1 < u_2 < \cdots < u_{k-1} < \min(x_1, y_1), \quad \max(x_n, y_n) < v_{k+1} < v_{k+2} < \cdots < v_n,$$

and

$$x_k < v_1 < v_2 < \cdots < v_k < w_1 < w_2 < \cdots < w_n < u_k < u_{k+1} < \cdots < u_n < y_k,$$

and set

$$U = \{u_1, u_2, \ldots, u_n\},\ V = \{v_1, v_2, \ldots, v_n\},\ W = \{w_1, w_2, \ldots, w_n\}.$$

Then

- $u_i < y_i$ and $x_i < v_i$ for all i, so $U \prec Y$ and $X \prec V$ by the second condition.

- The elements of $U \cup W$ are ordered in the same way as those of $A_{k-1} \cup B_{k-1}$, and since $A_{k-1} \succ B_{k-1}$ by our choice of k, we also have $U \succ W$ (if $k = 1$, this is trivial).
- The elements of $V \cup W$ are ordered in the same way as those of $A_k \cup B_k$, and since $A_k \prec B_k$ by our choice of k, we also have $V \prec W$.

It follows that

$$X \prec V \prec W \prec U \prec Y,$$

so $X \prec Y$ by the third condition, which is what we wanted to prove.

Solution 2. Another possible approach to Part II of this problem is induction on n. For $n = 1$, there is trivially only one rule in view of the second condition.

In the following, we assume that our claim (namely, that there are no possible rules other than those given in Part I) holds for $n-1$ in place of n. We start with the following observation:

Claim. At least one of the two relations

$$\big(\{2\} \cup \{2i-1 \,|\, 2 \leqslant i \leqslant n\}\big) \prec \big(\{1\} \cup \{2i \,|\, 2 \leqslant i \leqslant n\}\big)$$

and

$$\big(\{2i-1 \,|\, 1 \leqslant i \leqslant n-1\} \cup \{2n\}\big) \prec \big(\{2i \,|\, 1 \leqslant i \leqslant n-1\} \cup \{2n-1\}\big)$$

holds.

Proof. Suppose that the first relation does not hold. Since our rule may only depend on the relative order, we must also have

$$\big(\{2\} \cup \{3i-2 \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n-2\}\big) \succ \big(\{1\} \cup \{3i-1 \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n\}\big).$$

Likewise, if the second relation does not hold, then we must also have

$$\big(\{1\} \cup \{3i-1 \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n\}\big) \succ \big(\{3\} \cup \{3i \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n-1\}\big).$$

Now condition 3 implies that

$$\big(\{2\} \cup \{3i-2 \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n-2\}\big) \succ \big(\{3\} \cup \{3i \,|\, 2 \leqslant i \leqslant n-1\} \cup \{3n-1\}\big),$$

which contradicts the second condition. □

Now we distinguish two cases, depending on which of the two relations actually holds:

First case: $\big(\{2\} \cup \{2i-1 \,|\, 2 \leqslant i \leqslant n\}\big) \prec \big(\{1\} \cup \{2i \,|\, 2 \leqslant i \leqslant n\}\big)$.

Let $A = \{a_1, a_2, \ldots, a_n\}$ and $B = \{b_1, b_2, \ldots, b_n\}$ be two disjoint sets, both in increasing order. We claim that the winner can be decided only from the values of $a_2, \ldots, a_n$ and $b_2, \ldots, b_n$, while a_1 and b_1 are actually irrelevant. Suppose that this was not the case, and assume without loss of generality that $a_2 < b_2$. Then the relative order of $a_1, a_2, \ldots, a_n, b_2, \ldots, b_n$ is fixed, and the position of b_1 has to decide the winner. Suppose that for some value $b_1 = x$, B wins, while for some other value $b_1 = y$, A wins.

Write $B_x = \{x, b_2, \ldots, b_n\}$ and $B_y = \{y, b_2, \ldots, b_n\}$, and let $\varepsilon > 0$ be smaller than half the distance between any two of the numbers in $B_x \cup B_y \cup A$. For any set M, let $M \pm \varepsilon$ be the set obtained by adding/subtracting ε to all elements of M. By our choice of ε, the relative order of the elements of $(B_y + \varepsilon) \cup A$ is still the same as for $B_y \cup A$, while the relative order of the elements of $(B_x - \varepsilon) \cup A$ is still the same as for $B_x \cup A$. Thus $A \prec B_x - \varepsilon$, but $A \succ B_y + \varepsilon$. Moreover, if $y > x$, then $B_x - \varepsilon \prec B_y + \varepsilon$ by condition 2, while otherwise the relative order of

the elements in $(B_x - \varepsilon) \cup (B_y + \varepsilon)$ is the same as for the two sets $\{2\} \cup \{2i - 1 \,|\, 2 \leqslant i \leqslant n\}$ and $\{1\} \cup \{2i \,|\, 2 \leqslant i \leqslant n\}$, so that $B_x - \varepsilon \prec B_y + \varepsilon$. In either case, we obtain

$$A \prec B_x - \varepsilon \prec B_y + \varepsilon \prec A,$$

which contradicts condition 3.

So we know now that the winner does not depend on a_1, b_1. Therefore, we can define a new rule $\prec^*$ on sets of cardinality $n-1$ by saying that $A \prec^* B$ if and only if $A \cup \{a\} \prec B \cup \{b\}$ for some a, b (or equivalently, all a, b) such that $a < \min A$, $b < \min B$ and $A \cup \{a\}$ and $B \cup \{b\}$ are disjoint. The rule $\prec^*$ satisfies all conditions again, so by the induction hypothesis, there exists an index i such that $A \prec^* B$ if and only if the ith smallest element of A is less than the ith smallest element of B. This implies that $C \prec D$ if and only if the $(i+1)^{\text{th}}$ smallest element of C is less than the $(i+1)^{\text{th}}$ smallest element of D, which completes our induction.

Second case: $\big(\{2i - 1 \,|\, 1 \leqslant i \leqslant n-1\} \cup \{2n\}\big) \prec \big(\{2i \,|\, 1 \leqslant i \leqslant n-1\} \cup \{2n-1\}\big)$.

Set $-A = \{-a \,|\, a \in A\}$ for any $A \subseteq \mathbb{R}$. For any two disjoint sets $A, B \subseteq \mathbb{R}$ of cardinality n, we write $A \prec^\circ B$ to mean $(-B) \prec (-A)$. It is easy to see that $\prec^\circ$ defines a rule to determine a winner that satisfies the three conditions of our problem as well as the relation of the first case. So it follows in the same way as in the first case that for some i, $A \prec^\circ B$ if and only if the ith smallest element of A is less than the ith smallest element of B, which is equivalent to the condition that the ith largest element of $-A$ is greater than the ith largest element of $-B$. This proves that the original rule $\prec$ also has the desired form.

Comment. The problem asks for all possible partial orders on the set of n-element subsets of $\mathbb{R}$ such that any two disjoint sets are comparable, the order relation only depends on the relative order of the elements, and $\{a_1, a_2, \ldots, a_n\} \prec \{b_1, b_2, \ldots, b_n\}$ whenever $a_i < b_i$ for all i.

As the proposer points out, one may also ask for all **total** orders on all n-element subsets of $\mathbb{R}$ (dropping the condition of disjointness and requiring that $\{a_1, a_2, \ldots, a_n\} \preceq \{b_1, b_2, \ldots, b_n\}$ whenever $a_i \leqslant b_i$ for all i). It turns out that the number of possibilities in this case is $n!$, and all possible total orders are obtained in the following way. Fix a permutation $\sigma \in S_n$. Let $A = \{a_1, a_2, \ldots, a_n\}$ and $B = \{b_1, b_2, \ldots, b_n\}$ be two subsets of $\mathbb{R}$ with $a_1 < a_2 < \cdots < a_n$ and $b_1 < b_2 < \cdots < b_n$. Then we say that $A \succ_\sigma B$ if and only if $(a_{\sigma(1)}, \ldots, a_{\sigma(n)})$ is lexicographically greater than $(b_{\sigma(1)}, \ldots, b_{\sigma(n)})$.

It seems, however, that this formulation adds rather more technicalities to the problem than additional ideas.

This page is intentionally left blank

C7. Let M be a set of $n \geqslant 4$ points in the plane, no three of which are collinear. Initially these points are connected with n segments so that each point in M is the endpoint of exactly two segments. Then, at each step, one may choose two segments AB and CD sharing a common interior point and replace them by the segments AC and BD if none of them is present at this moment. Prove that it is impossible to perform $n^3/4$ or more such moves.

(Russia)

Solution. A line is said to be *red* if it contains two points of M. As no three points of M are collinear, each red line determines a unique pair of points of M. Moreover, there are precisely $\binom{n}{2} < \frac{n^2}{2}$ red lines. By the *value of a segment* we mean the number of red lines intersecting it in its interior, and the *value of a set of segments* is defined to be the sum of the values of its elements. We will prove that (i) the value of the initial set of segments is smaller than $n^3/2$ and that (ii) each step decreases the value of the set of segments present by at least 2. Since such a value can never be negative, these two assertions imply the statement of the problem.

To show (i) we just need to observe that each segment has a value that is smaller than $n^2/2$. Thus the combined value of the n initial segments is indeed below $n \cdot n^2/2 = n^3/2$.

It remains to establish (ii). Suppose that at some moment we have two segments AB and CD sharing an interior point S, and that at the next moment we have the two segments AC and BD instead. Let X_{AB} denote the set of red lines intersecting the segment AB in its interior and let the sets X_{AC}, X_{BD}, and X_{CD} be defined similarly. We are to prove that $|X_{AC}| + |X_{BD}| + 2 \leqslant |X_{AB}| + |X_{CD}|$.

As a first step in this direction, we claim that

$$|X_{AC} \cup X_{BD}| + 2 \leqslant |X_{AB} \cup X_{CD}|\,. \tag{1}$$

Indeed, if g is a red line intersecting, e.g. the segment AC in its interior, then it has to intersect the triangle ACS once again, either in the interior of its side AS, or in the interior of its side CS, or at S, meaning that it belongs to X_{AB} or to X_{CD} (see Figure 1). Moreover, the red lines AB and CD contribute to $X_{AB} \cup X_{CD}$ but not to $X_{AC} \cup X_{BD}$. Thereby (1) is proved.

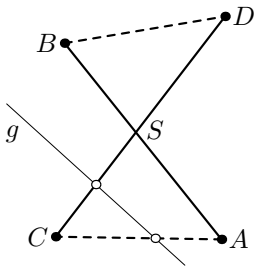


Figure 1

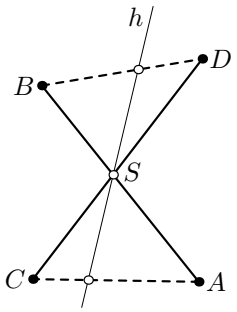


Figure 2

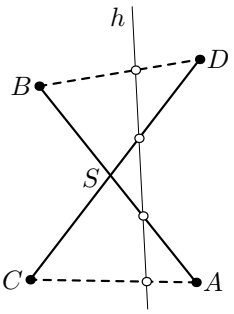


Figure 3

Similarly but more easily one obtains

$$|X_{AC} \cap X_{BD}| \leqslant |X_{AB} \cap X_{CD}|\,. \tag{2}$$

Indeed, a red line h appearing in $X_{AC} \cap X_{BD}$ belongs, for similar reasons as above, also to $X_{AB} \cap X_{CD}$. To make the argument precise, one may just distinguish the cases $S \in h$ (see Figure 2) and $S \notin h$ (see Figure 3). Thereby (2) is proved.

Adding (1) and (2) we obtain the desired conclusion, thus completing the solution of this problem.

Comment 1. There is a problem belonging to the folklore, in the solution of which one may use the same kind of operation:

Given n red and n green points in the plane, prove that one may draw n nonintersecting segments each of which connects a red point with a green point.

A standard approach to this problem consists in taking n arbitrary segments connecting the red points with the green points, and to perform the same operation as in the above proposal whenever an intersection occurs. Now each time one performs such a step, the total length of the segments that are present decreases due to the triangle inequality. So, as there are only finitely many possibilities for the set of segments present, the process must end at some stage.

In the above proposal, however, considering the sum of the Euclidean lengths of the segment that are present does not seem to help much, for even though it shows that the process must necessarily terminate after some finite number of steps, it does not seem to easily yield any upper bound on the number of these steps that grows polynomially with n.

One may regard the concept of the value of a segment introduced in the above solution as an appropriately discretised version of Euclidean length suitable for obtaining such a bound.

The Problem Selection Committee still believes the problem to be sufficiently original for the competition.

Comment 2. There are some other essentially equivalent ways of presenting the same solution. E.g., put $M = \{A_1, A_2, \ldots, A_n\}$, denote the set of segments present at any moment by $\{e_1, e_2, \ldots, e_n\}$, and called a triple (i, j, k) of indices with $i \neq j$ *intersecting*, if the line A_iA_j intersects the segment e_k. It may then be shown that the number S of intersecting triples satisfies $0 \leqslant S < n^3$ at the beginning and decreases by at least 4 in each step.

Comment 3. It is not difficult to construct an example where cn^2 moves are possible (for some absolute constant $c > 0$). It would be interesting to say more about the gap between cn^2 and cn^3.

C8. A card deck consists of 1024 cards. On each card, a set of distinct decimal digits is written in such a way that no two of these sets coincide (thus, one of the cards is empty). Two players alternately take cards from the deck, one card per turn. After the deck is empty, each player checks if he can throw out one of his cards so that each of the ten digits occurs on an even number of his remaining cards. If one player can do this but the other one cannot, the one who can is the winner; otherwise a draw is declared.

Determine all possible first moves of the first player after which he has a winning strategy. *(Russia)*

Answer. All the moves except for taking the empty card.

Solution. Let us identify each card with the set of digits written on it. For any collection of cards $C_1, C_2, \ldots, C_k$ denote by their *sum* the set $C_1 \triangle C_2 \triangle \cdots \triangle C_k$ consisting of all elements belonging to an odd number of the C_i's. Denote the first and the second player by $\mathcal{F}$ and $\mathcal{S}$, respectively.

Since each digit is written on exactly 512 cards, the sum of all the cards is $\varnothing$. Therefore, at the end of the game the sum of all the cards of $\mathcal{F}$ will be the same as that of $\mathcal{S}$; denote this sum by C. Then the player who took C can throw it out and get the desired situation, while the other one cannot. Thus, the player getting card C wins, and no draw is possible.

Now, given a nonempty card B, one can easily see that all the cards can be split into 512 pairs of the form $(X, X \triangle B)$ because $(X \triangle B) \triangle B = X$. The following lemma shows a property of such a partition that is important for the solution.

Lemma. Let $B \neq \varnothing$ be some card. Let us choose 512 cards so that exactly one card is chosen from every pair $(X, X \triangle B)$. Then the sum of all chosen cards is either $\varnothing$ or B.

Proof. Let b be some element of B. Enumerate the pairs; let X_i be the card not containing b in the ith pair, and let Y_i be the other card in this pair. Then the sets X_i are exactly all the sets not containing b, therefore each digit $a \neq b$ is written on exactly 256 of these cards, so $X_1 \triangle X_2 \triangle \cdots \triangle X_{512} = \varnothing$. Now, if we replace some summands in this sum by the other elements from their pairs, we will simply add B several times to this sum, thus the sum will either remain unchanged or change by B, as required. □

Now we consider two cases.

Case 1. Assume that $\mathcal{F}$ takes the card $\varnothing$ on his first move. In this case, we present a winning strategy for $\mathcal{S}$.

Let $\mathcal{S}$ take an arbitrary card A. Assume that $\mathcal{F}$ takes card B after that; then $\mathcal{S}$ takes $A \triangle B$. Split all 1024 cards into 512 pairs of the form $(X, X \triangle B)$; we call two cards in one pair *partners*. Then the four cards taken so far form two pairs $(\varnothing, B)$ and $(A, A \triangle B)$ belonging to $\mathcal{F}$ and $\mathcal{S}$, respectively. On each of the subsequent moves, when $\mathcal{F}$ takes some card, $\mathcal{S}$ should take the partner of this card in response.

Consider the situation at the end of the game. Let us for a moment replace card A belonging to $\mathcal{S}$ by $\varnothing$. Then he would have one card from each pair; by our lemma, the sum of all these cards would be either $\varnothing$ or B. Now, replacing $\varnothing$ back by A we get that the actual sum of the cards of $\mathcal{S}$ is either A or $A \triangle B$, and he has both these cards. Thus $\mathcal{S}$ wins.

Case 2. Now assume that $\mathcal{F}$ takes some card $A \neq \varnothing$ on his first move. Let us present a winning strategy for $\mathcal{F}$ in this case.

Assume that $\mathcal{S}$ takes some card $B \neq \varnothing$ on his first move; then $\mathcal{F}$ takes $A \triangle B$. Again, let us split all the cards into pairs of the form $(X, X \triangle B)$; then the cards which have not been taken yet form several complete pairs and one extra element (card $\varnothing$ has not been taken while its partner B has). Now, on each of the subsequent moves, if $\mathcal{S}$ takes some element from a

complete pair, then $\mathcal{F}$ takes its partner. If $\mathcal{S}$ takes the extra element, then $\mathcal{F}$ takes an arbitrary card Y, and the partner of Y becomes the new extra element.

Thus, on his last move $\mathcal{S}$ is forced to take the extra element. After that player $\mathcal{F}$ has cards A and $A \triangle B$, player $\mathcal{S}$ has cards B and $\varnothing$, and $\mathcal{F}$ has exactly one element from every other pair. Thus the situation is the same as in the previous case with roles reversed, and $\mathcal{F}$ wins.

Finally, if $\mathcal{S}$ takes $\varnothing$ on his first move then $\mathcal{F}$ denotes any card which has not been taken yet by B and takes $A \triangle B$. After that, the same strategy as above is applicable.

Comment 1. If one wants to avoid the unusual question about the first move, one may change the formulation as follows. (The difficulty of the problem would decrease somewhat.)

A card deck consists of 1023 cards; on each card, a nonempty set of distinct decimal digits is written in such a way that no two of these sets coincide. Two players alternately take cards from the deck, one card per turn. When the deck is empty, each player checks if he can throw out one of his cards so that for each of the ten digits, he still holds an even number of cards with this digit. If one player can do this but the other one cannot, the one who can is the winner; otherwise a draw is declared.

Determine which of the players (if any) has a winning strategy.

The winner in this version is the first player. The analysis of the game from the first two paragraphs of the previous solution applies to this version as well, except for the case $C = \varnothing$ in which the result is a draw. Then the strategy for $\mathcal{S}$ in Case 1 works for $\mathcal{F}$ in this version: the sum of all his cards at the end is either A or $A \triangle B$, thus nonempty in both cases.

Comment 2. Notice that all the cards form a vector space over $\mathbb{F}_2$, with $\triangle$ the operation of addition. Due to the automorphisms of this space, all possibilities for $\mathcal{F}$'s first move except $\varnothing$ are equivalent. The same holds for the response by $\mathcal{S}$ if $\mathcal{F}$ takes the card $\varnothing$ on his first move.

Comment 3. It is not that hard to show that in the initial game, $\mathcal{F}$ has a winning move, by the idea of "strategy stealing".

Namely, assume that $\mathcal{S}$ has a winning strategy. Let us take two card decks and start two games, in which $\mathcal{S}$ will act by his strategy. In the first game, $\mathcal{F}$ takes an arbitrary card A_1; assume that $\mathcal{S}$ takes some B_1 in response. Then $\mathcal{F}$ takes the card B_1 at the second game; let the response by $\mathcal{S}$ be A_2. Then $\mathcal{F}$ takes A_2 in the first game and gets a response B_2, and so on.

This process stops at some moment when in the second game $\mathcal{S}$ takes $A_i = A_1$. At this moment the players hold the same sets of cards in both games, but with roles reversed. Now, if some cards remain in the decks, $\mathcal{F}$ takes an arbitrary card from the first deck starting a similar cycle.

At the end of the game, player $\mathcal{F}$'s cards in the first game are exactly player $\mathcal{S}$'s cards in the second game, and vice versa. Thus in one of the games $\mathcal{F}$ will win, which is impossible by our assumption.

One may notice that the strategy in Case 2 is constructed exactly in this way from the strategy in Case 1. This is possible since every response by $\mathcal{S}$ wins if $\mathcal{F}$ takes the card $\varnothing$ on his first move.

C9. There are n circles drawn on a piece of paper in such a way that any two circles intersect in two points, and no three circles pass through the same point. Turbo the snail slides along the circles in the following fashion. Initially he moves on one of the circles in clockwise direction. Turbo always keeps sliding along the current circle until he reaches an intersection with another circle. Then he continues his journey on this new circle and also changes the direction of moving, i.e. from clockwise to anticlockwise or *vice versa*.

Suppose that Turbo's path entirely covers all circles. Prove that n must be odd.

(India)

Solution 1. Replace every cross (i.e. intersection of two circles) by two small circle arcs that indicate the direction in which the snail should leave the cross (see Figure 1.1). Notice that the placement of the small arcs does not depend on the direction of moving on the curves; no matter which direction the snail is moving on the circle arcs, he will follow the same curves (see Figure 1.2). In this way we have a set of curves, that are the possible paths of the snail. Call these curves *snail orbits* or just *orbits*. Every snail orbit is a simple closed curve that has no intersection with any other orbit.

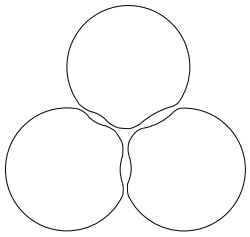


Figure 1.1

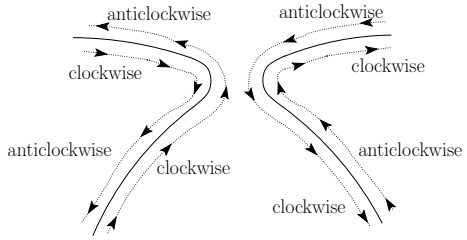


Figure 1.2

We prove the following, more general statement.

> (∗) *In any configuration of n circles such that no two of them are tangent, the number of snail orbits has the same parity as the number n.* (Note that it is not assumed that all circle pairs intersect.)

This immediately solves the problem.

Let us introduce the following operation that will be called *flipping a cross*. At a cross, remove the two small arcs of the orbits, and replace them by the other two arcs. Hence, when the snail arrives at a flipped cross, he will continue on the other circle as before, but he will preserve the orientation in which he goes along the circle arcs (see Figure 2).

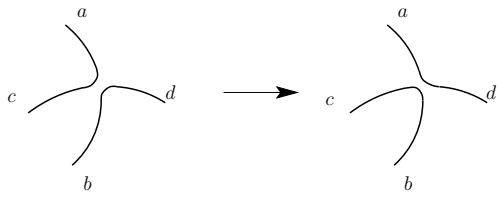


Figure 2

Consider what happens to the number of orbits when a cross is flipped. Denote by a, b, c, and d the four arcs that meet at the cross such that a and b belong to the same circle. Before the flipping a and b were connected to c and d, respectively, and after the flipping a and b are connected to d and c, respectively.

The orbits passing through the cross are closed curves, so each of the arcs a, b, c, and d is connected to another one by orbits outside the cross. We distinguish three cases.

Case 1: a is connected to b and c is connected to d by the orbits outside the cross (see Figure 3.1).

We show that this case is impossible. Remove the two small arcs at the cross, connect a to b, and connect c to d at the cross. Let γ be the new closed curve containing a and b, and let δ be the new curve that connects c and d. These two curves intersect at the cross. So one of c and d is inside γ and the other one is outside γ. Then the two closed curves have to meet at least one more time, but this is a contradiction, since no orbit can intersect itself.

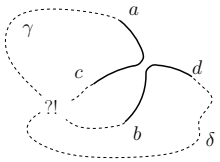


Figure 3.1

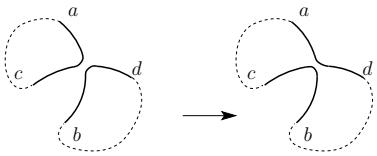


Figure 3.2

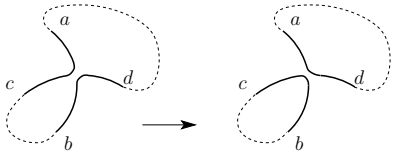


Figure 3.3

Case 2: a is connected to c and b is connected to d (see Figure 3.2).

Before the flipping a and c belong to one orbit and b and d belong to another orbit. Flipping the cross merges the two orbits into a single orbit. Hence, the number of orbits decreases by 1.

Case 3: a is connected to d and b is connected to c (see Figure 3.3).

Before the flipping the arcs a, b, c, and d belong to a single orbit. Flipping the cross splits that orbit in two. The number of orbits increases by 1.

As can be seen, every flipping decreases or increases the number of orbits by one, thus changes its parity.

Now flip every cross, one by one. Since every pair of circles has 0 or 2 intersections, the number of crosses is even. Therefore, when all crosses have been flipped, the original parity of the number of orbits is restored. So it is sufficient to prove $(*)$ for the new configuration, where all crosses are flipped. Of course also in this new configuration the (modified) orbits are simple closed curves not intersecting each other.

Orient the orbits in such a way that the snail always moves anticlockwise along the circle arcs. Figure 4 shows the same circles as in Figure 1 after flipping all crosses and adding orientation. (Note that this orientation may be different from the orientation of the orbit as a planar curve; the orientation of every orbit may be negative as well as positive, like the middle orbit in Figure 4.) If the snail moves around an orbit, the total angle change in his moving direction, the *total curvature*, is either $+2\pi$ or -2π, depending on the orientation of the orbit. Let P and N be the number of orbits with positive and negative orientation, respectively. Then the total curvature of all orbits is $(P - N) \cdot 2\pi$.

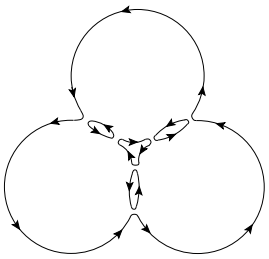


Figure 4

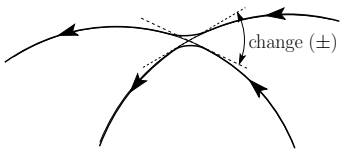


Figure 5

Double-count the total curvature of all orbits. Along every circle the total curvature is 2π. At every cross, the two turnings make two changes with some angles having the same absolute value but opposite signs, as depicted in Figure 5. So the changes in the direction at the crosses cancel out. Hence, the total curvature is $n \cdot 2\pi$.

Now we have $(P - N) \cdot 2\pi = n \cdot 2\pi$, so $P - N = n$. The number of (modified) orbits is $P + N$, that has a same parity as $P - N = n$.

Solution 2. We present a different proof of $(*)$.

We perform a sequence of small modification steps on the configuration of the circles in such a way that at the end they have no intersection at all (see Figure 6.1). We use two kinds of local changes to the structure of the orbits (see Figure 6.2):

- *Type-1 step:* An arc of a circle is moved over an arc of another circle; such a step creates or removes two intersections.

- *Type-2 step:* An arc of a circle is moved through the intersection of two other circles.

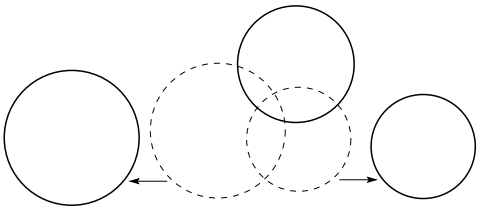


Figure 6.1

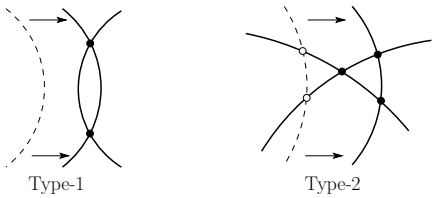


Figure 6.2

We assume that in every step only one circle is moved, and that this circle is moved over at most one arc or intersection point of other circles.

We will show that the parity of the number of orbits does not change in any step. As every circle becomes a separate orbit at the end of the procedure, this fact proves $(*)$.

Consider what happens to the number of orbits when a Type-1 step is performed. The two intersection points are created or removed in a small neighbourhood. Denote some points of the two circles where they enter or leave this neighbourhood by a, b, c, and d in this order around the neighbourhood; let a and b belong to one circle and let c and d belong to the other circle. The two circle arcs may have the same or opposite orientations. Moreover, the four end-points of the two arcs are connected by the other parts of the orbits. This can happen in two ways without intersection: either a is connected to d and b is connected to c, or a is connected to b and c is connected to d. Altogether we have four cases, as shown in Figure 7.

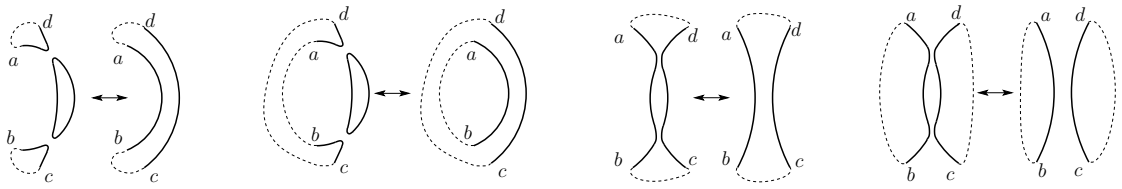


Figure 7

We can see that the number of orbits is changed by -2 or $+2$ in the leftmost case when the arcs have the same orientation, a is connected to d, and b is connected to c. In the other three cases the number of orbits is not changed. Hence, Type-1 steps do not change the parity of the number of orbits.

Now consider a Type-2 step. The three circles enclose a small, triangular region; by the step, this triangle is replaced by another triangle. Again, the modification of the orbits is done in some small neighbourhood; the structure does not change outside. Each side of the triangle shaped region can be convex or concave; the number of concave sides can be 0, 1, 2 or 3, so there are 4 possible arrangements of the orbits inside the neighbourhood, as shown in Figure 8.

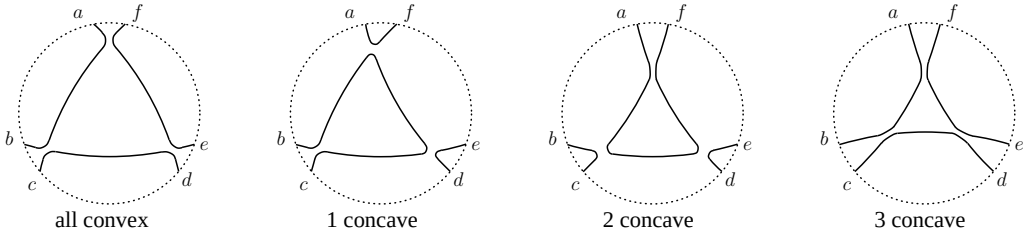


Figure 8

Denote the points where the three circles enter or leave the neighbourhood by a, b, c, d, e, and f in this order around the neighbourhood. As can be seen in Figure 8, there are only two essentially different cases; either a, c, e are connected to b, d, f, respectively, or a, c, e are connected to f, b, d, respectively. The step either preserves the set of connections or switches to the other arrangement. Obviously, in the earlier case the number of orbits is not changed; therefore we have to consider only the latter case.

The points a, b, c, d, e, and f are connected by the orbits outside, without intersection. If a was connected to c, say, then this orbit would isolate b, so this is impossible. Hence, each of a, b, c, d, e and f must be connected either to one of its neighbours or to the opposite point. If say a is connected to d, then this orbit separates b and c from e and f, therefore b must be connected to c and e must be connected to f. Altogether there are only two cases and their reverses: either each point is connected to one of its neighbours or two opposite points are connected and the the remaining neigh boring pairs are connected to each other. See Figure 9.

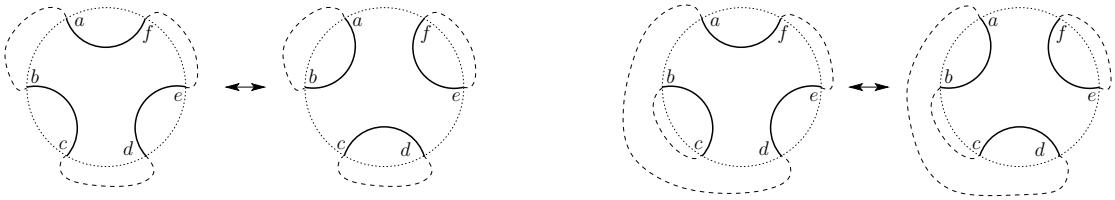


Figure 9

We can see that if only neighbouring points are connected, then the number of orbits is changed by $+2$ or -2. If two opposite points are connected (a and d in the figure), then the orbits are re-arranged, but their number is unchanged. Hence, Type-2 steps also preserve the parity. This completes the proof of $(*)$.

Solution 3. Like in the previous solutions, we do not need all circle pairs to intersect but we assume that the circles form a connected set. Denote by $\mathcal{C}$ and $\mathcal{P}$ the sets of circles and their intersection points, respectively.

The circles divide the plane into several simply connected, bounded regions and one unbounded region. Denote the set of these regions by $\mathcal{R}$. We say that an intersection point or a region is *odd* or *even* if it is contained inside an odd or even number of circles, respectively. Let $\mathcal{P}_{\text{odd}}$ and $\mathcal{R}_{\text{odd}}$ be the sets of odd intersection points and odd regions, respectively.

Claim.

$$|\mathcal{R}_{\text{odd}}| - |\mathcal{P}_{\text{odd}}| \equiv n \pmod 2. \tag{1}$$

Proof. For each circle $c \in \mathcal{C}$, denote by R_c, P_c, and X_c the number of regions inside c, the number of intersection points inside c, and the number of circles intersecting c, respectively. The circles divide each other into several arcs; denote by A_c the number of such arcs inside c. By double counting the regions and intersection points inside the circles we get

$$|\mathcal{R}_{\text{odd}}| \equiv \sum_{c\in\mathcal{C}} R_c \pmod 2 \quad \text{and} \quad |\mathcal{P}_{\text{odd}}| \equiv \sum_{c\in\mathcal{C}} P_c \pmod 2.$$

For each circle c, apply Euler's polyhedron theorem to the (simply connected) regions in c. There are $2X_c$ intersection points on c; they divide the circle into $2X_c$ arcs. The polyhedron theorem yields $(R_c + 1) + (P_c + 2X_c) = (A_c + 2X_c) + 2$, considering the exterior of c as a single region. Therefore,

$$R_c + P_c = A_c + 1. \tag{2}$$

Moreover, we have four arcs starting from every interior points inside c and a single arc starting into the interior from each intersection point on the circle. By double-counting the end-points of the interior arcs we get $2A_c = 4P_c + 2X_c$, so

$$A_c = 2P_c + X_c. \tag{3}$$

The relations (2) and (3) together yield

$$R_c - P_c = X_c + 1. \tag{4}$$

By summing up (4) for all circles we obtain

$$\sum_{c\in\mathcal{C}} R_c - \sum_{c\in\mathcal{C}} P_c = \sum_{c\in\mathcal{C}} X_c + |\mathcal{C}|,$$

which yields

$$|\mathcal{R}_{\text{odd}}| - |\mathcal{P}_{\text{odd}}| \equiv \sum_{c\in\mathcal{C}} X_c + n \pmod 2. \tag{5}$$

Notice that in $\sum_{c\in\mathcal{C}} X_c$ each intersecting circle pair is counted twice, i.e., for both circles in the pair, so

$$\sum_{c\in\mathcal{C}} X_c \equiv 0 \pmod 2,$$

which finishes the proof of the Claim. □

Now insert the same small arcs at the intersections as in the first solution, and suppose that there is a single snail orbit b.

First we show that the odd regions are inside the curve b, while the even regions are outside. Take a region $r \in \mathcal{R}$ and a point x in its interior, and draw a ray y, starting from x, that does not pass through any intersection point of the circles and is neither tangent to any of the circles. As is well-known, x is inside the curve b if and only if y intersects b an odd number of times (see Figure 10). Notice that if an arbitrary circle c contains x in its interior, then c intersects y at a single point; otherwise, if x is outside c, then c has 2 or 0 intersections with y. Therefore, y intersects b an odd number of times if and only if x is contained in an odd number of circles, so if and only if r is odd.

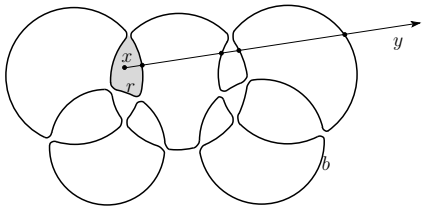


Figure 10

Now consider an intersection point p of two circles c_1 and c_2 and a small neighbourhood around p. Suppose that p is contained inside k circles.

We have four regions that meet at p. Let r_1 be the region that lies outside both c_1 and c_2, let r_2 be the region that lies inside both c_1 and c_2, and let r_3 and r_4 be the two remaining regions, each lying inside exactly one of c_1 and c_2. The region r_1 is contained inside the same k circles as p; the region r_2 is contained also by c_1 and c_2, so by $k+2$ circles in total; each of the regions r_3 and r_4 is contained inside $k+1$ circles. After the small arcs have been inserted at p, the regions r_1 and r_2 get connected, and the regions r_3 and r_4 remain separated at p (see Figure 11). If p is an odd point, then r_1 and r_2 are odd, so two odd regions are connected at p. Otherwise, if p is even, then we have two even regions connected at p.

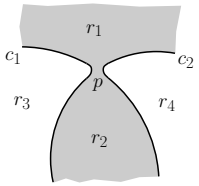


Figure 11

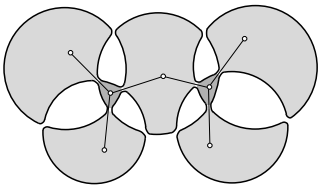


Figure 12

Consider the system of odd regions and their connections at the odd points as a graph. In this graph the odd regions are the vertices, and each odd point establishes an edge that connects two vertices (see Figure 12). As b is a single closed curve, this graph is connected and contains no cycle, so the graph is a tree. Then the number of vertices must be by one greater than the number of edges, so

$$|\mathcal{R}_{\text{odd}}| - |\mathcal{P}_{\text{odd}}| = 1. \tag{9}$$

The relations (1) and (9) together prove that n must be odd.

Comment. For every odd n there exists at least one configuration of n circles with a single snail orbit. Figure 13 shows a possible configuration with 5 circles. In general, if a circle is rotated by $k \cdot \frac{360^\circ}{n}$ $(k = 1, 2, \ldots, n-1)$ around an interior point other than the centre, the circle and its rotated copies together provide a single snail orbit.

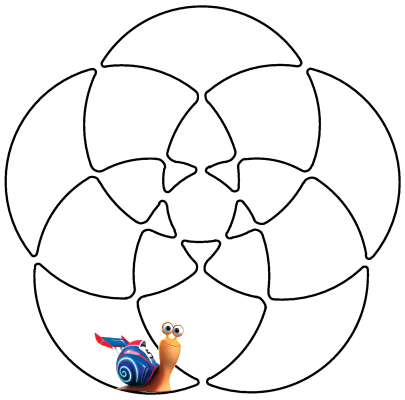


Figure 13

Geometry

G1. The points P and Q are chosen on the side BC of an acute-angled triangle ABC so that $\angle PAB = \angle ACB$ and $\angle QAC = \angle CBA$. The points M and N are taken on the rays AP and AQ, respectively, so that $AP = PM$ and $AQ = QN$. Prove that the lines BM and CN intersect on the circumcircle of the triangle ABC.

(Georgia)

Solution 1. Denote by S the intersection point of the lines BM and CN. Let moreover $\beta = \angle QAC = \angle CBA$ and $\gamma = \angle PAB = \angle ACB$. From these equalities it follows that the triangles ABP and CAQ are similar (see Figure 1). Therefore we obtain

$$\frac{BP}{PM} = \frac{BP}{PA} = \frac{AQ}{QC} = \frac{NQ}{QC}\,.$$

Moreover,

$$\angle BPM = \beta + \gamma = \angle CQN\,.$$

Hence the triangles BPM and NQC are similar. This gives $\angle BMP = \angle NCQ$, so the triangles BPM and BSC are also similar. Thus we get

$$\angle CSB = \angle BPM = \beta + \gamma = 180^\circ - \angle BAC\,,$$

which completes the solution.

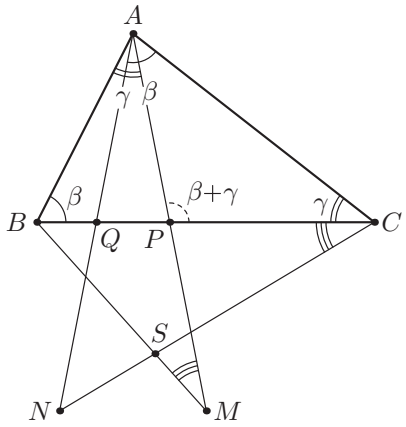


Figure 1

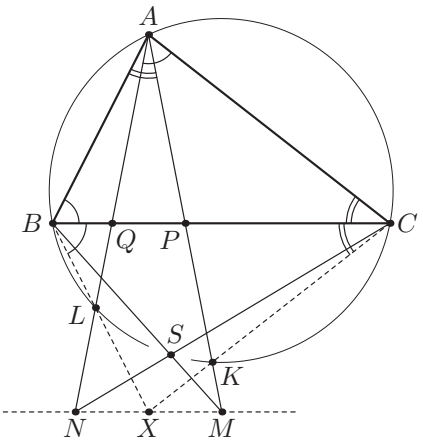


Figure 2

Solution 2. As in the previous solution, denote by S the intersection point of the lines BM and NC. Let moreover the circumcircle of the triangle ABC intersect the lines AP and AQ again at K and L, respectively (see Figure 2).

Note that $\angle LBC = \angle LAC = \angle CBA$ and similarly $\angle KCB = \angle KAB = \angle BCA$. It implies that the lines BL and CK meet at a point X, being symmetric to the point A with respect to the line BC. Since $AP = PM$ and $AQ = QN$, it follows that X lies on the line MN. Therefore, using Pascal's theorem for the hexagon $ALBSCK$, we infer that S lies on the circumcircle of the triangle ABC, which finishes the proof.

Comment. Both solutions can be modified to obtain a more general result, with the equalities

$$AP = PM \quad \text{and} \quad AQ = QN$$

replaced by

$$\frac{AP}{PM} = \frac{QN}{AQ}\,.$$

G2. Let ABC be a triangle. The points K, L, and M lie on the segments BC, CA, and AB, respectively, such that the lines AK, BL, and CM intersect in a common point. Prove that it is possible to choose two of the triangles ALM, BMK, and CKL whose inradii sum up to at least the inradius of the triangle ABC.

(Estonia)

Solution. Denote

$$a = \frac{BK}{KC}, \qquad b = \frac{CL}{LA}, \qquad c = \frac{AM}{MB}.$$

By Ceva's theorem, $abc = 1$, so we may, without loss of generality, assume that $a \geqslant 1$. Then at least one of the numbers b or c is not greater than 1. Therefore at least one of the pairs (a, b), (b, c) has its first component not less than 1 and the second one not greater than 1. Without loss of generality, assume that $1 \leqslant a$ and $b \leqslant 1$.

Therefore, we obtain $bc \leqslant 1$ and $1 \leqslant ca$, or equivalently

$$\frac{AM}{MB} \leqslant \frac{LA}{CL} \qquad \text{and} \qquad \frac{MB}{AM} \leqslant \frac{BK}{KC}.$$

The first inequality implies that the line passing through M and parallel to BC intersects the *segment* AL at a point X (see Figure 1). Therefore the inradius of the triangle ALM is not less than the inradius r_1 of triangle AMX.

Similarly, the line passing through M and parallel to AC intersects the *segment* BK at a point Y, so the inradius of the triangle BMK is not less than the inradius r_2 of the triangle BMY. Thus, to complete our solution, it is enough to show that $r_1 + r_2 \geqslant r$, where r is the inradius of the triangle ABC. We prove that in fact $r_1 + r_2 = r$.

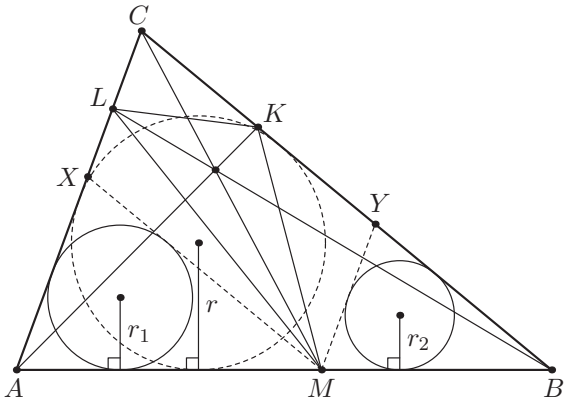


Figure 1

Since $MX \parallel BC$, the dilation with centre A that takes M to B takes the incircle of the triangle AMX to the incircle of the triangle ABC. Therefore

$$\frac{r_1}{r} = \frac{AM}{AB}, \qquad \text{and similarly} \quad \frac{r_2}{r} = \frac{MB}{AB}.$$

Adding these equalities gives $r_1 + r_2 = r$, as required.

Comment. Alternatively, one can use Desargues' theorem instead of Ceva's theorem, as follows: The lines AB, BC, CA dissect the plane into seven regions. One of them is bounded, and amongst the other six, three are two-sided and three are three-sided. Now define the points $P = BC \cap LM$, $Q = CA \cap MK$, and $R = AB \cap KL$ (in the projective plane). By Desargues' theorem, the points P, Q, R lie on a common line ℓ. This line intersects only unbounded regions. If we now assume (without loss of generality) that P, Q and R lie on ℓ in that order, then one of the segments PQ or QR lies inside a two-sided region. If, for example, this segment is PQ, then the triangles ALM and BMK will satisfy the statement of the problem for the same reason.

G3. Let Ω and O be the circumcircle and the circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM. The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q, respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that $BR \parallel AC$. (Here we always assume that an angle bisector is a ray.)

(Russia)

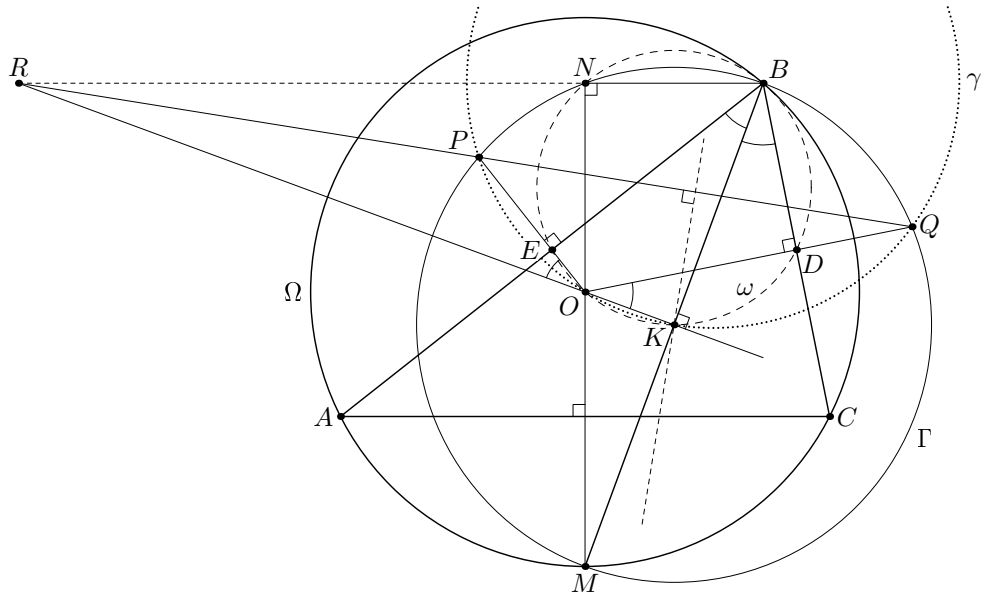
Solution. Let K be the midpoint of BM, i.e., the centre of Γ. Notice that $AB \neq BC$ implies $K \neq O$. Clearly, the lines OM and OK are the perpendicular bisectors of AC and BM, respectively. Therefore, R is the intersection point of PQ and OK.

Let N be the second point of intersection of Γ with the line OM. Since BM is a diameter of Γ, the lines BN and AC are both perpendicular to OM. Hence $BN \parallel AC$, and it suffices to prove that BN passes through R. Our plan for doing this is to interpret the lines BN, OK, and PQ as the radical axes of three appropriate circles.

Let ω be the circle with diameter BO. Since $\angle BNO = \angle BKO = 90^\circ$, the points N and K lie on ω.

Next we show that the points O, K, P, and Q are concyclic. To this end, let D and E be the midpoints of BC and AB, respectively. Clearly, D and E lie on the rays OQ and OP, respectively. By our assumptions about the triangle ABC, the points B, E, O, K, and D lie in this order on ω. It follows that $\angle EOR = \angle EBK = \angle KBD = \angle KOD$, so the line KO externally bisects the angle POQ. Since the point K is the centre of Γ, it also lies on the perpendicular bisector of PQ. So K coincides with the midpoint of the arc POQ of the circumcircle γ of triangle POQ.

Thus the lines OK, BN, and PQ are pairwise radical axes of the circles ω, γ, and Γ. Hence they are concurrent at R, as required.



G4. Consider a fixed circle Γ with three fixed points A, B, and C on it. Also, let us fix a real number $\lambda \in (0,1)$. For a variable point $P \notin \{A, B, C\}$ on Γ, let M be the point on the segment CP such that $CM = \lambda \cdot CP$. Let Q be the second point of intersection of the circumcircles of the triangles AMP and BMC. Prove that as P varies, the point Q lies on a fixed circle.

(United Kingdom)

Solution 1. Throughout the solution, we denote by $\measuredangle(a, b)$ the directed angle between the lines a and b.

Let D be the point on the segment AB such that $BD = \lambda \cdot BA$. We will show that either $Q = D$, or $\measuredangle(DQ, QB) = \measuredangle(AB, BC)$; this would mean that the point Q varies over the constant circle through D tangent to BC at B, as required.

Denote the circumcircles of the triangles AMP and BMC by ω_A and ω_B, respectively. The lines AP, BC, and MQ are pairwise radical axes of the circles Γ, ω_A, and ω_B, thus either they are parallel, or they share a common point X.

Assume that these lines are parallel (see Figure 1). Then the segments AP, QM, and BC have a common perpendicular bisector; the reflection in this bisector maps the segment CP to BA, and maps M to Q. Therefore, in this case Q lies on AB, and $BQ/AB = CM/CP = BD/AB$; so we have $Q = D$.

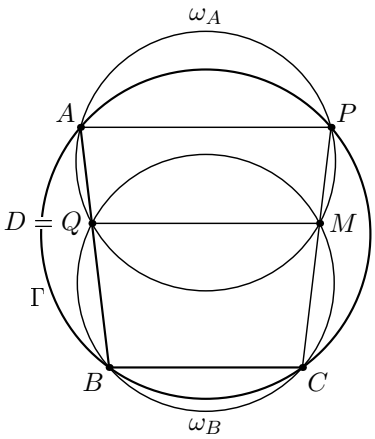


Figure 1

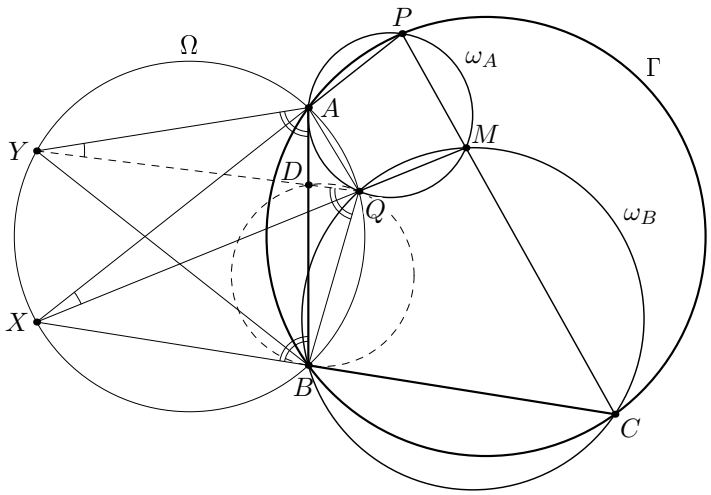


Figure 2

Now assume that the lines AP, QM, and BC are concurrent at some point X (see Figure 2). Notice that the points A, B, Q, and X lie on a common circle Ω by Miquel's theorem applied to the triangle XPC. Let us denote by Y the symmetric image of X about the perpendicular bisector of AB. Clearly, Y lies on Ω, and the triangles YAB and $\triangle XBA$ are congruent. Moreover, the triangle XPC is similar to the triangle XBA, so it is also similar to the triangle YAB.

Next, the points D and M correspond to each other in similar triangles YAB and XPC, since $BD/BA = CM/CP = \lambda$. Moreover, the triangles YAB and XPC are equi-oriented, so $\measuredangle(MX, XP) = \measuredangle(DY, YA)$. On the other hand, since the points A, Q, X, and Y lie on Ω, we have $\measuredangle(QY, YA) = \measuredangle(MX, XP)$. Therefore, $\measuredangle(QY, YA) = \measuredangle(DY, YA)$, so the points Y, D, and Q are collinear.

Finally, we have $\measuredangle(DQ, QB) = \measuredangle(YQ, QB) = \measuredangle(YA, AB) = \measuredangle(AB, BX) = \measuredangle(AB, BC)$, as desired.

Comment. In the original proposal, λ was supposed to be an arbitrary real number distinct from 0 and 1, and the point M was defined by $\overrightarrow{CM} = \lambda \cdot \overrightarrow{CP}$. The Problem Selection Committee decided to add the restriction $\lambda \in (0, 1)$ in order to avoid a large case distinction.

Solution 2. As in the previous solution, we introduce the radical centre $X = AP \cap BC \cap MQ$ of the circles ω_A, ω_B, and Γ. Next, we also notice that the points A, Q, B, and X lie on a common circle Ω.

If the point P lies on the arc BAC of Γ, then the point X is outside Γ, thus the point Q belongs to the ray XM, and therefore the points P, A, and Q lie on the same side of BC. Otherwise, if P lies on the arc BC not containing A, then X lies inside Γ, so M and Q lie on different sides of BC; thus again Q and A lie on the same side of BC. So, in each case the points Q and A lie on the same side of BC.

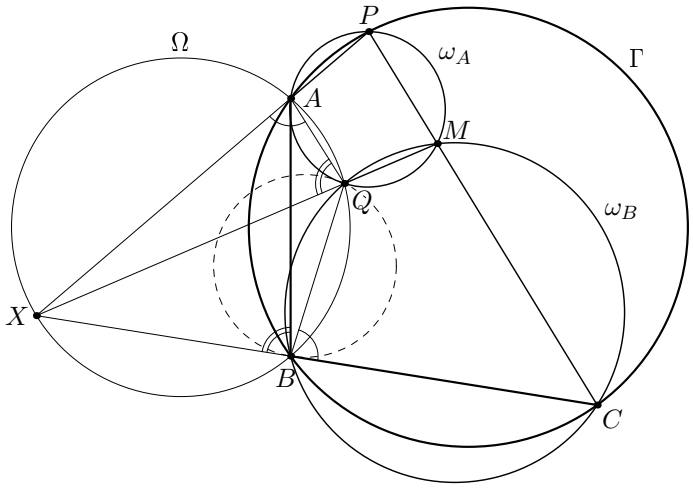


Figure 3

Now we prove that the ratio

$$\frac{QB}{\sin\angle QBC} = \frac{QB}{QX} \cdot \frac{QX}{\sin\angle QBX}$$

is constant. Since the points A, Q, B, and X are concyclic, we have

$$\frac{QX}{\sin\angle QBX} = \frac{AX}{\sin\angle ABC}.$$

Next, since the points B, Q, M, and C are concyclic, the triangles XBQ and XMC are similar, so

$$\frac{QB}{QX} = \frac{CM}{CX} = \lambda \cdot \frac{CP}{CX}.$$

Analogously, the triangles XCP and XAB are also similar, so

$$\frac{CP}{CX} = \frac{AB}{AX}.$$

Therefore, we obtain

$$\frac{QB}{\sin\angle QBC} = \lambda \cdot \frac{AB}{AX} \cdot \frac{AX}{\sin\angle ABC} = \lambda \cdot \frac{AB}{\sin\angle ABC},$$

so this ratio is indeed constant. Thus the circle passing through Q and tangent to BC at B is also constant, and Q varies over this fixed circle.

Comment. It is not hard to guess that the desired circle should be tangent to BC at B. Indeed, the second paragraph of this solution shows that this circle lies on one side of BC; on the other hand, in the limit case $P = B$, the point Q also coincides with B.

Solution 3. Let us perform an inversion centred at C. Denote by X' the image of a point X under this inversion.

The circle Γ maps to the line Γ' passing through the constant points A' and B', and containing the variable point P'. By the problem condition, the point M varies over the circle γ which is the homothetic image of Γ with centre C and coefficient λ. Thus M' varies over the constant line $\gamma' \parallel A'B'$ which is the homothetic image of $A'B'$ with centre C and coefficient $1/\lambda$, and $M = \gamma' \cap CP'$. Next, the circumcircles ω_A and ω_B of the triangles AMP and BMC map to the circumcircle ω'_A of the triangle $A'M'P'$ and to the line $B'M'$, respectively; the point Q thus maps to the second point of intersection of $B'M'$ with ω'_A (see Figure 4).

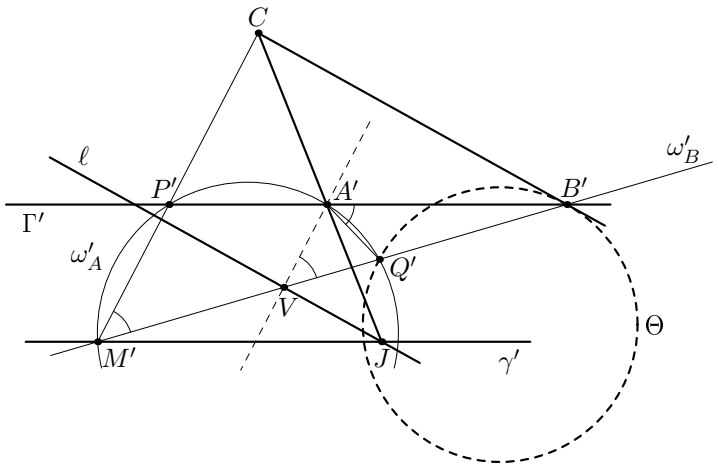


Figure 4

Let J be the (constant) common point of the lines γ' and CA', and let ℓ be the (constant) line through J parallel to CB'. Let V be the common point of the lines ℓ and $B'M'$. Applying Pappus' theorem to the triples (C, J, A') and (V, B', M') we get that the points $CB' \cap JV$, $JM' \cap A'B'$, and $CM' \cap A'V$ are collinear. The first two of these points are ideal, hence so is the third, which means that $CM' \parallel A'V$.

Now we have $\measuredangle(Q'A', A'P') = \measuredangle(Q'M', M'P') = \angle(VM', A'V)$, which means that the triangles $B'Q'A'$ and $B'A'V$ are similar, and $(B'A')^2 = B'Q' \cdot B'V$. Thus Q' is the image of V under the second (fixed) inversion with centre B' and radius $B'A'$. Since V varies over the constant line ℓ, Q' varies over some constant circle Θ. Thus, applying the first inversion back we get that Q also varies over some fixed circle.

One should notice that this last circle is not a line; otherwise Θ would contain C, and thus ℓ would contain the image of C under the second inversion. This is impossible, since $CB' \parallel \ell$.

G5. Let $ABCD$ be a convex quadrilateral with $\angle B = \angle D = 90^\circ$. Point H is the foot of the perpendicular from A to BD. The points S and T are chosen on the sides AB and AD, respectively, in such a way that H lies inside triangle SCT and

$$\angle SHC - \angle BSC = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ .$$

Prove that the circumcircle of triangle SHT is tangent to the line BD.

(Iran)

Solution. Let the line passing through C and perpendicular to the line SC intersect the line AB at Q (see Figure 1). Then

$$\angle SQC = 90^\circ - \angle BSC = 180^\circ - \angle SHC ,$$

which implies that the points C, H, S, and Q lie on a common circle. Moreover, since SQ is a diameter of this circle, we infer that the circumcentre K of triangle SHC lies on the line AB. Similarly, we prove that the circumcentre L of triangle CHT lies on the line AD.

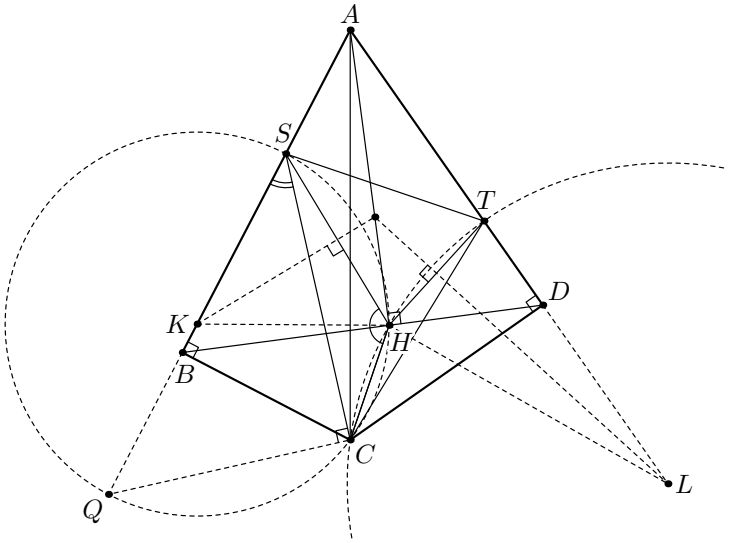


Figure 1

In order to prove that the circumcircle of triangle SHT is tangent to BD, it suffices to show that the perpendicular bisectors of HS and HT intersect on the line AH. However, these two perpendicular bisectors coincide with the angle bisectors of angles AKH and ALH. Therefore, in order to complete the solution, it is enough (by the bisector theorem) to show that

$$\frac{AK}{KH} = \frac{AL}{LH} . \tag{1}$$

We present two proofs of this equality.

First proof. Let the lines KL and HC intersect at M (see Figure 2). Since $KH = KC$ and $LH = LC$, the points H and C are symmetric to each other with respect to the line KL. Therefore M is the midpoint of HC. Denote by O the circumcentre of quadrilateral $ABCD$. Then O is the midpoint of AC. Therefore we have $OM \parallel AH$ and hence $OM \perp BD$. This together with the equality $OB = OD$ implies that OM is the perpendicular bisector of BD and therefore $BM = DM$.

Since $CM \perp KL$, the points B, C, M, and K lie on a common circle with diameter KC. Similarly, the points L, C, M, and D lie on a circle with diameter LC. Thus, using the sine law, we obtain

$$\frac{AK}{AL} = \frac{\sin\angle ALK}{\sin\angle AKL} = \frac{DM}{CL} \cdot \frac{CK}{BM} = \frac{CK}{CL} = \frac{KH}{LH} ,$$

which finishes the proof of (1).

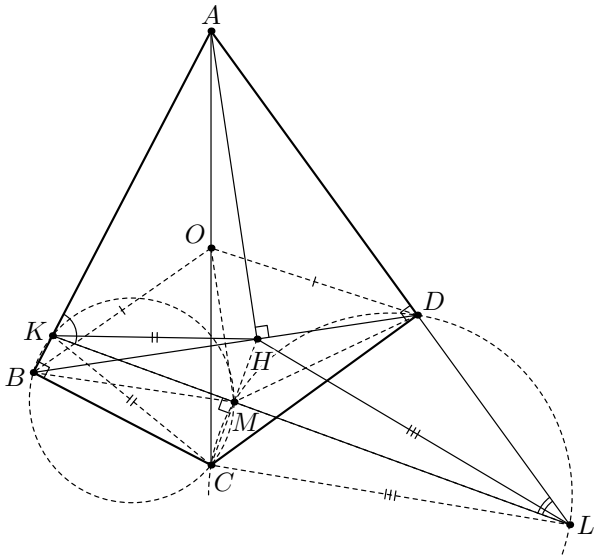


Figure 2

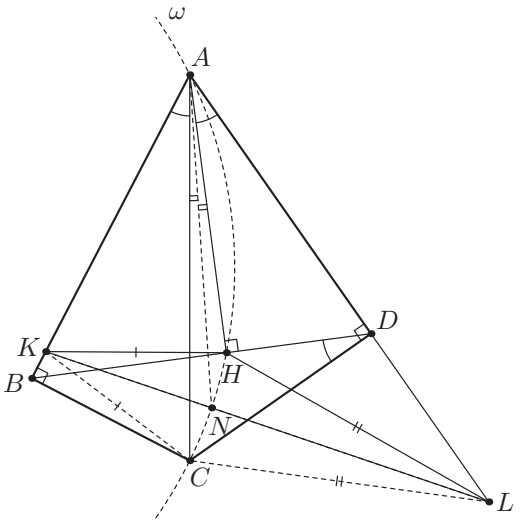


Figure 3

Second proof. If the points A, H, and C are collinear, then $AK = AL$ and $KH = LH$, so the equality (1) follows. Assume therefore that the points A, H, and C do not lie in a line and consider the circle ω passing through them (see Figure 3). Since the quadrilateral $ABCD$ is cyclic,

$$\angle BAC = \angle BDC = 90^\circ - \angle ADH = \angle HAD\,.$$

Let $N \neq A$ be the intersection point of the circle ω and the angle bisector of $\angle CAH$. Then AN is also the angle bisector of $\angle BAD$. Since H and C are symmetric to each other with respect to the line KL and $HN = NC$, it follows that both N and the centre of ω lie on the line KL. This means that the circle ω is an Apollonius circle of the points K and L. This immediately yields (1).

Comment. Either proof can be used to obtain the following generalised result:

Let $ABCD$ be a convex quadrilateral and let H be a point in its interior with $\angle BAC = \angle DAH$. The points S and T are chosen on the sides AB and AD, respectively, in such a way that H lies inside triangle SCT and

$$\angle SHC - \angle BSC = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ\,.$$

Then the circumcentre of triangle SHT lies on the line AH (and moreover the circumcentre of triangle SCT lies on AC).

G6. Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB, respectively, and let M be the midpoint of EF. Let the perpendicular bisector of EF intersect the line BC at K, and let the perpendicular bisector of MK intersect the lines AC and AB at S and T, respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic.

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that

$$\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}.$$

(Iran)

Solution 1. For any interesting pair (E, F), we will say that the corresponding triangle EFK is also *interesting*.

Let EFK be an interesting triangle. Firstly, we prove that $\angle KEF = \angle KFE = \angle A$, which also means that the circumcircle ω_1 of the triangle AEF is tangent to the lines KE and KF.

Denote by ω the circle passing through the points K, S, A, and T. Let the line AM intersect the line ST and the circle ω (for the second time) at N and L, respectively (see Figure 1).

Since $EF \parallel TS$ and M is the midpoint of EF, N is the midpoint of ST. Moreover, since K and M are symmetric to each other with respect to the line ST, we have $\angle KNS = \angle MNS = \angle LNT$. Thus the points K and L are symmetric to each other with respect to the perpendicular bisector of ST. Therefore $KL \parallel ST$.

Let G be the point symmetric to K with respect to N. Then G lies on the line EF, and we may assume that it lies on the ray MF. One has

$$\angle KGE = \angle KNS = \angle SNM = \angle KLA = 180^\circ - \angle KSA$$

(if $K = L$, then the angle KLA is understood to be the angle between AL and the tangent to ω at L). This means that the points K, G, E, and S are concyclic. Now, since $KSGT$ is a parallelogram, we obtain $\angle KEF = \angle KSG = 180^\circ - \angle TKS = \angle A$. Since $KE = KF$, we also have $\angle KFE = \angle KEF = \angle A$.

After having proved this fact, one may finish the solution by different methods.

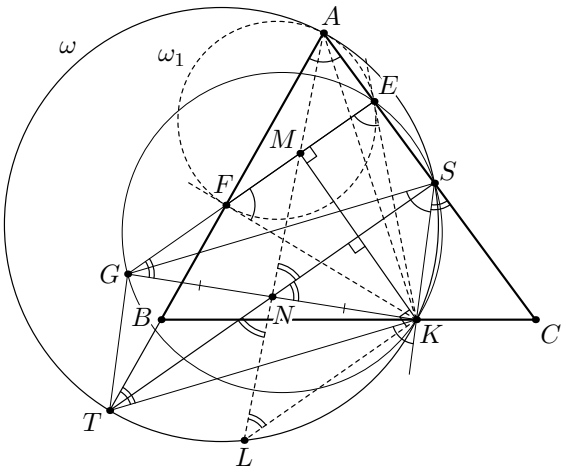


Figure 1

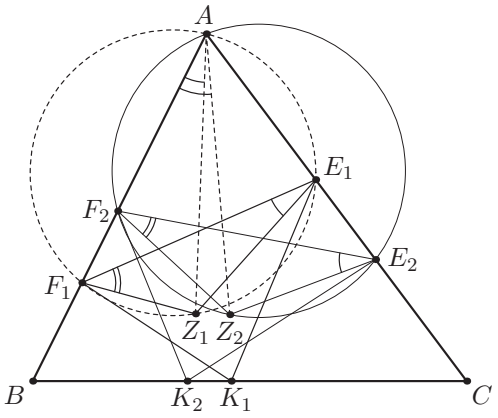


Figure 2

First method. We have just proved that all interesting triangles are similar to each other. This allows us to use the following lemma.

Lemma. Let ABC be an arbitrary triangle. Choose two points E_1 and E_2 on the side AC, two points F_1 and F_2 on the side AB, and two points K_1 and K_2 on the side BC, in a way that the triangles $E_1F_1K_1$ and $E_2F_2K_2$ are similar. Then the six circumcircles of the triangles AE_iF_i, BF_iK_i, and CE_iK_i $(i = 1, 2)$ meet at a common point Z. Moreover, Z is the centre of the spiral similarity that takes the triangle $E_1F_1K_1$ to the triangle $E_2F_2K_2$.

Proof. Firstly, notice that for each $i = 1, 2$, the circumcircles of the triangles AE_iF_i, BF_iK_i, and CK_iE_i have a common point Z_i by Miquel's theorem. Moreover, we have

$$\measuredangle(Z_iF_i, Z_iE_i) = \measuredangle(AB, CA)\,, \quad \measuredangle(Z_iK_i, Z_iF_i) = \measuredangle(BC, AB)\,, \quad \measuredangle(Z_iE_i, Z_iK_i) = \measuredangle(CA, BC)\,.$$

This yields that the points Z_1 and Z_2 correspond to each other in similar triangles $E_1F_1K_1$ and $E_2F_2K_2$. Thus, if they coincide, then this common point is indeed the desired centre of a spiral similarity.

Finally, in order to show that $Z_1 = Z_2$, one may notice that $\measuredangle(AB, AZ_1) = \measuredangle(E_1F_1, E_1Z_1) = \measuredangle(E_2F_2, E_2Z_2) = \measuredangle(AB, AZ_2)$ (see Figure 2). Similarly, one has $\measuredangle(BC, BZ_1) = \measuredangle(BC, BZ_2)$ and $\measuredangle(CA, CZ_1) = \measuredangle(CA, CZ_2)$. This yields $Z_1 = Z_2$. □

Now, let P and Q be the feet of the perpendiculars from B and C onto AC and AB, respectively, and let R be the midpoint of BC (see Figure 3). Then R is the circumcentre of the cyclic quadrilateral $BCPQ$. Thus we obtain $\angle APQ = \angle B$ and $\angle RPC = \angle C$, which yields $\angle QPR = \angle A$. Similarly, we show that $\angle PQR = \angle A$. Thus, all interesting triangles are similar to the triangle PQR.

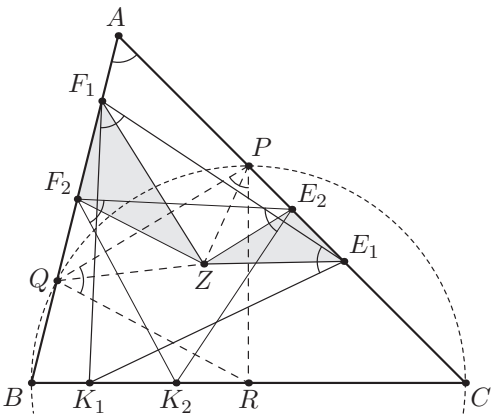


Figure 3

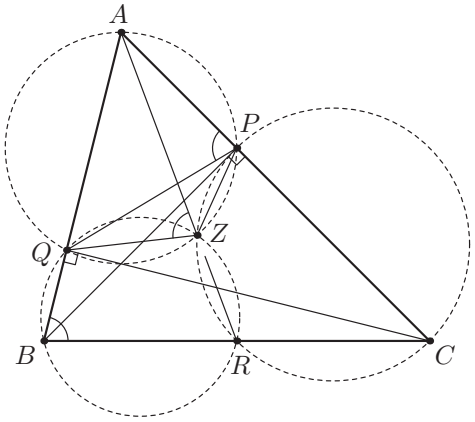


Figure 4

Denote now by Z the common point of the circumcircles of APQ, BQR, and CPR. Let $E_1F_1K_1$ and $E_2F_2K_2$ be two interesting triangles. By the lemma, Z is the centre of any spiral similarity taking one of the triangles $E_1F_1K_1$, $E_2F_2K_2$, and PQR to some other of them. Therefore the triangles ZE_1E_2 and ZF_1F_2 are similar, as well as the triangles ZE_1F_1 and ZPQ. Hence

$$\frac{E_1E_2}{F_1F_2} = \frac{ZE_1}{ZF_1} = \frac{ZP}{ZQ}\,.$$

Moreover, the equalities $\angle AZQ = \angle APQ = \angle ABC = 180^\circ - \angle QZR$ show that the point Z lies on the line AR (see Figure 4). Therefore the triangles AZP and ACR are similar, as well as the triangles AZQ and ABR. This yields

$$\frac{ZP}{ZQ} = \frac{ZP}{RC} \cdot \frac{RB}{ZQ} = \frac{AZ}{AC} \cdot \frac{AB}{AZ} = \frac{AB}{AC}\,,$$

which completes the solution.

Second method. Now we will start from the fact that ω_1 is tangent to the lines KE and KF (see Figure 5). We prove that if (E, F) is an interesting pair, then

$$\frac{AE}{AB} + \frac{AF}{AC} = 2\cos\angle A. \tag{1}$$

Let Y be the intersection point of the segments BE and CF. The points B, K, and C are collinear, hence applying Pascal's theorem to the degenerated hexagon $AFFYEE$, we infer that Y lies on the circle ω_1.

Denote by Z the second intersection point of the circumcircle of the triangle BFY with the line BC (see Figure 6). By Miquel's theorem, the points C, Z, Y, and E are concyclic. Therefore we obtain

$$BF \cdot AB + CE \cdot AC = BY \cdot BE + CY \cdot CF = BZ \cdot BC + CZ \cdot BC = BC^2\,.$$

On the other hand, $BC^2 = AB^2 + AC^2 - 2AB \cdot AC\cos\angle A$, by the cosine law. Hence

$$(AB - AF) \cdot AB + (AC - AE) \cdot AC = AB^2 + AC^2 - 2AB \cdot AC\cos\angle A\,,$$

which simplifies to the desired equality (1).

Let now (E_1, F_1) and (E_2, F_2) be two interesting pairs of points. Then we get

$$\frac{AE_1}{AB} + \frac{AF_1}{AC} = \frac{AE_2}{AB} + \frac{AF_2}{AC}\,,$$

which gives the desired result.

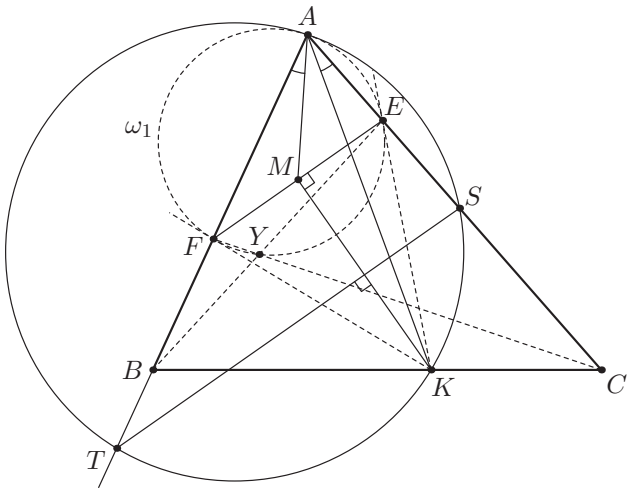


Figure 5

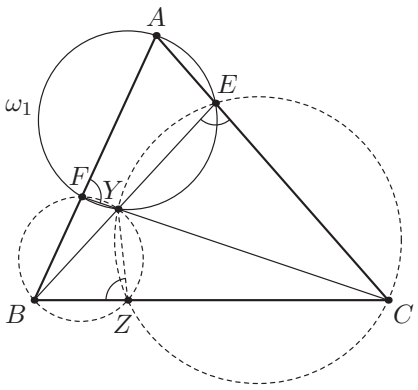


Figure 6

Third method. Again, we make use of the fact that all interesting triangles are similar (and equi-oriented). Let us put the picture onto a complex plane such that A is at the origin, and identify each point with the corresponding complex number.

Let EFK be any interesting triangle. The equalities $\angle KEF = \angle KFE = \angle A$ yield that the ratio $\nu = \frac{K-E}{F-E}$ is the same for all interesting triangles. This in turn means that the numbers E, F, and K satisfy the linear equation

$$K = \mu E + \nu F, \qquad \text{where} \quad \mu = 1 - \nu. \tag{2}$$

Now let us choose the points X and Y on the rays AB and AC, respectively, so that $\angle CXA = \angle AYB = \angle A = \angle KEF$ (see Figure 7). Then each of the triangles AXC and YAB is similar to any interesting triangle, which also means that

$$C = \mu A + \nu X = \nu X \quad \text{and} \quad B = \mu Y + \nu A = \mu Y. \tag{3}$$

Moreover, one has $X/Y = \overline{C/B}$.

Since the points E, F, and K lie on AC, AB, and BC, respectively, one gets

$$E = \rho Y, \quad F = \sigma X, \quad \text{and} \quad K = \lambda B + (1-\lambda)C$$

for some real ρ, σ, and λ. In view of (3), the equation (2) now reads $\lambda B + (1-\lambda)C = K = \mu E + \nu F = \rho B + \sigma C$, or

$$(\lambda - \rho)B = (\sigma + \lambda - 1)C.$$

Since the nonzero complex numbers B and C have different arguments, the coefficients in the brackets vanish, so $\rho = \lambda$ and $\sigma = 1 - \lambda$. Therefore,

$$\frac{E}{Y} + \frac{F}{X} = \rho + \sigma = 1. \tag{4}$$

Now, if (E_1, F_1) and (E_2, F_2) are two distinct interesting pairs, one may apply (4) to both pairs. Subtracting, we get

$$\frac{E_1 - E_2}{Y} = \frac{F_2 - F_1}{X}, \qquad \text{so} \quad \frac{E_1 - E_2}{F_2 - F_1} = \frac{Y}{X} = \frac{\overline{B}}{\overline{C}}.$$

Taking absolute values provides the required result.

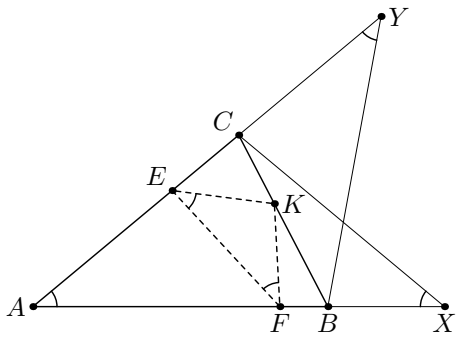


Figure 7

Comment 1. One may notice that the triangle PQR is also interesting.

Comment 2. In order to prove that $\angle KEF = \angle KFE = \angle A$, one may also use the following well-known fact:

Let AEF be a triangle with $AE \neq AF$, and let K be the common point of the symmedian taken from A and the perpendicular bisector of EF. Then the lines KE and KF are tangent to the circumcircle ω_1 of the triangle AEF.

In this case, however, one needs to deal with the case $AE = AF$ separately.

Solution 2. Let (E, F) be an interesting pair. This time we prove that

$$\frac{AM}{AK} = \cos\angle A\,. \tag{5}$$

As in Solution 1, we introduce the circle ω passing through the points K, S, A, and T, together with the points N and L at which the line AM intersect the line ST and the circle ω for the second time, respectively. Let moreover O be the centre of ω (see Figures 8 and 9). As in Solution 1, we note that N is the midpoint of ST and show that $KL \parallel ST$, which implies $\angle FAM = \angle EAK$.

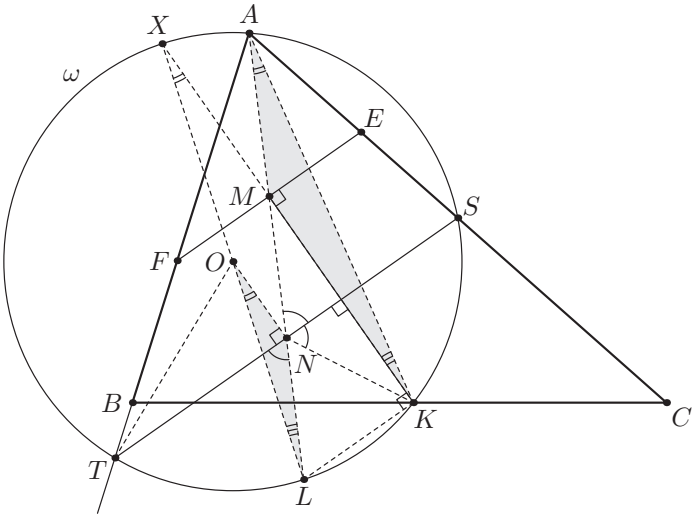


Figure 8

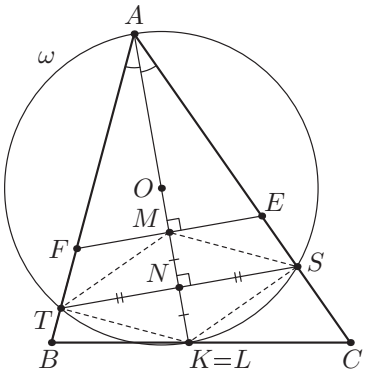


Figure 9

Suppose now that $K \neq L$ (see Figure 8). Then $KL \parallel ST$, and consequently the lines KM and KL are perpendicular. It implies that the lines LO and KM meet at a point X lying on the circle ω. Since the lines ON and XM are both perpendicular to the line ST, they are parallel to each other, and hence $\angle LON = \angle LXK = \angle MAK$. On the other hand, $\angle OLN = \angle MKA$, so we infer that triangles NOL and MAK are similar. This yields

$$\frac{AM}{AK} = \frac{ON}{OL} = \frac{ON}{OT} = \cos\angle TON = \cos\angle A\,.$$

If, on the other hand, $K = L$, then the points A, M, N, and K lie on a common line, and this line is the perpendicular bisector of ST (see Figure 9). This implies that AK is a diameter of ω, which yields $AM = 2OK - 2NK = 2ON$. So also in this case we obtain

$$\frac{AM}{AK} = \frac{2ON}{2OT} = \cos\angle TON = \cos\angle A\,.$$

Thus (5) is proved.

Let P and Q be the feet of the perpendiculars from B and C onto AC and AB, respectively (see Figure 10). We claim that the point M lies on the line PQ. Consider now the composition of the dilatation with factor $\cos\angle A$ and centre A, and the reflection with respect to the angle bisector of $\angle BAC$. This transformation is a similarity that takes B, C, and K to P, Q, and M, respectively. Since K lies on the line BC, the point M lies on the line PQ.

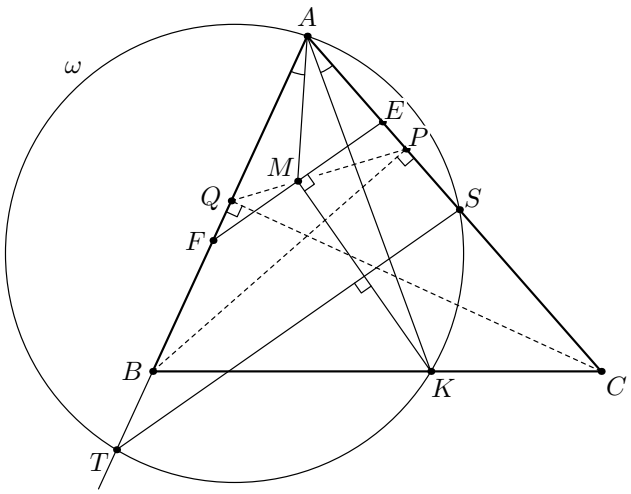


Figure 10

Suppose that $E \neq P$. Then also $F \neq Q$, and by MENELAUS' theorem, we obtain

$$\frac{AQ}{FQ} \cdot \frac{FM}{EM} \cdot \frac{EP}{AP} = 1\,.$$

Using the similarity of the triangles APQ and ABC, we infer that

$$\frac{EP}{FQ} = \frac{AP}{AQ} = \frac{AB}{AC}\,, \quad \text{and hence} \quad \frac{EP}{AB} = \frac{FQ}{AC}\,.$$

The last equality holds obviously also in case $E = P$, because then $F = Q$. Moreover, since the line PQ intersects the *segment* EF, we infer that the point E lies on the segment AP if and only if the point F lies outside of the segment AQ.

Let now (E_1, F_1) and (E_2, F_2) be two interesting pairs. Then we obtain

$$\frac{E_1P}{AB} = \frac{F_1Q}{AC} \qquad \text{and} \qquad \frac{E_2P}{AB} = \frac{F_2Q}{AC}\,.$$

If P lies between the points E_1 and E_2, we add the equalities above, otherwise we subtract them. In any case we obtain

$$\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}\,,$$

which completes the solution.

G7. Let ABC be a triangle with circumcircle Ω and incentre I. Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at points U and V, respectively. Let the line passing through U and parallel to AI intersect AV at X, and let the line passing through V and parallel to AI intersect AB at Y. Let W and Z be the midpoints of AX and BC, respectively. Prove that if the points I, X, and Y are collinear, then the points I, W, and Z are also collinear.

(U.S.A.)

Solution 1. We start with some general observations. Set $\alpha = \angle A/2$, $\beta = \angle B/2$, $\gamma = \angle C/2$. Then obviously $\alpha + \beta + \gamma = 90^\circ$. Since $\angle UIC = 90^\circ$, we obtain $\angle IUC = \alpha + \beta$. Therefore $\angle BIV = \angle IUC - \angle IBC = \alpha = \angle BAI = \angle BYV$, which implies that the points B, Y, I, and V lie on a common circle (see Figure 1).

Assume now that the points I, X and Y are collinear. We prove that $\angle YIA = 90^\circ$.

Let the line XU intersect AB at N. Since the lines AI, UX, and VY are parallel, we get

$$\frac{NX}{AI} = \frac{YN}{YA} = \frac{VU}{VI} = \frac{XU}{AI},$$

implying $NX = XU$. Moreover, $\angle BIU = \alpha = \angle BNU$. This implies that the quadrilateral $BUIN$ is cyclic, and since BI is the angle bisector of $\angle UBN$, we infer that $NI = UI$. Thus in the isosceles triangle NIU, the point X is the midpoint of the base NU. This gives $\angle IXN = 90^\circ$, i.e., $\angle YIA = 90^\circ$.

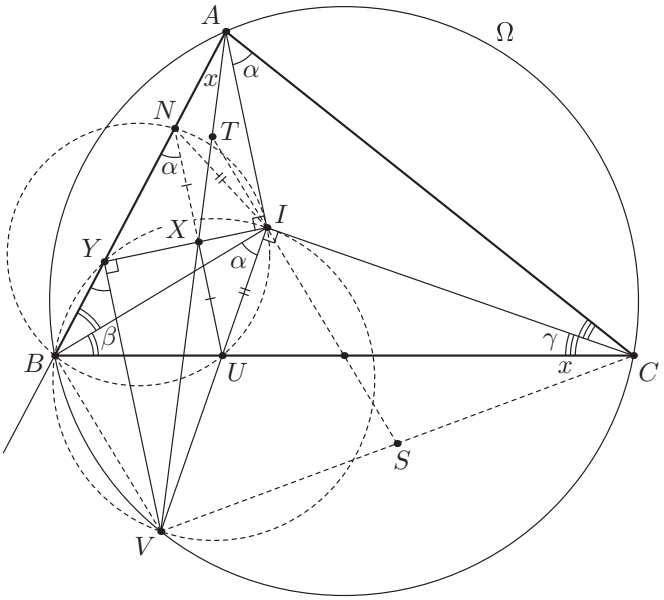


Figure 1

Let S be the midpoint of the segment VC. Let moreover T be the intersection point of the lines AX and SI, and set $x = \angle BAV = \angle BCV$. Since $\angle CIA = 90^\circ + \beta$ and $SI = SC$, we obtain

$$\angle TIA = 180^\circ - \angle AIS = 90^\circ - \beta - \angle CIS = 90^\circ - \beta - \gamma - x = \alpha - x = \angle TAI,$$

which implies that $TI = TA$. Therefore, since $\angle XIA = 90^\circ$, the point T is the midpoint of AX, i.e., $T = W$.

To complete our solution, it remains to show that the intersection point of the lines IS and BC coincide with the midpoint of the segment BC. But since S is the midpoint of the segment VC, it suffices to show that the lines BV and IS are parallel.

Since the quadrilateral $BYIV$ is cyclic, $\angle VBI = \angle VYI = \angle YIA = 90^\circ$. This implies that BV is the external angle bisector of the angle ABC, which yields $\angle VAC = \angle VCA$. Therefore $2\alpha - x = 2\gamma + x$, which gives $\alpha = \gamma + x$. Hence $\angle SCI = \alpha$, so $\angle VSI = 2\alpha$.

On the other hand, $\angle BVC = 180^\circ - \angle BAC = 180^\circ - 2\alpha$, which implies that the lines BV and IS are parallel. This completes the solution.

Solution 2. As in Solution 1, we first prove that the points B, Y, I, V lie on a common circle and $\angle YIA = 90^\circ$. The remaining part of the solution is based on the following lemma, which holds true for any triangle ABC, not necessarily with the property that I, X, Y are collinear.

Lemma. Let ABC be the triangle inscribed in a circle Γ and let I be its incentre. Assume that the line passing through I and perpendicular to the line AI intersects the side AB at the point Y. Let the circumcircle of the triangle BYI intersect the circle Γ for the second time at V, and let the excircle of the triangle ABC opposite to the vertex A be tangent to the side BC at E. Then

$$\angle BAV = \angle CAE\,.$$

Proof. Let ρ be the composition of the inversion with centre A and radius $\sqrt{AB \cdot AC}$, and the symmetry with respect to AI. Clearly, ρ interchanges B and C.

Let J be the excentre of the triangle ABC opposite to A (see Figure 2). Then we have $\angle JAC = \angle BAI$ and $\angle JCA = 90^\circ + \gamma = \angle BIA$, so the triangles ACJ and AIB are similar, and therefore $AB \cdot AC = AI \cdot AJ$. This means that ρ interchanges I and J. Moreover, since Y lies on AB and $\angle AIY = 90^\circ$, the point $Y' = \rho(Y)$ lies on AC, and $\angle JY'A = 90^\circ$. Thus ρ maps the circumcircle γ of the triangle BYI to a circle γ' with diameter JC.

Finally, since V lies on both Γ and γ, the point $V' = \rho(V)$ lies on the line $\rho(\Gamma) = AB$ as well as on γ', which in turn means that $V' = E$. This implies the desired result. □

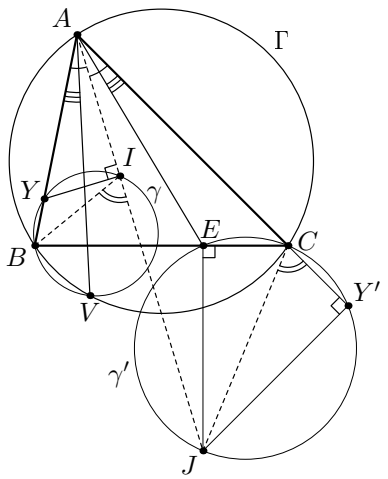


Figure 2

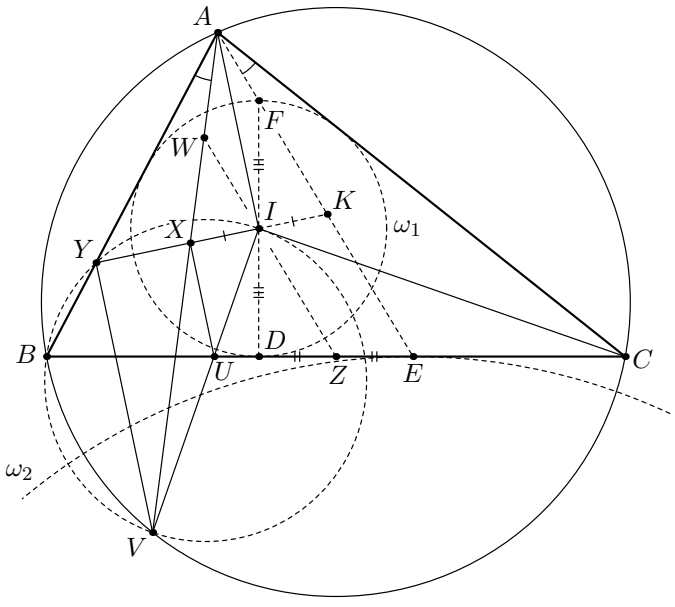


Figure 3

Now we turn to the solution of the problem.

Assume that the incircle ω_1 of the triangle ABC is tangent to BC at D, and let the excircle ω_2 of the triangle ABC opposite to the vertex A touch the side BC at E (see Figure 3). The homothety with centre A that takes ω_2 to ω_1 takes the point E to some point F, and the

tangent to ω_1 at F is parallel to BC. Therefore DF is a diameter of ω_1. Moreover, Z is the midpoint of DE. This implies that the lines IZ and FE are parallel.

Let $K = YI \cap AE$. Since $\angle YIA = 90^\circ$, the lemma yields that I is the midpoint of XK. This implies that the segments IW and AK are parallel. Therefore, the points W, I and Z are collinear.

Comment 1. The properties $\angle YIA = 90^\circ$ and $VA = VC$ can be established in various ways. The main difficulty of the problem seems to find out how to use these properties in connection to the points W and Z.

In Solution 2 this principal part is more or less covered by the lemma, for which we have presented a direct proof. On the other hand, this lemma appears to be a combination of two well-known facts; let us formulate them in terms of the lemma statement.

Let the line IY intersect AC at P (see Figure 4). The first fact states that the circumcircle ω of the triangle VYP is tangent to the segments AB and AC, as well as to the circle Γ. The second fact states that for such a circle, the angles BAV and CAE are equal.

The awareness of this lemma may help a lot in solving this problem; so the Jury might also consider a variation of the proposed problem, for which the lemma does not seem to be useful; see Comment 3.

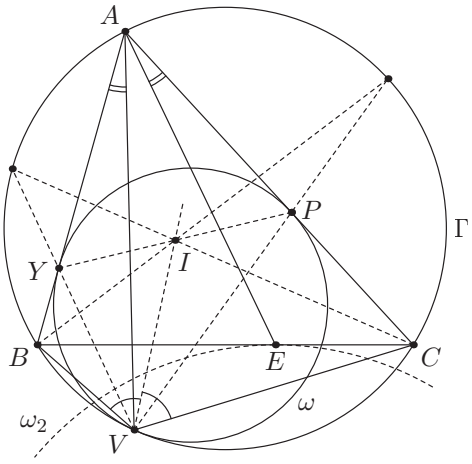


Figure 4

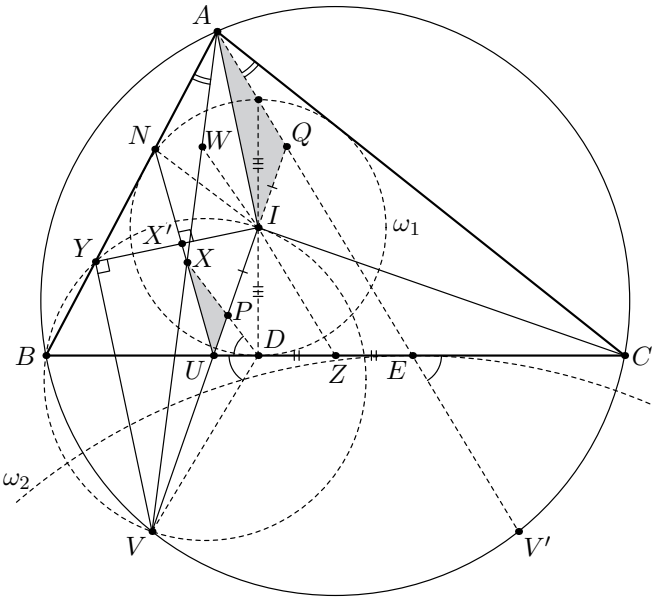


Figure 5

Comment 2. The proposed problem stated the equivalence: the point I lies on the line XY if and only if I lies on the line WZ. Here we sketch the proof of the "if" part (see Figure 5).

As in Solution 2, let BC touch the circles ω_1 and ω_2 at D and E, respectively. Since $IZ \parallel AE$ and W lies on IZ, the line DX is also parallel to AE. Therefore, the triangles XUP and AIQ are similar. Moreover, the line DX is symmetric to AE with respect to I, so $IP = IQ$, where $P = UV \cap XD$ and $Q = UV \cap AE$. Thus we obtain

$$\frac{UV}{VI} = \frac{UX}{IA} = \frac{UP}{IQ} = \frac{UP}{IP}.$$

So the pairs IU and PV are harmonic conjugates, and since $\angle UDI = 90^\circ$, we get $\angle VDB = \angle BDX = \angle BEA$. Therefore the point V' symmetric to V with respect to the perpendicular bisector of BC lies on the line AE. So we obtain $\angle BAV = \angle CAE$.

The rest can be obtained by simply reversing the arguments in Solution 2. The points B, V, I, and Y are concyclic. The lemma implies that $\angle YIA = 90^\circ$. Moreover, the points B, U, I, and N, where $N = UX \cap AB$, lie on a common circle, so $IN = IU$. Since $IY \perp UN$, the point $X' = IY \cap UN$ is the midpoint of UN. But in the trapezoid $AYVI$, the line XU is parallel to the sides AI and YV, so $NX = UX'$. This yields $X = X'$.

The reasoning presented in Solution 1 can also be reversed, but it requires a lot of technicalities. Therefore the Problem Selection Committee proposes to consider only the "only if" part of the original proposal, which is still challenging enough.

Comment 3. The Jury might also consider the following variation of the proposed problem.

Let ABC be a triangle with circumcircle Ω and incentre I. Let the line through I perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at U and V, respectively. Let the line through U parallel to AI intersect AV at X. Prove that if the lines XI and AI are perpendicular, then the midpoint of the segment AC lies on the line XI (see Figure 6).

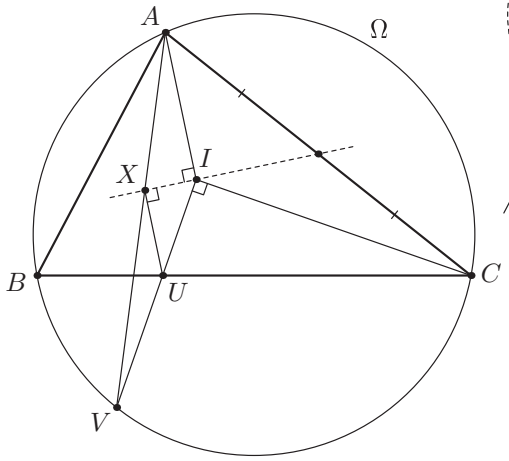


Figure 6

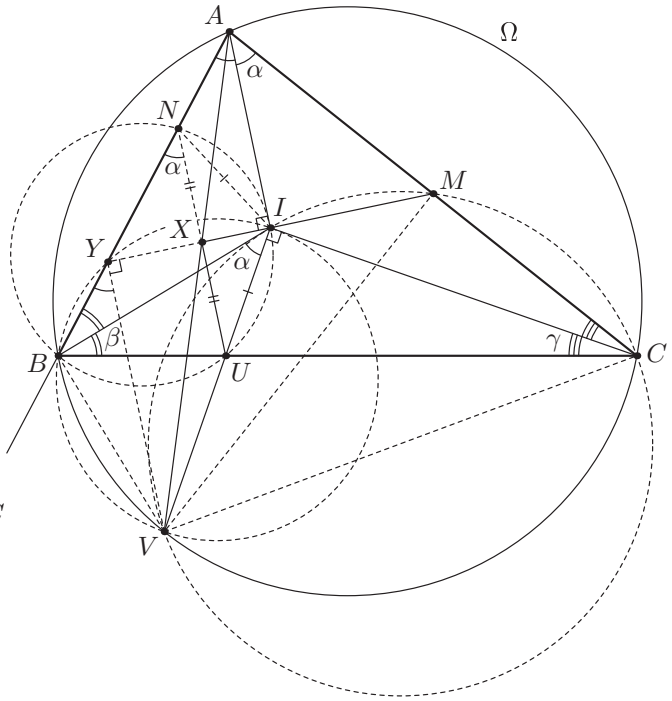


Figure 7

Since the solution contains the arguments used above, we only sketch it.

Let $N = XU \cap AB$ (see Figure 7). Then $\angle BNU = \angle BAI = \angle BIU$, so the points B, U, I, and N lie on a common circle. Therefore $IU = IN$, and since $IX \perp NU$, it follows that $NX = XU$.

Now set $Y = XI \cap AB$. The equality $NX = XU$ implies that

$$\frac{VX}{VA} = \frac{XU}{AI} = \frac{NX}{AI} = \frac{YX}{YI},$$

and therefore $YV \parallel AI$. Hence $\angle BYV = \angle BAI = \angle BIV$, so the points B, V, I, Y are concyclic. Next we have $IY \perp YV$, so $\angle IBV = 90^\circ$. This implies that BV is the external angle bisector of the angle ABC, which gives $\angle VAC = \angle VCA$.

So in order to show that $M = XI \cap AC$ is the midpoint of AC, it suffices to prove that $\angle VMC = 90^\circ$. But this follows immediately from the observation that the points V, C, M, and I are concyclic, as $\angle MIV = \angle YBV = 180^\circ - \angle ACV$.

The converse statement is also true, but its proof requires some technicalities as well.

Number Theory

N1. Let $n \geqslant 2$ be an integer, and let A_n be the set

$$A_n = \{2^n - 2^k \mid k \in \mathbb{Z},\ 0 \leqslant k < n\}.$$

Determine the largest positive integer that cannot be written as the sum of one or more (not necessarily distinct) elements of A_n.

(Serbia)

Answer. $(n-2)2^n + 1$.

Solution 1.

Part I. First we show that every integer greater than $(n-2)2^n + 1$ can be represented as such a sum. This is achieved by induction on n.

For $n = 2$, the set A_n consists of the two elements 2 and 3. Every positive integer m except for 1 can be represented as the sum of elements of A_n in this case: as $m = 2 + 2 + \cdots + 2$ if m is even, and as $m = 3 + 2 + 2 + \cdots + 2$ if m is odd.

Now consider some $n > 2$, and take an integer $m > (n-2)2^n + 1$. If m is even, then consider

$$\frac{m}{2} \geqslant \frac{(n-2)2^n + 2}{2} = (n-2)2^{n-1} + 1 > (n-3)2^{n-1} + 1.$$

By the induction hypothesis, there is a representation of the form

$$\frac{m}{2} = (2^{n-1} - 2^{k_1}) + (2^{n-1} - 2^{k_2}) + \cdots + (2^{n-1} - 2^{k_r})$$

for some k_i with $0 \leqslant k_i < n-1$. It follows that

$$m = (2^n - 2^{k_1+1}) + (2^n - 2^{k_2+1}) + \cdots + (2^n - 2^{k_r+1}),$$

giving us the desired representation as a sum of elements of A_n. If m is odd, we consider

$$\frac{m - (2^n - 1)}{2} > \frac{(n-2)2^n + 1 - (2^n - 1)}{2} = (n-3)2^{n-1} + 1.$$

By the induction hypothesis, there is a representation of the form

$$\frac{m - (2^n - 1)}{2} = (2^{n-1} - 2^{k_1}) + (2^{n-1} - 2^{k_2}) + \cdots + (2^{n-1} - 2^{k_r})$$

for some k_i with $0 \leqslant k_i < n-1$. It follows that

$$m = (2^n - 2^{k_1+1}) + (2^n - 2^{k_2+1}) + \cdots + (2^n - 2^{k_r+1}) + (2^n - 1),$$

giving us the desired representation of m once again.

Part II. It remains to show that there is no representation for $(n-2)2^n + 1$. Let N be the smallest positive integer that satisfies $N \equiv 1 \pmod{2^n}$, and which can be represented as a sum of elements of A_n. Consider a representation of N, i.e.,

$$N = (2^n - 2^{k_1}) + (2^n - 2^{k_2}) + \cdots + (2^n - 2^{k_r}), \tag{1}$$

where $0 \leqslant k_1, k_2, \ldots, k_r < n$. Suppose first that two of the terms in the sum are the same, i.e., $k_i = k_j$ for some $i \neq j$. If $k_i = k_j = n-1$, then we can simply remove these two terms to get a representation for

$$N - 2(2^n - 2^{n-1}) = N - 2^n$$

as a sum of elements of A_n, which contradicts our choice of N. If $k_i = k_j = k < n-1$, replace the two terms by $2^n - 2^{k+1}$, which is also an element of A_n, to get a representation for

$$N - 2(2^n - 2^k) + 2^n - 2^{k+1} = N - 2^n.$$

This is a contradiction once again. Therefore, all k_i have to be distinct, which means that

$$2^{k_1} + 2^{k_2} + \cdots + 2^{k_r} \leqslant 2^0 + 2^1 + 2^2 + \cdots + 2^{n-1} = 2^n - 1.$$

On the other hand, taking (1) modulo 2^n, we find

$$2^{k_1} + 2^{k_2} + \cdots + 2^{k_r} \equiv -N \equiv -1 \pmod{2^n}.$$

Thus we must have $2^{k_1} + 2^{k_2} + \cdots + 2^{k_r} = 2^n - 1$, which is only possible if each element of $\{0, 1, \ldots, n-1\}$ occurs as one of the k_i. This gives us

$$N = n2^n - (2^0 + 2^1 + \cdots + 2^{n-1}) = (n-1)2^n + 1.$$

In particular, this means that $(n-2)2^n + 1$ cannot be represented as a sum of elements of A_n.

Solution 2. The fact that $m = (n-2)2^n + 1$ cannot be represented as a sum of elements of A_n can also be shown in other ways. We prove the following statement by induction on n:

Claim. If a, b are integers with $a \geqslant 0$, $b \geqslant 1$, and $a + b < n$, then $a2^n + b$ cannot be written as a sum of elements of A_n.

Proof. The claim is clearly true for $n = 2$ (since $a = 0$, $b = 1$ is the only possibility). For $n > 2$, assume that there exist integers a, b with $a \geqslant 0$, $b \geqslant 1$ and $a + b < n$ as well as elements $m_1, m_2, \ldots, m_r$ of A_n such that

$$a2^n + b = m_1 + m_2 + \cdots + m_r.$$

We can suppose, without loss of generality, that $m_1 \geqslant m_2 \geqslant \cdots \geqslant m_r$. Let ℓ be the largest index for which $m_\ell = 2^n - 1$ ($\ell = 0$ if $m_1 \neq 2^n - 1$). Clearly, ℓ and b must have the same parity. Now

$$(a - \ell)2^n + (b + \ell) = m_{\ell+1} + m_{\ell+2} + \cdots + m_r$$

and thus

$$(a - \ell)2^{n-1} + \frac{b+\ell}{2} = \frac{m_{\ell+1}}{2} + \frac{m_{\ell+2}}{2} + \cdots + \frac{m_r}{2}.$$

Note that $m_{\ell+1}/2, m_{\ell+2}/2, \ldots, m_r/2$ are elements of A_{n-1}. Moreover, $a - \ell$ and $(b + \ell)/2$ are integers, and $(b + \ell)/2 \geqslant 1$. If $a - \ell$ was negative, then we would have

$$a2^n + b \geqslant \ell(2^n - 1) \geqslant (a + 1)(2^n - 1) = a2^n + 2^n - a - 1,$$

thus $n \geqslant a + b + 1 \geqslant 2^n$, which is impossible. So $a - \ell \geqslant 0$. By the induction hypothesis, we must have $a - \ell + \frac{b+\ell}{2} \geqslant n - 1$, which gives us a contradiction, since

$$a - \ell + \frac{b+\ell}{2} \leqslant a - \ell + b + \ell - 1 = a + b - 1 < n - 1. \qquad \square$$

Considering the special case $a = n - 2$, $b = 1$ now completes the proof.

Solution 3. Denote by B_n the set of all positive integers that can be written as a sum of elements of A_n. In this solution, we explicitly describe all the numbers in B_n by an argument similar to the first solution.

For a positive integer n, we denote by $\sigma_2(n)$ the sum of its digits in the binary representation. Notice that every positive integer m has a unique representation of the form $m = s2^n - t$ with some positive integer s and $0 \leqslant t \leqslant 2^n - 1$.

Lemma. For any two integers $s \geqslant 1$ and $0 \leqslant t \leqslant 2^n - 1$, the number $m = s2^n - t$ belongs to B_n if and only if $s \geqslant \sigma_2(t)$.

Proof. For $t = 0$, the statement of the Lemma is obvious, since $m = 2s \cdot (2^n - 2^{n-1})$.

Now suppose that $t \geqslant 1$, and let

$$t = 2^{k_1} + \cdots + 2^{k_\sigma} \qquad (0 \leqslant k_1 < \cdots < k_\sigma \leqslant n-1, \quad \sigma = \sigma_2(t))$$

be its binary expansion. If $s \geqslant \sigma$, then $m \in B_n$ since

$$m = (s-\sigma)2^n + (\sigma 2^n - t) = 2(s-\sigma) \cdot (2^n - 2^{n-1}) + \sum_{i=1}^{\sigma} (2^n - 2^{k_i}).$$

Assume now that there exist integers s and t with $1 \leqslant s < \sigma_2(t)$ and $0 \leqslant t \leqslant 2^n - 1$ such that the number $m = s2^n - t$ belongs to B_n. Among all such instances, choose the one for which m is smallest, and let

$$m = \sum_{i=1}^{d} (2^n - 2^{\ell_i}) \qquad (0 \leqslant \ell_i \leqslant n-1)$$

be the corresponding representation. If all the ℓ_i's are distinct, then $\sum_{i=1}^{d} 2^{\ell_i} \leqslant \sum_{j=0}^{n-1} 2^j = 2^n - 1$, so one has $s = d$ and $t = \sum_{i=1}^{d} 2^{\ell_i}$, whence $s = d = \sigma_2(t)$; this is impossible. Therefore, two of the ℓ_i's must be equal, say $\ell_{d-1} = \ell_d$. Then $m \geqslant 2(2^n - 2^{\ell_d}) \geqslant 2^n$, so $s \geqslant 2$.

Now we claim that the number $m' = m - 2^n = (s-1)2^n - t$ also belongs to B_n, which contradicts the minimality assumption. Indeed, one has

$$(2^n - 2^{\ell_{d-1}}) + (2^n - 2^{\ell_d}) = 2(2^n - 2^{\ell_d}) = 2^n + (2^n - 2^{\ell_d + 1}),$$

so

$$m' = \sum_{i=1}^{d-2} (2^n - 2^{\ell_i}) + (2^n - 2^{\ell_d + 1})$$

is the desired representation of m' (if $\ell_d = n-1$, then the last summand is simply omitted). This contradiction finishes the proof. □

By our lemma, the largest number M which does not belong to B_n must have the form

$$m_t = (\sigma_2(t) - 1)2^n - t$$

for some t with $1 \leqslant t \leqslant 2^n - 1$, so M is just the largest of these numbers. For $t_0 = 2^n - 1$ we have $m_{t_0} = (n-1)2^n - (2^n - 1) = (n-2)2^n + 1$; for every other value of t one has $\sigma_2(t) \leqslant n-1$, thus $m_t \leqslant (\sigma(t) - 1)2^n \leqslant (n-2)2^n < m_{t_0}$. This means that $M = m_{t_0} = (n-2)2^n + 1$.

N2. Determine all pairs (x, y) of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1 \,. \tag{1}$$

(U.S.A.)

Answer. Either $(x, y) = (1, 1)$ or $\{x, y\} = \{m^3 + m^2 - 2m - 1, m^3 + 2m^2 - m - 1\}$ for some positive integer $m \geqslant 2$.

Solution. Let (x, y) be any pair of positive integers solving (1). We shall prove that it appears in the list displayed above. The converse assertion that all these pairs do actually satisfy (1) either may be checked directly by means of a somewhat laborious calculation, or it can be seen by going in reverse order through the displayed equations that follow.

In case $x = y$ the given equation reduces to $x^{2/3} = 1$, which is equivalent to $x = 1$, whereby he have found the first solution.

To find the solutions with $x \neq y$ we may assume $x > y$ due to symmetry. Then the integer $n = x - y$ is positive and (1) may be rewritten as

$$\sqrt[3]{7(y + n)^2 - 13(y + n)y + 7y^2} = n + 1 \,.$$

Raising this to the third power and simplifying the result one obtains

$$y^2 + yn = n^3 - 4n^2 + 3n + 1 \,.$$

To complete the square on the left hand side, we multiply by 4 and add n^2, thus getting

$$(2y + n)^2 = 4n^3 - 15n^2 + 12n + 4 = (n - 2)^2(4n + 1) \,.$$

This shows that the cases $n = 1$ and $n = 2$ are impossible, whence $n > 2$, and $4n + 1$ is the square of the rational number $\frac{2y+n}{n-2}$. Consequently, it has to be a perfect square, and, since it is odd as well, there has to exist some nonnegative integer m such that $4n + 1 = (2m + 1)^2$, i.e.

$$n = m^2 + m \,.$$

Notice that $n > 2$ entails $m \geqslant 2$. Substituting the value of n just found into the previous displayed equation we arrive at

$$(2y + m^2 + m)^2 = (m^2 + m - 2)^2(2m + 1)^2 = (2m^3 + 3m^2 - 3m - 2)^2 \,.$$

Extracting square roots and taking $2m^3 + 3m^2 - 3m - 2 = (m - 1)(2m^2 + 5m + 2) > 0$ into account we derive $2y + m^2 + m = 2m^3 + 3m^2 - 3m - 2$, which in turn yields

$$y = m^3 + m^2 - 2m - 1 \,.$$

Notice that $m \geqslant 2$ implies that $y = (m^3 - 1) + (m - 2)m$ is indeed positive, as it should be. In view of $x = y + n = y + m^2 + m$ it also follows that

$$x = m^3 + 2m^2 - m - 1 \,,$$

and that this integer is positive as well.

Comment. Alternatively one could ask to find all pairs (x, y) of – not necessarily positive – integers solving (1). The answer to that question is a bit nicer than the answer above: the set of solutions are now described by

$$\{x, y\} = \{m^3 + m^2 - 2m - 1, m^3 + 2m^2 - m - 1\} \,,$$

where m varies through $\mathbb{Z}$. This may be shown using essentially the same arguments as above. We finally observe that the pair $(x, y) = (1, 1)$, that appears to be sporadic above, corresponds to $m = -1$.

N3. A coin is called a *Cape Town coin* if its value is $1/n$ for some positive integer n. Given a collection of Cape Town coins of total value at most $99 + \frac{1}{2}$, prove that it is possible to split this collection into at most 100 groups each of total value at most 1.

(Luxembourg)

Solution. We will show that for every positive integer N any collection of Cape Town coins of total value at most $N - \frac{1}{2}$ can be split into N groups each of total value at most 1. The problem statement is a particular case for $N = 100$.

We start with some preparations. If several given coins together have a total value also of the form $\frac{1}{k}$ for a positive integer k, then we may merge them into one new coin. Clearly, if the resulting collection can be split in the required way then the initial collection can also be split.

After each such merging, the total number of coins decreases, thus at some moment we come to a situation when no more merging is possible. At this moment, for every even k there is at most one coin of value $\frac{1}{k}$ (otherwise two such coins may be merged), and for every odd $k > 1$ there are at most $k - 1$ coins of value $\frac{1}{k}$ (otherwise k such coins may also be merged).

Now, clearly, each coin of value 1 should form a single group; if there are d such coins then we may remove them from the collection and replace N by $N - d$. So from now on we may assume that there are no coins of value 1.

Finally, we may split all the coins in the following way. For each $k = 1, 2, \ldots, N$ we put all the coins of values $\frac{1}{2k-1}$ and $\frac{1}{2k}$ into a group G_k; the total value of G_k does not exceed

$$(2k-2) \cdot \frac{1}{2k-1} + \frac{1}{2k} < 1.$$

It remains to distribute the "small" coins of values which are less than $\frac{1}{2N}$; we will add them one by one. In each step, take any remaining small coin. The total value of coins in the groups at this moment is at most $N - \frac{1}{2}$, so there exists a group of total value at most $\frac{1}{N}\left(N - \frac{1}{2}\right) = 1 - \frac{1}{2N}$; thus it is possible to put our small coin into this group. Acting so, we will finally distribute all the coins.

Comment 1. The algorithm may be modified, at least the step where one distributes the coins of values $\geqslant \frac{1}{2N}$. One different way is to put into G_k all the coins of values $\frac{1}{(2k-1)2^s}$ for all integer $s \geqslant 0$. One may easily see that their total value also does not exceed 1.

Comment 2. The original proposal also contained another part, suggesting to show that a required splitting may be impossible if the total value of coins is at most 100. There are many examples of such a collection, e.g. one may take 98 coins of value 1, one coin of value $\frac{1}{2}$, two coins of value $\frac{1}{3}$, and four coins of value $\frac{1}{5}$.

The Problem Selection Committee thinks that this part is less suitable for the competition.

N4. Let $n > 1$ be a given integer. Prove that infinitely many terms of the sequence $(a_k)_{k\geqslant 1}$, defined by

$$a_k = \left\lfloor \frac{n^k}{k} \right\rfloor,$$

are odd. (For a real number x, $\lfloor x \rfloor$ denotes the largest integer not exceeding x.)

(Hong Kong)

Solution 1. If n is odd, let $k = n^m$ for $m = 1, 2, \ldots$. Then $a_k = n^{n^m - m}$, which is odd for each m.

Henceforth, assume that n is even, say $n = 2t$ for some integer $t \geqslant 1$. Then, for any $m \geqslant 2$, the integer $n^{2^m} - 2^m = 2^m(2^{2^m - m} \cdot t^{2^m} - 1)$ has an odd prime divisor p, since $2^m - m > 1$. Then, for $k = p \cdot 2^m$, we have

$$n^k = (n^{2^m})^p \equiv (2^m)^p = (2^p)^m \equiv 2^m,$$

where the congruences are taken modulo p (recall that $2^p \equiv 2 \pmod p$, by Fermat's little theorem). Also, from $n^k - 2^m < n^k < n^k + 2^m(p-1)$, we see that the fraction $\dfrac{n^k}{k}$ lies strictly between the consecutive integers $\dfrac{n^k - 2^m}{p \cdot 2^m}$ and $\dfrac{n^k + 2^m(p-1)}{p \cdot 2^m}$, which gives

$$\left\lfloor \frac{n^k}{k} \right\rfloor = \frac{n^k - 2^m}{p \cdot 2^m}.$$

We finally observe that $\dfrac{n^k - 2^m}{p \cdot 2^m} = \dfrac{\frac{n^k}{2^m} - 1}{p}$ is an odd integer, since the integer $\dfrac{n^k}{2^m} - 1$ is odd (recall that $k > m$). Note that for different values of m, we get different values of k, due to the different powers of 2 in the prime factorisation of k.

Solution 2. Treat the (trivial) case when n is odd as in Solution 1.

Now assume that n is even and $n > 2$. Let p be a prime divisor of $n - 1$.

Proceed by induction on i to prove that p^{i+1} is a divisor of $n^{p^i} - 1$ for every $i \geqslant 0$. The case $i = 0$ is true by the way in which p is chosen. Suppose the result is true for some $i \geqslant 0$. The factorisation

$$n^{p^{i+1}} - 1 = (n^{p^i} - 1)[n^{p^i(p-1)} + n^{p^i(p-2)} + \cdots + n^{p^i} + 1],$$

together with the fact that each of the p terms between the square brackets is congruent to 1 modulo p, implies that the result is also true for $i + 1$.

Hence $\left\lfloor \dfrac{n^{p^i}}{p^i} \right\rfloor = \dfrac{n^{p^i} - 1}{p^i}$, an odd integer for each $i \geqslant 1$.

Finally, we consider the case $n = 2$. We observe that $3 \cdot 4^i$ is a divisor of $2^{3 \cdot 4^i} - 4^i$ for every $i \geqslant 1$: Trivially, 4^i is a divisor of $2^{3 \cdot 4^i} - 4^i$, since $3 \cdot 4^i > 2i$. Furthermore, since $2^{3 \cdot 4^i}$ and 4^i are both congruent to 1 modulo 3, we have $3 \mid 2^{3 \cdot 4^i} - 4^i$. Hence, $\left\lfloor \dfrac{2^{3 \cdot 4^i}}{3 \cdot 4^i} \right\rfloor = \dfrac{2^{3 \cdot 4^i} - 4^i}{3 \cdot 4^i} = \dfrac{2^{3 \cdot 4^i - 2i} - 1}{3}$, which is odd for every $i \geqslant 1$.

Comment. The case n even and $n > 2$ can also be solved by recursively defining the sequence $(k_i)_{i\geqslant 1}$ by $k_1 = 1$ and $k_{i+1} = n^{k_i} - 1$ for $i \geqslant 1$. Then (k_i) is strictly increasing and it follows (by induction on i) that $k_i \mid n^{k_i} - 1$ for all $i \geqslant 1$, so the k_i are as desired.

The case $n = 2$ can also be solved as follows: Let $i \geqslant 2$. By Bertrand's postulate, there exists a prime number p such that $2^{2^i - 1} < p \cdot 2^i < 2^{2^i}$. This gives

$$p \cdot 2^i < 2^{2^i} < 2p \cdot 2^i. \tag{1}$$

Also, we have that $p \cdot 2^i$ is a divisor of $2^{p \cdot 2^i} - 2^{2^i}$, hence, using (1), we get that

$$\left\lfloor \frac{2^{p \cdot 2^i}}{p \cdot 2^i} \right\rfloor = \frac{2^{p \cdot 2^i} - 2^{2^i} + p \cdot 2^i}{p \cdot 2^i} = \frac{2^{p \cdot 2^i - i} - 2^{2^i - i} + p}{p},$$

which is an odd integer.

Solution 3. Treat the (trivial) case when n is odd as in Solution 1.

Let n be even, and let p be a prime divisor of $n+1$. Define the sequence $(a_i)_{i \geqslant 1}$ by

$$a_i = \min\bigl\{a \in \mathbb{Z}_{>0} \colon 2^i \text{ divides } ap + 1\bigr\}.$$

Recall that there exists a with $1 \leqslant a < 2^i$ such that $ap \equiv -1 \pmod{2^i}$, so each a_i satisfies $1 \leqslant a_i < 2^i$. This implies that $a_i p + 1 < p \cdot 2^i$. Also, $a_i \to \infty$ as $i \to \infty$, whence there are infinitely many i such that $a_i < a_{i+1}$. From now on, we restrict ourselves only to these i.

Notice that p is a divisor of $n^p + 1$, which, in turn, divides $n^{p \cdot 2^i} - 1$. It follows that $p \cdot 2^i$ is a divisor of $n^{p \cdot 2^i} - (a_i p + 1)$, and we consequently see that the integer $\left\lfloor \dfrac{n^{p \cdot 2^i}}{p \cdot 2^i} \right\rfloor = \dfrac{n^{p \cdot 2^i} - (a_i p + 1)}{p \cdot 2^i}$ is odd, since 2^{i+1} divides $n^{p \cdot 2^i}$, but not $a_i p + 1$.

N5. Find all triples (p, x, y) consisting of a prime number p and two positive integers x and y such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p.

(Belgium)

Answer. $(p, x, y) \in \{(3, 2, 5), (3, 5, 2)\} \cup \{(2, n, 2^k - n) \mid 0 < n < 2^k\}$.

Solution 1. For $p = 2$, clearly all pairs of two positive integers x and y whose sum is a power of 2 satisfy the condition. Thus we assume in the following that $p > 2$, and we let a and b be positive integers such that $x^{p-1} + y = p^a$ and $x + y^{p-1} = p^b$. Assume further, without loss of generality, that $x \leqslant y$, so that $p^a = x^{p-1} + y \leqslant x + y^{p-1} = p^b$, which means that $a \leqslant b$ (and thus $p^a \mid p^b$).

Now we have

$$p^b = y^{p-1} + x = (p^a - x^{p-1})^{p-1} + x.$$

We take this equation modulo p^a and take into account that $p - 1$ is even, which gives us

$$0 \equiv x^{(p-1)^2} + x \pmod{p^a}.$$

If $p \mid x$, then $p^a \mid x$, since $x^{(p-1)^2-1} + 1$ is not divisible by p in this case. However, this is impossible, since $x \leqslant x^{p-1} < p^a$. Thus we know that $p \nmid x$, which means that

$$p^a \mid x^{(p-1)^2-1} + 1 = x^{p(p-2)} + 1.$$

By Fermat's little theorem, $x^{(p-1)^2} \equiv 1 \pmod{p}$, thus p divides $x+1$. Let p^r be the highest power of p that divides $x + 1$. By the binomial theorem, we have

$$x^{p(p-2)} = \sum_{k=0}^{p(p-2)} \binom{p(p-2)}{k} (-1)^{p(p-2)-k} (x+1)^k.$$

Except for the terms corresponding to $k = 0$, $k = 1$ and $k = 2$, all terms in the sum are clearly divisible by p^{3r} and thus by p^{r+2}. The remaining terms are

$$-\frac{p(p-2)(p^2-2p-1)}{2}(x+1)^2,$$

which is divisible by p^{2r+1} and thus also by p^{r+2},

$$p(p-2)(x+1),$$

which is divisible by p^{r+1}, but not p^{r+2} by our choice of r, and the final term -1 corresponding to $k = 0$. It follows that the highest power of p that divides $x^{p(p-2)} + 1$ is p^{r+1}.

On the other hand, we already know that p^a divides $x^{p(p-2)} + 1$, which means that $a \leqslant r + 1$. Moreover,

$$p^r \leqslant x + 1 \leqslant x^{p-1} + y = p^a.$$

Hence we either have $a = r$ or $a = r + 1$.

If $a = r$, then $x = y = 1$ needs to hold in the inequality above, which is impossible for $p > 2$. Thus $a = r + 1$. Now since $p^r \leqslant x + 1$, we get

$$x = \frac{x^2 + x}{x + 1} \leqslant \frac{x^{p-1} + y}{x + 1} = \frac{p^a}{x + 1} \leqslant \frac{p^a}{p^r} = p,$$

so we must have $x = p - 1$ for p to divide $x + 1$.

It follows that $r = 1$ and $a = 2$. If $p \geqslant 5$, we obtain

$$p^a = x^{p-1} + y > (p-1)^4 = (p^2 - 2p + 1)^2 > (3p)^2 > p^2 = p^a,$$

a contradiction. So the only case that remains is $p = 3$, and indeed $x = 2$ and $y = p^a - x^{p-1} = 5$ satisfy the conditions.

Comment 1. In this solution, we are implicitly using a special case of the following lemma known as "lifting the exponent":

Lemma. Let n be a positive integer, let p be an odd prime, and let $v_p(m)$ denote the exponent of the highest power of p that divides m.

If x and y are integers not divisible by p such that $p \mid x-y$, then we have

$$v_p(x^n - y^n) = v_p(x-y) + v_p(n).$$

Likewise, if x and y are integers not divisible by p such that $p \mid x+y$, then we have

$$v_p(x^n + y^n) = v_p(x+y) + v_p(n).$$

Comment 2. There exist various ways of solving the problem involving the "lifting the exponent" lemma. Let us sketch another one.

The cases $x = y$ and $p \mid x$ are ruled out easily, so we assume that $p > 2$, $x < y$, and $p \nmid x$. In this case we also have $p^a < p^b$ and $p \mid x+1$.

Now one has

$$y^p - x^p \equiv y(y^{p-1}+x) - x(x^{p-1}+y) \equiv 0 \pmod{p^a},$$

so by the lemma mentioned above one has $p^{a-1} \mid y-x$ and hence $y = x + tp^{a-1}$ for some positive integer t. Thus one gets

$$x(x^{p-2}+1) = x^{p-1} + x = (x^{p-1}+y) - (y-x) = p^{a-1}(p-t).$$

The factors on the left-hand side are coprime. So if $p \mid x$, then $x^{p-2}+1 \mid p-t$, which is impossible since $x < x^{p-2}+1$. Therefore, $p \nmid x$, and thus $x \mid p-t$. Since $p \mid x+1$, the only remaining case is $x = p-1$, $t = 1$, and $y = p^{a-1} + p - 1$. Now the solution can be completed in the same way as before.

Solution 2. Again, we can focus on the case that $p > 2$. If $p \mid x$, then also $p \mid y$. In this case, let p^k and p^ℓ be the highest powers of p that divide x and y respectively, and assume without loss of generality that $k \leqslant \ell$. Then p^k divides $x + y^{p-1}$ while p^{k+1} does not, but $p^k < x + y^{p-1}$, which yields a contradiction. So x and y are not divisible by p. Fermat's little theorem yields $0 \equiv x^{p-1} + y \equiv 1 + y \pmod{p}$, so $y \equiv -1 \pmod{p}$ and for the same reason $x \equiv -1 \pmod{p}$.

In particular, $x, y \geqslant p-1$ and thus $x^{p-1} + y \geqslant 2(p-1) > p$, so $x^{p-1} + y$ and $y^{p-1} + x$ are both at least equal to p^2. Now we have

$$x^{p-1} \equiv -y \pmod{p^2} \qquad \text{and} \qquad y^{p-1} \equiv -x \pmod{p^2}.$$

These two congruences, together with the Euler–Fermat theorem, give us

$$1 \equiv x^{p(p-1)} \equiv (-y)^p \equiv -y^p \equiv xy \pmod{p^2}.$$

Since $x \equiv y \equiv -1 \pmod{p}$, $x-y$ is divisible by p, so $(x-y)^2$ is divisible by p^2. This means that

$$(x+y)^2 = (x-y)^2 + 4xy \equiv 4 \pmod{p^2},$$

so p^2 divides $(x+y-2)(x+y+2)$. We already know that $x+y \equiv -2 \pmod{p}$, so $x+y-2 \equiv -4 \not\equiv 0 \pmod{p}$. This means that p^2 divides $x+y+2$.

Using the same notation as in the first solution, we subtract the two original equations to obtain

$$p^b - p^a = y^{p-1} - x^{p-1} + x - y = (y-x)(y^{p-2} + y^{p-3}x + \cdots + x^{p-2} - 1). \tag{1}$$

The second factor is symmetric in x and y, so it can be written as a polynomial of the elementary symmetric polynomials $x+y$ and xy with integer coefficients. In particular, its value modulo

p^2 is characterised by the two congruences $xy \equiv 1 \pmod{p^2}$ and $x + y \equiv -2 \pmod{p^2}$. Since both congruences are satisfied when $x = y = -1$, we must have

$$y^{p-2} + y^{p-3}x + \cdots + x^{p-2} - 1 \equiv (-1)^{p-2} + (-1)^{p-3}(-1) + \cdots + (-1)^{p-2} - 1 \pmod{p^2},$$

which simplifies to $y^{p-2} + y^{p-3}x + \cdots + x^{p-2} - 1 \equiv -p \pmod{p^2}$. Thus the second factor in (1) is divisible by p, but not p^2.

This means that p^{a-1} has to divide the other factor $y - x$. It follows that

$$0 \equiv x^{p-1} + y \equiv x^{p-1} + x \equiv x(x+1)(x^{p-3} - x^{p-4} + \cdots + 1) \pmod{p^{a-1}}.$$

Since $x \equiv -1 \pmod{p}$, the last factor is $x^{p-3} - x^{p-4} + \cdots + 1 \equiv p - 2 \pmod{p}$ and in particular not divisible by p. We infer that $p^{a-1} \mid x + 1$ and continue as in the first solution.

Comment. Instead of reasoning by means of elementary symmetric polynomials, it is possible to provide a more direct argument as well. For odd r, $(x+1)^2$ divides $(x^r+1)^2$, and since p divides $x + 1$, we deduce that p^2 divides $(x^r + 1)^2$. Together with the fact that $xy \equiv 1 \pmod{p^2}$, we obtain

$$0 \equiv y^r(x^r+1)^2 \equiv x^{2r}y^r + 2x^r y^r + y^r \equiv x^r + 2 + y^r \pmod{p^2}.$$

We apply this congruence with $r = p - 2 - 2k$ (where $0 \leqslant k < (p-2)/2$) to find that

$$x^k y^{p-2-k} + x^{p-2-k}y^k \equiv (xy)^k(x^{p-2-2k} + y^{p-2-2k}) \equiv 1^k \cdot (-2) \equiv -2 \pmod{p^2}.$$

Summing over all k yields

$$y^{p-2} + y^{p-3}x + \cdots + x^{p-2} - 1 \equiv \tfrac{p-1}{2} \cdot (-2) - 1 \equiv -p \pmod{p^2}$$

once again.

N6. Let $a_1 < a_2 < \cdots < a_n$ be pairwise coprime positive integers with a_1 being prime and $a_1 \geqslant n+2$. On the segment $I = [0, a_1 a_2 \cdots a_n]$ of the real line, mark all integers that are divisible by at least one of the numbers $a_1, \ldots, a_n$. These points split I into a number of smaller segments. Prove that the sum of the squares of the lengths of these segments is divisible by a_1.
(Serbia)

Solution 1. Let $A = a_1 \cdots a_n$. Throughout the solution, all intervals will be nonempty and have integer end-points. For any interval X, the length of X will be denoted by $|X|$.

Define the following two families of intervals:

$$\mathcal{S} = \big\{[x,y]\colon x < y \text{ are consecutive marked points}\big\}$$
$$\mathcal{T} = \big\{[x,y]\colon x < y \text{ are integers, } 0 \leqslant x \leqslant A-1, \text{ and no point is marked in } (x,y)\big\}$$

We are interested in computing $\sum\limits_{X\in\mathcal{S}} |X|^2$ modulo a_1.

Note that the number A is marked, so in the definition of $\mathcal{T}$ the condition $y \leqslant A$ is enforced without explicitly prescribing it.

Assign weights to the intervals in $\mathcal{T}$, depending only on their lengths. The weight of an arbitrary interval $Y \in \mathcal{T}$ will be $w\big(|Y|\big)$, where

$$w(k) = \begin{cases} 1 & \text{if } k = 1, \\ 2 & \text{if } k \geqslant 2\,. \end{cases}$$

Consider an arbitrary interval $X \in \mathcal{S}$ and its sub-intervals $Y \in \mathcal{T}$. Clearly, X has one sub-interval of length $|X|$, two sub-intervals of length $|X|-1$ and so on; in general X has $|X|-d+1$ sub-intervals of length d for every $d = 1, 2, \ldots, |X|$. The sum of the weights of the sub-intervals of X is

$$\sum_{Y\in\mathcal{T},\, Y\subseteq X} w\big(|Y|\big) = \sum_{d=1}^{|X|} (|X|-d+1)\cdot w(d) = |X|\cdot 1 + \big((|X|-1)+(|X|-2)+\cdots+1\big)\cdot 2 = |X|^2.$$

Since the intervals in $\mathcal{S}$ are non-overlapping, every interval $Y \in \mathcal{T}$ is a sub-interval of a single interval $X \in \mathcal{S}$. Therefore,

$$\sum_{X\in\mathcal{S}} |X|^2 = \sum_{X\in\mathcal{S}} \left(\sum_{Y\in\mathcal{T},\, Y\subseteq X} w\big(|Y|\big) \right) = \sum_{Y\in\mathcal{T}} w\big(|Y|\big). \tag{1}$$

For every $d = 1, 2, \ldots, a_1$, we count how many intervals in $\mathcal{T}$ are of length d. Notice that the multiples of a_1 are all marked, so the lengths of the intervals in $\mathcal{S}$ and $\mathcal{T}$ cannot exceed a_1. Let x be an arbitrary integer with $0 \leqslant x \leqslant A-1$ and consider the interval $[x, x+d]$. Let r_1, $\ldots$, r_n be the remainders of x modulo a_1, $\ldots$, a_n, respectively. Since $a_1, \ldots, a_n$ are pairwise coprime, the number x is uniquely identified by the sequence $(r_1, \ldots, r_n)$, due to the Chinese remainder theorem.

For every $i = 1, \ldots, n$, the property that the interval $(x, x+d)$ does not contain any multiple of a_i is equivalent with $r_i + d \leqslant a_i$, i.e. $r_i \in \{0, 1, \ldots, a_i - d\}$, so there are $a_i - d + 1$ choices for the number r_i for each i. Therefore, the number of the remainder sequences $(r_1, \ldots, r_n)$ that satisfy $[x, x+d] \in \mathcal{T}$ is precisely $(a_1 + 1 - d)\cdots(a_n + 1 - d)$. Denote this product by $f(d)$.

Now we can group the last sum in (1) by length of the intervals. As we have seen, for every $d = 1, \ldots, a_1$ there are $f(d)$ intervals $Y \in \mathcal{T}$ with $|Y| = d$. Therefore, (1) can be continued as

$$\sum_{X\in\mathcal{S}} |X|^2 = \sum_{Y\in\mathcal{T}} w\big(|Y|\big) = \sum_{d=1}^{a_1} f(d)\cdot w(d) = 2\sum_{d=1}^{a_1} f(d) - f(1). \tag{2}$$

Having the formula (2), the solution can be finished using the following well-known fact:

Lemma. If p is a prime, $F(x)$ is a polynomial with integer coefficients, and $\deg F \leqslant p-2$, then $\sum\limits_{x=1}^{p} F(x)$ is divisible by p.

Proof. Obviously, it is sufficient to prove the lemma for monomials of the form x^k with $k \leqslant p-2$. Apply induction on k. If $k = 0$ then $F = 1$, and the statement is trivial.

Let $1 \leqslant k \leqslant p-2$, and assume that the lemma is proved for all lower degrees. Then

$$\begin{aligned} 0 \equiv p^{k+1} &= \sum_{x=1}^{p}\big(x^{k+1} - (x-1)^{k+1}\big) = \sum_{x=1}^{p}\left(\sum_{\ell=0}^{k}(-1)^{k-\ell}\binom{k+1}{\ell}x^\ell\right) \\ &= (k+1)\sum_{x=1}^{p} x^k + \sum_{\ell=0}^{k-1}(-1)^{k-\ell}\binom{k+1}{\ell}\sum_{x=1}^{p} x^\ell \equiv (k+1)\sum_{x=1}^{p} x^k \pmod p. \end{aligned}$$

Since $0 < k+1 < p$, this proves $\sum\limits_{x=1}^{p} x^k \equiv 0 \pmod p$. □

In (2), by applying the lemma to the polynomial f and the prime a_1, we obtain that $\sum\limits_{d=1}^{a_1} f(d)$ is divisible by a_1. The term $f(1) = a_1 \cdots a_n$ is also divisible by a_1; these two facts together prove that $\sum\limits_{X\in\mathcal{S}} |X|^2$ is divisible by a_1.

Comment 1. With suitable sets of weights, the same method can be used to sum up other expressions on the lengths of the segments. For example, $w(1) = 1$ and $w(k) = 6(k-1)$ for $k \geqslant 2$ can be used to compute $\sum\limits_{X\in\mathcal{S}} |X|^3$ and to prove that this sum is divisible by a_1 if a_1 is a prime with $a_1 \geqslant n+3$. See also Comment 2 after the second solution.

Solution 2. The conventions from the first paragraph of the first solution are still in force. We shall prove the following more general statement:

> (⊞) *Let p denote a prime number, let $p = a_1 < a_2 < \cdots < a_n$ be n pairwise coprime positive integers, and let d be an integer with $1 \leqslant d \leqslant p-n$. Mark all integers that are divisible by at least one of the numbers $a_1, \ldots, a_n$ on the interval $I = [0, a_1a_2\cdots a_n]$ of the real line. These points split I into a number of smaller segments, say of lengths $b_1, \ldots, b_k$. Then the sum $\sum\limits_{i=1}^{k}\binom{b_i}{d}$ is divisible by p.*

Applying (⊞) to $d = 1$ and $d = 2$ and using the equation $x^2 = 2\binom{x}{2} + \binom{x}{1}$, one easily gets the statement of the problem.

To prove (⊞) itself, we argue by induction on n. The base case $n = 1$ follows from the known fact that the binomial coefficient $\binom{p}{d}$ is divisible by p whenever $1 \leqslant d \leqslant p-1$.

Let us now assume that $n \geqslant 2$, and that the statement is known whenever $n-1$ rather than n coprime integers are given together with some integer $d \in [1, p-n+1]$. Suppose that

the numbers $p = a_1 < a_2 < \cdots < a_n$ and d are as above. Write $A' = \prod_{i=1}^{n-1} a_i$ and $A = A'\,a_n$. Mark the points on the real axis divisible by one of the numbers $a_1, \ldots, a_{n-1}$ *green* and those divisible by a_n *red*. The green points divide $[0, A']$ into certain sub-intervals, say J_1, J_2, $\ldots$, and J_ℓ.

To translate intervals we use the notation $[a, b] + m = [a + m, b + m]$ whenever $a, b, m \in \mathbb{Z}$.

For each $i \in \{1, 2, \ldots, \ell\}$ let $\mathcal{F}_i$ be the family of intervals into which the red points partition the intervals J_i, $J_i + A'$, $\ldots$, and $J_i + (a_n - 1)A'$. We are to prove that

$$\sum_{i=1}^{\ell} \sum_{X \in \mathcal{F}_i} \binom{|X|}{d}$$

is divisible by p.

Let us fix any index i with $1 \leqslant i \leqslant \ell$ for a while. Since the numbers A' and a_n are coprime by hypothesis, the numbers $0, A', \ldots, (a_n - 1)A'$ form a complete system of residues modulo a_n. Moreover, we have $|J_i| \leqslant p < a_n$, as in particular all multiples of p are green. So each of the intervals J_i, $J_i + A'$, $\ldots$, and $J_i + (a_n - 1)A'$ contains at most one red point. More precisely, for each $j \in \{1, \ldots, |J_i| - 1\}$ there is exactly one amongst those intervals containing a red point splitting it into an interval of length j followed by an interval of length $|J_i| - j$, while the remaining $a_n - |J_i| + 1$ such intervals have no red points in their interiors. For these reasons

$$\begin{aligned}
\sum_{X \in \mathcal{F}_i} \binom{|X|}{d} &= 2\left(\binom{1}{d} + \cdots + \binom{|J_i| - 1}{d}\right) + (a_n - |J_i| + 1)\binom{|J_i|}{d} \\
&= 2\binom{|J_i|}{d+1} + (a_n - d + 1)\binom{|J_i|}{d} - (d+1)\binom{|J_i|}{d+1} \\
&= (1 - d)\binom{|J_i|}{d+1} + (a_n - d + 1)\binom{|J_i|}{d}.
\end{aligned}$$

So it remains to prove that

$$(1 - d)\sum_{i=1}^{\ell} \binom{|J_i|}{d+1} + (a_n - d + 1)\sum_{i=1}^{\ell} \binom{|J_i|}{d}$$

is divisible by p. By the induction hypothesis, however, it is even true that both summands are divisible by p, for $1 \leqslant d < d + 1 \leqslant p - (n - 1)$. This completes the proof of ($\boxplus$) and hence the solution of the problem.

Comment 2. The statement ($\boxplus$) can also be proved by the method of the first solution, using the weights $w(x) = \binom{x-2}{d-2}$.

This page is intentionally left blank

N7. Let $c \geqslant 1$ be an integer. Define a sequence of positive integers by $a_1 = c$ and

$$a_{n+1} = a_n^3 - 4c \cdot a_n^2 + 5c^2 \cdot a_n + c$$

for all $n \geqslant 1$. Prove that for each integer $n \geqslant 2$ there exists a prime number p dividing a_n but none of the numbers $a_1, \ldots, a_{n-1}$.

(Austria)

Solution. Let us define $x_0 = 0$ and $x_n = a_n/c$ for all integers $n \geqslant 1$. It is easy to see that the sequence (x_n) thus obtained obeys the recursive law

$$x_{n+1} = c^2(x_n^3 - 4x_n^2 + 5x_n) + 1 \tag{1}$$

for all integers $n \geqslant 0$. In particular, all of its terms are positive integers; notice that $x_1 = 1$ and $x_2 = 2c^2 + 1$. Since

$$x_{n+1} = c^2 x_n (x_n - 2)^2 + c^2 x_n + 1 > x_n \tag{2}$$

holds for all integers $n \geqslant 0$, it is also strictly increasing. Since x_{n+1} is by (1) coprime to c for any $n \geqslant 0$, it suffices to prove that for each $n \geqslant 2$ there exists a prime number p dividing x_n but none of the numbers $x_1, \ldots, x_{n-1}$. Let us begin by establishing three preliminary claims.

Claim 1. If $i \equiv j \pmod{m}$ holds for some integers $i, j \geqslant 0$ and $m \geqslant 1$, then $x_i \equiv x_j \pmod{x_m}$ holds as well.

Proof. Evidently, it suffices to show $x_{i+m} \equiv x_i \pmod{x_m}$ for all integers $i \geqslant 0$ and $m \geqslant 1$. For this purpose we may argue for fixed m by induction on i using $x_0 = 0$ in the base case $i = 0$. Now, if we have $x_{i+m} \equiv x_i \pmod{x_m}$ for some integer i, then the recursive equation (1) yields

$$x_{i+m+1} \equiv c^2(x_{i+m}^3 - 4x_{i+m}^2 + 5x_{i+m}) + 1 \equiv c^2(x_i^3 - 4x_i^2 + 5x_i) + 1 \equiv x_{i+1} \pmod{x_m},$$

which completes the induction. □

Claim 2. If the integers $i, j \geqslant 2$ and $m \geqslant 1$ satisfy $i \equiv j \pmod{m}$, then $x_i \equiv x_j \pmod{x_m^2}$ holds as well.

Proof. Again it suffices to prove $x_{i+m} \equiv x_i \pmod{x_m^2}$ for all integers $i \geqslant 2$ and $m \geqslant 1$. As above, we proceed for fixed m by induction on i. The induction step is again easy using (1), but this time the base case $i = 2$ requires some calculation. Set $L = 5c^2$. By (1) we have $x_{m+1} \equiv Lx_m + 1 \pmod{x_m^2}$, and hence

$$\begin{aligned} x_{m+1}^3 - 4x_{m+1}^2 + 5x_{m+1} &\equiv (Lx_m + 1)^3 - 4(Lx_m + 1)^2 + 5(Lx_m + 1) \\ &\equiv (3Lx_m + 1) - 4(2Lx_m + 1) + 5(Lx_m + 1) \equiv 2 \pmod{x_m^2}, \end{aligned}$$

which in turn gives indeed $x_{m+2} \equiv 2c^2 + 1 \equiv x_2 \pmod{x_m^2}$. □

Claim 3. For each integer $n \geqslant 2$, we have $x_n > x_1 \cdot x_2 \cdots x_{n-2}$.

Proof. The cases $n = 2$ and $n = 3$ are clear. Arguing inductively, we assume now that the claim holds for some $n \geqslant 3$. Recall that $x_2 \geqslant 3$, so by monotonicity and (2) we get $x_n \geqslant x_3 \geqslant x_2(x_2 - 2)^2 + x_2 + 1 \geqslant 7$. It follows that

$$x_{n+1} > x_n^3 - 4x_n^2 + 5x_n > 7x_n^2 - 4x_n^2 > x_n^2 > x_n x_{n-1},$$

which by the induction hypothesis yields $x_{n+1} > x_1 \cdot x_2 \cdots x_{n-1}$, as desired. □

Now we direct our attention to the problem itself: let any integer $n \geqslant 2$ be given. By Claim 3 there exists a prime number p appearing with a higher exponent in the prime factorisation of x_n than in the prime factorisation of $x_1 \cdots x_{n-2}$. In particular, $p \mid x_n$, and it suffices to prove that p divides none of $x_1, \ldots, x_{n-1}$.

Otherwise let $k \in \{1, \ldots, n-1\}$ be minimal such that p divides x_k. Since x_{n-1} and x_n are coprime by (1) and $x_1 = 1$, we actually have $2 \leqslant k \leqslant n-2$. Write $n = qk + r$ with some integers $q \geqslant 0$ and $0 \leqslant r < k$. By Claim 1 we have $x_n \equiv x_r \pmod{x_k}$, whence $p \mid x_r$. Due to the minimality of k this entails $r = 0$, i.e. $k \mid n$.

Thus from Claim 2 we infer

$$x_n \equiv x_k \pmod{x_k^2}.$$

Now let $\alpha \geqslant 1$ be maximal with the property $p^\alpha \mid x_k$. Then x_k^2 is divisible by $p^{\alpha+1}$ and by our choice of p so is x_n. So by the previous congruence x_k is a multiple of $p^{\alpha+1}$ as well, contrary to our choice of α. This is the final contradiction concluding the solution.

N8. For every real number x, let $\|x\|$ denote the distance between x and the nearest integer. Prove that for every pair (a, b) of positive integers there exist an odd prime p and a positive integer k satisfying

$$\left\|\frac{a}{p^k}\right\| + \left\|\frac{b}{p^k}\right\| + \left\|\frac{a+b}{p^k}\right\| = 1. \tag{1}$$

(Hungary)

Solution. Notice first that $\left\lfloor x + \frac{1}{2}\right\rfloor$ is an integer nearest to x, so $\|x\| = \left|\left\lfloor x + \frac{1}{2}\right\rfloor - x\right|$. Thus we have

$$\left\lfloor x + \frac{1}{2}\right\rfloor = x \pm \|x\|. \tag{2}$$

For every rational number r and every prime number p, denote by $v_p(r)$ the exponent of p in the prime factorisation of r. Recall the notation $(2n-1)!!$ for the product of all odd positive integers not exceeding $2n-1$, i.e., $(2n-1)!! = 1 \cdot 3 \cdots (2n-1)$.

Lemma. For every positive integer n and every odd prime p, we have

$$v_p\big((2n-1)!!\big) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} + \frac{1}{2}\right\rfloor.$$

Proof. For every positive integer k, let us count the multiples of p^k among the factors $1, 3, \ldots, 2n-1$. If ℓ is an arbitrary integer, the number $(2\ell-1)p^k$ is listed above if and only if

$$0 < (2\ell-1)p^k \leqslant 2n \quad\Longleftrightarrow\quad \frac{1}{2} < \ell \leqslant \frac{n}{p^k} + \frac{1}{2} \quad\Longleftrightarrow\quad 1 \leqslant \ell \leqslant \left\lfloor \frac{n}{p^k} + \frac{1}{2}\right\rfloor.$$

Hence, the number of multiples of p^k among the factors is precisely $m_k = \left\lfloor \frac{n}{p^k} + \frac{1}{2}\right\rfloor$. Thus we obtain

$$v_p\big((2n-1)!!\big) = \sum_{i=1}^{n} v_p(2i-1) = \sum_{i=1}^{n} \sum_{k=1}^{v_p(2i-1)} 1 = \sum_{k=1}^{\infty} \sum_{\ell=1}^{m_k} 1 = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} + \frac{1}{2}\right\rfloor. \qquad \square$$

In order to prove the problem statement, consider the rational number

$$N = \frac{(2a+2b-1)!!}{(2a-1)!! \cdot (2b-1)!!} = \frac{(2a+1)(2a+3)\cdots(2a+2b-1)}{1 \cdot 3 \cdots (2b-1)}.$$

Obviously, $N > 1$, so there exists a prime p with $v_p(N) > 0$. Since N is a fraction of two odd numbers, p is odd.

By our lemma,

$$0 < v_p(N) = \sum_{k=1}^{\infty} \left(\left\lfloor \frac{a+b}{p^k} + \frac{1}{2}\right\rfloor - \left\lfloor \frac{a}{p^k} + \frac{1}{2}\right\rfloor - \left\lfloor \frac{b}{p^k} + \frac{1}{2}\right\rfloor \right).$$

Therefore, there exists some positive integer k such that the integer number

$$d_k = \left\lfloor \frac{a+b}{p^k} + \frac{1}{2}\right\rfloor - \left\lfloor \frac{a}{p^k} + \frac{1}{2}\right\rfloor - \left\lfloor \frac{b}{p^k} + \frac{1}{2}\right\rfloor$$

is positive, so $d_k \geqslant 1$. By (2) we have

$$1 \leqslant d_k = \frac{a+b}{p^k} - \frac{a}{p^k} - \frac{b}{p^k} \pm \left\|\frac{a+b}{p^k}\right\| \pm \left\|\frac{a}{p^k}\right\| \pm \left\|\frac{b}{p^k}\right\| = \pm \left\|\frac{a+b}{p^k}\right\| \pm \left\|\frac{a}{p^k}\right\| \pm \left\|\frac{b}{p^k}\right\|. \tag{3}$$

Since $\|x\| < \frac{1}{2}$ for every rational x with odd denominator, the relation (3) can only be satisfied if all three signs on the right-hand side are positive and $d_k = 1$. Thus we get

$$\left\|\frac{a}{p^k}\right\| + \left\|\frac{b}{p^k}\right\| + \left\|\frac{a+b}{p^k}\right\| = d_k = 1,$$

as required.

Comment 1. There are various choices for the number N in the solution. Here we sketch such a version.

Let x and y be two rational numbers with odd denominators. It is easy to see that the condition $\|x\| + \|y\| + \|x+y\| = 1$ is satisfied if and only if

$$\text{either} \quad \{x\} < \tfrac{1}{2}, \quad \{y\} < \tfrac{1}{2}, \quad \{x+y\} > \tfrac{1}{2}, \qquad \text{or} \quad \{x\} > \tfrac{1}{2}, \quad \{y\} > \tfrac{1}{2}, \quad \{x+y\} < \tfrac{1}{2},$$

where $\{x\}$ denotes the fractional part of x.

In the context of our problem, the first condition seems easier to deal with. Also, one may notice that

$$\{x\} < \tfrac{1}{2} \iff \varkappa(x) = 0 \qquad \text{and} \qquad \{x\} \geqslant \tfrac{1}{2} \iff \varkappa(x) = 1, \tag{4}$$

where

$$\varkappa(x) = \lfloor 2x \rfloor - 2\lfloor x \rfloor.$$

Now it is natural to consider the number

$$M = \frac{\binom{2a+2b}{a+b}}{\binom{2a}{a}\binom{2b}{b}},$$

since

$$v_p(M) = \sum_{k=1}^{\infty} \left(\varkappa\left(\frac{2(a+b)}{p^k}\right) - \varkappa\left(\frac{2a}{p^k}\right) - \varkappa\left(\frac{2b}{p^k}\right) \right).$$

One may see that $M > 1$, and that $v_2(M) \leqslant 0$. Thus, there exist an odd prime p and a positive integer k with

$$\varkappa\left(\frac{2(a+b)}{p^k}\right) - \varkappa\left(\frac{2a}{p^k}\right) - \varkappa\left(\frac{2b}{p^k}\right) > 0.$$

In view of (4), the last inequality yields

$$\left\{\frac{a}{p^k}\right\} < \frac{1}{2}, \quad \left\{\frac{b}{p^k}\right\} < \frac{1}{2}, \quad \text{and} \quad \left\{\frac{a+b}{p^k}\right\} > \frac{1}{2}, \tag{5}$$

which is what we wanted to obtain.

Comment 2. Once one tries to prove the existence of suitable p and k satisfying (5), it seems somehow natural to suppose that $a \leqslant b$ and to add the restriction $p^k > a$. In this case the inequalities (5) can be rewritten as

$$2a < p^k, \quad 2mp^k < 2b < (2m+1)p^k, \quad \text{and} \quad (2m+1)p^k < 2(a+b) < (2m+2)p^k$$

for some positive integer m. This means exactly that one of the numbers $2a+1, 2a+3, \ldots, 2a+2b-1$ is divisible by some number of the form p^k which is greater than $2a$.

Using more advanced techniques, one can show that such a number p^k exists even with $k = 1$. This was shown in 2004 by Laishram and Shorey; the methods used for this proof are elementary but still quite involved. In fact, their result generalises a theorem by Sylvester which states that for every pair of integers (n, k) with $n \geqslant k \geqslant 1$, the product $(n+1)(n+2)\cdots(n+k)$ is divisible by some prime $p > k$. We would like to mention here that Sylvester's theorem itself does not seem to suffice for solving the problem.

Shortlisted Problems with Solutions

54th International Mathematical Olympiad

Santa Marta, Colombia 2013

Note of Confidentiality

The Shortlisted Problems should be kept strictly confidential until IMO 2014.

Contributing Countries

The Organizing Committee and the Problem Selection Committee of IMO 2013 thank the following 50 countries for contributing 149 problem proposals.

Argentina, Armenia, Australia, Austria, Belgium, Belarus, Brazil, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, El Salvador, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, India, Indonesia, Iran, Ireland, Israel, Italy, Japan, Latvia, Lithuania, Luxembourg, Malaysia, Mexico, Netherlands, Nicaragua, Pakistan, Panama, Poland, Romania, Russia, Saudi Arabia, Serbia, Slovenia, Sweden, Switzerland, Tajikistan, Thailand, Turkey, U.S.A., Ukraine, United Kingdom

Problem Selection Committee

Federico Ardila (chairman)
Ilya I. Bogdanov
Géza Kós
Carlos Gustavo Tamm de Araújo Moreira (Gugu)
Christian Reiher

Problems

Algebra

A1. Let n be a positive integer and let $a_1, \ldots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \ldots, u_n$ and $v_0, \ldots, v_n$ inductively by $u_0 = u_1 = v_0 = v_1 = 1$, and

$$u_{k+1} = u_k + a_k u_{k-1}, \qquad v_{k+1} = v_k + a_{n-k} v_{k-1} \qquad \text{for } k = 1, \ldots, n-1.$$

Prove that $u_n = v_n$.

(France)

A2. Prove that in any set of 2000 distinct real numbers there exist two pairs $a > b$ and $c > d$ with $a \neq c$ or $b \neq d$, such that

$$\left| \frac{a-b}{c-d} - 1 \right| < \frac{1}{100000}.$$

(Lithuania)

A3. Let $\mathbb{Q}_{>0}$ be the set of positive rational numbers. Let $f\colon \mathbb{Q}_{>0} \to \mathbb{R}$ be a function satisfying the conditions

$$f(x)f(y) \geqslant f(xy) \quad \text{and} \quad f(x+y) \geqslant f(x) + f(y)$$

for all $x, y \in \mathbb{Q}_{>0}$. Given that $f(a) = a$ for some rational $a > 1$, prove that $f(x) = x$ for all $x \in \mathbb{Q}_{>0}$.

(Bulgaria)

A4. Let n be a positive integer, and consider a sequence $a_1, a_2, \ldots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \ldots$ by defining $a_{n+i} = a_i$ for all $i \geqslant 1$. If

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n \leqslant a_1 + n$$

and

$$a_{a_i} \leqslant n + i - 1 \qquad \text{for } i = 1, 2, \ldots, n,$$

prove that

$$a_1 + \cdots + a_n \leqslant n^2.$$

(Germany)

A5. Let $\mathbb{Z}_{\geqslant 0}$ be the set of all nonnegative integers. Find all the functions $f\colon \mathbb{Z}_{\geqslant 0} \to \mathbb{Z}_{\geqslant 0}$ satisfying the relation

$$f(f(f(n))) = f(n+1) + 1$$

for all $n \in \mathbb{Z}_{\geqslant 0}$.

(Serbia)

A6. Let $m \neq 0$ be an integer. Find all polynomials $P(x)$ with real coefficients such that

$$(x^3 - mx^2 + 1)P(x+1) + (x^3 + mx^2 + 1)P(x-1) = 2(x^3 - mx + 1)P(x)$$

for all real numbers x.

(Serbia)

Combinatorics

C1. Let n be a positive integer. Find the smallest integer k with the following property: Given any real numbers $a_1, \ldots, a_d$ such that $a_1 + a_2 + \cdots + a_d = n$ and $0 \leqslant a_i \leqslant 1$ for $i = 1, 2, \ldots, d$, it is possible to partition these numbers into k groups (some of which may be empty) such that the sum of the numbers in each group is at most 1.

(Poland)

C2. In the plane, 2013 red points and 2014 blue points are marked so that no three of the marked points are collinear. One needs to draw k lines not passing through the marked points and dividing the plane into several regions. The goal is to do it in such a way that no region contains points of both colors.

Find the minimal value of k such that the goal is attainable for every possible configuration of 4027 points.

(Australia)

C3. A crazy physicist discovered a new kind of particle which he called an *imon*, after some of them mysteriously appeared in his lab. Some pairs of imons in the lab can be *entangled*, and each imon can participate in many entanglement relations. The physicist has found a way to perform the following two kinds of operations with these particles, one operation at a time.

(i) If some imon is entangled with an odd number of other imons in the lab, then the physicist can destroy it.

(ii) At any moment, he may double the whole family of imons in his lab by creating a copy I' of each imon I. During this procedure, the two copies I' and J' become entangled if and only if the original imons I and J are entangled, and each copy I' becomes entangled with its original imon I; no other entanglements occur or disappear at this moment.

Prove that the physicist may apply a sequence of such operations resulting in a family of imons, no two of which are entangled.

(Japan)

C4. Let n be a positive integer, and let A be a subset of $\{1, \ldots, n\}$. An *A-partition of n into k parts* is a representation of n as a sum $n = a_1 + \cdots + a_k$, where the parts $a_1, \ldots, a_k$ belong to A and are not necessarily distinct. The *number of different parts* in such a partition is the number of (distinct) elements in the set $\{a_1, a_2, \ldots, a_k\}$.

We say that an A-partition of n into k parts is *optimal* if there is no A-partition of n into r parts with $r < k$. Prove that any optimal A-partition of n contains at most $\sqrt[3]{6n}$ different parts.

(Germany)

C5. Let r be a positive integer, and let $a_0, a_1, \ldots$ be an infinite sequence of real numbers. Assume that for all nonnegative integers m and s there exists a positive integer $n \in [m+1, m+r]$ such that

$$a_m + a_{m+1} + \cdots + a_{m+s} = a_n + a_{n+1} + \cdots + a_{n+s}.$$

Prove that the sequence is periodic, i. e. there exists some $p \geqslant 1$ such that $a_{n+p} = a_n$ for all $n \geqslant 0$.

(India)

C6. In some country several pairs of cities are connected by direct two-way flights. It is possible to go from any city to any other by a sequence of flights. The *distance* between two cities is defined to be the least possible number of flights required to go from one of them to the other. It is known that for any city there are at most 100 cities at distance exactly three from it. Prove that there is no city such that more than 2550 other cities have distance exactly four from it.

(Russia)

C7. Let $n \geqslant 2$ be an integer. Consider all circular arrangements of the numbers $0, 1, \ldots, n$; the $n + 1$ rotations of an arrangement are considered to be equal. A circular arrangement is called *beautiful* if, for any four distinct numbers $0 \leqslant a, b, c, d \leqslant n$ with $a + c = b + d$, the chord joining numbers a and c does not intersect the chord joining numbers b and d.

Let M be the number of beautiful arrangements of $0, 1, \ldots, n$. Let N be the number of pairs (x, y) of positive integers such that $x + y \leqslant n$ and $\gcd(x, y) = 1$. Prove that

$$M = N + 1.$$

(Russia)

C8. Players A and B play a paintful game on the real line. Player A has a pot of paint with four units of black ink. A quantity p of this ink suffices to blacken a (closed) real interval of length p. In every round, player A picks some positive integer m and provides $1/2^m$ units of ink from the pot. Player B then picks an integer k and blackens the interval from $k/2^m$ to $(k + 1)/2^m$ (some parts of this interval may have been blackened before). The goal of player A is to reach a situation where the pot is empty and the interval $[0, 1]$ is not completely blackened.

Decide whether there exists a strategy for player A to win in a finite number of moves.

(Austria)

Geometry

G1. Let ABC be an acute-angled triangle with orthocenter H, and let W be a point on side BC. Denote by M and N the feet of the altitudes from B and C, respectively. Denote by ω_1 the circumcircle of BWN, and let X be the point on ω_1 which is diametrically opposite to W. Analogously, denote by ω_2 the circumcircle of CWM, and let Y be the point on ω_2 which is diametrically opposite to W. Prove that X, Y and H are collinear.

(Thaliand)

G2. Let ω be the circumcircle of a triangle ABC. Denote by M and N the midpoints of the sides AB and AC, respectively, and denote by T the midpoint of the arc BC of ω not containing A. The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y, respectively; assume that X and Y lie inside the triangle ABC. The lines MN and XY intersect at K. Prove that $KA = KT$.

(Iran)

G3. In a triangle ABC, let D and E be the feet of the angle bisectors of angles A and B, respectively. A rhombus is inscribed into the quadrilateral $AEDB$ (all vertices of the rhombus lie on different sides of $AEDB$). Let φ be the non-obtuse angle of the rhombus. Prove that $\varphi \leqslant \max\{\angle BAC, \angle ABC\}$.

(Serbia)

G4. Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C. Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.

(Georgia)

G5. Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD, BE, and CF are concurrent.

(Ukraine)

G6. Let the excircle of the triangle ABC lying opposite to A touch its side BC at the point A_1. Define the points B_1 and C_1 analogously. Suppose that the circumcentre of the triangle $A_1B_1C_1$ lies on the circumcircle of the triangle ABC. Prove that the triangle ABC is right-angled.

(Russia)

Number Theory

N1. Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all positive integers m and n.

(Malaysia)

N2. Prove that for any pair of positive integers k and n there exist k positive integers $m_1, m_2, \ldots, m_k$ such that

$$1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right)\left(1 + \frac{1}{m_2}\right)\cdots\left(1 + \frac{1}{m_k}\right).$$

(Japan)

N3. Prove that there exist infinitely many positive integers n such that the largest prime divisor of $n^4 + n^2 + 1$ is equal to the largest prime divisor of $(n+1)^4 + (n+1)^2 + 1$.

(Belgium)

N4. Determine whether there exists an infinite sequence of nonzero digits $a_1, a_2, a_3, \ldots$ and a positive integer N such that for every integer $k > N$, the number $\overline{a_k a_{k-1} \ldots a_1}$ is a perfect square.

(Iran)

N5. Fix an integer $k \geqslant 2$. Two players, called Ana and Banana, play the following *game of numbers*: Initially, some integer $n \geqslant k$ gets written on the blackboard. Then they take moves in turn, with Ana beginning. A player making a move erases the number m just written on the blackboard and replaces it by some number m' with $k \leqslant m' < m$ that is coprime to m. The first player who cannot move anymore loses.

An integer $n \geqslant k$ is called *good* if Banana has a winning strategy when the initial number is n, and *bad* otherwise.

Consider two integers $n, n' \geqslant k$ with the property that each prime number $p \leqslant k$ divides n if and only if it divides n'. Prove that either both n and n' are good or both are bad.

(Italy)

N6. Determine all functions $f\colon \mathbb{Q} \longrightarrow \mathbb{Z}$ satisfying

$$f\left(\frac{f(x) + a}{b}\right) = f\left(\frac{x + a}{b}\right)$$

for all $x \in \mathbb{Q}$, $a \in \mathbb{Z}$, and $b \in \mathbb{Z}_{>0}$. (Here, $\mathbb{Z}_{>0}$ denotes the set of positive integers.)

(Israel)

N7. Let ν be an irrational positive number, and let m be a positive integer. A pair (a, b) of positive integers is called *good* if

$$a\lceil b\nu \rceil - b\lfloor a\nu \rfloor = m.$$

A good pair (a, b) is called *excellent* if neither of the pairs $(a-b, b)$ and $(a, b-a)$ is good. (As usual, by $\lfloor x \rfloor$ and $\lceil x \rceil$ we denote the integer numbers such that $x - 1 < \lfloor x \rfloor \leqslant x$ and $x \leqslant \lceil x \rceil < x + 1$.)

Prove that the number of excellent pairs is equal to the sum of the positive divisors of m.

(U.S.A.)

Solutions

Algebra

A1. Let n be a positive integer and let $a_1, \ldots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \ldots, u_n$ and $v_0, \ldots, v_n$ inductively by $u_0 = u_1 = v_0 = v_1 = 1$, and

$$u_{k+1} = u_k + a_k u_{k-1}, \qquad v_{k+1} = v_k + a_{n-k} v_{k-1} \qquad \text{for } k = 1, \ldots, n-1.$$

Prove that $u_n = v_n$.

(France)

Solution 1. We prove by induction on k that

$$u_k = \sum_{\substack{0<i_1<\ldots<i_t<k,\\ i_{j+1}-i_j \geqslant 2}} a_{i_1} \ldots a_{i_t}. \tag{1}$$

Note that we have one trivial summand equal to 1 (which corresponds to $t = 0$ and the empty sequence, whose product is 1).

For $k = 0, 1$ the sum on the right-hand side only contains the empty product, so (1) holds due to $u_0 = u_1 = 1$. For $k \geqslant 1$, assuming the result is true for $0, 1, \ldots, k$, we have

$$\begin{aligned} u_{k+1} &= \sum_{\substack{0<i_1<\ldots<i_t<k,\\ i_{j+1}-i_j \geqslant 2}} a_{i_1} \ldots a_{i_t} + \sum_{\substack{0<i_1<\ldots i_t<k-1,\\ i_{j+1}-i_j \geqslant 2}} a_{i_1} \ldots a_{i_t} \cdot a_k \\ &= \sum_{\substack{0<i_1<\ldots<i_t<k+1,\\ i_{j+1}-i_j \geqslant 2,\\ k \notin \{i_1,\ldots,i_t\}}} a_{i_1} \ldots a_{i_t} + \sum_{\substack{0<i_1<\ldots<i_t<k+1,\\ i_{j+1}-i_j \geqslant 2,\\ k \in \{i_1,\ldots,i_t\}}} a_{i_1} \ldots a_{i_t} \\ &= \sum_{\substack{0<i_1<\ldots<i_t<k+1,\\ i_{j+1}-i_j \geqslant 2}} a_{i_1} \ldots a_{i_t}, \end{aligned}$$

as required.

Applying (1) to the sequence $b_1, \ldots, b_n$ given by $b_k = a_{n-k}$ for $1 \leqslant k \leqslant n$, we get

$$v_k = \sum_{\substack{0<i_1<\ldots<i_t<k,\\ i_{j+1}-i_j \geqslant 2}} b_{i_1} \ldots b_{i_t} = \sum_{\substack{n>i_1>\ldots>i_t>n-k,\\ i_j-i_{j+1} \geqslant 2}} a_{i_1} \ldots a_{i_t}. \tag{2}$$

For $k = n$ the expressions (1) and (2) coincide, so indeed $u_n = v_n$.

Solution 2. Define recursively a sequence of multivariate polynomials by

$$P_0 = P_1 = 1, \qquad P_{k+1}(x_1, \ldots, x_k) = P_k(x_1, \ldots, x_{k-1}) + x_k P_{k-1}(x_1, \ldots, x_{k-2}),$$

so P_n is a polynomial in $n-1$ variables for each $n \geqslant 1$. Two easy inductive arguments show that

$$u_n = P_n(a_1, \ldots, a_{n-1}), \qquad v_n = P_n(a_{n-1}, \ldots, a_1),$$

so we need to prove $P_n(x_1,\dots,x_{n-1}) = P_n(x_{n-1},\dots,x_1)$ for every positive integer n. The cases $n = 1, 2$ are trivial, and the cases $n = 3, 4$ follow from $P_3(x,y) = 1 + x + y$ and $P_4(x,y,z) = 1 + x + y + z + xz$.

Now we proceed by induction, assuming that $n \geqslant 5$ and the claim hold for all smaller cases. Using $F(a,b)$ as an abbreviation for $P_{|a-b|+1}(x_a,\dots,x_b)$ (where the indices $a,\dots,b$ can be either in increasing or decreasing order),

$$\begin{aligned} F(n,1) &= F(n,2) + x_1F(n,3) = F(2,n) + x_1F(3,n) \\ &= (F(2,n-1) + x_nF(2,n-2)) + x_1(F(3,n-1) + x_nF(3,n-2)) \\ &= (F(n-1,2) + x_1F(n-1,3)) + x_n(F(n-2,2) + x_1F(n-2,3)) \\ &= F(n-1,1) + x_nF(n-2,1) = F(1,n-1) + x_nF(1,n-2) \\ &= F(1,n), \end{aligned}$$

as we wished to show.

Solution 3. Using matrix notation, we can rewrite the recurrence relation as

$$\begin{pmatrix} u_{k+1} \\ u_{k+1} - u_k \end{pmatrix} = \begin{pmatrix} u_k + a_k u_{k-1} \\ a_k u_{k-1} \end{pmatrix} = \begin{pmatrix} 1 + a_k & -a_k \\ a_k & -a_k \end{pmatrix} \begin{pmatrix} u_k \\ u_k - u_{k-1} \end{pmatrix}$$

for $1 \leqslant k \leqslant n-1$, and similarly

$$(v_{k+1}; v_k - v_{k+1}) = \left(v_k + a_{n-k}v_{k-1}; -a_{n-k}v_{k-1}\right) = (v_k; v_{k-1} - v_k) \begin{pmatrix} 1 + a_{n-k} & -a_{n-k} \\ a_{n-k} & -a_{n-k} \end{pmatrix}$$

for $1 \leqslant k \leqslant n-1$. Hence, introducing the 2×2 matrices $A_k = \begin{pmatrix} 1 + a_k & -a_k \\ a_k & -a_k \end{pmatrix}$ we have

$$\begin{pmatrix} u_{k+1} \\ u_{k+1} - u_k \end{pmatrix} = A_k \begin{pmatrix} u_k \\ u_k - u_{k-1} \end{pmatrix} \quad \text{and} \quad (v_{k+1}; v_k - v_{k+1}) = (v_k; v_{k-1} - v_k) A_{n-k}.$$

for $1 \leqslant k \leqslant n-1$. Since $\binom{u_1}{u_1 - u_0} = \binom{1}{0}$ and $(v_1; v_0 - v_1) = (1; 0)$, we get

$$\begin{pmatrix} u_n \\ u_n - u_{n-1} \end{pmatrix} = A_{n-1}A_{n-2}\cdots A_1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad (v_n; v_{n-1} - v_n) = (1; 0) \cdot A_{n-1}A_{n-2}\cdots A_1.$$

It follows that

$$(u_n) = (1;0) \begin{pmatrix} u_n \\ u_n - u_{n-1} \end{pmatrix} = (1;0) \cdot A_{n-1}A_{n-2}\cdots A_1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (v_n; v_{n-1} - v_n) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (v_n).$$

Comment 1. These sequences are related to the Fibonacci sequence; when $a_1 = \cdots = a_{n-1} = 1$, we have $u_k = v_k = F_{k+1}$, the $(k+1)$st Fibonacci number. Also, for every positive integer k, the polynomial $P_k(x_1,\dots,x_{k-1})$ from Solution 2 is the sum of F_{k+1} monomials.

Comment 2. One may notice that the condition is equivalent to

$$\frac{u_{k+1}}{u_k} = 1 + \cfrac{a_k}{1 + \cfrac{a_{k-1}}{1 + \ldots + \cfrac{a_2}{1 + a_1}}} \qquad \text{and} \qquad \frac{v_{k+1}}{v_k} = 1 + \cfrac{a_{n-k}}{1 + \cfrac{a_{n-k+1}}{1 + \ldots + \cfrac{a_{n-2}}{1 + a_{n-1}}}}$$

so the problem claims that the corresponding continued fractions for u_n/u_{n-1} and v_n/v_{n-1} have the same numerator.

Comment 3. An alternative variant of the problem is the following.

Let n be a positive integer and let $a_1, \ldots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \ldots, u_n$ and $v_0, \ldots, v_n$ inductively by $u_0 = v_0 = 0$, $u_1 = v_1 = 1$, and

$$u_{k+1} = a_k u_k + u_{k-1}, \qquad v_{k+1} = a_{n-k} v_k + v_{k-1} \qquad \text{for } k = 1, \ldots, n-1.$$

Prove that $u_n = v_n$.

All three solutions above can be reformulated to prove this statement; one may prove

$$u_n = v_n = \sum_{\substack{0=i_0<i_1<\ldots<i_t=n,\\ i_{j+1}-i_j \text{ is odd}}} a_{i_1} \ldots a_{i_{t-1}} \qquad \text{for } n > 0$$

or observe that

$$\begin{pmatrix} u_{k+1} \\ u_k \end{pmatrix} = \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_k \\ u_{k-1} \end{pmatrix} \quad \text{and} \quad (v_{k+1}; v_k) = (v_k; v_{k-1}) \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix}.$$

Here we have

$$\frac{u_{k+1}}{u_k} = a_k + \cfrac{1}{a_{k-1} + \cfrac{1}{a_{k-2} + \ldots + \cfrac{1}{a_1}}} = [a_k; a_{k-1}, \ldots, a_1]$$

and

$$\frac{v_{k+1}}{v_k} = a_{n-k} + \cfrac{1}{a_{n-k+1} + \cfrac{1}{a_{n-k+2} + \ldots + \cfrac{1}{a_{n-1}}}} = [a_{n-k}; a_{n-k+1}, \ldots, a_{n-1}],$$

so this alternative statement is equivalent to the known fact that the continued fractions $[a_{n-1}; a_{n-2}, \ldots, a_1]$ and $[a_1; a_2, \ldots, a_{n-1}]$ have the same numerator.

A2. Prove that in any set of 2000 distinct real numbers there exist two pairs $a > b$ and $c > d$ with $a \neq c$ or $b \neq d$, such that

$$\left|\frac{a-b}{c-d} - 1\right| < \frac{1}{100000}.$$

(Lithuania)

Solution. For any set S of $n = 2000$ distinct real numbers, let $D_1 \leqslant D_2 \leqslant \cdots \leqslant D_m$ be the distances between them, displayed with their multiplicities. Here $m = n(n-1)/2$. By rescaling the numbers, we may assume that the smallest distance D_1 between two elements of S is $D_1 = 1$. Let $D_1 = 1 = y - x$ for $x, y \in S$. Evidently $D_m = v - u$ is the difference between the largest element v and the smallest element u of S.

If $D_{i+1}/D_i < 1 + 10^{-5}$ for some $i = 1, 2, \ldots, m-1$ then the required inequality holds, because $0 \leqslant D_{i+1}/D_i - 1 < 10^{-5}$. Otherwise, the reverse inequality

$$\frac{D_{i+1}}{D_i} \geqslant 1 + \frac{1}{10^5}$$

holds for each $i = 1, 2, \ldots, m-1$, and therefore

$$v - u = D_m = \frac{D_m}{D_1} = \frac{D_m}{D_{m-1}} \cdots \frac{D_3}{D_2} \cdot \frac{D_2}{D_1} \geqslant \left(1 + \frac{1}{10^5}\right)^{m-1}.$$

From $m - 1 = n(n-1)/2 - 1 = 1000 \cdot 1999 - 1 > 19 \cdot 10^5$, together with the fact that for all $n \geqslant 1$, $\left(1 + \frac{1}{n}\right)^n \geqslant 1 + \binom{n}{1} \cdot \frac{1}{n} = 2$, we get

$$\left(1 + \frac{1}{10^5}\right)^{19 \cdot 10^5} = \left(\left(1 + \frac{1}{10^5}\right)^{10^5}\right)^{19} \geqslant 2^{19} = 2^9 \cdot 2^{10} > 500 \cdot 1000 > 2 \cdot 10^5,$$

and so $v - u = D_m > 2 \cdot 10^5$.

Since the distance of x to at least one of the numbers u, v is at least $(u - v)/2 > 10^5$, we have

$$|x - z| > 10^5.$$

for some $z \in \{u, v\}$. Since $y - x = 1$, we have either $z > y > x$ (if $z = v$) or $y > x > z$ (if $z = u$). If $z > y > x$, selecting $a = z, b = y, c = z$ and $d = x$ (so that $b \neq d$), we obtain

$$\left|\frac{a-b}{c-d} - 1\right| = \left|\frac{z-y}{z-x} - 1\right| = \left|\frac{x-y}{z-x}\right| = \frac{1}{z-x} < 10^{-5}.$$

Otherwise, if $y > x > z$, we may choose $a = y, b = z, c = x$ and $d = z$ (so that $a \neq c$), and obtain

$$\left|\frac{a-b}{c-d} - 1\right| = \left|\frac{y-z}{x-z} - 1\right| = \left|\frac{y-x}{x-z}\right| = \frac{1}{x-z} < 10^{-5}.$$

The desired result follows.

Comment. As the solution shows, the numbers 2000 and $\frac{1}{100000}$ appearing in the statement of the problem may be replaced by any $n \in \mathbb{Z}_{>0}$ and $\delta > 0$ satisfying

$$\delta(1 + \delta)^{n(n-1)/2 - 1} > 2.$$

A3. Let $\mathbb{Q}_{>0}$ be the set of positive rational numbers. Let $f\colon \mathbb{Q}_{>0} \to \mathbb{R}$ be a function satisfying the conditions

$$f(x)f(y) \geqslant f(xy), \tag{1}$$
$$f(x+y) \geqslant f(x) + f(y) \tag{2}$$

for all $x, y \in \mathbb{Q}_{>0}$. Given that $f(a) = a$ for some rational $a > 1$, prove that $f(x) = x$ for all $x \in \mathbb{Q}_{>0}$.

(Bulgaria)

Solution. Denote by $\mathbb{Z}_{>0}$ the set of positive integers.

Plugging $x = 1$, $y = a$ into (1) we get $f(1) \geqslant 1$. Next, by an easy induction on n we get from (2) that

$$f(nx) \geqslant nf(x) \quad \text{for all } n \in \mathbb{Z}_{>0} \text{ and } x \in \mathbb{Q}_{>0}. \tag{3}$$

In particular, we have

$$f(n) \geqslant nf(1) \geqslant n \quad \text{for all } n \in \mathbb{Z}_{>0}. \tag{4}$$

From (1) again we have $f(m/n)f(n) \geqslant f(m)$, so $f(q) > 0$ for all $q \in \mathbb{Q}_{>0}$.

Now, (2) implies that f is strictly increasing; this fact together with (4) yields

$$f(x) \geqslant f(\lfloor x \rfloor) \geqslant \lfloor x \rfloor > x - 1 \quad \text{for all } x \geqslant 1.$$

By an easy induction we get from (1) that $f(x)^n \geqslant f(x^n)$, so

$$f(x)^n \geqslant f(x^n) > x^n - 1 \quad \Longrightarrow \quad f(x) \geqslant \sqrt[n]{x^n - 1} \quad \text{for all } x > 1 \text{ and } n \in \mathbb{Z}_{>0}.$$

This yields

$$f(x) \geqslant x \quad \text{for every } x > 1. \tag{5}$$

(Indeed, if $x > y > 1$ then $x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + \cdots + y^n) > n(x-y)$, so for a large n we have $x^n - 1 > y^n$ and thus $f(x) > y$.)

Now, (1) and (5) give $a^n = f(a)^n \geqslant f(a^n) \geqslant a^n$, so $f(a^n) = a^n$. Now, for $x > 1$ let us choose $n \in \mathbb{Z}_{>0}$ such that $a^n - x > 1$. Then by (2) and (5) we get

$$a^n = f(a^n) \geqslant f(x) + f(a^n - x) \geqslant x + (a^n - x) = a^n$$

and therefore $f(x) = x$ for $x > 1$. Finally, for every $x \in \mathbb{Q}_{>0}$ and every $n \in \mathbb{Z}_{>0}$, from (1) and (3) we get

$$nf(x) = f(n)f(x) \geqslant f(nx) \geqslant nf(x),$$

which gives $f(nx) = nf(x)$. Therefore $f(m/n) = f(m)/n = m/n$ for all $m, n \in \mathbb{Z}_{>0}$.

Comment. The condition $f(a) = a > 1$ is essential. Indeed, for $b \geqslant 1$ the function $f(x) = bx^2$ satisfies (1) and (2) for all $x, y \in \mathbb{Q}_{>0}$, and it has a unique fixed point $1/b \leqslant 1$.

A4. Let n be a positive integer, and consider a sequence $a_1, a_2, \ldots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \ldots$ by defining $a_{n+i} = a_i$ for all $i \geqslant 1$. If

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n \leqslant a_1 + n \tag{1}$$

and

$$a_{a_i} \leqslant n + i - 1 \qquad \text{for } i = 1, 2, \ldots, n, \tag{2}$$

prove that

$$a_1 + \cdots + a_n \leqslant n^2.$$

(Germany)

Solution 1. First, we claim that

$$a_i \leqslant n + i - 1 \qquad \text{for } i = 1, 2, \ldots, n. \tag{3}$$

Assume contrariwise that i is the smallest counterexample. From $a_n \geqslant a_{n-1} \geqslant \cdots \geqslant a_i \geqslant n + i$ and $a_{a_i} \leqslant n + i - 1$, taking into account the periodicity of our sequence, it follows that

$$a_i \text{ cannot be congruent to } i, i+1, \ldots, n-1, \text{or } n \pmod{n}. \tag{4}$$

Thus our assumption that $a_i \geqslant n + i$ implies the stronger statement that $a_i \geqslant 2n + 1$, which by $a_1 + n \geqslant a_n \geqslant a_i$ gives $a_1 \geqslant n + 1$. The minimality of i then yields $i = 1$, and (4) becomes contradictory. This establishes our first claim.

In particular we now know that $a_1 \leqslant n$. If $a_n \leqslant n$, then $a_1 \leqslant \cdots \leqslant \cdots a_n \leqslant n$ and the desired inequality holds trivially. Otherwise, consider the number t with $1 \leqslant t \leqslant n - 1$ such that

$$a_1 \leqslant a_2 \leqslant \ldots \leqslant a_t \leqslant n < a_{t+1} \leqslant \ldots \leqslant a_n. \tag{5}$$

Since $1 \leqslant a_1 \leqslant n$ and $a_{a_1} \leqslant n$ by (2), we have $a_1 \leqslant t$ and hence $a_n \leqslant n + t$. Therefore if for each positive integer i we let b_i be the number of indices $j \in \{t+1, \ldots, n\}$ satisfying $a_j \geqslant n + i$, we have

$$b_1 \geqslant b_2 \geqslant \ldots \geqslant b_t \geqslant b_{t+1} = 0.$$

Next we claim that $a_i + b_i \leqslant n$ for $1 \leqslant i \leqslant t$. Indeed, by $n + i - 1 \geqslant a_{a_i}$ and $a_i \leqslant n$, each j with $a_j \geqslant n + i$ (thus $a_j > a_{a_i}$) belongs to $\{a_i + 1, \ldots, n\}$, and for this reason $b_i \leqslant n - a_i$.

It follows from the definition of the b_is and (5) that

$$a_{t+1} + \ldots + a_n \leqslant n(n - t) + b_1 + \ldots + b_t.$$

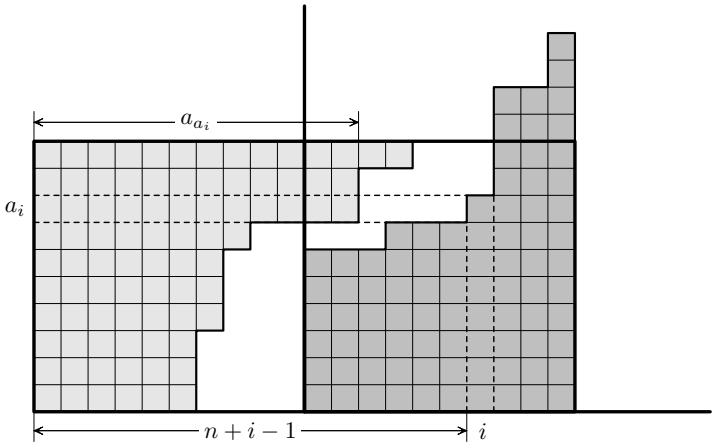
Adding $a_1 + \ldots + a_t$ to both sides and using that $a_i + b_i \leqslant n$ for $1 \leqslant i \leqslant t$, we get

$$a_1 + a_2 + \cdots + a_n \leqslant n(n - t) + nt = n^2$$

as we wished to prove.

Solution 2. In the first quadrant of an infinite grid, consider the increasing "staircase" obtained by shading in dark the bottom a_i cells of the ith column for $1 \leqslant i \leqslant n$. We will prove that there are at most n^2 dark cells.

To do it, consider the $n \times n$ square S in the first quadrant with a vertex at the origin. Also consider the $n \times n$ square directly to the left of S. Starting from its lower left corner, shade in light the leftmost a_j cells of the jth row for $1 \leqslant j \leqslant n$. Equivalently, the light shading is obtained by reflecting the dark shading across the line $x = y$ and translating it n units to the left. The figure below illustrates this construction for the sequence $6, 6, 6, 7, 7, 7, 8, 12, 12, 14$.



We claim that there is no cell in S which is both dark and light. Assume, contrariwise, that there is such a cell in column i. Consider the highest dark cell in column i which is inside S. Since it is above a light cell and inside S, it must be light as well. There are two cases:

Case 1. $a_i \leqslant n$

If $a_i \leqslant n$ then this dark and light cell is (i, a_i), as highlighted in the figure. However, this is the $(n+i)$-th cell in row a_i, and we only shaded $a_{a_i} < n + i$ light cells in that row, a contradiction.

Case 2. $a_i \geqslant n + 1$

If $a_i \geqslant n + 1$, this dark and light cell is (i, n). This is the $(n+i)$-th cell in row n and we shaded $a_n \leqslant a_1 + n$ light cells in this row, so we must have $i \leqslant a_1$. But $a_1 \leqslant a_{a_1} \leqslant n$ by (1) and (2), so $i \leqslant a_1$ implies $a_i \leqslant a_{a_1} \leqslant n$, contradicting our assumption.

We conclude that there are no cells in S which are both dark and light. It follows that the number of shaded cells in S is at most n^2.

Finally, observe that if we had a light cell to the right of S, then by symmetry we would have a dark cell above S, and then the cell (n, n) would be dark and light. It follows that the number of light cells in S equals the number of dark cells outside of S, and therefore the number of shaded cells in S equals $a_1 + \cdots + a_n$. The desired result follows.

Solution 3. As in Solution 1, we first establish that $a_i \leqslant n + i - 1$ for $1 \leqslant i \leqslant n$. Now define $c_i = \max(a_i, i)$ for $1 \leqslant i \leqslant n$ and extend the sequence $c_1, c_2, \ldots$ periodically modulo n. We claim that this sequence also satisfies the conditions of the problem.

For $1 \leqslant i < j \leqslant n$ we have $a_i \leqslant a_j$ and $i < j$, so $c_i \leqslant c_j$. Also $a_n \leqslant a_1 + n$ and $n < 1 + n$ imply $c_n \leqslant c_1 + n$. Finally, the definitions imply that $c_{c_i} \in \{a_{a_i}, a_i, a_i - n, i\}$ so $c_{c_i} \leqslant n + i - 1$ by (2) and (3). This establishes (1) and (2) for $c_1, c_2, \ldots$.

Our new sequence has the additional property that

$$c_i \geqslant i \qquad \text{for } i = 1, 2, \ldots, n, \tag{6}$$

which allows us to construct the following visualization: Consider n equally spaced points on a circle, sequentially labelled 1, 2, $\ldots, n \pmod n$, so point k is also labelled $n + k$. We draw arrows from vertex i to vertices $i + 1, \ldots, c_i$ for $1 \leqslant i \leqslant n$, keeping in mind that $c_i \geqslant i$ by (6). Since $c_i \leqslant n + i - 1$ by (3), no arrow will be drawn twice, and there is no arrow from a vertex to itself. The total number of arrows is

$$\text{number of arrows } = \sum_{i=1}^{n}(c_i - i) = \sum_{i=1}^{n} c_i - \binom{n+1}{2}$$

Now we show that we never draw both arrows $i \to j$ and $j \to i$ for $1 \leqslant i < j \leqslant n$. Assume contrariwise. This means, respectively, that

$$i < j \leqslant c_i \qquad \text{and} \qquad j < n + i \leqslant c_j.$$

We have $n + i \leqslant c_j \leqslant c_1 + n$ by (1), so $i \leqslant c_1$. Since $c_1 \leqslant n$ by (3), this implies that $c_i \leqslant c_{c_1} \leqslant n$ using (1) and (3). But then, using (1) again, $j \leqslant c_i \leqslant n$ implies $c_j \leqslant c_{c_i}$, which combined with $n + i \leqslant c_j$ gives us that $n + i \leqslant c_{c_i}$. This contradicts (2).

This means that the number of arrows is at most $\binom{n}{2}$, which implies that

$$\sum_{i=1}^{n} c_i \leqslant \binom{n}{2} + \binom{n+1}{2} = n^2.$$

Recalling that $a_i \leqslant c_i$ for $1 \leqslant i \leqslant n$, the desired inequality follows.

Comment 1. We sketch an alternative proof by induction. Begin by verifying the initial case $n = 1$ and the simple cases when $a_1 = 1$, $a_1 = n$, or $a_n \leqslant n$. Then, as in Solution 1, consider the index t such that $a_1 \leqslant \cdots \leqslant a_t \leqslant n < a_{t+1} \leqslant \cdots \leqslant a_n$. Observe again that $a_1 \leqslant t$. Define the sequence $d_1, \ldots, d_{n-1}$ by

$$d_i = \begin{cases} a_{i+1} - 1 & \text{if } i \leqslant t - 1 \\ a_{i+1} - 2 & \text{if } i \geqslant t \end{cases}$$

and extend it periodically modulo $n - 1$. One may verify that this sequence also satisfies the hypotheses of the problem. The induction hypothesis then gives $d_1 + \cdots + d_{n-1} \leqslant (n-1)^2$, which implies that

$$\sum_{i=1}^{n} a_i = a_1 + \sum_{i=2}^{t}(d_{i-1} + 1) + \sum_{i=t+1}^{n}(d_{i-1} + 2) \leqslant t + (t-1) + 2(n-t) + (n-1)^2 = n^2.$$

Comment 2. One unusual feature of this problem is that there are many different sequences for which equality holds. The discovery of such *optimal sequences* is not difficult, and it is useful in guiding the steps of a proof.

In fact, Solution 2 gives a complete description of the optimal sequences. Start with any lattice path P from the lower left to the upper right corner of the $n \times n$ square S using only steps up and right, such that the total number of steps along the left and top edges of S is at least n. Shade the cells of S below P dark, and the cells of S above P light. Now reflect the light shape across the line $x = y$ and shift it up n units, and shade it dark. As Solution 2 shows, the dark region will then correspond to an optimal sequence, and every optimal sequence arises in this way.

A5. Let $\mathbb{Z}_{\geqslant 0}$ be the set of all nonnegative integers. Find all the functions $f\colon \mathbb{Z}_{\geqslant 0} \to \mathbb{Z}_{\geqslant 0}$ satisfying the relation

$$f(f(f(n))) = f(n+1) + 1 \qquad (*)$$

for all $n \in \mathbb{Z}_{\geqslant 0}$.

(Serbia)

Answer. There are two such functions: $f(n) = n+1$ for all $n \in \mathbb{Z}_{\geqslant 0}$, and

$$f(n) = \begin{cases} n+1, & n \equiv 0 \pmod 4 \text{ or } n \equiv 2 \pmod 4, \\ n+5, & n \equiv 1 \pmod 4, \\ n-3, & n \equiv 3 \pmod 4 \end{cases} \qquad \text{for all } n \in \mathbb{Z}_{\geqslant 0}. \qquad (1)$$

Throughout all the solutions, we write $h^k(x)$ to abbreviate the kth iteration of function h, so h^0 is the identity function, and $h^k(x) = \underbrace{h(\ldots h}_{k \text{ times}}(x)\ldots))$ for $k \geqslant 1$.

Solution 1. To start, we get from $(*)$ that

$$f^4(n) = f(f^3(n)) = f\big(f(n+1)+1\big) \quad \text{and} \quad f^4(n+1) = f^3(f(n+1)) = f\big(f(n+1)+1\big) + 1,$$

thus

$$f^4(n) + 1 = f^4(n+1). \qquad (2)$$

I. Let us denote by R_i the range of f^i; note that $R_0 = \mathbb{Z}_{\geqslant 0}$ since f^0 is the identity function. Obviously, $R_0 \supseteq R_1 \supseteq \ldots$. Next, from (2) we get that if $a \in R_4$ then also $a+1 \in R_4$. This implies that $\mathbb{Z}_{\geqslant 0} \backslash R_4$ — and hence $\mathbb{Z}_{\geqslant 0} \backslash R_1$ — is finite. In particular, R_1 is unbounded.

Assume that $f(m) = f(n)$ for some distinct m and n. Then from $(*)$ we obtain $f(m+1) = f(n+1)$; by an easy induction we then get that $f(m+c) = f(n+c)$ for every $c \geqslant 0$. So the function $f(k)$ is periodic with period $|m-n|$ for $k \geqslant m$, and thus R_1 should be bounded, which is false. So, f is injective.

II. Denote now $S_i = R_{i-1} \backslash R_i$; all these sets are finite for $i \leqslant 4$. On the other hand, by the injectivity we have $n \in S_i \iff f(n) \in S_{i+1}$. By the injectivity again, f implements a bijection between S_i and S_{i+1}, thus $|S_1| = |S_2| = \ldots$; denote this common cardinality by k. If $0 \in R_3$ then $0 = f(f(f(n)))$ for some n, thus from $(*)$ we get $f(n+1) = -1$ which is impossible. Therefore $0 \in R_0 \backslash R_3 = S_1 \cup S_2 \cup S_3$, thus $k \geqslant 1$.

Next, let us describe the elements b of $R_0 \backslash R_3 = S_1 \cup S_2 \cup S_3$. We claim that each such element satisfies at least one of three conditions (i) $b = 0$, (ii) $b = f(0)+1$, and (iii) $b-1 \in S_1$. Otherwise $b-1 \in \mathbb{Z}_{\geqslant 0}$, and there exists some $n > 0$ such that $f(n) = b-1$; but then $f^3(n-1) = f(n)+1 = b$, so $b \in R_3$.

This yields

$$3k = |S_1 \cup S_2 \cup S_3| \leqslant 1 + 1 + |S_1| = k + 2,$$

or $k \leqslant 1$. Therefore $k = 1$, and the inequality above comes to equality. So we have $S_1 = \{a\}$, $S_2 = \{f(a)\}$, and $S_3 = \{f^2(a)\}$ for some $a \in \mathbb{Z}_{\geqslant 0}$, and each one of the three options (i), (ii), and (iii) should be realized exactly once, which means that

$$\{a, f(a), f^2(a)\} = \{0, a+1, f(0)+1\}. \qquad (3)$$

III. From (3), we get $a+1 \in \{f(a), f^2(a)\}$ (the case $a+1=a$ is impossible). If $a+1=f^2(a)$ then we have $f(a+1)=f^3(a)=f(a+1)+1$ which is absurd. Therefore

$$f(a)=a+1. \tag{4}$$

Next, again from (3) we have $0 \in \{a, f^2(a)\}$. Let us consider these two cases separately.

Case 1. Assume that $a=0$, then $f(0)=f(a)=a+1=1$. Also from (3) we get $f(1)=f^2(a)=f(0)+1=2$. Now, let us show that $f(n)=n+1$ by induction on n; the base cases $n \leqslant 1$ are established. Next, if $n \geqslant 2$ then the induction hypothesis implies

$$n+1=f(n-1)+1=f^3(n-2)=f^2(n-1)=f(n),$$

establishing the step. In this case we have obtained the first of two answers; checking that is satisfies $(*)$ is straightforward.

Case 2. Assume now that $f^2(a)=0$; then by (3) we get $a=f(0)+1$. By (4) we get $f(a+1)=f^2(a)=0$, then $f(0)=f^3(a)=f(a+1)+1=1$, hence $a=f(0)+1=2$ and $f(2)=3$ by (4). To summarize,

$$f(0)=1, \quad f(2)=3, \quad f(3)=0.$$

Now let us prove by induction on m that (1) holds for all $n=4k, 4k+2, 4k+3$ with $k \leqslant m$ and for all $n=4k+1$ with $k<m$. The base case $m=0$ is established above. For the step, assume that $m \geqslant 1$. From $(*)$ we get $f^3(4m-3)=f(4m-2)+1=4m$. Next, by (2) we have

$$f(4m)=f^4(4m-3)=f^4(4m-4)+1=f^3(4m-3)+1=4m+1.$$

Then by the induction hypothesis together with $(*)$ we successively obtain

$$\begin{aligned} f(4m-3)&=f^3(4m-1)=f(4m)+1=4m+2,\\ f(4m+2)&=f^3(4m-4)=f(4m-3)+1=4m+3,\\ f(4m+3)&=f^3(4m-3)=f(4m-2)+1=4m, \end{aligned}$$

thus finishing the induction step.

Finally, it is straightforward to check that the constructed function works:

$$\begin{aligned} f^3(4k)&=4k+7=f(4k+1)+1, & f^3(4k+1)&=4k+4=f(4k+2)+1,\\ f^3(4k+2)&=4k+1=f(4k+3)+1, & f^3(4k+3)&=4k+6=f(4k+4)+1. \end{aligned}$$

Solution 2. I. For convenience, let us introduce the function $g(n)=f(n)+1$. Substituting $f(n)$ instead of n into $(*)$ we obtain

$$f^4(n)=f\big(f(n)+1\big)+1, \quad \text{or} \quad f^4(n)=g^2(n). \tag{5}$$

Applying f to both parts of $(*)$ and using (5) we get

$$f^4(n)+1=f\big(f(n+1)+1\big)+1=f^4(n+1). \tag{6}$$

Thus, if $g^2(0)=f^4(0)=c$ then an easy induction on n shows that

$$g^2(n)=f^4(n)=n+c, \qquad n \in \mathbb{Z}_{\geqslant 0}. \tag{7}$$

This relation implies that both f and g are injective: if, say, $f(m) = f(n)$ then $m + c = f^4(m) = f^4(n) = n + c$. Next, since $g(n) \geqslant 1$ for every n, we have $c = g^2(0) \geqslant 1$. Thus from (7) again we obtain $f(n) \neq n$ and $g(n) \neq n$ for all $n \in \mathbb{Z}_{\geqslant 0}$.

II. Next, application of f and g to (7) yields

$$f(n + c) = f^5(n) = f^4(f(n)) = f(n) + c \quad \text{and} \quad g(n + c) = g^3(n) = g(n) + c. \tag{8}$$

In particular, this means that if $m \equiv n \pmod{c}$ then $f(m) \equiv f(n) \pmod{c}$. Conversely, if $f(m) \equiv f(n) \pmod{c}$ then we get $m + c = f^4(m) \equiv f^4(n) = n + c \pmod{c}$. Thus,

$$m \equiv n \pmod{c} \iff f(m) \equiv f(n) \pmod{c} \iff g(m) \equiv g(n) \pmod{c}. \tag{9}$$

Now, let us introduce the function $\delta(n) = f(n) - n = g(n) - n - 1$. Set

$$S = \sum_{n=0}^{c-1} \delta(n).$$

Using (8), we get that for every complete residue system $n_1, \ldots, n_c$ modulo c we also have

$$S = \sum_{i=1}^{c} \delta(n_i).$$

By (9), we get that $\{f^k(n) \colon n = 0, \ldots, c-1\}$ and $\{g^k(n) \colon n = 0, \ldots, c-1\}$ are complete residue systems modulo c for all k. Thus we have

$$c^2 = \sum_{n=0}^{c-1} \big(f^4(n) - n\big) = \sum_{k=0}^{3} \sum_{n=0}^{c-1} \big(f^{k+1}(n) - f^k(n)\big) = \sum_{k=0}^{3} \sum_{n=0}^{c-1} \delta(f^k(n)) = 4S$$

and similarly

$$c^2 = \sum_{n=0}^{c-1} \big(g^2(n) - n\big) = \sum_{k=0}^{1} \sum_{n=0}^{c-1} \big(g^{k+1}(n) - g^k(n)\big) = \sum_{k=0}^{1} \sum_{n=0}^{c-1} \big(\delta(g^k(n)) + 1\big) = 2S + 2c.$$

Therefore $c^2 = 4S = 2 \cdot 2S = 2(c^2 - 2c)$, or $c^2 = 4c$. Since $c \neq 0$, we get $c = 4$. Thus, in view of (8) it is sufficient to determine the values of f on the numbers $0, 1, 2, 3$.

III. Let $d = g(0) \geqslant 1$. Then $g(d) = g^2(0) = 0 + c = 4$. Now, if $d \geqslant 4$, then we would have $g(d - 4) = g(d) - 4 = 0$ which is impossible. Thus $d \in \{1, 2, 3\}$. If $d = 1$ then we have $f(0) = g(0) - 1 = 0$ which is impossible since $f(n) \neq n$ for all n. If $d = 3$ then $g(3) = g^2(0) = 4$ and hence $f(3) = 3$ which is also impossible. Thus $g(0) = 2$ and hence $g(2) = g^2(0) = 4$.

Next, if $g(1) = 1 + 4k$ for some integer k, then $5 = g^2(1) = g(1 + 4k) = g(1) + 4k = 1 + 8k$ which is impossible. Thus, since $\{g(n) \colon n = 0, 1, 2, 3\}$ is a complete residue system modulo 4, we get $g(1) = 3 + 4k$ and hence $g(3) = g^2(1) - 4k = 5 - 4k$, leading to $k = 0$ or $k = 1$. So, we obtain iether

$$f(0) = 1,\ f(1) = 2,\ f(2) = 3,\ f(3) = 4, \quad \text{or} \quad f(0) = 1,\ f(1) = 6,\ f(2) = 3,\ f(3) = 0,$$

thus arriving to the two functions listed in the answer.

Finally, one can check that these two function work as in Solution 1. One may simplify the checking by noticing that (8) allows us to reduce it to $n = 0, 1, 2, 3$.

A6. Let $m \neq 0$ be an integer. Find all polynomials $P(x)$ with real coefficients such that

$$(x^3 - mx^2 + 1)P(x+1) + (x^3 + mx^2 + 1)P(x-1) = 2(x^3 - mx + 1)P(x) \tag{1}$$

for all real numbers x.

(Serbia)

Answer. $P(x) = tx$ for any real number t.

Solution. Let $P(x) = a_n x^n + \cdots + a_0 x^0$ with $a_n \neq 0$. Comparing the coefficients of x^{n+1} on both sides gives $a_n(n-2m)(n-1) = 0$, so $n = 1$ or $n = 2m$.

If $n = 1$, one easily verifies that $P(x) = x$ is a solution, while $P(x) = 1$ is not. Since the given condition is linear in P, this means that the linear solutions are precisely $P(x) = tx$ for $t \in \mathbb{R}$.

Now assume that $n = 2m$. The polynomial $xP(x+1) - (x+1)P(x) = (n-1)a_n x^n + \cdots$ has degree n, and therefore it has at least one (possibly complex) root r. If $r \notin \{0, -1\}$, define $k = P(r)/r = P(r+1)/(r+1)$. If $r = 0$, let $k = P(1)$. If $r = -1$, let $k = -P(-1)$. We now consider the polynomial $S(x) = P(x) - kx$. It also satisfies (1) because $P(x)$ and kx satisfy it. Additionally, it has the useful property that r and $r+1$ are roots.

Let $A(x) = x^3 - mx^2 + 1$ and $B(x) = x^3 + mx^2 + 1$. Plugging in $x = s$ into (1) implies that:

> If $s-1$ and s are roots of S and s is not a root of A, then $s+1$ is a root of S.
>
> If s and $s+1$ are roots of S and s is not a root of B, then $s-1$ is a root of S.

Let $a \geqslant 0$ and $b \geqslant 1$ be such that $r-a, r-a+1, \ldots, r, r+1, \ldots, r+b-1, r+b$ are roots of S, while $r-a-1$ and $r+b+1$ are not. The two statements above imply that $r-a$ is a root of B and $r+b$ is a root of A.

Since $r-a$ is a root of $B(x)$ and of $A(x+a+b)$, it is also a root of their greatest common divisor $C(x)$ as integer polynomials. If $C(x)$ was a non-trivial divisor of $B(x)$, then B would have a rational root α. Since the first and last coefficients of B are 1, α can only be 1 or -1; but $B(-1) = m > 0$ and $B(1) = m + 2 > 0$ since $n = 2m$.

Therefore $B(x) = A(x+a+b)$. Writing $c = a + b \geqslant 1$ we compute

$$0 = A(x+c) - B(x) = (3c - 2m)x^2 + c(3c - 2m)x + c^2(c - m).$$

Then we must have $3c - 2m = c - m = 0$, which gives $m = 0$, a contradiction. We conclude that $f(x) = tx$ is the only solution.

Solution 2. Multiplying (1) by x, we rewrite it as

$$x(x^3 - mx^2 + 1)P(x+1) + x(x^3 + mx^2 + 1)P(x-1) = [(x+1) + (x-1)]\,(x^3 - mx + 1)P(x).$$

After regrouping, it becomes

$$(x^3 - mx^2 + 1)Q(x) = (x^3 + mx^2 + 1)Q(x-1), \tag{2}$$

where $Q(x) = xP(x+1) - (x+1)P(x)$. If $\deg P \geqslant 2$ then $\deg Q = \deg P$, so $Q(x)$ has a finite multiset of complex roots, which we denote R_Q. Each root is taken with its multiplicity. Then the multiset of complex roots of $Q(x-1)$ is $R_Q + 1 = \{z + 1 : z \in R_Q\}$.

Let $\{x_1, x_2, x_3\}$ and $\{y_1, y_2, y_3\}$ be the multisets of roots of the polynomials $A(x) = x^3 - mx^2 + 1$ and $B(x) = x^3 + mx^2 + 1$, respectively. From (2) we get the equality of multisets

$$\{x_1, x_2, x_3\} \cup R_Q = \{y_1, y_2, y_3\} \cup (R_Q + 1).$$

For every $r \in R_Q$, since $r + 1$ is in the set of the right hand side, we must have $r + 1 \in R_Q$ or $r + 1 = x_i$ for some i. Similarly, since r is in the set of the left hand side, either $r - 1 \in R_Q$ or $r = y_i$ for some i. This implies that, possibly after relabelling y_1, y_2, y_3, all the roots of (2) may be partitioned into three chains of the form $\{y_i, y_i + 1, \ldots, y_i + k_i = x_i\}$ for $i = 1, 2, 3$ and some integers $k_1, k_2, k_3 \geqslant 0$.

Now we analyze the roots of the polynomial $A_a(x) = x^3 + ax^2 + 1$. Using calculus or elementary methods, we find that the local extrema of $A_a(x)$ occur at $x = 0$ and $x = -2a/3$; their values are $A_a(0) = 1 > 0$ and $A_a(-2a/3) = 1 + 4a^3/27$, which is positive for integers $a \geqslant -1$ and negative for integers $a \leqslant -2$. So when $a \in \mathbb{Z}$, A_a has three real roots if $a \leqslant -2$ and one if $a \geqslant -1$.

Now, since $y_i - x_i \in \mathbb{Z}$ for $i = 1, 2, 3$, the cubics A_m and A_{-m} must have the same number of real roots. The previous analysis then implies that $m = 1$ or $m = -1$. Therefore the real root α of $A_1(x) = x^3 + x^2 + 1$ and the real root β of $A_{-1}(x) = x^3 - x^2 + 1$ must differ by an integer. But this is impossible, because $A_1\left(-\frac{3}{2}\right) = -\frac{1}{8}$ and $A_1(-1) = 1$ so $-1.5 < \alpha < -1$, while $A_{-1}(-1) = -1$ and $A_{-1}\left(-\frac{1}{2}\right) = \frac{5}{8}$, so $-1 < \beta < -0.5$.

It follows that $\deg P \leqslant 1$. Then, as shown in Solution 1, we conclude that the solutions are $P(x) = tx$ for all real numbers t.

Combinatorics

C1. Let n be a positive integer. Find the smallest integer k with the following property: Given any real numbers $a_1, \ldots, a_d$ such that $a_1 + a_2 + \cdots + a_d = n$ and $0 \leqslant a_i \leqslant 1$ for $i = 1, 2, \ldots, d$, it is possible to partition these numbers into k groups (some of which may be empty) such that the sum of the numbers in each group is at most 1.

(Poland)

Answer. $k = 2n - 1$.

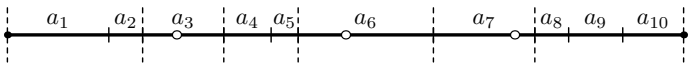
Solution 1. If $d = 2n-1$ and $a_1 = \cdots = a_{2n-1} = n/(2n-1)$, then each group in such a partition can contain at most one number, since $2n/(2n-1) > 1$. Therefore $k \geqslant 2n - 1$. It remains to show that a suitable partition into $2n - 1$ groups always exists.

We proceed by induction on d. For $d \leqslant 2n - 1$ the result is trivial. If $d \geqslant 2n$, then since

$$(a_1 + a_2) + \ldots + (a_{2n-1} + a_{2n}) \leqslant n$$

we may find two numbers a_i, a_{i+1} such that $a_i + a_{i+1} \leqslant 1$. We "merge" these two numbers into one new number $a_i + a_{i+1}$. By the induction hypothesis, a suitable partition exists for the $d - 1$ numbers $a_1, \ldots, a_{i-1}, a_i + a_{i+1}, a_{i+2}, \ldots, a_d$. This induces a suitable partition for $a_1, \ldots, a_d$.

Solution 2. We will show that it is even possible to split the sequence $a_1, \ldots, a_d$ into $2n - 1$ *contiguous* groups so that the sum of the numbers in each groups does not exceed 1. Consider a segment S of length n, and partition it into segments $S_1, \ldots, S_d$ of lengths $a_1, \ldots, a_d$, respectively, as shown below. Consider a second partition of S into n equal parts by $n - 1$ "empty dots".



Assume that the $n - 1$ empty dots are in segments $S_{i_1}, \ldots, S_{i_{n-1}}$. (If a dot is on the boundary of two segments, we choose the right segment). These $n - 1$ segments are distinct because they have length at most 1. Consider the partition:

$$\{a_1, \ldots, a_{i_1-1}\}, \{a_{i_1}\}, \{a_{i_1+1}, \ldots, a_{i_2-1}\}, \{a_{i_2}\}, \ldots \{a_{i_{n-1}}\}, \{a_{i_{n-1}+1}, \ldots, a_d\}.$$

In the example above, this partition is $\{a_1, a_2\}, \{a_3\}, \{a_4, a_5\}, \{a_6\}, \varnothing, \{a_7\}, \{a_8, a_9, a_{10}\}$. We claim that in this partition, the sum of the numbers in this group is at most 1.

For the sets $\{a_{i_t}\}$ this is obvious since $a_{i_t} \leqslant 1$. For the sets $\{a_{i_t} + 1, \ldots, a_{i_{t+1}-1}\}$ this follows from the fact that the corresponding segments lie between two neighboring empty dots, or between an endpoint of S and its nearest empty dot. Therefore the sum of their lengths cannot exceed 1.

Solution 3. First put all numbers greater than $\frac{1}{2}$ in their own groups. Then, form the remaining groups as follows: For each group, add new a_is one at a time until their sum exceeds $\frac{1}{2}$. Since the last summand is at most $\frac{1}{2}$, this group has sum at most 1. Continue this procedure until we have used all the a_is. Notice that the last group may have sum less than $\frac{1}{2}$. If the sum of the numbers in the last two groups is less than or equal to 1, we merge them into one group. In the end we are left with m groups. If $m = 1$ we are done. Otherwise the first $m - 2$ have sums greater than $\frac{1}{2}$ and the last two have total sum greater than 1. Therefore $n > (m-2)/2 + 1$ so $m \leqslant 2n - 1$ as desired.

Comment 1. The original proposal asked for the minimal value of k when $n = 2$.

Comment 2. More generally, one may ask the same question for real numbers between 0 and 1 whose sum is a real number r. In this case the smallest value of k is $k = \lceil 2r \rceil - 1$, as Solution 3 shows.

Solutions 1 and 2 lead to the slightly weaker bound $k \leqslant 2\lceil r \rceil - 1$. This is actually the optimal bound for partitions into *consecutive* groups, which are the ones contemplated in these two solutions. To see this, assume that r is not an integer and let $c = (r + 1 - \lceil r \rceil)/(1 + \lceil r \rceil)$. One easily checks that $0 < c < \frac{1}{2}$ and $\lceil r \rceil(2c) + (\lceil r \rceil - 1)(1 - c) = r$, so the sequence

$$2c,\ 1 - c,\ 2c,\ 1 - c,\ \ldots,\ 1 - c,\ 2c$$

of $2\lceil r \rceil - 1$ numbers satisfies the given conditions. For this sequence, the only suitable partition into consecutive groups is the trivial partition, which requires $2\lceil r \rceil - 1$ groups.

C2. In the plane, 2013 red points and 2014 blue points are marked so that no three of the marked points are collinear. One needs to draw k lines not passing through the marked points and dividing the plane into several regions. The goal is to do it in such a way that no region contains points of both colors.

Find the minimal value of k such that the goal is attainable for every possible configuration of 4027 points.

(Australia)

Answer. $k = 2013$.

Solution 1. Firstly, let us present an example showing that $k \geqslant 2013$. Mark 2013 red and 2013 blue points on some circle alternately, and mark one more blue point somewhere in the plane. The circle is thus split into 4026 arcs, each arc having endpoints of different colors. Thus, if the goal is reached, then each arc should intersect some of the drawn lines. Since any line contains at most two points of the circle, one needs at least $4026/2 = 2013$ lines.

It remains to prove that one can reach the goal using 2013 lines. First of all, let us mention that for every two points A and B having the same color, one can draw two lines separating these points from all other ones. Namely, it suffices to take two lines parallel to AB and lying on different sides of AB sufficiently close to it: the only two points between these lines will be A and B.

Now, let P be the convex hull of all marked points. Two cases are possible.

Case 1. Assume that P has a red vertex A. Then one may draw a line separating A from all the other points, pair up the other 2012 red points into 1006 pairs, and separate each pair from the other points by two lines. Thus, 2013 lines will be used.

Case 2. Assume now that all the vertices of P are blue. Consider any two consecutive vertices of P, say A and B. One may separate these two points from the others by a line parallel to AB. Then, as in the previous case, one pairs up all the other 2012 blue points into 1006 pairs, and separates each pair from the other points by two lines. Again, 2013 lines will be used.

Comment 1. Instead of considering the convex hull, one may simply take a line containing two marked points A and B such that all the other marked points are on one side of this line. If one of A and B is red, then one may act as in Case 1; otherwise both are blue, and one may act as in Case 2.

Solution 2. Let us present a different proof of the fact that $k = 2013$ suffices. In fact, we will prove a more general statement:

If n points in the plane, no three of which are collinear, are colored in red and blue arbitrarily, then it suffices to draw $\lfloor n/2 \rfloor$ lines to reach the goal.

We proceed by induction on n. If $n \leqslant 2$ then the statement is obvious. Now assume that $n \geqslant 3$, and consider a line ℓ containing two marked points A and B such that all the other marked points are on one side of ℓ; for instance, any line containing a side of the convex hull works.

Remove for a moment the points A and B. By the induction hypothesis, for the remaining configuration it suffices to draw $\lfloor n/2 \rfloor - 1$ lines to reach the goal. Now return the points A and B back. Three cases are possible.

Case 1. If A and B have the same color, then one may draw a line parallel to ℓ and separating A and B from the other points. Obviously, the obtained configuration of $\lfloor n/2 \rfloor$ lines works.

Case 2. If A and B have different colors, but they are separated by some drawn line, then again the same line parallel to ℓ works.

Case 3. Finally, assume that A and B have different colors and lie in one of the regions defined by the drawn lines. By the induction assumption, this region contains no other points of one of the colors — without loss of generality, the only blue point it contains is A. Then it suffices to draw a line separating A from all other points.

Thus the step of the induction is proved.

Comment 2. One may ask a more general question, replacing the numbers 2013 and 2014 by any positive integers m and n, say with $m \leqslant n$. Denote the answer for this problem by $f(m, n)$.

One may show along the lines of Solution 1 that $m \leqslant f(m, n) \leqslant m + 1$; moreover, if m is even then $f(m, n) = m$. On the other hand, for every odd m there exists an N such that $f(m, n) = m$ for all $m \leqslant n \leqslant N$, and $f(m, n) = m + 1$ for all $n > N$.

C3. A crazy physicist discovered a new kind of particle which he called an *imon*, after some of them mysteriously appeared in his lab. Some pairs of imons in the lab can be *entangled*, and each imon can participate in many entanglement relations. The physicist has found a way to perform the following two kinds of operations with these particles, one operation at a time.

(i) If some imon is entangled with an odd number of other imons in the lab, then the physicist can destroy it.

(ii) At any moment, he may double the whole family of imons in his lab by creating a copy I' of each imon I. During this procedure, the two copies I' and J' become entangled if and only if the original imons I and J are entangled, and each copy I' becomes entangled with its original imon I; no other entanglements occur or disappear at this moment.

Prove that the physicist may apply a sequence of such operations resulting in a family of imons, no two of which are entangled.

(Japan)

Solution 1. Let us consider a graph with the imons as vertices, and two imons being connected if and only if they are entangled. Recall that a *proper coloring* of a graph G is a coloring of its vertices in several colors so that every two connected vertices have different colors.

Lemma. Assume that a graph G admits a proper coloring in n colors ($n > 1$). Then one may perform a sequence of operations resulting in a graph which admits a proper coloring in $n - 1$ colors.

Proof. Let us apply repeatedly operation (i) to any appropriate vertices while it is possible. Since the number of vertices decreases, this process finally results in a graph where all the degrees are even. Surely this graph also admits a proper coloring in n colors $1, \ldots, n$; let us fix this coloring.

Now apply the operation (ii) to this graph. A proper coloring of the resulting graph in n colors still exists: one may preserve the colors of the original vertices and color the vertex I' in a color $k + 1 \pmod n$ if the vertex I has color k. Then two connected original vertices still have different colors, and so do their two connected copies. On the other hand, the vertices I and I' have different colors since $n > 1$.

All the degrees of the vertices in the resulting graph are odd, so one may apply operation (i) to delete consecutively all the vertices of color n one by one; no two of them are connected by an edge, so their degrees do not change during the process. Thus, we obtain a graph admitting a proper coloring in $n - 1$ colors, as required. The lemma is proved. □

Now, assume that a graph G has n vertices; then it admits a proper coloring in n colors. Applying repeatedly the lemma we finally obtain a graph admitting a proper coloring in one color, that is — a graph with no edges, as required.

Solution 2. Again, we will use the graph language.

I. We start with the following observation.

Lemma. Assume that a graph G contains an isolated vertex A, and a graph G° is obtained from G by deleting this vertex. Then, if one can apply a sequence of operations which makes a graph with no edges from G°, then such a sequence also exists for G.

Proof. Consider any operation applicable to G° resulting in a graph G_1°; then there exists a sequence of operations applicable to G and resulting in a graph G_1 differing from G_1° by an addition of an isolated vertex A. Indeed, if this operation is of type (i), then one may simply repeat it in G.

Otherwise, the operation is of type (ii), and one may apply it to G and then delete the vertex A' (it will have degree 1).

Thus one may change the process for G° into a corresponding process for G step by step. □

In view of this lemma, if at some moment a graph contains some isolated vertex, then we may simply delete it; let us call this operation (iii).

II. Let $V = \{A_1^0, \dots, A_n^0\}$ be the vertices of the initial graph. Let us describe which graphs can appear during our operations. Assume that operation (ii) was applied m times. If these were the only operations applied, then the resulting graph G_n^m has the set of vertices which can be enumerated as

$$V_n^m = \{A_i^j \colon 1 \leqslant i \leqslant n,\ 0 \leqslant j \leqslant 2^m - 1\},$$

where A_i^0 is the common "ancestor" of all the vertices A_i^j, and the binary expansion of j (adjoined with some zeroes at the left to have m digits) "keeps the history" of this vertex: the dth digit from the right is 0 if at the dth doubling the ancestor of A_i^j was in the original part, and this digit is 1 if it was in the copy.

Next, the two vertices A_i^j and A_k^ℓ in G_n^m are connected with an edge exactly if either (1) $j = \ell$ and there was an edge between A_i^0 and A_k^0 (so these vertices appeared at the same application of operation (ii)); or (2) $i = k$ and the binary expansions of j and ℓ differ in exactly one digit (so their ancestors became connected as a copy and the original vertex at some application of (ii)).

Now, if some operations (i) were applied during the process, then simply some vertices in G_n^m disappeared. So, in any case the resulting graph is some induced subgraph of G_n^m.

III. Finally, we will show that from each (not necessarily induced) subgraph of G_n^m one can obtain a graph with no vertices by applying operations (i), (ii) and (iii). We proceed by induction on n; the base case $n = 0$ is trivial.

For the induction step, let us show how to apply several operations so as to obtain a graph containing no vertices of the form A_n^j for $j \in \mathbb{Z}$. We will do this in three steps.

Step 1. We apply repeatedly operation (i) to any appropriate vertices while it is possible. In the resulting graph, all vertices have even degrees.

Step 2. Apply operation (ii) obtaining a subgraph of G_n^{m+1} with all degrees being odd. In this graph, we delete one by one all the vertices A_n^j where the sum of the binary digits of j is even; it is possible since there are no edges between such vertices, so all their degrees remain odd. After that, we delete all isolated vertices.

Step 3. Finally, consider any remaining vertex A_n^j (then the sum of digits of j is odd). If its degree is odd, then we simply delete it. Otherwise, since A_n^j is not isolated, we consider any vertex adjacent to it. It has the form A_k^j for some $k < n$ (otherwise it would have the form A_n^ℓ, where ℓ has an even digit sum; but any such vertex has already been deleted at Step 2). No neighbor of A_k^j was deleted at Steps 2 and 3, so it has an odd degree. Then we successively delete A_k^j and A_n^j.

Notice that this deletion does not affect the applicability of this step to other vertices, since no two vertices A_i^j and A_k^ℓ for different j, ℓ with odd digit sum are connected with an edge. Thus we will delete all the remaining vertices of the form A_n^j, obtaining a subgraph of G_{n-1}^{m+1}. The application of the induction hypothesis finishes the proof.

Comment. In fact, the graph G_n^m is a Cartesian product of G and the graph of an m-dimensional hypercube.

C4. Let n be a positive integer, and let A be a subset of $\{1, \ldots, n\}$. An *A-partition* of n *into* k *parts* is a representation of n as a sum $n = a_1 + \cdots + a_k$, where the parts $a_1, \ldots, a_k$ belong to A and are not necessarily distinct. The *number of different parts* in such a partition is the number of (distinct) elements in the set $\{a_1, a_2, \ldots, a_k\}$.

We say that an A-partition of n into k parts is *optimal* if there is no A-partition of n into r parts with $r < k$. Prove that any optimal A-partition of n contains at most $\sqrt[3]{6n}$ different parts.

(Germany)

Solution 1. If there are no A-partitions of n, the result is vacuously true. Otherwise, let $k_{\min}$ be the minimum number of parts in an A-partition of n, and let $n = a_1 + \cdots + a_{k_{\min}}$ be an optimal partition. Denote by s the number of different parts in this partition, so we can write $S = \{a_1, \ldots, a_{k_{\min}}\} = \{b_1, \ldots, b_s\}$ for some pairwise different numbers $b_1 < \cdots < b_s$ in A.

If $s > \sqrt[3]{6n}$, we will prove that there exist subsets X and Y of S such that $|X| < |Y|$ and $\sum_{x\in X} x = \sum_{y\in Y} y$. Then, deleting the elements of Y from our partition and adding the elements of X to it, we obtain an A-partition of n into less than $k_{\min}$ parts, which is the desired contradiction.

For each positive integer $k \leqslant s$, we consider the k-element subset

$$S^k_{1,0} := \{b_1, \ldots, b_k\}$$

as well as the following k-element subsets $S^k_{i,j}$ of S:

$$S^k_{i,j} := \big\{b_1, \ldots, b_{k-i}, b_{k-i+j+1}, b_{s-i+2}, \ldots, b_s\big\}, \quad i = 1, \ldots, k, \quad j = 1, \ldots, s-k.$$

Pictorially, if we represent the elements of S by a sequence of dots in increasing order, and represent a subset of S by shading in the appropriate dots, we have:

$$S^k_{i,j} = \underbrace{\bullet\bullet\bullet\bullet\bullet\bullet}_{k-i}\underbrace{\circ\circ\circ\circ\circ}_{j}\bullet\underbrace{\circ\circ\circ\circ\circ\circ\circ}_{s-k-j}\underbrace{\bullet\bullet\bullet\bullet\bullet\bullet\bullet\bullet}_{i-1}$$

Denote by $\Sigma^k_{i,j}$ the sum of elements in $S^k_{i,j}$. Clearly, $\Sigma^k_{1,0}$ is the minimum sum of a k-element subset of S. Next, for all appropriate indices i and j we have

$$\Sigma^k_{i,j} = \Sigma^k_{i,j+1} + b_{k-i+j+1} - b_{k-i+j+2} < \Sigma^k_{i,j+1} \quad \text{and} \quad \Sigma^k_{i,s-k} = \Sigma^k_{i+1,1} + b_{k-i} - b_{k-i+1} < \Sigma^k_{i+1,1}.$$

Therefore

$$1 \leqslant \Sigma^k_{1,0} < \Sigma^k_{1,1} < \Sigma^k_{1,2} < \cdots < \Sigma^k_{1,s-k} < \Sigma^k_{2,1} < \cdots < \Sigma^k_{2,s-k} < \Sigma^k_{3,1} < \cdots < \Sigma^k_{k,s-k} \leqslant n.$$

To see this in the picture, we start with the k leftmost points marked. At each step, we look for the rightmost point which can move to the right, and move it one unit to the right. We continue until the k rightmost points are marked. As we do this, the corresponding sums clearly increase.

For each k we have found $k(s-k)+1$ different integers of the form $\Sigma^k_{i,j}$ between 1 and n. As we vary k, the total number of integers we are considering is

$$\sum_{k=1}^{s}\big(k(s-k)+1\big) = s\cdot\frac{s(s+1)}{2} - \frac{s(s+1)(2s+1)}{6} + s = \frac{s(s^2+5)}{6} > \frac{s^3}{6} > n.$$

Since they are between 1 and n, at least two of these integers are equal. Consequently, there exist $1 \leqslant k < k' \leqslant s$ and $X = S^k_{i,j}$ as well as $Y = S^{k'}_{i',j'}$ such that

$$\sum_{x\in X} x = \sum_{y\in Y} y, \quad \text{but} \quad |X| = k < k' = |Y|,$$

as required. The result follows.

Solution 2. Assume, to the contrary, that the statement is false, and choose the minimum number n for which it fails. So there exists a set $A \subseteq \{1, \ldots, n\}$ together with an optimal A-partition $n = a_1 + \cdots + a_{k_{\min}}$ of n refuting our statement, where, of course, $k_{\min}$ is the minimum number of parts in an A-partition of n. Again, we define $S = \{a_1, \ldots, a_{k_{\min}}\} = \{b_1, \ldots, b_s\}$ with $b_1 < \cdots < b_s$; by our assumption we have $s > \sqrt[3]{6n} > 1$. Without loss of generality we assume that $a_{k_{\min}} = b_s$. Let us distinguish two cases.

Case 1. $b_s \geqslant \frac{s(s-1)}{2} + 1$.

Consider the partition $n - b_s = a_1 + \cdots + a_{k_{\min}-1}$, which is clearly a minimum A-partition of $n - b_s$ with at least $s - 1 \geqslant 1$ different parts. Now, from $n < \frac{s^3}{6}$ we obtain

$$n - b_s \leqslant n - \frac{s(s-1)}{2} - 1 < \frac{s^3}{6} - \frac{s(s-1)}{2} - 1 < \frac{(s-1)^3}{6},$$

so $s - 1 > \sqrt[3]{6(n - b_s)}$, which contradicts the choice of n.

Case 2. $b_s \leqslant \frac{s(s-1)}{2}$.

Set $b_0 = 0$, $\Sigma_{0,0} = 0$, and $\Sigma_{i,j} = b_1 + \cdots + b_{i-1} + b_j$ for $1 \leqslant i \leqslant j < s$. There are $\frac{s(s-1)}{2} + 1 > b_s$ such sums; so at least two of them, say $\Sigma_{i,j}$ and $\Sigma_{i',j'}$, are congruent modulo b_s (where $(i, j) \neq (i', j')$). This means that $\Sigma_{i,j} - \Sigma_{i',j'} = r b_s$ for some integer r. Notice that for $i \leqslant j < k < s$ we have

$$0 < \Sigma_{i,k} - \Sigma_{i,j} = b_k - b_j < b_s,$$

so the indices i and i' are distinct, and we may assume that $i > i'$. Next, we observe that $\Sigma_{i,j} - \Sigma_{i',j'} = (b_{i'} - b_{j'}) + b_j + b_{i'+1} + \cdots + b_{i-1}$ and $b_{i'} \leqslant b_{j'}$ imply

$$-b_s < -b_{j'} < \Sigma_{i,j} - \Sigma_{i',j'} < (i - i')b_s,$$

so $0 \leqslant r \leqslant i - i' - 1$.

Thus, we may remove the i terms of $\Sigma_{i,j}$ in our A-partition, and replace them by the i' terms of $\Sigma_{i',j'}$ and r terms equal to b_s, for a total of $r + i' < i$ terms. The result is an A-partition of n into a smaller number of parts, a contradiction.

Comment. The original proposal also contained a second part, showing that the estimate appearing in the problem has the correct order of magnitude:

For every positive integer n, there exist a set A and an optimal A-partition of n that contains $\lfloor \sqrt[3]{2n} \rfloor$ different parts.

The Problem Selection Committee removed this statement from the problem, since it seems to be less suitable for the competiton; but for completeness we provide an outline of its proof here.

Let $k = \lfloor \sqrt[3]{2n} \rfloor - 1$. The statement is trivial for $n < 4$, so we assume $n \geqslant 4$ and hence $k \geqslant 1$. Let $h = \lfloor \frac{n-1}{k} \rfloor$. Notice that $h \geqslant \frac{n}{k} - 1$.

Now let $A = \{1, \ldots, h\}$, and set $a_1 = h$, $a_2 = h - 1$, $\ldots$, $a_k = h - k + 1$, and $a_{k+1} = n - (a_1 + \cdots + a_k)$. It is not difficult to prove that $a_k > a_{k+1} \geqslant 1$, which shows that

$$n = a_1 + \ldots + a_{k+1}$$

is an A-partition of n into $k+1$ different parts. Since $kh < n$, any A-partition of n has at least $k+1$ parts. Therefore our A-partition is optimal, and it has $\lfloor \sqrt[3]{2n} \rfloor$ distinct parts, as desired.

C5. Let r be a positive integer, and let $a_0, a_1, \ldots$ be an infinite sequence of real numbers. Assume that for all nonnegative integers m and s there exists a positive integer $n \in [m+1, m+r]$ such that

$$a_m + a_{m+1} + \cdots + a_{m+s} = a_n + a_{n+1} + \cdots + a_{n+s}.$$

Prove that the sequence is periodic, i. e. there exists some $p \geqslant 1$ such that $a_{n+p} = a_n$ for all $n \geqslant 0$.

(India)

Solution. For every indices $m \leqslant n$ we will denote $S(m,n) = a_m + a_{m+1} + \cdots + a_{n-1}$; thus $S(n,n) = 0$. Let us start with the following lemma.

Lemma. Let $b_0, b_1, \ldots$ be an infinite sequence. Assume that for every nonnegative integer m there exists a nonnegative integer $n \in [m+1, m+r]$ such that $b_m = b_n$. Then for every indices $k \leqslant \ell$ there exists an index $t \in [\ell, \ell + r - 1]$ such that $b_t = b_k$. Moreover, there are at most r distinct numbers among the terms of (b_i).

Proof. To prove the first claim, let us notice that there exists an infinite sequence of indices $k_1 = k, k_2, k_3, \ldots$ such that $b_{k_1} = b_{k_2} = \cdots = b_k$ and $k_i < k_{i+1} \leqslant k_i + r$ for all $i \geqslant 1$. This sequence is unbounded from above, thus it hits each segment of the form $[\ell, \ell+r-1]$ with $\ell \geqslant k$, as required.

To prove the second claim, assume, to the contrary, that there exist $r+1$ distinct numbers $b_{i_1}, \ldots, b_{i_{r+1}}$. Let us apply the first claim to $k = i_1, \ldots, i_{r+1}$ and $\ell = \max\{i_1, \ldots, i_{r+1}\}$; we obtain that for every $j \in \{1, \ldots, r+1\}$ there exists $t_j \in [s, s+r-1]$ such that $b_{t_j} = b_{i_j}$. Thus the segment $[s, s+r-1]$ should contain $r+1$ distinct integers, which is absurd. □

Setting $s = 0$ in the problem condition, we see that the sequence (a_i) satisfies the condition of the lemma, thus it attains at most r distinct values. Denote by A_i the ordered r-tuple $(a_i, \ldots, a_{i+r-1})$; then among A_i's there are at most r^r distinct tuples, so for every $k \geqslant 0$ two of the tuples $A_k, A_{k+1}, \ldots, A_{k+r^r}$ are identical. This means that there exists a positive integer $p \leqslant r^r$ such that the equality $A_d = A_{d+p}$ holds infinitely many times. Let D be the set of indices d satisfying this relation.

Now we claim that D coincides with the set of all nonnegative integers. Since D is unbounded, it suffices to show that $d \in D$ whenever $d+1 \in D$. For that, denote $b_k = S(k, p+k)$. The sequence $b_0, b_1, \ldots$ satisfies the lemma conditions, so there exists an index $t \in [d+1, d+r]$ such that $S(t, t+p) = S(d, d+p)$. This last relation rewrites as $S(d,t) = S(d+p, t+p)$. Since $A_{d+1} = A_{d+p+1}$, we have $S(d+1, t) = S(d+p+1, t+p)$, therefore we obtain

$$a_d = S(d,t) - S(d+1,t) = S(d+p, t+p) - S(d+p+1, t+p) = a_{d+p}$$

and thus $A_d = A_{d+p}$, as required.

Finally, we get $A_d = A_{d+p}$ for all d, so in particular $a_d = a_{d+p}$ for all d, QED.

Comment 1. In the present proof, the upper bound for the minimal period length is r^r. This bound is not sharp; for instance, one may improve it to $(r-1)^r$ for $r \geqslant 3$..

On the other hand, this minimal length may happen to be greater than r. For instance, it is easy to check that the sequence with period $(3, -3, 3, -3, 3, -1, -1, -1)$ satisfies the problem condition for $r = 7$.

Comment 2. The conclusion remains true even if the problem condition only holds for every $s \geqslant N$ for some positive integer N. To show that, one can act as follows. Firstly, the sums of the form $S(i, i+N)$ attain at most r values, as well as the sums of the form $S(i, i+N+1)$. Thus the terms $a_i = S(i, i+N+1) - S(i+1, i+N+1)$ attain at most r^2 distinct values. Then, among the tuples $A_k, A_{k+N}, \ldots, A_{k+r^{2r}N}$ two

are identical, so for some $p \leqslant r^{2r}$ the set $D = \{d\colon A_d = A_{d+Np}\}$ is infinite. The further arguments apply almost literally, with p being replaced by Np.

After having proved that such a sequence is also necessarily periodic, one may reduce the bound for the minimal period length to r^r — essentially by verifying that the sequence satisfies the original version of the condition.

C6. In some country several pairs of cities are connected by direct two-way flights. It is possible to go from any city to any other by a sequence of flights. The *distance* between two cities is defined to be the least possible number of flights required to go from one of them to the other. It is known that for any city there are at most 100 cities at distance exactly three from it. Prove that there is no city such that more than 2550 other cities have distance exactly four from it.

(Russia)

Solution. Let us denote by $d(a, b)$ the distance between the cities a and b, and by

$$S_i(a) = \{c\colon d(a, c) = i\}$$

the set of cities at distance exactly i from city a.

Assume that for some city x the set $D = S_4(x)$ has size at least 2551. Let $A = S_1(x)$. A subset A' of A is said to be *substantial*, if every city in D can be reached from x with four flights while passing through some member of A'; in other terms, every city in D has distance 3 from some member of A', or $D \subseteq \bigcup_{a\in A'} S_3(a)$. For instance, A itself is substantial. Now let us fix some substantial subset A^* of A having the minimal cardinality $m = |A^*|$.

Since

$$m(101 - m) \leqslant 50 \cdot 51 = 2550,$$

there has to be a city $a \in A^*$ such that $|S_3(a) \cap D| \geqslant 102 - m$. As $|S_3(a)| \leqslant 100$, we obtain that $S_3(a)$ may contain at most $100 - (102 - m) = m - 2$ cities c with $d(c, x) \leqslant 3$. Let us denote by $T = \{c \in S_3(a)\colon d(x, c) \leqslant 3\}$ the set of all such cities, so $|T| \leqslant m - 2$. Now, to get a contradiction, we will construct $m - 1$ distinct elements in T, corresponding to $m - 1$ elements of the set $A_a = A^*\backslash\{a\}$.

Firstly, due to the minimality of A^*, for each $y \in A_a$ there exists some city $d_y \in D$ which can only be reached with four flights from x by passing through y. So, there is a way to get from x to d_y along x–y–b_y–c_y–d_y for some cities b_y and c_y; notice that $d(x, b_y) = 2$ and $d(x, c_y) = 3$ since this path has the minimal possible length.

Now we claim that all $2(m - 1)$ cities of the form b_y, c_y with $y \in A_a$ are distinct. Indeed, no b_y may coincide with any c_z since their distances from x are different. On the other hand, if one had $b_y = b_z$ for $y \neq z$, then there would exist a path of length 4 from x to d_z via y, namely x–y–b_z–c_z–d_z; this is impossible by the choice of d_z. Similarly, $c_y \neq c_z$ for $y \neq z$.

So, it suffices to prove that for every $y \in A_a$, one of the cities b_y and c_y has distance 3 from a (and thus belongs to T). For that, notice that $d(a, y) \leqslant 2$ due to the path a–x–y, while $d(a, d_y) \geqslant d(x, d_y) - d(x, a) = 3$. Moreover, $d(a, d_y) \neq 3$ by the choice of d_y; thus $d(a, d_y) > 3$. Finally, in the sequence $d(a, y)$, $d(a, b_y)$, $d(a, c_y)$, $d(a, d_y)$ the neighboring terms differ by at most 1, the first term is less than 3, and the last one is greater than 3; thus there exists one which is equal to 3, as required.

Comment 1. The upper bound 2550 is sharp. This can be seen by means of various examples; one of them is the "Roman Empire": it has one capital, called "Rome", that is connected to 51 semicapitals by internally disjoint paths of length 3. Moreover, each of these semicapitals is connected to 50 rural cities by direct flights.

Comment 2. Observe that, under the conditions of the problem, there exists no bound for the size of $S_1(x)$ or $S_2(x)$.

Comment 3. The numbers 100 and 2550 appearing in the statement of the problem may be replaced by n and $\left\lfloor \frac{(n+1)^2}{4} \right\rfloor$ for any positive integer n. Still more generally, one can also replace the pair $(3, 4)$ of distances under consideration by any pair (r, s) of positive integers satisfying $r < s \leqslant \frac{3}{2}r$.

To adapt the above proof to this situation, one takes $A = S_{s-r}(x)$ and defines the concept of substantiality as before. Then one takes A^* to be a minimal substantial subset of A, and for each $y \in A^*$ one fixes an element $d_y \in S_s(x)$ which is only reachable from x by a path of length s by passing through y. As before, it suffices to show that for distinct $a, y \in A^*$ and a path $y = y_0 - y_1 - \ldots - y_r = d_y$, at least one of the cities $y_0, \ldots, y_{r-1}$ has distance r from a. This can be done as above; the relation $s \leqslant \frac{3}{2}r$ is used here to show that $d(a, y_0) \leqslant r$.

Moreover, the estimate $\left\lfloor \frac{(n+1)^2}{4} \right\rfloor$ is also sharp for every positive integer n and every positive integers r, s with $r < s \leqslant \frac{3}{2}r$. This may be shown by an example similar to that in the previous comment.

C7. Let $n \geqslant 2$ be an integer. Consider all circular arrangements of the numbers $0, 1, \ldots, n$; the $n+1$ rotations of an arrangement are considered to be equal. A circular arrangement is called *beautiful* if, for any four distinct numbers $0 \leqslant a, b, c, d \leqslant n$ with $a + c = b + d$, the chord joining numbers a and c does not intersect the chord joining numbers b and d.

Let M be the number of beautiful arrangements of $0, 1, \ldots, n$. Let N be the number of pairs (x, y) of positive integers such that $x + y \leqslant n$ and $\gcd(x, y) = 1$. Prove that

$$M = N + 1.$$

(Russia)

Solution 1. Given a circular arrangement of $[0, n] = \{0, 1, \ldots, n\}$, we define a *$k$-chord* to be a (possibly degenerate) chord whose (possibly equal) endpoints add up to k. We say that three chords of a circle are *aligned* if one of them separates the other two. Say that $m \geqslant 3$ chords are aligned if any three of them are aligned. For instance, in Figure 1, A, B, and C are aligned, while B, C, and D are not.

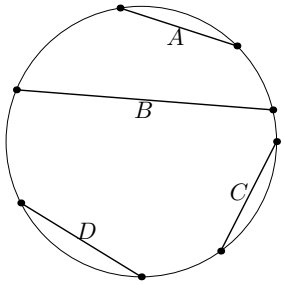


Figure 1

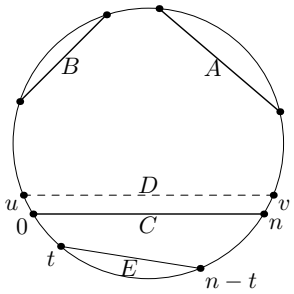


Figure 2

Claim. In a beautiful arrangement, the k–chords are aligned for any integer k.

Proof. We proceed by induction. For $n \leqslant 3$ the statement is trivial. Now let $n \geqslant 4$, and proceed by contradiction. Consider a beautiful arrangement S where the three k–chords A, B, C are not aligned. If n is not among the endpoints of A, B, and C, then by deleting n from S we obtain a beautiful arrangement $S\backslash\{n\}$ of $[0, n-1]$, where A, B, and C are aligned by the induction hypothesis. Similarly, if 0 is not among these endpoints, then deleting 0 and decreasing all the numbers by 1 gives a beautiful arrangement $S\backslash\{0\}$ where A, B, and C are aligned. Therefore both 0 and n are among the endpoints of these segments. If x and y are their respective partners, we have $n \geqslant 0 + x = k = n + y \geqslant n$. Thus 0 and n are the endpoints of one of the chords; say it is C.

Let D be the chord formed by the numbers u and v which are adjacent to 0 and n and on the same side of C as A and B, as shown in Figure 2. Set $t = u + v$. If we had $t = n$, the n–chords A, B, and D would not be aligned in the beautiful arrangement $S\backslash\{0, n\}$, contradicting the induction hypothesis. If $t < n$, then the t-chord from 0 to t cannot intersect D, so the chord C separates t and D. The chord E from t to $n - t$ does not intersect C, so t and $n - t$ are on the same side of C. But then the chords A, B, and E are not aligned in $S\backslash\{0, n\}$, a contradiction. Finally, the case $t > n$ is equivalent to the case $t < n$ via the beauty-preserving relabelling $x \mapsto n - x$ for $0 \leqslant x \leqslant n$, which sends t-chords to $(2n - t)$–chords. This proves the Claim.

Having established the Claim, we prove the desired result by induction. The case $n = 2$ is trivial. Now assume that $n \geqslant 3$. Let S be a beautiful arrangement of $[0, n]$ and delete n to obtain

the beautiful arrangement T of $[0, n-1]$. The n–chords of T are aligned, and they contain every point except 0. Say T is of *Type 1* if 0 lies between two of these n–chords, and it is of *Type 2* otherwise; *i.e.*, if 0 is aligned with these n–chords. We will show that each Type 1 arrangement of $[0, n-1]$ arises from a unique arrangement of $[0, n]$, and each Type 2 arrangement of $[0, n-1]$ arises from exactly two beautiful arrangements of $[0, n]$.

If T is of Type 1, let 0 lie between chords A and B. Since the chord from 0 to n must be aligned with A and B in S, n must be on the other arc between A and B. Therefore S can be recovered uniquely from T. In the other direction, if T is of Type 1 and we insert n as above, then we claim the resulting arrangement S is beautiful. For $0 < k < n$, the k–chords of S are also k–chords of T, so they are aligned. Finally, for $n < k < 2n$, notice that the n–chords of S are parallel by construction, so there is an antisymmetry axis ℓ such that x is symmetric to $n - x$ with respect to ℓ for all x. If we had two k–chords which intersect, then their reflections across ℓ would be two $(2n-k)$-chords which intersect, where $0 < 2n - k < n$, a contradiction.

If T is of Type 2, there are two possible positions for n in S, on either side of 0. As above, we check that both positions lead to beautiful arrangements of $[0, n]$.

Hence if we let M_n be the number of beautiful arrangements of $[0, n]$, and let L_n be the number of beautiful arrangements of $[0, n-1]$ of Type 2, we have

$$M_n = (M_{n-1} - L_{n-1}) + 2L_{n-1} = M_{n-1} + L_{n-1}.$$

It then remains to show that L_{n-1} is the number of pairs (x, y) of positive integers with $x + y = n$ and $\gcd(x, y) = 1$. Since $n \geqslant 3$, this number equals $\varphi(n) = \#\{x : 1 \leqslant x \leqslant n,\ \gcd(x, n) = 1\}$.

To prove this, consider a Type 2 beautiful arrangement of $[0, n-1]$. Label the positions $0, \ldots, n-1 \pmod n$ clockwise around the circle, so that number 0 is in position 0. Let $f(i)$ be the number in position i; note that f is a permutation of $[0, n-1]$. Let a be the position such that $f(a) = n - 1$.

Since the n–chords are aligned with 0, and every point is in an n–chord, these chords are all parallel and

$$f(i) + f(-i) = n \qquad \text{for all } i.$$

Similarly, since the $(n-1)$–chords are aligned and every point is in an $(n-1)$–chord, these chords are also parallel and

$$f(i) + f(a - i) = n - 1 \qquad \text{for all } i.$$

Therefore $f(a - i) = f(-i) - 1$ for all i; and since $f(0) = 0$, we get

$$f(-ak) = k \qquad \text{for all } k. \tag{1}$$

Recall that this is an equality modulo n. Since f is a permutation, we must have $(a, n) = 1$. Hence $L_{n-1} \leqslant \varphi(n)$.

To prove equality, it remains to observe that the labeling (1) is beautiful. To see this, consider four numbers w, x, y, z on the circle with $w + y = x + z$. Their positions around the circle satisfy $(-aw) + (-ay) = (-ax) + (-az)$, which means that the chord from w to y and the chord from x to z are parallel. Thus (1) is beautiful, and by construction it has Type 2. The desired result follows.

Solution 2. Notice that there are exactly N irreducible fractions $f_1 < \cdots < f_N$ in $(0,1)$ whose denominator is at most n, since the pair (x,y) with $x+y \leqslant n$ and $(x,y)=1$ corresponds to the fraction $x/(x+y)$. Write $f_i = \frac{a_i}{b_i}$ for $1 \leqslant i \leqslant N$.

We begin by constructing $N+1$ beautiful arrangements. Take any $\alpha \in (0,1)$ which is not one of the above N fractions. Consider a circle of perimeter 1. Successively mark points $0, 1, 2, \ldots, n$ where 0 is arbitrary, and the clockwise distance from i to $i+1$ is α. The point k will be at clockwise distance $\{k\alpha\}$ from 0, where $\{r\}$ denotes the fractional part of r. Call such a circular arrangement *cyclic* and denote it by $A(\alpha)$. If the clockwise order of the points is the same in $A(\alpha_1)$ and $A(\alpha_2)$, we regard them as the same circular arrangement. Figure 3 shows the cyclic arrangement $A(3/5+\epsilon)$ of $[0,13]$ where $\epsilon > 0$ is very small.

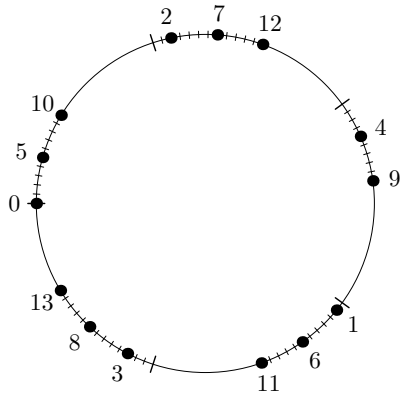


Figure 3

If $0 \leqslant a, b, c, d \leqslant n$ satisfy $a+c=b+d$, then $a\alpha + c\alpha = b\alpha + d\alpha$, so the chord from a to c is parallel to the chord from b to d in $A(\alpha)$. Hence in a cyclic arrangement all k—chords are parallel. In particular every cyclic arrangement is beautiful.

Next we show that there are exactly $N+1$ distinct cyclic arrangements. To see this, let us see how $A(\alpha)$ changes as we increase α from 0 to 1. The order of points p and q changes precisely when we cross a value $\alpha = f$ such that $\{pf\} = \{qf\}$; this can only happen if f is one of the N fractions $f_1, \ldots, f_N$. Therefore there are at most $N+1$ different cyclic arrangements.

To show they are all distinct, recall that $f_i = a_i/b_i$ and let $\epsilon > 0$ be a very small number. In the arrangement $A(f_i+\epsilon)$, point k lands at $\frac{ka_i \pmod{b_i}}{b_i} + k\epsilon$. Therefore the points are grouped into b_i clusters next to the points $0, \frac{1}{b_i}, \ldots, \frac{b_i-1}{b_i}$ of the circle. The cluster following $\frac{k}{b_i}$ contains the numbers congruent to ka_i^{-1} modulo b_i, listed clockwise in increasing order. It follows that the first number after 0 in $A(f_i+\epsilon)$ is b_i, and the first number after 0 which is less than b_i is $a_i^{-1} \pmod{b_i}$, which uniquely determines a_i. In this way we can recover f_i from the cyclic arrangement. Note also that $A(f_i+\epsilon)$ is not the trivial arrangement where we list $0, 1, \ldots, n$ in order clockwise. It follows that the $N+1$ cyclic arrangements $A(\epsilon), A(f_1+\epsilon), \ldots, A(f_N+\epsilon)$ are distinct.

Let us record an observation which will be useful later:

$$\text{if } f_i < \alpha < f_{i+1} \text{ then 0 is immediately after } b_{i+1} \text{ and before } b_i \text{ in } A(\alpha). \tag{2}$$

Indeed, we already observed that b_i is the first number after 0 in $A(f_i+\epsilon) = A(\alpha)$. Similarly we see that b_{i+1} is the last number before 0 in $A(f_{i+1}-\epsilon) = A(\alpha)$.

Finally, we show that any beautiful arrangement of $[0, n]$ is cyclic by induction on n. For $n \leqslant 2$ the result is clear. Now assume that all beautiful arrangements of $[0, n-1]$ are cyclic, and consider a beautiful arrangement A of $[0, n]$. The subarrangement $A_{n-1} = A\backslash\{n\}$ of $[0, n-1]$ obtained by deleting n is cyclic; say $A_{n-1} = A_{n-1}(\alpha)$.

Let α be between the consecutive fractions $\frac{p_1}{q_1} < \frac{p_2}{q_2}$ among the irreducible fractions of denominator at most $n-1$. There is at most one fraction $\frac{i}{n}$ in $(\frac{p_1}{q_1}, \frac{p_2}{q_2})$, since $\frac{i}{n} < \frac{i}{n-1} \leqslant \frac{i+1}{n}$ for $0 < i \leqslant n-1$.

Case 1. There is no fraction with denominator n between $\frac{p_1}{q_1}$ and $\frac{p_2}{q_2}$.

In this case the only cyclic arrangement extending $A_{n-1}(\alpha)$ is $A_n(\alpha)$. We know that A and $A_n(\alpha)$ can only differ in the position of n. Assume n is immediately after x and before y in $A_n(\alpha)$. Since the neighbors of 0 are q_1 and q_2 by (2), we have $x, y \geqslant 1$.

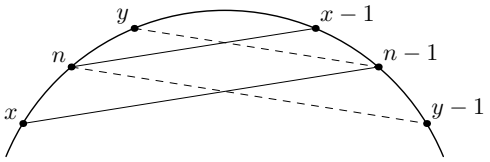


Figure 4

In $A_n(\alpha)$ the chord from $n-1$ to x is parallel and adjacent to the chord from n to $x-1$, so $n-1$ is between $x-1$ and x in clockwise order, as shown in Figure 4. Similarly, $n-1$ is between y and $y-1$. Therefore x, y, $x-1$, $n-1$, and $y-1$ occur in this order in $A_n(\alpha)$ and hence in A (possibly with $y = x-1$ or $x = y-1$).

Now, A may only differ from $A_n(\alpha)$ in the location of n. In A, since the chord from $n-1$ to x and the chord from n to $x-1$ do not intersect, n is between x and $n-1$. Similarly, n is between $n-1$ and y. Then n must be between x and y and $A = A_n(\alpha)$. Therefore A is cyclic as desired.

Case 2. There is exactly one i with $\frac{p_1}{q_1} < \frac{i}{n} < \frac{p_2}{q_2}$.

In this case there exist two cyclic arrangements $A_n(\alpha_1)$ and $A_n(\alpha_2)$ of the numbers $0, \ldots, n$ extending $A_{n-1}(\alpha)$, where $\frac{p_1}{q_1} < \alpha_1 < \frac{i}{n}$ and $\frac{i}{n} < \alpha_2 < \frac{p_2}{q_2}$. In $A_{n-1}(\alpha)$, 0 is the only number between q_2 and q_1 by (2). For the same reason, n is between q_2 and 0 in $A_n(\alpha_1)$, and between 0 and q_1 in $A_n(\alpha_2)$.

Letting $x = q_2$ and $y = q_1$, the argument of Case 1 tells us that n must be between x and y in A. Therefore A must equal $A_n(\alpha_1)$ or $A_n(\alpha_2)$, and therefore it is cyclic.

This concludes the proof that every beautiful arrangement is cyclic. It follows that there are exactly $N+1$ beautiful arrangements of $[0, n]$ as we wished to show.

C8. Players A and B play a paintful game on the real line. Player A has a pot of paint with four units of black ink. A quantity p of this ink suffices to blacken a (closed) real interval of length p. In every round, player A picks some positive integer m and provides $1/2^m$ units of ink from the pot. Player B then picks an integer k and blackens the interval from $k/2^m$ to $(k+1)/2^m$ (some parts of this interval may have been blackened before). The goal of player A is to reach a situation where the pot is empty and the interval $[0,1]$ is not completely blackened.

Decide whether there exists a strategy for player A to win in a finite number of moves.

(Austria)

Answer. No. Such a strategy for player A does not exist.

Solution. We will present a strategy for player B that guarantees that the interval $[0,1]$ is completely blackened, once the paint pot has become empty.

At the beginning of round r, let x_r denote the largest real number for which the interval between 0 and x_r has already been blackened; for completeness we define $x_1 = 0$. Let m be the integer picked by player A in this round; we define an integer y_r by

$$\frac{y_r}{2^m} \leqslant x_r < \frac{y_r+1}{2^m}.$$

Note that $I_0^r = [y_r/2^m, (y_r+1)/2^m]$ is the leftmost interval that may be painted in round r and that still contains some uncolored point.

Player B now looks at the *next* interval $I_1^r = [(y_r+1)/2^m, (y_r+2)/2^m]$. If I_1^r still contains an uncolored point, then player B blackens the interval I_1^r; otherwise he blackens the interval I_0^r. We make the convention that, at the beginning of the game, the interval $[1,2]$ is already blackened; thus, if $y_r + 1 = 2^m$, then B blackens I_0^r.

Our aim is to estimate the amount of ink used after each round. Firstly, we will prove by induction that, if before rth round the segment $[0,1]$ is not completely colored, then, before this move,

(i) the amount of ink used for the segment $[0, x_r]$ is at most $3x_r$; and

(ii) for every m, B has blackened at most one interval of length $1/2^m$ to the right of x_r.

Obviously, these conditions are satisfied for $r = 0$. Now assume that they were satisfied before the rth move, and consider the situation after this move; let m be the number A has picked at this move.

If B has blackened the interval I_1^r at this move, then $x_{r+1} = x_r$, and (i) holds by the induction hypothesis. Next, had B blackened before the rth move any interval of length $1/2^m$ to the right of x_r, this interval would necessarily coincide with I_1^r. By our strategy, this cannot happen. So, condition (ii) also remains valid.

Assume now that B has blackened the interval I_0^r at the rth move, but the interval $[0,1]$ still contains uncolored parts (which means that I_1^r is contained in $[0,1]$). Then condition (ii) clearly remains true, and we need to check (i) only. In our case, the intervals I_0^r and I_1^r are completely colored after the rth move, so x_{r+1} either reaches the right endpoint of I_1 or moves even further to the right. So, $x_{r+1} = x_r + \alpha$ for some $\alpha > 1/2^m$.

Next, any interval blackened by B before the rth move which intersects (x_r, x_{r+1}) should be contained in $[x_r, x_{r+1}]$; by (ii), all such intervals have different lengths not exceeding $1/2^m$, so the total amount of ink used for them is less than $2/2^m$. Thus, the amount of ink used for the segment $[0, x_{r+1}]$ does not exceed the sum of $2/2^m$, $3x_r$ (used for $[0, x_r]$), and $1/2^m$ used for the

segment I_0^r. In total it gives at most $3(x_r + 1/2^m) < 3(x_r + \alpha) = 3x_{r+1}$. Thus condition (i) is also verified in this case. The claim is proved.

Finally, we can perform the desired estimation. Consider any situation in the game, say after the $(r-1)$st move; assume that the segment $[0, 1]$ is not completely black. By (ii), in the segment $[x_r, 1]$ player B has colored several segments of different lengths; all these lengths are negative powers of 2 not exceeding $1 - x_r$; thus the total amount of ink used for this interval is at most $2(1 - x_r)$. Using (i), we obtain that the total amount of ink used is at most $3x_r + 2(1 - x_r) < 3$. Thus the pot is not empty, and therefore A never wins.

Comment 1. Notice that this strategy works even if the pot contains initially only 3 units of ink.

Comment 2. There exist other strategies for B allowing him to prevent emptying the pot before the whole interval is colored. On the other hand, let us mention some idea which *does not* work.

Player B could try a strategy in which the set of blackened points in each round is an interval of the type $[0, x]$. Such a strategy cannot work (even if there is more ink available). Indeed, under the assumption that B uses such a strategy, let us prove by induction on s the following statement:

For any positive integer s, player A has a strategy picking only positive integers $m \leqslant s$ in which, if player B ever paints a point $x \geqslant 1 - 1/2^s$ then after some move, exactly the interval $[0, 1 - 1/2^s]$ is blackened, and the amount of ink used up to this moment is at least $s/2$.

For the base case $s = 1$, player A just picks $m = 1$ in the first round. If for some positive integer k player A has such a strategy, for $s + 1$ he can first rescale his strategy to the interval $[0, 1/2]$ (sending in each round half of the amount of ink he would give by the original strategy). Thus, after some round, the interval $[0, 1/2 - 1/2^{s+1}]$ becomes blackened, and the amount of ink used is at least $s/4$. Now player A picks $m = 1/2$, and player B spends $1/2$ unit of ink to blacken the interval $[0, 1/2]$. After that, player A again rescales his strategy to the interval $[1/2, 1]$, and player B spends at least $s/4$ units of ink to blacken the interval $[1/2, 1 - 1/2^{s+1}]$, so he spends in total at least $s/4 + 1/2 + s/4 = (s + 1)/2$ units of ink.

Comment 3. In order to avoid finiteness issues, the statement could be replaced by the following one:

> *Players A and B play a paintful game on the real numbers. Player A has a paint pot with four units of black ink. A quantity p of this ink suffices to blacken a (closed) real interval of length p. In the beginning of the game, player A chooses (and announces) a positive integer N. In every round, player A picks some positive integer $m \leqslant N$ and provides $1/2^m$ units of ink from the pot. The player B picks an integer k and blackens the interval from $k/2^m$ to $(k + 1)/2^m$ (some parts of this interval may happen to be blackened before). The goal of player A is to reach a situation where the pot is empty and the interval $[0, 1]$ is not completely blackened.*
>
> *Decide whether there exists a strategy for player A to win.*

However, the Problem Selection Committee believes that this version may turn out to be harder than the original one.

Geometry

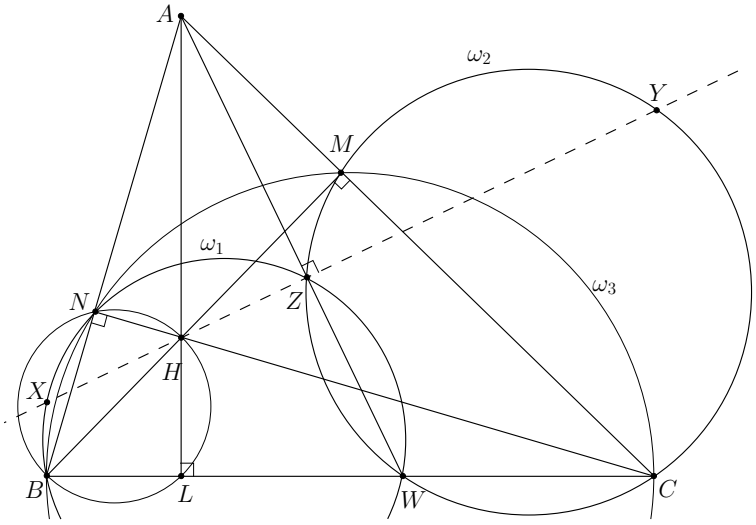
G1. Let ABC be an acute-angled triangle with orthocenter H, and let W be a point on side BC. Denote by M and N the feet of the altitudes from B and C, respectively. Denote by ω_1 the circumcircle of BWN, and let X be the point on ω_1 which is diametrically opposite to W. Analogously, denote by ω_2 the circumcircle of CWM, and let Y be the point on ω_2 which is diametrically opposite to W. Prove that X, Y and H are collinear.

(Thaliand)

Solution. Let L be the foot of the altitude from A, and let Z be the second intersection point of circles ω_1 and ω_2, other than W. We show that X, Y, Z and H lie on the same line.

Due to $\angle BNC = \angle BMC = 90^\circ$, the points B, C, N and M are concyclic; denote their circle by ω_3. Observe that the line WZ is the radical axis of ω_1 and ω_2; similarly, BN is the radical axis of ω_1 and ω_3, and CM is the radical axis of ω_2 and ω_3. Hence $A = BN \cap CM$ is the radical center of the three circles, and therefore WZ passes through A.

Since WX and WY are diameters in ω_1 and ω_2, respectively, we have $\angle WZX = \angle WZY = 90^\circ$, so the points X and Y lie on the line through Z, perpendicular to WZ.



The quadrilateral $BLHN$ is cyclic, because it has two opposite right angles. From the power of A with respect to the circles ω_1 and $BLHN$ we find $AL \cdot AH = AB \cdot AN = AW \cdot AZ$. If H lies on the line AW then this implies $H = Z$ immediately. Otherwise, by $\frac{AZ}{AH} = \frac{AL}{AW}$ the triangles AHZ and AWL are similar. Then $\angle HZA = \angle WLA = 90^\circ$, so the point H also lies on the line XYZ.

Comment. The original proposal also included a second statement:

> *Let P be the point on ω_1 such that WP is parallel to CN, and let Q be the point on ω_2 such that WQ is parallel to BM. Prove that P, Q and H are collinear if and only if $BW = CW$ or $AW \perp BC$.*

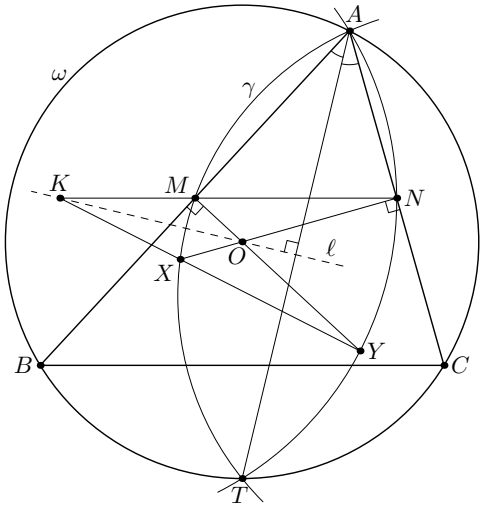
The Problem Selection Committee considered the first part more suitable for the competition.

G2. Let ω be the circumcircle of a triangle ABC. Denote by M and N the midpoints of the sides AB and AC, respectively, and denote by T the midpoint of the arc BC of ω not containing A. The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y, respectively; assume that X and Y lie inside the triangle ABC. The lines MN and XY intersect at K. Prove that $KA = KT$.

(Iran)

Solution 1. Let O be the center of ω, thus $O = MY \cap NX$. Let ℓ be the perpendicular bisector of AT (it also passes through O). Denote by r the operation of reflection about ℓ. Since AT is the angle bisector of $\angle BAC$, the line $r(AB)$ is parallel to AC. Since $OM \perp AB$ and $ON \perp AC$, this means that the line $r(OM)$ is parallel to the line ON and passes through O, so $r(OM) = ON$. Finally, the circumcircle γ of the triangle AMT is symmetric about ℓ, so $r(\gamma) = \gamma$. Thus the point M maps to the common point of ON with the arc AMT of γ — that is, $r(M) = X$.

Similarly, $r(N) = Y$. Thus, we get $r(MN) = XY$, and the common point K of MN nd XY lies on ℓ. This means exactly that $KA = KT$.



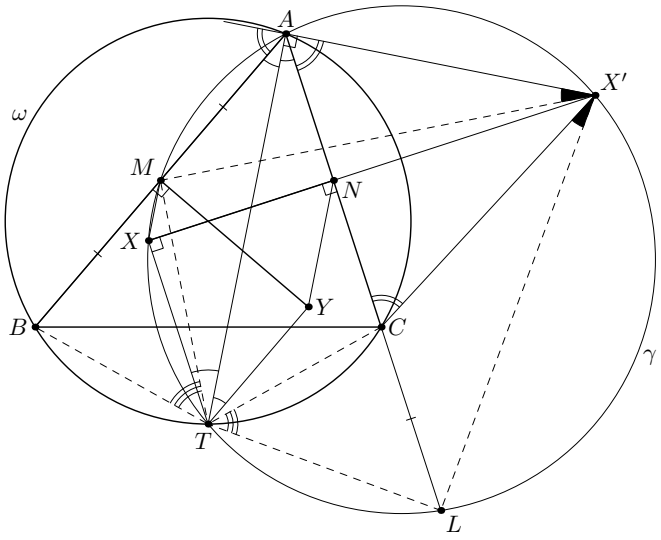
Solution 2. Let L be the second common point of the line AC with the circumcircle γ of the triangle AMT. From the cyclic quadrilaterals $ABTC$ and $AMTL$ we get $\angle BTC = 180^\circ - \angle BAC = \angle MTL$, which implies $\angle BTM = \angle CTL$. Since AT is an angle bisector in these quadrilaterals, we have $BT = TC$ and $MT = TL$. Thus the triangles BTM and CTL are congruent, so $CL = BM = AM$.

Let X' be the common point of the line NX with the external bisector of $\angle BAC$; notice that it lies outside the triangle ABC. Then we have $\angle TAX' = 90^\circ$ and $X'A = X'C$, so we get $\angle X'AM = 90^\circ + \angle BAC/2 = 180^\circ - \angle X'AC = 180^\circ - \angle X'CA = \angle X'CL$. Thus the triangles $X'AM$ and $X'CL$ are congruent, and therefore

$$\angle MX'L = \angle AX'C + (\angle CX'L - \angle AX'M) = \angle AX'C = 180^\circ - 2\angle X'AC = \angle BAC = \angle MAL.$$

This means that X' lies on γ.

Thus we have $\angle TXN = \angle TXX' = \angle TAX' = 90^\circ$, so $TX \parallel AC$. Then $\angle XTA = \angle TAC = \angle TAM$, so the cyclic quadrilateral $MATX$ is an isosceles trapezoid. Similarly, $NATY$ is an isosceles trapezoid, so again the lines MN and XY are the reflections of each other about the perpendicular bisector of AT. Thus K belongs to this perpendicular bisector.



Comment. There are several different ways of showing that the points X and M are symmetrical with respect to ℓ. For instance, one can show that the quadrilaterals $AMON$ and $TXOY$ are congruent. We chose Solution 1 as a simple way of doing it. On the other hand, Solution 2 shows some other interesting properties of the configuration.

Let us define Y', analogously to X', as the common point of MY and the external bisector of $\angle BAC$. One may easily see that in general the lines MN and $X'Y'$ (which is the external bisector of $\angle BAC$) do not intersect on the perpendicular bisector of AT. Thus, any solution should involve some argument using the choice of the intersection points X and Y.

G3. In a triangle ABC, let D and E be the feet of the angle bisectors of angles A and B, respectively. A rhombus is inscribed into the quadrilateral $AEDB$ (all vertices of the rhombus lie on different sides of $AEDB$). Let φ be the non-obtuse angle of the rhombus. Prove that $\varphi \leqslant \max\{\angle BAC, \angle ABC\}$.

(Serbia)

Solution 1. Let K, L, M, and N be the vertices of the rhombus lying on the sides AE, ED, DB, and BA, respectively. Denote by $d(X, YZ)$ the distance from a point X to a line YZ. Since D and E are the feet of the bisectors, we have $d(D, AB) = d(D, AC)$, $d(E, AB) = d(E, BC)$, and $d(D, BC) = d(E, AC) = 0$, which implies

$$d(D, AC) + d(D, BC) = d(D, AB) \quad \text{and} \quad d(E, AC) + d(E, BC) = d(E, AB).$$

Since L lies on the segment DE and the relation $d(X, AC) + d(X, BC) = d(X, AB)$ is linear in X inside the triangle, these two relations imply

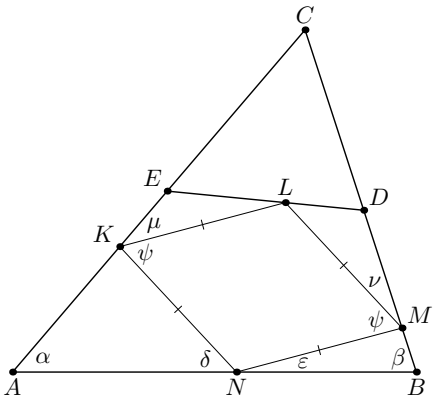
$$d(L, AC) + d(L, BC) = d(L, AB). \tag{1}$$

Denote the angles as in the figure below, and denote $a = KL$. Then we have $d(L, AC) = a \sin\mu$ and $d(L, BC) = a \sin\nu$. Since $KLMN$ is a parallelogram lying on one side of AB, we get

$$d(L, AB) = d(L, AB) + d(N, AB) = d(K, AB) + d(M, AB) = a(\sin\delta + \sin\varepsilon).$$

Thus the condition (1) reads

$$\sin\mu + \sin\nu = \sin\delta + \sin\varepsilon. \tag{2}$$



If one of the angles α and β is non-acute, then the desired inequality is trivial. So we assume that $\alpha, \beta < \pi/2$. It suffices to show then that $\psi = \angle NKL \leqslant \max\{\alpha, \beta\}$.

Assume, to the contrary, that $\psi > \max\{\alpha, \beta\}$. Since $\mu + \psi = \angle CKN = \alpha + \delta$, by our assumption we obtain $\mu = (\alpha - \psi) + \delta < \delta$. Similarly, $\nu < \varepsilon$. Next, since $KN \parallel ML$, we have $\beta = \delta + \nu$, so $\delta < \beta < \pi/2$. Similarly, $\varepsilon < \pi/2$. Finally, by $\mu < \delta < \pi/2$ and $\nu < \varepsilon < \pi/2$, we obtain

$$\sin\mu < \sin\delta \quad \text{and} \quad \sin\nu < \sin\varepsilon.$$

This contradicts (2).

Comment. One can see that the equality is achieved if $\alpha = \beta$ for every rhombus inscribed into the quadrilateral $AEDB$.

G4. Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C. Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.

(Georgia)

Solution 1. Denote by ω the circumcircle of the triangle ABC, and let $\angle ACB = \gamma$. Note that the condition $\gamma < \angle CBA$ implies $\gamma < 90^\circ$. Since $\angle PBA = \gamma$, the line PB is tangent to ω, so $PA \cdot PC = PB^2 = PD^2$. By $\frac{PA}{PD} = \frac{PD}{PC}$ the triangles PAD and PDC are similar, and $\angle ADP = \angle DCP$.

Next, since $\angle ABQ = \angle ACB$, the triangles ABC and AQB are also similar. Then $\angle AQB = \angle ABC = \angle ARC$, which means that the points D, R, C, and Q are concyclic. Therefore $\angle DRQ = \angle DCQ = \angle ADP$.

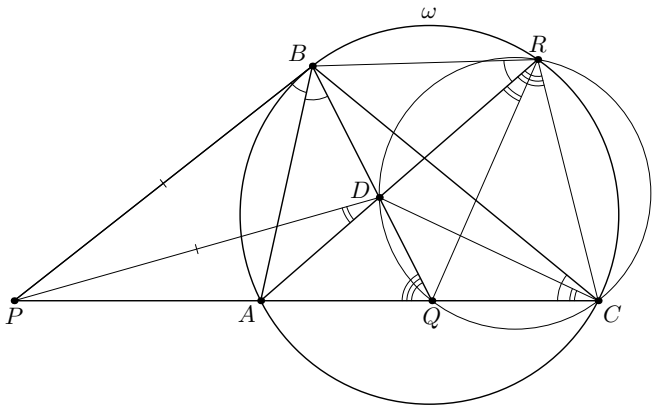


Figure 1

Now from $\angle ARB = \angle ACB = \gamma$ and $\angle PDB = \angle PBD = 2\gamma$ we get

$$\angle QBR = \angle ADB - \angle ARB = \angle ADP + \angle PDB - \angle ARB = \angle DRQ + \gamma = \angle QRB,$$

so the triangle QRB is isosceles, which yields $QB = QR$.

Solution 2. Again, denote by ω the circumcircle of the triangle ABC. Denote $\angle ACB = \gamma$. Since $\angle PBA = \gamma$, the line PB is tangent to ω.

Let E be the second intersection point of BQ with ω. If V' is any point on the ray CE beyond E, then $\angle BEV' = 180^\circ - \angle BEC = 180^\circ - \angle BAC = \angle PAB$; together with $\angle ABQ = \angle PBA$ this shows firstly, that the rays BA and CE intersect at some point V, and secondly that the triangle VEB is similar to the triangle PAB. Thus we have $\angle BVE = \angle BPA$. Next, $\angle AEV = \angle BEV - \gamma = \angle PAB - \angle ABQ = \angle AQB$; so the triangles PBQ and VAE are also similar.

Let PH be an altitude in the isosceles triangle PBD; then $BH = HD$. Let G be the intersection point of PH and AB. By the symmetry with respect to PH, we have $\angle BDG = \angle DBG = \gamma = \angle BEA$; thus $DG \parallel AE$ and hence $\frac{BG}{GA} = \frac{BD}{DE}$. Thus the points G and D correspond to each other in the similar triangles PAB and VEB, so $\angle DVB = \angle GPB = 90^\circ - \angle PBQ = 90^\circ - \angle VAE$. Thus $VD \perp AE$.

Let T be the common point of VD and AE, and let DS be an altitude in the triangle BDR. The points S and T are the feet of corresponding altitudes in the similar triangles ADE and BDR, so $\frac{BS}{SR} = \frac{AT}{TE}$. On the other hand, the points T and H are feet of corresponding altitudes in the similar triangles VAE and PBQ, so $\frac{AT}{TE} = \frac{BH}{HQ}$. Thus $\frac{BS}{SR} = \frac{AT}{TE} = \frac{BH}{HQ}$, and the triangles BHS and BQR are similar.

Finally, SH is a median in the right-angled triangle SBD; so $BH = HS$, and hence $BQ = QR$.

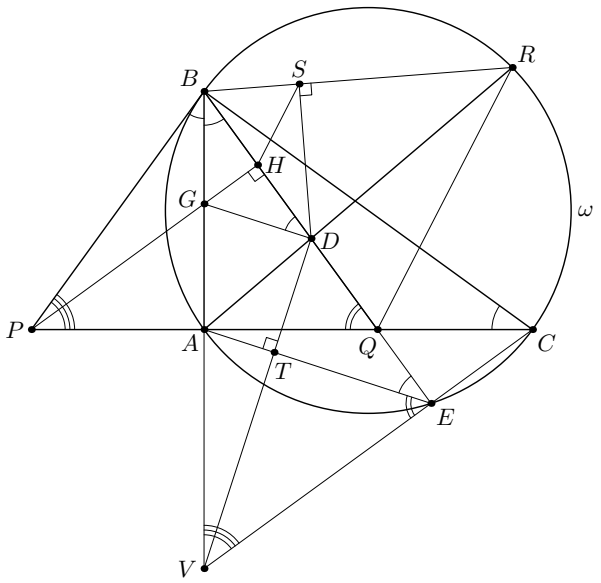


Figure 2

Solution 3. Denote by ω and O the circumcircle of the triangle ABC and its center, respectively. From the condition $\angle PBA = \angle BCA$ we know that BP is tangent to ω.

Let E be the second point of intersection of ω and BD. Due to the isosceles triangle BDP, the tangent of ω at E is parallel to DP and consequently it intersects BP at some point L. Of course, $PD \parallel LE$. Let M be the midpoint of BE, and let H be the midpoint of BR. Notice that $\angle AEB = \angle ACB = \angle ABQ = \angle ABE$, so A lies on the perpendicular bisector of BE; thus the points L, A, M, and O are collinear. Let ω_1 be the circle with diameter BO. Let $Q' = HO \cap BE$; since HO is the perpendicular bisector of BR, the statement of the problem is equivalent to $Q' = Q$.

Consider the following sequence of projections (see Fig. 3).

1. Project the line BE to the line LB through the center A. (This maps Q to P.)
2. Project the line LB to BE in parallel direction with LE. ($P \mapsto D$.)
3. Project the line BE to the circle ω through its point A. ($D \mapsto R$.)
4. Scale ω by the ratio $\frac{1}{2}$ from the point B to the circle ω_1. ($R \mapsto H$.)
5. Project ω_1 to the line BE through its point O. ($H \mapsto Q'$.)

We prove that the composition of these transforms, which maps the line BE to itself, is the identity. To achieve this, it suffices to show three fixed points. An obvious fixed point is B which is fixed by all the transformations above. Another fixed point is M, its path being $M \mapsto L \mapsto E \mapsto E \mapsto M \mapsto M$.

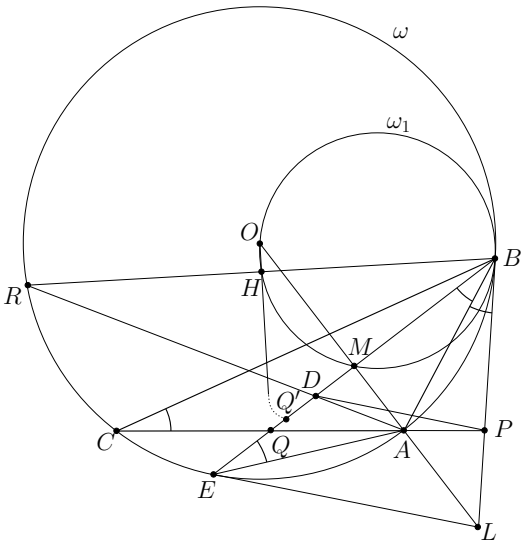


Figure 3

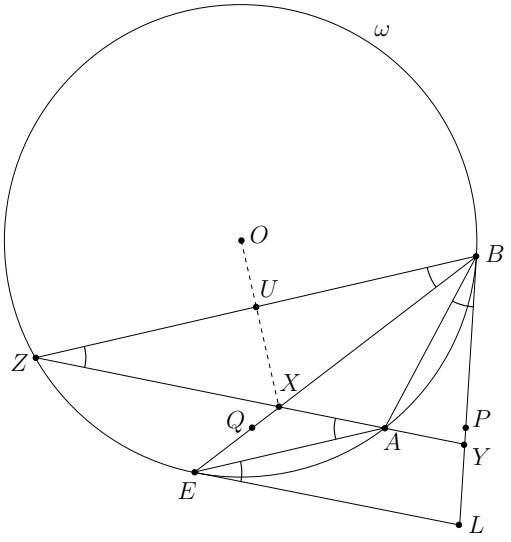


Figure 4

In order to show a third fixed point, draw a line parallel with LE through A; let that line intersect BE, LB and ω at X, Y and $Z \neq A$, respectively (see Fig. 4). We show that X is a fixed point. The images of X at the first three transformations are $X \mapsto Y \mapsto X \mapsto Z$. From $\angle XBZ = \angle EAZ = \angle AEL = \angle LBA = \angle BZX$ we can see that the triangle XBZ is isosceles. Let U be the midpoint of BZ; then the last two transformations do $Z \mapsto U \mapsto X$, and the point X is fixed.

Comment. Verifying that the point E is fixed seems more natural at first, but it appears to be less straightforward. Here we outline a possible proof.

Let the images of E at the first three transforms above be F, G and I. After comparing the angles depicted in Fig. 5 (noticing that the quadrilateral $AFBG$ is cyclic) we can observe that the tangent LE of ω is parallel to BI. Then, similarly to the above reasons, the point E is also fixed.

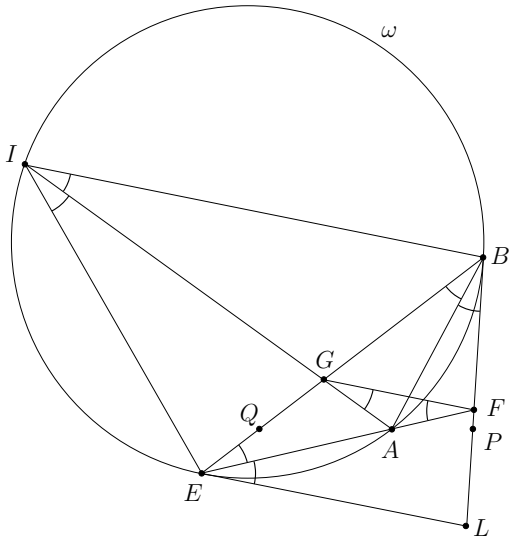


Figure 5

G5. Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD, BE, and CF are concurrent.

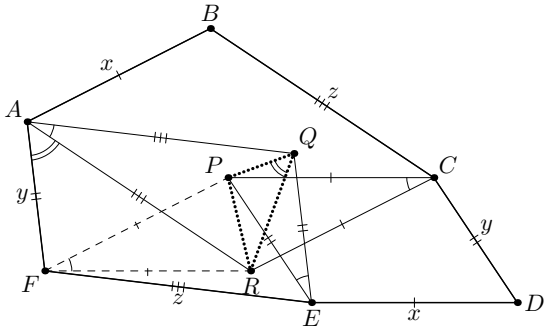
(Ukraine)

In all three solutions, we denote $\theta = \angle A - \angle D = \angle C - \angle F = \angle E - \angle B$ and assume without loss of generality that $\theta \geqslant 0$.

Solution 1. Let $x = AB = DE$, $y = CD = FA$, $z = EF = BC$. Consider the points P, Q, and R such that the quadrilaterals $CDEP$, $EFAQ$, and $ABCR$ are parallelograms. We compute

$$\begin{aligned}\angle PEQ &= \angle FEQ + \angle DEP - \angle E = (180^\circ - \angle F) + (180^\circ - \angle D) - \angle E \\ &= 360^\circ - \angle D - \angle E - \angle F = \tfrac{1}{2}\big(\angle A + \angle B + \angle C - \angle D - \angle E - \angle F\big) = \theta/2.\end{aligned}$$

Similarly, $\angle QAR = \angle RCP = \theta/2$.



If $\theta = 0$, since $\triangle RCP$ is isosceles, $R = P$. Therefore $AB \parallel RC = PC \parallel ED$, so $ABDE$ is a parallelogram. Similarly, $BCEF$ and $CDFA$ are parallelograms. It follows that AD, BE and CF meet at their common midpoint.

Now assume $\theta > 0$. Since $\triangle PEQ$, $\triangle QAR$, and $\triangle RCP$ are isosceles and have the same angle at the apex, we have $\triangle PEQ \sim \triangle QAR \sim \triangle RCP$ with ratios of similarity $y : z : x$. Thus

$$\triangle PQR \text{ is similar to the triangle with sidelengths } y, z, \text{ and } x. \tag{1}$$

Next, notice that

$$\frac{RQ}{QP} = \frac{z}{y} = \frac{RA}{AF}$$

and, using directed angles between rays,

$$\begin{aligned}\measuredangle(RQ, QP) &= \measuredangle(RQ, QE) + \measuredangle(QE, QP) \\ &= \measuredangle(RQ, QE) + \measuredangle(RA, RQ) = \measuredangle(RA, QE) = \measuredangle(RA, AF).\end{aligned}$$

Thus $\triangle PQR \sim \triangle FAR$. Since $FA = y$ and $AR = z$, (1) then implies that $FR = x$. Similarly $FP = x$. Therefore $CRFP$ is a rhombus.

We conclude that CF is the perpendicular bisector of PR. Similarly, BE is the perpendicular bisector of PQ and AD is the perpendicular bisector of QR. It follows that AD, BE, and CF are concurrent at the circumcenter of PQR.

Solution 2. Let $X = CD \cap EF$, $Y = EF \cap AB$, $Z = AB \cap CD$, $X' = FA \cap BC$, $Y' = BC \cap DE$, and $Z' = DE \cap FA$. From $\angle A + \angle B + \angle C = 360^\circ + \theta/2$ we get $\angle A + \angle B > 180^\circ$ and $\angle B + \angle C > 180^\circ$, so Z and X' are respectively on the opposite sides of BC and AB from the hexagon. Similar conclusions hold for X, Y, Y', and Z'. Then

$$\angle YZX = \angle B + \angle C - 180^\circ = \angle E + \angle F - 180^\circ = \angle Y'Z'X',$$

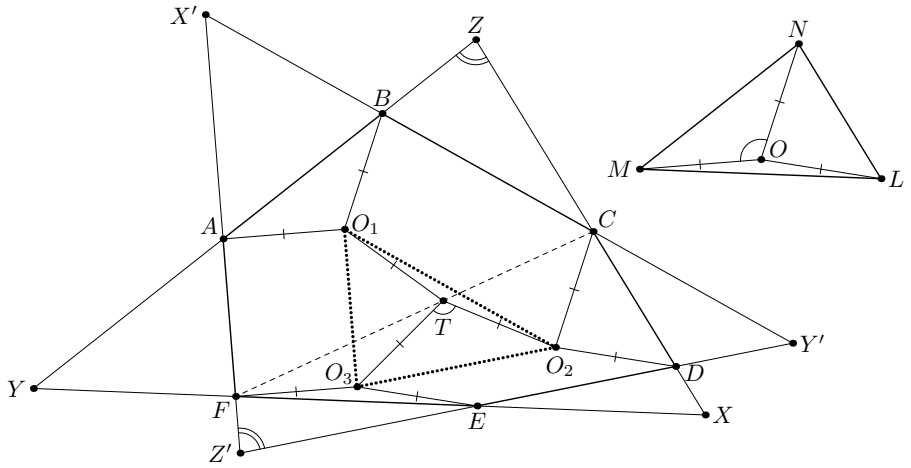
and similarly $\angle ZXY = \angle Z'X'Y'$ and $\angle XYZ = \angle X'Y'Z'$, so $\triangle XYZ \sim \triangle X'Y'Z'$. Thus there is a rotation R which sends $\triangle XYZ$ to a triangle with sides parallel to $\triangle X'Y'Z'$. Since $AB = DE$ we have $R\big(\overrightarrow{AB}\big) = \overrightarrow{DE}$. Similarly, $R\big(\overrightarrow{CD}\big) = \overrightarrow{FA}$ and $R\big(\overrightarrow{EF}\big) = \overrightarrow{BC}$. Therefore

$$\overrightarrow{0} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA} = \big(\overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{EF}\big) + R\big(\overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{EF}\big).$$

If R is a rotation by 180°, then any two opposite sides of our hexagon are equal and parallel, so the three diagonals meet at their common midpoint. Otherwise, we must have

$$\overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{EF} = \overrightarrow{0},$$

or else we would have two vectors with different directions whose sum is $\overrightarrow{0}$.



This allows us to consider a triangle LMN with $\overrightarrow{LM} = \overrightarrow{EF}$, $\overrightarrow{MN} = \overrightarrow{AB}$, and $\overrightarrow{NL} = \overrightarrow{CD}$. Let O be the circumcenter of $\triangle LMN$ and consider the points O_1, O_2, O_3 such that $\triangle AO_1B$, $\triangle CO_2D$, and $\triangle EO_3F$ are translations of $\triangle MON$, $\triangle NOL$, and $\triangle LOM$, respectively. Since FO_3 and AO_1 are translations of MO, quadrilateral AFO_3O_1 is a parallelogram and $O_3O_1 = FA = CD = NL$. Similarly, $O_1O_2 = LM$ and $O_2O_3 = MN$. Therefore $\triangle O_1O_2O_3 \cong \triangle LMN$. Moreover, by means of the rotation R one may check that these triangles have the same orientation.

Let T be the circumcenter of $\triangle O_1O_2O_3$. We claim that AD, BE, and CF meet at T. Let us show that C, T, and F are collinear. Notice that $CO_2 = O_2T = TO_3 = O_3F$ since they are all equal to the circumradius of $\triangle LMN$. Therefore $\triangle TO_3F$ and $\triangle CO_2T$ are isosceles. Using directed angles between rays again, we get

$$\measuredangle(TF, TO_3) = \measuredangle(FO_3, FT) \qquad \text{and} \qquad \measuredangle(TO_2, TC) = \measuredangle(CT, CO_2). \tag{2}$$

Also, T and O are the circumcenters of the congruent triangles $\triangle O_1O_2O_3$ and $\triangle LMN$ so we have $\measuredangle(TO_3, TO_2) = \measuredangle(ON, OM)$. Since CO_2 and FO_3 are translations of NO and MO respectively, this implies

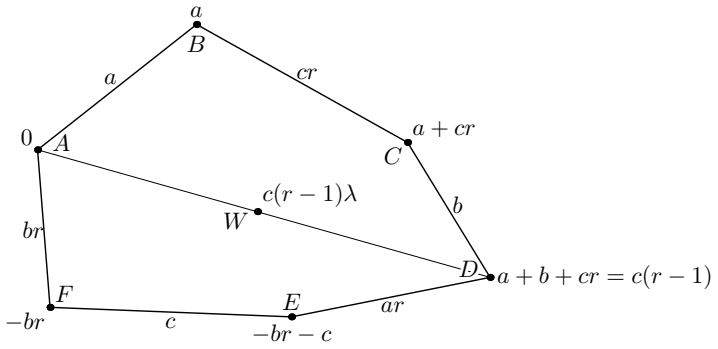
$$\measuredangle(TO_3, TO_2) = \measuredangle(CO_2, FO_3). \tag{3}$$

Adding the three equations in (2) and (3) gives

$$\measuredangle(TF, TC) = \measuredangle(CT, FT) = -\measuredangle(TF, TC)$$

which implies that T is on CF. Analogous arguments show that it is on AD and BE also. The desired result follows.

Solution 3. Place the hexagon on the complex plane, with A at the origin and vertices labelled clockwise. Now A, B, C, D, E, F represent the corresponding complex numbers. Also consider the complex numbers a, b, c, a', b', c' given by $B - A = a$, $D - C = b$, $F - E = c$, $E - D = a'$, $A - F = b'$, and $C - B = c'$. Let $k = |a|/|b|$. From $a/b' = -ke^{i\angle A}$ and $a'/b = -ke^{i\angle D}$ we get that $(a'/a)(b'/b) = e^{-i\theta}$ and similarly $(b'/b)(c'/c) = e^{-i\theta}$ and $(c'/c)(a'/a) = e^{-i\theta}$. It follows that $a' = ar$, $b' = br$, and $c' = cr$ for a complex number r with $|r| = 1$, as shown below.



We have

$$0 = a + cr + b + ar + c + br = (a + b + c)(1 + r).$$

If $r = -1$, then the hexagon is centrally symmetric and its diagonals intersect at its center of symmetry. Otherwise

$$a + b + c = 0.$$

Therefore

$$A = 0, \quad B = a, \quad C = a + cr, \quad D = c(r - 1), \quad E = -br - c, \quad F = -br.$$

Now consider a point W on AD given by the complex number $c(r-1)\lambda$, where λ is a real number with $0 < \lambda < 1$. Since $D \neq A$, we have $r \neq 1$, so we can define $s = 1/(r-1)$. From $r\overline{r} = |r|^2 = 1$ we get

$$1 + s = \frac{r}{r-1} = \frac{r}{r - r\overline{r}} = \frac{1}{1 - \overline{r}} = -\overline{s}.$$

Now,

$$\begin{aligned} W \text{ is on } BE &\iff c(r-1)\lambda - a \parallel a - (-br - c) = b(r-1) \iff c\lambda - as \parallel b \\ &\iff -a\lambda - b\lambda - as \parallel b \iff a(\lambda + s) \parallel b. \end{aligned}$$

One easily checks that $r \neq \pm 1$ implies that $\lambda + s \neq 0$ since s is not real. On the other hand,

$$\begin{aligned} W \text{ on } CF &\iff c(r-1)\lambda + br \parallel -br - (a + cr) = a(r-1) \iff c\lambda + b(1+s) \parallel a \\ &\iff -a\lambda - b\lambda - b\overline{s} \parallel a \iff b(\lambda + \overline{s}) \parallel a \iff b \parallel a(\lambda + s), \end{aligned}$$

where in the last step we use that $(\lambda + s)(\lambda + \overline{s}) = |\lambda + s|^2 \in \mathbb{R}_{>0}$. We conclude that $AD \cap BE = CF \cap BE$, and the desired result follows.

G6. Let the excircle of the triangle ABC lying opposite to A touch its side BC at the point A_1. Define the points B_1 and C_1 analogously. Suppose that the circumcentre of the triangle $A_1B_1C_1$ lies on the circumcircle of the triangle ABC. Prove that the triangle ABC is right-angled.

(Russia)

Solution 1. Denote the circumcircles of the triangles ABC and $A_1B_1C_1$ by Ω and Γ, respectively. Denote the midpoint of the arc CB of Ω containing A by A_0, and define B_0 as well as C_0 analogously. By our hypothesis the centre Q of Γ lies on Ω.

Lemma. One has $A_0B_1 = A_0C_1$. Moreover, the points A, A_0, B_1, and C_1 are concyclic. Finally, the points A and A_0 lie on the same side of B_1C_1. Similar statements hold for B and C.

Proof. Let us consider the case $A = A_0$ first. Then the triangle ABC is isosceles at A, which implies $AB_1 = AC_1$ while the remaining assertions of the Lemma are obvious. So let us suppose $A \neq A_0$ from now on.

By the definition of A_0, we have $A_0B = A_0C$. It is also well known and easy to show that $BC_1 = CB_1$. Next, we have $\angle C_1BA_0 = \angle ABA_0 = \angle ACA_0 = \angle B_1CA_0$. Hence the triangles A_0BC_1 and A_0CB_1 are congruent. This implies $A_0C_1 = A_0B_1$, establishing the first part of the Lemma. It also follows that $\angle A_0C_1A = \angle A_0B_1A$, as these are exterior angles at the corresponding vertices C_1 and B_1 of the congruent triangles A_0BC_1 and A_0CB_1. For that reason the points A, A_0, B_1, and C_1 are indeed the vertices of some cyclic quadrilateral two opposite sides of which are AA_0 and B_1C_1. □

Now we turn to the solution. Evidently the points A_1, B_1, and C_1 lie interior to some semicircle arc of Γ, so the triangle $A_1B_1C_1$ is obtuse-angled. Without loss of generality, we will assume that its angle at B_1 is obtuse. Thus Q and B_1 lie on different sides of A_1C_1; obviously, the same holds for the points B and B_1. So, the points Q and B are on the same side of A_1C_1.

Notice that the perpendicular bisector of A_1C_1 intersects Ω at two points lying on different sides of A_1C_1. By the first statement from the Lemma, both points B_0 and Q are among these points of intersection; since they share the same side of A_1C_1, they coincide (see Figure 1).

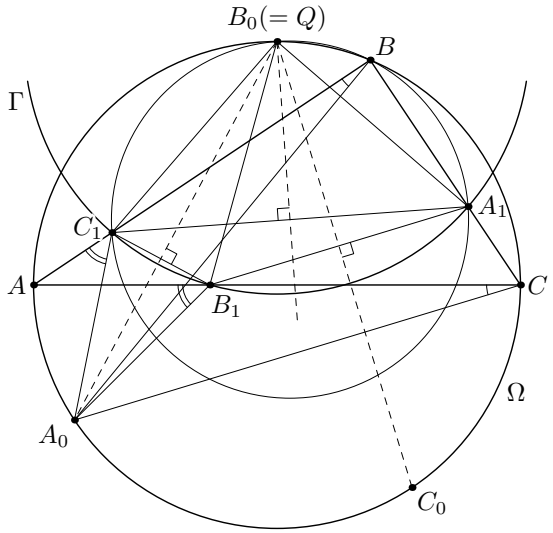


Figure 1

Now, by the first part of the Lemma again, the lines QA_0 and QC_0 are the perpendicular bisectors of B_1C_1 and A_1B_1, respectively. Thus

$$\angle C_1B_0A_1 = \angle C_1B_0B_1 + \angle B_1B_0A_1 = 2\angle A_0B_0B_1 + 2\angle B_1B_0C_0 = 2\angle A_0B_0C_0 = 180^\circ - \angle ABC,$$

recalling that A_0 and C_0 are the midpoints of the arcs CB and BA, respectively.

On the other hand, by the second part of the Lemma we have

$$\angle C_1B_0A_1 = \angle C_1BA_1 = \angle ABC.$$

From the last two equalities, we get $\angle ABC = 90^\circ$, whereby the problem is solved.

Solution 2. Let Q again denote the centre of the circumcircle of the triangle $A_1B_1C_1$, that lies on the circumcircle Ω of the triangle ABC. We first consider the case where Q coincides with one of the vertices of ABC, say $Q = B$. Then $BC_1 = BA_1$ and consequently the triangle ABC is isosceles at B. Moreover we have $BC_1 = B_1C$ in any triangle, and hence $BB_1 = BC_1 = B_1C$; similarly, $BB_1 = B_1A$. It follows that B_1 is the centre of Ω and that the triangle ABC has a right angle at B.

So from now on we may suppose $Q \notin \{A, B, C\}$. We start with the following well known fact.

Lemma. Let XYZ and $X'Y'Z'$ be two triangles with $XY = X'Y'$ and $YZ = Y'Z'$.

(*i*) If $XZ \neq X'Z'$ and $\angle YZX = \angle Y'Z'X'$, then $\angle ZXY + \angle Z'X'Y' = 180^\circ$.

(*ii*) If $\angle YZX + \angle X'Z'Y' = 180^\circ$, then $\angle ZXY = \angle Y'X'Z'$.

Proof. For both parts, we may move the triangle XYZ through the plane until $Y = Y'$ and $Z = Z'$. Possibly after reflecting one of the two triangles about YZ, we may also suppose that X and X' lie on the same side of YZ if we are in case (*i*) and on different sides if we are in case (*ii*). In both cases, the points X, Z, and X' are collinear due to the angle condition (see Fig. 2). Moreover we have $X \neq X'$, because in case (*i*) we assumed $XZ \neq X'Z'$ and in case (*ii*) these points even lie on different sides of YZ. Thus the triangle $XX'Y$ is isosceles at Y. The claim now follows by considering the equal angles at its base. □

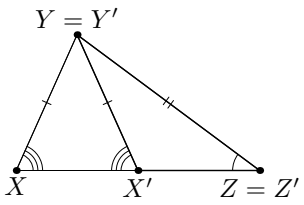


Figure 2(*i*)

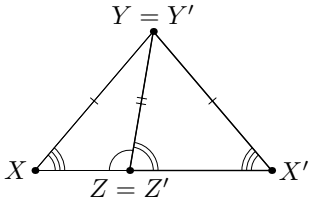


Figure 2(*ii*)

Relabeling the vertices of the triangle ABC if necessary we may suppose that Q lies in the interior of the arc AB of Ω not containing C. We will sometimes use tacitly that the six triangles QBA_1, QA_1C, QCB_1, QB_1A, QC_1A, and QBC_1 have the same orientation.

As Q cannot be the circumcentre of the triangle ABC, it is impossible that $QA = QB = QC$ and thus we may also suppose that $QC \neq QB$. Now the above Lemma (*i*) is applicable to the triangles QB_1C and QC_1B, since $QB_1 = QC_1$ and $B_1C = C_1B$, while $\angle B_1CQ = \angle C_1BQ$ holds as both angles appear over the same side of the chord QA in Ω (see Fig. 3). So we get

$$\angle CQB_1 + \angle BQC_1 = 180^\circ. \tag{1}$$

We claim that $QC = QA$. To see this, let us assume for the sake of a contradiction that $QC \neq QA$. Then arguing similarly as before but now with the triangles QA_1C and QC_1A we get

$$\angle A_1QC + \angle C_1QA = 180^\circ.$$

Adding this equation to (1), we get $\angle A_1QB_1 + \angle BQA = 360^\circ$, which is absurd as both summands lie in the interval $(0^\circ, 180^\circ)$.

This proves $QC = QA$; so the triangles QA_1C and QC_1A are congruent their sides being equal, which in turn yields

$$\angle A_1QC = \angle C_1QA. \tag{2}$$

Finally our Lemma (ii) is applicable to the triangles QA_1B and QB_1A. Indeed we have $QA_1 = QB_1$ and $A_1B = B_1A$ as usual, and the angle condition $\angle A_1BQ + \angle QAB_1 = 180^\circ$ holds as A and B lie on different sides of the chord QC in Ω. Consequently we have

$$\angle BQA_1 = \angle B_1QA. \tag{3}$$

From (1) and (3) we get

$$(\angle B_1QC + \angle B_1QA) + (\angle C_1QB - \angle BQA_1) = 180^\circ,$$

i.e. $\angle CQA + \angle A_1QC_1 = 180^\circ$. In light of (2) this may be rewritten as $2\angle CQA = 180^\circ$ and as Q lies on Ω this implies that the triangle ABC has a right angle at B.

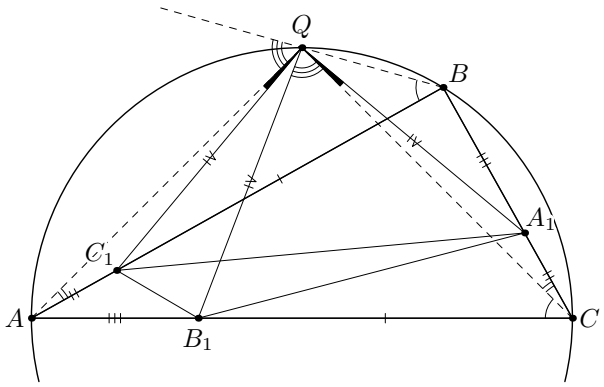


Figure 3

Comment 1. One may also check that Q is in the interior of Ω if and only if the triangle ABC is acute-angled.

Comment 2. The original proposal asked to prove the converse statement as well: if the triangle ABC is right-angled, then the point Q lies on its circumcircle. The Problem Selection Committee thinks that the above simplified version is more suitable for the competition.

Number Theory

N1. Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all positive integers m and n.

(Malaysia)

Answer. $f(n) = n$.

Solution 1. Setting $m = n = 2$ tells us that $4 + f(2) \mid 2f(2) + 2$. Since $2f(2) + 2 < 2(4 + f(2))$, we must have $2f(2) + 2 = 4 + f(2)$, so $f(2) = 2$. Plugging in $m = 2$ then tells us that $4 + f(n) \mid 4 + n$, which implies that $f(n) \leqslant n$ for all n.

Setting $m = n$ gives $n^2 + f(n) \mid nf(n) + n$, so $nf(n) + n \geqslant n^2 + f(n)$ which we rewrite as $(n-1)(f(n) - n) \geqslant 0$. Therefore $f(n) \geqslant n$ for all $n \geqslant 2$. This is trivially true for $n = 1$ also.

It follows that $f(n) = n$ for all n. This function obviously satisfies the desired property.

Solution 2. Setting $m = f(n)$ we get $f(n)(f(n)+1) \mid f(n)f(f(n)) + n$. This implies that $f(n) \mid n$ for all n.

Now let m be any positive integer, and let $p > 2m^2$ be a prime number. Note that $p > mf(m)$ also. Plugging in $n = p - mf(m)$ we learn that $m^2 + f(n)$ divides p. Since $m^2 + f(n)$ cannot equal 1, it must equal p. Therefore $p - m^2 = f(n) \mid n = p - mf(m)$. But $p - mf(m) < p < 2(p - m^2)$, so we must have $p - mf(m) = p - m^2$, i.e., $f(m) = m$.

Solution 3. Plugging $m = 1$ we obtain $1 + f(n) \leqslant f(1) + n$, so $f(n) \leqslant n + c$ for the constant $c = f(1) - 1$. Assume that $f(n) \neq n$ for some fixed n. When m is large enough (e.g. $m \geqslant \max(n, c+1)$) we have

$$mf(m) + n \leqslant m(m + c) + n \leqslant 2m^2 < 2(m^2 + f(n)),$$

so we must have $mf(m) + n = m^2 + f(n)$. This implies that

$$0 \neq f(n) - n = m(f(m) - m),$$

which is impossible for $m > |f(n) - n|$. It follows that f is the identity function.

N2. Prove that for any pair of positive integers k and n there exist k positive integers $m_1, m_2, \ldots, m_k$ such that

$$1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right)\left(1 + \frac{1}{m_2}\right)\cdots\left(1 + \frac{1}{m_k}\right).$$

(Japan)

Solution 1. We proceed by induction on k. For $k = 1$ the statement is trivial. Assuming we have proved it for $k = j - 1$, we now prove it for $k = j$.

Case 1. $n = 2t - 1$ for some positive integer t.

Observe that

$$1 + \frac{2^j - 1}{2t - 1} = \frac{2(t + 2^{j-1} - 1)}{2t} \cdot \frac{2t}{2t - 1} = \left(1 + \frac{2^{j-1} - 1}{t}\right)\left(1 + \frac{1}{2t - 1}\right).$$

By the induction hypothesis we can find $m_1, \ldots, m_{j-1}$ such that

$$1 + \frac{2^{j-1} - 1}{t} = \left(1 + \frac{1}{m_1}\right)\left(1 + \frac{1}{m_2}\right)\cdots\left(1 + \frac{1}{m_{j-1}}\right),$$

so setting $m_j = 2t - 1$ gives the desired expression.

Case 2. $n = 2t$ for some positive integer t.

Now we have

$$1 + \frac{2^j - 1}{2t} = \frac{2t + 2^j - 1}{2t + 2^j - 2} \cdot \frac{2t + 2^j - 2}{2t} = \left(1 + \frac{1}{2t + 2^j - 2}\right)\left(1 + \frac{2^{j-1} - 1}{t}\right),$$

noting that $2t + 2^j - 2 > 0$. Again, we use that

$$1 + \frac{2^{j-1} - 1}{t} = \left(1 + \frac{1}{m_1}\right)\left(1 + \frac{1}{m_2}\right)\cdots\left(1 + \frac{1}{m_{j-1}}\right).$$

Setting $m_j = 2t + 2^j - 2$ then gives the desired expression.

Solution 2. Consider the base 2 expansions of the residues of $n - 1$ and $-n$ modulo 2^k:

$$\begin{aligned} n - 1 &\equiv 2^{a_1} + 2^{a_2} + \cdots + 2^{a_r} \pmod{2^k} && \text{where } 0 \leqslant a_1 < a_2 < \ldots < a_r \leqslant k - 1, \\ -n &\equiv 2^{b_1} + 2^{b_2} + \cdots + 2^{b_s} \pmod{2^k} && \text{where } 0 \leqslant b_1 < b_2 < \ldots < b_s \leqslant k - 1. \end{aligned}$$

Since $-1 \equiv 2^0 + 2^1 + \cdots + 2^{k-1} \pmod{2^k}$, we have $\{a_1, \ldots, a_r\} \cup \{b_1 \ldots, b_s\} = \{0, 1, \ldots, k - 1\}$ and $r + s = k$. Write

$$\begin{aligned} S_p &= 2^{a_p} + 2^{a_{p+1}} + \cdots + 2^{a_r} && \text{for } 1 \leqslant p \leqslant r, \\ T_q &= 2^{b_1} + 2^{b_2} + \cdots + 2^{b_q} && \text{for } 1 \leqslant q \leqslant s. \end{aligned}$$

Also set $S_{r+1} = T_0 = 0$. Notice that $S_1 + T_s = 2^k - 1$ and $n + T_s \equiv 0 \pmod{2^k}$. We have

$$\begin{aligned} 1 + \frac{2^k - 1}{n} &= \frac{n + S_1 + T_s}{n} = \frac{n + S_1 + T_s}{n + T_s} \cdot \frac{n + T_s}{n} \\ &= \prod_{p=1}^{r} \frac{n + S_p + T_s}{n + S_{p+1} + T_s} \cdot \prod_{q=1}^{s} \frac{n + T_q}{n + T_{q-1}} \\ &= \prod_{p=1}^{r} \left(1 + \frac{2^{a_p}}{n + S_{p+1} + T_s}\right) \cdot \prod_{q=1}^{s} \left(1 + \frac{2^{b_q}}{n + T_{q-1}}\right), \end{aligned}$$

so if we define

$$m_p = \frac{n + S_{p+1} + T_s}{2^{a_p}} \quad \text{for } 1 \leqslant p \leqslant r \quad \text{and} \quad m_{r+q} = \frac{n + T_{q-1}}{2^{b_q}} \quad \text{for } 1 \leqslant q \leqslant s,$$

the desired equality holds. It remains to check that every m_i is an integer. For $1 \leqslant p \leqslant r$ we have

$$n + S_{p+1} + T_s \equiv n + T_s \equiv 0 \pmod{2^{a_p}}$$

and for $1 \leqslant q \leqslant r$ we have

$$n + T_{q-1} \equiv n + T_s \equiv 0 \pmod{2^{b_q}}.$$

The desired result follows.

N3. Prove that there exist infinitely many positive integers n such that the largest prime divisor of n^4+n^2+1 is equal to the largest prime divisor of $(n+1)^4+(n+1)^2+1$.

(Belgium)

Solution. Let p_n be the largest prime divisor of n^4+n^2+1 and let q_n be the largest prime divisor of n^2+n+1. Then $p_n = q_{n^2}$, and from

$$n^4+n^2+1 = (n^2+1)^2-n^2 = (n^2-n+1)(n^2+n+1) = ((n-1)^2+(n-1)+1)(n^2+n+1)$$

it follows that $p_n = \max\{q_n, q_{n-1}\}$ for $n \geqslant 2$. Keeping in mind that n^2-n+1 is odd, we have

$$\gcd(n^2+n+1, n^2-n+1) = \gcd(2n, n^2-n+1) = \gcd(n, n^2-n+1) = 1.$$

Therefore $q_n \neq q_{n-1}$.

To prove the result, it suffices to show that the set

$$S = \{n \in \mathbb{Z}_{\geqslant 2} \mid q_n > q_{n-1} \text{ and } q_n > q_{n+1}\}$$

is infinite, since for each $n \in S$ one has

$$p_n = \max\{q_n, q_{n-1}\} = q_n = \max\{q_n, q_{n+1}\} = p_{n+1}.$$

Suppose on the contrary that S is finite. Since $q_2 = 7 < 13 = q_3$ and $q_3 = 13 > 7 = q_4$, the set S is non-empty. Since it is finite, we can consider its largest element, say m.

Note that it is impossible that $q_m > q_{m+1} > q_{m+2} > \ldots$ because all these numbers are positive integers, so there exists a $k \geqslant m$ such that $q_k < q_{k+1}$ (recall that $q_k \neq q_{k+1}$). Next observe that it is impossible to have $q_k < q_{k+1} < q_{k+2} < \ldots$, because $q_{(k+1)^2} = p_{k+1} = \max\{q_k, q_{k+1}\} = q_{k+1}$, so let us take the smallest $\ell \geqslant k+1$ such that $q_\ell > q_{\ell+1}$. By the minimality of ℓ we have $q_{\ell-1} < q_\ell$, so $\ell \in S$. Since $\ell \geqslant k+1 > k \geqslant m$, this contradicts the maximality of m, and hence S is indeed infinite.

Comment. Once the factorization of n^4+n^2+1 is found and the set S is introduced, the problem is mainly about ruling out the case that

$$q_k < q_{k+1} < q_{k+2} < \ldots \tag{1}$$

might hold for some $k \in \mathbb{Z}_{>0}$. In the above solution, this is done by observing $q_{(k+1)^2} = \max(q_k, q_{k+1})$. Alternatively one may notice that (1) implies that $q_{j+2} - q_j \geqslant 6$ for $j \geqslant k+1$, since every prime greater than 3 is congruent to -1 or 1 modulo 6. Then there is some integer $C \geqslant 0$ such that $q_n \geqslant 3n - C$ for all $n \geqslant k$.

Now let the integer t be sufficiently large (e.g. $t = \max\{k+1, C+3\}$) and set $p = q_{t-1} \geqslant 2t$. Then $p \mid (t-1)^2+(t-1)+1$ implies that $p \mid (p-t)^2+(p-t)+1$, so p and q_{p-t} are prime divisors of $(p-t)^2+(p-t)+1$. But $p-t > t-1 \geqslant k$, so $q_{p-t} > q_{t-1} = p$ and $p \cdot q_{p-t} > p^2 > (p-t)^2+(p-t)+1$, a contradiction.

N4. Determine whether there exists an infinite sequence of nonzero digits $a_1, a_2, a_3, \ldots$ and a positive integer N such that for every integer $k > N$, the number $\overline{a_k a_{k-1} \ldots a_1}$ is a perfect square.

(Iran)

Answer. No.

Solution. Assume that $a_1, a_2, a_3, \ldots$ is such a sequence. For each positive integer k, let $y_k = \overline{a_k a_{k-1} \ldots a_1}$. By the assumption, for each $k > N$ there exists a positive integer x_k such that $y_k = x_k^2$.

I. For every n, let 5^{γ_n} be the greatest power of 5 dividing x_n. Let us show first that $2\gamma_n \geqslant n$ for every positive integer $n > N$.

Assume, to the contrary, that there exists a positive integer $n > N$ such that $2\gamma_n < n$, which yields

$$y_{n+1} = \overline{a_{n+1} a_n \ldots a_1} = 10^n a_{n+1} + \overline{a_n a_{n-1} \ldots a_1} = 10^n a_{n+1} + y_n = 5^{2\gamma_n} \left(2^n 5^{n-2\gamma_n} a_{n+1} + \frac{y_n}{5^{2\gamma_n}} \right).$$

Since $5 \nmid y_n / 5^{2\gamma_n}$, we obtain $\gamma_{n+1} = \gamma_n < n < n+1$. By the same arguments we obtain that $\gamma_n = \gamma_{n+1} = \gamma_{n+2} = \ldots$. Denote this common value by γ.

Now, for each $k \geqslant n$ we have

$$(x_{k+1} - x_k)(x_{k+1} + x_k) = x_{k+1}^2 - x_k^2 = y_{k+1} - y_k = a_{k+1} \cdot 10^k.$$

One of the numbers $x_{k+1} - x_k$ and $x_{k+1} + x_k$ is not divisible by $5^{\gamma+1}$ since otherwise one would have $5^{\gamma+1} \mid \big((x_{k+1} - x_k) + (x_{k+1} + x_k)\big) = 2x_{k+1}$. On the other hand, we have $5^k \mid (x_{k+1} - x_k)(x_{k+1} + x_k)$, so $5^{k-\gamma}$ divides one of these two factors. Thus we get

$$5^{k-\gamma} \leqslant \max\{x_{k+1} - x_k, x_{k+1} + x_k\} < 2x_{k+1} = 2\sqrt{y_{k+1}} < 2 \cdot 10^{(k+1)/2},$$

which implies $5^{2k} < 4 \cdot 5^{2\gamma} \cdot 10^{k+1}$, or $(5/2)^k < 40 \cdot 5^{2\gamma}$. The last inequality is clearly false for sufficiently large values of k. This contradiction shows that $2\gamma_n \geqslant n$ for all $n > N$.

II. Consider now any integer $k > \max\{N/2, 2\}$. Since $2\gamma_{2k+1} \geqslant 2k+1$ and $2\gamma_{2k+2} \geqslant 2k+2$, we have $\gamma_{2k+1} \geqslant k+1$ and $\gamma_{2k+2} \geqslant k+1$. So, from $y_{2k+2} = a_{2k+2} \cdot 10^{2k+1} + y_{2k+1}$ we obtain $5^{2k+2} \mid y_{2k+2} - y_{2k+1} = a_{2k+2} \cdot 10^{2k+1}$ and thus $5 \mid a_{2k+2}$, which implies $a_{2k+2} = 5$. Therefore,

$$(x_{2k+2} - x_{2k+1})(x_{2k+2} + x_{2k+1}) = x_{2k+2}^2 - x_{2k+1}^2 = y_{2k+2} - y_{2k+1} = 5 \cdot 10^{2k+1} = 2^{2k+1} \cdot 5^{2k+2}.$$

Setting $A_k = x_{2k+2}/5^{k+1}$ and $B_k = x_{2k+1}/5^{k+1}$, which are integers, we obtain

$$(A_k - B_k)(A_k + B_k) = 2^{2k+1}. \tag{1}$$

Both A_k and B_k are odd, since otherwise y_{2k+2} or y_{2k+1} would be a multiple of 10 which is false by $a_1 \neq 0$; so one of the numbers $A_k - B_k$ and $A_k + B_k$ is not divisible by 4. Therefore (1) yields $A_k - B_k = 2$ and $A_k + B_k = 2^{2k}$, hence $A_k = 2^{2k-1} + 1$ and thus

$$x_{2k+2} = 5^{k+1} A_k = 10^{k+1} \cdot 2^{k-2} + 5^{k+1} > 10^{k+1},$$

since $k \geqslant 2$. This implies that $y_{2k+2} > 10^{2k+2}$ which contradicts the fact that y_{2k+2} contains $2k+2$ digits. The desired result follows.

Solution 2. Again, we assume that a sequence $a_1, a_2, a_3, \ldots$ satisfies the problem conditions, introduce the numbers x_k and y_k as in the previous solution, and notice that

$$y_{k+1} - y_k = (x_{k+1} - x_k)(x_{k+1} + x_k) = 10^k a_{k+1} \tag{2}$$

for all $k > N$. Consider any such k. Since $a_1 \neq 0$, the numbers x_k and x_{k+1} are not multiples of 10, and therefore the numbers $p_k = x_{k+1} - x_k$ and $q_k = x_{k+1} + x_k$ cannot be simultaneously multiples of 20, and hence one of them is not divisible either by 4 or by 5. In view of (2), this means that the other one is divisible by either 5^k or by 2^{k-1}. Notice also that p_k and q_k have the same parity, so both are even.

On the other hand, we have $x_{k+1}^2 = x_k^2 + 10^k a_{k+1} \geqslant x_k^2 + 10^k > 2x_k^2$, so $x_{k+1}/x_k > \sqrt{2}$, which implies that

$$1 < \frac{q_k}{p_k} = 1 + \frac{2}{x_{k+1}/x_k - 1} < 1 + \frac{2}{\sqrt{2} - 1} < 6. \tag{3}$$

Thus, if one of the numbers p_k and q_k is divisible by 5^k, then we have

$$10^{k+1} > 10^k a_{k+1} = p_k q_k \geqslant \frac{(5^k)^2}{6}$$

and hence $(5/2)^k < 60$ which is false for sufficiently large k. So, assuming that k is large, we get that 2^{k-1} divides one of the numbers p_k and q_k. Hence

$$\{p_k,\, q_k\} = \{2^{k-1} \cdot 5^{r_k} b_k,\, 2 \cdot 5^{k-r_k} c_k\} \quad \text{with nonnegative integers } b_k,\, c_k,\, r_k \text{ such that } b_k c_k = a_{k+1}.$$

Moreover, from (3) we get

$$6 > \frac{2^{k-1} \cdot 5^{r_k} b_k}{2 \cdot 5^{k-r_k} c_k} \geqslant \frac{1}{36} \cdot \left(\frac{2}{5}\right)^k \cdot 5^{2r_k} \quad \text{and} \quad 6 > \frac{2 \cdot 5^{k-r_k} c_k}{2^{k-1} \cdot 5^{r_k} b_k} \geqslant \frac{4}{9} \cdot \left(\frac{5}{2}\right)^k \cdot 5^{-2r_k},$$

so

$$\alpha k + c_1 < r_k < \alpha k + c_2 \qquad \text{for } \alpha = \tfrac{1}{2}\log_5\left(\tfrac{5}{2}\right) < 1 \text{ and some constants } c_2 > c_1. \tag{4}$$

Consequently, for $C = c_2 - c_1 + 1 - \alpha > 0$ we have

$$(k+1) - r_{k+1} \leqslant k - r_k + C. \tag{5}$$

Next, we will use the following easy lemma.

Lemma. Let s be a positive integer. Then $5^{s+2^s} \equiv 5^s \pmod{10^s}$.

Proof. Euler's theorem gives $5^{2^s} \equiv 1 \pmod{2^s}$, so $5^{s+2^s} - 5^s = 5^s(5^{2^s} - 1)$ is divisible by 2^s and 5^s.

Now, for every large k we have

$$x_{k+1} = \frac{p_k + q_k}{2} = 5^{r_k} \cdot 2^{k-2} b_k + 5^{k-r_k} c_k \equiv 5^{k-r_k} c_k \pmod{10^{r_k}} \tag{6}$$

since $r_k \leqslant k - 2$ by (4); hence $y_{k+1} \equiv 5^{2(k-r_k)} c_k^2 \pmod{10^{r_k}}$. Let us consider some large integer s, and choose the minimal k such that $2(k - r_k) \geqslant s + 2^s$; it exists by (4). Set $d = 2(k - r_k) - (s + 2^s)$. By (4) we have $2^s < 2(k - r_k) < \left(\frac{2}{\alpha} - 2\right) r_k - \frac{2c_1}{\alpha}$; if s is large this implies $r_k > s$, so (6) also holds modulo 10^s. Then (6) and the lemma give

$$y_{k+1} \equiv 5^{2(k-r_k)} c_k^2 = 5^{s+2^s} \cdot 5^d c_k^2 \equiv 5^s \cdot 5^d c_k^2 \pmod{10^s}. \tag{7}$$

By (5) and the minimality of k we have $d \leqslant 2C$, so $5^d c_k^2 \leqslant 5^{2C} \cdot 81 = D$. Using $5^4 < 10^3$ we obtain

$$5^s \cdot 5^d c_k^2 < 10^{3s/4} D < 10^{s-1}$$

for sufficiently large s. This, together with (7), shows that the sth digit from the right in y_{k+1}, which is a_s, is zero. This contradicts the problem condition.

N5. Fix an integer $k \geqslant 2$. Two players, called Ana and Banana, play the following *game of numbers*: Initially, some integer $n \geqslant k$ gets written on the blackboard. Then they take moves in turn, with Ana beginning. A player making a move erases the number m just written on the blackboard and replaces it by some number m' with $k \leqslant m' < m$ that is coprime to m. The first player who cannot move anymore loses.

An integer $n \geqslant k$ is called *good* if Banana has a winning strategy when the initial number is n, and *bad* otherwise.

Consider two integers $n, n' \geqslant k$ with the property that each prime number $p \leqslant k$ divides n if and only if it divides n'. Prove that either both n and n' are good or both are bad.

(Italy)

Solution 1. Let us first observe that the number appearing on the blackboard decreases after every move; so the game necessarily ends after at most n steps, and consequently there always has to be some player possessing a winning strategy. So if some $n \geqslant k$ is bad, then Ana has a winning strategy in the game with starting number n.

More precisely, if $n \geqslant k$ is such that there is a good integer m with $n > m \geqslant k$ and $\gcd(m, n) = 1$, then n itself is bad, for Ana has the following winning strategy in the game with initial number n: She proceeds by first playing m and then using Banana's strategy for the game with starting number m.

Otherwise, if some integer $n \geqslant k$ has the property that every integer m with $n > m \geqslant k$ and $\gcd(m, n) = 1$ is bad, then n is good. Indeed, if Ana can make a first move at all in the game with initial number n, then she leaves it in a position where the first player has a winning strategy, so that Banana can defeat her.

In particular, this implies that any two good numbers have a non–trivial common divisor. Also, k itself is good.

For brevity, we say that $n \longrightarrow x$ is a *move* if n and x are two coprime integers with $n > x \geqslant k$.

Claim 1. If n is good and n' is a multiple of n, then n' is also good.

Proof. If n' were bad, there would have to be some move $n' \longrightarrow x$, where x is good. As n' is a multiple of n this implies that the two good numbers n and x are coprime, which is absurd. □

Claim 2. If r and s denote two positive integers for which $rs \geqslant k$ is bad, then $r^2 s$ is also bad.

Proof. Since rs is bad, there is a move $rs \longrightarrow x$ for some good x. Evidently x is coprime to $r^2 s$ as well, and hence the move $r^2 s \longrightarrow x$ shows that $r^2 s$ is indeed bad. □

Claim 3. If $p > k$ is prime and $n \geqslant k$ is bad, then np is also bad.

Proof. Otherwise we choose a counterexample with n being as small as possible. In particular, np is good. Since n is bad, there is a move $n \longrightarrow x$ for some good x. Now $np \longrightarrow x$ cannot be a valid move, which tells us that x has to be divisible by p. So we can write $x = p^r y$, where r and y denote some positive integers, the latter of which is not divisible by p.

Note that $y = 1$ is impossible, for then we would have $x = p^r$ and the move $x \longrightarrow k$ would establish that x is bad. In view of this, there is a least power y^α of y that is at least as large as k. Since the numbers np and y^α are coprime and the former is good, the latter has to be bad. Moreover, the minimality of α implies $y^\alpha < ky < py = \frac{x}{p^{r-1}} < \frac{n}{p^{r-1}}$. So $p^{r-1} \cdot y^\alpha < n$ and consequently all the numbers $y^\alpha, py^\alpha, \ldots, p^r \cdot y^\alpha = p(p^{r-1} \cdot y^\alpha)$ are bad due to the minimal choice of n. But now by Claim 1 the divisor x of $p^r \cdot y^\alpha$ cannot be good, whereby we have reached a contradiction that proves Claim 3. □

We now deduce the statement of the problem from these three claims. To this end, we call two integers $a, b \geqslant k$ *similar* if they are divisible by the same prime numbers not exceeding k. We are to prove that if a and b are similar, then either both of them are good or both are bad. As in this case the product ab is similar to both a and b, it suffices to show the following: if $c \geqslant k$ is similar to some of its multiples d, then either both c and d are good or both are bad.

Assuming that this is not true in general, we choose a counterexample (c_0, d_0) with d_0 being as small as possible. By Claim 1, c_0 is bad whilst d_0 is good. Plainly d_0 is strictly greater than c_0 and hence the quotient $\frac{d_0}{c_0}$ has some prime factor p. Clearly p divides d_0. If $p \leqslant k$, then p divides c_0 as well due to the similarity, and hence d_0 is actually divisible by p^2. So $\frac{d_0}{p}$ is good by the contrapositive of Claim 2. Since $c_0 \mid \frac{d_0}{p}$, the pair $(c_0, \frac{d_0}{p})$ contradicts the supposed minimality of d_0. This proves $p > k$, but now we get the same contradiction using Claim 3 instead of Claim 2. Thereby the problem is solved.

Solution 2. We use the same analysis of the game of numbers as in the first five paragraphs of the first solution. Let us call a prime number p *small* in case $p \leqslant k$ and *big* otherwise. We again call two integers *similar* if their sets of small prime factors coincide.

Claim 4. For each integer $b \geqslant k$ having some small prime factor, there exists an integer x similar to it with $b \geqslant x \geqslant k$ and having no big prime factors.

Proof. Unless b has a big prime factor we may simply choose $x = b$. Now let p and q denote a small and a big prime factor of b, respectively. Let a be the product of all small prime factors of b. Further define n to be the least non–negative integer for which the number $x = p^n a$ is at least as large as k. It suffices to show that $b > x$. This is clear in case $n = 0$, so let us assume $n > 0$ from now on. Then we have $x < pk$ due to the minimality of n, $p \leqslant a$ because p divides a by construction, and $k < q$. Therefore $x < aq$ and, as the right hand side is a product of distinct prime factors of b, this implies indeed $x < b$. □

Let us now assume that there is a pair (a, b) of similar numbers such that a is bad and b is good. Take such a pair with $\max(a, b)$ being as small as possible. Since a is bad, there exists a move $a \longrightarrow r$ for some good r. Since the numbers k and r are both good, they have a common prime factor, which necessarily has to be small. Thus Claim 4 is applicable to r, which yields an integer r' similar to r containing small prime factors only and satisfying $r \geqslant r' \geqslant k$. Since $\max(r, r') = r < a \leqslant \max(a, b)$ the number r' is also good. Now let p denote a common prime factor of the good numbers r' and b. By our construction of r', this prime is small and due to the similarities it consequently divides a and r, contrary to $a \longrightarrow r$ being a move. Thereby the problem is solved.

Comment 1. Having reached Claim 4 of Solution 2, there are various other ways to proceed. For instance, one may directly obtain the following fact, which seems to be interesting in its own right:

Claim 5. Any two good numbers have a common small prime factor.

Proof. Otherwise there exists a pair (b, b') of good numbers with $b' \geqslant b \geqslant k$ all of whose common prime factors are big. Choose such a pair with b' being as small as possible. Since b and k are both good, there has to be a common prime factor p of b and k. Evidently p is small and thus it cannot divide b', which in turn tells us $b' > b$. Applying Claim 4 to b we get an integer x with $b \geqslant x \geqslant k$ that is similar to b and has no big prime divisors at all. By our assumption, b' and x are coprime, and as b' is good this implies that x is bad. Consequently there has to be some move $x \longrightarrow b^*$ such that b^* is good. But now all the small prime factors of b also appear in x and thus they cannot divide b^*. Therefore the pair (b^*, b) contradicts the supposed minimality of b'. □

From that point, it is easy to complete the solution: assume that there are two similar integers a and b such that a is bad and b is good. Since a is bad, there is a move $a \longrightarrow b'$ for some good b'. By Claim 5, there is a small prime p dividing b and b'. Due to the similarity of a and b, the prime p has to divide a as well, but this contradicts the fact that $a \longrightarrow b'$ is a valid move. Thereby the problem is solved.

Comment 2. There are infinitely many good numbers, e.g. all multiples of k. The increasing sequence $b_0, b_1, \ldots$, of all good numbers may be constructed recursively as follows:

- Start with $b_0 = k$.
- If b_n has just been defined for some $n \geqslant 0$, then b_{n+1} is the smallest number $b > b_n$ that is coprime to none of $b_0, \ldots, b_n$.

This construction can be used to determine the set of good numbers for any specific k as explained in the next comment. It is already clear that if $k = p^\alpha$ is a prime power, then a number $b \geqslant k$ is good if and only if it is divisible by p.

Comment 3. Let $P > 1$ denote the product of all small prime numbers. Then any two integers $a, b \geqslant k$ that are congruent modulo P are similar. Thus the infinite word $W_k = (X_k, X_{k+1}, \ldots)$ defined by

$$X_i = \begin{cases} A & \text{if } i \text{ is bad} \\ B & \text{if } i \text{ is good} \end{cases}$$

for all $i \geqslant k$ is periodic and the length of its period divides P. As the prime power example shows, the true period can sometimes be much smaller than P. On the other hand, there are cases where the period is rather large; e.g., if $k = 15$, the sequence of good numbers begins with $15, 18, 20, 24, 30, 36, 40, 42, 45$ and the period of W_{15} is 30.

Comment 4. The original proposal contained two questions about the game of numbers, namely (a) to show that if two numbers have the same prime factors then either both are good or both are bad, and (b) to show that the word W_k introduced in the previous comment is indeed periodic. The Problem Selection Committee thinks that the above version of the problem is somewhat easier, even though it demands to prove a stronger result.

N6. Determine all functions $f\colon \mathbb{Q} \longrightarrow \mathbb{Z}$ satisfying

$$f\left(\frac{f(x)+a}{b}\right) = f\left(\frac{x+a}{b}\right) \tag{1}$$

for all $x \in \mathbb{Q}$, $a \in \mathbb{Z}$, and $b \in \mathbb{Z}_{>0}$. (Here, $\mathbb{Z}_{>0}$ denotes the set of positive integers.)

(Israel)

Answer. There are three kinds of such functions, which are: all constant functions, the floor function, and the ceiling function.

Solution 1. I. We start by verifying that these functions do indeed satisfy (1). This is clear for all constant functions. Now consider any triple $(x, a, b) \in \mathbb{Q} \times \mathbb{Z} \times \mathbb{Z}_{>0}$ and set

$$q = \left\lfloor \frac{x+a}{b} \right\rfloor.$$

This means that q is an integer and $bq \leqslant x + a < b(q+1)$. It follows that $bq \leqslant \lfloor x \rfloor + a < b(q+1)$ holds as well, and thus we have

$$\left\lfloor \frac{\lfloor x \rfloor + a}{b} \right\rfloor = \left\lfloor \frac{x+a}{b} \right\rfloor,$$

meaning that the floor function does indeed satisfy (1). One can check similarly that the ceiling function has the same property.

II. Let us now suppose conversely that the function $f\colon \mathbb{Q} \longrightarrow \mathbb{Z}$ satisfies (1) for all $(x, a, b) \in \mathbb{Q} \times \mathbb{Z} \times \mathbb{Z}_{>0}$. According to the behaviour of the restriction of f to the integers we distinguish two cases.

Case 1: There is some $m \in \mathbb{Z}$ such that $f(m) \neq m$.

Write $f(m) = C$ and let $\eta \in \{-1, +1\}$ and b denote the sign and absolute value of $f(m) - m$, respectively. Given any integer r, we may plug the triple $(m, rb - C, b)$ into (1), thus getting $f(r) = f(r - \eta)$. Starting with m and using induction in both directions, we deduce from this that the equation $f(r) = C$ holds for all integers r. Now any rational number y can be written in the form $y = \frac{p}{q}$ with $(p, q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$, and substituting $(C - p, p - C, q)$ into (1) we get $f(y) = f(0) = C$. Thus f is the constant function whose value is always C.

Case 2: One has $f(m) = m$ for all integers m.

Note that now the special case $b = 1$ of (1) takes a particularly simple form, namely

$$f(x) + a = f(x + a) \qquad \text{for all } (x, a) \in \mathbb{Q} \times \mathbb{Z}. \tag{2}$$

Defining $f\left(\frac{1}{2}\right) = \omega$ we proceed in three steps.

Step A. We show that $\omega \in \{0, 1\}$.

If $\omega \leqslant 0$, we may plug $\left(\frac{1}{2}, -\omega, 1 - 2\omega\right)$ into (1), obtaining $0 = f(0) = f\left(\frac{1}{2}\right) = \omega$. In the contrary case $\omega \geqslant 1$ we argue similarly using the triple $\left(\frac{1}{2}, \omega - 1, 2\omega - 1\right)$.

Step B. We show that $f(x) = \omega$ for all rational numbers x with $0 < x < 1$.

Assume that this fails and pick some rational number $\frac{a}{b} \in (0, 1)$ with minimal b such that $f\left(\frac{a}{b}\right) \neq \omega$. Obviously, $\gcd(a, b) = 1$ and $b \geqslant 2$. If b is even, then a has to be odd and we can substitute $\left(\frac{1}{2}, \frac{a-1}{2}, \frac{b}{2}\right)$ into (1), which yields

$$f\left(\frac{\omega + (a-1)/2}{b/2}\right) = f\left(\frac{a}{b}\right) \neq \omega. \tag{3}$$

Recall that $0 \leqslant (a-1)/2 < b/2$. Thus, in both cases $\omega = 0$ and $\omega = 1$, the left-hand part of (3) equals ω either by the minimality of b, or by $f(\omega) = \omega$. A contradiction.

Thus b has to be odd, so $b = 2k+1$ for some $k \geqslant 1$. Applying (1) to $\left(\frac{1}{2}, k, b\right)$ we get

$$f\left(\frac{\omega+k}{b}\right) = f\left(\frac{1}{2}\right) = \omega. \tag{4}$$

Since a and b are coprime, there exist integers $r \in \{1, 2, \ldots, b\}$ and m such that $ra - mb = k + \omega$. Note that we actually have $1 \leqslant r < b$, since the right hand side is not a multiple of b. If m is negative, then we have $ra - mb > b \geqslant k + \omega$, which is absurd. Similarly, $m \geqslant r$ leads to $ra - mb < br - br = 0$, which is likewise impossible; so we must have $0 \leqslant m \leqslant r - 1$.

We finally substitute $\left(\frac{k+\omega}{b}, m, r\right)$ into (1) and use (4) to learn

$$f\left(\frac{\omega+m}{r}\right) = f\left(\frac{a}{b}\right) \neq \omega.$$

But as above one may see that the left hand side has to equal ω due to the minimality of b. This contradiction concludes our step B.

Step C. Now notice that if $\omega = 0$, then $f(x) = \lfloor x \rfloor$ holds for all rational x with $0 \leqslant x < 1$ and hence by (2) this even holds for all rational numbers x. Similarly, if $\omega = 1$, then $f(x) = \lceil x \rceil$ holds for all $x \in \mathbb{Q}$. Thereby the problem is solved.

Comment 1. An alternative treatment of Steps B and C from the second case, due to the proposer, proceeds as follows. Let square brackets indicate the floor function in case $\omega = 0$ and the ceiling function if $\omega = 1$. We are to prove that $f(x) = [x]$ holds for all $x \in \mathbb{Q}$, and because of Step A and (2) we already know this in case $2x \in \mathbb{Z}$. Applying (1) to $(2x, 0, 2)$ we get

$$f(x) = f\left(\frac{f(2x)}{2}\right),$$

and by the previous observation this yields

$$f(x) = \left[\frac{f(2x)}{2}\right] \qquad \text{for all } x \in \mathbb{Q}. \tag{5}$$

An easy induction now shows

$$f(x) = \left[\frac{f(2^n x)}{2^n}\right] \qquad \text{for all } (x, n) \in \mathbb{Q} \times \mathbb{Z}_{>0}. \tag{6}$$

Now suppose first that x is not an integer but can be written in the form $\frac{p}{q}$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$ both being odd. Let d denote the multiplicative order of 2 modulo q and let m be any large integer. Plugging $n = dm$ into (6) and using (2) we get

$$f(x) = \left[\frac{f(2^{dm}x)}{2^{dm}}\right] = \left[\frac{f(x) + (2^{dm}-1)x}{2^{dm}}\right] = \left[x + \frac{f(x)-x}{2^{dm}}\right].$$

Since x is not an integer, the square bracket function is continuous at x; hence as m tends to infinity the above fomula gives $f(x) = [x]$. To complete the argument we just need to observe that if some $y \in \mathbb{Q}$ satisfies $f(y) = [y]$, then (5) yields $f\left(\frac{y}{2}\right) = f\left(\frac{[y]}{2}\right) = \left[\frac{[y]}{2}\right] = \left[\frac{y}{2}\right]$.

Solution 2. Here we just give another argument for the second case of the above solution. Again we use equation (2). It follows that the set S of all zeros of f contains for each $x \in \mathbb{Q}$ exactly one term from the infinite sequence $\ldots, x-2, x-1, x, x+1, x+2, \ldots$.

Next we claim that

$$\text{if } (p,q) \in \mathbb{Z} \times \mathbb{Z}_{>0} \text{ and } \tfrac{p}{q} \in S, \text{ then } \tfrac{p}{q+1} \in S \text{ holds as well.} \tag{7}$$

To see this we just plug $\left(\frac{p}{q}, p, q+1\right)$ into (1), thus getting $f\left(\frac{p}{q+1}\right) = f\left(\frac{p}{q}\right) = 0$.

From this we get that

$$\text{if } x, y \in \mathbb{Q},\ x > y > 0, \text{ and } x \in S, \text{ then } y \in S. \tag{8}$$

Indeed, if we write $x = \frac{p}{q}$ and $y = \frac{r}{s}$ with $p, q, r, s \in \mathbb{Z}_{>0}$, then $ps > qr$ and (7) tells us

$$0 = f\left(\frac{p}{q}\right) = f\left(\frac{pr}{qr}\right) = f\left(\frac{pr}{qr+1}\right) = \ldots = f\left(\frac{pr}{ps}\right) = f\left(\frac{r}{s}\right).$$

Essentially the same argument also establishes that

$$\text{if } x, y \in \mathbb{Q},\ x < y < 0, \text{ and } x \in S, \text{ then } y \in S. \tag{9}$$

From (8) and (9) we get $0 \in S \subseteq (-1, +1)$ and hence the real number $\alpha = \sup(S)$ exists and satisfies $0 \leqslant \alpha \leqslant 1$.

Let us assume that we actually had $0 < \alpha < 1$. Note that $f(x) = 0$ if $x \in (0, \alpha) \cap \mathbb{Q}$ by (8), and $f(x) = 1$ if $x \in (\alpha, 1) \cap \mathbb{Q}$ by (9) and (2). Let K denote the unique positive integer satisfying $K\alpha < 1 \leqslant (K+1)\alpha$. The first of these two inequalities entails $\alpha < \frac{1+\alpha}{K+1}$, and thus there is a rational number $x \in \left(\alpha, \frac{1+\alpha}{K+1}\right)$. Setting $y = (K+1)x - 1$ and substituting $(y, 1, K+1)$ into (1) we learn

$$f\left(\frac{f(y)+1}{K+1}\right) = f\left(\frac{y+1}{K+1}\right) = f(x).$$

Since $\alpha < x < 1$ and $0 < y < \alpha$, this simplifies to

$$f\left(\frac{1}{K+1}\right) = 1.$$

But, as $0 < \frac{1}{K+1} \leqslant \alpha$, this is only possible if $\alpha = \frac{1}{K+1}$ and $f(\alpha) = 1$. From this, however, we get the contradiction

$$0 = f\left(\frac{1}{(K+1)^2}\right) = f\left(\frac{\alpha+0}{K+1}\right) = f\left(\frac{f(\alpha)+0}{K+1}\right) = f(\alpha) = 1.$$

Thus our assumption $0 < \alpha < 1$ has turned out to be wrong and it follows that $\alpha \in \{0, 1\}$. If $\alpha = 0$, then we have $S \subseteq (-1, 0]$, whence $S = (-1, 0] \cap \mathbb{Q}$, which in turn yields $f(x) = \lceil x \rceil$ for all $x \in \mathbb{Q}$ due to (2). Similarly, $\alpha = 1$ entails $S = [0, 1) \cap \mathbb{Q}$ and $f(x) = \lfloor x \rfloor$ for all $x \in \mathbb{Q}$. Thereby the solution is complete.

Comment 2. It seems that all solutions to this problems involve some case distinction separating the constant solutions from the unbounded ones, though the "descriptions" of the cases may be different depending on the work that has been done at the beginning of the solution. For instance, these two cases can also be "f is periodic on the integers" and "f is not periodic on the integers". The case leading to the unbounded solutions appears to be the harder one.

In most approaches, the cases leading to the two functions $x \longmapsto \lfloor x \rfloor$ and $x \longmapsto \lceil x \rceil$ can easily be treated parallelly, but sometimes it may be useful to know that there is some symmetry in the problem interchanging these two functions. Namely, if a function $f\colon \mathbb{Q} \longrightarrow \mathbb{Z}$ satisfies (1), then so does the function $g\colon \mathbb{Q} \longrightarrow \mathbb{Z}$ defined by $g(x) = -f(-x)$ for all $x \in \mathbb{Q}$. For that reason, we could have restricted our attention to the case $\omega = 0$ in the first solution and, once $\alpha \in \{0, 1\}$ had been obtained, to the case $\alpha = 0$ in the second solution.

N7. Let ν be an irrational positive number, and let m be a positive integer. A pair (a, b) of positive integers is called *good* if

$$a\lceil b\nu\rceil - b\lfloor a\nu\rfloor = m. \qquad (*)$$

A good pair (a, b) is called *excellent* if neither of the pairs $(a-b, b)$ and $(a, b-a)$ is good. (As usual, by $\lfloor x\rfloor$ and $\lceil x\rceil$ we denote the integer numbers such that $x - 1 < \lfloor x\rfloor \leqslant x$ and $x \leqslant \lceil x\rceil < x + 1$.)

Prove that the number of excellent pairs is equal to the sum of the positive divisors of m.

(U.S.A.)

Solution. For positive integers a and b, let us denote

$$f(a, b) = a\lceil b\nu\rceil - b\lfloor a\nu\rfloor.$$

We will deal with various values of m; thus it is convenient to say that a pair (a, b) is *m-good* or *m-excellent* if the corresponding conditions are satisfied.

To start, let us investigate how the values $f(a + b, b)$ and $f(a, b + a)$ are related to $f(a, b)$. If $\{a\nu\} + \{b\nu\} < 1$, then we have $\lfloor(a + b)\nu\rfloor = \lfloor a\nu\rfloor + \lfloor b\nu\rfloor$ and $\lceil(a + b)\nu\rceil = \lceil a\nu\rceil + \lceil b\nu\rceil - 1$, so

$$f(a + b, b) = (a + b)\lceil b\nu\rceil - b(\lfloor a\nu\rfloor + \lfloor b\nu\rfloor) = f(a, b) + b(\lceil b\nu\rceil - \lfloor b\nu\rfloor) = f(a, b) + b$$

and

$$f(a, b + a) = a(\lceil b\nu\rceil + \lceil a\nu\rceil - 1) - (b + a)\lfloor a\nu\rfloor = f(a, b) + a(\lceil a\nu\rceil - 1 - \lfloor a\nu\rfloor) = f(a, b).$$

Similarly, if $\{a\nu\} + \{b\nu\} \geqslant 1$ then one obtains

$$f(a + b, b) = f(a, b) \quad \text{and} \quad f(a, b + a) = f(a, b) + a.$$

So, in both cases one of the numbers $f(a + b, a)$ and $f(a, b + a)$ is equal to $f(a, b)$ while the other is greater than $f(a, b)$ by one of a and b. Thus, exactly one of the pairs $(a + b, b)$ and $(a, b + a)$ is excellent (for an appropriate value of m).

Now let us say that the pairs $(a + b, b)$ and $(a, b + a)$ are the *children* of the pair (a, b), while this pair is their *parent*. Next, if a pair (c, d) can be obtained from (a, b) by several passings from a parent to a child, we will say that (c, d) is a *descendant* of (a, b), while (a, b) is an *ancestor* of (c, d) (a pair is neither an ancestor nor a descendant of itself). Thus each pair (a, b) has two children, it has a unique parent if $a \neq b$, and no parents otherwise. Therefore, each pair of distinct positive integers has a unique ancestor of the form (a, a); our aim is now to find how many m-excellent descendants each such pair has.

Notice now that if a pair (a, b) is m-excellent then $\min\{a, b\} \leqslant m$. Indeed, if $a = b$ then $f(a, a) = a = m$, so the statement is valid. Otherwise, the pair (a, b) is a child of some pair (a', b'). If $b = b'$ and $a = a' + b'$, then we should have $m = f(a, b) = f(a', b') + b'$, so $b = b' = m - f(a', b') < m$. Similarly, if $a = a'$ and $b = b' + a'$ then $a < m$.

Let us consider the set S_m of all pairs (a, b) such that $f(a, b) \leqslant m$ and $\min\{a, b\} \leqslant m$. Then all the ancestors of the elements in S_m are again in S_m, and each element in S_m either is of the form (a, a) with $a \leqslant m$, or has a unique ancestor of this form. From the arguments above we see that all m-excellent pairs lie in S_m.

We claim now that the set S_m is finite. Indeed, assume, for instance, that it contains infinitely many pairs (c, d) with $d > 2m$. Such a pair is necessarily a child of $(c, d-c)$, and thus a descendant of some pair (c, d') with $m < d' \leqslant 2m$. Therefore, one of the pairs $(a, b) \in S_m$ with $m < b \leqslant 2m$

has infinitely many descendants in S_m, and all these descendants have the form $(a, b+ka)$ with k a positive integer. Since $f(a, b+ka)$ does not decrease as k grows, it becomes constant for $k \geqslant k_0$, where k_0 is some positive integer. This means that $\{a\nu\} + \{(b+ka)\nu\} < 1$ for all $k \geqslant k_0$. But this yields $1 > \{(b+ka)\nu\} = \{(b+k_0a)\nu\} + (k-k_0)\{a\nu\}$ for all $k > k_0$, which is absurd.

Similarly, one can prove that S_m contains finitely many pairs (c, d) with $c > 2m$, thus finitely many elements at all.

We are now prepared for proving the following crucial lemma.

Lemma. Consider any pair (a, b) with $f(a, b) \neq m$. Then the number $g(a, b)$ of its m-excellent descendants is equal to the number $h(a, b)$ of ways to represent the number $t = m - f(a, b)$ as $t = ka + \ell b$ with k and ℓ being some nonnegative integers.

Proof. We proceed by induction on the number N of descendants of (a, b) in S_m. If $N = 0$ then clearly $g(a, b) = 0$. Assume that $h(a, b) > 0$; without loss of generality, we have $a \leqslant b$. Then, clearly, $m - f(a, b) \geqslant a$, so $f(a, b+a) \leqslant f(a, b) + a \leqslant m$ and $a \leqslant m$, hence $(a, b+a) \in S_m$ which is impossible. Thus in the base case we have $g(a, b) = h(a, b) = 0$, as desired.

Now let $N > 0$. Assume that $f(a+b, b) = f(a, b) + b$ and $f(a, b+a) = f(a, b)$ (the other case is similar). If $f(a, b) + b \neq m$, then by the induction hypothesis we have

$$g(a, b) = g(a+b, b) + g(a, b+a) = h(a+b, b) + h(a, b+a).$$

Notice that both pairs $(a+b, b)$ and $(a, b+a)$ are descendants of (a, b) and thus each of them has strictly less descendants in S_m than (a, b) does.

Next, each one of the $h(a+b, b)$ representations of $m - f(a+b, b) = m - b - f(a, b)$ as the sum $k'(a+b) + \ell' b$ provides the representation $m - f(a, b) = ka + \ell b$ with $k = k' < k' + \ell' + 1 = \ell$. Similarly, each one of the $h(a, b+a)$ representations of $m - f(a, b+a) = m - f(a, b)$ as the sum $k'a + \ell'(b+a)$ provides the representation $m - f(a, b) = ka + \ell b$ with $k = k' + \ell' \geqslant \ell' = \ell$. This correspondence is obviously bijective, so

$$h(a, b) = h(a+b, b) + h(a, b+a) = g(a, b),$$

as required.

Finally, if $f(a, b) + b = m$ then $(a+b, b)$ is m-excellent, so $g(a, b) = 1 + g(a, b+a) = 1 + h(a, b+a)$ by the induction hypothesis. On the other hand, the number $m - f(a, b) = b$ has a representation $0 \cdot a + 1 \cdot b$ and sometimes one more representation as $ka + 0 \cdot b$; this last representation exists simultaneously with the representation $m - f(a, b+a) = ka + 0 \cdot (b+a)$, so $h(a, b) = 1 + h(a, b+a)$ as well. Thus in this case the step is also proved. □

Now it is easy to finish the solution. There exists a unique m-excellent pair of the form (a, a), and each other m-excellent pair (a, b) has a unique ancestor of the form (x, x) with $x < m$. By the lemma, for every $x < m$ the number of its m-excellent descendants is $h(x, x)$, which is the number of ways to represent $m - f(x, x) = m - x$ as $kx + \ell x$ (with nonnegative integer k and ℓ). This number is 0 if $x \nmid m$, and m/x otherwise. So the total number of excellent pairs is

$$1 + \sum_{x \mid m,\ x < m} \frac{m}{x} = 1 + \sum_{d \mid m,\ d > 1} d = \sum_{d \mid m} d,$$

as required.

Comment. Let us present a sketch of an outline of a different solution. The plan is to check that the number of excellent pairs does not depend on the (irrational) number ν, and to find this number for some appropriate value of ν. For that, we first introduce some geometrical language. We deal only with the excellent pairs (a, b) with $a \neq b$.

Part I. Given an irrational positive ν, for every positive integer n we introduce two integral points $F_\nu(n) = (n, \lfloor n\nu \rfloor)$ and $C_\nu(n) = (n, \lceil n\nu \rceil)$ on the coordinate plane Oxy. Then $(*)$ reads as $[OF_\nu(a)C_\nu(b)] = m/2$; here $[\cdot]$ stands for the signed area. Next, we rewrite in these terms the condition on a pair (a, b) to be excellent. Let ℓ_ν, ℓ_ν^+, and ℓ_ν^- be the lines determined by the equations $y = \nu x$, $y = \nu x + 1$, and $y = \nu x - 1$, respectively.

a). Firstly, we deal with all excellent pairs (a, b) with $a < b$. Given some value of a, all the points C such that $[OF_\nu(a)C] = m/2$ lie on some line $f_\nu(a)$; if there exist any good pairs (a, b) at all, this line has to contain at least one integral point, which happens exactly when $\gcd(a, \lfloor a\nu \rfloor) \mid m$.

Let $P_\nu(a)$ be the point of intersection of ℓ_ν^+ and $f_\nu(a)$, and let $p_\nu(a)$ be its abscissa; notice that $p_\nu(a)$ is irrational if it is nonzero. Now, if (a, b) is good, then the point $C_\nu(b)$ lies on $f_\nu(a)$, which means that the point of $f_\nu(a)$ with abscissa b lies between ℓ_ν and ℓ_ν^+ and is integral. If in addition the pair $(a, b-a)$ is not good, then the point of $f_\nu(a)$ with abscissa $b-a$ lies above ℓ_ν^+ (see Fig. 1). Thus, the pair (a, b) with $b > a$ is excellent exactly when $p_\nu(a)$ lies between $b-a$ and b, and the point of $f_\nu(a)$ with abscissa b is integral (which means that this point is $C_\nu(b)$).

Notice now that, if $p_\nu(a) > a$, then the number of excellent pairs of the form (a, b) (with $b > a$) is $\gcd(a, \lfloor a\nu \rfloor)$.

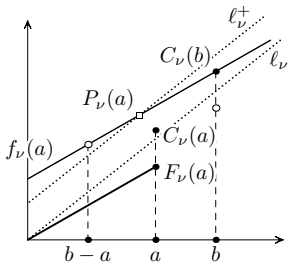


Figure 1

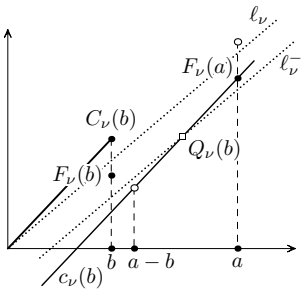


Figure 2

b). Analogously, considering the pairs (a, b) with $a > b$, we fix the value of b, introduce the line $c_\nu(b)$ containing all the points F with $[OFC_\nu(b)] = m/2$, assume that this line contains an integral point (which means $\gcd(b, \lfloor b\nu \rfloor) \mid m$), and denote the common point of $c_\nu(b)$ and ℓ_ν^- by $Q_\nu(b)$, its abscissa being $q_\nu(b)$. Similarly to the previous case, we obtain that the pair (a, b) is excellent exactly when $q_\nu(a)$ lies between $a-b$ and a, and the point of $c_\nu(b)$ with abscissa a is integral (see Fig. 2). Again, if $q_\nu(b) > b$, then the number of excellent pairs of the form (a, b) (with $a > b$) is $\gcd(b, \lfloor b\nu \rfloor)$.

Part II, sketchy. Having obtained such a description, one may check how the number of excellent pairs changes as ν grows. (Having done that, one may find this number for one appropriate value of ν; for instance, it is relatively easy to make this calculation for $\nu \in \left(1, 1 + \frac{1}{m}\right)$.)

Consider, for the initial value of ν, some excellent pair (a, t) with $a > t$. As ν grows, this pair eventually stops being excellent; this happens when the point $Q_\nu(t)$ passes through $F_\nu(a)$. At the same moment, the pair $(a + t, t)$ becomes excellent instead.

This process halts when the point $Q_\nu(t)$ eventually disappears, i.e. when ν passes through the ratio of the coordinates of the point $T = C_\nu(t)$. Hence, the point T afterwards is regarded as $F_\nu(t)$. Thus, all the old excellent pairs of the form (a, t) with $a > t$ disappear; on the other hand, the same number of excellent pairs with the first element being t just appear.

Similarly, if some pair (t, b) with $t < b$ is initially ν-excellent, then at some moment it stops being excellent when $P_\nu(t)$ passes through $C_\nu(b)$; at the same moment, the pair $(t, b-t)$ becomes excellent. This process eventually stops when $b - t < t$. At this moment, again the second element of the pair becomes fixed, and the first one starts to increase.

These ideas can be made precise enough to show that the number of excellent pairs remains unchanged, as required.

We should warn the reader that the rigorous elaboration of Part II is technically quite involved, mostly by the reason that the set of moments when the collection of excellent pairs changes is infinite. Especially much care should be applied to the limit points of this set, which are exactly the points when the line ℓ_ν passes through some point of the form $C_\nu(b)$.

The same ideas may be explained in an algebraic language instead of a geometrical one; the same technicalities remain in this way as well.

Shortlisted Problems with Solutions

53rd International Mathematical Olympiad

Mar del Plata, Argentina 2012

Note of Confidentiality

The shortlisted problems should be kept strictly confidential until IMO 2013

Contributing Countries

The Organizing Committee and the Problem Selection Committee of IMO 2012 thank the following 40 countries for contributing 136 problem proposals:

Australia, Austria, Belarus, Belgium, Bulgaria, Canada, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hong Kong, India, Iran, Ireland, Israel, Japan, Kazakhstan, Luxembourg, Malaysia, Montenegro, Netherlands, Norway, Pakistan, Romania, Russia, Serbia, Slovakia, Slovenia, South Africa, South Korea, Sweden, Thailand, Ukraine, United Kingdom, United States of America, Uzbekistan

Problem Selection Committee

Martín Avendaño
Carlos di Fiore
Géza Kós
Svetoslav Savchev

4

Algebra

A1. Find all the functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$

for all integers a, b, c satisfying $a + b + c = 0$.

A2. Let $\mathbb{Z}$ and $\mathbb{Q}$ be the sets of integers and rationals respectively.

a) Does there exist a partition of $\mathbb{Z}$ into three non-empty subsets A, B, C such that the sets $A + B$, $B + C$, $C + A$ are disjoint?

b) Does there exist a partition of $\mathbb{Q}$ into three non-empty subsets A, B, C such that the sets $A + B$, $B + C$, $C + A$ are disjoint?

Here $X + Y$ denotes the set $\{x + y \mid x \in X, y \in Y\}$, for $X, Y \subseteq \mathbb{Z}$ and $X, Y \subseteq \mathbb{Q}$.

A3. Let $a_2, \ldots, a_n$ be $n - 1$ positive real numbers, where $n \geq 3$, such that $a_2 a_3 \cdots a_n = 1$. Prove that

$$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

A4. Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

A5. Find all functions $f \colon \mathbb{R} \to \mathbb{R}$ that satisfy the conditions

$$f(1 + xy) - f(x + y) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R}$$

and $f(-1) \neq 0$.

A6. Let $f : \mathbb{N} \to \mathbb{N}$ be a function, and let f^m be f applied m times. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k. Prove that the sequence $k_1, k_2, \ldots$ is unbounded.

A7. We say that a function $f : \mathbb{R}^k \to \mathbb{R}$ is a metapolynomial if, for some positive integers m and n, it can be represented in the form

$$f(x_1, \ldots, x_k) = \max_{i=1,\ldots,m} \min_{j=1,\ldots,n} P_{i,j}(x_1, \ldots, x_k)$$

where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

Combinatorics

C1. Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y, and replaces the pair (x, y) by either $(y + 1, x)$ or $(x - 1, x)$. Prove that she can perform only finitely many such iterations.

C2. Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \ldots, n\}$ such that the sums of the different pairs are different integers not exceeding n?

C3. In a 999×999 square table some cells are white and the remaining ones are red. Let T be the number of triples (C_1, C_2, C_3) of cells, the first two in the same row and the last two in the same column, with C_1 and C_3 white and C_2 red. Find the maximum value T can attain.

C4. Players A and B play a game with $N \geq 2012$ coins and 2012 boxes arranged around a circle. Initially A distributes the coins among the boxes so that there is at least 1 coin in each box. Then the two of them make moves in the order $B, A, B, A, \ldots$ by the following rules:

- On every move of his B passes 1 coin from every box to an adjacent box.
- On every move of hers A chooses several coins that were *not* involved in B's previous move and are in different boxes. She passes every chosen coin to an adjacent box.

Player A's goal is to ensure at least 1 coin in each box after every move of hers, regardless of how B plays and how many moves are made. Find the least N that enables her to succeed.

C5. The columns and the rows of a $3n \times 3n$ square board are numbered $1, 2, \ldots, 3n$. Every square (x, y) with $1 \leq x, y \leq 3n$ is colored asparagus, byzantium or citrine according as the modulo 3 remainder of $x + y$ is 0, 1 or 2 respectively. One token colored asparagus, byzantium or citrine is placed on each square, so that there are $3n^2$ tokens of each color.

Suppose that one can permute the tokens so that each token is moved to a distance of at most d from its original position, each asparagus token replaces a byzantium token, each byzantium token replaces a citrine token, and each citrine token replaces an asparagus token. Prove that it is possible to permute the tokens so that each token is moved to a distance of at most $d + 2$ from its original position, and each square contains a token with the same color as the square.

C6. Let k and n be fixed positive integers. In the liar's guessing game, Amy chooses integers x and N with $1 \leq x \leq N$. She tells Ben what N is, but not what x is. Ben may then repeatedly ask Amy whether $x \in S$ for arbitrary sets S of integers. Amy will always answer with *yes* or *no*, but she might lie. The only restriction is that she can lie at most k times in a row. After he has asked as many questions as he wants, Ben must specify a set of at most n positive integers. If x is in this set he wins; otherwise, he loses. Prove that:

a) If $n \geq 2^k$ then Ben can always win.

b) For sufficiently large k there exist $n \geq 1.99^k$ such that Ben cannot guarantee a win.

C7. There are given 2^{500} points on a circle labeled $1, 2, \ldots, 2^{500}$ in some order. Prove that one can choose 100 pairwise disjoint chords joining some of these points so that the 100 sums of the pairs of numbers at the endpoints of the chosen chords are equal.

6

Geometry

G1. In the triangle ABC the point J is the center of the excircle opposite to A. This excircle is tangent to the side BC at M, and to the lines AB and AC at K and L respectively. The lines LM and BJ meet at F, and the lines KM and CJ meet at G. Let S be the point of intersection of the lines AF and BC, and let T be the point of intersection of the lines AG and BC. Prove that M is the midpoint of ST.

G2. Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E. The extensions of the sides AD and BC beyond A and B meet at F. Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD. Prove that D, H, F, G are concyclic.

G3. In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

G4. Let ABC be a triangle with $AB \neq AC$ and circumcenter O. The bisector of $\angle BAC$ intersects BC at D. Let E be the reflection of D with respect to the midpoint of BC. The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.

G5. Let ABC be a triangle with $\angle BCA = 90^\circ$, and let C_0 be the foot of the altitude from C. Choose a point X in the interior of the segment CC_0, and let K, L be the points on the segments AX, BX for which $BK = BC$ and $AL = AC$ respectively. Denote by M the intersection of AL and BK. Show that $MK = ML$.

G6. Let ABC be a triangle with circumcenter O and incenter I. The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

G7. Let $ABCD$ be a convex quadrilateral with non-parallel sides BC and AD. Assume that there is a point E on the side BC such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point F on the side AD such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if AB is parallel to CD.

G8. Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω. Denote by P the foot of the perpendicular from the center of ω to ℓ. The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P. Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P.

Number Theory

N1. Call admissible a set A of integers that has the following property:

$$\text{If } x, y \in A \text{ (possibly } x = y) \text{ then } x^2 + kxy + y^2 \in A \text{ for every integer } k.$$

Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

N2. Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

N3. Determine all integers $m \geq 2$ such that every n with $\frac{m}{3} \leq n \leq \frac{m}{2}$ divides the binomial coefficient $\binom{n}{m-2n}$.

N4. An integer a is called friendly if the equation $(m^2 + n)(n^2 + m) = a(m - n)^3$ has a solution over the positive integers.

a) Prove that there are at least 500 friendly integers in the set $\{1, 2, \ldots, 2012\}$.

b) Decide whether $a = 2$ is friendly.

N5. For a nonnegative integer n define $rad(n) = 1$ if $n = 0$ or $n = 1$, and $rad(n) = p_1 p_2 \cdots p_k$ where $p_1 < p_2 < \cdots < p_k$ are all prime factors of n. Find all polynomials $f(x)$ with nonnegative integer coefficients such that $rad(f(n))$ divides $rad(f(n^{rad(n)}))$ for every nonnegative integer n.

N6. Let x and y be positive integers. If $x^{2^n} - 1$ is divisible by $2^n y + 1$ for every positive integer n, prove that $x = 1$.

N7. Find all $n \in \mathbb{N}$ for which there exist nonnegative integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

N8. Prove that for every prime $p > 100$ and every integer r there exist two integers a and b such that p divides $a^2 + b^5 - r$.

8

Algebra

A1. Find all the functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$

for all integers a, b, c satisfying $a + b + c = 0$.

Solution. The substitution $a = b = c = 0$ gives $3f(0)^2 = 6f(0)^2$, hence

$$f(0) = 0. \tag{1}$$

The substitution $b = -a$ and $c = 0$ gives $((f(a) - f(-a))^2 = 0$. Hence f is an even function:

$$f(a) = f(-a) \quad \text{for all } a \in \mathbb{Z}. \tag{2}$$

Now set $b = a$ and $c = -2a$ to obtain $2f(a)^2 + f(2a)^2 = 2f(a)^2 + 4f(a)f(2a)$. Hence

$$f(2a) = 0 \ \text{ or } \ f(2a) = 4f(a) \quad \text{for all } a \in \mathbb{Z}. \tag{3}$$

If $f(r) = 0$ for some $r \geq 1$ then the substitution $b = r$ and $c = -a-r$ gives $(f(a+r)-f(a))^2 = 0$. So f is periodic with period r, i. e.

$$f(a + r) = f(a) \quad \text{for all } a \in \mathbb{Z}.$$

In particular, if $f(1) = 0$ then f is constant, thus $f(a) = 0$ for all $a \in \mathbb{Z}$. This function clearly satisfies the functional equation. For the rest of the analysis, we assume $f(1) = k \neq 0$.

By (3) we have $f(2) = 0$ or $f(2) = 4k$. If $f(2) = 0$ then f is periodic of period 2, thus $f(\text{even}) = 0$ and $f(\text{odd}) = k$. This function is a solution for every k. We postpone the verification; for the sequel assume $f(2) = 4k \neq 0$.

By (3) again, we have $f(4) = 0$ or $f(4) = 16k$. In the first case f is periodic of period 4, and $f(3) = f(-1) = f(1) = k$, so we have $f(4n) = 0$, $f(4n+1) = f(4n+3) = k$, and $f(4n+2) = 4k$ for all $n \in \mathbb{Z}$. This function is a solution too, which we justify later. For the rest of the analysis, we assume $f(4) = 16k \neq 0$.

We show now that $f(3) = 9k$. In order to do so, we need two substitutions:

$$a = 1, b = 2, c = -3 \Longrightarrow f(3)^2 - 10kf(3) + 9k^2 = 0 \Longrightarrow f(3) \in \{k, 9k\},$$
$$a = 1, b = 3, c = -4 \Longrightarrow f(3)^2 - 34kf(3) + 225k^2 = 0 \Longrightarrow f(3) \in \{9k, 25k\}.$$

Therefore $f(3) = 9k$, as claimed. Now we prove inductively that the only remaining function is $f(x) = kx^2$, $x \in \mathbb{Z}$. We proved this for $x = 0, 1, 2, 3, 4$. Assume that $n \geq 4$ and that $f(x) = kx^2$ holds for all integers $x \in [0, n]$. Then the substitutions $a = n$, $b = 1$, $c = -n-1$ and $a = n-1$, $b = 2$, $c = -n-1$ lead respectively to

$$f(n+1) \in \{k(n+1)^2, k(n-1)^2\} \qquad \text{and} \qquad f(n+1) \in \{k(n+1)^2, k(n-3)^2\}.$$

Since $k(n-1)^2 \neq k(n-3)^2$ for $n \neq 2$, the only possibility is $f(n+1) = k(n+1)^2$. This completes the induction, so $f(x) = kx^2$ for all $x \geq 0$. The same expression is valid for negative values of x since f is even. To verify that $f(x) = kx^2$ is actually a solution, we need to check the identity $a^4 + b^4 + (a+b)^4 = 2a^2b^2 + 2a^2(a+b)^2 + 2b^2(a+b)^2$, which follows directly by expanding both sides.

Therefore the only possible solutions of the functional equation are the constant function $f_1(x) = 0$ and the following functions:

$$f_2(x) = kx^2 \qquad f_3(x) = \begin{cases} 0 & x \text{ even} \\ k & x \text{ odd} \end{cases} \qquad f_4(x) = \begin{cases} 0 & x \equiv 0 \pmod 4 \\ k & x \equiv 1 \pmod 2 \\ 4k & x \equiv 2 \pmod 4 \end{cases}$$

for any non-zero integer k. The verification that they are indeed solutions was done for the first two. For f_3 note that if $a + b + c = 0$ then either a, b, c are all even, in which case $f(a) = f(b) = f(c) = 0$, or one of them is even and the other two are odd, so both sides of the equation equal $2k^2$. For f_4 we use similar parity considerations and the symmetry of the equation, which reduces the verification to the triples $(0, k, k)$, $(4k, k, k)$, $(0, 0, 0)$, $(0, 4k, 4k)$. They all satisfy the equation.

Comment. We used several times the same fact: For any $a, b \in \mathbb{Z}$ the functional equation is a quadratic equation in $f(a + b)$ whose coefficients depend on $f(a)$ and $f(b)$:

$$f(a + b)^2 - 2(f(a) + f(b))f(a + b) + (f(a) - f(b))^2 = 0.$$

Its discriminant is $16f(a)f(b)$. Since this value has to be non-negative for any $a, b \in \mathbb{Z}$, we conclude that either f or $-f$ is always non-negative. Also, if f is a solution of the functional equation, then $-f$ is also a solution. Therefore we can assume $f(x) \geq 0$ for all $x \in \mathbb{Z}$. Now, the two solutions of the quadratic equation are

$$f(a + b) \in \left\{ \left(\sqrt{f(a)} + \sqrt{f(b)} \right)^2, \left(\sqrt{f(a)} - \sqrt{f(b)} \right)^2 \right\} \quad \text{for all } a, b \in \mathbb{Z}.$$

The computation of $f(3)$ from $f(1)$, $f(2)$ and $f(4)$ that we did above follows immediately by setting $(a, b) = (1, 2)$ and $(a, b) = (1, -4)$. The inductive step, where $f(n + 1)$ is derived from $f(n)$, $f(n - 1)$, $f(2)$ and $f(1)$, follows immediately using $(a, b) = (n, 1)$ and $(a, b) = (n - 1, 2)$.

A2. Let $\mathbb{Z}$ and $\mathbb{Q}$ be the sets of integers and rationals respectively.

a) Does there exist a partition of $\mathbb{Z}$ into three non-empty subsets A, B, C such that the sets $A+B$, $B+C$, $C+A$ are disjoint?

b) Does there exist a partition of $\mathbb{Q}$ into three non-empty subsets A, B, C such that the sets $A+B$, $B+C$, $C+A$ are disjoint?

Here $X+Y$ denotes the set $\{x+y \mid x \in X, y \in Y\}$, for $X, Y \subseteq \mathbb{Z}$ and $X, Y \subseteq \mathbb{Q}$.

Solution 1. a) The residue classes modulo 3 yield such a partition:

$$A = \{3k \mid k \in \mathbb{Z}\}, \quad B = \{3k+1 \mid k \in \mathbb{Z}\}, \quad C = \{3k+2 \mid k \in \mathbb{Z}\}.$$

b) The answer is *no*. Suppose that $\mathbb{Q}$ can be partitioned into non-empty subsets A, B, C as stated. Note that for all $a \in A$, $b \in B$, $c \in C$ one has

$$a+b-c \in C, \quad b+c-a \in A, \quad c+a-b \in B. \tag{1}$$

Indeed $a+b-c \notin A$ as $(A+B) \cap (A+C) = \emptyset$, and similarly $a+b-c \notin B$, hence $a+b-c \in C$. The other two relations follow by symmetry. Hence $A+B \subset C+C$, $B+C \subset A+A$, $C+A \subset B+B$.

The opposite inclusions also hold. Let $a, a' \in A$ and $b \in B$, $c \in C$ be arbitrary. By (1) $a'+c-b \in B$, and since $a \in A$, $c \in C$, we use (1) again to obtain

$$a+a'-b = a+(a'+c-b)-c \in C.$$

So $A+A \subset B+C$ and likewise $B+B \subset C+A$, $C+C \subset A+B$. In summary

$$B+C = A+A, \qquad C+A = B+B, \qquad A+B = C+C.$$

Furthermore suppose that $0 \in A$ without loss of generality. Then $B = \{0\}+B \subset A+B$ and $C = \{0\}+C \subset A+C$. So, since $B+C$ is disjoint with $A+B$ and $A+C$, it is also disjoint with B and C. Hence $B+C$ is contained in $\mathbb{Z} \setminus (B \cup C) = A$. Because $B+C = A+A$, we obtain $A+A \subset A$. On the other hand $A = \{0\}+A \subset A+A$, implying $A = A+A = B+C$.

Therefore $A+B+C = A+A+A = A$, and now $B+B = C+A$ and $C+C = A+B$ yield $B+B+B = A+B+C = A$, $C+C+C = A+B+C = A$. In particular if $r \in \mathbb{Q} = A \cup B \cup C$ is arbitrary then $3r \in A$.

However such a conclusion is impossible. Take any $b \in B$ ($B \neq \emptyset$) and let $r = b/3 \in \mathbb{Q}$. Then $b = 3r \in A$ which is a contradiction.

Solution 2. We prove that the example for $\mathbb{Z}$ from the first solution is unique, and then use this fact to solve part b).

Let $\mathbb{Z} = A \cup B \cup C$ be a partition of $\mathbb{Z}$ with $A, B, C \neq \emptyset$ and $A+B$, $B+C$, $C+A$ disjoint. We need the relations (1) which clearly hold for $\mathbb{Z}$. Fix two consecutive integers from different sets, say $b \in B$ and $c = b+1 \in C$. For every $a \in A$ we have, in view of (1), $a-1 = a+b-c \in C$ and $a+1 = a+c-b \in B$. So every $a \in A$ is preceded by a number from C and followed by a number from B.

In particular there are pairs of the form $c, c+1$ with $c \in C$, $c+1 \in A$. For such a pair and any $b \in B$ analogous reasoning shows that each $b \in B$ is preceded by a number from A and followed by a number from C. There are also pairs $b, b-1$ with $b \in B$, $b-1 \in A$. We use them in a similar way to prove that each $c \in C$ is preceded by a number from B and followed by a number from A.

By putting the observations together we infer that A, B, C are the three congruence classes modulo 3. Observe that all multiples of 3 are in the set of the partition that contains 0.

Now we turn to part b). Suppose that there is a partition of $\mathbb{Q}$ with the given properties. Choose three rationals $r_i = p_i/q_i$ from the three sets A, B, C, $i = 1, 2, 3$, and set $N = 3q_1q_2q_3$.

Let $S \subset \mathbb{Q}$ be the set of fractions with denominators N (irreducible or not). It is obtained through multiplication of every integer by the constant $1/N$, hence closed under sums and differences. Moreover, if we identify each $k \in \mathbb{Z}$ with $k/N \in S$ then S is essentially the set $\mathbb{Z}$ with respect to addition. The numbers r_i belong to S because

$$r_1 = \frac{3p_1q_2q_3}{N}, \qquad r_2 = \frac{3p_2q_3q_1}{N}, \qquad r_3 = \frac{3p_3q_1q_2}{N}.$$

The partition $\mathbb{Q} = A \cup B \cup C$ of $\mathbb{Q}$ induces a partition $S = A' \cup B' \cup C'$ of S, with $A' = A \cap S$, $B' = B \cap S$, $C' = C \cap S$. Clearly $A' + B'$, $B' + C'$, $C' + A'$ are disjoint, so this partition has the properties we consider.

By the uniqueness of the example for $\mathbb{Z}$ the sets A', B', C' are the congruence classes modulo 3, multiplied by $1/N$. Also all multiples of $3/N$ are in the same set, A', B' or C'. This holds for r_1, r_2, r_3 in particular as they are all multiples of $3/N$. However r_1, r_2, r_3 are in different sets A', B', C' since they were chosen from different sets A, B, C. The contradiction ends the proof.

Comment. The uniqueness of the example for $\mathbb{Z}$ can also be deduced from the argument in the first solution.

12

A3. Let $a_2, \ldots, a_n$ be $n-1$ positive real numbers, where $n \geq 3$, such that $a_2 a_3 \cdots a_n = 1$. Prove that

$$(1+a_2)^2(1+a_3)^3 \cdots (1+a_n)^n > n^n.$$

Solution. The substitution $a_2 = \dfrac{x_2}{x_1}$, $a_3 = \dfrac{x_3}{x_2}$, $\ldots$, $a_n = \dfrac{x_1}{x_{n-1}}$ transforms the original problem into the inequality

$$(x_1+x_2)^2(x_2+x_3)^3 \cdots (x_{n-1}+x_1)^n > n^n x_1^2 x_2^3 \cdots x_{n-1}^n \qquad (*)$$

for all $x_1, \ldots, x_{n-1} > 0$. To prove this, we use the AM-GM inequality for each factor of the left-hand side as follows:

$$\begin{array}{rcccl}
(x_1+x_2)^2 & & & & \geq 2^2 x_1 x_2 \\
(x_2+x_3)^3 & = & \left(2\left(\frac{x_2}{2}\right)+x_3\right)^3 & & \geq 3^3\left(\frac{x_2}{2}\right)^2 x_3 \\
(x_3+x_4)^4 & = & \left(3\left(\frac{x_3}{3}\right)+x_4\right)^4 & & \geq 4^4\left(\frac{x_3}{3}\right)^3 x_4 \\
 & \vdots & \vdots & & \vdots \\
(x_{n-1}+x_1)^n & = & \left((n-1)\left(\frac{x_{n-1}}{n-1}\right)+x_1\right)^n & & \geq n^n\left(\frac{x_{n-1}}{n-1}\right)^{n-1} x_1.
\end{array}$$

Multiplying these inequalities together gives (*), with inequality sign $\geq$ instead of $>$. However for the equality to occur it is necessary that $x_1 = x_2$, $x_2 = 2x_3$, $\ldots$, $x_{n-1} = (n-1)x_1$, implying $x_1 = (n-1)!x_1$. This is impossible since $x_1 > 0$ and $n \geq 3$. Therefore the inequality is strict.

Comment. One can avoid the substitution $a_i = x_i/x_{i-1}$. Apply the weighted AM-GM inequality to each factor $(1+a_k)^k$, with the same weights like above, to obtain

$$(1+a_k)^k = \left((k-1)\frac{1}{k-1} + a_k\right)^k \geq \frac{k^k}{(k-1)^{k-1}} a_k.$$

Multiplying all these inequalities together gives

$$(1+a_2)^2(1+a_3)^3 \cdots (1+a_n)^n \geq n^n a_2 a_3 \cdots a_n = n^n.$$

The same argument as in the proof above shows that the equality cannot be attained.

A4. Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

Solution 1. Since $\deg f > \deg g$, we have $|g(x)/f(x)| < 1$ for sufficiently large x; more precisely, there is a real number R such that $|g(x)/f(x)| < 1$ for all x with $|x| > R$. Then for all such x and all primes p we have

$$\big|pf(x) + g(x)\big| \geq \big|f(x)\big| \left(p - \frac{|g(x)|}{|f(x)|}\right) > 0.$$

Hence all real roots of the polynomials $pf + g$ lie in the interval $[-R, R]$.

Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ and $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$ where $n > m$, $a_n \neq 0$ and $b_m \neq 0$. Upon replacing $f(x)$ and $g(x)$ by $a_n^{n-1} f(x/a_n)$ and $a_n^{n-1} g(x/a_n)$ respectively, we reduce the problem to the case $a_n = 1$. In other words one can assume that f is monic. Then the leading coefficient of $pf + g$ is p, and if $r = u/v$ is a rational root of $pf + g$ with $(u, v) = 1$ and $v > 0$, then either $v = 1$ or $v = p$.

First consider the case when $v = 1$ infinitely many times. If $v = 1$ then $|u| \leq R$, so there are only finitely many possibilities for the integer u. Therefore there exist distinct primes p and q for which we have the same value of u. Then the polynomials $pf + g$ and $qf + g$ share this root, implying $f(u) = g(u) = 0$. So in this case f and g have an integer root in common.

Now suppose that $v = p$ infinitely many times. By comparing the exponent of p in the denominators of $pf(u/p)$ and $g(u/p)$ we get $m = n - 1$ and $pf(u/p) + g(u/p) = 0$ reduces to an equation of the form

$$\left(u^n + a_{n-1}pu^{n-1} + \ldots + a_0 p^n\right) + \left(b_{n-1}u^{n-1} + b_{n-2}pu^{n-2} + \ldots + b_0 p^{n-1}\right) = 0.$$

The equation above implies that $u^n + b_{n-1}u^{n-1}$ is divisible by p and hence, since $(u, p) = 1$, we have $u + b_{n-1} = pk$ with some integer k. On the other hand all roots of $pf + g$ lie in the interval $[-R, R]$, so that

$$\frac{|pk - b_{n-1}|}{p} = \frac{|u|}{p} < R,$$
$$|k| < R + \frac{|b_{n-1}|}{p} < R + |b_{n-1}|.$$

Therefore the integer k can attain only finitely many values. Hence there exists an integer k such that the number $\frac{pk - b_{n-1}}{p} = k - \frac{b_{n-1}}{p}$ is a root of $pf + g$ for infinitely many primes p. For these primes we have

$$f\left(k - b_{n-1}\frac{1}{p}\right) + \frac{1}{p} g\left(k - b_{n-1}\frac{1}{p}\right) = 0.$$

So the equation

$$f\left(k - b_{n-1}x\right) + xg\left(k - b_{n-1}x\right) = 0 \tag{1}$$

has infinitely many solutions of the form $x = 1/p$. Since the left-hand side is a polynomial, this implies that (1) is a polynomial identity, so it holds for all real x. In particular, by substituting $x = 0$ in (1) we get $f(k) = 0$. Thus the integer k is a root of f.

In summary the monic polynomial f obtained after the initial reduction always has an integer root. Therefore the original polynomial f has a rational root.

Solution 2. Analogously to the first solution, there exists a real number R such that the complex roots of all polynomials of the form $pf + g$ lie in the disk $|z| \le R$.

For each prime p such that $pf + g$ has a rational root, by Gauss' lemma $pf + g$ is the product of two integer polynomials, one with degree 1 and the other with degree $\deg f - 1$. Since p is a prime, the leading coefficient of one of these factors divides the leading coefficient of f. Denote that factor by h_p.

By narrowing the set of the primes used we can assume that all polynomials h_p have the same degree and the same leading coefficient. Their complex roots lie in the disk $|z| \le R$, hence Vieta's formulae imply that all coefficients of all polynomials h_p form a bounded set. Since these coefficients are integers, there are only finitely many possible polynomials h_p. Hence there is a polynomial h such that $h_p = h$ for infinitely many primes p.

Finally, if p and q are distinct primes with $h_p = h_q = h$ then h divides $(p - q)f$. Since $\deg h = 1$ or $\deg h = \deg f - 1$, in both cases f has a rational root.

Comment. Clearly the polynomial h is a common factor of f and g. If $\deg h = 1$ then f and g share a rational root. Otherwise $\deg h = \deg f - 1$ forces $\deg g = \deg f - 1$ and g divides f over the rationals.

Solution 3. Like in the first solution, there is a real number R such that the real roots of all polynomials of the form $pf + g$ lie in the interval $[-R, R]$.

Let $p_1 < p_2 < \cdots$ be an infinite sequence of primes so that for every index k the polynomial $p_k f + g$ has a rational root r_k. The sequence $r_1, r_2, \dots$ is bounded, so it has a convergent subsequence $r_{k_1}, r_{k_2}, \dots$. Now replace the sequences $(p_1, p_2, \dots)$ and $(r_1, r_2, \dots)$ by $(p_{k_1}, p_{k_2}, \dots)$ and $(r_{k_1}, r_{k_2}, \dots)$; after this we can assume that the sequence $r_1, r_2, \dots$ is convergent. Let $\alpha = \lim\limits_{k\to\infty} r_k$. We show that α is a rational root of f.

Over the interval $[-R, R]$, the polynomial g is bounded, $|g(x)| \le M$ with some fixed M. Therefore

$$|f(r_k)| = \left| f(r_k) - \frac{p_k f(r_k) + g(r_k)}{p_k} \right| = \frac{|g(r_k)|}{p_k} \le \frac{M}{p_k} \to 0,$$

and

$$f(\alpha) = f\Big(\lim_{k\to\infty} r_k \Big) = \lim_{k\to\infty} f(r_k) = 0.$$

So α is a root of f indeed.

Now let u_k, v_k be relative prime integers for which $r_k = \frac{u_k}{v_k}$. Let a be the leading coefficient of f, let $b = f(0)$ and $c = g(0)$ be the constant terms of f and g, respectively. The leading coefficient of the polynomial $p_k f + g$ is $p_k a$, its constant term is $p_k b + c$. So v_k divides $p_k a$ and u_k divides $p_k b + c$. Let $p_k b + c = u_k e_k$ (if $p_k b + c = u_k = 0$ then let $e_k = 1$).

We prove that α is rational by using the following fact. *Let (p_n) and (q_n) be sequences of integers such that the sequence (p_n/q_n) converges. If (p_n) or (q_n) is bounded then $\lim(p_n/q_n)$ is rational.*

Case 1: There is an infinite subsequence (k_n) of indices such that v_{k_n} divides a. Then (v_{k_n}) is bounded, so $\alpha = \lim_{n\to\infty}(u_{k_n}/v_{k_n})$ is rational.

Case 2: There is an infinite subsequence (k_n) of indices such that v_{k_n} does not divide a. For such indices we have $v_{k_n} = p_{k_n} d_{k_n}$ where d_{k_n} is a divisor of a. Then

$$\alpha = \lim_{n\to\infty} \frac{u_{k_n}}{v_{k_n}} = \lim_{n\to\infty} \frac{p_{k_n} b + c}{p_{k_n} d_{k_n} e_{k_n}} = \lim_{n\to\infty} \frac{b}{d_{k_n} e_{k_n}} + \lim_{n\to\infty} \frac{c}{p_{k_n} d_{k_n} e_{k_n}} = \lim_{n\to\infty} \frac{b}{d_{k_n} e_{k_n}}.$$

Because the numerator b in the last limit is bounded, α is rational.

A5. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ that satisfy the conditions

$$f(1+xy) - f(x+y) = f(x)f(y) \quad \text{for all } x, y \in \mathbb{R}$$

and $f(-1) \neq 0$.

Solution. The only solution is the function $f(x) = x - 1$, $x \in \mathbb{R}$.

We set $g(x) = f(x) + 1$ and show that $g(x) = x$ for all real x. The conditions take the form

$$g(1+xy) - g(x+y) = \big(g(x) - 1\big)\big(g(y) - 1\big) \quad \text{for all } x, y \in \mathbb{R} \text{ and } g(-1) \neq 1. \tag{1}$$

Denote $C = g(-1) - 1 \neq 0$. Setting $y = -1$ in (1) gives

$$g(1-x) - g(x-1) = C(g(x) - 1). \tag{2}$$

Set $x = 1$ in (2) to obtain $C(g(1) - 1) = 0$. Hence $g(1) = 1$ as $C \neq 0$. Now plugging in $x = 0$ and $x = 2$ yields $g(0) = 0$ and $g(2) = 2$ respectively.

We pass on to the key observations

$$g(x) + g(2-x) = 2 \quad \text{for all } x \in \mathbb{R}, \tag{3}$$

$$g(x+2) - g(x) = 2 \quad \text{for all } x \in \mathbb{R}. \tag{4}$$

Replace x by $1 - x$ in (2), then change x to $-x$ in the resulting equation. We obtain the relations $g(x) - g(-x) = C(g(1-x) - 1)$, $g(-x) - g(x) = C(g(1+x) - 1)$. Then adding them up leads to $C(g(1-x) + g(1+x) - 2) = 0$. Thus $C \neq 0$ implies (3).

Let u, v be such that $u + v = 1$. Apply (1) to the pairs (u, v) and $(2-u, 2-v)$:

$$g(1+uv) - g(1) = \big(g(u) - 1\big)\big(g(v) - 1\big), \quad g(3+uv) - g(3) = \big(g(2-u) - 1\big)\big(g(2-v) - 1\big).$$

Observe that the last two equations have equal right-hand sides by (3). Hence $u+v = 1$ implies

$$g(uv+3) - g(uv+1) = g(3) - g(1).$$

Each $x \leq 5/4$ is expressible in the form $x = uv + 1$ with $u + v = 1$ (the quadratic function $t^2 - t + (x-1)$ has real roots for $x \leq 5/4$). Hence $g(x+2) - g(x) = g(3) - g(1)$ whenever $x \leq 5/4$. Because $g(x) = x$ holds for $x = 0, 1, 2$, setting $x = 0$ yields $g(3) = 3$. This proves (4) for $x \leq 5/4$. If $x > 5/4$ then $-x < 5/4$ and so $g(2-x) - g(-x) = 2$ by the above. On the other hand (3) gives $g(x) = 2 - g(2-x)$, $g(x+2) = 2 - g(-x)$, so that $g(x+2) - g(x) = g(2-x) - g(-x) = 2$. Thus (4) is true for all $x \in \mathbb{R}$.

Now replace x by $-x$ in (3) to obtain $g(-x) + g(2+x) = 2$. In view of (4) this leads to $g(x) + g(-x) = 0$, i. e. $g(-x) = -g(x)$ for all x. Taking this into account, we apply (1) to the pairs $(-x, y)$ and $(x, -y)$:

$$g(1-xy) - g(-x+y) = \big(g(x) + 1\big)\big(1 - g(y)\big), \quad g(1-xy) - g(x-y) = \big(1 - g(x)\big)\big(g(y) + 1\big).$$

Adding up yields $g(1-xy) = 1 - g(x)g(y)$. Then $g(1+xy) = 1 + g(x)g(y)$ by (3). Now the original equation (1) takes the form $g(x+y) = g(x) + g(y)$. Hence g is additive.

By additvity $g(1+xy) = g(1) + g(xy) = 1 + g(xy)$; since $g(1+xy) = 1 + g(x)g(y)$ was shown above, we also have $g(xy) = g(x)g(y)$ (g is multiplicative). In particular $y = x$ gives $g(x^2) = g(x)^2 \geq 0$ for all x, meaning that $g(x) \geq 0$ for $x \geq 0$. Since g is additive and bounded from below on $[0, +\infty)$, it is linear; more exactly $g(x) = g(1)x = x$ for all $x \in \mathbb{R}$.

In summary $f(x) = x - 1$, $x \in \mathbb{R}$. It is straightforward that this function satisfies the requirements.

Comment. There are functions that satisfy the given equation but vanish at -1, for instance the constant function 0 and $f(x) = x^2 - 1$, $x \in \mathbb{R}$.

16

A6. Let $f : \mathbb{N} \to \mathbb{N}$ be a function, and let f^m be f applied m times. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k. Prove that the sequence $k_1, k_2, \ldots$ is unbounded.

Solution. We restrict attention to the set

$$S = \{1, f(1), f^2(1), \ldots\}.$$

Observe that S is unbounded because for every number n in S there exists a $k > 0$ such that $f^{2k}(n) = n + k$ is in S. Clearly f maps S into itself; moreover f is injective on S. Indeed if $f^i(1) = f^j(1)$ with $i \neq j$ then the values $f^m(1)$ start repeating periodically from some point on, and S would be finite.

Define $g : S \to S$ by $g(n) = f^{2k_n}(n) = n + k_n$. We prove that g is injective too. Suppose that $g(a) = g(b)$ with $a < b$. Then $a + k_a = f^{2k_a}(a) = f^{2k_b}(b) = b + k_b$ implies $k_a > k_b$. So, since f is injective on S, we obtain

$$f^{2(k_a - k_b)}(a) = b = a + (k_a - k_b).$$

However this contradicts the minimality of k_a as $0 < k_a - k_b < k_a$.

Let T be the set of elements of S that are not of the form $g(n)$ with $n \in S$. Note that $1 \in T$ by $g(n) > n$ for $n \in S$, so T is non-empty. For each $t \in T$ denote $C_t = \{t, g(t), g^2(t), \ldots\}$; call C_t the chain starting at t. Observe that distinct chains are disjoint because g is injective. Each $n \in S \backslash T$ has the form $n = g(n')$ with $n' < n$, $n' \in S$. Repeated applications of the same observation show that $n \in C_t$ for some $t \in T$, i. e. S is the disjoint union of the chains C_t.

If $f^n(1)$ is in the chain C_t starting at $t = f^{n_t}(1)$ then $n = n_t + 2a_1 + \cdots + 2a_j$ with

$$f^n(1) = g^j(f^{n_t}(1)) = f^{2a_j}(f^{2a_{j-1}}(\cdots f^{2a_1}(f^{n_t}(1)))) = f^{n_t}(1) + a_1 + \cdots + a_j.$$

Hence

$$f^n(1) = f^{n_t}(1) + \frac{n - n_t}{2} = t + \frac{n - n_t}{2}. \tag{1}$$

Now we show that T is infinite. We argue by contradiction. Suppose that there are only finitely many chains $C_{t_1}, \ldots, C_{t_r}$, starting at $t_1 < \cdots < t_r$. Fix N. If $f^n(1)$ with $1 \leq n \leq N$ is in C_t then $f^n(1) = t + \frac{n - n_t}{2} \leq t_r + \frac{N}{2}$ by (1). But then the $N + 1$ distinct natural numbers $1, f(1), \ldots, f^N(1)$ are all less than $t_r + \frac{N}{2}$ and hence $N + 1 \leq t_r + \frac{N}{2}$. This is a contradiction if N is sufficiently large, and hence T is infinite.

To complete the argument, choose any k in $\mathbb{N}$ and consider the $k + 1$ chains starting at the first $k + 1$ numbers in T. Let t be the greatest one among these numbers. Then each of the chains in question contains a number not exceeding t, and at least one of them does not contain any number among $t+1, \ldots, t+k$. So there is a number n in this chain such that $g(n) - n > k$, i. e. $k_n > k$. In conclusion $k_1, k_2, \ldots$ is unbounded.

A7. We say that a function $f : \mathbb{R}^k \to \mathbb{R}$ is a metapolynomial if, for some positive integers m and n, it can be represented in the form

$$f(x_1, \ldots, x_k) = \max_{i=1,\ldots,m} \min_{j=1,\ldots,n} P_{i,j}(x_1, \ldots, x_k)$$

where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

Solution. We use the notation $f(x) = f(x_1, \ldots, x_k)$ for $x = (x_1, \ldots, x_k)$ and $[m] = \{1, 2, \ldots, m\}$. Observe that if a metapolynomial $f(x)$ admits a representation like the one in the statement for certain positive integers m and n, then they can be replaced by any $m' \geq m$ and $n' \geq n$. For instance, if we want to replace m by $m+1$ then it is enough to define $P_{m+1,j}(x) = P_{m,j}(x)$ and note that repeating elements of a set do not change its maximum nor its minimum. So one can assume that any two metapolynomials are defined with the same m and n. We reserve letters P and Q for polynomials, so every function called $P, P_{i,j}, Q, Q_{i,j}, \ldots$ is a polynomial function.

We start with a lemma that is useful to change expressions of the form $\min \max f_{i,j}$ to ones of the form $\max \min g_{i,j}$.

Lemma. Let $\{a_{i,j}\}$ be real numbers, for all $i \in [m]$ and $j \in [n]$. Then

$$\min_{i\in[m]} \max_{j\in[n]} a_{i,j} = \max_{j_1,\ldots,j_m\in[n]} \min_{i\in[m]} a_{i,j_i},$$

where the max in the right-hand side is over all vectors $(j_1, \ldots, j_m)$ with $j_1, \ldots, j_m \in [n]$.

Proof. We can assume for all i that $a_{i,n} = \max\{a_{i,1}, \ldots, a_{i,n}\}$ and $a_{m,n} = \min\{a_{1,n}, \ldots, a_{m,n}\}$. The left-hand side is $= a_{m,n}$ and hence we need to prove the same for the right-hand side. If $(j_1, j_2, \ldots, j_m) = (n, n, \ldots, n)$ then $\min\{a_{1,j_1}, \ldots, a_{m,j_m}\} = \min\{a_{1,n}, \ldots, a_{m,n}\} = a_{m,n}$ which implies that the right-hand side is $\geq a_{m,n}$. It remains to prove the opposite inequality and this is equivalent to $\min\{a_{1,j_1}, \ldots, a_{m,j_m}\} \leq a_{m,n}$ for all possible $(j_1, j_2, \ldots, j_m)$. This is true because $\min\{a_{1,j_1}, \ldots, a_{m,j_m}\} \leq a_{m,j_m} \leq a_{m,n}$. □

We need to show that the family $\mathcal{M}$ of metapolynomials is closed under multiplication, but it turns out easier to prove more: that it is also closed under addition, maxima and minima.

First we prove the assertions about the maxima and the minima. If $f_1, \ldots, f_r$ are metapolynomials, assume them defined with the same m and n. Then

$$f = \max\{f_1, \ldots, f_r\} = \max\{\max_{i\in[m]} \min_{j\in[n]} P^1_{i,j}, \ldots, \max_{i\in[m]} \min_{j\in[n]} P^r_{i,j}\} = \max_{s\in[r],i\in[m]} \min_{j\in[n]} P^s_{i,j}.$$

It follows that $f = \max\{f_1, \ldots, f_r\}$ is a metapolynomial. The same argument works for the minima, but first we have to replace *min max* by *max min*, and this is done via the lemma.

Another property we need is that if $f = \max \min P_{i,j}$ is a metapolynomial then so is $-f$. Indeed, $-f = \min(-\min P_{i,j}) = \min \max P_{i,j}$.

To prove $\mathcal{M}$ is closed under addition let $f = \max \min P_{i,j}$ and $g = \max \min Q_{i,j}$. Then

$$f(x) + g(x) = \max_{i\in[m]} \min_{j\in[n]} P_{i,j}(x) + \max_{i\in[m]} \min_{j\in[n]} Q_{i,j}(x)$$

$$= \max_{i_1,i_2\in[m]} \big(\min_{j\in[n]} P_{i_1,j}(x) + \min_{j\in[n]} Q_{i_2,j}(x)\big) = \max_{i_1,i_2\in[m]} \min_{j_1,j_2\in[n]} \big(P_{i_1,j_1}(x) + Q_{i_2,j_2}(x)\big),$$

and hence $f(x) + g(x)$ is a metapolynomial.

We proved that $\mathcal{M}$ is closed under sums, maxima and minima, in particular any function that can be expressed by sums, max, min, polynomials or even metapolynomials is in $\mathcal{M}$.

We would like to proceed with multiplication along the same lines like with addition, but there is an essential difference. In general the product of the maxima of two sets is not equal

to the maximum of the product of the sets. We need to deal with the fact that $a < b$ and $c < d$ do not imply $ac < bd$. However this is true for $a, b, c, d \geq 0$.

In view of this we decompose each function $f(x)$ into its positive part $f^+(x) = \max\{f(x), 0\}$ and its negative part $f^-(x) = \max\{0, -f(x)\}$. Note that $f = f^+ - f^-$ and $f^+, f^- \in \mathcal{M}$ if $f \in \mathcal{M}$. The whole problem reduces to the claim that if f and g are metapolynomials with $f, g \geq 0$ then fg it is also a metapolynomial.

Assuming this claim, consider arbitrary $f, g \in \mathcal{M}$. We have

$$fg = (f^+ - f^-)(g^+ - g^-) = f^+g^+ - f^+g^- - f^-g^+ + f^-g^-,$$

and hence $fg \in \mathcal{M}$. Indeed, $\mathcal{M}$ is closed under addition, also $f^+g^+, f^+g^-, f^-g^+, f^-g^- \in \mathcal{M}$ because $f^+, f^-, g^+, g^- \geq 0$.

It remains to prove the claim. In this case $f, g \geq 0$, and one can try to repeat the argument for the sum. More precisely, let $f = \max\min P_{ij} \geq 0$ and $g = \max\min Q_{ij} \geq 0$. Then

$$fg = \max\min P_{i,j} \cdot \max\min Q_{i,j} = \max\min P^+_{i,j} \cdot \max\min Q^+_{i,j} = \max\min P^+_{i_1,j_1} \cdot Q^+_{i_2,j_2}.$$

Hence it suffices to check that $P^+Q^+ \in \mathcal{M}$ for any pair of polynomials P and Q. This reduces to the identity

$$u^+v^+ = \max\{0, \min\{uv, u, v\}, \min\{uv, uv^2, u^2v\}, \min\{uv, u, u^2v\}, \min\{uv, uv^2, v\}\},$$

with u replaced by $P(x)$ and v replaced by $Q(x)$. The formula is proved by a case-by-case analysis. If $u \leq 0$ or $v \leq 0$ then both sides equal 0. In case $u, v \geq 0$, the right-hand side is clearly $\leq uv$. To prove the opposite inequality we use that uv equals

$$\begin{array}{ll} \min\{uv, u, v\} & \text{if } 0 \leq u, v \leq 1, \\ \min\{uv, uv^2, u^2v\} & \text{if } 1 \leq u, v, \\ \min\{uv, u, u^2v\} & \text{if } 0 \leq v \leq 1 \leq u, \\ \min\{uv, uv^2, v\} & \text{if } 0 \leq u \leq 1 \leq v. \end{array}$$

Comment. The case $k = 1$ is simpler and can be solved by proving that a function $f : \mathbb{R} \to \mathbb{R}$ is a metapolynomial if and only if it is a piecewise polinomial (and continuos) function.

It is enough to prove that all such functions are metapolynomials, and this easily reduces to the following case. Given a polynomial $P(x)$ with $P(0) = 0$, the function f defined by $f(x) = P(x)$ for $x \geq 0$ and 0 otherwise is a metapolynomial. For this last claim, it suffices to prove that $(x^+)^n$ is a metapolynomial, and this follows from the formula $(x^+)^n = \max\{0, \min\{x^{n-1}, x^n\}, \min\{x^n, x^{n+1}\}\}$.

Combinatorics

C1. Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y, and replaces the pair (x, y) by either $(y + 1, x)$ or $(x - 1, x)$. Prove that she can perform only finitely many such iterations.

Solution 1. Note first that the allowed operation does not change the maximum M of the initial sequence. Let $a_1, a_2, \ldots, a_n$ be the numbers obtained at some point of the process. Consider the sum

$$S = a_1 + 2a_2 + \cdots + na_n.$$

We claim that S increases by a positive integer amount with every operation. Let the operation replace the pair (a_i, a_{i+1}) by a pair (c, a_i), where $a_i > a_{i+1}$ and $c = a_{i+1}+1$ or $c = a_i-1$. Then the new and the old value of S differ by $d = (ic+(i+1)a_i)-(ia_i+(i+1)a_{i+1}) = a_i-a_{i+1}+i(c-a_{i+1})$. The integer d is positive since $a_i - a_{i+1} \geq 1$ and $c - a_{i+1} \geq 0$.

On the other hand $S \leq (1 + 2 + \cdots + n)M$ as $a_i \leq M$ for all $i = 1, \ldots, n$. Since S increases by at least 1 at each step and never exceeds the constant $(1 + 2 + \cdots + n)M$, the process stops after a finite number of iterations.

Solution 2. Like in the first solution note that the operations do not change the maximum M of the initial sequence. Now consider the reverse lexicographical order for n-tuples of integers. We say that $(x_1, \ldots, x_n) < (y_1, \ldots, y_n)$ if $x_n < y_n$, or if $x_n = y_n$ and $x_{n-1} < y_{n-1}$, or if $x_n = y_n$, $x_{n-1} = y_{n-1}$ and $x_{n-2} < y_{n-2}$, etc. Each iteration creates a sequence that is greater than the previous one with respect to this order, and no sequence occurs twice during the process. On the other hand there are finitely many possible sequences because their terms are always positive integers not exceeding M. Hence the process cannot continue forever.

Solution 3. Let the current numbers be $a_1, a_2, \ldots, a_n$. Define the *score* s_i of a_i as the number of a_j's that are less than a_i. Call the sequence $s_1, s_2, \ldots, s_n$ the score sequence of $a_1, a_2, \ldots, a_n$.

Let us say that a sequence $x_1, \ldots, x_n$ dominates a sequence $y_1, \ldots, y_n$ if the first index i with $x_i \neq y_i$ is such that $x_i < y_i$. We show that after each operation the new score sequence dominates the old one. Score sequences do not repeat, and there are finitely many possibilities for them, no more than $(n - 1)^n$. Hence the process will terminate.

Consider an operation that replaces (x, y) by (a, x), with $a = y + 1$ or $a = x - 1$. Suppose that x was originally at position i. For each $j < i$ the score s_j does not increase with the change because $y \leq a$ and $x \leq x$. If s_j decreases for some $j < i$ then the new score sequence dominates the old one. Assume that s_j stays the same for all $j < i$ and consider s_i. Since $x > y$ and $y \leq a \leq x$, we see that s_i decreases by at least 1. This concludes the proof.

Comment. All three proofs work if x and y are not necessarily adjacent, and if the pair (x, y) is replaced by any pair (a, x), with a an integer satisfying $y \leq a \leq x$. There is nothing special about the "weights" $1, 2, \ldots, n$ in the definition of $S = \sum_{i=1}^{n} ia_i$ from the first solution. For any sequence $w_1 < w_2 < \cdots < w_n$ of positive integers, the sum $\sum_{i=1}^{n} w_i a_i$ increases by at least 1 with each operation.

Consider the same problem, but letting Alice replace the pair (x, y) by (a, x), where a is any positive integer less than x. The same conclusion holds in this version, i. e. the process stops eventually. The solution using the reverse lexicographical order works without any change. The first solution would require a special set of weights like $w_i = M^i$ for $i = 1, \ldots, n$.

Comment. The first and the second solutions provide upper bounds for the number of possible operations, respectively of order Mn^2 and M^n where M is the maximum of the original sequence. The upper bound $(n - 1)^n$ in the third solution does not depend on M.

C2. Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \ldots, n\}$ such that the sums of the different pairs are different integers not exceeding n?

Solution. Consider x such pairs in $\{1, 2, \ldots, n\}$. The sum S of the $2x$ numbers in them is at least $1+2+\cdots+2x$ since the pairs are disjoint. On the other hand $S \leq n+(n-1)+\cdots+(n-x+1)$ because the sums of the pairs are different and do not exceed n. This gives the inequality

$$\frac{2x(2x+1)}{2} \leq nx - \frac{x(x-1)}{2},$$

which leads to $x \leq \frac{2n-1}{5}$. Hence there are at most $\left\lfloor \frac{2n-1}{5} \right\rfloor$ pairs with the given properties.

We show a construction with exactly $\left\lfloor \frac{2n-1}{5} \right\rfloor$ pairs. First consider the case $n = 5k+3$ with $k \geq 0$, where $\left\lfloor \frac{2n-1}{5} \right\rfloor = 2k+1$. The pairs are displayed in the following table.

Pairs	$3k+1$ 2	$3k$ 4	$\cdots$ $\cdots$	$2k+2$ $2k$	$4k+2$ 1	$4k+1$ 3	$\cdots$ $\cdots$	$3k+3$ $2k-1$	$3k+2$ $2k+1$
Sums	$3k+3$	$3k+4$	$\cdots$	$4k+2$	$4k+3$	$4k+4$	$\cdots$	$5k+2$	$5k+3$

The $2k+1$ pairs involve all numbers from 1 to $4k+2$; their sums are all numbers from $3k+3$ to $5k+3$. The same construction works for $n = 5k+4$ and $n = 5k+5$ with $k \geq 0$. In these cases the required number $\left\lfloor \frac{2n-1}{5} \right\rfloor$ of pairs equals $2k+1$ again, and the numbers in the table do not exceed $5k+3$. In the case $n = 5k+2$ with $k \geq 0$ one needs only $2k$ pairs. They can be obtained by ignoring the last column of the table (thus removing $5k+3$). Finally, $2k$ pairs are also needed for the case $n = 5k+1$ with $k \geq 0$. Now it suffices to ignore the last column of the table and then subtract 1 from each number in the first row.

Comment. The construction above is not unique. For instance, the following table shows another set of $2k+1$ pairs for the cases $n = 5k+3$, $n = 5k+4$, and $n = 5k+5$.

Pairs	1 $4k+1$	2 $4k-1$	$\cdots$ $\cdots$	k $2k+3$	$k+1$ $4k+2$	$k+2$ $4k$	$\cdots$ $\cdots$	$2k+1$ $2k+2$
Sums	$4k+2$	$4k+1$	$\cdots$	$3k+3$	$5k+3$	$5k+2$	$\cdots$	$4k+3$

The table for the case $n = 5k+2$ would be the same, with the pair $(k+1, 4k+2)$ removed. For the case $n = 5k+1$ remove the last column and subtract 2 from each number in the second row.

C3. In a 999×999 square table some cells are white and the remaining ones are red. Let T be the number of triples (C_1, C_2, C_3) of cells, the first two in the same row and the last two in the same column, with C_1 and C_3 white and C_2 red. Find the maximum value T can attain.

Solution. We prove that in an $n \times n$ square table there are at most $\frac{4n^4}{27}$ such triples.

Let row i and column j contain a_i and b_j white cells respectively, and let R be the set of red cells. For every red cell (i, j) there are $a_i b_j$ admissible triples (C_1, C_2, C_3) with $C_2 = (i, j)$, therefore

$$T = \sum_{(i,j)\in R} a_i b_j.$$

We use the inequality $2ab \le a^2 + b^2$ to obtain

$$T \le \frac{1}{2} \sum_{(i,j)\in R} (a_i^2 + b_j^2) = \frac{1}{2} \sum_{i=1}^{n} (n - a_i) a_i^2 + \frac{1}{2} \sum_{j=1}^{n} (n - b_j) b_j^2.$$

This is because there are $n - a_i$ red cells in row i and $n - b_j$ red cells in column j. Now we maximize the right-hand side.

By the AM-GM inequality we have

$$(n - x)x^2 = \frac{1}{2}(2n - 2x) \cdot x \cdot x \le \frac{1}{2} \left(\frac{2n}{3} \right)^3 = \frac{4n^3}{27},$$

with equality if and only if $x = \frac{2n}{3}$. By putting everything together, we get

$$T \le \frac{n}{2} \frac{4n^3}{27} + \frac{n}{2} \frac{4n^3}{27} = \frac{4n^4}{27}.$$

If $n = 999$ then any coloring of the square table with $x = \frac{2n}{3} = 666$ white cells in each row and column attains the maximum as all inequalities in the previous argument become equalities. For example color a cell (i, j) white if $i - j \equiv 1, 2, \ldots, 666 \pmod{999}$, and red otherwise.

Therefore the maximum value T can attain is $T = \frac{4 \cdot 999^4}{27}$.

Comment. One can obtain a better preliminary estimate with the Cauchy-Schwarz inequality:

$$T = \sum_{(i,j)\in R} a_i b_j \le \left(\sum_{(i,j)\in R} a_i^2 \right)^{\frac{1}{2}} \cdot \left(\sum_{(i,j)\in R} b_j^2 \right)^{\frac{1}{2}} = \left(\sum_{i=1}^{n} (n - a_i) a_i^2 \right)^{\frac{1}{2}} \cdot \left(\sum_{j=1}^{n} (n - b_j) b_j^2 \right)^{\frac{1}{2}}.$$

It can be used to reach the same conclusion.

22

C4. Players A and B play a game with $N \geq 2012$ coins and 2012 boxes arranged around a circle. Initially A distributes the coins among the boxes so that there is at least 1 coin in each box. Then the two of them make moves in the order $B, A, B, A, \ldots$ by the following rules:

- On every move of his B passes 1 coin from every box to an adjacent box.
- On every move of hers A chooses several coins that were *not* involved in B's previous move and are in different boxes. She passes every chosen coin to an adjacent box.

Player A's goal is to ensure at least 1 coin in each box after every move of hers, regardless of how B plays and how many moves are made. Find the least N that enables her to succeed.

Solution. We argue for a general $n \geq 7$ instead of 2012 and prove that the required minimum N is $2n-2$. For $n = 2012$ this gives $N_{\min} = 4022$.

a) If $N = 2n-2$ player A can achieve her goal. Let her start the game with a *regular* distribution: $n-2$ boxes with 2 coins and 2 boxes with 1 coin. Call the boxes of the two kinds *red* and *white* respectively. We claim that on her first move A can achieve a regular distribution again, regardless of B's first move M. She acts according as the following situation S occurs after M or not: *The initial distribution contains a red box R with 2 white neighbors, and R receives no coins from them on move M.*

Suppose that S does not occur. Exactly one of the coins c_1 and c_2 in a given red box X is involved in M, say c_1. If M passes c_1 to the right neighbor of X, let A pass c_2 to its left neighbor, and vice versa. By doing so with all red boxes A performs a legal move M'. Thus M and M' combined move the 2 coins of every red box in opposite directions. Hence after M and M' are complete each neighbor of a red box X contains exactly 1 coin that was initially in X. So each box with a red neighbor is non-empty after M'. If initially there is a box X with 2 white neighbors (X is red and unique) then X receives a coin from at least one of them on move M since S does not occur. Such a coin is not involved in M', so X is also non-empty after M'. Furthermore each box Y has given away its initial content after M and M'. A red neighbor of Y adds 1 coin to it; a white neighbor adds at most 1 coin because it is not involved in M'. Hence each box contains 1 or 2 coins after M'. Because $N = 2n-2$, such a distribution is regular.

Now let S occur after move M. Then A leaves untouched the exceptional red box R. With all remaining red boxes she proceeds like in the previous case, thus making a legal move M''. Box R receives no coins from its neighbors on either move, so there is 1 coin in it after M''. Like above M and M'' combined pass exactly 1 coin from every red box different from R to each of its neighbors. Every box except R has a red neighbor different from R, hence all boxes are non-empty after M''. Next, each box Y except R loses its initial content after M and M''. A red neighbor of Y adds at most 1 coin to it; a white neighbor also adds at most 1 coin as it does not participate in M''. Thus each box has 1 or 2 coins after M'', and the obtained distribution is regular.

Player A can apply the described strategy indefinitely, so $N = 2n-2$ enables her to succeed.

b) For $N \leq 2n-3$ player B can achieve an empty box after some move of A. Let α be a set of ℓ consecutive boxes containing a total of $N(\alpha)$ coins. We call α an *arc* if $\ell \leq n-2$ and $N(\alpha) \leq 2\ell - 3$. Note that $\ell \geq 2$ by the last condition. Moreover if both extremes of α are non-empty boxes then $N(\alpha) \geq 2$, so that $N(\alpha) \leq 2\ell - 3$ implies $\ell \geq 3$. Observe also that if an extreme X of α has more than 1 coin then ignoring X yields a shorter arc. It follows that every arc contains an arc whose extremes have at most 1 coin each.

Given a clockwise labeling $1, 2, \ldots, n$ of the boxes, suppose that boxes $1, 2, \ldots, \ell$ form an arc α, with $\ell \leq n-2$ and $N(\alpha) \leq 2\ell - 3$. Suppose also that all $n \geq 7$ boxes are non-empty. Then B can move so that an arc α' with $N(\alpha') < N(\alpha)$ will appear after any response of A.

One may assume exactly 1 coin in boxes 1 and ℓ by a previous remark. Let B pass 1 coin in counterclockwise direction from box 1 and box n, and in clockwise direction from each remaining box. This leaves $N(\alpha)-2$ coins in the boxes of α. In addition, due to $3 \le \ell \le n-2$, box ℓ has exactly 1 coin c, the one received from box $\ell - 1$.

Let player A's next move M pass $k \le 2$ coins to boxes $1, 2, \ldots, \ell$ from the remaining ones. Only boxes 1 and ℓ can receive such coins, at most 1 each. If $k < 2$ then after move M boxes $1, 2, \ldots, \ell$ form an arc α' with $N(\alpha') < N(\alpha)$. If $k = 2$ then M adds a coin to box ℓ. Also M does not move coin c from ℓ because c is involved in the previous move of B. In summary boxes $1, 2, \ldots, \ell$ contain $N(\alpha)$ coins like before, so they form an arc. However there are 2 coins now in the extreme ℓ of the arc. Ignore ℓ to obtain a shorter arc α' with $N(\alpha') < N(\alpha)$.

Consider any initial distribution without empty boxes. Since $N \le 2n-3$, there are at least 3 boxes in it with exactly 1 coin. It follows from $n \ge 7$ that some 2 of them are the extremes of an arc α. Hence B can make the move described above, which leads to an arc α' with $N(\alpha') < N(\alpha)$ after A's response. If all boxes in the new distribution are non-empty he can repeat the same, and so on. Because $N(\alpha)$ cannot decrease indefinitely, an empty box will occur after some move of A.

C5. The columns and the rows of a $3n \times 3n$ square board are numbered $1, 2, \ldots, 3n$. Every square (x, y) with $1 \leq x, y \leq 3n$ is colored asparagus, byzantium or citrine according as the modulo 3 remainder of $x + y$ is 0, 1 or 2 respectively. One token colored asparagus, byzantium or citrine is placed on each square, so that there are $3n^2$ tokens of each color.

Suppose that one can permute the tokens so that each token is moved to a distance of at most d from its original position, each asparagus token replaces a byzantium token, each byzantium token replaces a citrine token, and each citrine token replaces an asparagus token. Prove that it is possible to permute the tokens so that each token is moved to a distance of at most $d + 2$ from its original position, and each square contains a token with the same color as the square.

Solution. Without loss of generality it suffices to prove that the A-tokens can be moved to distinct A-squares in such a way that each A-token is moved to a distance at most $d + 2$ from its original place. This means we need a perfect matching between the $3n^2$ A-squares and the $3n^2$ A-tokens such that the distance in each pair of the matching is at most $d + 2$.

To find the matching, we construct a bipartite graph. The A-squares will be the vertices in one class of the graph; the vertices in the other class will be the A-tokens.

Split the board into 3×1 horizontal triminos; then each trimino contains exactly one A-square. Take a permutation π of the tokens which moves A-tokens to B-tokens, B-tokens to C-tokens, and C-tokens to A-tokens, in each case to a distance at most d. For each A-square S, and for each A-token T, connect S and T by an edge if T, $\pi(T)$ or $\pi^{-1}(T)$ is on the trimino containing S. We allow multiple edges; it is even possible that the same square and the same token are connected with three edges. Obviously the lengths of the edges in the graph do not exceed $d + 2$. By length of an edge we mean the distance between the A-square and the A-token it connects.

Each A-token T is connected with the three A-squares whose triminos contain T, $\pi(T)$ and $\pi^{-1}(T)$. Therefore in the graph all tokens are of degree 3. We show that the same is true for the A-squares. Let S be an arbitrary A-square, and let T_1, T_2, T_3 be the three tokens on the trimino containing S. For $i = 1, 2, 3$, if T_i is an A-token, then S is connected with T_i; if T_i is a B-token then S is connected with $\pi^{-1}(T_i)$; finally, if T_i is a C-token then S is connected with $\pi(T_i)$. Hence in the graph the A-squares also are of degree 3.

Since the A-squares are of degree 3, from every set $\mathcal{S}$ of A-squares exactly $3|\mathcal{S}|$ edges start. These edges end in at least $|\mathcal{S}|$ tokens because the A-tokens also are of degree 3. Hence every set $\mathcal{S}$ of A-squares has at least $|\mathcal{S}|$ neighbors among the A-tokens.

Therefore, by Hall's marriage theorem, the graph contains a perfect matching between the two vertex classes. So there is a perfect matching between the A-squares and A-tokens with edges no longer than $d + 2$. It follows that the tokens can be permuted as specified in the problem statement.

Comment 1. In the original problem proposal the board was infinite and there were only two colors. Having n colors for some positive integer n was an option; we chose $n = 3$. Moreover, we changed the board to a finite one to avoid dealing with infinite graphs (although Hall's theorem works in the infinite case as well).

With only two colors Hall's theorem is not needed. In this case we split the board into 2×1 dominos, and in the resulting graph all vertices are of degree 2. The graph consists of disjoint cycles with even length and infinite paths, so the existence of the matching is trivial.

Having more than three colors would make the problem statement more complicated, because we need a matching between every two color classes of tokens. However, this would not mean a significant increase in difficulty.

Comment 2. According to Wikipedia, the color *asparagus* (hexadecimal code `#87A96B`) is a tone of green that is named after the vegetable. Crayola created this color in 1993 as one of the 16 to be named in the Name The Color Contest. *Byzantium* (`#702963`) is a dark tone of purple. Its first recorded use as a color name in English was in 1926. *Citrine* (`#E4D00A`) is variously described as yellow, greenish-yellow, brownish-yellow or orange. The first known use of citrine as a color name in English was in the 14th century.

C6. Let k and n be fixed positive integers. In the liar's guessing game, Amy chooses integers x and N with $1 \le x \le N$. She tells Ben what N is, but not what x is. Ben may then repeatedly ask Amy whether $x \in S$ for arbitrary sets S of integers. Amy will always answer with *yes* or *no*, but she might lie. The only restriction is that she can lie at most k times in a row. After he has asked as many questions as he wants, Ben must specify a set of at most n positive integers. If x is in this set he wins; otherwise, he loses. Prove that:

a) If $n \ge 2^k$ then Ben can always win.

b) For sufficiently large k there exist $n \ge 1.99^k$ such that Ben cannot guarantee a win.

Solution. Consider an answer $A \in \{yes, no\}$ to a question of the kind "Is x in the set S?" We say that A is inconsistent with a number i if $A = yes$ and $i \notin S$, or if $A = no$ and $i \in S$. Observe that an answer inconsistent with the target number x is a lie.

a) Suppose that Ben has determined a set T of size m that contains x. This is true initially with $m = N$ and $T = \{1, 2, \ldots, N\}$. For $m > 2^k$ we show how Ben can find a number $y \in T$ that is different from x. By performing this step repeatedly he can reduce T to be of size $2^k \le n$ and thus win.

Since only the size $m > 2^k$ of T is relevant, assume that $T = \{0, 1, \ldots, 2^k, \ldots, m-1\}$. Ben begins by asking repeatedly whether x is 2^k. If Amy answers *no* $k+1$ times in a row, one of these answers is truthful, and so $x \ne 2^k$. Otherwise Ben stops asking about 2^k at the first answer *yes*. He then asks, for each $i = 1, \ldots, k$, if the binary representation of x has a 0 in the ith digit. Regardless of what the k answers are, they are all inconsistent with a certain number $y \in \{0, 1, \ldots, 2^k - 1\}$. The preceding answer *yes* about 2^k is also inconsistent with y. Hence $y \ne x$. Otherwise the last $k+1$ answers are not truthful, which is impossible.

Either way, Ben finds a number in T that is different from x, and the claim is proven.

b) We prove that if $1 < \lambda < 2$ and $n = \lfloor (2-\lambda)\lambda^{k+1} \rfloor - 1$ then Ben cannot guarantee a win. To complete the proof, then it suffices to take λ such that $1.99 < \lambda < 2$ and k large enough so that

$$n = \lfloor (2-\lambda)\lambda^{k+1} \rfloor - 1 \ge 1.99^k.$$

Consider the following strategy for Amy. First she chooses $N = n+1$ and $x \in \{1, 2, \ldots, n+1\}$ arbitrarily. After every answer of hers Amy determines, for each $i = 1, 2, \ldots, n+1$, the number m_i of consecutive answers she has given by that point that are inconsistent with i. To decide on her next answer, she then uses the quantity

$$\phi = \sum_{i=1}^{n+1} \lambda^{m_i}.$$

No matter what Ben's next question is, Amy chooses the answer which minimizes ϕ.

We claim that with this strategy ϕ will always stay less than λ^{k+1}. Consequently no exponent m_i in ϕ will ever exceed k, hence Amy will never give more than k consecutive answers inconsistent with some i. In particular this applies to the target number x, so she will never lie more than k times in a row. Thus, given the claim, Amy's strategy is legal. Since the strategy does not depend on x in any way, Ben can make no deductions about x, and therefore he cannot guarantee a win.

It remains to show that $\phi < \lambda^{k+1}$ at all times. Initially each m_i is 0, so this condition holds in the beginning due to $1 < \lambda < 2$ and $n = \lfloor (2-\lambda)\lambda^{k+1} \rfloor - 1$. Suppose that $\phi < \lambda^{k+1}$ at some point, and Ben has just asked if $x \in S$ for some set S. According as Amy answers *yes* or *no*, the new value of ϕ becomes

$$\phi_1 = \sum_{i \in S} 1 + \sum_{i \notin S} \lambda^{m_i+1} \qquad \text{or} \qquad \phi_2 = \sum_{i \in S} \lambda^{m_i+1} + \sum_{i \notin S} 1.$$

Since Amy chooses the option minimizing ϕ, the new ϕ will equal $\min(\phi_1, \phi_2)$. Now we have

$$\min(\phi_1, \phi_2) \le \frac{1}{2}(\phi_1 + \phi_2) = \frac{1}{2}\left(\sum_{i \in S}\left(1 + \lambda^{m_i+1}\right) + \sum_{i \notin S}\left(\lambda^{m_i+1} + 1\right)\right) = \frac{1}{2}(\lambda\phi + n + 1).$$

Because $\phi < \lambda^{k+1}$, the assumptions $\lambda < 2$ and $n = \left\lfloor (2-\lambda)\lambda^{k+1} \right\rfloor - 1$ lead to

$$\min(\phi_1, \phi_2) < \frac{1}{2}(\lambda^{k+2} + (2-\lambda)\lambda^{k+1}) = \lambda^{k+1}.$$

The claim follows, which completes the solution.

Comment. Given a fixed k, let $f(k)$ denote the minimum value of n for which Ben can guarantee a victory. The problem asks for a proof that for large k

$$1.99^k \le f(k) \le 2^k.$$

A computer search shows that $f(k) = 2, 3, 4, 7, 11, 17$ for $k = 1, 2, 3, 4, 5, 6$.

28

C7. There are given 2^{500} points on a circle labeled $1, 2, \ldots, 2^{500}$ in some order. Prove that one can choose 100 pairwise disjoint chords joining some of these points so that the 100 sums of the pairs of numbers at the endpoints of the chosen chords are equal.

Solution. The proof is based on the following general fact.

Lemma. In a graph G each vertex v has degree d_v. Then G contains an independent set S of vertices such that $|S| \geq f(G)$ where

$$f(G) = \sum_{v \in G} \frac{1}{d_v + 1}.$$

Proof. Induction on $n = |G|$. The base $n = 1$ is clear. For the inductive step choose a vertex v_0 in G of minimum degree d. Delete v_0 and all of its neighbors $v_1, \ldots, v_d$ and also all edges with endpoints $v_0, v_1, \ldots, v_d$. This gives a new graph G'. By the inductive assumption G' contains an independent set S' of vertices such that $|S'| \geq f(G')$. Since no vertex in S' is a neighbor of v_0 in G, the set $S = S' \cup \{v_0\}$ is independent in G.

Let d'_v be the degree of a vertex v in G'. Clearly $d'_v \leq d_v$ for every such vertex v, and also $d_{v_i} \geq d$ for all $i = 0, 1, \ldots, d$ by the minimal choice of v_0. Therefore

$$f(G') = \sum_{v \in G'} \frac{1}{d'_v + 1} \geq \sum_{v \in G'} \frac{1}{d_v + 1} = f(G) - \sum_{i=0}^{d} \frac{1}{d_{v_i} + 1} \geq f(G) - \frac{d+1}{d+1} = f(G) - 1.$$

Hence $|S| = |S'| + 1 \geq f(G') + 1 \geq f(G)$, and the induction is complete. □

We pass on to our problem. For clarity denote $n = 2^{499}$ and draw all chords determined by the given $2n$ points. Color each chord with one of the colors $3, 4, \ldots, 4n - 1$ according to the sum of the numbers at its endpoints. Chords with a common endpoint have different colors. For each color c consider the following graph G_c. Its vertices are the chords of color c, and two chords are neighbors in G_c if they intersect. Let $f(G_c)$ have the same meaning as in the lemma for all graphs G_c.

Every chord ℓ divides the circle into two arcs, and one of them contains $m(\ell) \leq n - 1$ given points. (In particular $m(\ell) = 0$ if ℓ joins two consecutive points.) For each $i = 0, 1, \ldots, n - 2$ there are $2n$ chords ℓ with $m(\ell) = i$. Such a chord has degree at most i in the respective graph. Indeed let $A_1, \ldots, A_i$ be all points on either arc determined by a chord ℓ with $m(\ell) = i$ and color c. Every A_j is an endpoint of at most 1 chord colored c, $j = 1, \ldots, i$. Hence at most i chords of color c intersect ℓ.

It follows that for each $i = 0, 1, \ldots, n - 2$ the $2n$ chords ℓ with $m(\ell) = i$ contribute at least $\frac{2n}{i+1}$ to the sum $\sum_c f(G_c)$. Summation over $i = 0, 1, \ldots, n - 2$ gives

$$\sum_c f(G_c) \geq 2n \sum_{i=1}^{n-1} \frac{1}{i}.$$

Because there are $4n - 3$ colors in all, averaging yields a color c such that

$$f(G_c) \geq \frac{2n}{4n-3} \sum_{i=1}^{n-1} \frac{1}{i} > \frac{1}{2} \sum_{i=1}^{n-1} \frac{1}{i}.$$

By the lemma there are at least $\frac{1}{2} \sum_{i=1}^{n-1} \frac{1}{i}$ pairwise disjoint chords of color c, i. e. with the same sum c of the pairs of numbers at their endpoints. It remains to show that $\frac{1}{2} \sum_{i=1}^{n-1} \frac{1}{i} \geq 100$ for $n = 2^{499}$. Indeed we have

$$\sum_{i=1}^{n-1} \frac{1}{i} > \sum_{i=1}^{2^{400}} \frac{1}{i} = 1 + \sum_{k=1}^{400} \sum_{i=2^{k-1}+1}^{2^k} \frac{1}{i} > 1 + \sum_{k=1}^{400} \frac{2^{k-1}}{2^k} = 201 > 200.$$

This completes the solution.

Geometry

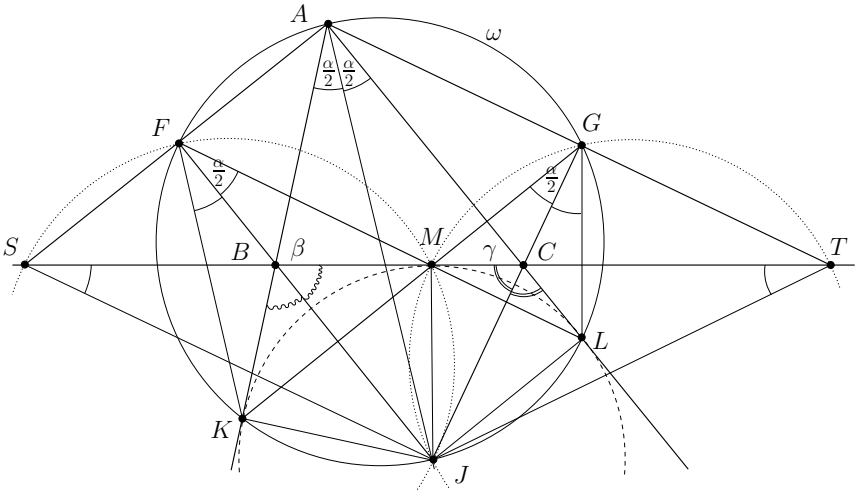
G1. In the triangle ABC the point J is the center of the excircle opposite to A. This excircle is tangent to the side BC at M, and to the lines AB and AC at K and L respectively. The lines LM and BJ meet at F, and the lines KM and CJ meet at G. Let S be the point of intersection of the lines AF and BC, and let T be the point of intersection of the lines AG and BC. Prove that M is the midpoint of ST.

Solution. Let $\alpha = \angle CAB$, $\beta = \angle ABC$ and $\gamma = \angle BCA$. The line AJ is the bisector of $\angle CAB$, so $\angle JAK = \angle JAL = \frac{\alpha}{2}$. By $\angle AKJ = \angle ALJ = 90^\circ$ the points K and L lie on the circle ω with diameter AJ.

The triangle KBM is isosceles as BK and BM are tangents to the excircle. Since BJ is the bisector of $\angle KBM$, we have $\angle MBJ = 90^\circ - \frac{\beta}{2}$ and $\angle BMK = \frac{\beta}{2}$. Likewise $\angle MCJ = 90^\circ - \frac{\gamma}{2}$ and $\angle CML = \frac{\gamma}{2}$. Also $\angle BMF = \angle CML$, therefore

$$\angle LFJ = \angle MBJ - \angle BMF = \left(90^\circ - \frac{\beta}{2}\right) - \frac{\gamma}{2} = \frac{\alpha}{2} = \angle LAJ.$$

Hence F lies on the circle ω. (By the angle computation, F and A are on the same side of BC.) Analogously, G also lies on ω. Since AJ is a diameter of ω, we obtain $\angle AFJ = \angle AGJ = 90^\circ$.



The lines AB and BC are symmetric with respect to the external bisector BF. Because $AF \perp BF$ and $KM \perp BF$, the segments SM and AK are symmetric with respect to BF, hence $SM = AK$. By symmetry $TM = AL$. Since AK and AL are equal as tangents to the excircle, it follows that $SM = TM$, and the proof is complete.

Comment. After discovering the circle $AFKJLG$, there are many other ways to complete the solution. For instance, from the cyclic quadrilaterals $JMFS$ and $JMGT$ one can find $\angle TSJ = \angle STJ = \frac{\alpha}{2}$. Another possibility is to use the fact that the lines AS and GM are parallel (both are perpendicular to the external angle bisector BJ), so $\frac{MS}{MT} = \frac{AG}{GT} = 1$.

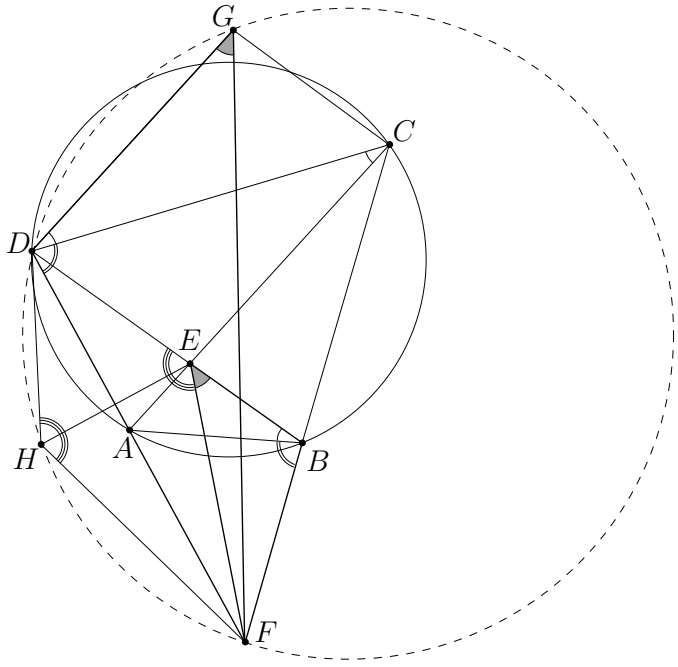
30

G2. Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E. The extensions of the sides AD and BC beyond A and B meet at F. Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD. Prove that D, H, F, G are concyclic.

Solution. We show first that the triangles FDG and FBE are similar. Since $ABCD$ is cyclic, the triangles EAB and EDC are similar, as well as FAB and FCD. The parallelogram $ECGD$ yields $GD = EC$ and $\angle CDG = \angle DCE$; also $\angle DCE = \angle DCA = \angle DBA$ by inscribed angles. Therefore

$$\angle FDG = \angle FDC + \angle CDG = \angle FBA + \angle ABD = \angle FBE,$$
$$\frac{GD}{EB} = \frac{CE}{EB} = \frac{CD}{AB} = \frac{FD}{FB}.$$

It follows that FDG and FBE are similar, and so $\angle FGD = \angle FEB$.



Since H is the reflection of E with respect to FD, we conclude that

$$\angle FHD = \angle FED = 180^\circ - \angle FEB = 180^\circ - \angle FGD.$$

This proves that D, H, F, G are concyclic.

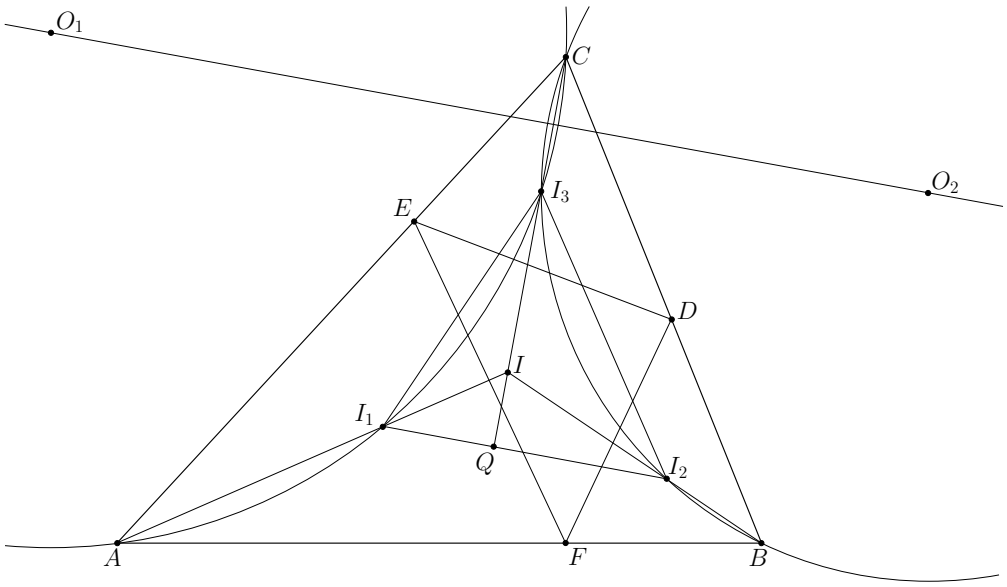
Comment. Points E and G are always in the half-plane determined by the line FD that contains B and C, but H is always in the other half-plane. In particular, $DHFG$ is cyclic if and only if $\angle FHD + \angle FGD = 180^\circ$.

G3. In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

Solution. Let $\angle CAB = \alpha$, $\angle ABC = \beta$, $\angle BCA = \gamma$. We start by showing that A, B, I_1 and I_2 are concyclic. Since AI_1 and BI_2 bisect $\angle CAB$ and $\angle ABC$, their extensions beyond I_1 and I_2 meet at the incenter I of the triangle. The points E and F are on the circle with diameter BC, so $\angle AEF = \angle ABC$ and $\angle AFE = \angle ACB$. Hence the triangles AEF and ABC are similar with ratio of similitude $\frac{AE}{AB} = \cos\alpha$. Because I_1 and I are their incenters, we obtain $I_1A = IA\cos\alpha$ and $II_1 = IA - I_1A = 2IA\sin^2\frac{\alpha}{2}$. By symmetry $II_2 = 2IB\sin^2\frac{\beta}{2}$. The law of sines in the triangle ABI gives $IA\sin\frac{\alpha}{2} = IB\sin\frac{\beta}{2}$. Hence

$$II_1 \cdot IA = 2\left(IA\sin\tfrac{\alpha}{2}\right)^2 = 2\left(IB\sin\tfrac{\beta}{2}\right)^2 = II_2 \cdot IB.$$

Therefore A, B, I_1 and I_2 are concyclic, as claimed.



In addition $II_1 \cdot IA = II_2 \cdot IB$ implies that I has the same power with respect to the circles (ACI_1), (BCI_2) and (ABI_1I_2). Then CI is the radical axis of (ACI_1) and (BCI_2); in particular CI is perpendicular to the line of centers O_1O_2.

Now it suffices to prove that $CI \perp I_1I_2$. Let CI meet I_1I_2 at Q, then it is enough to check that $\angle II_1Q + \angle I_1IQ = 90^\circ$. Since $\angle I_1IQ$ is external for the triangle ACI, we have

$$\angle II_1Q + \angle I_1IQ = \angle II_1Q + (\angle ACI + \angle CAI) = \angle II_1I_2 + \angle ACI + \angle CAI.$$

It remains to note that $\angle II_1I_2 = \frac{\beta}{2}$ from the cyclic quadrilateral ABI_1I_2, and $\angle ACI = \frac{\gamma}{2}$, $\angle CAI = \frac{\alpha}{2}$. Therefore $\angle II_1Q + \angle I_1IQ = \frac{\alpha}{2} + \frac{\beta}{2} + \frac{\gamma}{2} = 90^\circ$, completing the proof.

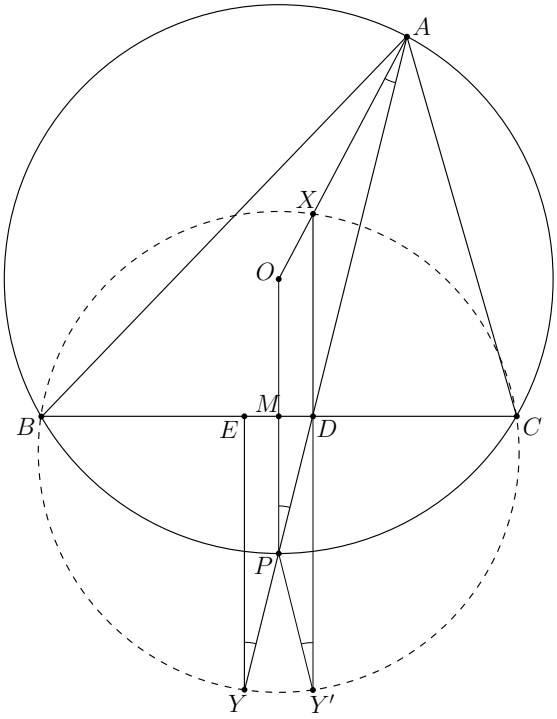
Comment. It follows from the first part of the solution that the common point $I_3 \neq C$ of the circles (ACI_1) and (BCI_2) is the incenter of the triangle CDE.

32

G4. Let ABC be a triangle with $AB \neq AC$ and circumcenter O. The bisector of $\angle BAC$ intersects BC at D. Let E be the reflection of D with respect to the midpoint of BC. The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.

Solution. The bisector of $\angle BAC$ and the perpendicular bisector of BC meet at P, the midpoint of the minor arc $\widehat{BC}$ (they are different lines as $AB \neq AC$). In particular OP is perpendicular to BC and intersects it at M, the midpoint of BC.

Denote by Y' the reflexion of Y with respect to OP. Since $\angle BYC = \angle BY'C$, it suffices to prove that $BXCY'$ is cyclic.



We have

$$\angle XAP = \angle OPA = \angle EYP.$$

The first equality holds because $OA = OP$, and the second one because EY and OP are both perpendicular to BC and hence parallel. But $\{Y, Y'\}$ and $\{E, D\}$ are pairs of symmetric points with respect to OP, it follows that $\angle EYP = \angle DY'P$ and hence

$$\angle XAP = \angle DY'P = \angle XY'P.$$

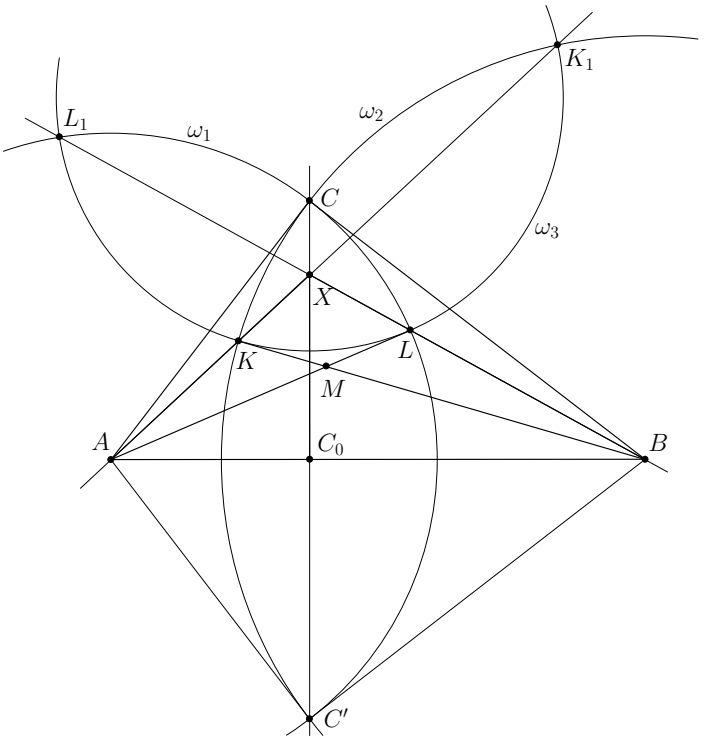
The last equation implies that $XAY'P$ is cyclic. By the powers of D with respect to the circles $(XAY'P)$ and $(ABPC)$ we obtain

$$XD \cdot DY' = AD \cdot DP = BD \cdot DC.$$

It follows that $BXCY'$ is cyclic, as desired.

G5. Let ABC be a triangle with $\angle BCA = 90°$, and let C_0 be the foot of the altitude from C. Choose a point X in the interior of the segment CC_0, and let K, L be the points on the segments AX, BX for which $BK = BC$ and $AL = AC$ respectively. Denote by M the intersection of AL and BK. Show that $MK = ML$.

Solution. Let C' be the reflection of C in the line AB, and let ω_1 and ω_2 be the circles with centers A and B, passing through L and K respectively. Since $AC' = AC = AL$ and $BC' = BC = BK$, both ω_1 and ω_2 pass through C and C'. By $\angle BCA = 90°$, AC is tangent to ω_2 at C, and BC is tangent to ω_1 at C. Let $K_1 \neq K$ be the second intersection of AX and ω_2, and let $L_1 \neq L$ be the second intersection of BX and ω_1.



By the powers of X with respect to ω_2 and ω_1,

$$XK \cdot XK_1 = XC \cdot XC' = XL \cdot XL_1,$$

so the points K_1, L, K, L_1 lie on a circle ω_3.

The power of A with respect to ω_2 gives

$$AL^2 = AC^2 = AK \cdot AK_1,$$

indicating that AL is tangent to ω_3 at L. Analogously, BK is tangent to ω_3 at K. Hence MK and ML are the two tangents from M to ω_3 and therefore $MK = ML$.

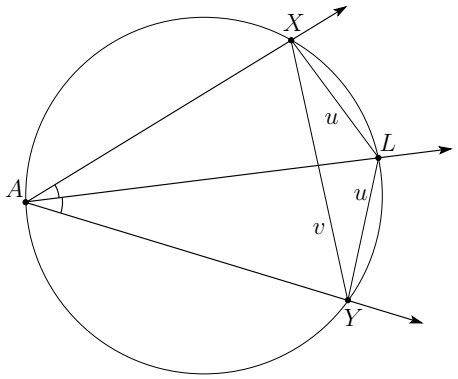
G6. Let ABC be a triangle with circumcenter O and incenter I. The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

Solution. By MIQUEL's theorem the circles $(AEF) = \omega_A$, $(BFD) = \omega_B$ and $(CDE) = \omega_C$ have a common point, for arbitrary points D, E and F on BC, CA and AB. So ω_A passes through the common point $P \neq D$ of ω_B and ω_C.

Let ω_A, ω_B and ω_C meet the bisectors AI, BI and CI at $A \neq A'$, $B \neq B'$ and $C \neq C'$ respectively. The key observation is that A', B' and C' do not depend on the particular choice of D, E and F, provided that $BD + BF = CA$, $CD + CE = AB$ and $AE + AF = BC$ hold true (the last equality follows from the other two). For a proof we need the following fact.

Lemma. Given is an angle with vertex A and measure α. A circle ω through A intersects the angle bisector at L and sides of the angle at X and Y. Then $AX + AY = 2AL\cos\frac{\alpha}{2}$.

Proof. Note that L is the midpoint of arc $\widehat{XLY}$ in ω and set $XL = YL = u$, $XY = v$. By PTOLEMY's theorem $AX \cdot YL + AY \cdot XL = AL \cdot XY$, which rewrites as $(AX + AY)u = AL \cdot v$. Since $\angle LXY = \frac{\alpha}{2}$ and $\angle XLY = 180^\circ - \alpha$, we have $v = 2\cos\frac{\alpha}{2}u$ by the law of sines, and the claim follows. □



Apply the lemma to $\angle BAC = \alpha$ and the circle $\omega = \omega_A$, which intersects AI at A'. This gives $2AA'\cos\frac{\alpha}{2} = AE + AF = BC$; by symmetry analogous relations hold for BB' and CC'. It follows that A', B' and C' are independent of the choice of D, E and F, as stated.

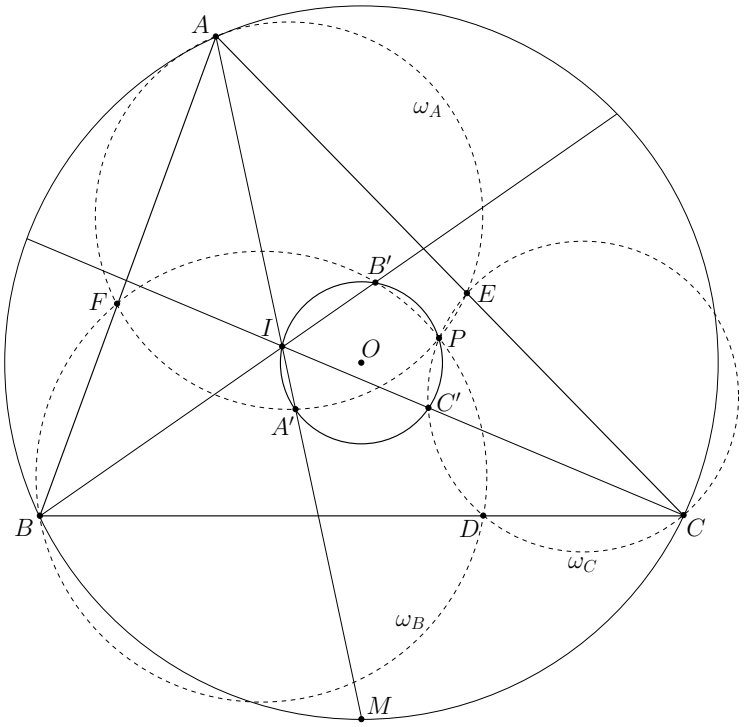
We use the lemma two more times with $\angle BAC = \alpha$. Let ω be the circle with diameter AI. Then X and Y are the tangency points of the incircle of ABC with AB and AC, and hence $AX = AY = \frac{1}{2}(AB + AC - BC)$. So the lemma yields $2AI\cos\frac{\alpha}{2} = AB + AC - BC$. Next, if ω is the circumcircle of ABC and AI intersects ω at $M \neq A$ then $\{X, Y\} = \{B, C\}$, and so $2AM\cos\frac{\alpha}{2} = AB + AC$ by the lemma. To summarize,

$$2AA'\cos\tfrac{\alpha}{2} = BC, \qquad 2AI\cos\tfrac{\alpha}{2} = AB + AC - BC, \qquad 2AM\cos\tfrac{\alpha}{2} = AB + AC. \qquad (*)$$

These equalities imply $AA' + AI = AM$, hence the segments AM and IA' have a common midpoint. It follows that I and A' are equidistant from the circumcenter O. By symmetry $OI = OA' = OB' = OC'$, so I, A', B', C' are on a circle centered at O.

To prove $OP = OI$, now it suffices to show that I, A', B', C' and P are concyclic. Clearly one can assume $P \neq I, A', B', C'$.

We use oriented angles to avoid heavy case distinction. The oriented angle between the lines l and m is denoted by $\angle(l, m)$. We have $\angle(l, m) = -\angle(m, l)$ and $\angle(l, m) + \angle(m, n) = \angle(l, n)$ for arbitrary lines l, m and n. Four distinct non-collinear points U, V, X, Y are concyclic if and only if $\angle(UX, VX) = \angle(UY, VY)$.



Suppose for the moment that A', B', P, I are distinct and noncollinear; then it is enough to check the equality $\angle(A'P, B'P) = \angle(A'I, B'I)$. Because A, F, P, A' are on the circle ω_A, we have $\angle(A'P, FP) = \angle(A'A, FA) = \angle(A'I, AB)$. Likewise $\angle(B'P, FP) = \angle(B'I, AB)$. Therefore

$$\angle(A'P, B'P) = \angle(A'P, FP) + \angle(FP, B'P) = \angle(A'I, AB) - \angle(B'I, AB) = \angle(A'I, B'I).$$

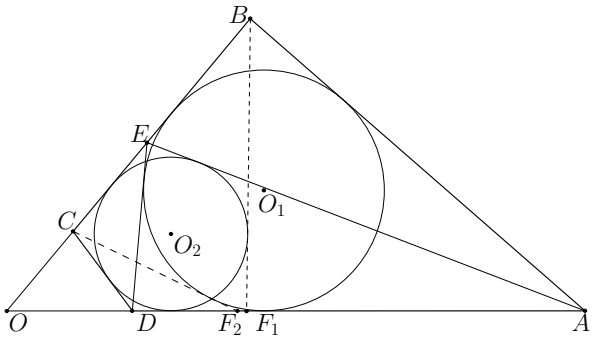
Here we assumed that $P \neq F$. If $P = F$ then $P \neq D, E$ and the conclusion follows similarly (use $\angle(A'F, B'F) = \angle(A'F, EF) + \angle(EF, DF) + \angle(DF, B'F)$ and inscribed angles in $\omega_A, \omega_B, \omega_C$).

There is no loss of generality in assuming A', B', P, I distinct and noncollinear. If ABC is an equilateral triangle then the equalities (*) imply that A', B', C', I, O and P coincide, so $OP = OI$. Otherwise at most one of A', B', C' coincides with I. If say $C' = I$ then $OI \perp CI$ by the previous reasoning. It follows that $A', B' \neq I$ and hence $A' \neq B'$. Finally A', B' and I are noncollinear because I, A', B', C' are concyclic.

Comment. The proposer remarks that the locus γ of the points P is an arc of the circle $(A'B'C'I)$. The reflection I' of I in O belongs to γ; it is obtained by choosing D, E and F to be the tangency points of the three excircles with their respective sides. The rest of the circle $(A'B'C'I)$, except I, can be included in γ by letting D, E and F vary on the extensions of the sides and assuming signed lengths. For instance if B is between C and D then the length BD must be taken with a negative sign. The incenter I corresponds to the limit case where D tends to infinity.

G7. Let $ABCD$ be a convex quadrilateral with non-parallel sides BC and AD. Assume that there is a point E on the side BC such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point F on the side AD such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if AB is parallel to CD.

Solution. Let ω_1 and ω_2 be the incircles and O_1 and O_2 the incenters of the quadrilaterals $ABED$ and $AECD$ respectively. A point F with the stated property exists only if ω_1 and ω_2 are also the incircles of the quadrilaterals $ABCF$ and $BCDF$.



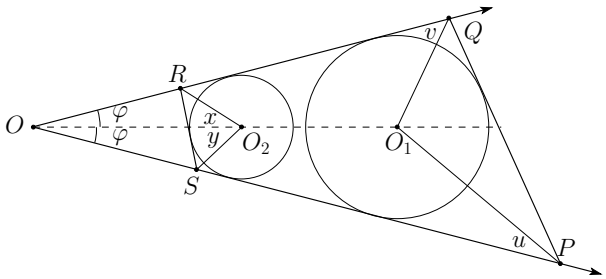
Let the tangents from B to ω_2 and from C to ω_1 (other than BC) meet AD at F_1 and F_2 respectively. We need to prove that $F_1 = F_2$ if and only if $AB \parallel CD$.

Lemma. The circles ω_1 and ω_2 with centers O_1 and O_2 are inscribed in an angle with vertex O. The points P, S on one side of the angle and Q, R on the other side are such that ω_1 is the incircle of the triangle PQO, and ω_2 is the excircle of the triangle RSO opposite to O. Denote $p = OO_1 \cdot OO_2$. Then exactly one of the following relations holds:

$$OP \cdot OR < p < OQ \cdot OS, \qquad OP \cdot OR > p > OQ \cdot OS, \qquad OP \cdot OR = p = OQ \cdot OS.$$

Proof. Denote $\angle OPO_1 = u$, $\angle OQO_1 = v$, $\angle OO_2R = x$, $\angle OO_2S = y$, $\angle POQ = 2\varphi$. Because PO_1, QO_1, RO_2, SO_2 are internal or external bisectors in the triangles PQO and RSO, we have

$$u + v = x + y \ (= 90^\circ - \varphi). \tag{1}$$



By the law of sines

$$\frac{OP}{OO_1} = \frac{\sin(u+\varphi)}{\sin u} \quad \text{and} \quad \frac{OO_2}{OR} = \frac{\sin(x+\varphi)}{\sin x}.$$

Therefore, since x, u and φ are acute,

$$OP \cdot OR \geq p \Leftrightarrow \frac{OP}{OO_1} \geq \frac{OO_2}{OR} \Leftrightarrow \sin x \sin(u+\varphi) \geq \sin u \sin(x+\varphi) \Leftrightarrow \sin(x-u) \geq 0 \Leftrightarrow x \geq u.$$

Thus $OP \cdot OR \geq p$ is equivalent to $x \geq u$, with $OP \cdot OR = p$ if and only if $x = u$.

Analogously, $p \geq OQ \cdot OS$ is equivalent to $v \geq y$, with $p = OQ \cdot OS$ if and only if $v = y$. On the other hand $x \geq u$ and $v \geq y$ are equivalent by (1), with $x = u$ if and only if $v = y$. The conclusion of the lemma follows from here. $\square$

Going back to the problem, apply the lemma to the quadruples $\{B, E, D, F_1\}$, $\{A, B, C, D\}$ and $\{A, E, C, F_2\}$. Assuming $OE \cdot OF_1 > p$, we obtain

$$OE \cdot OF_1 > p \;\Rightarrow\; OB \cdot OD < p \;\Rightarrow\; OA \cdot OC > p \;\Rightarrow\; OE \cdot OF_2 < p.$$

In other words, $OE \cdot OF_1 > p$ implies

$$OB \cdot OD < p < OA \cdot OC \quad \text{and} \quad OE \cdot OF_1 > p > OE \cdot OF_2.$$

Similarly, $OE \cdot OF_1 < p$ implies

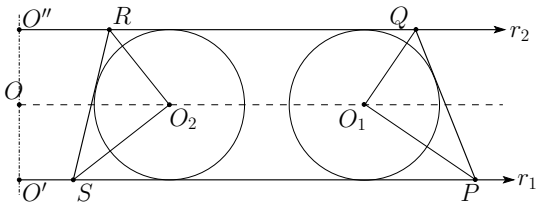
$$OB \cdot OD > p > OA \cdot OC \quad \text{and} \quad OE \cdot OF_1 < p < OE \cdot OF_2.$$

In these cases $F_1 \neq F_2$ and $OB \cdot OD \neq OA \cdot OC$, so the lines AB and CD are not parallel.

There remains the case $OE \cdot OF_1 = p$. Here the lemma leads to $OB \cdot OD = p = OA \cdot OC$ and $OE \cdot OF_1 = p = OE \cdot OF_2$. Therefore $F_1 = F_2$ and $AB \parallel CD$.

Comment. The conclusion is also true if BC and AD are parallel. One can prove a limit case of the lemma for the configuration shown in the figure below, where r_1 and r_2 are parallel rays starting at O' and O'', with $O'O'' \perp r_1, r_2$ and O the midpoint of $O'O''$. Two circles with centers O_1 and O_2 are inscribed in the strip between r_1 and r_2. The lines PQ and RS are tangent to the circles, with P, S on r_1, and Q, R on r_2, so that O, O_1 are on the same side of PQ and O, O_2 are on different sides of RS. Denote $s = OO_1 + OO_2$. Then exactly one of the following relations holds:

$$O'P + O''R < s < O''Q + O'S, \quad O'P + O''R > s > O''Q + O'S, \quad O'P + O''R = s = O''Q + O'S.$$

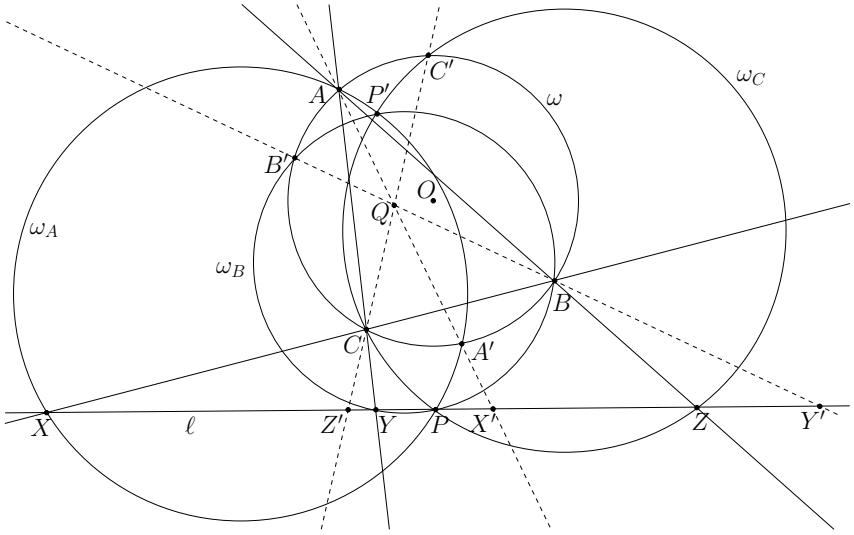


Once this is established, the proof of the original statement for $BC \parallel AD$ is analogous to the one in the intersecting case. One replaces products by sums of relevant segments.

G8. Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω. Denote by P the foot of the perpendicular from the center of ω to ℓ. The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P. Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P.

Solution 1. Let $\omega_A, \omega_B, \omega_C$ and ω be the circumcircles of triangles AXP, BYP, CZP and ABC respectively. The strategy of the proof is to construct a point Q with the same power with respect to the four circles. Then each of P and Q has the same power with respect to $\omega_A, \omega_B, \omega_C$ and hence the three circles are coaxial. In other words they have another common point P' or the three of them are tangent at P.

We first give a description of the point Q. Let $A' \neq A$ be the second intersection of ω and ω_A; define B' and C' analogously. We claim that AA', BB' and CC' have a common point. Once this claim is established, the point just constructed will be on the radical axes of the three pairs of circles $\{\omega, \omega_A\}, \{\omega, \omega_B\}, \{\omega, \omega_C\}$. Hence it will have the same power with respect to $\omega, \omega_A, \omega_B, \omega_C$.



We proceed to prove that AA', BB' and CC' intersect at one point. Let r be the circumradius of triangle ABC. Define the points X', Y', Z' as the intersections of AA', BB', CC' with ℓ. Observe that X', Y', Z' do exist. If AA' is parallel to ℓ then ω_A is tangent to ℓ; hence $X = P$ which is a contradiction. Similarly, BB' and CC' are not parallel to ℓ.

From the powers of the point X' with respect to the circles ω_A and ω we get

$$X'P \cdot (X'P + PX) = X'P \cdot X'X = X'A' \cdot X'A = X'O^2 - r^2,$$

hence

$$X'P \cdot PX = X'O^2 - r^2 - X'P^2 = OP^2 - r^2.$$

We argue analogously for the points Y' and Z', obtaining

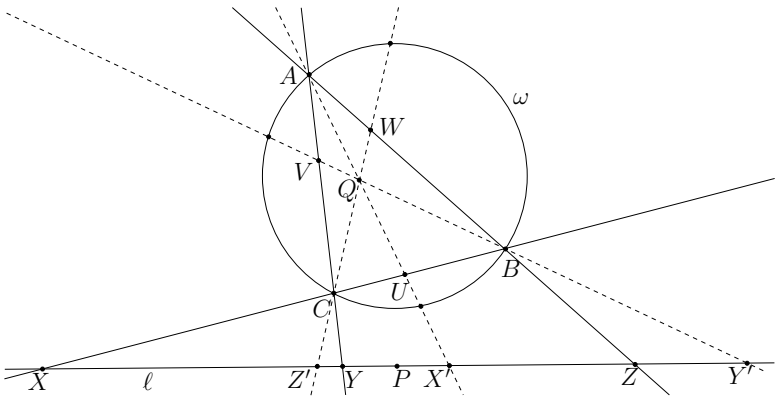
$$X'P \cdot PX = Y'P \cdot PY = Z'P \cdot PZ = OP^2 - r^2 = k^2. \tag{1}$$

In these computations all segments are regarded as directed segments. We keep the same convention for the sequel.

We prove that the lines AA', BB', CC' intersect at one point by Ceva's theorem. To avoid distracting remarks we interpret everything projectively, i. e. whenever two lines are parallel they meet at a point on the line at infinity.

Let U, V, W be the intersections of AA', BB', CC' with BC, CA, AB respectively. The idea is that although it is difficult to calculate the ratio $\frac{BU}{CU}$, it is easier to deal with the cross-ratio $\frac{BU}{CU}/\frac{BX}{CX}$ because we can send it to the line ℓ. With this in mind we apply Menelaus' theorem to the triangle ABC and obtain $\frac{BX}{CX} \cdot \frac{CY}{AY} \cdot \frac{AZ}{BZ} = 1$. Hence Ceva's ratio can be expressed as

$$\frac{BU}{CU} \cdot \frac{CV}{AV} \cdot \frac{AW}{BW} = \frac{BU}{CU}/\frac{BX}{CX} \cdot \frac{CV}{AV}/\frac{CY}{AY} \cdot \frac{AW}{BW}/\frac{AZ}{BZ}.$$



Project the line BC to ℓ from A. The cross-ratio between BC and UX equals the cross-ratio between ZY and $X'X$. Repeating the same argument with the lines CA and AB gives

$$\frac{BU}{CU} \cdot \frac{CV}{AV} \cdot \frac{AW}{BW} = \frac{ZX'}{YX'}/\frac{ZX}{YX} \cdot \frac{XY'}{ZY'}/\frac{XY}{ZY} \cdot \frac{YZ'}{XZ'}/\frac{YZ}{XZ}$$

and hence

$$\frac{BU}{CU} \cdot \frac{CV}{AV} \cdot \frac{AW}{BW} = (-1) \cdot \frac{ZX'}{YX'} \cdot \frac{XY'}{ZY'} \cdot \frac{YZ'}{XZ'}.$$

The equations (1) reduce the problem to a straightforward computation on the line ℓ. For instance, the transformation $t \mapsto -k^2/t$ preserves cross-ratio and interchanges the points X, Y, Z with the points X', Y', Z'. Then

$$\frac{BU}{CU} \cdot \frac{CV}{AV} \cdot \frac{AW}{BW} = (-1) \cdot \frac{ZX'}{YX'}/\frac{ZZ'}{YZ'} \cdot \frac{XY'}{ZY'}/\frac{XZ'}{ZZ'} = -1.$$

We proved that Ceva's ratio equals -1, so AA', BB', CC' intersect at one point Q.

Comment 1. There is a nice projective argument to prove that AX', BY', CZ' intersect at one point. Suppose that ℓ and ω intersect at a pair of complex conjugate points D and E. Consider a projective transformation that takes D and E to $[i; 1, 0]$ and $[-i, 1, 0]$. Then ℓ is the line at infinity, and ω is a conic through the special points $[i; 1, 0]$ and $[-i, 1, 0]$, hence it is a circle. So one can assume that AX, BY, CZ are parallel to BC, CA, AB. The involution on ℓ taking X, Y, Z to X', Y', Z' and leaving D, E fixed is the involution changing each direction to its perpendicular one. Hence AX, BY, CZ are also perpendicular to AX', BY', CZ'.

It follows from the above that AX', BY', CZ' intersect at the orthocenter of triangle ABC.

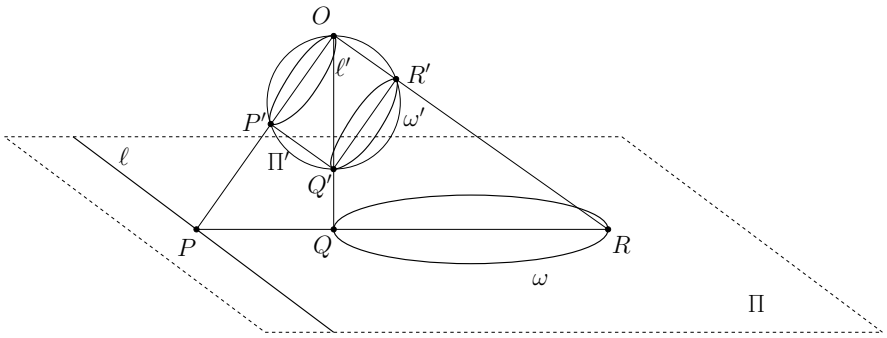
Comment 2. The restriction that the line ℓ does not intersect the circumcricle ω is unnecessary. The proof above works in general. In case ℓ intersects ω at D and E point P is the midpoint of DE, and some equations can be interpreted differently. For instance

$$X'P \cdot X'X = X'A' \cdot X'A = X'D \cdot X'E,$$

and hence the pairs $X'X$ and DE are harmonic conjugates. This means that X', Y', Z' are the harmonic conjugates of X, Y, Z with respect to the segment DE.

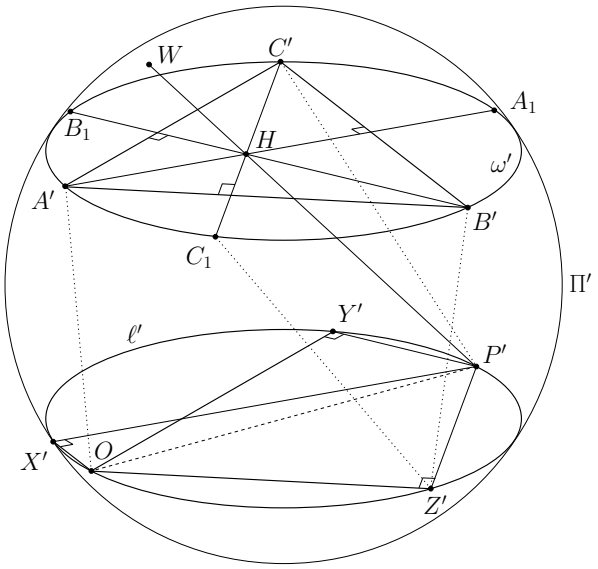
Solution 2. First we prove that there is an inversion in space that takes ℓ and ω to parallel circles on a sphere. Let QR be the diameter of ω whose extension beyond Q passes through P. Let Π be the plane carrying our objects. In space, choose a point O such that the line QO is perpendicular to Π and $\angle POR = 90°$, and apply an inversion with pole O (the radius of the inversion does not matter). For any object $\mathcal{T}$ denote by $\mathcal{T}'$ the image of $\mathcal{T}$ under this inversion.

The inversion takes the plane Π to a sphere Π'. The lines in Π are taken to circles through O, and the circles in Π also are taken to circles on Π'.



Since the line ℓ and the circle ω are perpendicular to the plane OPQ, the circles ℓ' and ω' also are perpendicular to this plane. Hence, the planes of the circles ℓ' and ω' are parallel.

Now consider the circles $A'X'P'$, $B'Y'P'$ and $C'Z'P'$. We want to prove that either they have a common point (on Π'), different from P', or they are tangent to each other.



The point X' is the second intersection of the circles $B'C'O$ and ℓ', other than O. Hence, the lines OX' and $B'C'$ are coplanar. Moreover, they lie in the parallel planes of ℓ' and ω'. Therefore, OX' and $B'C'$ are parallel. Analogously, OY' and OZ' are parallel to $A'C'$ and $A'B'$.

Let A_1 be the second intersection of the circles $A'X'P'$ and ω', other than A'. The segments $A'A_1$ and $P'X'$ are coplanar, and therefore parallel. Now we know that $B'C'$ and $A'A_1$ are parallel to OX' and $X'P'$ respectively, but these two segments are perpendicular because OP' is a diameter in ℓ'. We found that $A'A_1$ and $B'C'$ are perpendicular, hence $A'A_1$ is the altitude in the triangle $A'B'C'$, starting from A.

Analogously, let B_1 and C_1 be the second intersections of ω' with the circles $B'P'Y'$ and $C'P'Z'$, other than B' and C' respectively. Then $B'B_1$ and $C'C_1$ are the other two altitudes in the triangle $A'B'C'$.

Let H be the orthocenter of the triangle $A'B'C'$. Let W be the second intersection of the line $P'H$ with the sphere Π', other than P'. The point W lies on the sphere Π', in the plane of the circle $A'P'X'$, so W lies on the circle $A'P'X'$. Similarly, W lies on the circles $B'P'Y'$ and $C'P'Z'$ as well; indeed W is the second common point of the three circles.

If the line $P'H$ is tangent to the sphere then W coincides with P', and $P'H$ is the common tangent of the three circles.

Number Theory

N1. Call admissible a set A of integers that has the following property:

$$\text{If } x, y \in A \text{ (possibly } x = y) \text{ then } x^2 + kxy + y^2 \in A \text{ for every integer } k.$$

Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

Solution. A pair of integers m, n fulfills the condition if and only if $gcd(m, n) = 1$. Suppose that $gcd(m, n) = d > 1$. The set

$$A = \{\ldots, -2d, -d, 0, d, 2d, \ldots\}$$

is admissible, because if d divides x and y then it divides $x^2 + kxy + y^2$ for every integer k. Also $m, n \in A$ and $A \neq \mathbb{Z}$.

Now let $gcd(m, n) = 1$, and let A be an admissible set containing m and n. We use the following observations to prove that $A = \mathbb{Z}$:

(i) $kx^2 \in A$ for every $x \in A$ and every integer k.

(ii) $(x + y)^2 \in A$ for all $x, y \in A$.

To justify (i) let $y = x$ in the definition of an admissible set; to justify (ii) let $k = 2$.

Since $gcd(m, n) = 1$, we also have $gcd(m^2, n^2) = 1$. Hence one can find integers a, b such that $am^2 + bn^2 = 1$. It follows from (i) that $am^2 \in A$ and $bn^2 \in A$. Now we deduce from (ii) that $1 = (am^2 + bn^2)^2 \in A$. But if $1 \in A$ then (i) implies $k \in A$ for every integer k.

N2. Find all triples (x, y, z) of positive integers such that $x \le y \le z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

Solution. First note that x divides $2012 \cdot 2 = 2^3 \cdot 503$. If $503 \mid x$ then the right-hand side of the equation is divisible by 503^3, and it follows that $503^2 \mid xyz + 2$. This is false as $503 \mid x$. Hence $x = 2^m$ with $m \in \{0, 1, 2, 3\}$. If $m \ge 2$ then $2^6 \mid 2012(xyz + 2)$. However the highest powers of 2 dividing 2012 and $xyz + 2 = 2^m yz + 2$ are 2^2 and 2^1 respectively. So $x = 1$ or $x = 2$, yielding the two equations

$$y^3 + z^3 = 2012(yz + 2), \qquad \text{and} \qquad y^3 + z^3 = 503(yz + 1).$$

In both cases the prime $503 = 3 \cdot 167 + 2$ divides $y^3 + z^3$. We claim that $503 \mid y + z$. This is clear if $503 \mid y$, so let $503 \nmid y$ and $503 \nmid z$. Then $y^{502} \equiv z^{502} \pmod{503}$ by Fermat's little theorem. On the other hand $y^3 \equiv -z^3 \pmod{503}$ implies $y^{3 \cdot 167} \equiv -z^{3 \cdot 167} \pmod{503}$, i. e. $y^{501} \equiv -z^{501} \pmod{503}$. It follows that $y \equiv -z \pmod{503}$ as claimed.

Therefore $y + z = 503k$ with $k \ge 1$. In view of $y^3 + z^3 = (y + z)\big((y - z)^2 + yz\big)$ the two equations take the form

$$k(y - z)^2 + (k - 4)yz = 8, \tag{1}$$
$$k(y - z)^2 + (k - 1)yz = 1. \tag{2}$$

In (1) we have $(k - 4)yz \le 8$, which implies $k \le 4$. Indeed if $k > 4$ then $1 \le (k - 4)yz \le 8$, so that $y \le 8$ and $z \le 8$. This is impossible as $y + z = 503k \ge 503$. Note next that $y^3 + z^3$ is even in the first equation. Hence $y + z = 503k$ is even too, meaning that k is even. Thus $k = 2$ or $k = 4$. Clearly (1) has no integer solutions for $k = 4$. If $k = 2$ then (1) takes the form $(y + z)^2 - 5yz = 4$. Since $y + z = 503k = 503 \cdot 2$, this leads to $5yz = 503^2 \cdot 2^2 - 4$. However $503^2 \cdot 2^2 - 4$ is not a multiple of 5. Therefore (1) has no integer solutions.

Equation (2) implies $0 \le (k - 1)yz \le 1$, so that $k = 1$ or $k = 2$. Also $0 \le k(y - z)^2 \le 1$, hence $k = 2$ only if $y = z$. However then $y = z = 1$, which is false in view of $y + z \ge 503$. Therefore $k = 1$ and (2) takes the form $(y - z)^2 = 1$, yielding $z - y = |y - z| = 1$. Combined with $k = 1$ and $y + z = 503k$, this leads to $y = 251$, $z = 252$.

In summary the triple $(2, 251, 252)$ is the only solution.

N3. Determine all integers $m \geq 2$ such that every n with $\frac{m}{3} \leq n \leq \frac{m}{2}$ divides the binomial coefficient $\binom{n}{m-2n}$.

Solution. The integers in question are all prime numbers.

First we check that all primes satisfy the condition, and even a stronger one. Namely, if p is a prime then every n with $1 \leq n \leq \frac{p}{2}$ divides $\binom{n}{p-2n}$. This is true for $p = 2$ where $n = 1$ is the only possibility. For an odd prime p take $n \in [1, \frac{p}{2}]$ and consider the following identity of binomial coefficients:

$$(p-2n) \cdot \binom{n}{p-2n} = n \cdot \binom{n-1}{p-2n-1}.$$

Since $p \geq 2n$ and p is odd, all factors are non-zero. If $d = gcd(p-2n, n)$ then d divides p, but $d \leq n < p$ and hence $d = 1$. It follows that $p - 2n$ and n are relatively prime, and so the factor n in the right-hand side divides the binomial coefficient $\binom{n}{p-2n}$.

Next we show that no composite number m has the stated property. Consider two cases.

- If $m = 2k$ with $k > 1$, pick $n = k$. Then $\frac{m}{3} \leq n \leq \frac{m}{2}$ but $\binom{n}{m-2n} = \binom{k}{0} = 1$ is not divisible by $k > 1$.
- If m is odd then there exist an odd prime p and an integer $k \geq 1$ with $m = p(2k+1)$. Pick $n = pk$, then $\frac{m}{3} \leq n \leq \frac{m}{2}$ by $k \geq 1$. However

$$\frac{1}{n}\binom{n}{m-2n} = \frac{1}{pk}\binom{pk}{p} = \frac{(pk-1)(pk-2)\cdots(pk-(p-1))}{p!}$$

is not an integer, because p divides the denominator but not the numerator.

N4. An integer a is called friendly if the equation $(m^2+n)(n^2+m) = a(m-n)^3$ has a solution over the positive integers.

a) Prove that there are at least 500 friendly integers in the set $\{1, 2, \ldots, 2012\}$.

b) Decide whether $a = 2$ is friendly.

Solution. a) Every a of the form $a = 4k-3$ with $k \geq 2$ is friendly. Indeed the numbers $m = 2k-1 > 0$ and $n = k-1 > 0$ satisfy the given equation with $a = 4k-3$:

$$(m^2+n)(n^2+m) = \big((2k-1)^2+(k-1)\big)\big((k-1)^2+(2k-1)\big) = (4k-3)k^3 = a(m-n)^3.$$

Hence $5, 9, \ldots, 2009$ are friendly and so $\{1, 2, \ldots, 2012\}$ contains at least 502 friendly numbers.

b) We show that $a = 2$ is not friendly. Consider the equation with $a = 2$ and rewrite its left-hand side as a difference of squares:

$$\frac{1}{4}\left((m^2+n+n^2+m)^2 - (m^2+n-n^2-m)^2\right) = 2(m-n)^3.$$

Since $m^2+n-n^2-m = (m-n)(m+n-1)$, we can further reformulate the equation as

$$(m^2+n+n^2+m)^2 = (m-n)^2\left(8(m-n)+(m+n-1)^2\right).$$

It follows that $8(m-n)+(m+n-1)^2$ is a perfect square. Clearly $m > n$, hence there is an integer $s \geq 1$ such that

$$(m+n-1+2s)^2 = 8(m-n)+(m+n-1)^2.$$

Subtracting the squares gives $s(m+n-1+s) = 2(m-n)$. Since $m+n-1+s > m-n$, we conclude that $s < 2$. Therefore the only possibility is $s = 1$ and $m = 3n$. However then the left-hand side of the given equation (with $a = 2$) is greater than $m^3 = 27n^3$, whereas its right-hand side equals $16n^3$. The contradiction proves that $a = 2$ is not friendly.

Comment. A computer search shows that there are 561 friendly numbers in $\{1, 2, \ldots, 2012\}$.

N5. For a nonnegative integer n define $rad(n) = 1$ if $n = 0$ or $n = 1$, and $rad(n) = p_1 p_2 \cdots p_k$ where $p_1 < p_2 < \cdots < p_k$ are all prime factors of n. Find all polynomials $f(x)$ with nonnegative integer coefficients such that $rad(f(n))$ divides $rad(f(n^{rad(n)}))$ for every nonnegative integer n.

Solution 1. We are going to prove that $f(x) = ax^m$ for some nonnegative integers a and m. If $f(x)$ is the zero polynomial we are done, so assume that $f(x)$ has at least one positive coefficient. In particular $f(1) > 0$.

Let p be a prime number. The condition is that $f(n) \equiv 0 \pmod p$ implies

$$f(n^{rad(n)}) \equiv 0 \pmod p. \tag{1}$$

Since $rad(n^{rad(n)^k}) = rad(n)$ for all k, repeated applications of the preceding implication show that if p divides $f(n)$ then

$$f(n^{rad(n)^k}) \equiv 0 \pmod p \quad \text{for all } k.$$

The idea is to construct a prime p and a positive integer n such that $p-1$ divides n and p divides $f(n)$. In this case, for k large enough $p-1$ divides $rad(n)^k$. Hence if $(p, n) = 1$ then $n^{rad(n)^k} \equiv 1 \pmod p$ by Fermat's little theorem, so that

$$f(1) \equiv f(n^{rad(n)^k}) \equiv 0 \pmod p. \tag{2}$$

Suppose that $f(x) = g(x)x^m$ with $g(0) \neq 0$. Let t be a positive integer, p any prime factor of $g(-t)$ and $n = (p-1)t$. So $p-1$ divides n and $f(n) = f((p-1)t) \equiv f(-t) \equiv 0 \pmod p$, hence either $(p, n) > 1$ or (2) holds. If $(p, (p-1)t) > 1$ then p divides t and $g(0) \equiv g(-t) \equiv 0 \pmod p$, meaning that p divides $g(0)$.

In conclusion we proved that each prime factor of $g(-t)$ divides $g(0)f(1) \neq 0$, and thus the set of prime factors of $g(-t)$ when t ranges through the positive integers is finite. This is known to imply that $g(x)$ is a constant polynomial, and so $f(x) = ax^m$.

Solution 2. Let $f(x)$ be a polynomial with integer coefficients (not necessarily nonnegative) such that $rad(f(n))$ divides $rad(f(n^{rad(n)}))$ for any nonnegative integer n. We give a complete description of all polynomials with this property. More precisely, we claim that if $f(x)$ is such a polynomial and ξ is a root of $f(x)$ then so is ξ^d for every positive integer d.

Therefore each root of $f(x)$ is zero or a root of unity. In particular, if a root of unity ξ is a root of $f(x)$ then $1 = \xi^d$ is a root too (for some positive integer d). In the original problem $f(x)$ has nonnegative coefficients. Then either $f(x)$ is the zero polynomial or $f(1) > 0$ and $\xi = 0$ is the only possible root. In either case $f(x) = ax^m$ with a and m nonnegative integers.

To prove the claim let ξ be a root of $f(x)$, and let $g(x)$ be an irreducible factor of $f(x)$ such that $g(\xi) = 0$. If 0 or 1 are roots of $g(x)$ then either $\xi = 0$ or $\xi = 1$ (because $g(x)$ is irreducible) and we are done. So assume that $g(0), g(1) \neq 0$. By decomposing d as a product of prime numbers, it is enough to consider the case $d = p$ prime. We argue for $p = 2$. Since $rad(2^k) = 2$ for every k, we have

$$rad(f(2^k)) \mid rad(f(2^{2k})).$$

Now we prove that $g(x)$ divides $f(x^2)$. Suppose that this is not the case. Then, since $g(x)$ is irreducible, there are integer-coefficient polynomials $a(x)$, $b(x)$ and an integer N such that

$$a(x)g(x) + b(x)f(x^2) = N. \tag{3}$$

Each prime factor p of $g(2^k)$ divides $f(2^k)$, so by $rad(f(2^k))|rad(f(2^{2k}))$ it also divides $f(2^{2k})$. From the equation above with $x = 2^k$ it follows that p divides N.

In summary, each prime divisor of $g(2^k)$ divides N, for all $k \geq 0$. Let $p_1, \ldots, p_n$ be the odd primes dividing N, and suppose that

$$g(1) = 2^{\alpha} p_1^{\alpha_1} \cdots p_n^{\alpha_n}.$$

If k is divisible by $\varphi(p_1^{\alpha_1+1} \cdots p_n^{\alpha_n+1})$ then

$$2^k \equiv 1 \pmod{p_1^{\alpha_1+1} \cdots p_n^{\alpha_n+1}},$$

yielding

$$g(2^k) \equiv g(1) \pmod{p_1^{\alpha_1+1} \cdots p_n^{\alpha_n+1}}.$$

It follows that for each i the maximal power of p_i dividing $g(2^k)$ and $g(1)$ is the same, namely $p_i^{\alpha_i}$. On the other hand, for large enough k, the maximal power of 2 dividing $g(2^k)$ and $g(0) \neq 0$ is the same. From the above, for k divisible by $\varphi(p_1^{\alpha_1+1} \cdots p_n^{\alpha_n+1})$ and large enough, we obtain that $g(2^k)$ divides $g(0) \cdot g(1)$. This is impossible because $g(0), g(1) \neq 0$ are fixed and $g(2^k)$ is arbitrarily large.

In conclusion, $g(x)$ divides $f(x^2)$. Recall that ξ is a root of $f(x)$ such that $g(\xi) = 0$; then $f(\xi^2) = 0$, i. e. ξ^2 is a root of $f(x)$.

Likewise if ξ is a root of $f(x)$ and p an arbitrary prime then ξ^p is a root too. The argument is completely analogous, in the proof above just replace 2 by p and "odd prime" by "prime different from p."

Comment. The claim in the second solution can be proved by varying $n \pmod p$ in (1). For instance, we obtain

$$f(n^{rad(n+pk)}) \equiv 0 \pmod p$$

for every positive integer k. One can prove that if $(n, p) = 1$ then $rad(n + pk)$ runs through all residue classes $r \pmod{p-1}$ with $(r, p-1)$ squarefree. Hence if $f(n) \equiv 0 \pmod p$ then $f(n^r) \equiv 0 \pmod p$ for all integers r. This implies the claim by an argument leading to the identity (3).

N6. Let x and y be positive integers. If $x^{2^n}-1$ is divisible by $2^n y+1$ for every positive integer n, prove that $x=1$.

Solution. First we prove the following fact: For every positive integer y there exist infinitely many primes $p\equiv 3 \pmod 4$ such that p divides some number of the form $2^n y+1$.

Clearly it is enough to consider the case y odd. Let

$$2y+1=p_1^{e_1}\cdots p_r^{e_r}$$

be the prime factorization of $2y+1$. Suppose on the contrary that there are finitely many primes $p_{r+1},\ldots,p_{r+s}\equiv 3 \pmod 4$ that divide some number of the form $2^n y+1$ but do not divide $2y+1$.

We want to find an n such that $p_i^{e_i}||2^n y+1$ for $1\le i\le r$ and $p_i\nmid 2^n y+1$ for $r+1\le i\le r+s$. For this it suffices to take

$$n=1+\varphi(p_1^{e_1+1}\cdots p_r^{e_r+1}p_{r+1}^1\cdots p_{r+s}^1),$$

because then

$$2^n y+1\equiv 2y+1 \pmod{p_1^{e_1+1}\cdots p_r^{e_r+1}p_{r+1}^1\cdots p_{r+s}^1}.$$

The last congruence means that $p_1^{e_1},\ldots,p_r^{e_r}$ divide exactly $2^n y+1$ and no prime $p_{r+1},\ldots,p_{r+s}$ divides $2^n y+1$. It follows that the prime factorization of $2^n y+1$ consists of the prime powers $p_1^{e_1},\ldots,p_r^{e_r}$ and powers of primes $\equiv 1 \pmod 4$. Because y is odd, we obtain

$$2^n y+1\equiv p_1^{e_1}\cdots p_r^{e_r}\equiv 2y+1\equiv 3 \pmod 4.$$

This is a contradiction since $n>1$, and so $2^n y+1\equiv 1 \pmod 4$.

Now we proceed to the problem. If p is a prime divisor of $2^n y+1$ the problem statement implies that $x^d\equiv 1 \pmod p$ for $d=2^n$. By Fermat's little theorem the same congruence holds for $d=p-1$, so it must also hold for $d=(2^n,p-1)$. For $p\equiv 3 \pmod 4$ we have $(2^n,p-1)=2$, therefore in this case $x^2\equiv 1 \pmod p$.

In summary, we proved that every prime $p\equiv 3 \pmod 4$ that divides some number of the form $2^n y+1$ also divides x^2-1. This is possible only if $x=1$, otherwise by the above x^2-1 would be a positive integer with infinitely many prime factors.

Comment. For each x and each odd prime p the maximal power of p dividing $x^{2^n}-1$ for some n is bounded and hence the same must be true for the numbers $2^n y+1$. We infer that p^2 divides $2^{p-1}-1$ for each prime divisor p of $2^n y+1$. However trying to reach a contradiction with this conclusion alone seems hopeless, since it is not even known if there are infinitely many primes p *without* this property.

N7. Find all $n \in \mathbb{N}$ for which there exist nonnegative integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

Solution. Such numbers $a_1, a_2, \ldots, a_n$ exist if and only if $n \equiv 1 \pmod 4$ or $n \equiv 2 \pmod 4$.

Let $\sum_{k=1}^{n} \frac{k}{3^{a_k}} = 1$ with $a_1, a_2, \ldots, a_n$ nonnegative integers. Then $1 \cdot x_1 + 2 \cdot x_2 + \cdots + n \cdot x_n = 3^a$ with $x_1, \ldots, x_n$ powers of 3 and $a \geq 0$. The right-hand side is odd, and the left-hand side has the same parity as $1 + 2 + \cdots + n$. Hence the latter sum is odd, which implies $n \equiv 1, 2 \pmod 4$. Now we prove the converse.

Call *feasible* a sequence $b_1, b_2, \ldots, b_n$ if there are nonnegative integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{b_1}{3^{a_1}} + \frac{b_2}{3^{a_2}} + \cdots + \frac{b_n}{3^{a_n}} = 1.$$

Let b_k be a term of a feasible sequence $b_1, b_2, \ldots, b_n$ with exponents $a_1, a_2, \ldots, a_n$ like above, and let u, v be nonnegative integers with sum $3b_k$. Observe that

$$\frac{1}{2^{a_k+1}} + \frac{1}{2^{a_k+1}} = \frac{1}{2^{a_k}} \qquad \text{and} \qquad \frac{u}{3^{a_k+1}} + \frac{v}{3^{a_k+1}} = \frac{b_k}{3^{a_k}}.$$

It follows that the sequence $b_1, \ldots, b_{k-1}, u, v, b_{k+1}, \ldots, b_n$ is feasible. The exponents a_i are the same for the unchanged terms b_i, $i \neq k$; the new terms u, v have exponents $a_k + 1$.

We state the conclusion in reverse. If two terms u, v of a sequence are replaced by one term $\frac{u+v}{3}$ and the obtained sequence is feasible, then the original sequence is feasible too. Denote by α_n the sequence $1, 2, \ldots, n$. To show that α_n is feasible for $n \equiv 1, 2 \pmod 4$, we transform it by $n - 1$ replacements $\{u, v\} \mapsto \frac{u+v}{3}$ to the one-term sequence α_1. The latter is feasible, with $a_1 = 0$. Note that if m and $2m$ are terms of a sequence then $\{m, 2m\} \mapsto m$, so $2m$ can be ignored if necessary.

Let $n \geq 16$. We prove that α_n can be reduced to α_{n-12} by 12 operations. Write $n = 12k + r$ where $k \geq 1$ and $0 \leq r \leq 11$. If $0 \leq r \leq 5$ then the last 12 terms of α_n can be partitioned into 2 singletons $\{12k - 6\}$, $\{12k\}$ and the following 5 pairs:

$$\{12k - 6 - i, 12k - 6 + i\}, i = 1, \ldots, 5 - r; \qquad \{12k - j, 12k + j\}, j = 1, \ldots, r.$$

(There is only one kind of pairs if $r \in \{0, 5\}$.) One can ignore $12k - 6$ and $12k$ since α_n contains $6k - 3$ and $6k$. Furthermore the 5 operations $\{12k - 6 - i, 12k - 6 + i\} \mapsto 8k - 4$ and $\{12k - j, 12k + j\} \mapsto 8k$ remove the 10 terms in the pairs and bring in 5 new terms equal to $8k - 4$ or $8k$. All of these can be ignored too as $4k - 2$ and $4k$ are still present in the sequence. Indeed $4k \leq n - 12$ is equivalent to $8k \geq 12 - r$, which is true for $r \in \{4, 5\}$. And if $r \in \{0, 1, 2, 3\}$ then $n \geq 16$ implies $k \geq 2$, so $8k \geq 12 - r$ also holds. Thus α_n reduces to α_{n-12}.

The case $6 \leq r \leq 11$ is analogous. Consider the singletons $\{12k\}$, $\{12k + 6\}$ and the 5 pairs

$$\{12k - i, 12k + i\}, i = 1, \ldots, 11 - r; \qquad \{12k + 6 - j, 12k + 6 + j\}, j = 1, \ldots, r - 6.$$

Ignore the singletons like before, then remove the pairs via operations $\{12k - i, 12k + i\} \mapsto 8k$ and $\{12k + 6 - j, 12k + 6 + j\} \mapsto 8k + 4$. The 5 newly-appeared terms $8k$ and $8k + 4$ can be ignored too since $4k + 2 \leq n - 12$ (this follows from $k \geq 1$ and $r \geq 6$). We obtain α_{n-12} again.

The problem reduces to $2 \leq n \leq 15$. In fact $n \in \{2, 5, 6, 9, 10, 13, 14\}$ by $n \equiv 1, 2 \pmod 4$. The cases $n = 2, 6, 10, 14$ reduce to $n = 1, 5, 9, 13$ respectively because the last even term of α_n can be ignored. For $n = 5$ apply $\{4, 5\} \mapsto 3$, then $\{3, 3\} \mapsto 2$, then ignore the 2 occurrences of 2. For $n = 9$ ignore 6 first, then apply $\{5, 7\} \mapsto 4$, $\{4, 8\} \mapsto 4$, $\{3, 9\} \mapsto 4$. Now ignore the 3 occurrences of 4, then ignore 2. Finally $n = 13$ reduces to $n = 10$ by $\{11, 13\} \mapsto 8$ and ignoring 8 and 12. The proof is complete.

N8. Prove that for every prime $p > 100$ and every integer r there exist two integers a and b such that p divides $a^2 + b^5 - r$.

Solution 1. Throughout the solution, all congruence relations are meant modulo p.

Fix p, and let $\mathcal{P} = \{0, 1, \ldots, p-1\}$ be the set of residue classes modulo p. For every $r \in \mathcal{P}$, let $S_r = \big\{(a, b) \in \mathcal{P} \times \mathcal{P} \colon a^2 + b^5 \equiv r\big\}$, and let $s_r = |S_r|$. Our aim is to prove $s_r > 0$ for all $r \in \mathcal{P}$.

We will use the well-known fact that for every residue class $r \in \mathcal{P}$ and every positive integer k, there are at most k values $x \in \mathcal{P}$ such that $x^k \equiv r$.

Lemma. Let N be the number of quadruples $(a, b, c, d) \in \mathcal{P}^4$ for which $a^2 + b^5 \equiv c^2 + d^5$. Then

$$N = \sum_{r \in \mathcal{P}} s_r^2 \tag{a}$$

and

$$N \le p(p^2 + 4p - 4). \tag{b}$$

Proof. (a) For each residue class r there exist exactly s_r pairs (a, b) with $a^2 + b^5 \equiv r$ and s_r pairs (c, d) with $c^2 + d^5 \equiv r$. So there are s_r^2 quadruples with $a^2 + b^5 \equiv c^2 + d^5 \equiv r$. Taking the sum over all $r \in \mathcal{P}$, the statement follows.

(b) Choose an arbitrary pair $(b, d) \in \mathcal{P}$ and look for the possible values of a, c.

1. Suppose that $b^5 \equiv d^5$, and let k be the number of such pairs (b, d). The value b can be chosen in p different ways. For $b \equiv 0$ only $d = 0$ has this property; for the nonzero values of b there are at most 5 possible values for d. So we have $k \le 1 + 5(p-1) = 5p - 4$.

The values a and c must satisfy $a^2 \equiv c^2$, so $a \equiv \pm c$, and there are exactly $2p - 1$ such pairs (a, c).

2. Now suppose $b^5 \not\equiv d^5$. In this case a and c must be distinct. By $(a - c)(a + c) = d^5 - b^5$, the value of $a - c$ uniquely determines $a + c$ and thus a and c as well. Hence, there are $p - 1$ suitable pairs (a, c).

Thus, for each of the k pairs (b, d) with $b^5 \equiv d^5$ there are $2p - 1$ pairs (a, c), and for each of the other $p^2 - k$ pairs (b, d) there are $p - 1$ pairs (a, c). Hence,

$$N = k(2p-1) + (p^2 - k)(p-1) = p^2(p-1) + kp \le p^2(p-1) + (5p-4)p = p(p^2 + 4p - 4). \quad \square$$

To prove the statement of the problem, suppose that $S_r = \emptyset$ for some $r \in \mathcal{P}$; obviously $r \not\equiv 0$. Let $T = \big\{x^{10} \colon x \in \mathcal{P} \setminus \{0\}\big\}$ be the set of nonzero 10th powers modulo p. Since each residue class is the 10th power of at most 10 elements in $\mathcal{P}$, we have $|T| \ge \frac{p-1}{10} \ge 4$ by $p > 100$.

For every $t \in T$, we have $S_{tr} = \emptyset$. Indeed, if $(x, y) \in S_{tr}$ and $t \equiv z^{10}$ then

$$(z^{-5}x)^2 + (z^{-2}y)^5 \equiv t^{-1}(x^2 + y^5) \equiv r,$$

so $(z^{-5}x, z^{-2}y) \in S_r$. So, there are at least $\frac{p-1}{10} \ge 4$ empty sets among $S_1, \ldots, S_{p-1}$, and there are at most $p - 4$ nonzero values among $s_0, s_2, \ldots, s_{p-1}$. Then by the AM-QM inequality we obtain

$$N = \sum_{r \in \mathcal{P} \setminus rT} s_r^2 \ge \frac{1}{p-4} \left(\sum_{r \in \mathcal{P} \setminus rT} s_r \right)^2 = \frac{|\mathcal{P} \times \mathcal{P}|^2}{p-4} = \frac{p^4}{p-4} > p(p^2 + 4p - 4),$$

which is impossible by the lemma.

Solution 2. If $5 \nmid p-1$, then all modulo p residue classes are complete fifth powers and the statement is trivial. So assume that $p = 10k+1$ where $k \geq 10$. Let g be a primitive root modulo p.

We will use the following facts:

(F1) If some residue class x is not quadratic then $x^{(p-1)/2} \equiv -1 \pmod{p}$.

(F2) For every integer d, as a simple corollary of the summation formula for geometric progressions,

$$\sum_{i=0}^{2k-1} g^{5di} \equiv \begin{cases} 2k & \text{if } 2k \mid d \\ 0 & \text{if } 2k \nmid d \end{cases} \pmod{p}.$$

Suppose that, contrary to the statement, some modulo p residue class r cannot be expressed as a^2+b^5. Of course $r \not\equiv 0 \pmod{p}$. By (F1) we have $(r-b^5)^{(p-1)/2} = (r-b^5)^{5k} \equiv -1 \pmod{p}$ for all residue classes b.

For $t = 1, 2 \ldots, k-1$ consider the sums

$$S(t) = \sum_{i=0}^{2k-1} \left(r - g^{5i}\right)^{5k} g^{5ti}.$$

By the indirect assumption and (F2),

$$S(t) = \sum_{i=0}^{2k-1} \left(r - (g^i)^5\right)^{5k} g^{5ti} \equiv \sum_{i=0}^{2k-1} (-1) g^{5ti} \equiv -\sum_{i=0}^{2k-1} g^{5ti} \equiv 0 \pmod{p}$$

because $2k$ cannot divide t.

On the other hand, by the binomial theorem,

$$S(t) = \sum_{i=0}^{2k-1} \left(\sum_{j=0}^{5k} \binom{5k}{j} r^{5k-j} \left(-g^{5i}\right)^j \right) g^{5ti} = \sum_{j=0}^{5k} (-1)^j \binom{5k}{j} r^{5k-j} \left(\sum_{i=0}^{2k-1} g^{5(j+t)i} \right) \equiv$$

$$\equiv \sum_{j=0}^{5k} (-1)^j \binom{5k}{j} r^{5k-j} \begin{cases} 2k & \text{if } 2k \mid j+t \\ 0 & \text{if } 2k \nmid j+t \end{cases} \pmod{p}.$$

Since $1 \leq j+t < 6k$, the number $2k$ divides $j+t$ only for $j = 2k-t$ and $j = 4k-t$. Hence,

$$0 \equiv S(t) \equiv (-1)^t \left(\binom{5k}{2k-t} r^{3k+t} + \binom{5k}{4k-t} r^{k+t} \right) \cdot 2k \pmod{p},$$

$$\binom{5k}{2k-t} r^{2k} + \binom{5k}{4k-t} \equiv 0 \pmod{p}.$$

Taking this for $t = 1, 2$ and eliminating r, we get

$$0 \equiv \binom{5k}{2k-2} \left(\binom{5k}{2k-1} r^{2k} + \binom{5k}{4k-1} \right) - \binom{5k}{2k-1} \left(\binom{5k}{2k-2} r^{2k} + \binom{5k}{4k-2} \right)$$

$$= \binom{5k}{2k-2} \binom{5k}{4k-1} - \binom{5k}{2k-1} \binom{5k}{4k-2}$$

$$= \frac{(5k)!^2}{(2k-1)!(3k+2)!(4k-1)!(k+2)!} \Big((2k-1)(k+2) - (3k+2)(4k-1) \Big)$$

$$= \frac{-(5k)!^2 \cdot 2k(5k+1)}{(2k-1)!(3k+2)!(4k-1)!(k+2)!} \pmod{p}.$$

But in the last expression none of the numbers is divisible by $p = 10k+1$, a contradiction.

Comment 1. The argument in the second solution is valid whenever $k \geq 3$, that is for all primes $p = 10k + 1$ except $p = 11$. This is an exceptional case when the statement is not true; $r = 7$ cannot be expressed as desired.

Comment 2. The statement is true in a more general setting: for every positive integer n, for all sufficiently large p, each residue class modulo p can be expressed as $a^2 + b^n$. Choosing $t = 3$ would allow using the Cauchy-Davenport theorem (together with some analysis on the case of equality).

In the literature more general results are known. For instance, the statement easily follows from the *Hasse-Weil bound*.